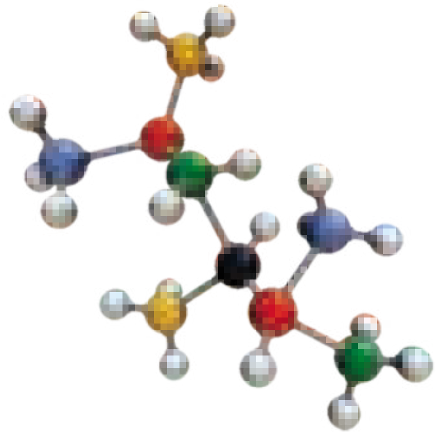
श्री BalaJi



Advanced Theory in

ORGANIC CHEMISTRY

for JEE &

All other Competitive Examinations

VOLUME-1

Basic to Advanced Level

The Beginning

by:

M.S. Chouhan

Director

Vibrant Academy, Kota

SHRI BALAJI PUBLICATIONS

(EDUCATIONAL PUBLISHERS & DISTRIBUTORS)

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CHAPTER

1

Introduction

WHAT IS ORGANIC CHEMISTRY

What is organic chemistry, and why should you study it (other then scoring good marks in exams)☺ The answers to these questions are all around you. Every living organism is made of organic chemicals. The proteins that make up your hair, skin, and muscles; the DNA that controls your genetic heritage; the foods that nourish you; and the medicines that heal you are all organic chemicals. Anyone with a curiosity about life and living things, and anyone who wants to be a part of the remarkable advances now occurring in medicine and the biological sciences, must first understand organic chemistry.

Organic chemistry, then, is the study of carbon compounds.

But why is carbon special? Why, of the more than 50 million presently known chemical compounds, do most of them contain carbon?

From the simple methane, with one carbon atom, to the staggeringly complex DNA, which can have more than 100 million carbons.

At the time of writing there were about 16.5 million organic compounds known. How many more are possible? There is no limit. Imagine you've just made the longest hydrocarbon ever made—you just have to add another carbon atom and you've made another.

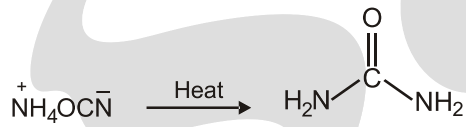
The Wohler synthesis is the conversion of ammonium cynate into urea. This chemical reaction was discovered in 1828 by Friedrich Wohler in an attempt to synthesize ammonium cyanate.

Ammonium cyanate decomposes to ammonia and cyanic acid which in turn react to produce urea in a nucleophilic addition followed by tautomeric isomerization :

The **Wohler synthesis** is of great historical significance because for the first time an organic compound was produced from inorganic reactants. This finding went against the mainstream theory of that time called vitalism which stated that organic matter possessed a special force or vital force inherent to all things living. For this reason a sharp boundary existed between organic and inorganic compounds. Urea was discovered in 1799 and could until then only be obtained from biological sources such as urine Wohler reported to his teacher Berzelius

"I cannot, so to say, hold my chemical water and must tell you that I can make urea without thereby needing to have kidneys, or anyhow, an animal, be it human or dog".

Little more than a decade later, the vitalistic theory suffered still further when Friedrich Wöhler discovered in 1828 that it was possible to convert the "inorganic" salt ammonium cyanate into the "organic" substance urea, which had previously been found in human urine.

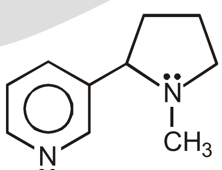


Ammonium cyanate Urea

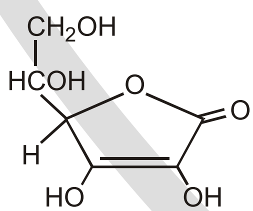
ORGANIC CHEMISTRY AND DRUGS

Four examples of organic compound in living organisms.

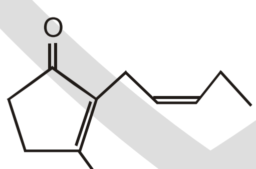
Nicotine



Tobacco contains nicotine, an addictive alkaloid.



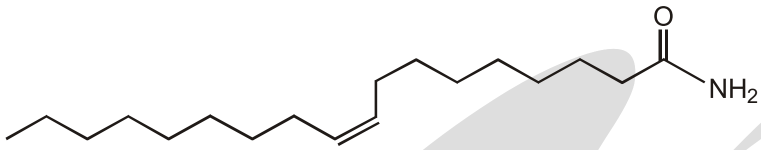
Vitamin C (ascorbic acid) Rose hips contain vitamin C, essential for preventing scurvy.



cis-jasmone

cis-jasmone an example of a perfume distilled from jasmine flowers.

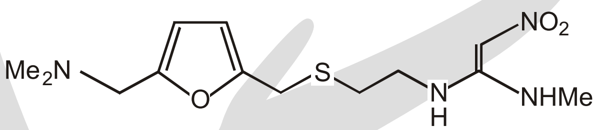
Again, let's not forget other creatures. Cats seem to be able to go to sleep at any time and recently a compound was isolated from the cerebrospinal fluid of cats that makes them, or rats, or humans go off to sleep quickly. It is a surprisingly simple compound.



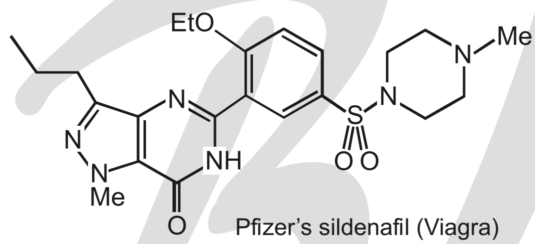
a sleep-inducing fatty acid derivative
cis-9,10-octadecenoamide

The pharmaceutical businesses produce drugs and medicinal products of many kinds. One of the great revolutions of modern life has been the expectation that humans will survive diseases because of a treatment designed to deal specifically with that disease. The most successful drug ever is ranitidine (Zantac), the Glaxo–Wellcome ulcer treatment, and one of the fastest-growing is Pfizer’s sildenafil (Viagra). ‘Success’ refers both to human health and to profit!

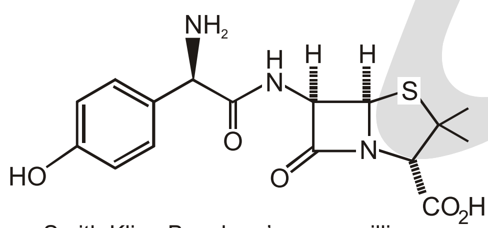
One of the most successful of these is Smith Kline Beecham’s amoxycillin. The four-membered ring at the heart of the molecule is the ‘ -lactam’.



Glaxo-Wellcome’s ranitidine
the most successful drug to date
world wide sales peaked >£1,000,000,000 per annum



Pfizer’s sildenafil (Viagra)
three million satisfied customers in 1998

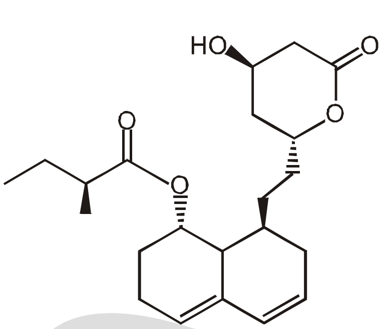


Smith Kline Beecham’s amoxycillin
β-lactam antibiotic
for treatment of bacterial infections

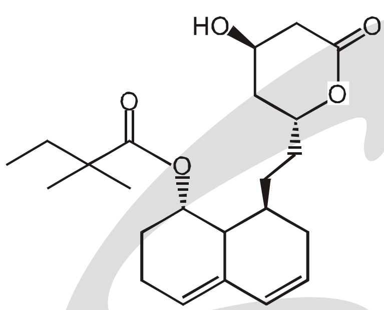
☑ SPECIAL TOPIC

HOW HIGH CHOLESTEROL IS TREATED CLINICALLY

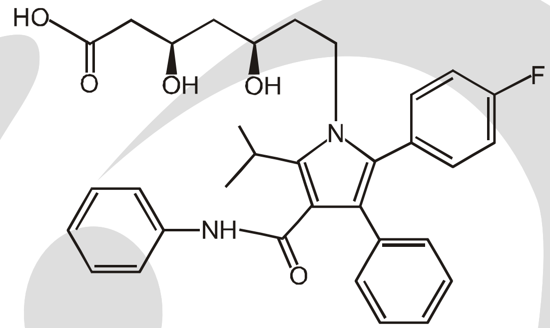
Statins are drugs that reduce serum cholesterol levels by inhibiting the enzyme that catalyzes the formation of a compound needed for the synthesis of cholesterol. As a consequence of diminished cholesterol synthesis in the liver, the liver forms more LDL receptors—the receptors that help clear LDL (the so-called “bad” cholesterol) from the bloodstream. Studies show that for every 10% that cholesterol is reduced, deaths from coronary heart disease are reduced by 15% and total death risk is reduced by 11%.☺



lovastatin
Mevacor



simvastatin
Zocor



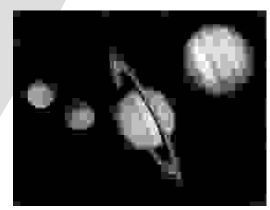
atorvastatin
Lipitor

Lovastatin and simvastatin are natural statins used clinically under the trade names Mevacor and Zocor. Atorvastatin (Lipitor), a synthetic statin, is the most popular statin. It has greater potency and lasts longer in the body than natural statins because the products of its breakdown are as active as the parent drug in reducing cholesterol levels. Therefore, smaller doses of the drug may be administered.

In addition, Lipitor is less polar than lovastatin and simvastatin, so it persists longer in liver cells, where it is needed. Lipitor has been one of the most widely prescribed drugs in the United States for the past several years.

☑ SPECIAL TOPIC

ORGANIC CHEMISTRY AND COLOURS



The blue colors of Uranus and Neptune are caused by the presence of methane, a colorless and odorless gas, in their atmospheres. Natural gas—called a fossil fuel because it is formed from the decomposition of plant and animal material in the Earth's crust—is approximately 75% methane. The methane in Uranus' upper atmosphere absorbs the red light from the Sun but reflects the blue light from the Sun back into space. This is why Uranus appears blue.

ORGANIC FOODS

Contrary to what you may hear in supermarkets or on television, all foods are organic—that is, complex mixtures of organic molecules. Even so, when applied to food, the word organic has come to mean an absence of synthetic chemicals, typically pesticides, antibiotics, and preservatives. How concerned should we be about traces of pesticides in the food we eat? Or toxins in the water we drink? Or pollutants in the air we breathe?☹

Life is not risk-free—we all take many risks each day without even thinking about it. We decide to ride a bike rather than drive, even though there is a ten times greater likelihood per mile of dying in a bicycling accident than in a car. Some of us decide to smoke cigarettes, even though it increases our chance of getting cancer by 50%. But what about risks from chemicals like pesticides?

One thing is certain: without pesticides, whether they target weeds (herbicides), insects (insecticides), or molds and fungi (fungicides), crop production would drop significantly, food prices would increase, and famines would occur in less developed parts of the world. Take the herbicide atrazine, for instance. In the United States alone, approximately 100 million pounds of atrazine are used each year to kill weeds in corn, sorghum, and sugarcane fields, greatly improving the yields of these crops. The results obtained in animal tests are then distilled into a single number called an LD_{50}, the amount of substance per kilogram body weight that is a lethal dose for 50% of the test animals. For atrazine, the LD_{50} value is between 1 and 4 g/kg depending on the animal species. Aspirin, for comparison, has an LD_{50} of 1.1 g/kg, and ethanol (ethyl alcohol) has an LD_{50} of 10.6 g/kg.

Table-1 : lists values for some other familiar substances. The lower the value, the more toxic the substance. Note, though, that LD_{50} values tell only about the effects of heavy exposure for a relatively short time.

Table - 1 : Some LD_{50} Values

Substance	LD_{50}(g/kg)	Substance	LD_{50}(g/kg)
Strychnine	0.005	Chloroform	1.2
Arsenic trioxide	0.015	Iron (II) sulfate	1.5
DDT	0.115	Ethyl alcohol	10.6
Aspirin	1.1	Sodium cyclamate	17

They say nothing about the risks of long-term exposure, such as whether the substance can cause cancer or interfere with development in the unborn.

So, should we still use atrazine? All decisions involve tradeoffs, and the answer is rarely obvious. Does the benefit of increased food production outweigh possible health risks of a pesticide? Do the beneficial effects of a new drug outweigh a potentially dangerous side effect in a small number of users? Different people will have different opinions, but an honest evaluation of facts is surely the best way to start. At present, atrazine is approved for continued use in the United States because the EPA believes that the benefits of increased food production outweigh possible health risks. At the same time, though, the use of atrazine is being phased out in Europe.

❑❑❑

CHAPTER

2

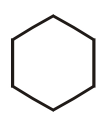
Representation of Organic Compounds

BOND-LINE DRAWINGS

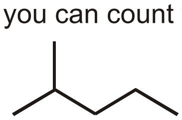
To do well in organic chemistry, you must first learn to interpret the drawings that organic chemists use. When you see a drawing of a molecule, it is absolutely critical that you can read all of the information contained in that drawing. Without this skill, it will be impossible to master even the most basic reactions and concepts.

HOW TO READ BOND-LINE DRAWINGS

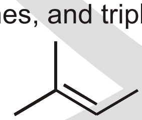
For example, the following compounds has 6 carbon atoms:



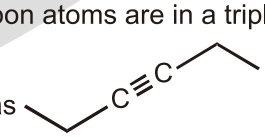
It is a common mistake to forget that the ends of lines represent carbon atoms as well. For example, the following molecule has six carbon atoms (Make sure you can count them)



Double bonds are shown with two lines, and triple bonds are shown with three lines :

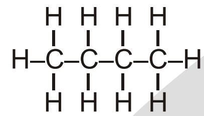
 

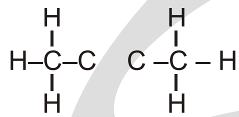
When drawings triple bonds, be sure to draw them in a straight line rather than zigzag, because triple bonds are linear (There will be more about this in the chapter on geometry). This can be quite confusing at first, because it can get hard to see just how many carbon atoms are in a triple bond, so let's make it clear:

 is the same as 

so this compound has 6 carbon atoms

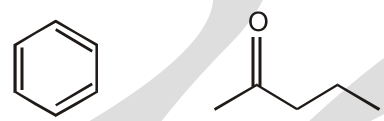
Don't let triple bonds confuse you. The two carbon atoms of the triple bond and the two carbons connected to them are drawn in a straight line. All other bonds are drawn as a zigzag:

 is drawn like this 

But  is drawn like this 

Solved Example

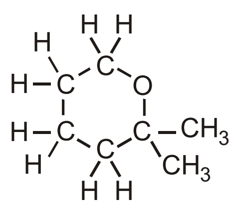
- Count the number of carbon atoms in each of the following drawings:



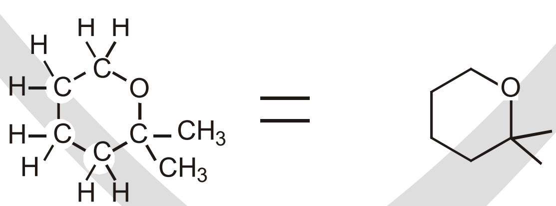
Ans. The first compound has six carbon atoms, and the second compound has five carbon atoms.

HOW TO DRAW BOND-LINE DRAWINGS

Now that we know how to read these drawings, we need to learn how to draw them. Take the following molecule as an example:

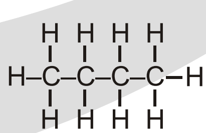
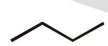


To draw this as a bond-line drawing, we focus on the carbon skeleton, making sure to draw any atoms other than C and H. All atoms other than carbon and hydrogen must be drawn. So the example above would look like this:

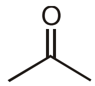


Points to Remember

1. Don't forget that carbon atoms in a straight chain are drawn in a zigzag format:

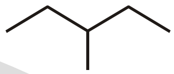
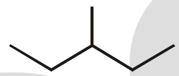
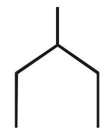
 is drawn like this 

2. When drawing double bonds, try to draw the other bonds as far away from the double bond as possible:

☺  is much better than  ☹

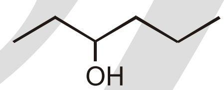
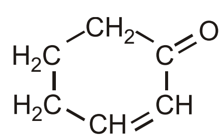
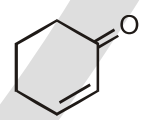
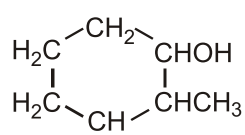
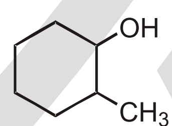
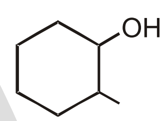
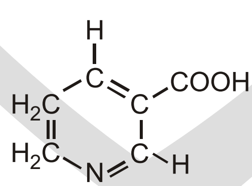
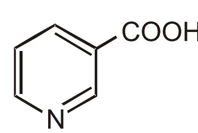
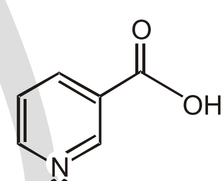
BAD

3. When drawing zigzags, it does not matter in which direction you start drawing:

 Is the same as  Is the same as 

LINE-ANGLE FORMULAS

Another kind of shorthand used for organic structures is the line-angle formula, sometimes called a skeletal structure or a stick figure. Line-angle formulas are often used for cyclic compounds and occasionally for noncyclic ones. In a stick figure, bonds are represented by lines, and carbon atoms are assumed to be present wherever two lines meet or a line begins or ends. Nitrogen, oxygen, and halogen atoms are shown, but hydrogen atoms are not usually drawn unless they are bonded to an atom that is drawn. Each carbon atom is assumed to have enough hydrogen atoms to give it a total of four bonds. Nonbonding electrons are rarely shown.\

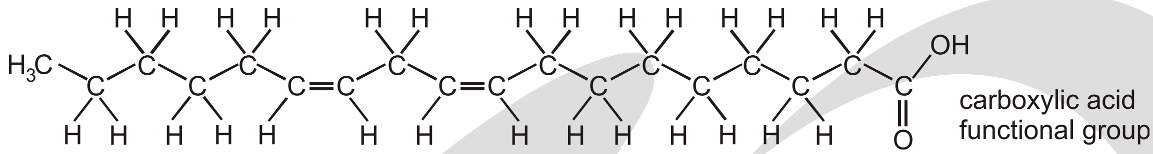
Compound	Condensed Structure	Line-angle Formula
hexane	$CH_3(CH_2)_4CH_3$	
hex-2-ene	$CH_3CH = CHCH_2CH_2CH_3$	
hexan-3-ol	$CH_3CH_2CH(OH)CH_2CH_2CH_3$	
cyclohex-2-en-1-one		
2-methylcyclohexan-1-ol		 or 
nicotinic acid (a vitamin,also called niacin)		 or 

❑ **NOTE:** IUPAC names will be discussed in next chapter.

DRAWING MOLECULES

Be realistic

Below is another organic structure—again, you may be familiar with the molecule it represents; it is a fatty acid commonly called linoleic acid.



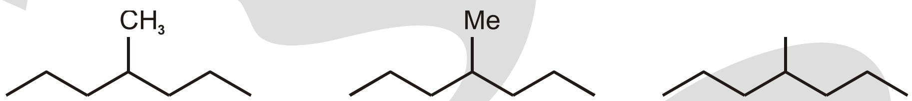
linoleic acid

We could also depict linoleic acid as

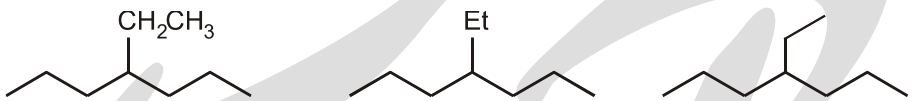
$CH_3CH_2CH_2CH_2CH$ $CHCH_2CH$ $CHCH_2CH_2CH_2CH_2CH_2CH_2CH_2CO_2H$

(Condensed formula)

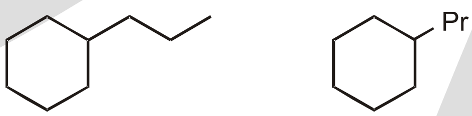
Methyl groups can be shown in a numbers of ways, and all of them are acceptable :



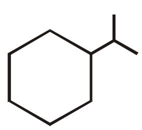
Ethyl groups can also be shown in a number of ways:



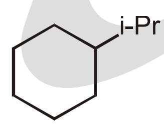
Propyl groups are usually just drawn, but sometimes you will see the term Pr (which stands for propyl):



Look at the propyl group above and you will notice that it is a small chain of 3 carbon atoms that is attached to the parent chain by the first carbon of the small chain. But what if it is attached by the middle carbon? Then it is not called propyl anymore :



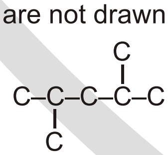
or



It is called as iso-Propyl or i-Pr.

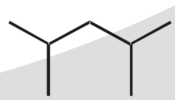
MISTAKES TO AVOID

Drawing where the C's and H's are not drawn. You cannot draw the C's without also drawing the H's:

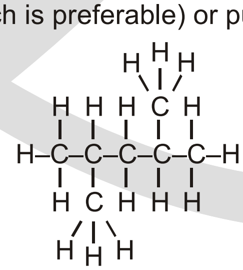


Never do this ☹

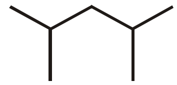
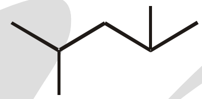
This drawing is no good. Either leave out the C's (which is preferable) or put in the H's:



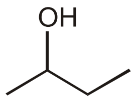
or

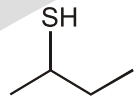


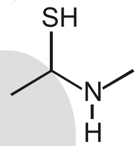
When drawing each carbon atom in a zigzag, try to draw all of the bonds as far apart as possible:

 is better than 

In bond-line drawings, we do draw any H's that are connected to atoms other than carbon. For example,

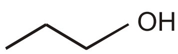


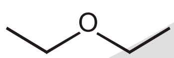


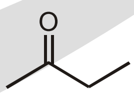
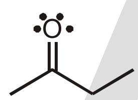


FINDING LONE PAIRS THAT ARE NOT DRAWN

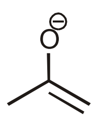
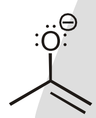
When oxygen has no formal charge, it will have two bonds and two lone pairs:

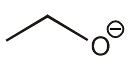
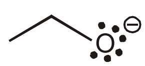
 Is the same as 

 Is the same as 

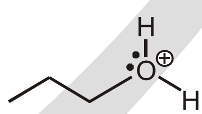
 Is the same as 

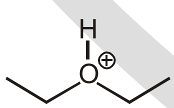
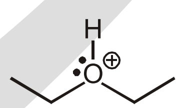
If oxygen has a negative formal charge, then it must have one bond and three lone pairs:

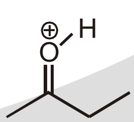
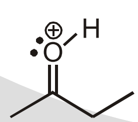
 is the same as 

 Is the same as 

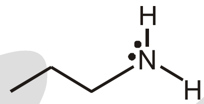
If oxygen has a positive charge, then it must have three bonds and one lone pair:

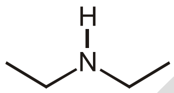
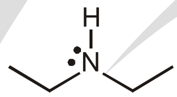
 Is the same as 

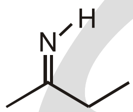
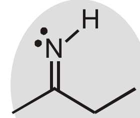
 Is the same as 

 Is the same as 

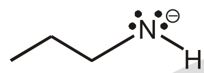
Now let's look at the common situations for nitrogen atoms. When nitrogen has no formal charge, it will have three bonds and one lone pair:

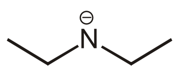
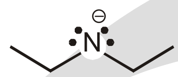
 Is the same as 

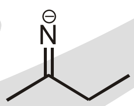
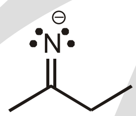
 Is the same as 

 Is the same as 

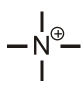
If nitrogen has a negative formal charge, then it must have two bonds and two lone pairs:

 Is the same as 

 Is the same as 

 Is the same as 

If nitrogen has a positive charge, then it must have four bonds and no lone pairs:

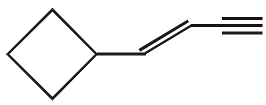
 has no lone pairs

 has no lone pairs

 has no lone pairs

Solved Example

- The number of hydrogen atoms associated with the molecule shown below is ?

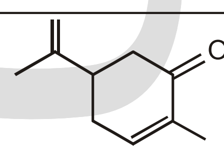


Ans. 10 hydrogens

INTERPRETING A BOND-LINE STRUCTURE

Solved Example

- Carvone, a substance responsible for the odor of spearmint, spearmint, has the following structure. Tell how many hydrogens are bonded to each carbon, and give the molecular formula of carvone.

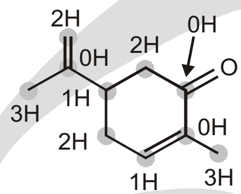


Carvone

Strategy

The end of a line represents a carbon atom with 3 hydrogens, CH_3; a two-way intersection is a carbon atom with 2 hydrogens, CH_2; a three-way intersection is a carbon atom with 1 hydrogen, CH; and a four-way intersection is a carbon atom with no attached hydrogens.

Sol.

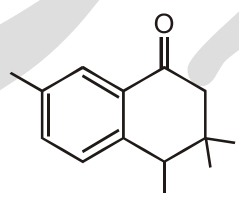


Carvone ($C_{10}H_{14}O$)

COUNTING THE NUMBER OF HYDROGEN ATOMS

Now that we know to count carbon atoms, we must learn how to count the hydrogen atoms in a bond-line drawing of a molecule. The hydrogen atoms are not shown, and this is why it is so easy and fast to draw bond-line drawings. Neutral carbon atoms always have a total of four bonds.

So you only need to count the number that you can see on a carbon atom, and then you know that there should be enough hydrogen atoms to give a total of four bonds to the carbon atom.

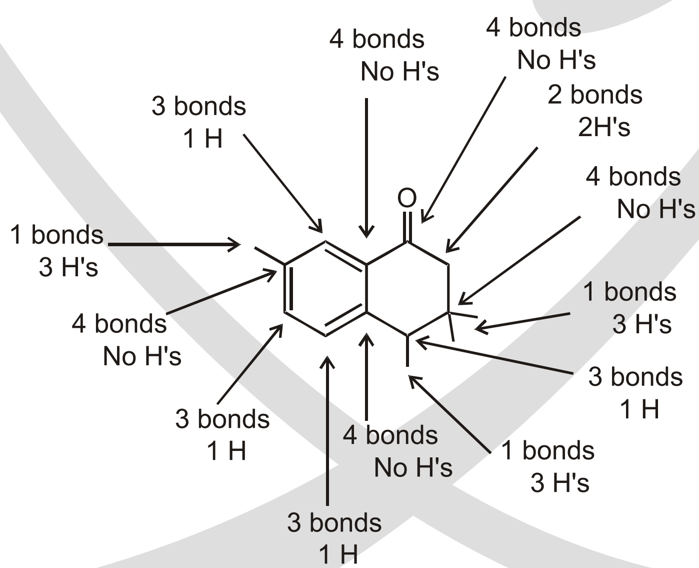


Solved Example

- The following molecule has 14 carbon atoms. Count the number of hydrogen atoms connected to each carbon atom

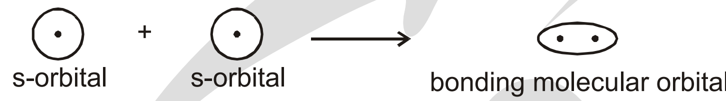
BONDING IN ORGANIC CHEMISTRY

Ans.

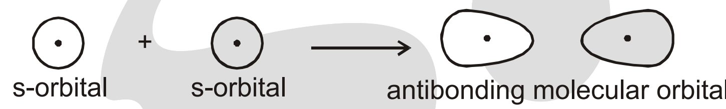


SIGMA (–) AND PI (–) BONDS

The electrons shared in a covalent bond result from overlap of atomic orbitals to give a new molecular orbital. Electrons in 1s and 2s orbitals combine to give sigma (–) bonds. When two 1s orbitals combine in phase, this produces a bonding molecular orbital.



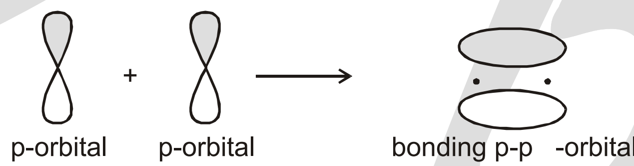
When two 1s orbitals combine out-of-phase, this produces an antibonding molecular orbital.

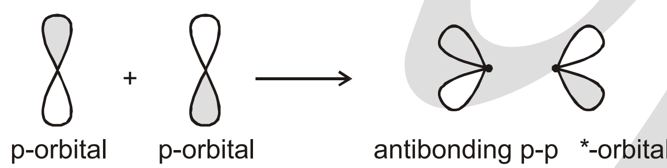


Electrons in p orbitals can combine to give sigma ( ) or pi ( ) bonds.

Sigma (-) bonds are strong bonds formed by head-on overlap of two atomic orbitals.

Pi(–) bonds are weaker bonds formed by side-on overlap of two -orbitals.



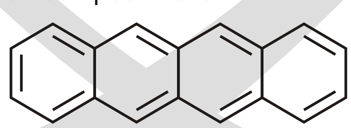


Only - or -bonds are present in organic compounds. All single bonds are s-bonds while all multiple (double or triple) bonds are composed of one s-bond and one or two p-bonds.

EXERCISE

SINGLE CHOICE QUESTIONS

1. Number of p-bonds present in given compound are:



(A) 8 (B) 9 (C) 10 (D) 12

2. How many Hydrogens does an alkane with 17 carbons have?

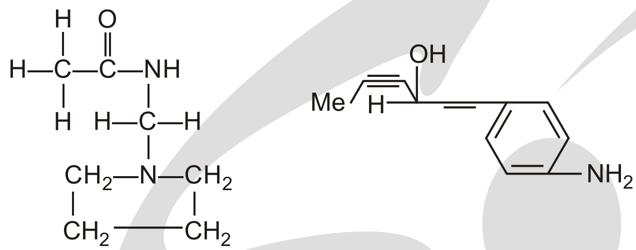
(A) 32 (B) 34 (C) 36 (D) 38

3. How many carbons does an alkane with 34 hydrogens have?

(A) 16 (B) 14 (C) 15 (D) 17

SUBJECTIVE TYPE QUESTIONS

1. What is wrong with these structures ? Suggest better ways of representing these molecules.

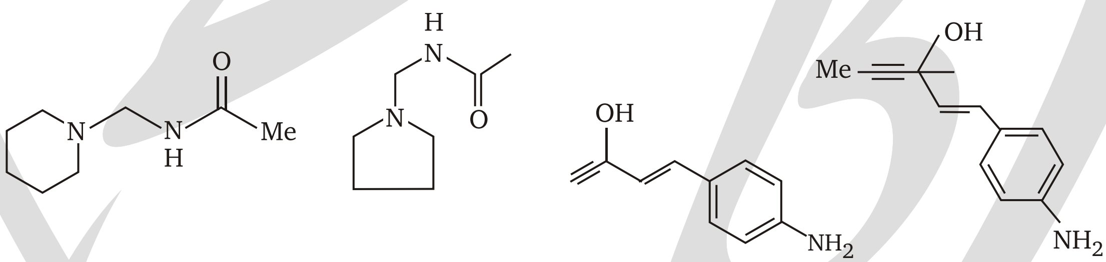


Purpose of the Problem

To shock you with two dreadful structure and to try and convince you that well drawn realistic structures are more attractive to the eye as well as easier to understand.

Suggested solution

The bond angles are grotesque with square planar saturated carbon, alkynes at 120°, alkenes at 180°, bonds coming off benzene rings at the wrong angle, and so on, The left-hand structure would be clear if most of the hydrogens were omitted. Hence there are two possible better structure for each molecule. There are many other correct possibilities.



Answers

Single Choice Questions

1. (B) **2.** (C) **3.** (A)

❑❑❑

Degree of Carbon & Hydrogen, Alcohol & Amine

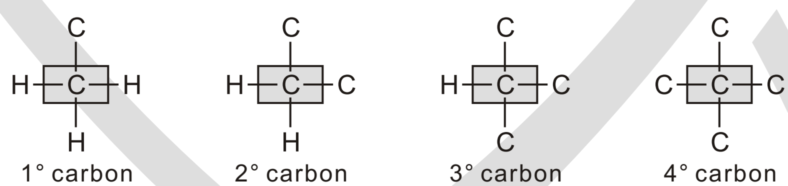
DEGREE OF CARBON AND HYDROGEN IN HYDROCARBON

DEGREE OF CARBON

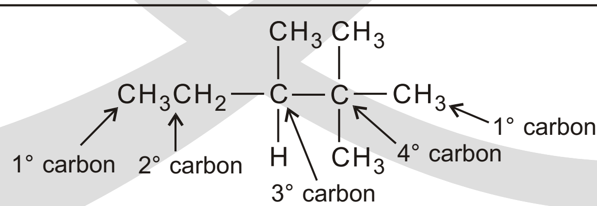
Carbon atoms in alkanes and other organic compounds are classified by the number of other carbons directly bonded to them.

1º or Primary	2º or Secondary
3º or Tertiary	4º or Quaternary

CLASSIFICATION OF CARBON ATOMS

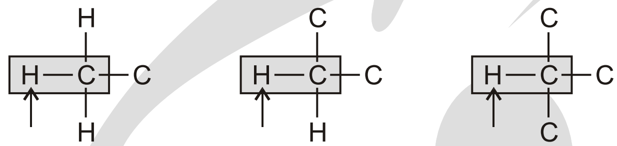


Solved Example

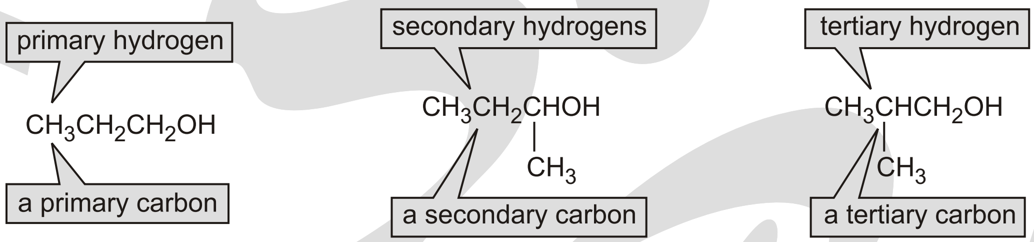


CLASSIFICATION OF HYDROGEN ATOMS

Like the carbons, the hydrogens in a molecule are also referred to as primary, secondary, and tertiary. **Primary hydrogens** are attached to a primary carbon, **secondary hydrogens** are attached to a secondary carbon, and **tertiary hydrogens** are attached to a tertiary carbon.



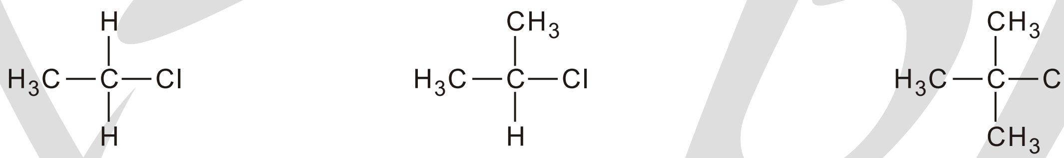
1° H 2° H 3° H



DEGREE OF CARBON IN ALKYLHALIDE

Alkyl halides are classified as primary, secondary and tertiary alkyl halides depending on whether the halogen atom is attached to a primary, secondary or tertiary carbon atom respectively.

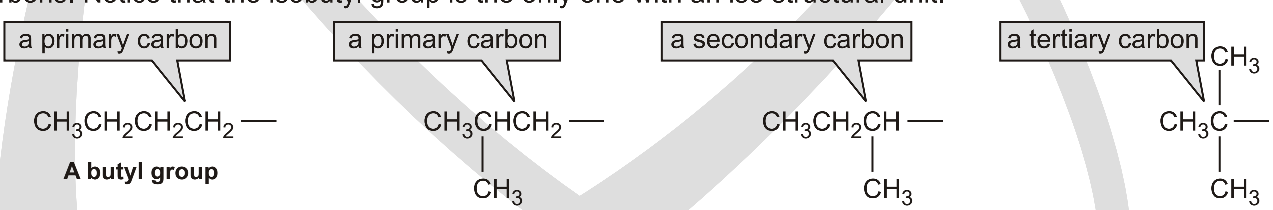
For example



Chloro ethane (Primary) 2-Chloro propane (secondary) 2-Chloro-2-methyl propane (tertiary)

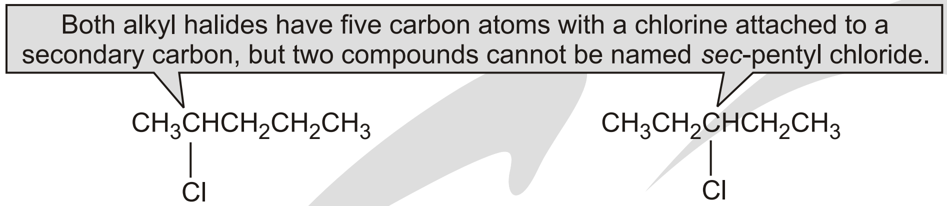
Aromatic halogen compounds or halo arenes are the halogen compounds which contain atleast one aromatic ring.

There are four alkyl groups that have four carbons. Two of them, the butyl and isobutyl groups, have a hydrogen removed from a primary carbon. A *sec* -butyl group has a hydrogen removed from a secondary carbon (*sec*- , sometimes abbreviated *s*- , stands for secondary), and a *tert* -butyl group has a hydrogen removed from a tertiary carbon (*tert*-, often abbreviated *t*- , stands for tertiary). **A tertiary carbon** is a carbon that is bonded to three other carbons. Notice that the isobutyl group is the only one with an iso structural unit.



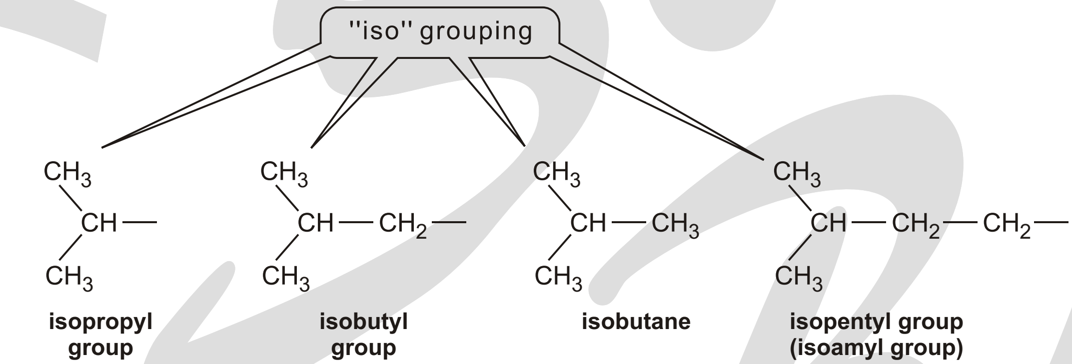
A butyl group **an isobutyl group** **a sec-butyl group** **a tert-butyl group**

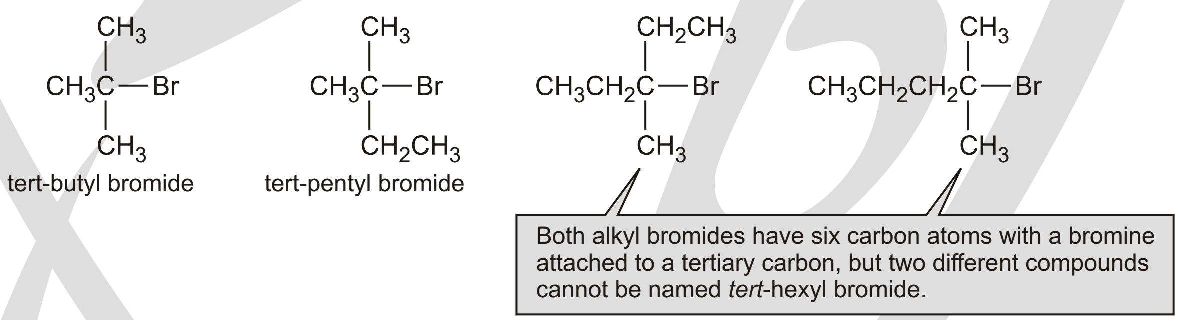
A chemical name must specify one compound only. The prefix "*sec*, " therefore, can be used only for *sec* -butyl compounds. **The name "*sec*-pentyl" cannot be used because pentane has two different secondary carbons.** Thus, removing a hydrogen from a secondary carbon of pentane produces one of two different alkyl groups, depending on which hydrogen is removed. As a result, *sec*-pentyl chloride would specify two different alkyl chlorides, so it is *not* a correct name.



PROBLEM-SOLVING HINT

When looking for the longest continuous chain (to give the base name), look to find all the different chains of that length, Often, the longest chain with the most substituents is not obvious.

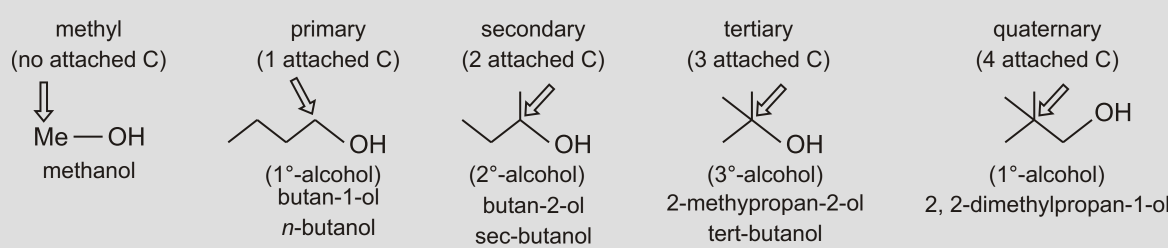




DEGREE OF ALCOHOL

- **Primary, secondary, and tertiary**

The prefixes *sec* and *tert* are really short for secondary and tertiary, terms that refer to the carbon atom that attaches these groups to the rest of the molecular structure.

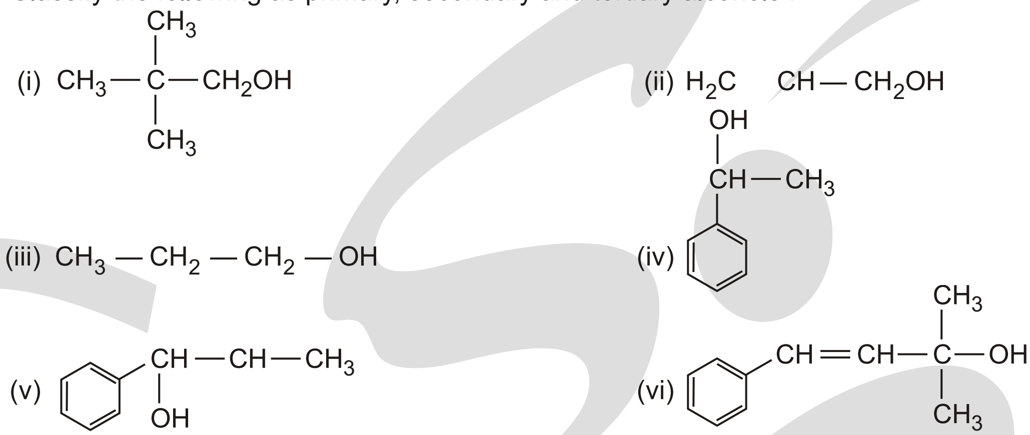


A primary carbon atom is attached to only one other C atom, a secondary to two other C atoms, and so on. This means there are five types of carbon atom.

These names for bits of hydrocarbon framework are more than just useful ways of writing or talking about chemistry. They tell us something fundamental about the molecule and we shall use them when we describe reactions.

Solved Example

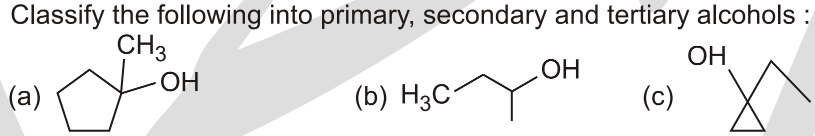
- Classify the following as primary, secondary and tertiary alcohols :



Ans. Primary alcohol (i), (ii) and (iii), Secondary alcohol (iv) and (v), Tertiary alcohol (vi)

Solved Example

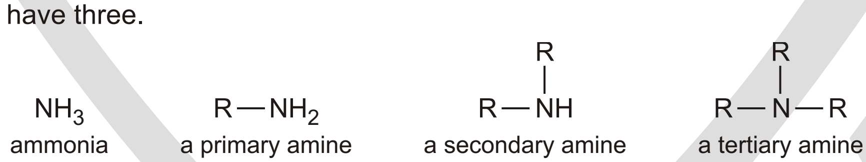
- Classify the following into primary, secondary and tertiary alcohols :



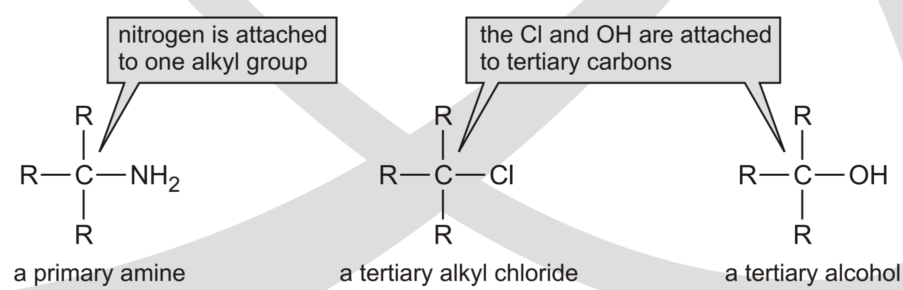
Ans. (a) Tertiary (b) Secondary (c) Tertiary

DEGREE OF AMINE

An amine is a compound in which one or more hydrogens of ammonia have been replaced by alkyl groups. Amines are classified as **primary**, **secondary**, and **tertiary**, depending on how many alkyl groups are attached to the nitrogen. Primary amines have one alkyl group attached to the nitrogen, secondary amines have two, and tertiary amines have three.



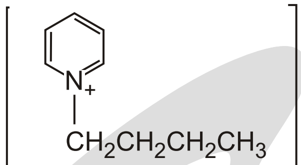
Be sure to note that the number of alkyl groups attached to the nitrogen determines whether an amine is primary, secondary,



Quaternary ammonium salts have four alkyl or aryl bonds to a nitrogen atom. The nitrogen atom bears a positive charge, just as it does in simple ammonium salts such as ammonium chloride. The following are examples of quaternary (4°) ammonium salts:

$$\left[\begin{array}{c} \quad CH_3—CH_2I^- \\ CH_3CH_2—\overset{+}{N}—CH_2CH_3 \\ CH_2CH_3 \end{array}\right]^+ I^{\ominus}$$

tertraethylammonium iodide



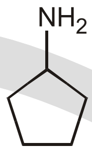
N-butylpyridinium bromide

$$CH_3—\overset{O}{\overset{\|}{C}}—O—CH_2CH_2—\overset{\oplus}{N}(CH_3)_2—CH_3$$

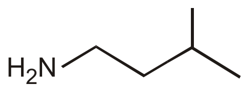
acetylcholine, a neurotransmitter

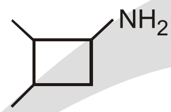
Solved Example

- Which among the following compound(s) is a primary amine with the molecular formula $C_5H_{11}N$?

(A) 

(B) 

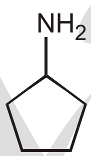
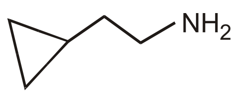
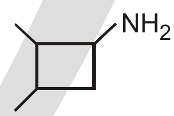
(C) 

(D) 

Ans. (A, B)

Sol. $C_5H_{11}N$; D.B.E. $(5 \quad 1) \quad \frac{(11 \quad 1)}{2} \quad 1$

Thus, amine either be a cyclic or having double bond

$C_5H_{11}N$	$C_5H_{11}N$	$C_5H_{13}N$	$C_6H_{13}N$
✓	✓	×	×

Solved Example

- Give a systematic name and a common name (if it has one) for each of the following amines and indicate whether each is a primary, secondary, or tertiary amine :

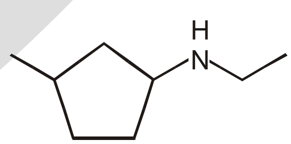
(a) $CH_3CH_2CH_2CH_2CH_2CH_2NH_2$

(b) $CH_3\underset{CH_3}{\underset{|}{C}}HCH_2NH\underset{CH_3}{\underset{|}{C}}HCH_2CH_3$

(c) $(CH_3CH_2)_2NCH_3$

(d) $CH_3CH_2CH_2NHCH_2CH_2CH_2CH_3$

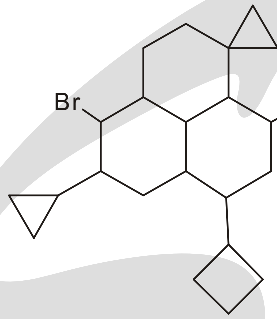
(e) $CH_3CH_2CH_2\underset{CH_2CH_3}{\underset{|}{N}}CH_2CH_3$

(f) 

Sol. (a) Hexan-1-amine (1)
(b) N-(2-methylpropyl)butan-2-amine (2)
(c) N-ethyl-N-methylethan-1-amine (3)
(d) N-propylbutanamine (2)
(e) N,N-diethylpropan-1-amine (3)
(f) N-ethyl-3-methylcyclopentanamine (2)

Solved Example

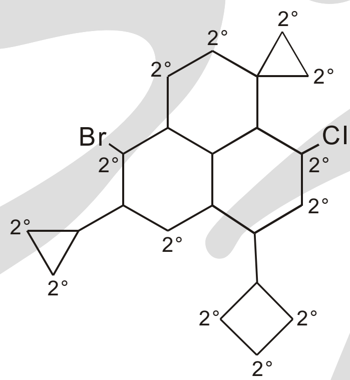
- Total number of 2° carbon present in given compound is , so the value of



is :

Ans. 13

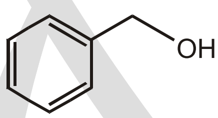
Sol.

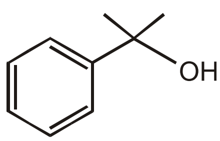


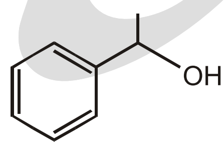
2º carbons are present = 13

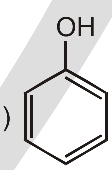
Solved Example

- Which of the following compounds is a secondary alcohol?

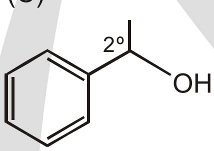
(A) 

(B) 

(C) 

(D) 

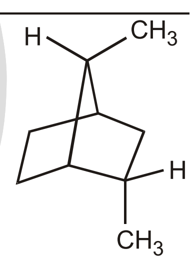
Ans. (C)

Sol. 

❑ **NOTE:** (Phenol is not alcohol).

Solved Example

- How many secondary hydrogens are present in the hydrocarbon below?



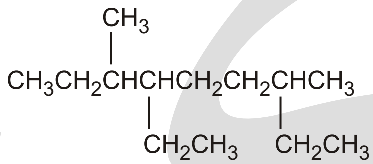
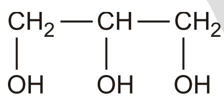
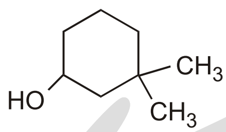
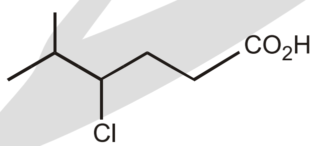
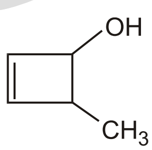
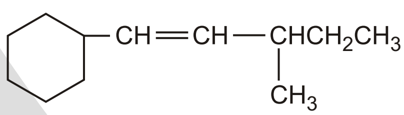
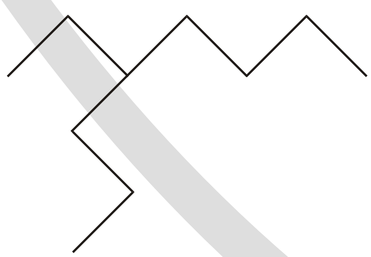
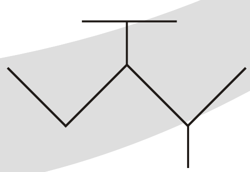
(A) 2 (B) 6
(C) 7 (D) 8
(E) 16

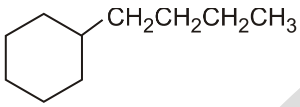
Ans. (B)

EXERCISE

WORK SHEET - 1

Count the number of primary, secondary, tertiary, quarternary carbon as well as hydrogen in given compound :

S.No.	Compound	1 C	2 C	3 C	4 C	1 H	2 H	3 H
1.								
2.								
3.								
4.								
5.								
6.								
7.								
8.								
9.								

10.								
11.								

Answers

Work Sheet-1

S.No.	1 C	2 C	3 C	4 C	1 H	2 H	3 H
1.	5	5	3	0	15	10	3
2.	2	1	0	0	4	1	0
3.	2	5	0	1	6	9	0
4.	3	3	1	0	6	5	1
5.	1	3	1	0	3	3	1
6.	2	8	2	0	6	14	2
7.	3	6	1	0	9	12	1
8.	3	4	1	0	7	4	1
9.	5	1	3	0	15	2	3
10.	0	4	0	0	0	4	0
11.	1	8	1	0	3	16	1

❑❑❑

CHAPTER 4

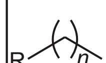
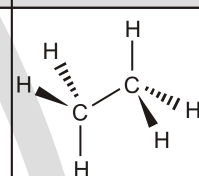
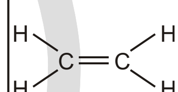
Functional Groups

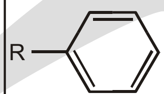
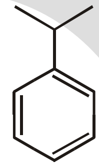
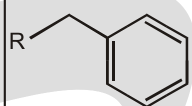
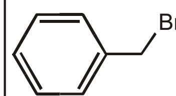
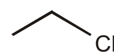
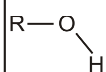
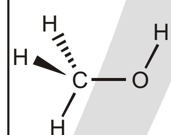
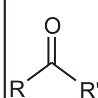
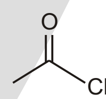
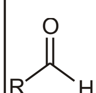
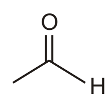
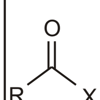
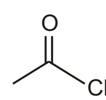
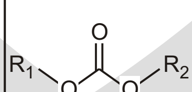
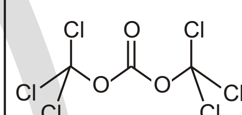
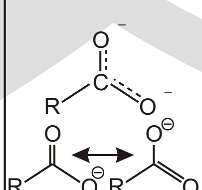
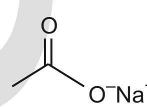
FUNCTIONAL GROUPS

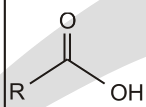
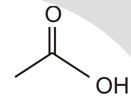
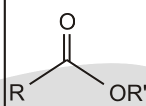
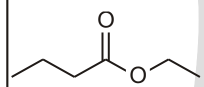
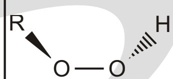
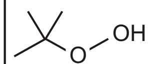
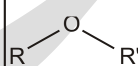
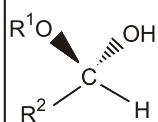
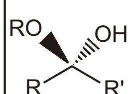
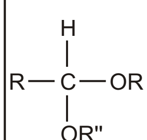
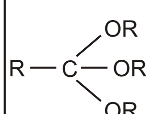
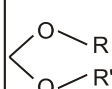
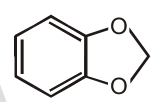
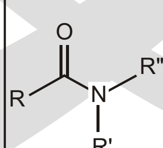
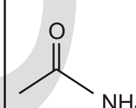
In organic chemistry, functional groups are specific groups of atoms or bonds within molecules that are responsible for the characteristic chemical reactions of those molecules. The same functional group will undergo the same or similar chemical reaction(s) regardless of the size of the molecule it is a part of.

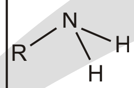
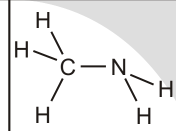
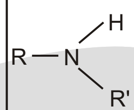
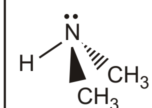
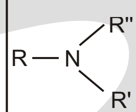
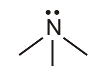
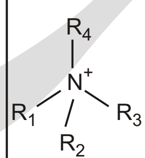
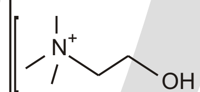
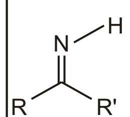
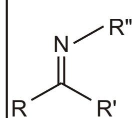
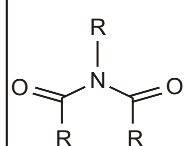
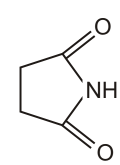
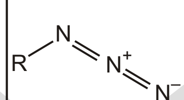
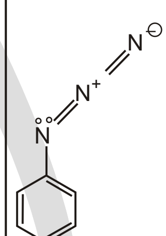
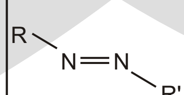
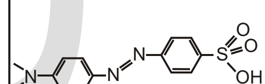
Combining the names of functional groups with the names of the parent alkanes generates a powerful systematic nomenclature for naming organic compounds.

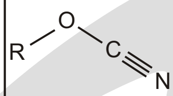
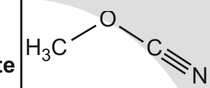
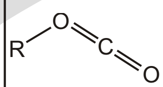
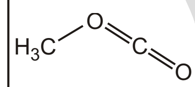
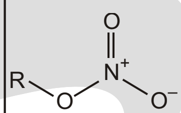
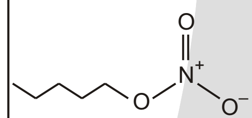
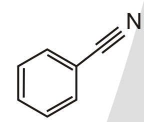
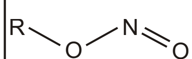
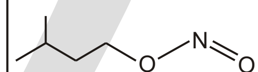
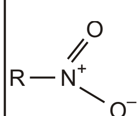
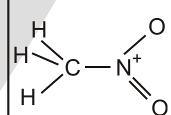
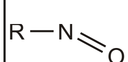
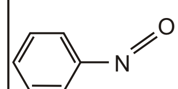
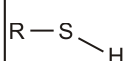
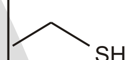
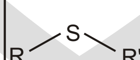
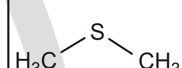
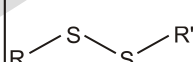
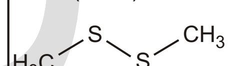
TABLE OF COMMON FUNCTIONAL GROUPS

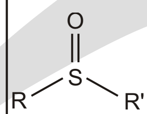
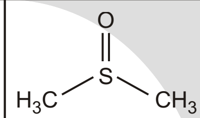
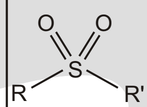
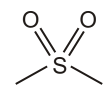
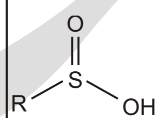
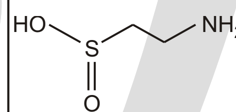
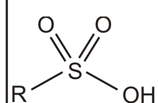
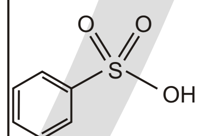
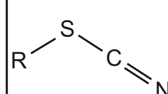
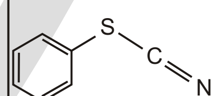
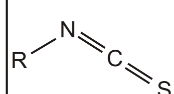
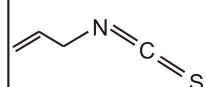
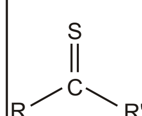
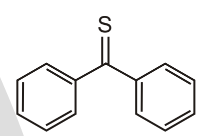
S.N.	Chemical class	Group	Formula	Structural Formula	Prefix	Suffix	Example
1.	Alkane	Alkyl	$R(CH_2)_nH$		alkyl	-ane	 Ethane
2.	alkene	alkenyl	R_2C CR_2		alkenyl-	-ene	
3.	Alkyne	Alkynyl	RC CR	R — C C — R	alkynyl-	-yne	H — C ≡ C — H Acetylene (Ethyne)

S.N.	Chemical class	Group	Formula	Structural Formula	Prefix	Suffix	Example
4.	Benzene derivative	Phenyl	RC_6H_5(RPh)		phenyl-	-benzene	 Cumene (2-phenylpropane)
5.	Toluene derivative	Benzyl	$RCH_2C_6H_5$ (R—Ph)		benzyl-	1-(substituent) toluene	 Benzyl bromide (α-Bromotoluene)
6.	haloalkane	halo	RX	R — X	halo-	alkyl halide	 Chloroethane (Ethylchloride)
7.	Alcohol	Hydroxyl	ROH		hydroxy-	-ol	 Methanol
8.	Ketone	Carbonyl	RCOR		-oyl-(-COR) or OXO or keto	-one	 Acetyl chloride (Ethanoyl chloride)
9.	Aldehyde	Aldehyde	RCHO		formyl-(—COH)	-al	 Acetaldehyde
10.	Acyl halide	Haloformyl	RCOX		carbonofluoridoyl- carbonochloridoyl- carbonobromidoyl- carbonoiodidoyl-	-oyl **halide**	 Acyl chloride (Ethanoyl chloride)
11.	Carbonate	Carbonate ester	ROCOOR		(alkoxycarbonyl) oxy-	alkyl **carbonate**	 Triphosgene (bis(trichloromethyl) carbonate)
12.	Carboxylate	Carboxylate	RCOO		carboxy-	-oate	 Sodium acetate (Sodium ethanoate)

S.N.	Chemical class	Group	Formula	Structural Formula	Prefix	Suffix	Example
13.	Carboxylic acid	Carboxyl	RCOOH		carboxy-	-oic **acid**	 Acetic acid (*Ethanoic acid*)
14.	Ester	Ester	RCOOR		alkanoyloxy- or alkoxycarbonyl	alkyl alkanoate	 Ethyl butyrate (*Ethyl butanoate*)
15.	Hydropero-xide	Hydroperoxy	ROOH		hydroperoxy-	alkyl **hydroperoxide**	 tert-Butyl hydroperoxide
16.	Ether	Ether	ROR		alkoxy-	alkyl **ether**	 Diethyl ether (Ethoxyethane)
17.	Hemiacetal	Hemiacetal	RCH(OR) (OH)		alkoxy-ol	-one alkyl **hemiacetal**	
18.	Hemiketal	Hemiketal	RC(OR)(OH)R		alkoxy-ol	-one alkyl **hemiketal**	
19.	Acetal	Acetal	RCH(OR)(OR)		dialkoxy-	-al dialkyl **acetal**	
20.	Orthoester	Orthoester	RC(OR)(OR) (OR)		trialkoxy-		
21.	Heterocycle	Methylenedioxy	ROCOR		methylene- dioxy-	-dioxole	 1,2-Methylenedioxy-benzene (1,3-Benzodioxole)
22.	Amide	Carboxamide	$RCONR_2$		Carboxamido- or carbamoyl-	-amide	 Acetamide (*Ethanamide*)

S.N.	Chemical class	Group	Formula	Structural Formula	Prefix	Suffix	Example
23.	Amines	Primary amine	RNH_2		amino-	-amine	 Methylamine (*Methanamine*)
24.	Amines	Secondary amine	R_2NH		amino-	-amine	 Dimethylamine
25.	Amines	Tertiary amine	R_3N		amino-	-amine	 Trimethylamine
26.	Amines	4 ammonium ion	R_4N		ammonio-	-ammonium	 Choline
27.	Imine	Primary ketimine	RC(NH) R				
28.	Imine	Secondary ketimine	RC(NR)R				
29.	Imide	Imide	$(RCO)_2NR$		imido-	imide	 Succinimide (Pyrrolidine-2,5-dione)
30.	Azide	Azide	RN_3		azido-	alkyl **azide**	 Phenyl azide (Azidobenzene)
31.	Azo compound	Azo (Diimide)	RN_2R		azo-	-diazene	 Methyl orange (*p*-dimethylamino-azobenzenesulfonic acid)

S.N.	Chemical class	Group	Formula	Structural Formula	Prefix	Suffix	Example
32.	Cyanates	Cyanate	ROCN		Cyanato-	alkyl **cyanate**	 Methyl cyanate
33.	Cyanates	Isocyanate	RNCO		isocyanato-	alkyl **isocyanate**	 Methyl isocyanate
34.	Nitrate	Nitrate	$RONO_2$		nitrooxy-, nitroxy-	alkyl **nitrate**	 Amyl nitrate (1-nitroxypentane)
35.	Nitrile	Nitrile	RCN		cyano-	alkanenitrile alkyl **cyanide**	 Benzonitrile (Phenyl cyanide)
36.	Isonitrile	Isonitrile	RNC	$R-N^+\equiv C^-$	isocyano-	alkaneisonitrile alkyl **isocyanide**	$H_3C-N^+\equiv C^-$ Methyl isocyanide
37.	Nitrite	Nitroxooxy	RONO		nitrosooxy-	alkyl **nitrite**	 Isoamyl nitrite (3-*methyl*-1- *nitrosoxybutane*)
38.	Nitro compound	Nitro	RNO_2		nitro-		 Nitromethane
39.	Nitroso	RNO		nitroso-(Nitrosyl-)			 Nitrosobenzene
40.	Thiol	Sulfhydryl	RSH		sulfanyl-(-SH)	**-thiol**	 Ethanethiol
41.	Sulfide (Thioether)	Sulfide	RSR		*substituent* sulfanyl- (-SSR)	di(substituent) **sulfide**	 (Methylsulfanyl) methane (prefix) or Dimethyl sulfide (sulfix)
42.	Disulfide	Disulfide	RSSR		*substituent* disulfanyl- (-SSR)	di(substituent) **dissulfide**	 (Methyldisulfanyl) methane (prefix) or Dimethyl disulfide (sulfix)

S.N.	Chemical class	Group	Formula	Structural Formula	Prefix	Suffix	Example
43.	Sulfoxide	Sulfinyl	RSOR		-sulfinyl-(-SOR)	di(substituent) **sulfoxide**	 (Methanesulfinyl) methane (prefix) or Dimethyl sulfoxide (suffix)
44.	Sulfone	Sulfonyl	RSO_2R		-sulfonyl- ($-SO_2R$)	di(substituent) **sulfone**	 (Methanesulfonyl) methane (prefix) or Dimethyl sulfone (suffix)
45.	Sulfinic acid	Sulfino	RSO_2H		sulfino- ($-SO_2H$)	**-sulfinic acid**	 2-Aminoethane sulfinic acid
46.	Sulfonic acid	Sulfo	RSO_3H		sulfo- ($-SO_3H$)	**-sulfonic acid**	 Benzenesulfonic acid
47.	Thiocyanate	Thiocyanate	RSCN		thiocyanato- (-SCN)	*substituent* **thiocyanate**	 Phenyl thiocyanate
48.	Isothio-cyanate	Isothiocyanate	RNCS		isothiocyanato- (-NCS)	*substituent* ***isothiocyanate***	 Allyl isothiocyanate
49.	Thione	Carbonothioyl	RCSR		-thioyl- (-CSR) or sulfanylidene- (=S)	-thione	 Diphenylmethanethione (Thiobenzophenone)

IDENTIFY FUNCTIONAL GROUPS

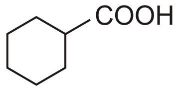
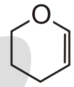
Solved Example

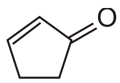
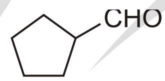
- Classify each of the following compounds. the possible classifications are as follows :

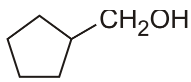
alcohol	ketone	carboxylic acid	ether
aldehyde	alkene		

(a) CH_2CH_2CHO **(b)** $CH_3CH_2CH(OH)CH_3$

(c) $CH_3COCH_2CH_3$ **(d)** $CH_3 — CH_2OCH_2CH_6$

(e)    **(f)** 

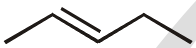
(g)    **(h)** 

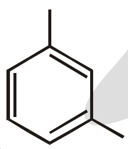
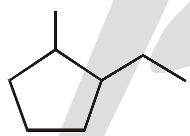
(i) 

Sol. (a) aldehyde (b) alcohol (c) ketone (d) ether
(e) carboxylic acid (f) ether, alkene (g) ketone, alkene (h) aldehyde
(i) alcohol

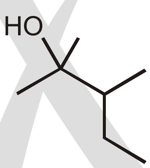
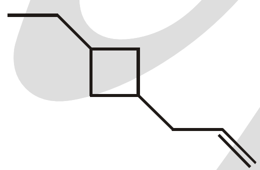
Solved Example

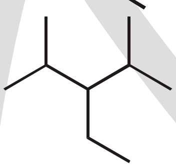
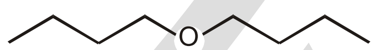
- For each molecule circle and name the functional group. If the functional group is an alcohol identify it as a primary (1°), secondary (2°), or tertiary (3°) alcohol. Some molecules will have more than one functional group; in those case circle and name all functional groups present. Functional groups: Alkane, alkene, alkyne, cyclic, aromatic, alcohol, ether.

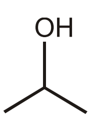
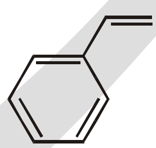
(a)    **(b)** 

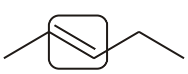
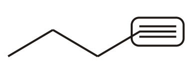
(c)    **(d)** 

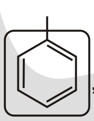
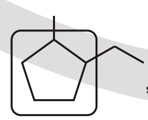
(e) CH_3CH_2OH **(f)** 

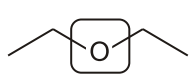
(g)    **(h)** 

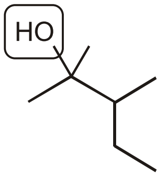
(i)    **(j)** 

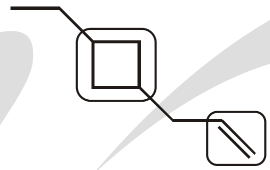
(k)    **(l)** 

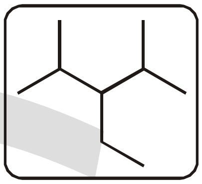
Ans. **(a)**  , alkene   **(b)**  , alkyne

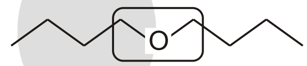
(c)  , aromatic   **(d)**  , cyclic

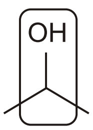
(e) $CH_3\boxed{CH_2OH}$ or $H_3C-\boxed{CH_2-OH}$, 1 Alcohol   **(f)**  , ether

(g)  , 3°alcohol

(h)  ,cyclic and alkene

(i)  , alkane

(j)  , ether

(k)  , 2° alcohol

(l)  ,aromatic and alkene

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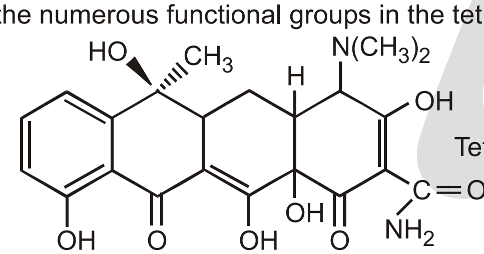
Solved Example

- The discovery of penicillin in 1928 marked the beginning of what has been called the "golden age of chemotherapy," in which previously life-threatening bacterial infections were transformed into little more than a source of discomfort. For those who are allergic to penicillin, a variety of antibiotics, including tetracycline, are available. Identify the numerous functional groups in the tetracycline molecule.



Tetracycline

Sol. The compound contains an aromatic ring fused to three six-membered rings. It is also an alcohol and phenol (with five — OH groups), a ketone (with $C{=}O$ groups at the bottom of the second and fourth rings), an amine [the — $N(CH_3)_2$ substituent at the top of the fourth ring], and an amide (the — $CONH_2$ group at the bottom right-hand corner of the fourth ring.)

HOMOLOGS

The family of alkanes shown in the table is an example of a homologous series. A **homologous series** (homos is Greek for "the same as") is a family of compounds in which each member differs from the one before it in the series by **one methylene (CH_2) group**. The members of a homologous series are called **homologs**

homologs

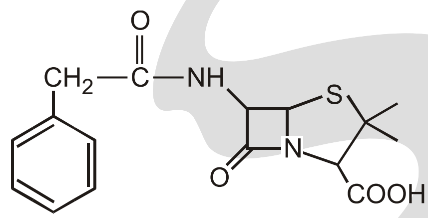
$CH_3CH_2CH_3$ $CH_3CH_2CH_2CH_3$

two different ways to draw isopropyl chloride

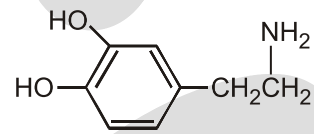
$$\underset{\displaystyle Cl}{CH_3\overset{}{C}HCH_3} \qquad\qquad \underset{\displaystyle CH_3}{CH_3CHCl}$$

Solved Example

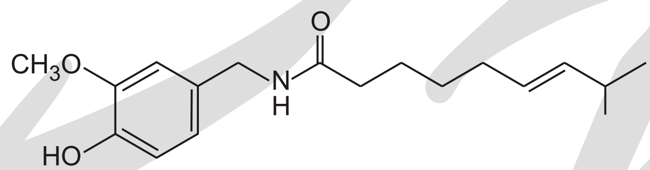
- Many naturally occurring compounds contain more than one functional group. Identify the functional groups in the following compounds:
  (a) Penicillin G is a naturally occurring antibiotic.
  (b) Dopamine is the neurotransmitter that is deficient in Parkinson's disease.
  (c) Capsaicin gives the fiery taste to chili peppers.
  (d) Thyroxine is the principal thyroid hormone.
  (e) Testosterone is a male sex hormone.



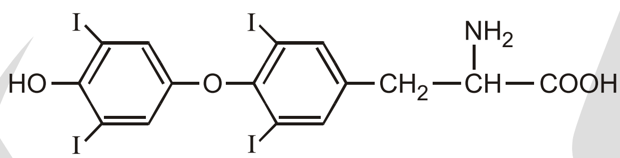
penicillin G



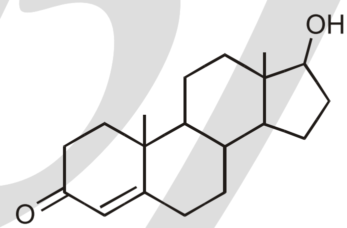
dopamine



capsaicin



thyroxine-T_4



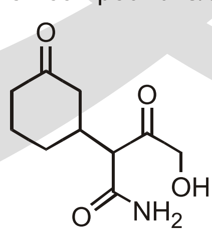
testosterone

Sol. **(a)** Penicillin-G: Carboxylic acid, thioether, amide
(b) Dopamine: Amine, aromatic alcohol (Phenol)
(c) Capsaicin: Phenol, ether, amide, alkene
(d) Thyroxine: Aryl iodide, phenol, ether, amine, carboxylic acid
(e) Testosterone: Alcohol, ketone, alkene

EXERCISE

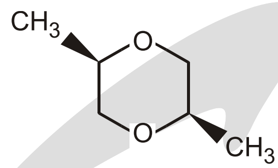
SINGLE CHOICE QUESTIONS

1. Functional group not present in given compound is/are?



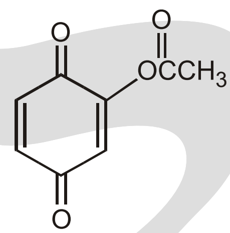
(A) Alcohol (B) Ketone (C) Carboxylic acid (D) Amide

2. Present functional group is :



(A) ketone (B) ester (C) ether (D) alcohol

3. Present functional group is/ are :



(A) ketone (B) ester (C) ether (D) A and B both

4. What is the lowest molecular weight possible for Ester?

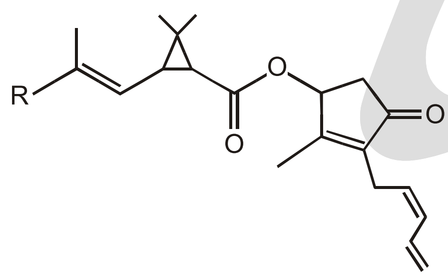
(A) 30 (B) 46 (C) 56 (D) 60

5. Which of the following compounds belong to the same homologous series ?

(1) 1-chloropropene (2) 1-chloropropane (3) 2-chlorobutane

(A) (1) and (2) only (B) (1) and (3) only (C) (2) and (3) only (D) (1), (2) and (3)

6. Pyrethrum flowers contain a natural insecticide called pyrethrin. Pyrethrin has the following structure:



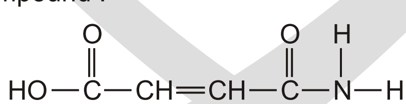
Which of the following functional groups are present in pyrethrin?

(1) Carbon-carbon double bond (2) Ester group

(3) Ketone group

(A) (1) and (2) only (B) (1) and (3) only (C) (2) and (3) only (D) (1), (2) and (3)

7. Consider the following compound :



Which of the following functional groups does it contain?

(1) Carboxyl group (2) Carbonyl group (3) Amide group

(A) (1) and (2) only (B) (1) and (3) only (C) (2) and (3) only (D) (1), (2) and (3)

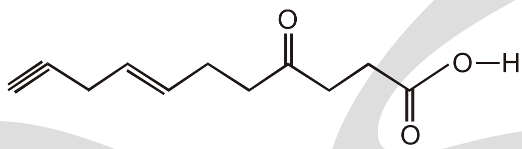
8. Which of the following statements is/are correct?

(1) Two organic compounds with the same general formula must belong to the same homologous series.

(2) Two organic compounds with main functional groups the same must belong to the same homologous series.

(3) Two organic compounds with the molecular mass differing by 14 must belong to the same homologous series.

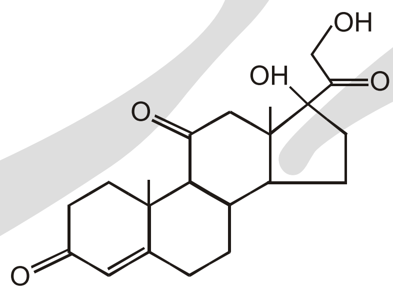
(A) (1) only (B) (2) only (C) (1) and (3) only (D) (2) and (3) only

9. 

Number of Functional group in above compound is

(A) 3 (B) 4 (C) 5 (D) 6

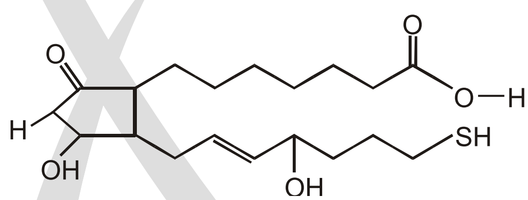
10. The functional groups in Cortisone are :



Cortisone

(A) Ether, alkene, alcohol (B) Alcohol, ketone, alkene

(C) Alcohol, ketone, amine (D) Ether, amine ketone

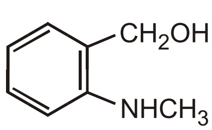
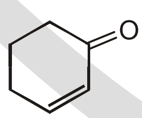
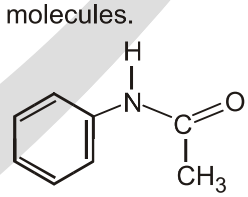
11. 

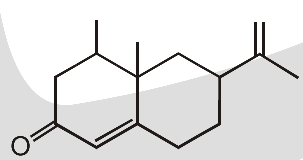
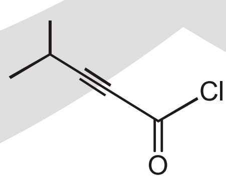
How many types of functional groups are present in given compound.

(A) 6 (B) 5 (C) 4 (D) 7

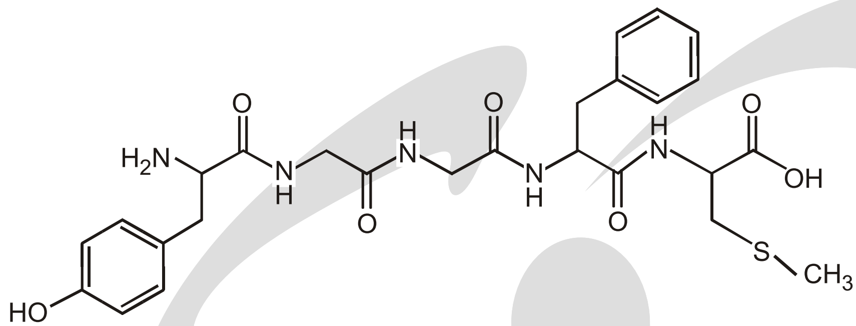
UNSOLVED EXAMPLE

1. Locate and identify the functional groups in the following molecules.

(a)  **(b)**  **(c)**  **(d)** $CH_3\underset{\underset{NH_2}{|}}{C}H\overset{\overset{O}{\|}}{C}OH$

(e)  **(f)** 

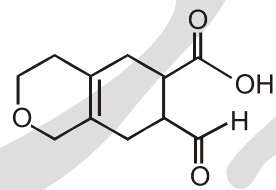
2. Met-enkephalin, an endorphin, serves as natural pain reliever that changes or removes the perception of nerve signals. Label all of the functional groups present in Met-enkephalin.



3. *x* Types of functional group

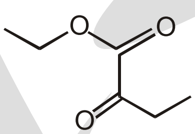
y Double bond equivalent

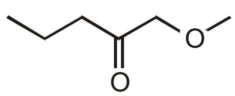
Value of (*x*   *y*) in given compound is :

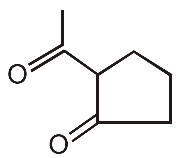


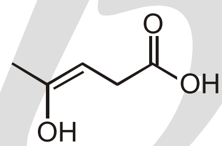
(A) 7 (B) 8 (C) 9 (D) 10

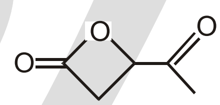
4. Which compound can be classified as an ester as well as a Ketone?


(A)


(B)


(C)

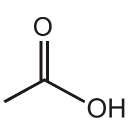

(D)

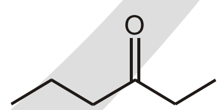

(E)

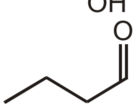
(A) A, B, E (B) E, B, C (C) A, E (D) C, E

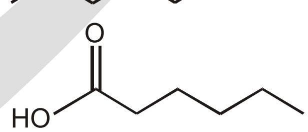
WORK SHEET

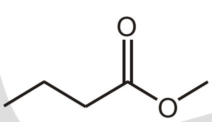
1. For each molecule circle and name the functional group. If the functional group is an amine identify it as a primary (1°), secondary (2°), or tertiary (3°) amine. some molecules will have more than one functional group; in those case circle and name all functional groups present. functional groups: Aldehyde, ketone, carboxylic acid, ester, amide, amine.

(a) 

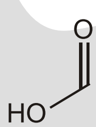
(b) 

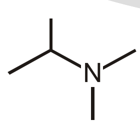
(c) 

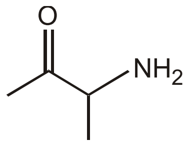
(d) 

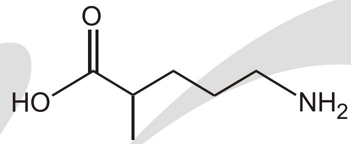
(e) 

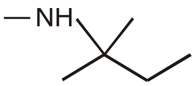
(f) CH_3COOH

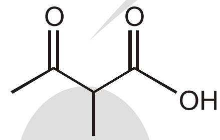
(g) 

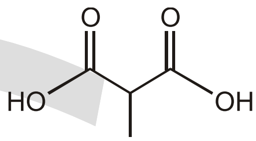
(h) 

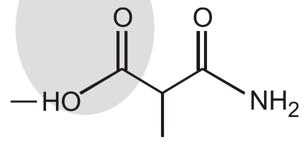
(i) 

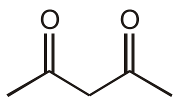
(j) 

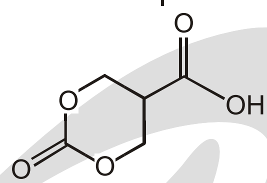
(k) 

(l) 

(m) 

(n) 

(o) 

(p) 

(q) HCHO

(r) HCOOH

(s) $CH_3(CO)CH_3$

SUBJECTIVE TYPE QUESTIONS

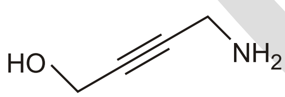
1. Suggest at least six different structures that would fit the formula C_4H_7NO. Make good realistic diagrams of each one and identify which functional groups(s) are present.

Purpose of the Problem

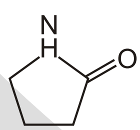
The identification and naming of functional groups is more important than the naming of compounds. This was your chance to experiment with different functional groups as well as different carbon skeletons.

Suggested solution

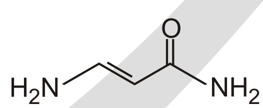
You will have found the carbonyl and amino groups very useful, but did you also use alkenes and alkynes, rings, ethers, alcohols, and cyanides? Here are twelve possibilities but there are many more. The functional group names in brackets are alternatives. Some you will not have known. You need not to have classify the alcohols and amines.



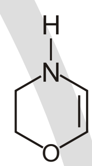
alkyne, primary alcohol, primary amine



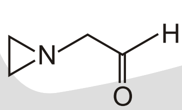
cyclic amide



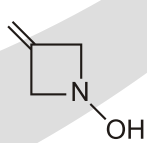
Amide, alkene, primary amine (enamine)



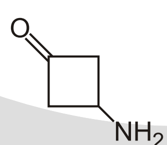
Ether, alkene, secondary amine



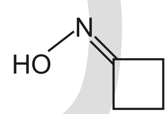
cyclic tertary amine, aldehyde



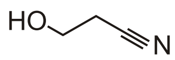
alkene, secondary amine, alcohol (cyclic hydroxylamine)



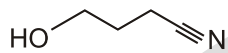
cyclic ketone, primary amine



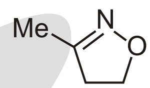
oxime, imine + alcohol



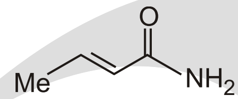
alcohol, nitrile (cyanide)



primary alcohol, nitrile (cyanide)



imine, ether (isoxazoine)



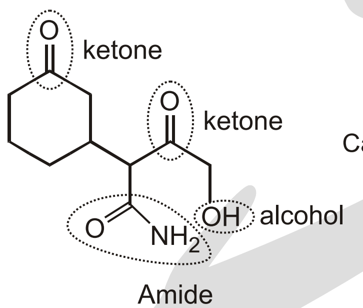
alkene, primary amide

Answers

Single Choice Questions

1. (C) **2.** (C) **3.** (D) **4.** (D) **5.** (C) **6.** (D) **7.** (B) **8.** (B)
9. (B) **10.** (B) **11.** (B)

1.

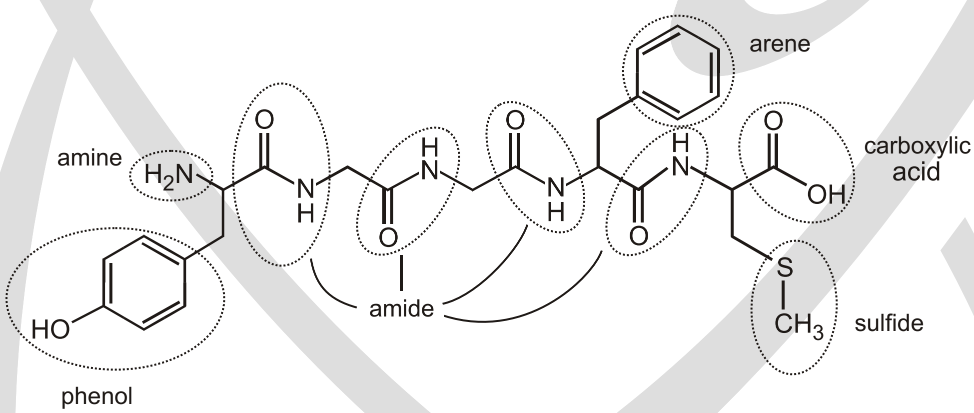


Carboxylic acid is not present

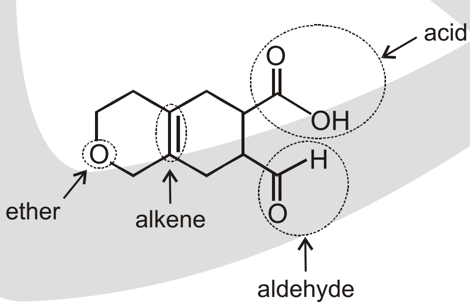
Unsolved Example

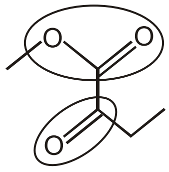
1. **(a)** alcohol, amine **(b)** ketone, alkene **(c)** amide
(d) carboxylic acid, amine **(e)** ketone,alkene **(f)** acyl halide, alkyne

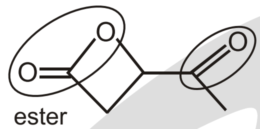
2.



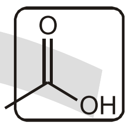
3.

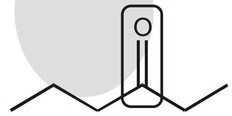


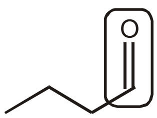
4.  ester, ketone

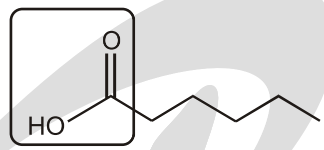
 ketone, ester

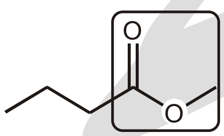
Work Sheet

1. (a)  , carboxylic acid

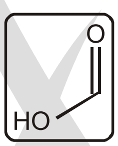
(b)  , ketone

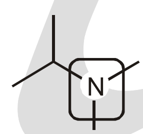
(c)  , aldehyde

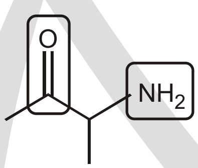
(d)  , carboxylic acid

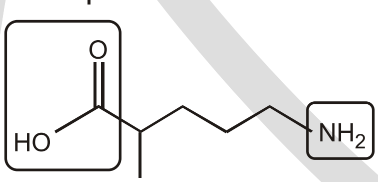
(e)  , ester

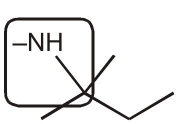
(f) $CH_3\boxed{COOH}$ or $H_3C-\boxed{\overset{O}{\overset{\|}{C}}-OH}$, carboxylic acid

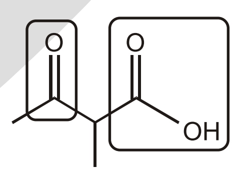
(g)  , carboxylic acid

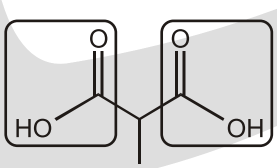
(h)  , 3° amine

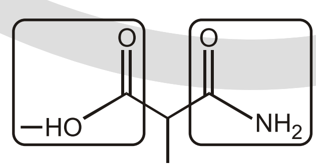
(i)  , ketone and 1º amine

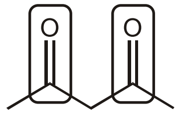
(j)  , carboxylic acid and 1°amine

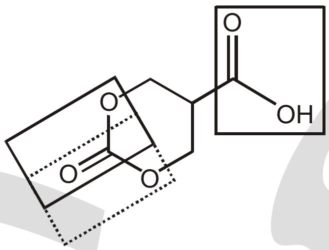
(k)  , 2° amine

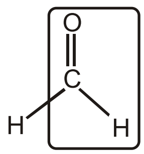
(l)  , ketone and carboxylic acid

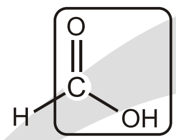
(m)  , 2 carboxylic acids

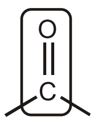
(n)  , Carboxylic acid and amide

(o)  , 2 ketones

(p)  , 2 esters and 1 carboxylic acid

(q)  , aldehyde

(r)  or , H COOH , carboxylic acid

(s)  , ketone

❑❑❑

CHAPTER

5

Double Bond Equivalent

DOUBLE BOND EQUIVALENTS (DBE) OR HYDROGEN DEFICIENCY INDEX OR DEGREES OF UNSATURATION

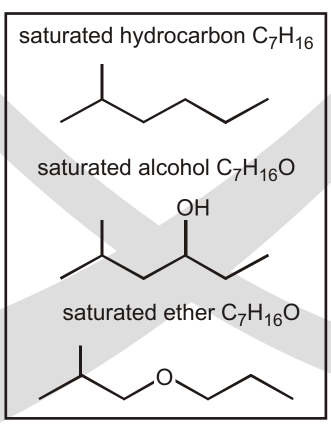
DBE help in the search for a structure

HOW TO CALCULATE DBE

Hello students! Have problems with calculating DBE ? No worries! Here is the tutorial which will help you step by step. Hopefully after reading this tutorial, you can calculate DBE faster and more accurately.

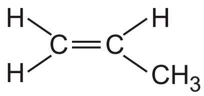
If DBE = 0

1. Ethylene, C_2H_6 is a saturated acyclic alkane and it does not have any bond or ring, so DBE = 0.

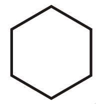


If DBE = 1

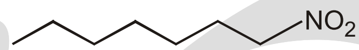
2. Propylene, C_3H_6 , contains a pi bond, so DBE = 1.



Propylene
DBE = 1



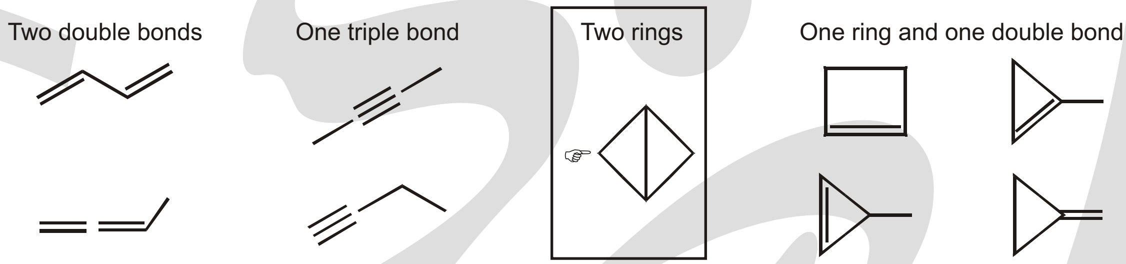
Cyclohexane
DBE = 1



$C_7H_{15}NO_2$ = one DBE

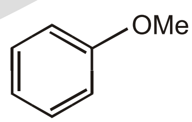
If DBE = 2

3. Propylene, C_4H_6 DBE = 2. There are several ways for a compound to possess two degrees of unsaturation : two double bonds, or two rings, or one double bond and one ring, or one triple bond. Let's explore all of these possibilities for C_4H_6 :

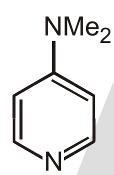


These are all of the possible constitutional isomers for C_6H_6. With this in mind, let's expand our skills set. Let's explore how to calculate the DBE when other elements are present in the molecular formula.

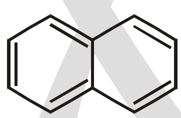
A benzene ring contains four DBE.



C_6H_8O = four DBE



$C_7H_{10}N_2$ = four DBE



only count two rings in this structure 5 pi bonds + 2 rings = > DBE = 5 + 2 = 7

HOW TO CALCULATE THE DBE IF WE DO NOT KNOW THE STRUCTURE OF THE CHEMICALS?

All the problems we have ever met talk about the organic chemicals which only contain carbon, oxygen, hydrogen, nitrogen, and halogens. Therefore, people summarized a DBE formula for our convenience.

$$\boxed{DBE \quad C \quad \frac{H}{2} \quad \frac{N}{2} \quad 1}$$

In this formula, C means the number of carbon. H means the number of hydrogen and X is number of halogen. N means the number of the nitrogen.

Let's apply the formula to the chemicals that we mentioned before.

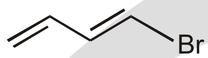
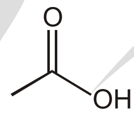
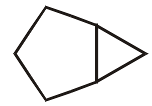
Ethylene (C_2H_6) : $DBE \quad C \quad \frac{H}{2} \quad \frac{N}{2} \quad 1 \quad 2 \quad \frac{6}{2} \quad \frac{0}{2} \quad 1 \quad 0$

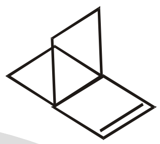
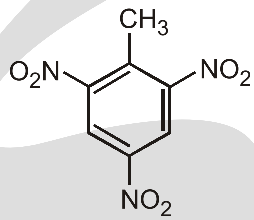
Propylene (C_3H_6) : $DBE \quad C \quad \frac{H}{2} \quad \frac{N}{2} \quad 1 \quad 3 \quad \frac{6}{2} \quad \frac{0}{2} \quad 1 \quad 1$

Cyclohexane (C_6H_{12}) : $DBE \quad C \quad \frac{H}{2} \quad \frac{N}{2} \quad 1 \quad 6 \quad \frac{12}{2} \quad \frac{0}{2} \quad 1 \quad 1$

Solved Example

- Look at the chemical structure below and calculate the DBE.

(1)  (2)  (3)  (4) 

(5)  (6) 

Ans. (1) One pi bond. DBE = 1 (2) Two pi bond. DBE = 2

(3) One pi bond. DBE = 1 (4) Two rings. DBE = 2

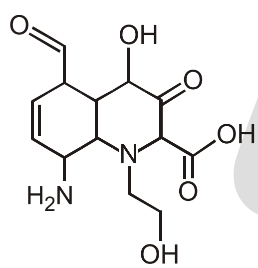
(5) One pi bonds and three rings. DBE = 4

(6) Three pi bonds and one ring in the middle and three pi bonds on substituents. DBE = 7 exercise

EXERCISE

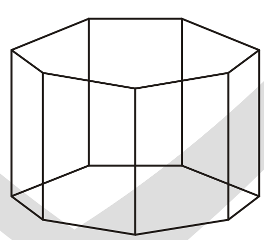
SINGLE CHOICE QUESTIONS

1. Find the sum of total number of different Functional groups and Double bond equivalent (DBE) value.



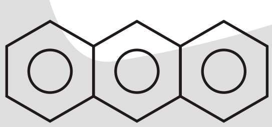
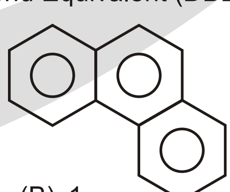
(A) 12 (B) 13 (C) 14 (D) 15

2. What is the Index of Hydrogen Deficiency (I.H.D) or Double Bond Equivalant (D.B.E.) for the following compound?



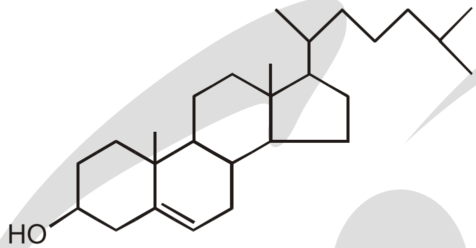
(A) 6 (B) 7 (C) 8 (D) 9

3. The difference in Double Bond Equivalent (DBE) value between

 and  is :

(A) 0 (B) 1 (C) 2 (D) 3

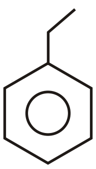
4. What is the correct molecular formula of following compound :

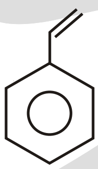


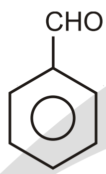
(Cholesterol)

(A) $C_{27}H_{46}O$ (B) $C_{25}H_{42}O$ (C) $C_{28}H_{46}O$ (D) $C_{23}H_{40}O$

5. Which of following compound. has D.B.E is 5 :

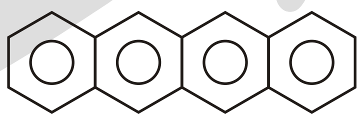
(A) 

(B) 

(C) 

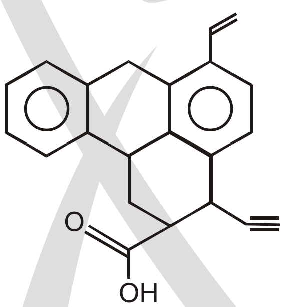
(D) Both (B) & (C)

6. Number of p-bond present in given compound is



(A) 8 (B) 9 (C) 10 (D) 12

7.



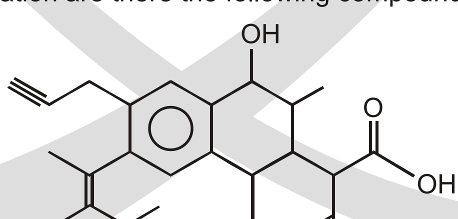
D.B. E of above compound is :

(A) 12 (B) 13 (C) 14 (D) 15

8. D.B.E of ($C_7H_5O_2$) is :

(A) 3 (B) 5 (C) 5.5 (D) 4.5

9. How many degrees of unsaturation are there the following compound?



(A) 6 (B) 7 (C) 10 (D) 11

10. How many elements of unsaturation are implied by the molecular formula C_6H_{12}?

(A) 0 (B) 1 (C) 2 (D) 3
(E) 4

11. How many elements of unsaturation are implied by the molecular formula C_5H_8O?

(A) 0 (B) 1 (C) 2 (D) 3
(E) 4

12. How many elements of unsaturation are implied by the molecular formula $C_7H_{11}Cl$?

(A) 0 (B) 1 (C) 2 (D) 3
(E) 4

13. How many elements of unsaturation are implied by the molecular formula $C_5H_5NO_2$?

(A) 0 (B) 1 (C) 2 (D) 3
(E) 4

14. How many elements of unsaturation are implied by the molecular formula $C_8H_{11}N$?

(A) 0 (B) 1 (C) 2 (D) 3
(E) 4

15. Consider molecules with the formula $C_{10}H_{16}$. Which of the following structural features are not possible within this set of molecules?

(A) 2 triple bonds
(B) 1 ring and 1 triple bond
(C) 2 rings and 1 double bond
(D) 2 double bonds and 1 ring
(E) 3 double bonds

16. A newly isolated natural product was found to have the molecular formula $C_{15}H_{28}O_2$. By hydrogenating a sample of the compound, it was determined to possess one -bond. How many rings are present in the compound?

(A) 0 (B) 1 (C) 2 (D) 3
(E) 4

17. Which of the following molecular formulas corresponds to a monocyclic saturated compound?

(A) C_6H_6 (B) C_3H_7Br (C) C_3H_7N (D) C_3H_8O
(E) C_3H_8O

MULTIPLE CHOICE QUESTIONS

1. Which of the following statements applies to $C_{10}H_{14}O_2$ compound?

(A) It may have 2 double bonds and 2 rings.
(B) It may have 3 double bond and Oxygen ring.
(C) It may have 1 triple bond and 2 rings.
(D) It may have zero double bond and 3 rings

UNSOLVED EXAMPLE

1. How many hydrogens does each of the following compounds have?

(a) $C_8H_?O_2$, has two rings and one double bond
(b) $C_7H_?N$, has two double bonds
(c) $C_9H_?NO$, has one ring and three double bonds

2. Calculate the degree of unsaturation in each of the following formulas :

(a) Cholesterol, $C_{27}H_{46}O$ (b) DDT, $C_{14}H_9Cl_5$
(c) Prostaglandin E_1, $C_{20}H_{34}O_5$ (d) Caffeine, $C_8H_{10}N_4O_2$
(e) Cortisone, $C_{21}H_{28}O_5$ (f) Atropine, $C_{17}H_{23}NO_3$

SUBJECTIVE TYPE QUESTIONS

1. The order of S_N1 reactivity in aqueous acetic acid solution for the compounds :

$\underset{(1)}{H_3C-\overset{\overset{\displaystyle O}{\|}}{C}-CH_2-Cl}$ $\quad \underset{(2)}{H_3C-CH_2-CH_2-Cl}$ $\quad \underset{(3)}{(H_3C)_3C-Cl}$

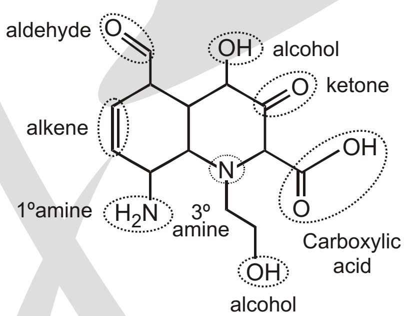
(a) 1 2 3 (b) 1 3 2 (c) 3 2 1 (d) 3 1 2

Answer

Single Choice Questions

1. (B)	**2.** (C)	**3.** (A)	**4.** (A)	**5.** (D)	**6.** (B)	**7.** (C)	**8.** (C)
9. (D)	**10.** (B)	**11.** (C)	**12.** (C)	**13.** (E)	**14.** (E)	**15.** (A)	**16.** (B)
17. (C)							

1. D.B.E. value = 6



Different functional group = 7

2. The molecular formula of the compound shown is $C_{14}H_{14}$ D.B.E. value (14 1) 14 / 2 8

3. D.B.E. of both anthracene & phenanthrene is 10.

4. Calculate DBE value of given compound DBE value of given compound is 5

Multiple Choice Questions

1. (A,B, C)

$C_{10}H_{14}O_2$, DBE (C 1) $\dfrac{H \quad X \quad N}{2}$

DBE (4) means = 2 double bonds + 2 rings
= 1 triple bond + 2 rings

Unsolved Example

1. (a) 12 (b) 13 (c) 13

2. (a) 5 (b) 8 (c) 4 (d) 6
(e) 8 (f) 7

Subjective Type Questions

1. (c)

❑❑❑

CHAPTER

6

Classification of Organic Compounds

CLASSIFICATION

The Organic compounds are classified as

- Organic Compounds
  - Open chain or acyclic
    - Saturated
    - Unsaturated
  - Cyclic
    - Homocyclic
      - Alicyclic
      - Aromatic
    - Heterocyclic

OPEN CHAINS

1. These compounds contain straight or branched chain of carbon atoms and are called as open chain or acyclic compounds.

   $CH_3-CH_2-CH_2-CH_3$ (open chain)

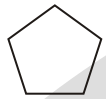
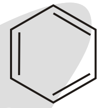
   n-butane

   $$\begin{array}{l} CH_3-CH-CH_3 \text{ (branched chain)} \\ \qquad\quad | \\ \qquad\ CH_3 \end{array}$$

   Isobutane

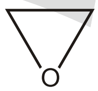
2. **Cyclic :** The compounds in which terminal carbon atoms join with each other to form ring like structures are called as cyclic or closed chain or ring compounds. These are of two types

(i) **Homocyclic compounds** where the atoms are all of similar type *e.g.*,

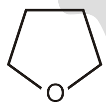



(ii) **Heterocyclic compounds** a wide variety of important organic compounds are derived from benzene, by replacing one of the hydrogens with a different functional group. They can have both common & systematic names.

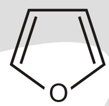
- Halogen-containing
- Hydrocarbon Derivatives
- Oxygen-containing
- Nitrogen-containing
- Sulfur-containing
- Polyaromatics



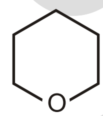
Ethylene Oxide
Oxirane
C_2H_4O



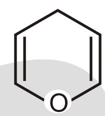
Tetrahydrofuran
Oxolane
C_4H_8O



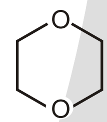
Furan
Oxole
C_2H_4O



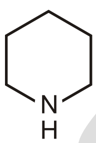
Tetrahydropyran
Oxane
$C_5H_{10}O$



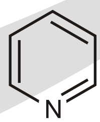
4H-Pyran
4H-oxine
C_5H_6O



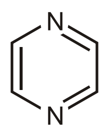
Ethylene Oxide
Oxirane
C_2H_4O



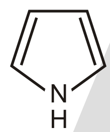
Piperidine
Azinane
$C_5H_{11}N$



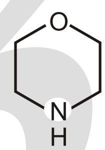
Pyridine
Azine
C_5H_5N



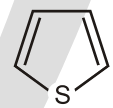
Pyrazine
1,4-diazine
$C_4H_4N_2$



Pyrrole
Azole
C_4H_5N



Morpholine

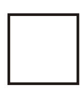


Thiophene
Thiole
C_4H_4S

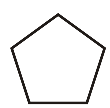
The cyclic compounds are further divided into two types :

(i) **Alicyclic compounds :** The cyclic compounds which resembles with open chains i.e. aliphatic compounds are called alicyclic compounds *e.g.*,

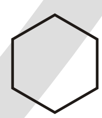
Cyclopropane



Cyclobutane

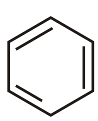


Cyclopentane

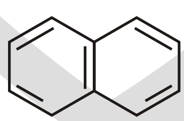


Cyclohexane

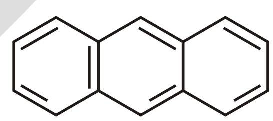
(ii) **Aromatic compounds :** In earlier days the compounds with pleasant smell were called aromatic compounds.



Benzene
(6π electrons)



Naphthalene
(10π electrons)



Anthralene
(14π electrons)

These are all aromatic compounds.

CHAPTER 7

Nomenclature of Alkanes

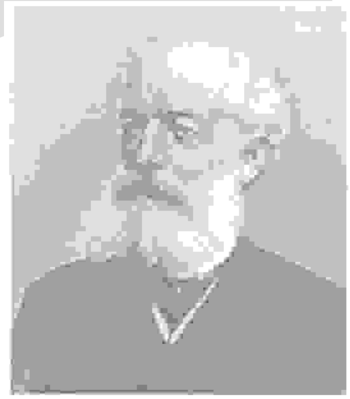
International Union of Pure and Applied Chemistry

I U P A C

Abbreviation	IUPAC
Motto	Advancing Chemistry Worldwide
Formation	1919
Type	International chemistry standards organization
Headquarters	Zurich, Switzerland
Region served	Worldwide
Official language	English
President	Mark Cesa
Website	www.iupac.org

The International Union of Pure and Applied Chemistry (IUPAC), is an international federation of National Adhering Organizations that represents chemists in individual countries. It is a member of the International Council for Science (ICSU). The international headquarters of IUPAC is in Zurich, Switzerland. The administrative office, known as the “IUPAC Secretariat”. Is in Research Triangle Park, North Carolina, United States. This administrative office is headed by the IUPAC executive director, currently Lynn Soby.

Creation and history



Friedrich August Kekule von Stradonitz

The need for an international standard for chemistry was first addressed in 1860 by a committee headed by German scientist Friedrich August Kekule von Stradonitz. This committee was the first internatioinal conference to create an international conference to create an international naming system for organic compounds. The ideas that were formulated in that conference evolved into the official IUPAC nomenclature of organic chemistry. The IUPAC stands as a legacy of this meeting, making it one of the most important historical international collaborations of chemistry societies. Since this time, IUPAC has been the official organization held with the responsibility of updating and maintaining official organic nomenclature. IUPAC as such was established in 1919. One notable country excluded from this early IUPAC was Germany. Germany's exclusion was a result of prejudice towards Germans by the allied powers after World War I. Germany was finally admitted into IUPAC during World War II.

During World war II, IUPAC was affiliated with the Allied powers, but had little involvement during the war effort itself. After the war, West Germany was allowed back into IUPAC. Since World War II, IUPAC has been focused on standardizing nomenclature and methods in science without interruption.

ALKANES

In organic chemistry, an **alkane**, or **paraffin** (a historical name that also has other meanings), is a saturated hydrocarbon. Alkanes consist only of hydrogen and carbon atoms and all bonds are single bonds. Alkanes (technically, always acyclic or open-chain compounds) have the general chemical formula $C_nH_{2n\ \ 2}$. For example, Methane is CH_4, in which $n\ \ \ 1$ (n being the number of carbon atoms).

☑ SPECIAL TOPIC

WHY ARE ALKANES CALLED PARRAFINS?

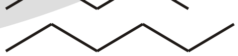
Paraffins is a latin word meaning (parum = little + affinis = reactivity). Alkanes are called paraffins because they have a little affinity towards a general reagent. In other words, alkanes are inert substances. They undergo reactions under drastic conditions.

A chemical name typically has four parts in the IUPAC system of nomenclature: prefix, parent, locant, and suffix. The prefix identifies the various substituent groups in the molecule, the parent selects a main part of the molecule and tells how many carbon atoms are in that part, the locants give the positions of the functional groups and substituents, and the suffix identifies the primary functional group.

Locant — Prefix — Parent — Suffix

Locant	Prefix	Parent	Suffix
Where are the substituents and functional groups?	What are the substituents?	How many carbons?	What is the primary functional group?

ACYCLIC HYDROCARBON

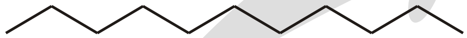
Methane	CH_4
Ethane	$H_3C—CH_3$
Propane	
Butane	
Pentane	
Hexane	

Heptane 

Octane 

Nonane 

Decane 

Undecane 

Dodecane 

NAMING CYCLOALKANES

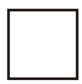
A ring is present → cyclo-

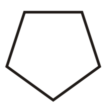
prefix + **cyclo-** + **parent** + **suffix**

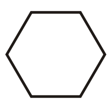
- prefix: What and where are the substituents?
- parent: How many C's are in the ring?
- suffix: What is the functional group?

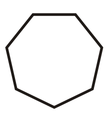
CYCLIC HYDROCARBON

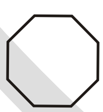
Cyclopropane 

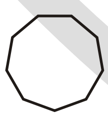
Cyclobutane 

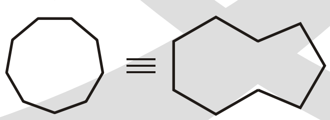
Cyclononane 

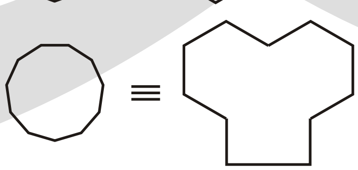
Cyclohexane 

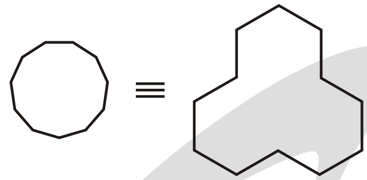
Cycloheptane 

Cyclooctane 

Cyclononane 

Cyclodecane 

Cycloundecane 

Cyclododecane 

IUPAC SYSTEM OF NAMING COMPOUNDS

The IUPAC name of any organic compound essentially consists of three parts:

1. Word root **2.** Suffix **3.** Prefix

WORD ROOT

Word root is the basic unit of the name denoting the number of C atoms present in the principal chain (longest possible continuous chain of C atoms including the functional group and multiple bonds).

For C_1 to C_4, the normal common root based on names like meth, -eth-, and prop- are used. For C_5 or more carbon atoms chain, an extra letter (a) is used only if the primary suffix to be added to word root begins with a consonant.

Straight-chain alkanes take the suffix "-ane" and are prefixed depending on the number of carbon atoms in the chain, following standard rules. The first few are :

Number of carbons	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	20	30	40	50
Prefix	meth	eth	Prop	But	Pent	Hex	Hept	Oct	Non	Dec	Undec	Dodec	Tridec	Tetradec	Pentadec	Eicos	Triacont	Tetracont	Pentacont

SUFFIX

Suffix are of two types.

(i) **Primary Suffix :** is always added to word root to indicate whether the carbon chain is saturated or unsaturated. For saturated the primary suffix is 'ane'. For unsaturated (one double bond) it is -'ene' and for unsaturated (one triple bond) it is - 'yne'. If the number of double bonds is two or three, then the primary suffix is 'diene' or 'triene'. If there are two triple bonds then it is 'diyne'.

Example:

HC   CH ethyne; CH_3 — CH CH_2 propene ; CH_2 CH — CH CH_2 Butadiene

('a' has been added to word root since primary suffix starts with a consonant 'd'.)

HC   C — C   CH Butadiyne

(ii) **Secondary Suffix:** is added to primary suffix to indicate the nature of the functional group present in an organic compound.

For alcohol (— OH), -ol is added; for aldehydes (— CHO), -al is added.

For ketones (>C = O), -one is added; for acids (— COOH) -oic acid is added.

While adding the secondary suffix (to represent the functional group), the terminal 'e' of primary suffix is dropped. For example, CH_3CH_2OH is ethanol (e-dropped) and CH_2 = CHCHO is prop-2-en-1-al (e-dropped). However, it is not always dropped. For example, CH_3CH_2CN is propanenitrile (e-not dropped)

It should also be noted that locants are to be placed immediately before the part of the name to which they relate. For example, $\overset{3}{\text{HC}}$ $\overset{2}{\text{C}}$ — $\overset{1}{\text{C}}$OOH is prop-2-yn-1-oic acid (working name 2-propynoic acid)

IUPAC Name

Cl$\overset{2}{\text{C}}H_2$ $\overset{1}{\text{C}}H_2$OH is 2-chloroethan-1-ol (working name 2-chloroethanol)

PREFIX

These are also of two types.

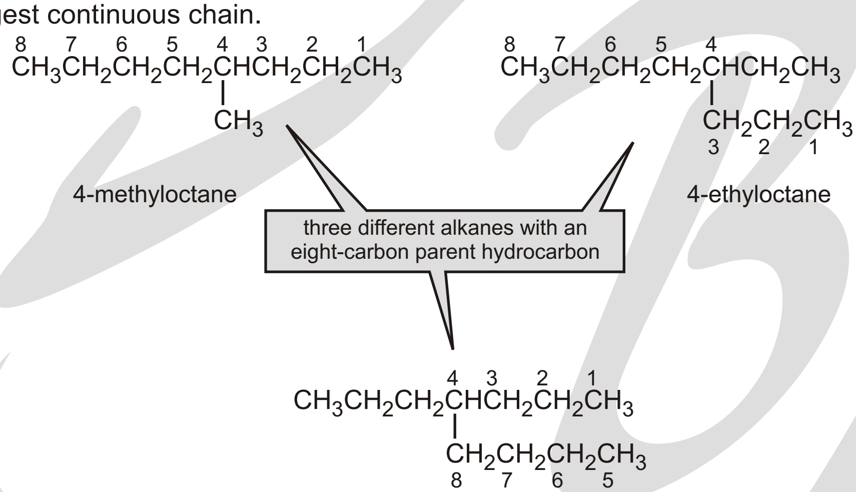
(i) Primary Prefix: It distinguishes between a cyclic and an acyclic compound. In a cyclic compound, the word 'cyclo' is used before the word root, for example,  is cyclopropane. If the prefix is not used, one can take the compound to be of open chain.

(ii) Secondary Prefix: Sometimes, certain groups are not considered functional groups. These are treated as substituents and added before the word root in an alphabetical order. For example, $C_2H_5 — O — C_2H_5$ is ethoxy ethane. In this, the secondary prefix is 'ethoxy', the word root is 'eth', and the primary suffix is 'ane'.

THE NOMENCLATURE OF ALKANES

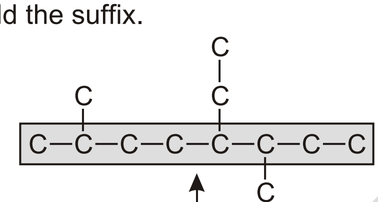
The systematic name of an alkane is obtained using the following rules:

Determine the number of carbons in the longest continuous carbon chain. This chain is called the parent hydrocarbon . The longest continuous chain is not always in a straight line; sometimes you have to "turn a corner" to obtain the longest continuous chain.



4-propyloctane

Find the parent carbon chain and add the suffix.

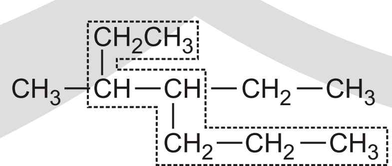


8 atoms in the longest chain

8 C's → octane

Rule 1 :

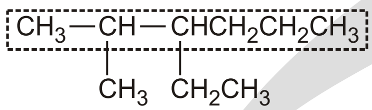
Longest chain rule : Select the longest continuous chain of carbon atoms. This is called the parent chain while all other carbon atoms which are not included in the parent chain are called branch chains or side chains or substituents.



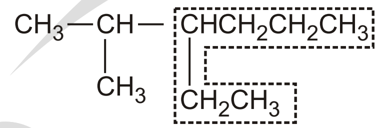
Longest chain contains seven carbon atoms and hence is named as a derivative of heptane

Rule 2 :

Rule for larger number of side chains : If two chains of equal lengths are possible, select the one with the larger number of side chains. For example,

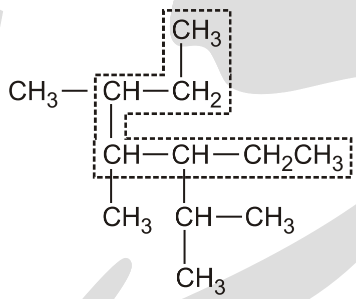


Named as hexane with **two** alkyl substituents (Correct)

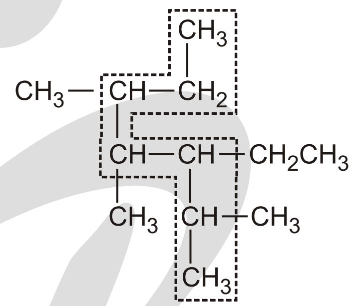


Named as hexane with **one** alkyl substituent (Wrong)

The series are called homologs. For example, butane is a homolog of propane, and both of these are homologs of hexane and decane.



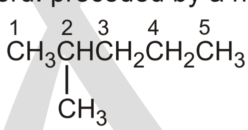
wrong
seven-carbon chain, but only three substituents



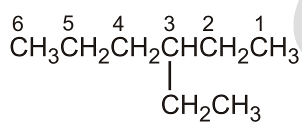
right
seven-carbon chain, four substituents

Rule 3 :

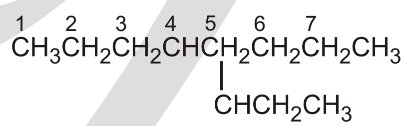
The name of any alkyl substituent that hangs off the parent hydrocarbon is placed in front of the name of the parent hydrocarbon, together with a number to designate the carbon to which the alkyl substituent is attached. The carbons in the parent chain are numbered in the direction that gives the substituent as low a number as possible. The substituents name and the name of the parent hydrocarbon are joined into one word. preceded by a hyphen that connects the substituent's number with its name.



2-methylpentane
not
4-methylpentane

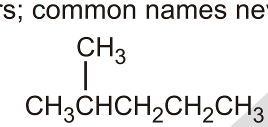


3-ethylhexane
not
4-ethylhexane



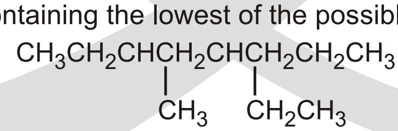
4-propyloctane
not
5-propyloctane

Only systematic names have numbers; common names never contain numbers



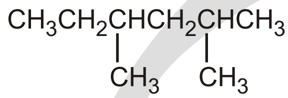
common name : isohexane
systematic names : 2-methypentane

If more than one substituent is attached to the parent hydrocarbon, the chain is numbered in the direction that will produce a name containing the lowest of the possible numbers.

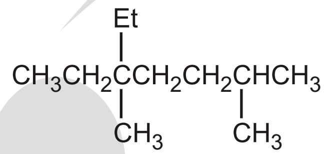


3-ethyl-5-methyloctane
not
4-ethyl-6-methyloctane
because 3 < 4

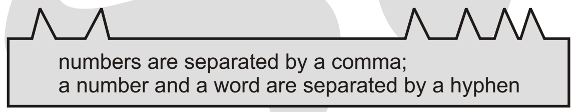
If two or more substituents are the same, the prefixes “di,” “tri,” and “tetra” are used to indicate how many identical substituents the compound has. The numbers indicating the locations of the identical substituents are listed together, separated by commas. There are no spaces on either side of a comma. There must be as many numbers in a name as there are substituents. The prefixes “di,” “tri,” “tetra,” “ sec ,” and “ tert ” are ignored in alphabetizing substituents.

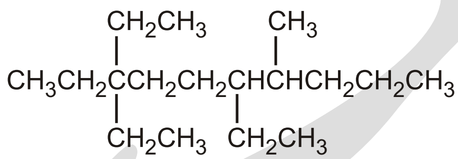


2,4-dimethylhexane

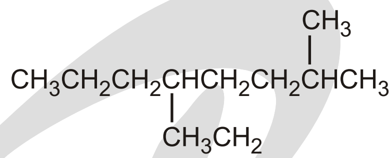


5-ethyl-2, 5-dimethylheptane





3,3,6-triethyl-7-methyl decane



5-ethyl-2-methyloctane

MISTAKES TO AVOID

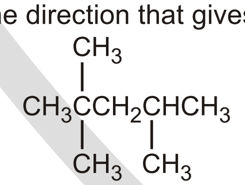
Adding of punctuation:

1. Commas are put between numbers (2, 5, 5 becomes 255)
2. Hyphens are put between a number and a letter (2,5,5 trimethylheptane becomes 2,5,5-trimethylheptane).
3. Successive words are merged into one word (trimethyl heptane becomes trimethylheptane)

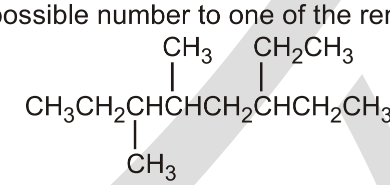
   ❑ **NOTE :** IUPAC uses one-word names throughout. This is why all parts are connected.
4. When assigning the numbers (*i.e.*, the locants) while naming an organic compound there is NO rule based on summing the numbers.

Rule 4 :

1. When numbering in either direction leads to the same lowest number for one of the substituents, the chain is numbered in the direction that gives the lowest possible number to one of the remaining substituents.



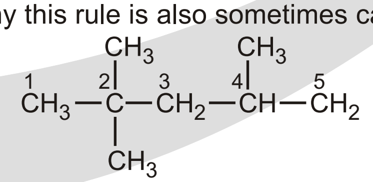
2,2,4-trimethylpentane
not
2,4,4-trimethylpentane
because 2 < 4



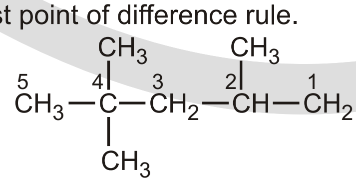
6-ethyl-3,4-dimethyloctane
not
3-ethyl-5,6-dimethyloctane
because 4 < 5

2. **Lowest set of locants rule :** *When two or more substituents are present, the lowest set of locants rule is applied*. According to this rule when two or more different sets of locants containing the same number of terms is possible, then that set of locants is the lowest which when compared term by term with other sets, each in order of increasing magnitude, has the lowest term at the first point of difference.

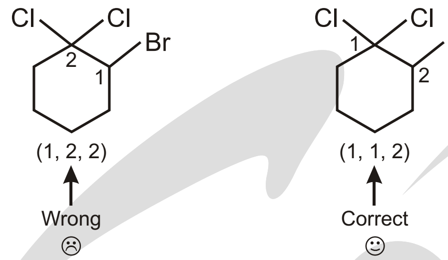
   That is why this rule is also sometimes called as first point of difference rule.



Set of locants = 2, 2, 4 (correct)

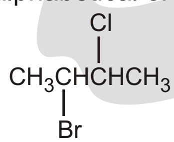


Set of locants = 2, 4, 4 (wrong)

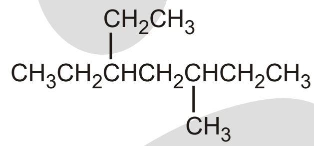


Rule 5 :

If the same substituent numbers are obtained in both directions, the first group listed receives the lower number (according to alphabetical order).



2-bromo-3-chlorobutane
not
3-bromo-2-chlorobutane



3-ethyl-5-methylheptane
not
5-ethyl-3-methylheptane

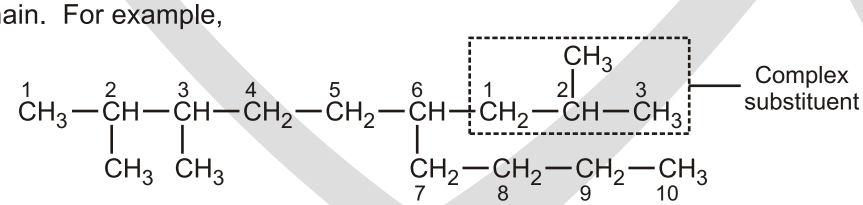
Rule 6 :

NAMING OF COMPLEX SUBSTITUENT

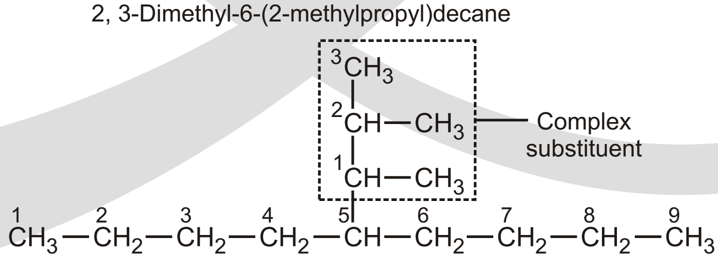
Alkyl group, R-	**Common name**	**Complex name** (if different)
CH_3-	methyl-	
CH_3CH_2-	ethyl-	
$CH_3CH_2CH_2$-	propyl-	
$(CH_3)_2CH$-	isopropyl-	(1-methylethyl)-
$CH_3CH_2CH_2CH_2$-	butyl-	
$CH_3 — CH_2 \underset{\mid}{C}H — CH_3$	sec-butyl- or s-butyl	(1-methylpropyl)-
$(CH_3)_2CHCH_2$-	isobutyl	(2-methylpropyl)-
$(CH_3)_3C$-	tert-butyl or t-butyl	(1,1-dimethylethyl)-

NUMBERING THE COMPLEX SUBSTITUENT

(a) In case the substituent on the parent chain is complex (*i.e.*, it has branched chain), it is named as a substituted alkyl group by numbering the carbon atom of this group attached to the parent chain as 1. The name of such a substituent is always enclosed in brackets to avoid confusion with the numbers of the parent chain. For example,



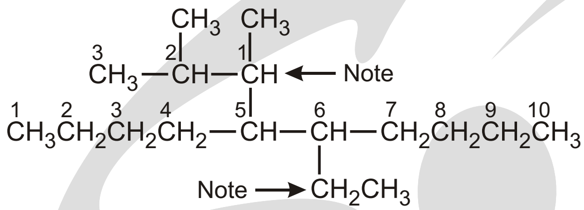
2, 3-Dimethyl-6-(2-methylpropyl)decane



5-(1, 2-Dimethylpropyl) nonane

(b) While deciding the alphabetical order of the various substituents, the name of the complex substituent is considered to begin with the first letter of the complete name, eventhough di, tri, iso, neo.

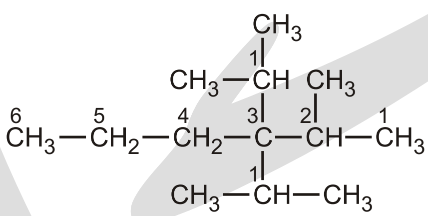
Solved Example



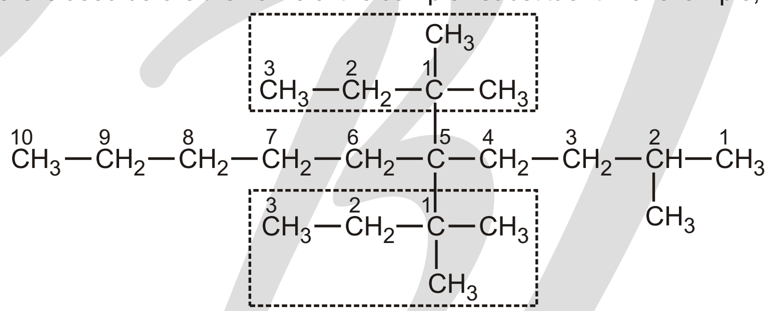
5-(1, 2-Dimethylpropyl)-6-ethyldecane

It may be noted here that the complete name of the complex substituent is dimethylbutyl. Since d of dimethylbutyl group comes first than e of the ethyl group in the alphabetical order, therefore, locant 5 is given to the complex substituent and 6 to the ethyl group.

(c) If the same complex substituent occurs more than once on the parent chain, prefixes bis (for two), tris (for three), tetrakis (for four), pentakis (for five) etc. are used before the name of the complex substituent. For example,



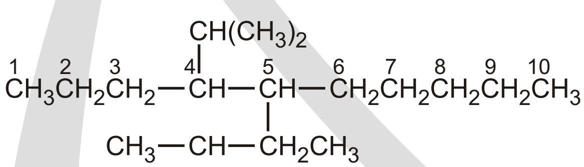
2-Methyl-3, 3-bis(1-methylethyl)hexane



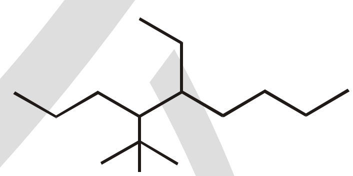
5, 5-Bis (1, 1-dimethylpropyl)-2-methyldecane

☞ Note that while deciding the alphabetical order of the various alkyl groups, **prefixes iso and neo are considered to be part of the fundamental name of the alkyl group while the prefixes sec, tert, di, tri are not.**

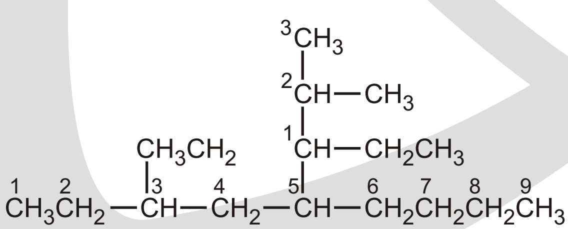
Solved Example



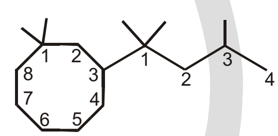
4-(Methylethyl)-5-(Methylpropyl)decane



4-(1,1-dimethylethyl)-5-ethylnonane
not 5-ethyl-4-(1,1-dimethylethyl) nonane



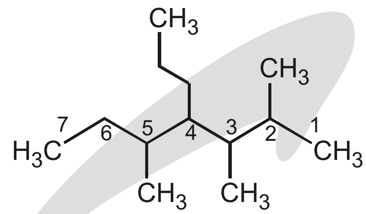
3-ethyl-5-(1-ethyl-2-methylpropyl)nonane



1,1 dimethyl-3-(1, 1,3-trimethylbutyl)cycloctane

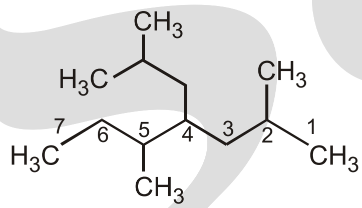
☞ If Chains of equal length are competing for selection as main chain in a saturated branched acyclic hydrocarbon, then the choice goes in series to :

(a) The chain which has the greatest number of side chains.



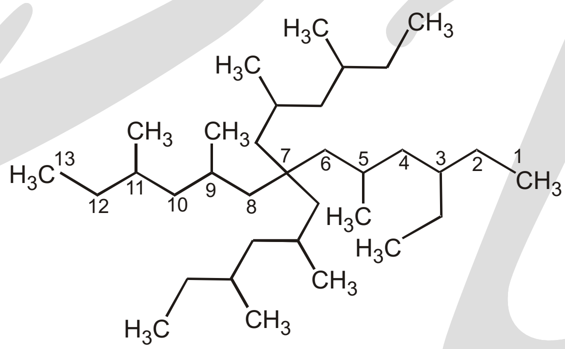
2,3,5-Trimethyl-4-propylheptane

(b) The chain whose side chains have the lowest-numbered locants.



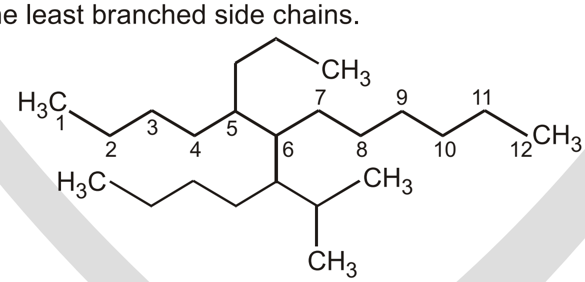
4-Isobutyl-2,5-dimethyl heptane

(c) The chain having the greatest number of carbon atoms in the smaller side chains.

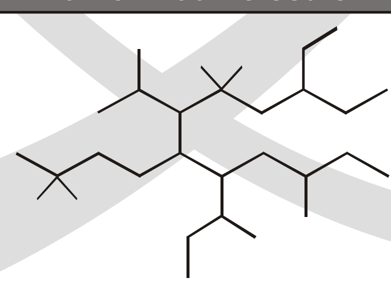


7,7,Bis(2,4-dimethylhexyl)-3-ethyl 5, 9, 11-trimethyl tridecane

(d) The chain having the least branched side chains.

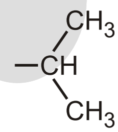
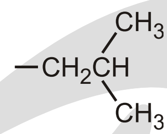
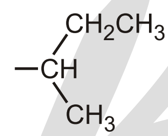
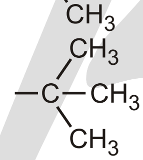
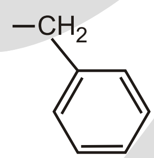
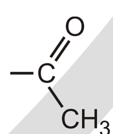
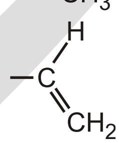
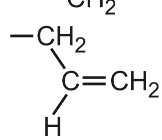


Name That Molecule !



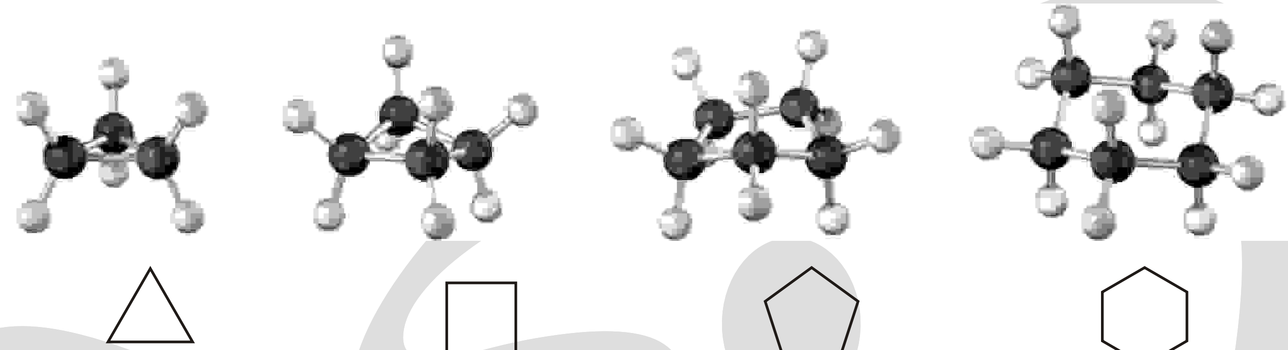
8-sec-butyl-3-ethyl-6-isopropyl-5,5, 10-trimethyl-7-neohexyldodecane
or
8-sec-butyl-7-(3, 3-dimethylbutyl)-3-ethyl-6-(1-methylethyl)-5,5,10-trimethyldodecane

COMMON NAMES AND STRUCTURAL ABBREVIATIONS

Group	Abbreviation	Structure
Alkyl	R	
Aryl	Ar	
Methyl	Me	$-CH_3$
Ethyl	Et	$-CH_2CH_3$
Propyl	Pr or n-Pr	$-CH_2CH_2CH_3$
Butyl	Bu or n-Bu	$-CH_2CH_2CH_2CH_3$
Isopropyl	i-Pr or ^{i}Pr	
Isobutyl	i-Bu or ^{i}Bu	
sec-Butyl	s-Bu or ^{s}Bu	
tert-Butyl	t-Bu or ^{t}Bu	
Phenyl	Ph	
Benzyl	Bn	
Acetyl	Ac	
Vinyl		
Allyl		
Halide	X	–F –Cl –Br –I

NOMENCLATURE OF CYCLIC ALKANE

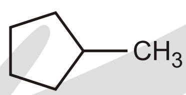
Saturated cyclic hydrocarbons are called cycloalkanes, or alicyclic compounds (aliphatic cyclic). Because cycloalkanes consist of rings of $-CH_2-$units, they have the general formula $(CH_2)_n$, or C_nH_{2n}, and can be represented by polygons in skeletal drawings.



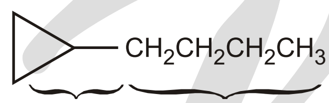
Cyclopropane   Cyclobutane   Cyclopentane   Cyclohexane

Find the parent.

Count the number of carbon atoms in the ring and the number in the largest substituent. If the number of carbon atoms in the ring is equal to or greater than the number in the substituent, the compound is named as an alkyl-substituted cycloalkane. If the number of carbon atoms in the largest substituent is greater than the number in the ring, the compound is named as a cycloalkyl-substituted alkane. For example:

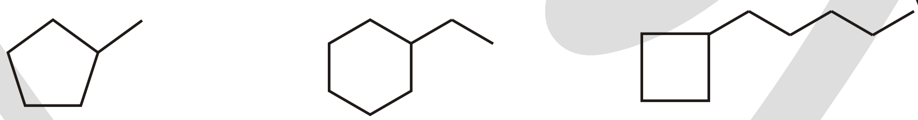


Methylcyclopentane



1-Cyclopropylbutane

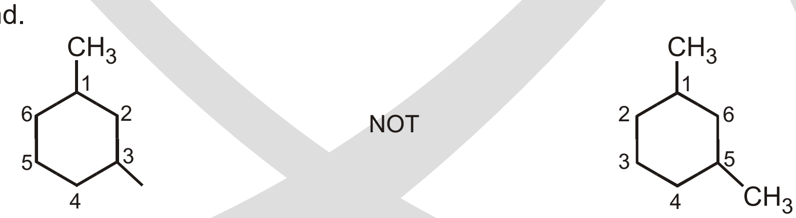
☞ In a cycloalkane with an attached alkyl substituent, the ring is the parent hydrocarbon unless the substituent has more carbons than the ring. In that case, the substituent is the parent hydrocarbon and the ring is named as a substituent. There is no need to number the position of a single substituent on a ring.



methylcyclopentane   ethylcyclohexane   1-cyclobutylpentane

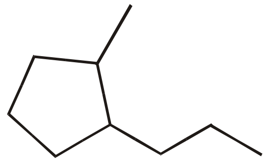
Number the substituents, and write the name.

☞ For an alkyl- or halo-substituted cycloalkane, choose a point of attachment as carbon 1 and number the substituents on the ring so that the second substituent has as low a number as possible. If ambiguity still exists, number so that the third or fourth substituent has as low a number as possible, until a point of difference is found.

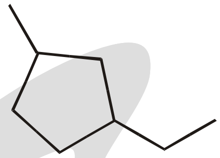


1,3-Dimethyl cyclohexane ↑ Lower   1,5-Dimethyl cyclohexane ↑ Higher

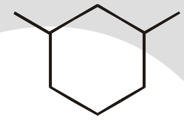
☞ If the ring has two different substituents, they are listed in alphabetical order and the number-1 position is given to the substituent listed first.



1-methyl-2-propylcyclopentane



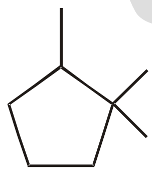
1-ethyl-3-methylcyclopentane



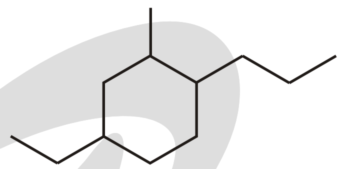
1,3-dimethylcyclohexane

☞ If there are more than two substituents on the ring, they are listed in alphabetical order, and the substituent given the number-1 position is the one that results in a second substituent getting as low a number as possible. If two substituents have the same low numbers, the ring is numbered—either clockwise or counterclockwise—in the direction that gives the third substituent the lowest possible number.

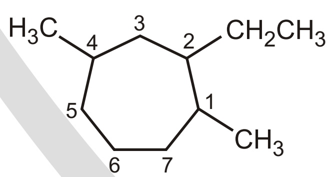
Solved Example



1,1,2-trimethylcyclopentane
not
1,2,2-trimethylcyclopentane
because 1 < 2
not
1,1,5-trimethylcyclopentane
because 2 < 5



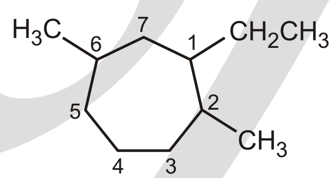
4-ethyl-2-methyl-1-propylcyclohexane
not
1-ethyl-3-methyl-4-propylcyclohexane
because 2 < 3
not
5-ethyl-1-methyl-2-propylcyclohexane
because 4 < 5



2-Ethyl-1,4-dimethyl cycloheptane

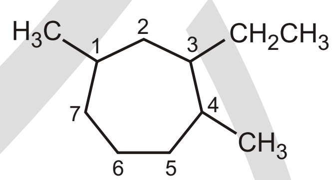
Lower Lower

NOT



1-Ethyl-2,6-dimethylcycloheptane

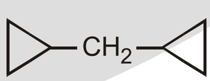
Higher



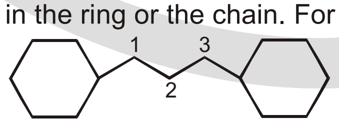
3-Ethyl-1,4-dimethylcycloheptane

Higher

☞ If more than one alicyclic ring is attached to a single chain, the compound is named as a derivative of alkane irrespective of the number of carbon atoms in the ring or the chain. For example,

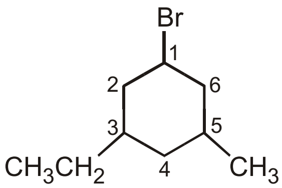


dicyclopropylmethane

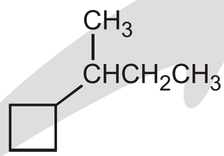


1,3-dicyclohexylpropane

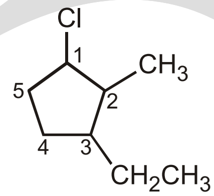
Solved Example



1-Bromo-3-ethyl-5-methyl cyclohexane



(1-Methylpropyl) cyclobutane or sec-butyl cyclobutane



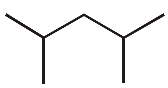
1-Chloro-3-ethyl-2-methyl-cyclopentane

Solved Problems

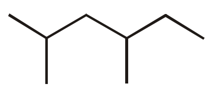
- Select the correct structure for the IUPAC names given below :

Part 1	Part 2	Part 3
4-methylheptane	2,4-dimethylhexane	2,3-dimethylpentane
Part 4	Part 5	Part 6
2,2-dimethylhexane	3-ethyl-1,1,2,2-tetramethylcyclopentane	3-ethyl-2-methylpentane
Part 7	Part 8	Part 9
2,5-dimethylhexane	1,1-dimethylcyclohexane	2,4,6-trimethylheptane

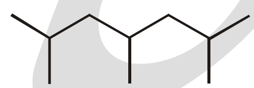
Structures:



A

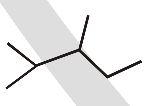


B

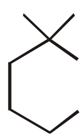


C

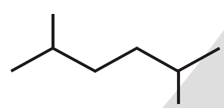
D



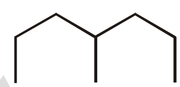
E



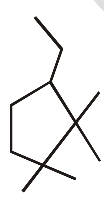
F



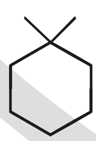
G



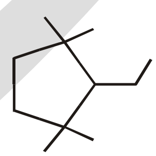
H



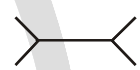
I



J



K



L

Sol.

Part - 1 H	Part - 2 B	Part - 3 E	Part - 4 F
Part - 5 I	Part - 6 D	Part - 7 G	Part - 8 J
Part - 9 C			

Solved Example

- Deduce structural formulas and give IUPAC names for the nine isomers of C_7H_{16}. (b) Why is 2-ethylpentane not among the nine ?

(a) seven-C chain	1. $CH_3CH_2CH_2CH_2CH_2CH_2CH_3$	Heptane
six-C chain	2. $CH_3\overset{\overset{CH_3}{|}}{C}HCH_2CH_2CH_2CH_3$	2-Methylhexane
	3. $CH_3CH_2\overset{\overset{CH_3}{|}}{C}HCH_2CH_2CH_3$	3-Methylhexane
five-C chain	4. $CH_3CH\overset{\overset{CH_3}{|}}{C}HCH_2CH_3$	2,3-Dimethylpentane
	5. $CH_3\overset{\overset{CH_3}{|}}{C}HCH_2\overset{\overset{CH_3}{|}}{C}HCH_3z$	2,4-Dimethylpentane
	6. $CH_3\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}CH_2CH_2CH_3$	2,2-Dimethylpentane

SPECIAL TOPIC

OCTANE NUMBER

In addition to being volatile, gasoline must resist the potentially damaging explosive combustion known as knocking. The antiknock properties of gasoline are rated by an octane number that is assigned by comparing the gasoline to a mixture of *n*-heptane (which knocks badly) and isooctane (2,2,4-trimethylpentane, which is not prone to knocking). The gasoline being tested is used in a test engine with a variable compression ratio. Higher compression ratios induce knocking, so the compression ratio is increased until knocking begins. Tables are available that show the percentage of isooctane in an isooctane/heptane blend that begins to knock at any given compression ratio. The octane number assigned to the gasoline is simply the percentage of isooctane in an isooctane/heptane mixture that begins to knock at that same compression ratio.

$$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$$

n-heptane (0 octane)
prone to knocking

$$CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_2-\overset{\overset{CH_3}{|}}{CH}-CH_3$$

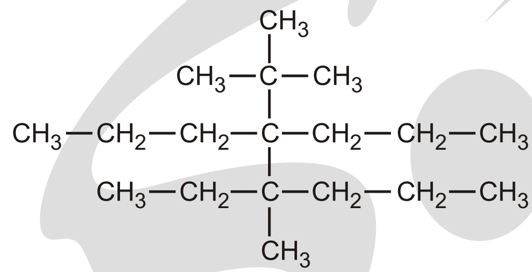
2,2,4-trimethylpentane (100 octane)
"isooctane," resists knocking

The octane number of a gasoline is determined by comparing its knocking with the knocking of mixtures of heptane and 2,2,4-trimethylpentane. The octane number given to the gasoline corresponds to the percent of 2,2,4-trimethylpentane in the matching mixture. Thus, a gasoline with an octane rating of 91 has the same "knocking" property as a mixture of 91% 2,2,4-trimethylpentane and 9% heptane. The term octane number originated from the fact that 2,2,4-trimethylpentane contains eight carbons. Because slightly different methods are used to determine the octane number, gasoline in Canada and the United States will have an octane number that is 4 to 5 points less than the same gasoline in Europe and Australia.

EXERCISE

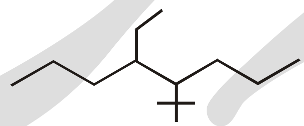
SINGLE CHOICE QUESTIONS

1. Total number of carbon atoms present in parent chain is :



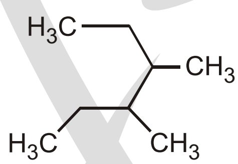
(A) 5 (B) 6 (C) 7 (D) None of these

2. Correct IUPAC name of the following compound is :

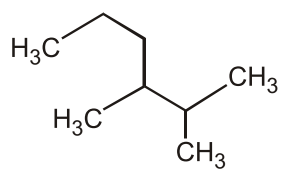


(A) 4-(1,1-dimethylethyl)-5-ethyloctane
(B) 5-ethyl-4-(1,1-dimethylethyl)octane
(C) 5-(1,1-dimethylethyl)-4-ethyloctane
(D) 5-ethyl-5-(1,1-dimethylethyl)octane

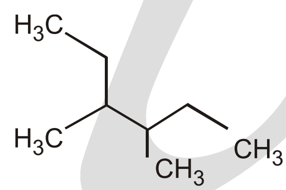
3. Among the following compounds



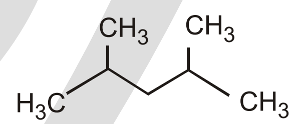
(i)



(ii)



(iii)

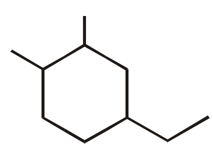


(iv)

2,3-dimethylhexane is :

(A) (i) (B) (ii) (C) (iii) (D) (iv)

4. What would be the best name for the following compound ? (Neglect any cis-trans isomerism that is possible.)



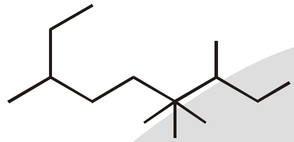
(A) 1-ethyl-3,4-dimethylcyclohexane
(B) 3-ethyl-1,6-dimethylcyclohexane
(C) 1-ethyl-4,5-dimethylcyclohexane
(D) 5-ethyl-1,2-dimethylcyclohexane
(E) 4-ethyl-1,2-dimethylcyclohexane

5. The correct IUPAC name of $CH_3CH_2CH(CH_3)CH(C_2H_5)_2$ is :

(A) 4-ethyl - 3-methyl hexane
(B) 3-ethyl - 4 methyl hexane
(C) 4-methyl - 3-ethyl hexane
(D) 2, 4-diethylpentane

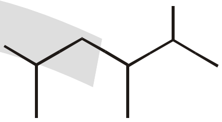
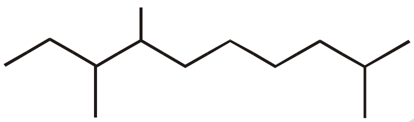
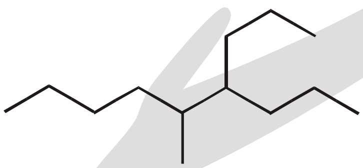
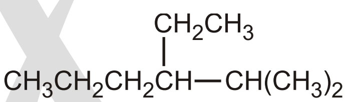
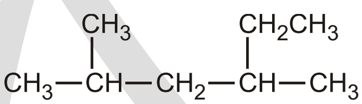
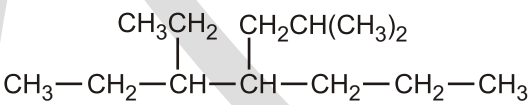
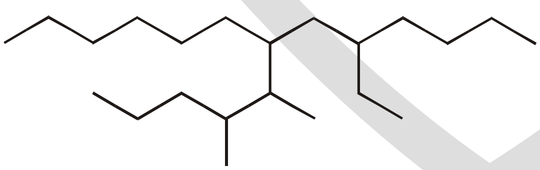
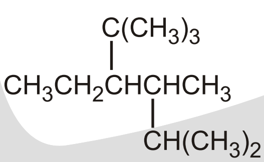
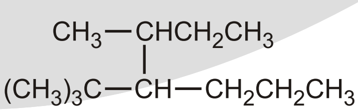
6. What is the parent name for the following alkane ?

(**Note :** The parent name corresponds to the longest continuous chain of carbon atoms.)



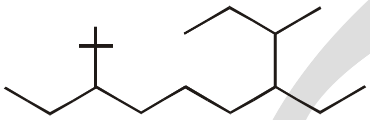
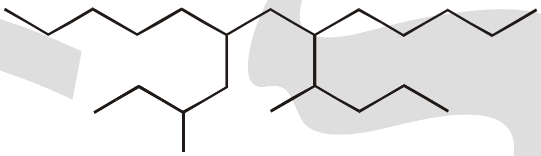
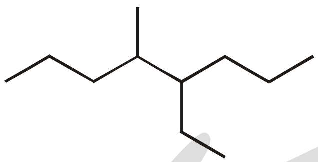
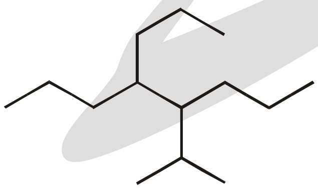
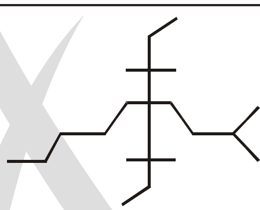
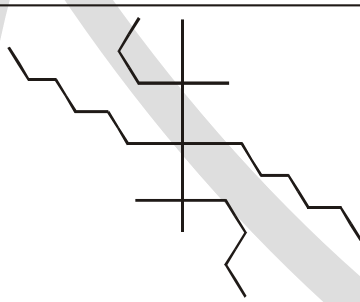
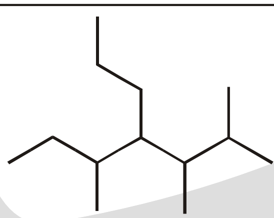
(A) Heptane (B) Octane (C) Nonane (D) Decane

WORK SHEET - 1

S.N.	Compound	Number of carbon atom in Parent chain
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		
10.		

11.	$CH_3C(CH_3)_2CH(CH_2CH_3)CH_2CH_2CH(CH_3)_2$

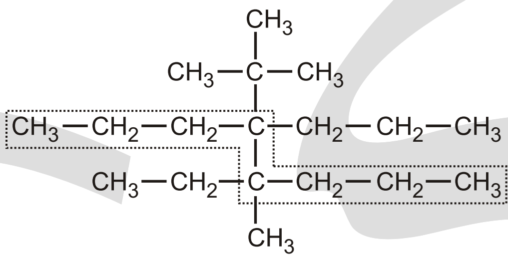
WORK SHEET - 2

S.N.	Compound	Write IUPAC - Name
1.		
2.		
3.		
4.		
5.		
6.		
7.		

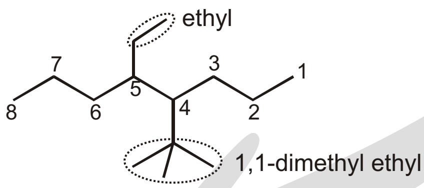
Answer

1. (D) **2.** (A) **3.** (B) **4.** (E) **5.** (B) **6.** (C)

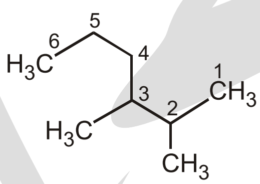
Single Choice Questions

1. 

8 Carbons in parent chain

2. 

4-(1,1-dimethylethyl)-5-ethyloctane

3. 

2, 3-Dimethyl hexane

Work Sheet - 1

1. 6 **2.** 10 **3.** 9 **4.** 7 **5.** 6 **6.** 6 **7.** 7 **8.** 13
9. 6 **10.** 7 **11.** 7

Work Sheet - 2

1. 3, 7-Diethyl - 2, 2,8-trimethyl decane
2. 6-(1-methylbutyl) - 8-(2-methylbutyl) tridecane
3. 4-ethyl - 5 - methyl octane
4. 4-(1-methyl ethyl) - 5-propyl octane **or** 4-isopropyl-5-propyl octane
5. 5, 5-Bis (1, 1-dimethyl propyl)-2-methyl- decane
6. 7-(1,1-dimethyl butyl)-7-(1,1-dimethyl pentyl) tridecane
7. 2, 3, 5-trimethyl - 4-propyl heptane

❑❑❑

CHAPTER

8

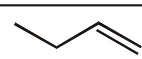
Nomenclature of Alkenes & Alkynes

ALKENES

Alkenes are named using a series of rules similar to those for alkanes with the suffix -ene used instead of -ane to identify the functional group. There are two steps.

1. **Name the parent hydrocarbon:** Find the longest carbon chain containing the double bond.
2. **Number the carbon atoms in the chain:** Begin at the end nearer the double bond or, if the double bond is equidistant from the two ends, begin at the end nearer the first branch point. This rule ensures that the doublebond carbons receive the lowest possible numbers.
   Functional group suffix = **–ene**
   Substituent name = **alkenyl**
   Structural unit : alkenes contain C = C bonds.

Solved Example



$CH_3CH_2CH{=}CH_2$

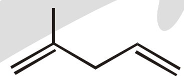
- Functional group is an alkene, therefore suffix = **–ene**
- The longest continuous chain is C4 therefore root = but
- In order to give the alkene the lowest number, number from the right as drawn.
- The C = C is between C1 and C2 so the locant is **but-1-ene** or **1-butene.**

$$\overset{1}{C}H_3\overset{2}{C}H{=}\overset{3}{C}H\overset{4}{C}H(CH_3)\overset{5}{C}H_3$$

4-methyl-2-pentene

☞ For a compound with two double bonds, the “ne” ending of the corresponding alkane is replaced with “diene.”

Solved Example

▸ 

$CH_3C(CH_3){=}CHCH{=}CH_2$

- Functional group is an alkene, therefore suffix = **–ene.**
- There are two alkenes, so insert the multiplier di.
- The longest continuous chain is C5 therefore root plus "a" = penta.
- The substituent is a C1 alkyl group i.e. a methyl group.
- The first point of difference doesn't distinguish the C = C.
- So, need to apply the first point of difference to the alkyl substituent.
- The first point of difference requires that we number from the left as drawn.
- The methyl group locant is 2-.
- Therefore the locants for C = C units are 1- and 4-

Name : 2-methylpenta-1,4-diene or 2-methyl-1,4-pentadiene

Solved Example

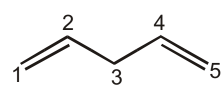
▸

$\overset{1}{C}H_3\overset{2}{C}H{=}\overset{3}{C}H{-}\overset{4}{C}H{=}\overset{5}{C}H\overset{6}{C}H_2\overset{7}{C}H_3$

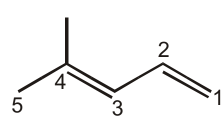
2, 4-heptadiene

$\overset{5}{C}H_3\overset{4}{C}H{=}\overset{3}{C}H{-}\overset{2}{C}H{=}\overset{1}{C}H_2$

1, 3-pentadiene



1,4-pentadiene



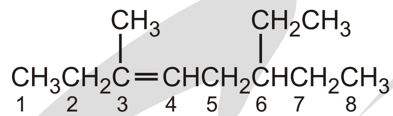
4-methyl-1,3-pentadiene

☞ Groups which always have less priority than multiple bonds :

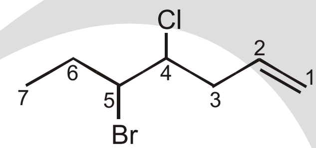
1.	– F	Fluoro	**2.**	– Cl	Chloro
3.	– Br	Bromo	**4.**	– I	Iodo
5.	– NO_2	Nitro	**6.**	– NO	Nitroso
7.	– OR	Alkoxy	**8.**	– OCH_3	Methoxy
9.	– OEt	ethoxy	**10.**	– OPh	Phenoxy
11.	– R	Alkyl	**12.**	– N_3	Azido

Solved Example

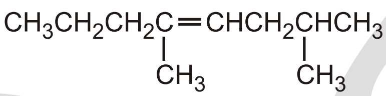
3-bromoprop-1-ene



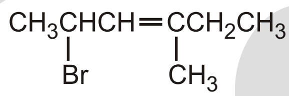
6-ethyl-3-methyl-3-octene



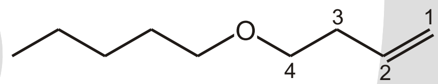
5-bromo-4-chloro-1-heptene



2,5-dimethyl-4-octene
not
4,7-dimethyl-4-octene
because 2 < 4



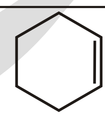
2-bromo-4-methyl-3-hexene
not
5-bromo-3-methyl-3-hexene
because 2 < 3



4-pentoxy-1-butene

NOMENCLATURE OF CYCLIC ALKENE

Solved Example

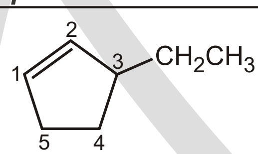


- Functional group is an alkene, therefore suffix = **–ene.**
- The longest continuous chain is C6 therefore root = **hex.**
- The C = C is unambiguously between C1 and C2 therefore the locant isn't required.

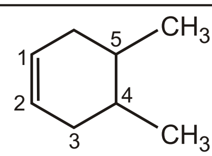
Name: Cyclohexene

☞ A number is not needed to denote the position of the double bond in a cyclic alkene because the ring is always numbered so that the double bond is between carbons 1 and 2. To assign numbers to any substituents, count around the ring in the direction (clockwise or counterclockwise) that puts the lowest number into the name.

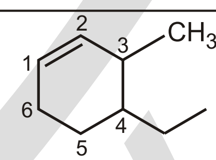
Solved Example



3-ethylcyclopentene



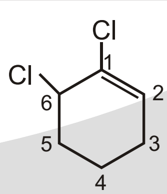
4,5-dimethylcyclohexene



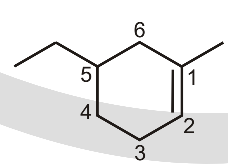
4-ethyl-3-methylcyclohexene

❑ **NOTE :** 1,6-dichlorocyclohexene is not called 2,3-dichlorocyclohexene because the former has the lowest substituent number (1), even though it does not have the lowest sum of substituent numbers (1 6 7 versus 3 2 5).

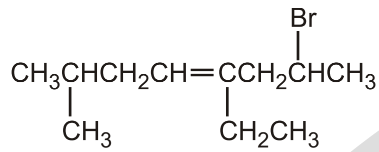
Solved Example



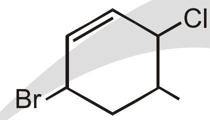
1,6-dichlorocyclohexene
not
2,3-dichlorocyclohexene
because 1 < 2



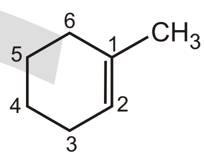
5-ethyl-1-methylcyclohexene
not
4-ethyl-2-methylcyclohexene
because 1 < 2



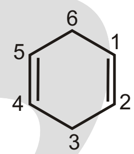
2-bromo-4-ethyl-7-methyl-4-octene
not
7-bromo-5-ethyl-2-methyl-4-octene
because 4 < 5



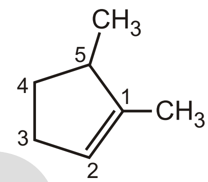
6-bromo-3-chloro-4-methylcyclohexene
not
3-bromo-6-chloro-5-methylcyclohexene
because 4 < 5



1-Methyl cyclohexene



1,4-Cyclohexadiene
(New: Cyclohexa-1,4-diene)

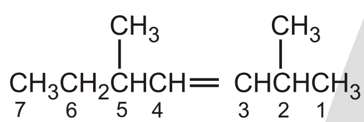


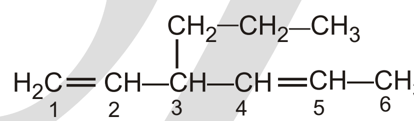
1,5-Dimethylene cyclopent

☞ We should also note that IUPAC changed their naming recommendations in 1993 to place the locant indicating the position of the double bond immediately before the -ene suffix rather than before the parent name: but-2-ene rather than 2-butene, for instance. This change has not been widely accepted by the chemical community in the United States, however, so we'll stay with the older but more commonly used names. Be aware, though, that you may occasionally encounter the newer system.

Solved Example

▶





Older naming system :	2,5-Dimethyl-3-heptene	3-Propyl-1,4-hexadiene
(Newer naming system:	2,5-Dimethylhept-3-ene	3-Propylhexa-1,4-diene

Common names of Some Alkenes

Compound	Systematic name	Common name
$H_2C \quad CH_2$	Ethene	Ethylene
$CH_3CH \quad CH_2$	Propene	Propylene
CH_3 | $CH_3C{=}CH_2$	2-Methylpropene	Isobutylene
CH_3 | $H_2C{=}C{-}CH{=}CH_2$	2-Methyl-1,3-butadiene	Isoprene

ALKENES AS SUBSTITUENTS

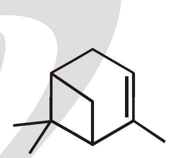
- In some cases, a group containing an alkene may need to be treated as a substituent.
- In these cases the substituent is named in a similar fashion to simple alkyl substituents.

Alkenyl group	Common name	Systematic name
CH_2 CH—	Vinyl-	ethenyl
CH_2 $CHCH_2$—	allyl-	2-propenyl
CH_3CH CH—	—	1-propenyl

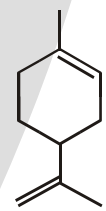
☑ SPECIAL TOPIC

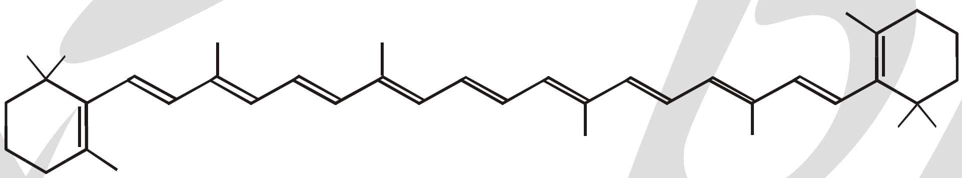
ALKENES (SOMETIMES CALLED OLEFINS) CONTAIN C C DOUBLE BONDS

It may seem strange to classify a type of bond as a functional group, but you will see later that C C double bonds impart reactivity to an organic molecule just as functional groups consisting of, say, oxygen or nitrogen atoms do. Some of the compounds produced by plants and used by perfumers are alkenes (see Chapter 1). For example, pinene has a smell evocative of pine forests, while limonene smells of citrus fruits.



α-pinene





β-carotene

ALLYLIC AND VINYLIC CARBON

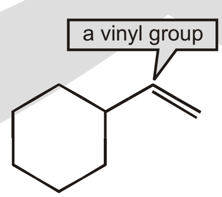
The sp^2 carbons of an alkene are called vinylic carbons . An sp^3 carbon that is adjacent to a vinylic carbon is called an allylic carbon. A hydrogen bonded to a vinylic carbon is called a vinylic hydrogen , and a hydrogen bonded to an allylic carbon is called an allylic hydrogen.

vinylic carbons

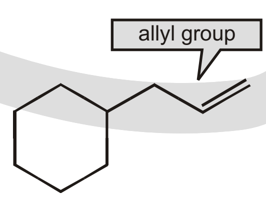
$$RCH_2—CH=CH—CH_2R$$

allylic carbons

	$CH_2=CHCl$	$CH_2=CHCH_2Br$
common name:	vinyl chloride	allyl bromide
systematic name:	chloroethene	3-bromopropene



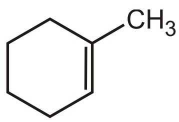
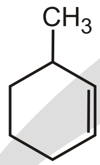
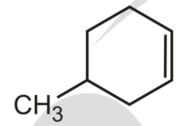
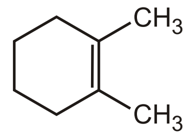
vinylcyclohexane



allylcyclohexane

Special Example

- How many carbons are in the planar double-bond system in each of the following compounds?

(a)  (b)  (c)  (d) 

Sol. (a) 5 (b) 4 (c) 4 (d) 6

THE NOMENCLATURE OF ALKYNES

Because of its triple bond, an alkyne has four fewer hydrogens than an alkane with the same number of carbons. Therefore, while the general molecular formula for an acyclic alkane is $C_nH_{2n\ 2}$, the general molecular formula for an acyclic alkyne is $C_nH_{2n\ 2}$ and that for a cyclic alkyne is $C_nH_{2n\ 4}$.

The systematic name of an alkyne is obtained by replacing the “ane” ending of the alkane name with “yne.” Analogous to the way compounds with other functional groups are named, the longest continuous chain containing the carbon–carbon triple bond is numbered in the direction that gives the functional group suffix as low a number as possible. If the triple bond is at the end of the chain, the alkyne is classified as a terminal alkyne . Alkynes with triple bonds located elsewhere along the chain are internal alkynes.

		an terminal alkyne	an internal alkyne	
	$HC{\equiv}CH$	$\overset{4}{C}H_3\overset{3}{C}H_2\overset{2}{C}{\equiv}\overset{1}{C}H$	$\overset{1}{C}H_3\overset{2}{C}{\equiv}\overset{3}{C}\overset{4}{C}H_2\overset{5}{C}H_3$	$CH_3\overset{4}{C}H(\overset{5}{C}H_2\overset{6}{C}H_3)\overset{3}{C}{\equiv}\overset{2}{C}\overset{1}{C}H_3$
Systematic:	ethyne	1-butyne	2-pentyne	4-methyl-2-hexyne
Common:	acetylene	ethylacetylene	ethylmethylacetylene	*sec*- butylmethylacetylene

☞ If counting from either direction leads to the same number for the functional group suffix, the correct systematic name is the one that contains the lowest substituent number. If the compound contains more than one substituent, the substituents are listed in alphabetical order.

Solved Example

- $\overset{1}{C}H_3\overset{2}{C}H(Cl)\overset{3}{C}H(Br)\overset{4}{C}{\equiv}\overset{5}{C}\overset{6}{C}H_2\overset{7}{C}H_2\overset{8}{C}H_3$

  3-bromo-2-chloro-4-octyne
  not 6-bromo-7-chloro-4-octyne
  because 2 < 6

  $\overset{6}{C}H_3\overset{5}{C}H(CH_3)\overset{4}{C}{\equiv}\overset{3}{C}\overset{2}{C}H_2\overset{1}{C}H_2Br$

  1-bromo-5-methyl-3-hexyne
  not 6-bromo-2-methyl-3-hexyne
  because 1 < 2

HOW TO NAME A COMPOUND THAT HAS MORE THAN ONE FUNCTIONAL GROUP

The rules for naming compounds with two triple bonds, using the ending ‘‘diyne’’, are similar to the rules for naming compounds with two double bonds.

	$CH_2{=}C{=}CH_2$	$\overset{6}{C}H_3\overset{5}{C}H{=}\overset{4}{C}H\overset{3}{C}H_2\overset{2}{C}(CH_3){=}\overset{1}{C}H_2$	$CH_3CH(CH_3)C{\equiv}CCH_2C{\equiv}CH$
Systematic:	propadiene	2-methyl-1,4-hexadiene or 2-methylhexa-1,4-diene	6-methyl-1,4-heptadiyne or 6-methylhepta-1,4-diyne
Common:	allene		

☞ If the two functional groups are a double bond and a triple bond , number the chain in the direction that produces a name containing the lower number. Thus, in the following examples, the lower number is given to the alkyne suffix in the compound on the left and to the alkene suffix in the compound on the right.

$$\overset{7}{C}H_3\overset{6}{C}H{=}\overset{5}{C}H\overset{4}{C}H_2\overset{3}{C}H_2\overset{2}{C}{\equiv}\overset{1}{C}H$$

5-hepten-1-yne *not* 2-hepten-6-yne
because 1 < 2

$$\overset{1}{C}H_2{=}\overset{2}{C}H\overset{3}{C}H_2\overset{4}{C}H_2\overset{5}{C}{\equiv}\overset{6}{C}\overset{7}{C}H_3$$

1-hepten-5-yne *not* 6-hepten-2-yne
because 1 < 2

$$\underset{1}{C}H_2{=}\underset{2}{C}H\underset{3}{\overset{\overset{\large CH_2CH_2CH_2CH_3}{|}}{C}}H\underset{4}{C}{\equiv}\underset{5}{C}\underset{6}{C}H_3$$

3-butyl-1-hexen-4-yne

the longest continuous chain has 8 carbons, but the 8-carbon chain does not contain both functional groups; therefore, the compound is named as a hexenyne because the longest continuous chain containing both functional groups has 6 carbons

Solved Example

$$\overset{1}{C}H_3\overset{2}{C}H{=}\overset{3}{C}H\overset{4}{C}{\equiv}\overset{5}{C}\overset{6}{C}H_3$$

2-hexen-4-yne
not 4-hexen-2-yne

$$H\overset{6}{C}{\equiv}\overset{5}{C}\overset{4}{C}H_2\overset{3}{C}H_2\overset{2}{C}H{=}\overset{1}{C}H_2$$

1-hexen-5-yne
not 5-hexen-1-yne

☞ If there is a tie between a double bond and a triple bond, the double bond gets the lower number.

☞ Compounds with more than one triple bond are called diynes, triynes, and so forth; compounds containing both double and triple bonds are called enynes (not ynenes). Numbering of an enyne chain starts from the end nearer the first multiple bond, whether double or triple. When there is a choice in numbering, double bonds receive lower numbers than triple bonds.

Solved Example

$$H\underset{7}{C}{\equiv}\underset{6}{C}\underset{5}{C}H_2\underset{4}{C}H_2\underset{3}{C}H_2\underset{2}{C}H{=}\underset{1}{C}H_2$$

1-Hepten-6-yne
(New : Hept-1-en-6-yne)

$$H\underset{1}{C}{\equiv}\underset{2}{C}\underset{3}{C}H_2\underset{4}{C}H\underset{5}{C}H_2\underset{6}{C}H_2\underset{7}{C}H{=}\underset{8}{C}H\underset{9}{C}H_2$$

4-Methyl-7-nonen-1-yne
(New : 4-Methyl-7-nonen-1-yne)

☞ As with alkyl and alkenyl substituents derived from alkanes and alkenes, respectively, alkynyl groups are also possible.

$CH_3CH_2CH_2CH_2-$

Butyl
(an alkyl group)

$CH_3CH_2CH{=}CH-$

1-Butenyl
(a vinylic group)
(New : But-1-enyl)

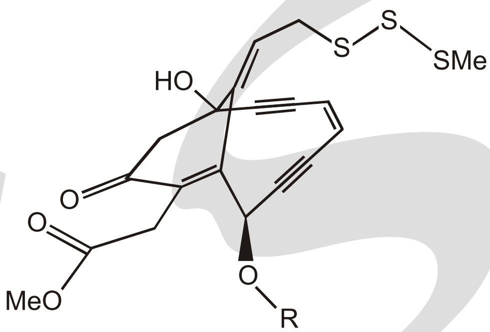
$CH_3CH_2C{\equiv}C-$

1-Butynyl
(an alkynyl group)
(New : But-1-ynyl)

☑ SPECIAL TOPIC

ALKYNES CONTAIN C C TRIPLE BONDS

Just like C C double bonds, C C triple bonds have a special type of reactivity associated with them, so it's useful to call a C C triple bond a functional group. Alkynes are linear so we draw them with four carbon atoms in a straight line. Alkynes are not as widespread in nature as alkenes, but one fascinating class of compounds containing C C triple bonds is a group of antitumour agents discovered during the 1980s. Calicheamicin is a member of this group. The highreactivity of this combination of functional groups enables calicheamicin to attack DNA and prevent cancer cells from proliferating. For the first time we have drawn a molecule in three dimensions, with two bonds crossing one another—can you see the shape?

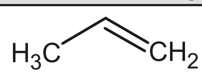
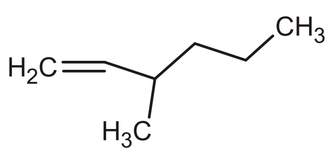
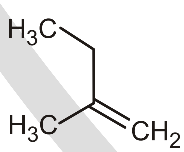
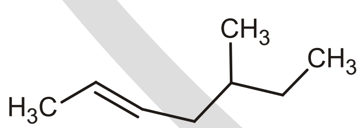
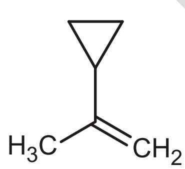
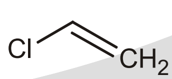
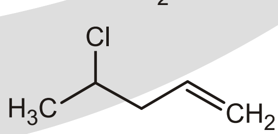


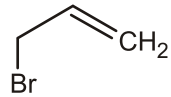
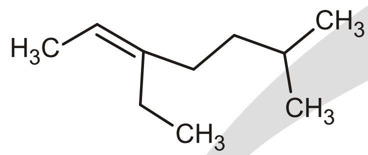
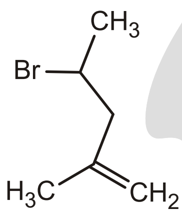
calicheamicin
(R = a string of sugar molecules)

► **Saturated and unsaturated carbon atoms**
In an alkane, each carbon atom is joined to four other atoms (C or H). It has no potential for forming more bonds and is therefore saturated. In alkenes, the carbon atoms making up the C=C double bond are attached to only three atoms each. They still have the potential to bond with one more atom, and are therefore unsaturated. In general, carbon atoms attached to four other atoms are saturated; those attached to three, two, or one are unsaturated.

EXERCISE

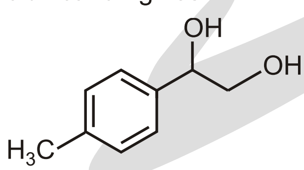
WORK SHEET - 1

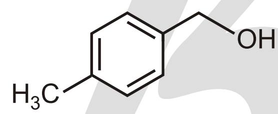
S.No.	Compounds	Write IUPAC - Name
1.		
2.		
3.		
4.		
5.		
6.		
7.		

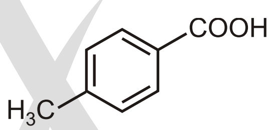
8.		
9.		
10.		

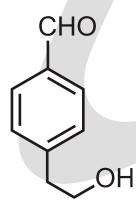
SUBJECTIVE TYPE QUESTIONS

1. The reaction of 50% aq. KOH on an equimolar mixture of 4 -methylbenzaldehyde and formaldehyde followed by acidification gives :

(a) 

(b)  + HCOOH

(c)  + CH_3OH

(d) 

Answers

Work Sheet - 1

1. propene
2. 3-methylhex-1-ene
3. 2-methylbut-1-ene
4. 5-methylhept-2-ene
5. 2-cyclopropylpropene
6. chloroethene
7. 4-chloropent-1-ene
8. 3-bromoprop-1-ene
9. 3-ethyl-6-methylhept-2-ene
10. 4-bromo-2-methylpent-1-ene

Subjective Type Questions

1. (b)

❑❑❑

CHAPTER

9

Nomenclature of Alcohol, Ether, Aldehyde and Ketone

PRIORITIES OF FUNCTIONAL GROUP SUFFIXES

highest priority $>C=O$ > $-OH$ > NH_2 > ($C=C$ or $C \equiv C$) lowest priority

the double bond is given priority over a triple bond only when there is a tie

The following rules are used to name a compound that has a functional group suffix :

1. The parent hydrocarbon is the longest continuous chain containing the functional group .
2. The parent hydrocarbon is numbered in the direction that gives the functional group suffix the lowest possible number .

☑ SPECIAL TOPIC

NAME ALCOHOL COMES FROM ?

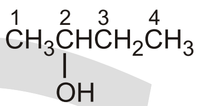
Arabic alchemy has given us a number of chemical terms; for example, alcohol is believed to derive from Arabic al-khwl or al-ghawl whose original meaning was a metallic powder used to darken women's eyelids (Kohl).

Alcohol entered the English language in the 17th Century with the meaning of a "sublimated" substance, then became the "pure spirit" of anything, and only became associated with "spirit of wine" in 1753. finally, in 1852, it become a part of chemical nomenclature that denoted a common class of organic compound. But it's still common practice to refer to the specific substance CH_3CH_2OH as "Alcohol" rather then its systematic name ethanol.

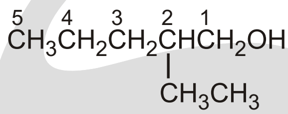
Solved Example

- $CH_3CH(OH)CH_3$
  - Functional groups is an alcohol, therefore suffix = -ol
  - Hydrocarbon structure is an alkane therefore -ane
  - The longest continuous chain is C3 therefore root = prop
  - It doesn't matter which end we number from, the alcohol group locant is 2-

Ans. Propan-2-ol

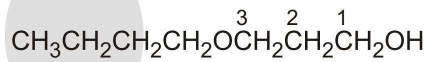


2-butanol
or
butan-2-ol



2-ethyl-1-pentanol
or
2-ethylpentan-1-ol

The longest continuous chain has six carbons, but the longest continuous chain containing the OH functional group has five carbons so the compound is named as a pentanol.

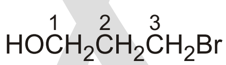


3-butoxy-1-propanol
or
3-butoxypropan-1-ol

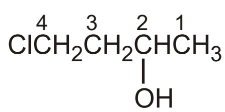
The longest continuous chain has four carbons, but the longest continuous chain containing the OH functional group has three carbons, so the compound is named as a propanol.

☞ If there is a functional group suffix and a substituent, the functional group suffix gets the lowest possible number.

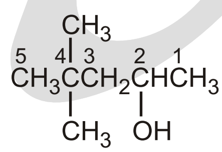
Solved Example



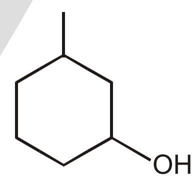
3-bromo-1-propanol



4-chloro-2-butanol



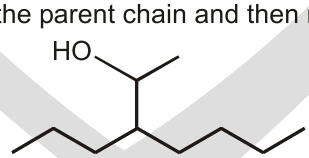
4,4-dimethyl-2-pentanol



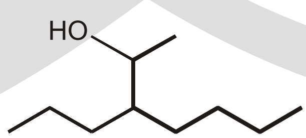
3-methylcyclohexanol
not
5-methylcyclohexanol

Solved Example

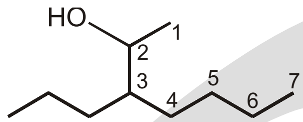
- For the compound below, choose the parent chain and then number it correctly :



Ans. To choose the parent chain, remember that we need to choose the longest chain containing the functional group:

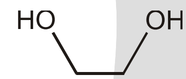


To number it correctly, we need to go in the direction that gives the functional group the lowest number:



NAME: 3-Propyl-2-heptanol

DIOLS (OR POLYOLS)

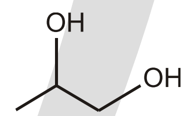


- Functional group is an alcohol, therefore suffix **= -ol**
- Hydrocarbon structure is an alkane therefore**-ane**
- There are two alcohols, so insert the multiplier **di**
- The longest continuous chain is C2 therefore root **= eth**

  ethane-1,2-diol

  or

  1,2-ethanediol

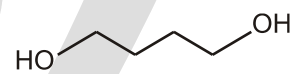


- Hydrocarbon structure is an alkane therefore**-ane**
- There are two alcohols, so insert the multiplier **di**
- The longest continuous chain is C3 therefore root **= prop**
- Locants for -OH units are **1-** and **2-**

  Propane-1, 2-diol

  or

  1,2-propanediol

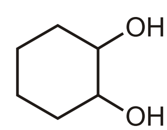


- Hydrocarbon structure is an alkane therefore **-ane**
- There are two alcohols, so insert the multiplier **di**
- The longest continuous chain is C4 therefore root **= but**

  butane-1,4-diol

  or

  1,4-butanediol



- Hydrocarbon structure is an alkane therefore **-ane**
- There are two alcohols, so insert the multiplier **di**
- The ring is C6 therefore root **= cyclohex**
- Locants for –OH units are **1-** and **2-**

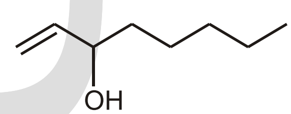
  Cyclohexane-1,2-diol

  or

  1,2-cyclohexanediol

☑ SPECIAL TOPIC

HOW A BANANA SLUG KNOWS WHAT TO EAT



1-octene-3-ol

Many species of mushrooms synthesize 1-octen-3-ol, a repellent that drives off predatory slugs. Such mushrooms can be recognized by small bite marks on their caps, where the slug started to nibble before the volatile compound was released. People are not put off by the release of this compound because to them it just smells like a mushroom. 1-Octen-3-ol also has antibacterial properties that may protect the mushroom from organisms that would otherwise invade the wound made by the slug. Not surprisingly, the species of mushroom that banana slugs commonly eat cannot synthesize 1-octen-3-ol.

Solved Example

▸ (a) $CH_3-CH_2-CH_2-CH_2-OH$
Butan-1-ol

(b) $CH_3-\underset{\underset{OH}{|}}{CH}-CH_3$
Propan-2-ol

(c) $CH_3-\overset{\overset{CH_3}{|}}{\underset{\underset{CH_3}{|}}{C}}-OH$
2-Methylpropan-2-ol

COMMON AND IUPAC NAMES OF SOME ALCOHOLS

Compound	Common name	IUPAC name
CH_3-OH	Methyl alcohol	Methanol
$CH_3-CH_2-CH_2-OH$	*n*-Propyl alcohol	Propan-1-ol
$CH_3-\underset{\underset{OH}{\vert}}{CH}-CH_3$	Isopropyl alcohol	Propan-2-ol
$CH_3-CH_2-CH_2-CH_2-OH$	*n*-Butyl alcohol	Butan-1-ol
$CH_3-\underset{\underset{OH}{\vert}}{CH}-CH_2-CH_3$	sec-Butyl alcohol	Butan-2-ol
$\underset{\underset{OH}{\vert}}{CH_2}-\overset{\overset{CH_3}{\vert}}{CH}-CH_3$	Isobutyl alcohol	2-Methylpropan-1-ol
$CH_3-\overset{\overset{CH_3}{\vert}}{\underset{\underset{CH_3}{\vert}}{C}}-OH$	tert-Butyl alcohol	2-Methylpropan-2-ol
$\underset{\underset{OH}{\vert}}{CH_2}-\underset{\underset{OH}{\vert}}{CH}-\underset{\underset{OH}{\vert}}{CH_2}$	Glycerol	Propan-1,2, 3-triol

☑ SPECIAL TOPIC

ETHERS (R^1–O–R^2) CONTAIN AN ALKOXY GROUP (–OR)

The name ether refers to any compound that has two alkyl groups linked through an oxygen atom. 'Ether' is also used as an everyday name for diethyl ether, Et_2O. You might compare this use of the word 'ether' with the common use of the word 'alcohol' to mean ethanol. Diethyl ether is a highly flammable solvent that boils at only 35 C. It used to be used as an anaesthetic. Tetrahydrofuran (THF) is another commonly used solvent and is a cyclic ether.

► Another common laboratory solvent is called 'petroleum ether'. Don't confuse this with diethyl ether! Petroleum ether is in fact not an ether, but a mixture of alkanes. 'Ether', according to the Oxford English Dictionary, means 'clear sky, upper region beyond the clouds', and hence used to be used for anything light, airy, and volatile.

Compound	IUPAC name
CH_3OCH_3	Methoxymethane
$C_2H_5OC_2H_5$	Ethoxyethane
$CH_3OCH_2CH_2CH_3$	1-Methoxypropane

ETHERS

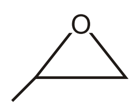
Common names of ethers are derived from the names of alkyl/aryl groups written as separate words in alphabetical order and adding the word 'ether' at the end. For example, $CH_3OC_2H_5$ is ethylmethyl ether.

COMMON AND IUPAC NAMES OF SOME ETHERS

Compound	Common name	IUPAC name
CH_3OCH_3	Dimethyl ether	Methoxymethane
$C_2H_5OC_2H_5$	Diethyl ether	Ethoxyethane
$CH_3OCH_2CH_2CH_3$	Methyl *n*-propyl ether	1-Methoxypropane
$C_6H_5OCH_3$	Methylphenyl ether (Anisole)	Methoxybenzene (Anisole)
$C_6H_5OCH_2CH_3$	Ethylphenyl ether (Phenetole)	Ethoxybenzene
$C_6H_5O(CH_2)_6 — CH_3$	Heptylphenyl ether	1-Phenoxyheptane
$CH_3O—\underset{\mid \atop CH_3}{CH}—CH_3$	methyl isopropyl ether	2-Methoxypropane

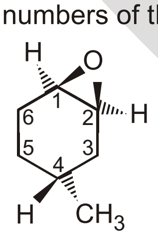
EPOXIDES

- Functional group is an epoxide, therefore suffix = –epoxide.
- The longest continuous chain is C3 therefore root **= prop.**
- Location of "alkene" is unambiguous, so no locant needed.

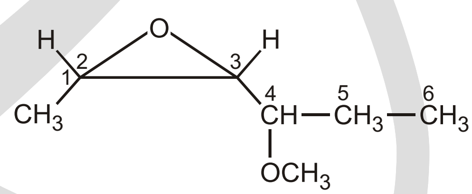


1,2-epoxypropane

One systematic method for naming epoxides is to name the rest of the molecule and use the term "epoxy" as a subsituent, giving the numbers of the two carbon atoms bonded to the epoxide oxygen.

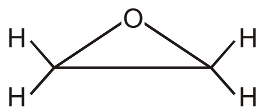


trans-1,2-epoxy-4-methylcyclohexane

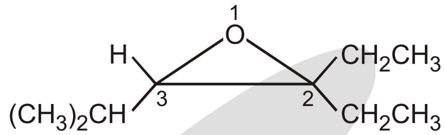


cis-2-3-epoxy-4-methoxyhexane

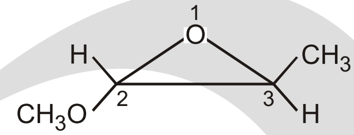
Another systematic method names epoxides as derivatives of the parent compound, ethylene oxide, using "oxirane" as the systematic name for ethylene oxide. In this system, the ring atoms of a heteroxyclic compound are numbered starting with the heteroatom and going the direction to give the lowest substituent numbers. The "epoxy" system names are also listed for comparison. Note that the numbering is different for the "epoxy" system names, which number the longest chain rather than the ring.



oxirane
1,2-epoxyethane



2,2-diethyl-3-isopropyloxirane
3,4-epoxy-4-ethyl-2-methylhexane

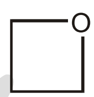


trans-2-methoxy-3-methyloxirane
1,2-epoxy-1-methoxypropane

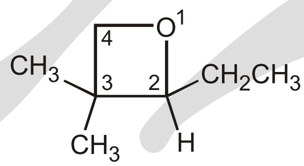
☑ SPECIAL TOPIC

OXETANES

The least common cyclic ethers are the four-membered oxetanes. Because these four-membered rings are strained, they are more reactive than larger cyclic ethers and open-chain ethers. However they are not as reactive as the highly strained oxiranes (epoxides).

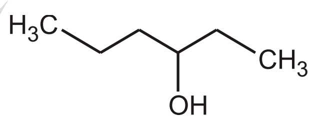
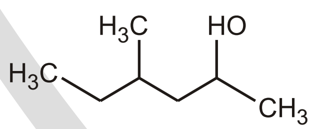
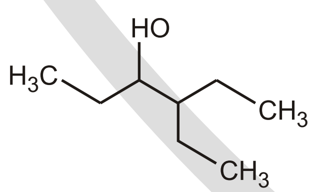
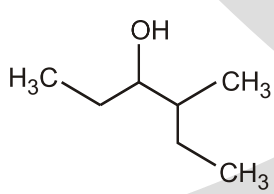
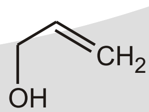


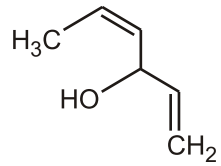
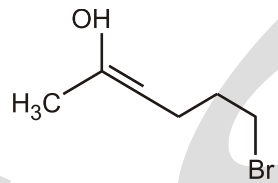
oxetane



2-ethyl-3,3-dimethyloxetane

WORK SHEET

S.No.	Compounds	Write IUPAC - Name
1.		
2.		
3.		
4.		
5.		

6.		
7.		
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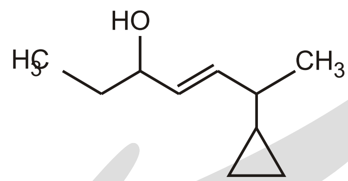
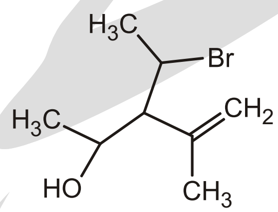
9.		
10.		

Answers

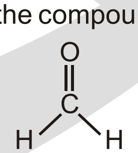
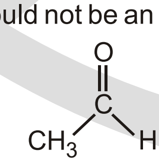
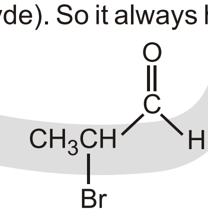
Work sheet

1. hexan-3-ol
2. 4-methylhexan-2-ol
3. 4-ethylhexan-3-ol
4. 4-methylhexan-3-ol
5. prop-2-en-1-ol
6. hexa-1,4-dien-3-ol
7. 6-bromohex-2-en-2-ol
8. hexa-1,5-dien-1-ol
9. 6-cyclopropylhept-4-en-3-ol
10. 3-(1-bromoethyl)-4-methylpent-4-en-2-ol

THE NOMENCLATURE OF ALDEHYDES

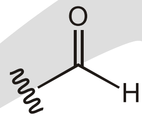
NAMING ALDEHYES

The systematic (IUPAC) name of an aldehyde is obtained by replacing in final “e” on the name of the parent hydrocarbon with “al.” For example. A one-carbon aldehyde is called methanal, and two-carbon aldehyde is called ethanal. The position of the carbonyl carbon does not have to be designated because it is always at the end of the parent hydrocarbon (or else the compound would not be an aldehyde). So it always has the 1-position.

			
systematic name:	methanal	ethanal	2-bromopropanal
common name :	formaldehyde	acetaldehyde	α-bromopropionaldehyde

Points to Remember

- –CHO represents:



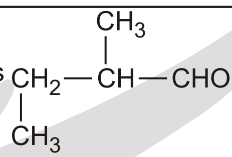
When we write aldehydes as R–CHO, we have no choice but to write in the C and H (because they're part of the functional group).

◎ Mistake to Avoid

- For drawing structures.

  Another point: always write R–CHO and never R–COH, which looks too much like an alcohol.

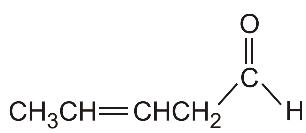
Solved Example

- The IUPAC name of is 

Ans. 2-methyl butanal

☞ If one of the functional groups is an alkene, suffix endings are used for both functional groups and the alkene functional group is stated first, with its "e" ending omitted to avoid two successive vowels.

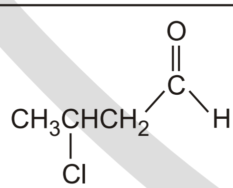
Solved Example

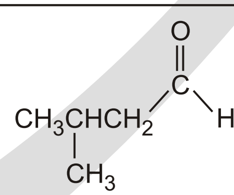


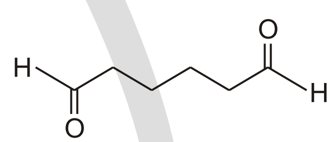
3-pentenal

☞ Note that the terminal "e" of the parent hydrocarbon is not removed in hexanedial (The "e" is removed only to avoid two successive vowels.)

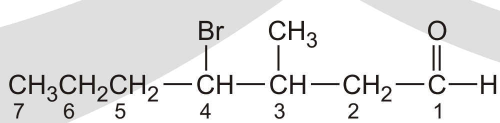
Solved Example



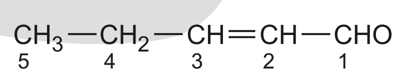




systematic name:	3-chlorobutanal	3-methylbutanal	hexanedial
common name :	β-chloropionaldehyde	isovaleraldehyde	



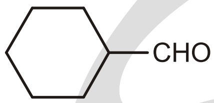
4-bromo-3-methylheptanal



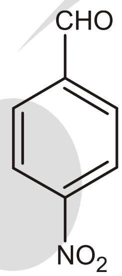
2-pentenal or pent-2-enal

☞ If the aldehydic group is attached to a ring, then the suffix carbaldehyde is added to the full name of cyclohexane.

Example



Cyclohexanecarbaldehyde



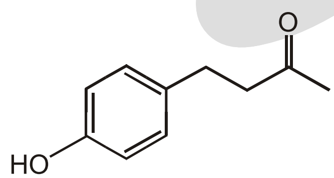
4-Nitrobenzenecarbaldehyde
or
4-Nitrobenzaldehyde

☑ SPECIAL TOPIC

SPECIAL TOPIC: ALDEHYDES (R–CHO) AND KETONES (R^1–CO–R^2) CONTAIN THE CARBONYL GROUP C O

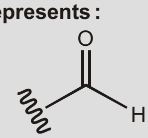
Aldehydes can be formed by oxidizing alcohols—in fact the liver detoxifies ethanol in the bloodstream by oxidizing it first to acetaldehyde (ethanal, CH_3CHO). Acetaldehyde in the blood is the cause of hangovers. Aldehydes often have pleasant smells—2-methylundecanal is a key component of the fragrance of Chanel No 5™, and 'raspberry ketone' is the major component of the flavour and smell of raspberries.

2-methylundecanal



"raspberry ketone"

► **—CHO represents :**



When we write aldehydes as R—CHO, we have no choice but to write in the C and H (because they're part of the functional group) — one important intance where you should ignore Guideline 3 for drawing structures.
Another point : always write R—CHO and never R—CHO, which looks too much like an alcohol.

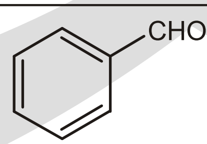
COMMON NAMES OF ALDEHYDE & KETONE ALDEHYDES :

- Often called by their common names instead of IUPAC names.
- Derived from the common names of the carboxylic acids by replacing the ending '–ic' of the acid with aldehyde.
- Location of the substituent in the carbon chain is indicated by the Greek letters , , , etc.

Example

CH_3CHO

Acetaldehyde



Benzaldehyde

$$\underset{\gamma}{H_3C}-\underset{\beta}{\overset{\overset{\large Br}{|}}{CH}}-\underset{\alpha}{CH_2}-\overset{\overset{\large O}{||}}{C}-H$$

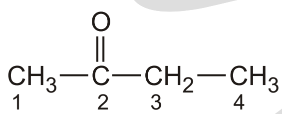
β-Bromobutyraldehyde

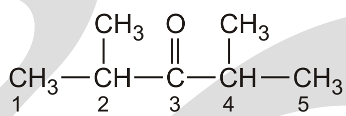
NOMENCLATURE OF KETONES

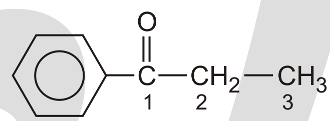
IUPAC NAMES

- For open-chain aliphatic aldehydes and ketones, IUPAC names are derived from the names of the corresponding alkanes by replacing the ending '–e' with '–al' and '–one' respectively.
- In the case of aldehydes, the longest chain is numbered starting from the carbon of the aldehydic group.
- In the case of ketones, the numbering begins from the end nearer to the carbonyl group.
- Substituents are prefixed in the alphabetical order along with the numerals indicating their positions in the carbon chain.
- Same rule is applicable to cyclic ketones.

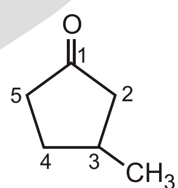
Solved Example



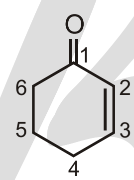




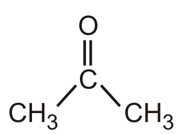
old IUPAC names:	2-butanone	2,4-dimethyl-3-pentanone	1-phenyl-1-propanone
new IUPAC names:	butan-2-one	2,4-dimethylpentan-3-one	1-phenylpropan-1-one

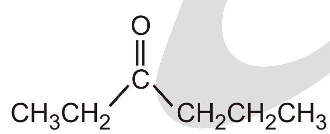


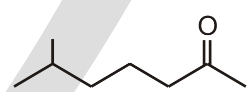
3-methylcyclopentanone



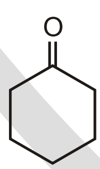
2-cyclohexenone
cyclohex-2-en-1-one

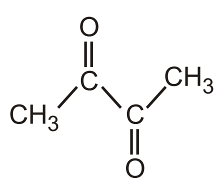


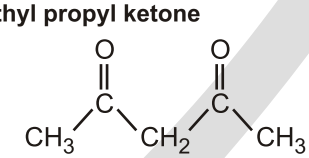


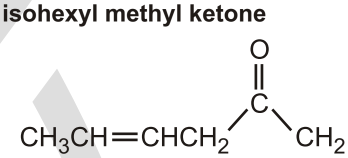


systematic name:	propanone	3-hexanone	6-methyl-2-heptanone
common name :	acetone		
derived name :	**dimethyl ketone**	**ethyl propyl ketone**	**isohexyl methyl ketone**





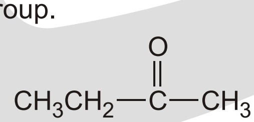




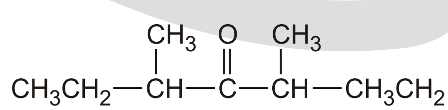
systematic name :	cyclohexanone	butanedione	2,4-pentanedione	4-hexen-2-one
common name :			acetylacetone	

COMMON NAMES

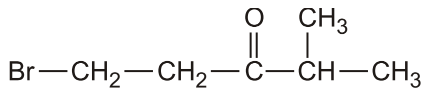
As with other classes of compounds, ketones and aldehydes are often called by common names instead of their systematic IUPAC names. Ketone common names are formed by naming the two alkyl groups bonded to the carbonyl group. Substituent locations are given using Greek letters, beginning with the carbon next to the carbonyl group.



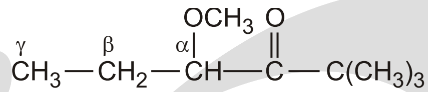
ethyl methylketone



di-sec-butyl ketone



bromoethyl isopropyl ketone

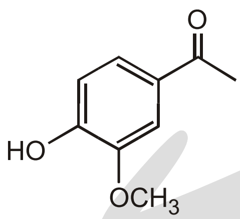


tert-butyl -methoxypropyl ketone

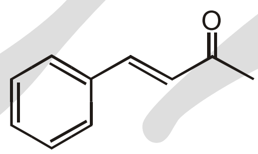
☑ SPECIAL TOPIC

SPECIAL TOPIC: SMELLS OF ALDEHYDE AND KETONES

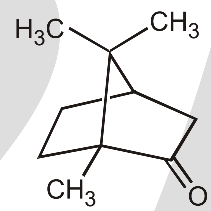
Many compounds found in nature have aldehyde or ketone functional groups. Aldehydes have pungent odors. Whereas ketones tend to smell sweet. Vanillin and cinnamaldehyde are examples of naturally occurring aldehydes. A whiff of vanilla extract will allow you to appreciate the pungent odor of vanillin. The ketones camphor and carvone are responsible for the characteristic sweet odoros of the leaves of camphor trees, spearmint leaves, and caraway seeds.



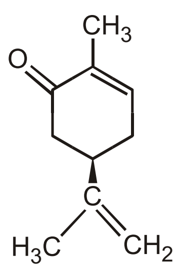
vanillin
vanilla flavoring



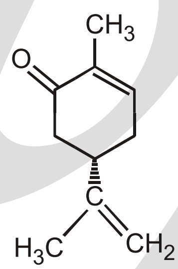
cinnamaldehyde
cinnamon flavoring



camphor

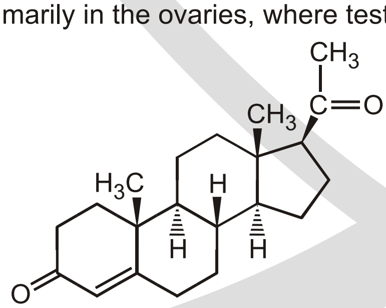


(R)-(–)-carvone
spearmint oil

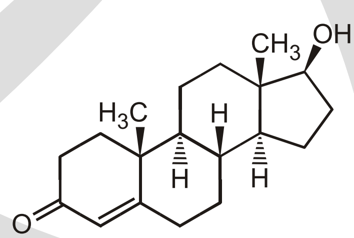


(S)-(+)-carvone
caraway speed oil

Progesterone and testosterone are two biologically important ketones that illustrate how a small difference in structure can be responsible for a large difference in biological activity. Both are sex hormones, but progesterone is synthesized primarily in the ovaries, where testosterone is synthesized primarily in the testes.



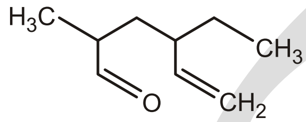
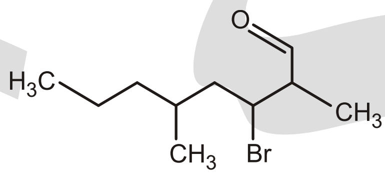
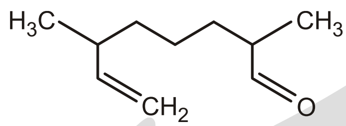
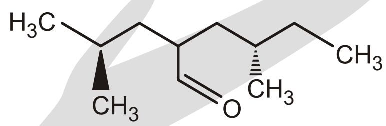
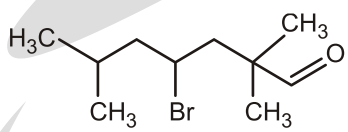
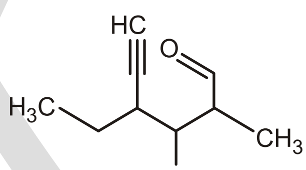
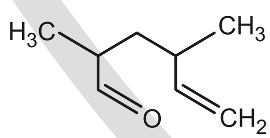
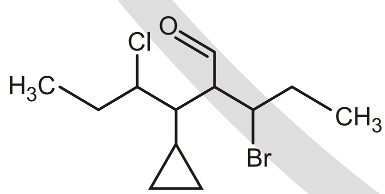
progesterone



testosterone

EXERCISE

WORK SHEET - 1

S.No.	Compounds	Write IUPAC - Name
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		

Answers

Work Sheet-1

1. 4-ethyl-2-methylhex-5-enal
2. 3-bromo-2,5-dimethyloctanal
3. 2,6-dimethyloct-7-enal
4. 4-methyl-2-(2-methylpropyl)hexanal
5. 4-bromo-2,2,6-trimethylheptanal
6. 4-ethyl-2,3-dimethylhex-5-ynal
7. 2,4-dimethylhex-5-enal
8. 2-(1-bromopropyl)-4-chloro-3-cyclopropylhexanal

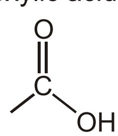
CHAPTER

10

Nomenclature of Carboxylic Acid, Ester, Cyanide, Amide, Amine and Anhydride

NAMING CARBOXYLIC ACIDS

The functional group of carboxylic acid is called a carboxyl group.

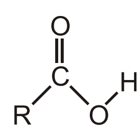


a carboxyl group

–COOH –CO_2H

carboxyl groups are frequently shown in abbreviated forms

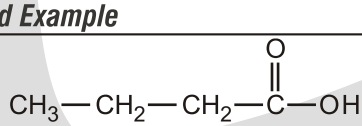
In systematic (IUPAC) nomenclature, a carboxylic acid is named by replacing the terminal "e" of the alkane name with "oic acid." For example, the one-carbon alkane is methane, so the one-carbon carboxylic acid is methanoic acid.



Substituent suffix = -oic acid *e.g.* ethanoic acid

Substituent prefix = carboxy

Solved Example

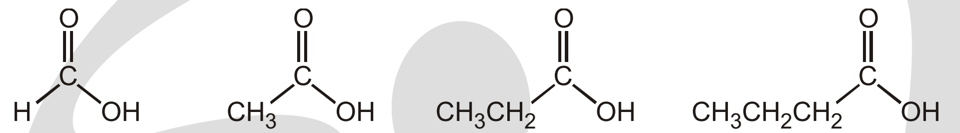


- The root name is based on the longest chain including the carboxylic acid group.
- Since the carboxylic acid group is at the end of the chain, it must be C1.
- The carboxylic acid suffix is appended after the hydrocarbon suffix minus the
- "e" : *e.g.* -ane + -oic acid = -anoic acid etc.

- Hydrocarbon structure is an alkane therefore-ane
- The longest continuous chain is C4 therefore root = but

Butanoic acid

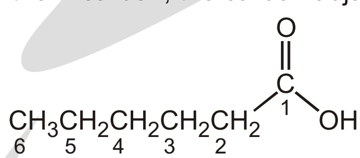
Solved Example



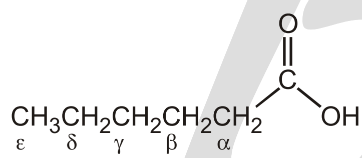
systematic name:	methanoic acid	ethanoic acid	propanoic acid	butanoic acid
common name:	formic acid	acetic acid	propionic acid	butyric acid

pentanoic acid	hexanoic acid	propenoic acid
valeric acid	caproic acid	acrylic acid

In systematic nomenclature, the position of a substituent is designated by a number. The carbonyl carbon is always the C-1 carbon. In common nomenclature, the position of a substituent is designated by a lowercase Greek letter, and the carbonyl carbon is not given a designation. Thus, the carbon adjacent to the carbonyl carbon is the -carbon, the carbon adjacent to the -carbon is the -carbon, and so on.



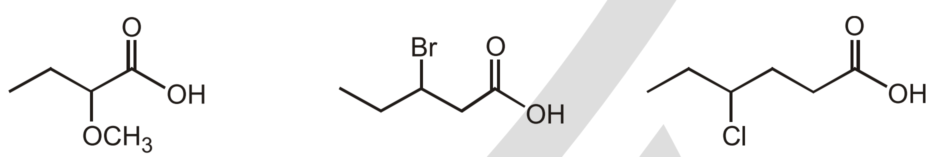
systematic nomenclature



common nomenclature

α = alpha
β = beta
γ = gamma
δ = delta
ε = epsilon

Take a care full look at the following examples to make sure that you understand the difference between systematic (IUPAC) and common nomenclature :



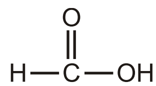
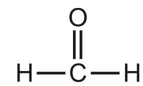
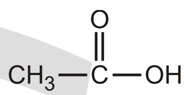
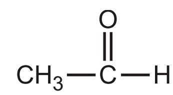
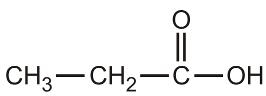
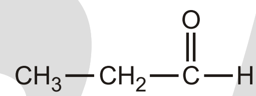
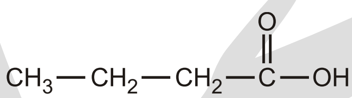
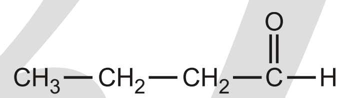
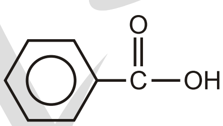
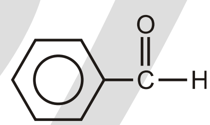
systematic name:	2-methoxybutanoic acid	3-bromopentanoic acid	4-chlorohexanoic acid
common name:	α-methoxybutyric acid	β-bromovaleric acid	γ-chlorocaproic acid

---

COMMON NAMES

Carboxylic acids containing six or fewer carbons are frequently called by their common names. These names were chosen by early chemists to describe some feature of the compound, usually its origin. For example, formic acid is found in ants, bees, and other stinging insects; its name comes from formica , which is Latin for "ant." Acetic acid— contained in vinegar—got its name from acetum, the Latin word for "vinegar." Propionic acid is the smallest acid that shows some of the characteristics of the larger fatty acids its name comes from the Greek words pro ("the first") and pion ("fat"). Butyric acid is found in rancid butter; the Latin word for " butter" is butyrum. Valeric acid got its name from valerian, an herb that has been used as a sedative since Greco/Roman times. Caproic acid is found in goat's milk. If you have ever smelled a goat, then you know what caproic acid smells like. Caper is the Latin word for "goat."

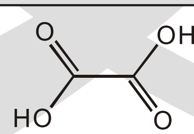
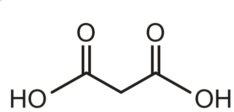
COMMON NAMES OF ACIDS AND ALDEHYDES

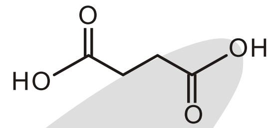
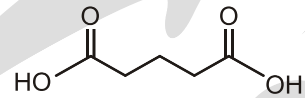
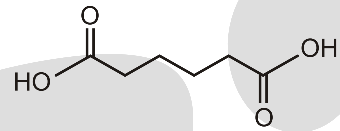
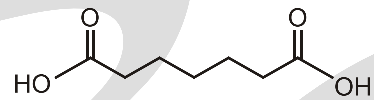
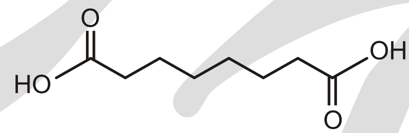
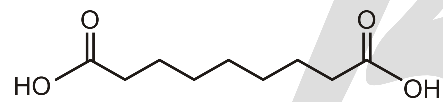
Carboxylic Acid	Derivation	Aldehyde
 formic acid (methanoic acid)	*formica,* "ants"	 formaldehyde (methanal)
 acetic acid (ethanoic acid)	*acetum,* "sour"	 acetaldehyde (ethanal)
 propionic acid (propanoic acid)	*protos pion,* "first fat"	 propionaldehyde (propanal)
 butyric acid (butanoic acid)	*butyrum,* "butter"	 butyraldehyde (butanal)
 benzoic acid	*gum benzoin,* "blending"	 benzaldehyde

DICARBOXYLIC ACIDS

A dicarboxylic acid is an organic compound containing two carboxyl functional groups (–COOH). The general molecular formula for dicarboxylic acids can be written as $HO_2C—R—CO_2H$, where R can be aliphatic or aromatic. In general, dicarboxylic acids show similar chemical behavior and reactivity to monocarboxylic acids. Dicarboxylic acids are also used in the preparation of copolymers such as polyamides and polyesters. The most widely used dicarboxylic acid in the industry is adipic acid, which is a precursor used in the production of nylon. Other examples of dicarboxylic acids include aspartic acid and glutamic acid, two essential amino acids in the human body.

General formula $HO_2C(CH_2)_nCO_2H$.

Common name	IUPAC name	Structure	pKa_1	pKa_2
Oxalic acid	Ethanedioic acid		1.27	4.27
Malonic acid	Propanedioic acid		2.85	5.05

Succinic acid	Butanedioic acid		4.21	5.41
Glutaric acid	Pentanedioic acid		4.34	5.41
Adipic acid	Hexanedoic acid		4.41	5.41
Pimelic acid	Heptanedioic acid		4.50	5.43
Suberic acid	Octanedioic acid		4.526	5.49
Azelaic acid	Nonanedioic acid		4.550	5.49
Sebacic acid	Decanedioic acid			

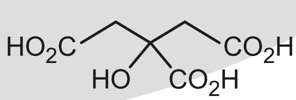
☞ **To memorize: OMSGAPSAS**
HOOC-$(CH_2)_n$-COOH

for *n* 0 Oxalic acid; *n* 1 malonic acid
n 2 Succinic acid; *n* 3 Glutaric acid
n 4 Adipic acid; *n* 5 Pimielic acid
n 6 Suberic acid; *n* 7 Azelaic acid
n 8 Sebacic acid

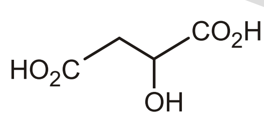
☑ SPECIAL TOPIC

SPECIAL TOPIC : CARBOXYLIC ACIDS ($R–CO_2H$) CONTAIN THE CARBOXYL GROUP CO_2H

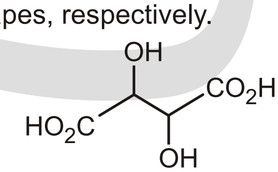
As their name implies, compounds containing the carboxylic acid (CO_2H) group can react with bases, losing a proton to form carboxylate salts. Edible carboxylic acids have sharp flavours and several are found in fruits—citric, malic, and tartaric acids are found in lemons, apples, and grapes, respectively.



citric acid



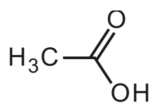
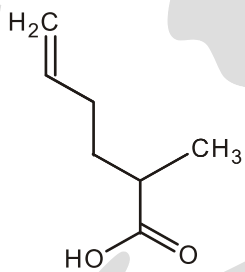
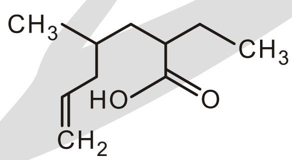
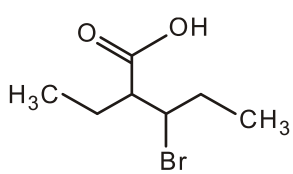
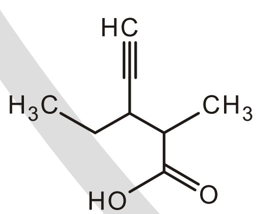
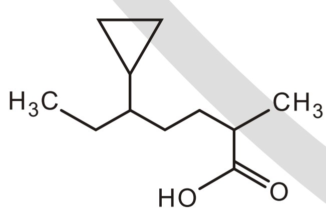
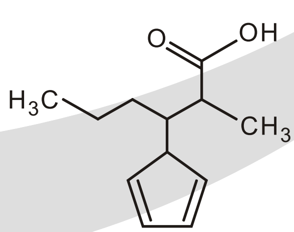
malic acid



tartaric acid

EXERCISE

WORK SHEET - 1

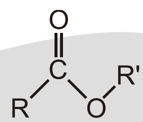
S.No.	Compounds	Write IUPAC - Name
1.		
2.		
3.		
4.		
5.		
6.		
7.		

Answers

Work Sheet-1

1. ethanoic acid
2. 2-methylhex-5-enoic acid
3. 2-ethyl-4-methylhept-6-enoic acid
4. 3-bromo-2-ethylpentanoic acid
5. 3-ethyl-2-methylpent-4-ynoic acid
6. 5-cyclopropyl-2-methylheptanoic acid
7. 3-(cyclopenta-2,4-dien-1-yl)-2-methylhexanoic acid

ESTERS



Functional class name **= alkyl alkanoate**

Substituent suffix **= –oate**

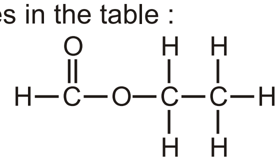
Solved Example

- $CH_3 - CH_2 - \overset{\overset{\large O}{\|}}{C} - O - CH_3$
  - The complete ester name is the alkyl alkanoate
  - Functional group is an ester
  - The alcohol component here is methanol, so the alkyl = methyl
  - The acid component here is propanoic acid, so propanoate

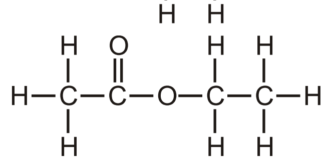
  Methyl propanoate

☞ The homologous series of linear esters that include an ethyl group attached (via a single covalent bond) to the oxygen atom in the chain of the ester molecule. This description is easier to follow in conjunction with viewing the molecular structures in the table :

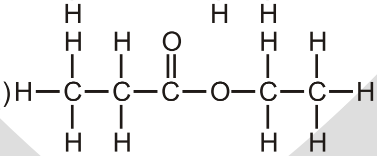
1. ethyl formate ($C_3H_6O_2$)



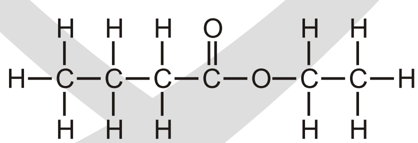
2. ethyl ethanoate ($C_4H_8O_2$)



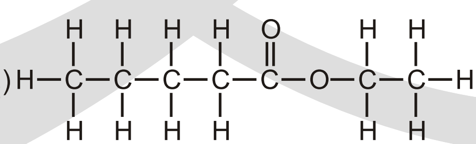
3. ethyl propanoate ($C_5H_{10}O_2$)



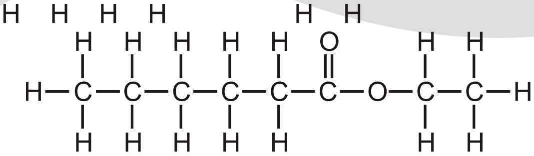
4. ethyl butanoate ($C_6H_{12}O_2$)



5. ethyl pentanoate ($C_7H_{14}O_2$)



6. ethyl hexanoate ($C_8H_{16}O_2$)

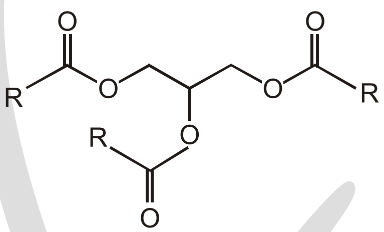


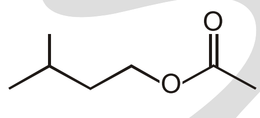
☑ SPECIAL TOPIC

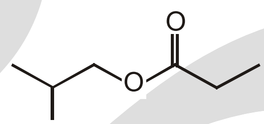
SPECIAL TOPIC : ESTERS ($R^1–CO_2R^2$) CONTAIN A CARBOXYL GROUP WITH AN EXTRA ALKYL GROUP (CO_2R)

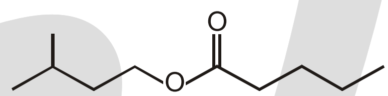
Fats are esters; in fact they contain three ester groups. They are formed in the body by condensing glycerol, a compound with three hydroxyl groups, with three fatty acid molecules. Other, more volatile esters, have pleasant, fruity smells and flavours. These three are components of the flavours of bananas, rum, and apples:

► The terms 'saturated fats' and 'unsaturated fats' are familiar—they refer to whether the R groups are saturated (no C=C double bonds) or unsaturated (contains C=C double bonds) —see the box on p. 000. Fats containing R groups with several double bonds (for example, those that are esters formed from linoleic acid, which we met at the beginning of this chapter) are known as'polyunsaturated'.


a fat molecule
(R = a long alkyl chain)


isopentyl acetate
(bananas)


isobutyl propionate
(rum)

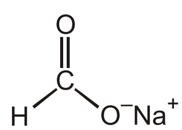
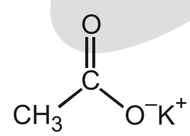
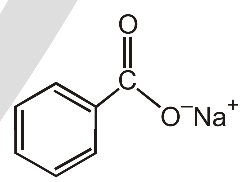

isopentyl valerate
(apples)

SALTS OF CARBOXYLIC ACIDS

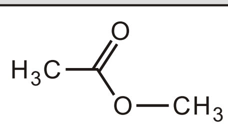
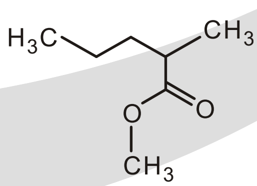
Salts of carboxylic acids are named in the same way. That is, the cation is named first, followed by the name of the acid, again with "ic acid" replaced by "ate."

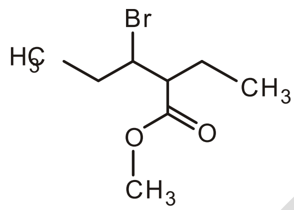
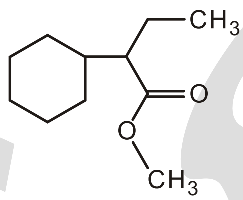
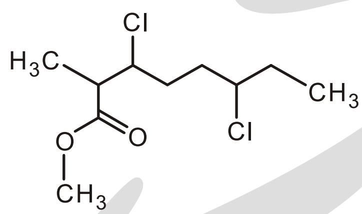
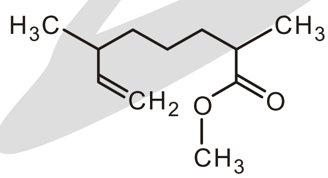
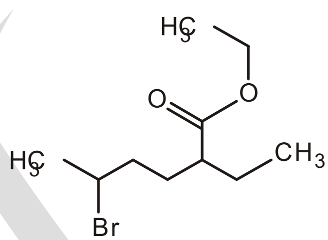
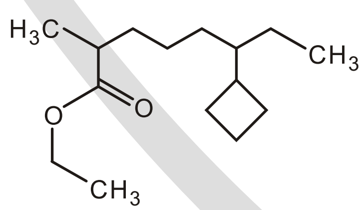
Solved Example

▸

			
systematic name:	sodium methanoate	potassium ethanoate	sodium benzenecarboxylate
common name:	sodium formate	potassium acetate	sodium benzoate

WORK SHEET

S.No.	Compounds	Write IUPAC - Name
1.		
2.		

3.		
4.		
5.		
6.		
7.		
8.		

Answers

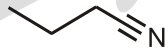
Work sheet

1. methyl acetate or methyl ethan oate
2. methyl-2-methylpentanoate
3. methyl-3-bromo-2-ethylpentanoate
4. methyl-2-cyclohexylbutanoate
5. methyl-3,6-dichloro-2-methyloctanoate
6. methyl-2,6-dimethyloct-7-enoate
7. ethyl-5-bromo-2-ethylhexanoate
8. ethyl-6-cyclobutyl-2-methyloctanoate

NITRILES

Functional group suffix = nitrile or **-onitrile**
Substituent prefix **= cyano-**

Solved Example

▸ 

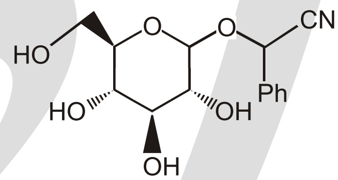
$CH_3CH_2CH_2C{\equiv}N$

- Functional group is a – C N, therefore suffix **= -nitrile**
- Hydrocarbon structure is an alkane therefore **-ane**
- The longest continuous chain is C4 therefore root **= but**

  butanenitrile

NITRILES OR CYANIDES (R–CN) CONTAIN THE CYANO GROUP – C N

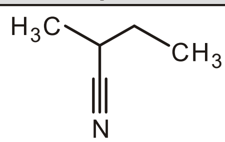
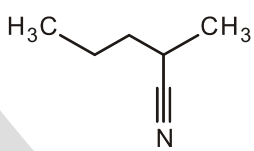
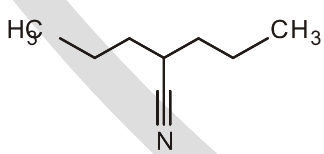
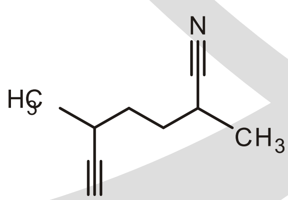
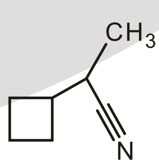
Nitrile groups can be introduced into molecules by reacting potassium cyanide with alkyl halides. The organic nitrile group has quite different properties associated with lethal inorganic cyanide: Laetrile, for example, is extracted from apricot kernels, and was once developed as an anticancer drug. It was later proposed that the name be spelt 'liar-trial' since the results of the clinical trials on laetrile turned out to have been falsified!



laetrile

EXERCISE

WORK SHEET

S.No.	Compounds	Write IUPAC - Name
1.		
2.		
3.		
4.		
5.		

Answers

Work sheet

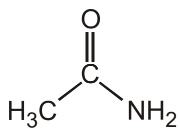
1. 2-methylbutanenitrile **2.** 2-methylpentanenitrile **3.** 2-propylpentanenitrile
4. 2,5-dimethylhept-6-ynenitrile **5.** 2-cyclobutylpropanenitrile

AMIDES, $RCONH_2$

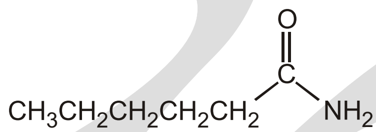
Functional class name **= alkyl alkanamide**

Substituent suffix **= -amide**

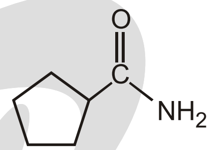
Amides with an unsubstituted $-CONH_2$ group are named by replacing the -oic acid or -ic acid ending with -amide, or by replacing the -carboxylic acid ending with -carboxamide.



Acetamide

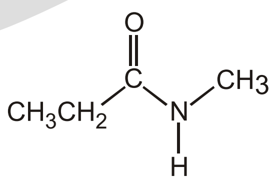


Hexanamide

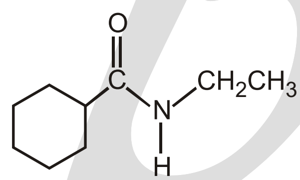


Cyclopentane-
carboxamide

If the nitrogen atom is further substituted, the compound is named by first identifying the substituent groups and then the parent amide. The substituents are preceded by the letter -N to identify them as being directly attached to nitrogen.



N-Methylpropanamide

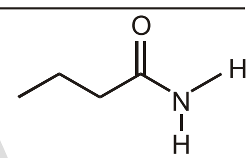


N-Ethylcyclohexanecarboxamide

Solved Example

- Functional group is an amide therefore suffix = -amide
- Hydrocarbon structure is an alkane therefore **-an-**
- The longest continuous chain is C4 therefore root = but

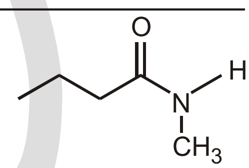
Butanamide



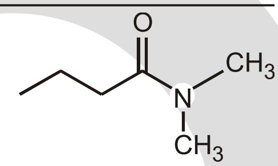
Solved Example

- Functional group is an amide therefore suffix **= -amide**
- Hydrocarbon structure is an alkane therefore-ane
- The longest continuous chain is C4 therefore root = but
- The nitrogen substituent is C1 *i.e.*, an **N-methyl** group

N-methylbutanamide



Solved Example



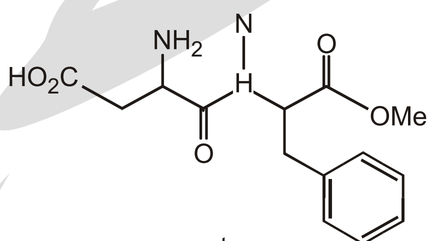
- Functional group is an amide therefore suffix **= -amide**
- Hydrocarbon structure is an alkane therefore**-ane**
- The longest continuous chain is C4 therefore root **= but**
- The two nitrogen substituents are C1 *i.e.* an **N-methyl** group
- There are two methyl groups, therefore multiplier **= di-**

  N,N-dimethylbutanamide

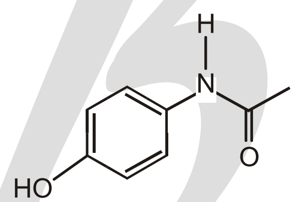
☑ SPECIAL TOPIC

AMIDES (R–CONH$_2$, R^1–CONHR^2, OR R^{1}CONR^{2}R^3)

Proteins are amides: they are formed when the carboxylic acid group of one amino acid condenses with the amino group of another to form an amide linkage (also known as a peptide bond). One protein molecule can contain hundreds of amide bonds. Aspartame, the artificial sweetener marketed as NutraSweet®, on the other hand contains just two amino acids, aspartic acid and phenylalanine, joined through one amide bond. Paracetamol is also an amide.



aspartame

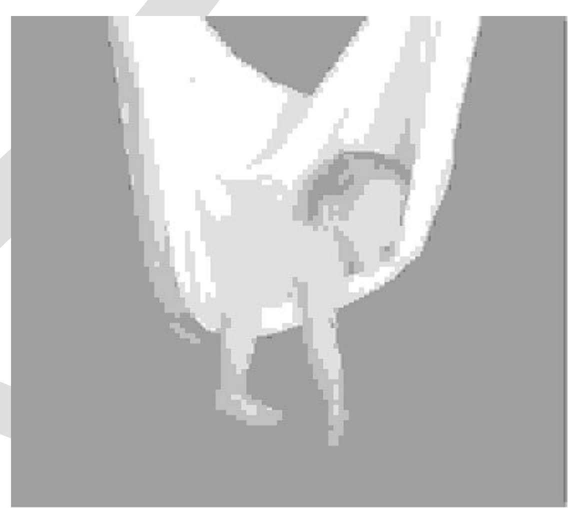


paracetamol

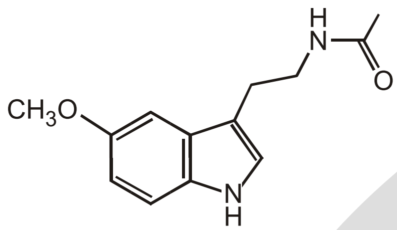
☑ SPECIAL TOPIC

AMIDES AND SLEEP

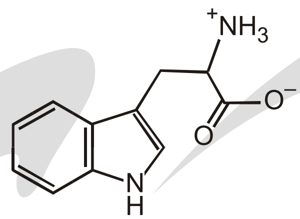
Melatonin, a naturally occurring amide, is a hormone synthesized by the pineal gland from the amino acid tryptophan. An amino acid is an a - aminocarboxylic acid. Melatonin regulates the dark–light clock in our brains that governs such things as the sleep–wake cycle, body temperature, and hormone production.



Melatonin levels increase from evening to night and then decrease as morning approaches. People with high levels of melatonin sleep longer and more soundly than those wi th low levels. The concentration of the hormone in our bodies varies with age—6-year-olds have more than five times the concentration that 80-year-olds have—which is one of the reasons young people have less trouble sleeping than older people. Melatonin supplements are used to treat insomnia, jet lag, and seasonal affective disorder.



melatonin

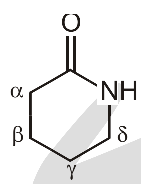


tryptophan

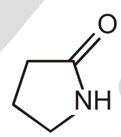
CYCLIC AMIDES

Cyclic amides are called lactams. Their nomenclature is similar to that of lactones(cyclic esters). In systematic nomenclature, they are named as “2-azacycloalkanones” (“aza” designates the nitrogen atom). For their common names, the length of the carbon chain is indicated by the common name of the carboxylic acid, and a Greek letter specifies the carbon to which the nitrogen is attached.

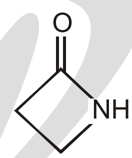
Solved Example



2-azacyclohexanone
δ-valerolactam
a δ-lactam



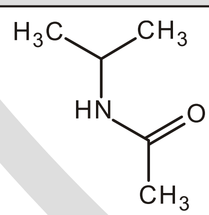
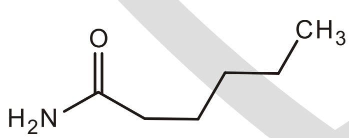
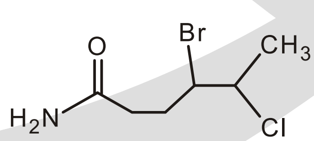
2-azacyclopentanone
γ-valerolactam
a γ-lactam

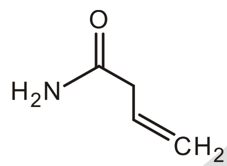
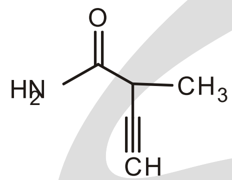
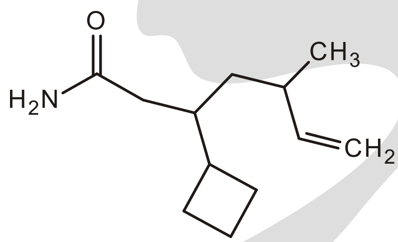


2-azacyclobutanone
β-propiolactam
a β-lactam

EXERCISE

WORK SHEET

S.No.	Compounds	Write IUPAC - Name
1.		
2.		
3.		

4.		
5.		
6.		

Answers

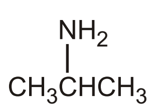
Work sheet

1. *N*-(1-methyl ethyl) ethanamide
2. hexanamide
3. 4-bromo-5-chlorohexanamide
4. but-3-enamide
5. 2-methylbut-3-ynamide
6. 3-cyclobutyl-5-methylhept-6-enamide

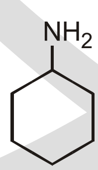
HOW DO WE NAME AMINES ?

IUPAC names for aliphatic amines are derived just as are for alcohols. The final -e of the parent alkane is dropped and replaced with -amine. Indicate the location of the amino group on the parent chain by a number.

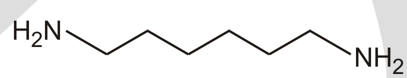
No. of C-atoms	Molecular formula	Parent	Common Name	IUPAC name
1.	CH_3NH_2	Methane	Methylamine	Methanamine
2.	$C_2H_5NH_2$	Ethane	Ethylamine	Ethanamine
3.	$CH_3H_7 — NH_2$	Propane	Propylamine	Propanamine
4.	$C_4H_{10}NH_2$	Butane	Butylamine	Butanamine



2-Propanamine

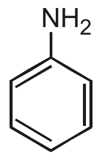


Cyclohexanamine

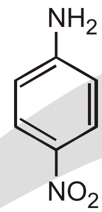


1,6-Hexanediamine

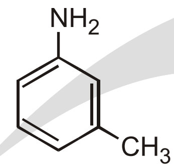
☞ IUPAC nomenclature retains the common name aniline for $C_6H_5NH_2$, the simplest aromatic amine. Its simple derivatives are named using numbers to locate substituents or, alternatively, using the locators ortho(o), meta(m), and para(p).



Aniline

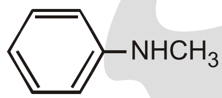


4-Nitroaniline
(p-Nitroniline)

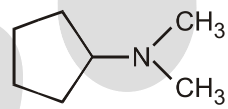


3-Methylaniline
(m-Toluidine)

Unsymmetrical secondary and tertiary amines are commonly named as N-substituted primary amines.

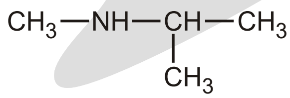


N-methylaniline

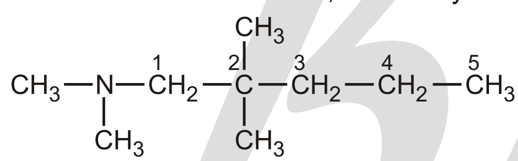


N, N-Dimethyl
cyclopentanamine

RNH_2	R_2NH	R_3N
Alkanamine	N-alkyl alkanamine	N,N-Di alkyl alkanamine
$CH_3CH_2CH_2NH_2$	$CH_3NHCH_2CH_3$	$(CH_3)_2NCH_2CH_2CH_2CH_3$
Propanamine	N-methyl ethanamine	N,N-Dimethyl Butanamine



N-methyl propan-2-amine



N,N,2,2-tetramethyl pentanamine

☞ Primary amines have one alkyl groups attached to the N.

- ★ The root name is based on the longest chain with the $-NH_2$ attached.
- ★ The chain is numbered so as to give the amine unit the lowest possible number.
- ★ The amine suffix is appended to the appropriate alkyl root or alkane-root.

Solved Example

▸ $CH_3CH_2CH(NH_2)CH_3$

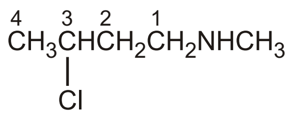
- ◆ Functional group is an amine, therefore suffix = -amine
- ◆ Hydrocarbon structure is an alkane therefore -ane
- ◆ The longest continuous chain is C4 therefore root = but
- ◆ The first point of difference rule requires numbering from the right as drawn to make the amine group locant **Name: butan-2-amine or 2-butylamine (or sec-butylamine)**

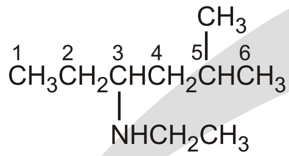
☞ Secondary amines have two alkyl groups attached to the N.

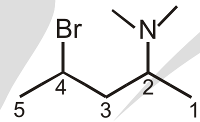
- ★ The root name is based on the longest chain with the -NH attached.
- ★ The chain is numbered so as to give the amine unit the lowest possible number.
- ★ The amine suffix is appended to the appropriate alkyl root or alkane-root.

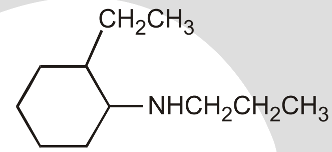
The substituents—regardless of whether they are attached to the nitrogen or to the parent hydrocarbon—are listed in alphabetical order, and then a number or an "*N*" is assigned to each one. The chain is numbered in the direction that gives the functional group suffix the lowest number.

Solved Example


3-chloro-*N*-methyl-1-butanamine


N-ethyl-5-methyl-3-hexanamine


4-bromo-*N*,*N*-dimethyl-2-pentanamine


2-ethyl-*N*-propylcyclohexanamine

Solved Example

- $CH_3NHCH_2CH_3$
  - Functional group is an amine, therefore suffix = -amine
  - Hydrocarbon structure is an alkane therefore -ane
  - The longest continuous chain is C2 therefore root =eth

  Name: N-methylethanmine

☞ Tertiary amines have three alkyl group attached to the N.
- ⋆ The root name is based on the longest chain with the -N attached.
- ⋆ The chain is numbered so as to give the amine unit the lowest possible number.
- ⋆ The other alkyl groups are treated as substituents, with N as the locant.
- ⋆ The amine suffix is appended to the appropriate alkyl root or alkana-root.

Solved Example

- $(CH_3)_3N$
  - Functional group is an amine, therefore suffix = -amine
  - Hydrocarbon structure is an alkane therefore -ane
  - The longest continuous chain is C1 therefore root =meth

  Name : N, N-dimethylmethylamine or N,N-dimethylmethanamine or Trimethylamine

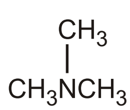
COMMON NAMES

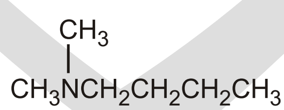
The common name of an amine consists of the names of the alkyl groups bonded to the nitrogen, in alphabetical order, followed by "amine."

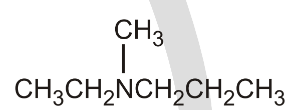
CH_3NH_2
methylamine

$CH_3NHCH_2CH_2CH_3$
methylpropylamine

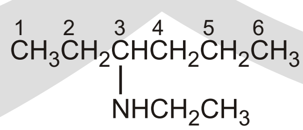
$CH_3CH_2NHCH_2CH_3$
diethylamine

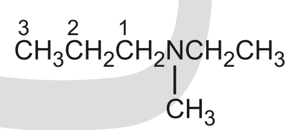

trimethylamine


butyldimethylamine


ethylmethylpropylamine

$\overset{4}{C}H_3\overset{3}{C}H_2\overset{2}{C}H_2\overset{1}{C}H_2NH_2$
1-butanamine
or
butan-1-amine

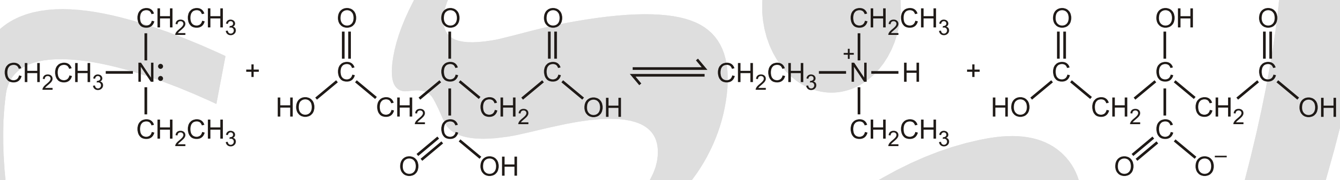

N-ethyl-3-hexanamine
or
N-ethylhexan-3-amine


N-ethyl-*N*-methyl-1-propanamine
or
N-ethyl-*N*-methylpropan-1-amine

☑ SPECIAL TOPIC

SPECIAL TOPIC : BAD-SMELLING COMPOUNDS

Amines are responsible for some of nature's unpleasant odors. Amines with relatively small alkyl groups have a fishy smell. For example, fermented shark, a traditional dish in Iceland, smells exactly like triethylamine. Fish is often served with lemon, because the citric acid in lemon protonates the amine, thereby converting it to its better smelling acidic form.



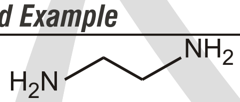
triethylamine bad smelling | citric acid | protonated triethylamine better smelling | conjugate base of citric acid

☑ SPECIAL TOPIC

SPECIAL TOPIC: SOURCE OF NAME AMMONIA

Most people can associate the name ammonia (NH_3) with a gas having a pungent odor; the systematic name "nitrogen trihydride" (which is rarely used) will tell you its formula. What it will not tell you is that smoke from burning camel dung (the staple fuel of North Africa) condenses on cool surfaces to form a crystalline deposit. The ancient Romans first noticed this on the walls and ceiling of the temple that the Egyptians had built to the Sun-god Amun in Thebes, and they named the material "sal ammoniac," meaning "salt of Amun". In 1774, Joseph Priestly (the discoverer of oxygen) found that heating sal ammoniac produced a gas with a pungent odor, which a T. Bergman named "ammonia" eight years later.

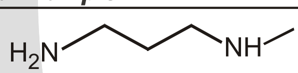
Solved Example

▸ 

- Functional group is an amine, therefore suffix = -amine
- There are two amines, so insert the multiple di
- The longest continuous chain is C2 therefore root = eth

Name: 1,2-ethyldiamine or ethane-1,2-diamine

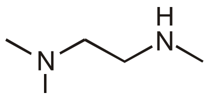
Solved Example

▸ 

- The longest continuous chain is C3 therefore root = prop
- The methyl group is located on the amine, so locant = N

Name: N-methyl-1,3-propyldiamine or N-methypropane-1,3-diamine

Solved Example

- 
  - Functional group is an amine, therefore suffix = -amine
  - The longest continuous chain is C2 therefore root = eth
  - There is are three C1 substituents = trimethyl

  Name: N,N,N'-trimethyl-1,2-ethyldiamine or N,N,N'-trimethylethane-1,2-diamine

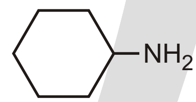
Solved Example

- Name the following amines (Common names) :

$CH_3-CH_2-CH_2-NH-CH_2CH_2CH_3$ (N bears H)
Dipropylamine

$CH_3CH_2-N(CH_2CH_3)-CH_2CH_3$
Triethylamine

$CH_3CH(CH_3)-NH_2$
Isopropylamine

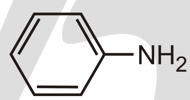

Cyclohexylamine

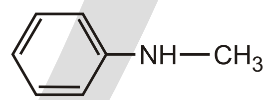
Solved Example

- Name of following amines (IUPAC names):

$CH_3CH_2-NH-CH_2CH_2CH_3$
N-Ethylpropanamine

$CH_3-N(CH_3)-CH_2CH_2CH_3$
N-N Dimethylpropanamine


Aniline


N-Methylaniline

Solved Example

- Name the following amine :

$H_2NCH_2CH_2COOH$

Sol. 3-Aminopropanoic acid

Solved Example

- IUPAC of the given amines :

$\overset{4}{CH_3}-\overset{3}{CH_2}-\overset{2}{CH}(NH_2)-\overset{1}{CH_3}$
2-butanamine

$\overset{4}{CH_3}-\overset{3}{CH}(CH_3)-\overset{2}{CH_2}-\overset{1}{CH_2}NH_2$
3-methyl-1-butanamine

$\overset{1}{CH_3}-\overset{2}{CH}(NHCH_3)\overset{3}{CH_2}\overset{4}{CH_3}$
N-methylbutan-2-amine

$\overset{6}{CH_3}-\overset{5}{CH_2}-\overset{4}{CH}(CH_3)-\overset{3}{CH}(NH_2)-\overset{2}{CH}(CH_3)-\overset{1}{CH_3}$
2,4-dimethyl-3-hexanamine

Solved Example

- Give IUPAC and common names and degrees of the given Amines:

$H_3C-NH-CH_2CH_3$	Ethylmethylamine	N-Methylethanamine	2º
$H_3C-N(CH_3)-CH_3$	Trimethylamine	N,N-Dimethylmethanamine	3º

Solved Example

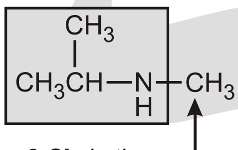
- Give the IUPAC name of $H_2N — CH_2 — CH_2 — CH \quad CH_2$.

Sol. But-3-en-1-amine

Solved Example

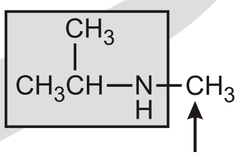
Step-1 Name the following 2º amine : $(CH_3)_2CHNHCH_3$

Designate the longest alkyl chain (or largest ring) bonded to the N atom as this parent amine and assign a common or systematic name.



3 C's in the longest chain → isopropylamine (common name) or 2-propanamine (systematic name)

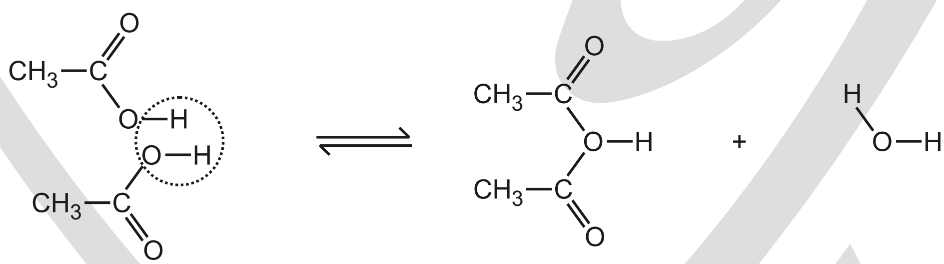
Step-2 Name the other groups on the N atom as alkyl groups, alphabetize the names, and put the prefix N-before the name.



one methyl substituent

Ans. N-methylisopropylamine (common name) or N-methyl-2-propanamine (systematic name)

FORMATION OF ANHYDRIDES



2x molecules of Ethanoic Acid (a carboxylic Acid)

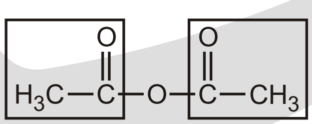
1x molecules of Ethanoic Anhydride (an Acid Anhydride)

1x molecules of Water

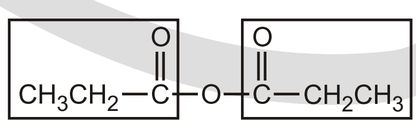
NOMENCLATURE OF ANHYDRIDES

The acid anhydride functional group results when two carboxylic acids combine and lose water (anhydride = without water). Symmetrical acid anhydrides are named like carboxylic acids except the ending-acid is replaced with -anhydride. This is true for both the IUPAC and Common nomenclature.

SYMMETRICAL ANHYDRIDES



IUPAC : ethanoic anhydride
Common : acetic anhydride

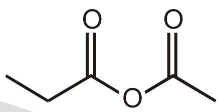


IUPAC : Propanoic anhydride
Common : Propionic anhydride

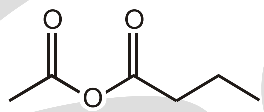
UNSYMMETRICAL ACID ANHYDRIDES

Unsymmetrical acid anhydrides are named by first naming each component carboxylic acid alphabetically arranged (without the word acid) followed by spaces and then the word anhydride.

Solved Example

▸ 

Ethanoic propanoic anhydride
(Acetic propionic anhydride)



Butanoic ethanoic anhydride
(Acetic butyric anhydride)

$$CH_3-CH_2-\overset{\overset{\displaystyle O}{||}}{C}-O-\overset{\overset{\displaystyle O}{||}}{C}-CH_2-CH_3$$

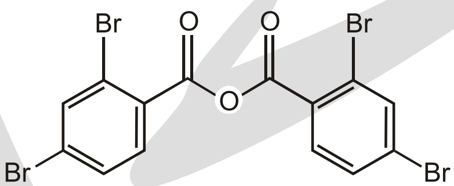
Propanoic anhydride

$$CH_3-\overset{\overset{\displaystyle O}{||}}{C}-O-\overset{\overset{\displaystyle O}{||}}{C}-CH_2-CH_3$$

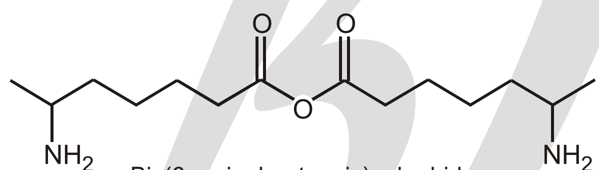
ethanoic propanoic anhydride

☞ Anhydrides of substituted monocarboxylic acids, if symmetrically substituted, are named by prefixing "bis-" to the name of the acid and replacing the word "acid" by "anhydride" The "bis" may, however, be omitted.

Solved Example

▸ 

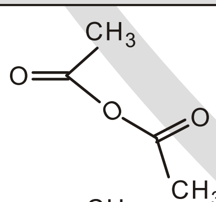
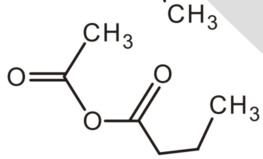
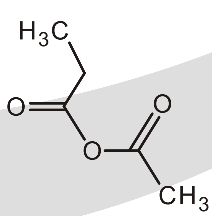
Bis(2,4-dibromobenzoic) anhydride

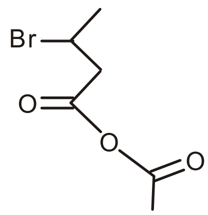
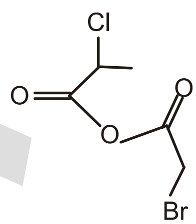
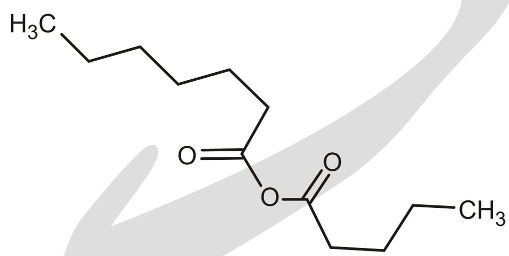


Bis(6-aminoheptanoic)anhydride

EXERCISE

WORK SHEET

S.No.	Compounds	Write IUPAC - Name
1.		
2.		
3.		

4.		
5.		
6.		

Answers

Work sheet

1. ethanoic anhydride
2. butanoic ethanoic anhydride
3. ethanoic propanoic anhydride
4. 3-bromobutanoic ethanoic anhydride
5. 2-bromoethanoic 2-chloropropanoic anhydride
6. heptanoic pentanoic anhydride

❑❑❑

CHAPTER

11

Nomenclature of Polyfunctional Groups

IUPAC SYSTEM OF NOMENCLATURE

Following is the priority to write the IUPAC names of different organic compounds having Polyfunctional groups.

PRIORITY LIST

S.No.	Functional group	Formula	Family name	Substitution
1.	Carboxylic acid	$-COOH$	Alkanoic acid or carboxylic acid	Carboxy
2.	Sulphonic acid	$-SO_3H$	Alkane Sulphonic acid	Sulpho
3.	Carboxylic acid anhydride	$-\underset{\underset{O}{\Vert}}{C}-O-\underset{\underset{O}{\Vert}}{C}-$	Alkanoic acid anhydride	
4.	Ester	$-\overset{*}{C}OOR$	Alkyl alkanoate	Carbalkoxy
5.	Acid halide	$-\overset{*}{C}OX$	Alkanoyl halide	Haloformyl, carbox halide, halocarbonyl
6.	Acid Amide	$-\overset{*}{C}ONH_2$	Alkanamide	Carbomyl
7.	Cyanide	$C\equiv N$	Alkanenitrile	Cyano
8.	Aldehyde	$-\overset{*}{C}HO$	Alkanal	Formyl. Aldo, Oxo
9.	Ketone	$-\overset{\overset{O}{\Vert}}{C}-$	Alkanone	Oxo, Keto
10.	Alcohol	$-OH$	Alkanol	Hydroxy
11.	Thiols	$-SH$	Alkane thiols	Sulphamyl

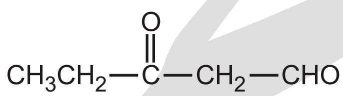
12.	Amines	— NH_2	Alkanamine	Amino
13.	Alkene, alkyne	C C C	Alkene, alkyne	
14.	Ethers	— OR	Alkoxyalkane	Alkoxy

❑ **NOTE:** 1 , 2 and 3 Amines are considered to be different funtional groups.

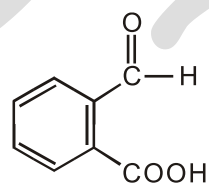
Step-1 : Identification of functional groups and classifying them into main, subsidiary and substituent groups.

1. **Main Functional group :** The functional group getting highest priority is called main functional group.
2. **Subsidiary group :** If the molecule contains C C or C C apart from Main functional group then the C C or C C are called subsidiary group.
3. **Substituent group :** Any other functional group apart from main or subsidiary.

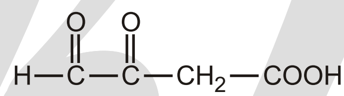
A ketone or aldehyde group can also be named as a substituent on a molecule with a higher priority functional group as its root. A ketone or aldehyde carbonyl is named by the prefix oxo- if it is included as part of the longest chain in the root name. When an aldehyde –CHO group is a substituent and not part of the longest chain, it is named by the prefix formyl. Carboxylic acids frequently contain ketone or aldehyde groups named as substituents.



3-oxopentanal



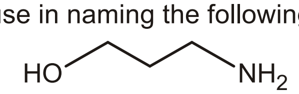
2-formylbenzoic acid



3,4-dioxobutanoic acid

Solved Example

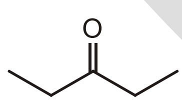
▸ Identify what suffix you would use in naming the following compound :



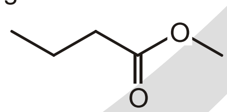
Sol. There are two functional groups in this compound, so we have to decide between calling this compound an amine or calling it an alcohol. If we look at the hierarchy above, we see that an alcohol outranks an amine. Therefore, we use the suffix -ol in naming this compound.

Solved Example

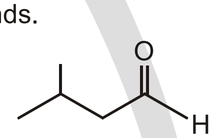
▸ Identify what suffix you would use in naming each of the following compounds.



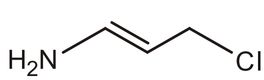
suffix : ....



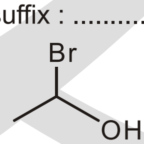
suffix : ....



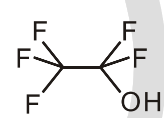
suffix : ....



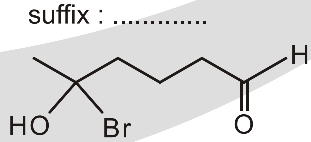
suffix : ....



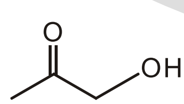
suffix : ....



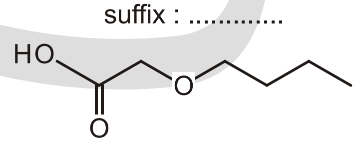
suffix : ....



suffix : ....



suffix : ....

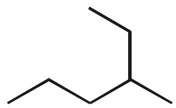


suffix : ....

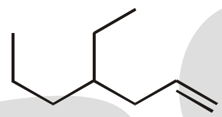
Ans.	-one	-oate	-al
	-amine	-ol	-ol
	-al	-one	-oic acid

Solved Example

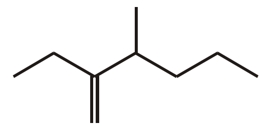
- Name the parent chain in the each of the following compounds.



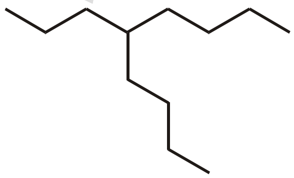
Parent: ....



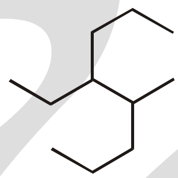
Parent: ....



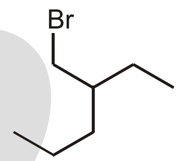
Parent: ....



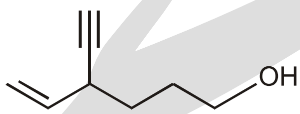
Parent: ....



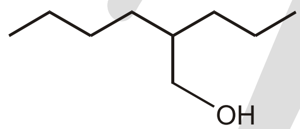
Parent: ....



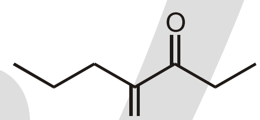
Parent: ....



Parent: ....



Parent: ....

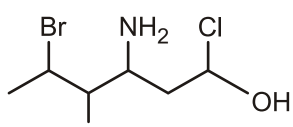


Parent: ....

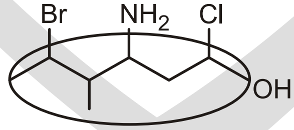
Ans.	Hexane	Heptene	Hexene
	Nonane	Octane	Hexane
	Hexene	Hexane	Pentene

Solved Example

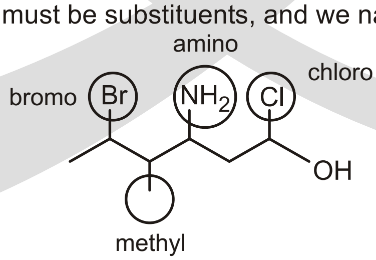
- In the following compound, identify all groups that would be considered substituents, and then indicate how you would name each substituent:



Ans. First we must locate the functional group that gets the priority. Alcohols outrank amines, so the OH group is the priority functional group. Then, we need to locate the parent chain. There are no double or triple bonds, so we choose the longest chain containing the OH group:

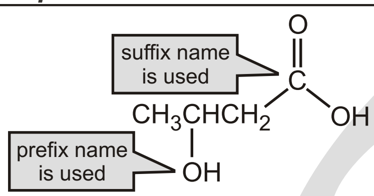


Now we know which groups must be substituents, and we name them accordingly:

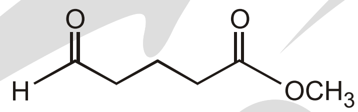


☞ If a compound has two functional groups, the one with the lower priority is indicated by a prefix and the one with the higher priority by a suffix (unless one of the functional groups is an alkene).

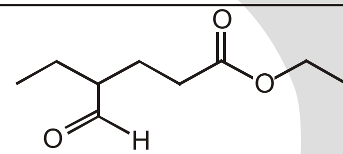
Solved Example



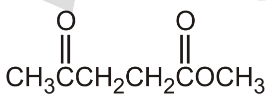
3-hydroxybutanoicacid



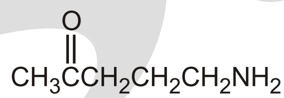
methyl 5-oxopentanoate



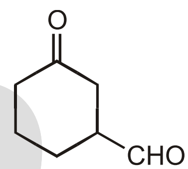
ethyl 4-formylhexanoate



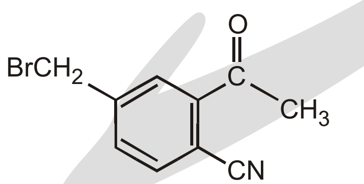
Methyl 4-oxopentanoate
(an ester with a ketone group)



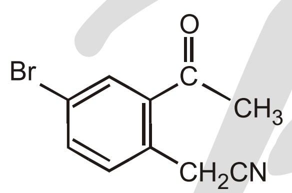
5-Aminopentan-2-one



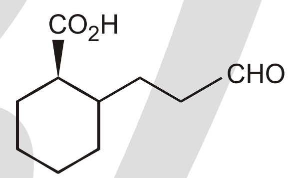
3-Oxocyclohexanecarbaldehyde
(an aldehyde with a ketone group)



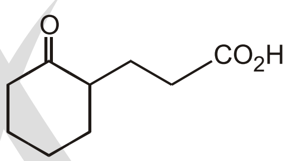
2-Acetyl(4-bromomethyl) benzonitrile



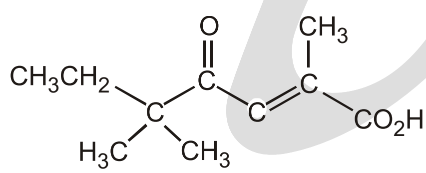
2-(2-Acetyl-4-bromophenyl) acetonitrile



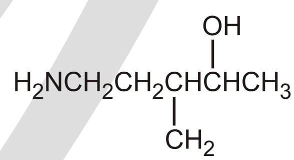
2-(3-oxopropyl)cyclo
-hexanecarboxylic acid



3-(2-Oxocyclohexyl) propanoic acid



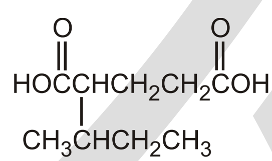
(E)-2,5,5-Trimethyl-4-oxohept-2-enoic acid



5-Amino-3-methylpentan-2-ol

$H_3C—Mg—CH_3$

Dimethylmagnesium
(one word, because magnesium is an element)



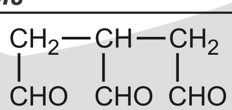
2-(1-methylpropyl) pentanedioic acid

CONDITIONS TO USE SPECIAL SECONDARY SUFFIXES

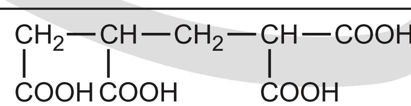
When carbon atom of carbon containing functional groups is not counted in parent chain (in word root) then special secondary suffixes are used.

☞ When three or more same carbon containing functional groups are present.

Solved Example



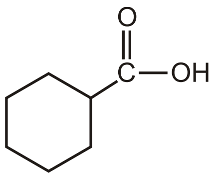
propane-1,2,3-tricarbaldehyde



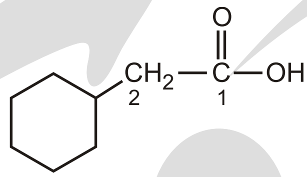
butane-1,1,3,4-tetracarboxylic acid

☞ When carbon containing functional group is directly attached with alicyclic ring.

Solved Example

▸ 

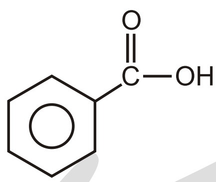
cyclohexanecarboxylic acid



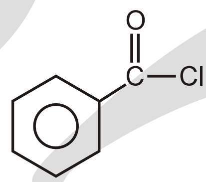
2-cyclohexylethanoic acid

☞ When carbon containg functions if is directly attached with benzene ring

Solved Example

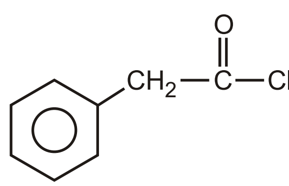
▸ 

benzenecarboxylic acid



benzenecarbonylchloride

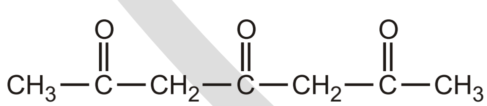
☞ But when benzene and carbon containing functional group are not directly atttached then benzene is treated as phenyl substituent.



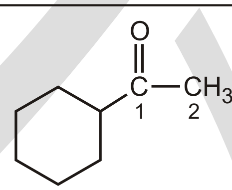
2-Phenylethanoylchloride

❑ **NOTE:** For Keto group above conditions are not applied because its carbon is always counted in parent chain (in word root).

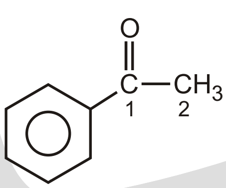
Solved Example

▸ 

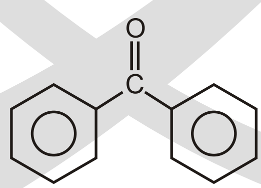
hept ane-2,4,6 trione



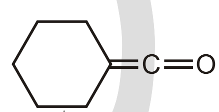
cyclohexyl**eth**anone



phenyl**eth**anone



diphenyl**meth**anone

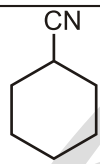


cyclohexylidene**meth**anone

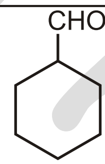
PREFIXES AND SUFFIXES FOR THE CARBON CONTAINING FUNCTIONAL GROUPS

Functional group	Prefix	Suffix
—CHO	Formyl	Carbaldehyde
—COOH	Carboxy	Carboxylic acid
—COX (X F, Cl, Br, I)	Halocarbonyl	Carbonyl halide
—COOR	Alkoxycarbonyl or Alkanoyloxy	Carboxylate
—$CONH_2$	Carbamoyl	Carboxamide
—CN	Cyano	Carbonitrile
$>C=O$	oxo/keto	—

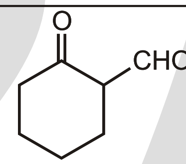
Solved Example



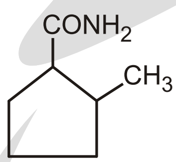
cyclohexanecarbonitrile



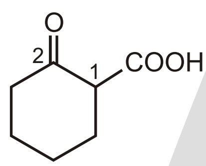
cyclohexanecarbaldehyde



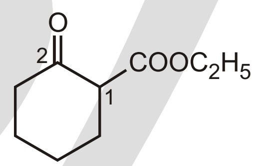
2-oxocyclohexane-1-carbaldehyde



2-methylcyclopentane-1-carboxamide



2-oxocyclohexane-1-carboxylic acid



Ethyl 2-oxocyclohexanecarboxylate

Solved Example

- Write the IUPAC name of the following compound:

$$CH_3CH_2-\overset{\displaystyle O}{\overset{\|}{C}}-CH_2CH_2CH_2-Cl$$

Sol. $\overset{1}{CH_3}\overset{2}{CH_2}-\overset{\displaystyle O}{\overset{3\|}{C}}-\overset{4}{CH_2}\overset{5}{CH_2}\overset{6}{CH_2}-Cl$

6-Chlorohexan-3-one

Solved Example

- Write the IUPAC name of the compound:

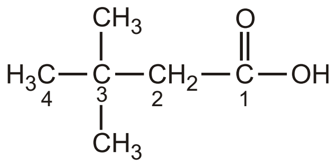
$$CH_3-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CO-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_3$$

Sol. $\overset{1}{CH_3}-\underset{\displaystyle CH_3}{\underset{|}{\overset{2}{CH}}}-\overset{3}{CO}-\underset{\displaystyle CH_3}{\underset{|}{\overset{4}{CH}}}-\overset{5}{CH_3}$

2,4-Dimethylpentan-3-one

Solved Example

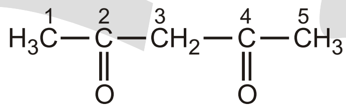
- Write the IUPAC name of the following compound : $(CH_3)_3CCH_2COOH$

Sol. 

The IUPAC name of the given compound is 3,3-dimethyl-butanoic acid.

Solved Example

- Write the IUPAC name of the following compound : $CH_3COCH_2COCH_3$

Sol. 

Pentane-2,4-dione

Solved Example

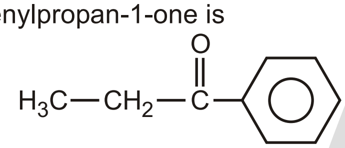
- Write the structure of 3-oxopentanal.

Sol. $CH_3 - CH_2CO - CH_2 - CHO$

Solved Example

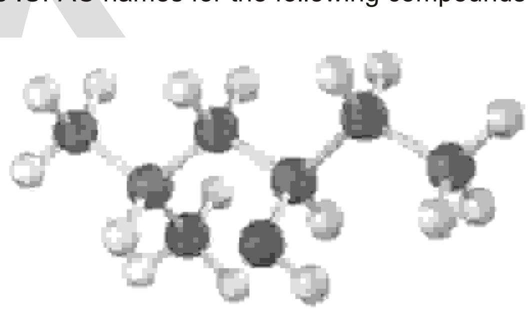
- Draw the structural formula of 1-phenylpropan-1-one molecule.

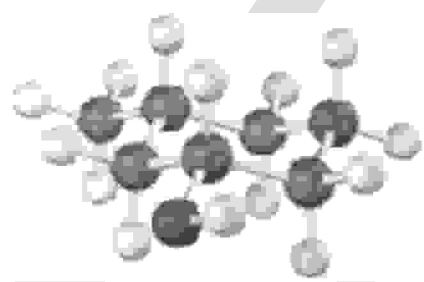
Sol. The structural formula of 1-phenylpropan-1-one is

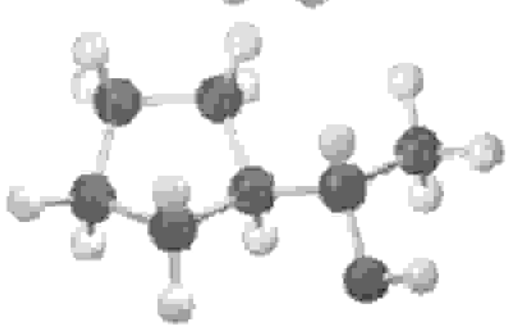


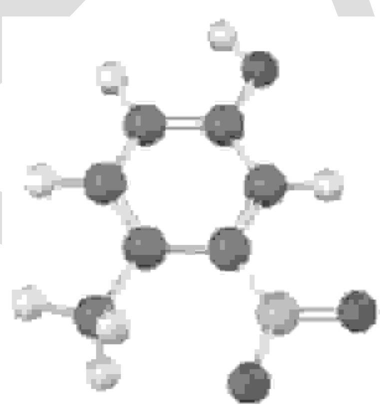
Solved Example

- Give IUPAC names for the following compounds :

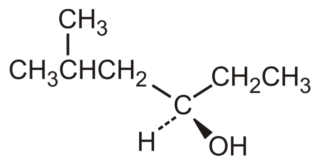
(a) 

(b) 

(c) 

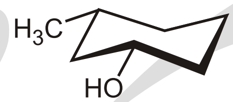
(d) 

Sol. (a)



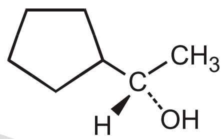
(R)-5-Methyl-3-hexanol

(b)



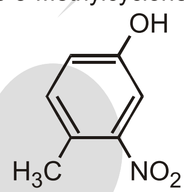
cis-3-Methylcyclohexanol

(c)



(S)-1-Cyclopentylethanol

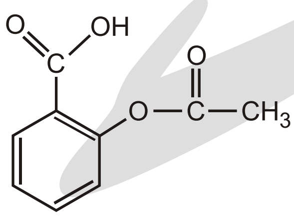
(d)



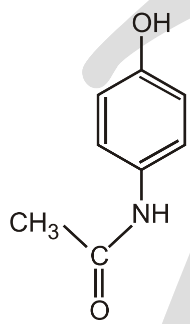
4-Methyl-3-nitrophenol

Solved Example

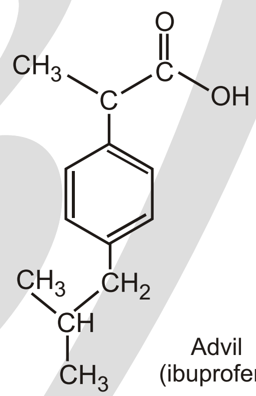
- The following compounds are the active ingredients in over-the-counter drugs used as analgestics (to relieve pain without decreasing sensibility or consciousness), antipyretics (to reduce the body temperature when it is elevated), and/or anti-inflammatory agents (to counteract swelling or inflammation of the joints, skin, and eyes). Identify the functional groups in each molecule.



Aspirin
(acetylsalicylic acid)



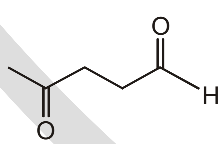
Tylenol
(acetaminophen)

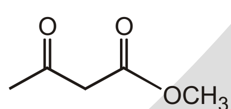


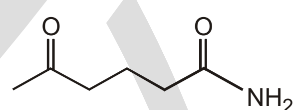
Advil
(ibuprofen)

Sol. All three compounds are aromatic. Aspirin is also a carboxylic acid ($-CO_2H$) and an ester ($-CO_2CH_3$). Tylenol is also an alcohol (–OH) and an amide (–CONH–). Ibuprofen contains alkane substituents and a carboxylic acid functional group.

☞ If a ketone has a second functional group of higher naming priority. The ketone oxygen is indicated by the prefix "oxo."







systematic name : 4-oxopentanal; methyl 3-oxobutanoate; 5-oxohexanamide

$\underset{}{CH_3}-\overset{O}{\overset{\|}{C}}-H$

Ethanal

$\underset{7}{CH_3}\underset{6}{CH_2}\underset{5}{CH_2}-\underset{4}{\overset{Br}{\overset{|}{CH}}}-\underset{3}{\overset{CH_3}{\overset{|}{CH}}}-\underset{2}{CH_2}-\underset{1}{\overset{O}{\overset{\|}{C}}}-H$

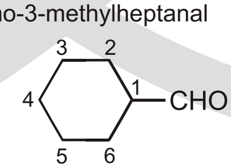
4-bromo-3-methylheptanal

$\underset{4}{CH_3}-\underset{3}{\overset{OH}{\overset{|}{CH}}}-\underset{2}{CH_2}-\underset{1}{\overset{O}{\overset{\|}{C}}}-H$

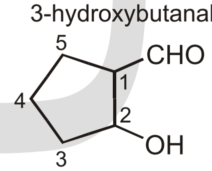
3-hydroxybutanal

$\underset{5}{CH_3}-\underset{4}{CH_2}-\underset{3}{CH}=\underset{2}{CH}-\underset{1}{CHO}$

2-pentenal
or
pent-2-enal



cyclohexanecarbaldehyde



2-hydroxycyclopentane-1-carbaldehyde

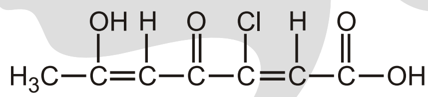
EXERCISE

SINGLE CHOICE QUESTIONS

1. Which of the following organic compounds does NOT have the molecular formula of $C_3H_6O_2$?
   (A) Propanoic acid (B) Methyl ethanoate (C) Ethyl ethanoate (D) Ethyl methanoate

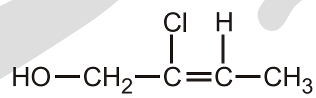
2. What is the IUPAC name for the organic compound with the condensed formula of $HCOCHBrCOCH_3$?
   (A) 3-bromo-4-formylpropan-2-one (B) 4-formyl-3-bromopropan-2-one
   (C) 2-bromo-3-oxobutanal (D) 3-oxo-2-bromobutanal

3. What is the IUPAC name of the following compound?



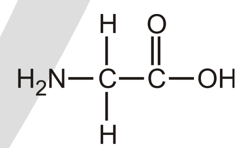
   (A) 5-chloro-2-hydroxy-4-oxohepta-2,5-dienoic acid (B) 2-hydroxy-4-oxo-5-chlorohepta-2,5-dienoic acid
   (C) 3-chloro-6-hydroxy-4-oxohepta-2,5-dienoic acid (D) 3-chloro-4-oxo-6-hydroxyhepta-2,5-dienoic acid

4. What is the IUPAC name of the following compound?



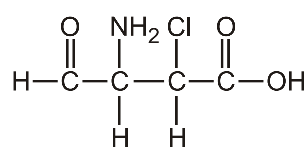
   (A) 2-chloro-l-hydroxybut-2-ene (B) 3-chloro-4-hydroxybut-2-ene
   (C) 2-chlorobut-2-en-1-ol (D) 3-chlorobut-2-en-4-ol

5. Consider the following organic compound :



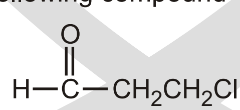
   Which of the following statements concerning the compound above are correct?
   (1) It has two functional groups namely amide group and carboxyl group.
   (2) It is soluble in water.
   (3) Its IUPAC name is 2-aminoethanoic acid.
   (A) (1) and (2) only (B) (1) and (3) only (C) (2) and (3) only (D) (1), (2) and (3)

6. What is the IUPAC name for the following compound?



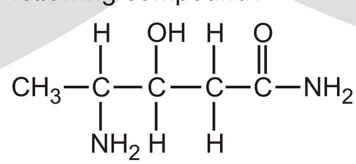
   (A) 2-chloro-3-amino-4-oxobutanoic acid (B) 3-amino-2-chloro-4-oxobutanoic acid
   (C) 2-amino-3 -carboxy-3-chloropropanal (D) 2-amino-3-chloro-3-carboxypropanal

7. What is the IUPAC name of the following compound ?



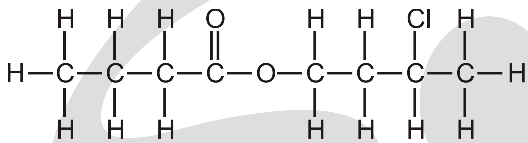
   (A) 1-chloropropan-3-al (B) Chloropropanal (C) 3-chloropropan-l-al (D) 3-chloropropanol

8. What is the IUPAC name for the following compound?



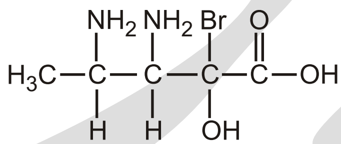
(A) 3-hydroxy-4-aminopentanamide
(B) 4-amino-3-hydroxypentanamide
(C) 4-amino-3-hydroxy-4-methylbutanamide
(D) 3-hydroxy-4-amino-4-methylbutanamide

9. What is the IUPAC name of the following compound ?



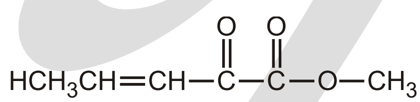
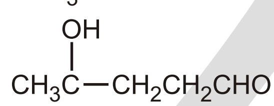
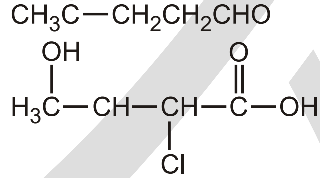
(A) 3-chlorobutyl butanoate
(B) 2-chlorobutyl butanoate
(C) Butyl 2-chlorobutanoate
(D) Propyl 4-chloropentanoate

10. What is the IUPAC name of the following compound?

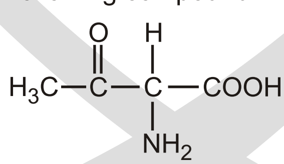


(A) 2,3-diamino-3-bromo-3-hydroxypentanoic acid
(B) 3,4-diamino-2-bromo-2-hydroxypentanoic acid
(C) 2-bromo-2-hydroxy-3,4-diaminopentanoicacid
(D) 2-bromo-2-hydroxy-3,4-diamino-4-methylbutanoic acid

11. Which of the following combinations about the structural formula for a compound correct?

	Compound	**Structural formula**
(A)	Diol	HOOC – COOH
(B)	Methyl 2-oxopent-3-enoate	
(C)	5-formylpent-2-ol	
(D)	3-carboxy-3-chlorobutan-2-ol	

12. What is the IUPAC name of the following compound?

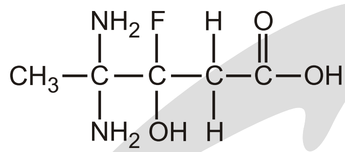


(A) 3-amino-4-carboxybutan-2-one
(B) 3-amino-4-carboxybutan-2-al
(C) 2-amino-3-oxobutanoic acid
(D) 2-amino-3-methyl-3-oxopropanoic acid

13. Which of the following is the condensed formula for 3-oxopentanal?

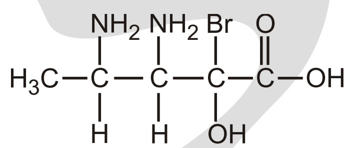
(A) $CH_3CH_2COCH_2COH$
(B) $CH_3CH_2COCH_2CHO$
(C) $CH_3CH_2CH_2COCHO$
(D) $CH_3CH_2CH_2COCOH$

14. What is the IUPAC name for the following compound?



(A) 3-fluoro-3-hydroxy-4,4-diaminopentanoic acid
(B) 4,4-diamino-3-fluoro-3-hydroxypentanoic acid
(C) 3-fluoro-3-hydroxy-4,4-diamino-4-methylbutanoic acid
(D) 4,4-diamino-3-fluoro-3-hydroxy-4-methylbutanoic acid

15. Consider the following compound:



Which of the following homologous series does the above compound belong to?

(A) Amines (B) Alcohols (C) Ketones (D) Carboxylic acids

16. Which of the following is the condensed formula for 4-aminobutanamide?

(A) $NH_2CH_2(CH_2)_2CONH_2$
(B) $NO_2CH_2(CH_2)_2CONH_2$
(C) $NH_2CO(CH_2)_2CONH_2$
(D) $CH_3(CH_2)_2CH(NH_2)_2$

17. Which of the following combinations is correct?

	IUPAC name	Trivial name	Common use
(A)	Propan-l-ol	Isopropyl alcohol	Solvent
(B)	Ethanoic acid	Acetic acid	Solvent
(C)	Methanal	Formaldehyde	Production of polymers
(D)	Trichloromethane	Chloroform	Fuel additive

18. Which of the following statements about ethanoic acid and methyl methanoate are correct?

(1) They are functional group isomers with the molecular formula $C_2H_4O_2$.
(2) They belong to different homologous series.
(3) They have different chemical properties.

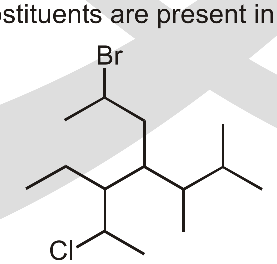
(A) (1) and (2) only (B) (1) and (3) only (C) (2) and (3) only (D) (1), (2) and (3)

19. Which of the following compound are functional group isomers of $C_4H_8O_2$?

(1) Methyl propanoate (2) 4-hydroxybutanal (3) Butane-1,4-diol

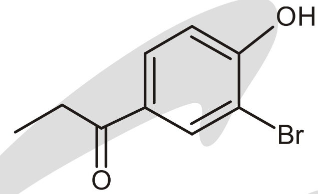
(A) (1) and (2) only (B) (1) and (3) only (C) (2) and (3) only (D) (1) (2) and (3)

20. How many total number of substituents are present in the following compound?



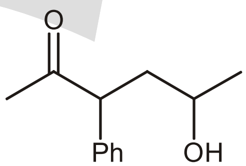
(A) 3 (B) 4 (C) 5 (D) 6

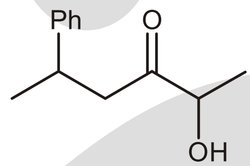
21. What is the IUPAC name for

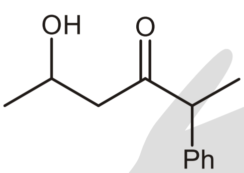


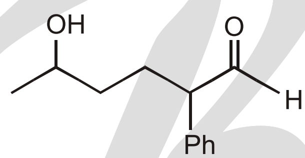
(A) m-Bromo-p-hydroxypropiophenone
(B) 2-Bromo-4-propanoyl phenol
(C) 3-Bromo-4-hydroxy phenyl propan-1-one
(D) 2-Hydroxy-5-propanoyl bromo benzene

22. What is the correct structure for 5-hydroxy-2-phenyl hexan-3-one?

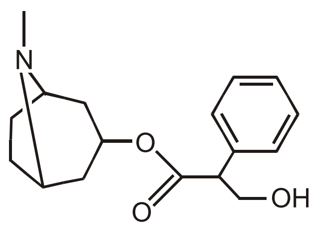
(A) 

(B) 

(C) 

(D) 

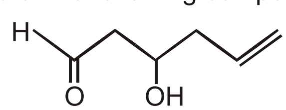
23. Which of the functional group is NOT present in Atropine?



Atropine

(A) Amine
(B) Phenol
(C) ester
(D) Benzene ring

24. What is the correct IUPAC name of the following compound.



(A) 4-Hydroxyhex-1-en-6-al
(B) 1-Oxohex-5-en-3-ol
(C) 3-Hydroxyhex-5-enal
(D) 6-Oxohex-1-en-4-ol

25. IUPAC name of Benzyl alcohol is :

(A) Phenol
(B) Hydroxymethyl Benzene
(C) Benzenol
(D) 1-Phenyl methanol

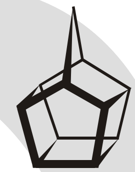
26. IUPAC name of T. N. T. is :

(A) Trinitrotoluene
(B) 1,2,3-Trinitrotoluene
(C) 2,4,6-Trinitrotoluene
(D) 2-methyl-1,3,5-Trinitrotoluene

27. What is the IUPAC name of Laughing gas?

(A) Nitrogen oxide
(B) Nitrogen dioxide
(C) Dinitrogen Oxide
(D) Nitrous oxide

28. The given compound is called Churchane. What is the Double Bond Equivalent value of Churchane?

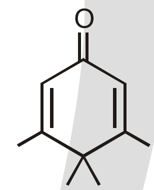


(A) 4 (B) 5
(C) 6 (D) 7

29. What is the IUPAC name of Natrium?

(A) Sodium (B) Nitride (C) Natride (D) Sodide

30. The common name of the given compound is Penguinone because it is a penguine shaped ketone. What is the IUPAC name of Penguinone?



(A) 6-oxo-2,3,3,4-tetramethylcyclohexa-1,4-diene
(B) 3,4,4,5-tetramethylcyclohexa-2,5-dien-1-one
(C) 6-formyl-2,3,3,4-tetramethylcyclohexa-1,4-diene
(D) 2,3,3,4-tetramethylcyclohexa-1,4-dien-6-one

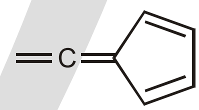
31. Which of the following is the correct IUPAC name of Unsymmetric Butylene?

(A) 1-Butene (B) 2-Butene (C) 2-MethylPropene (D) 2-Methylbutene

32. Which of the following Common names of Carboxylic acids and their sources is correctly matched?

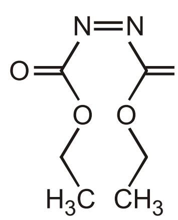
(A) Formic acid- Vinegar (B) Acetic acid- Ant (C) Butyric acid- Butter (D) Steric acid- Goat

33. What is correct IUPAC name of the given compound?



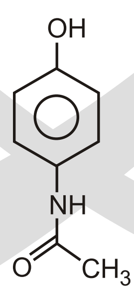
(A) 5-ethenylidene-1,3-cyclopentadiene
(B) 1,3-cyclopentadiene-5-ethenyl ketene
(C) cyclopentyl-2,4-dienylidene ethene
(D) 1-ethenylidenyl-cyclopenta-2,4-diene

34. The given compound is called as DEAD. What is the full-form of DEAD?



(A) Diethyl azodicaboxylic acid (B) Diethyl azodicaboxylic anhydride
(C) Diethyl azodicaboxylate (D) Diethyl azodione

35. What is the IUPAC name of Paracetamol?



(A) N-(4-hydroxyphenyl)acetamide (B) N-phenylacetamide
(C) N-phenolethanamide (D) N-(4-hydroxyphenyl)ethanamide

36. IUPAC name of Acetophenone is :

(A) Benzophenone
(B) 1,1-Biphenylmethanone
(C) Benzene Methyl methanone
(D) 1-Phenyl Ethanone

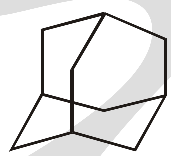
37. What is the name of the compound shown?

(A) 1,3-dimethylbicyclo[5.3.0]decane
(B) 1,3-dimethylbicyclo[5.3.1]decane
(C) 1,8-dimethylbicyclo[5.3.0]decane
(D) 1,9-dimethylbicyclo[5.3.0]decane

38. Which of the following is an allylic alcohol?

(A) CH_2 $CHCH_2OCH_3$
(B) CH_2 $CHCH(CH_3)OH$
(C) CH_3CH $CHCH_2OH$
(D) CH_2 $CHCH_2CH_3$

39. Number of 6-membered rings in Adamantane is :



(A) 4
(B) 5
(C) 6
(D) 7

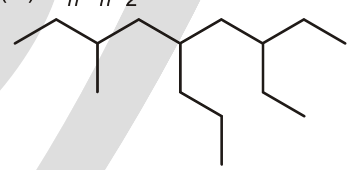
40. Which of the following hydrocarbons does not have isomers?

(A) C_3H_6
(B) C_4H_6
(C) C_4H_8
(D) C_3H_8

41. The general formula for non-cyclic alkenes is :

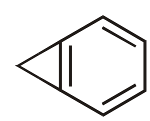
(A) $C_nH_{2n\ 2}$
(B) C_nH_{2n}
(C) C_nH_n
(D) $C_nH_{n\ 2}$

42. What is correct IUPAC name of the given compound?



(A) 7-ethyl-3-methyl-5-propyl Nonane
(B) 6-ethyl-4-(2-methylbutyl)
(C) 3-ethyl-7-methyl-5-propyl Nonane
(D) 3-methyl 5-(2-ethyl butyl)-octane

43. What is correct IUPAC name of the given compound?



(A) Bicyclo [4.1.0] hepta-1,2,4-triene
(B) Bicyclo [0.1.4] hepta-1,2,4-triene
(C) Bicyclo [4.1] hepta-1,2,4-triene
(D) Bicyclo [1.4] hepta-1,2,4-triene

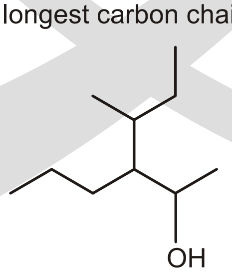
44. IUPAC name of Ethylformate is :

(A) Ethyl ethanoate
(B) Ethyl methanoate
(C) Methyl methanoate
(D) Methyl ethanoate

45. IUPAC name of Phenylacetate is :

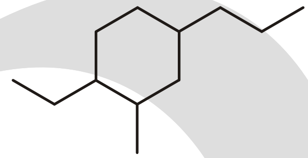
(A) Phenyl ethanoate
(B) Phenyl 1-Phenyl methanoate
(C) Phenyl methanoate
(D) 1-Phenyl Ethanone

46. How many carbon atoms are in the longest carbon chain (IUPAC)?



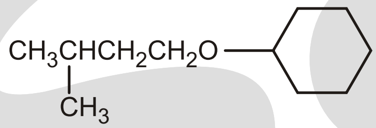
(A) 4
(B) 5
(C) 6
(D) 7

47. The correct IUPAC name of the given compound is :



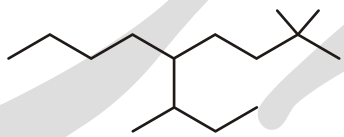
(A) 5-ethyl-1-methyl-2-propylcyclohexane
(B) 1-ethyl-3-methyl-4-propylcyclohexane
(C) 4-ethyl-2-methyl-1-propylcyclohexane
(D) 4-ethyl-1-methyl-3-propylcyclohexane

48. What is the common name of the given compound?



(A) Cyclohexyl secpentyl ether
(B) Cyclohexyl isopentyl ether
(C) Cyclohexyl isobutyl ether
(D) Cyclohexyl secbutyl ether

49. What is correct IUPAC name of the given compound?



(A) 4-Butyl-3,7,7-Trimethyloctane
(B) 5-Butyl-2,2,6-Trimethyloctane
(C) 6-methyl-5(3,3-dimethylbutyl)octane
(D) 2,2-Dimethyl-5-(1-methylpropyl)nonane

50. Number of -OH groups in Vitamin-C are :
(A) 2 (B) 3 (C) 4 (D) 5

51. Which of the following IUPAC names is correctly written?
(A) trans-1-tert-butylpropene
(B) 6-methylcycloheptene
(C) 3-butene
(D) (Z)-2-hexene

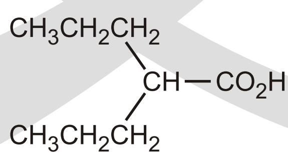
52. Without drawing the structures, correctly match the given compounds with fused bicyclic compound and bridged bicyclic compound:

P = bicyclo[2.1.1]hexane Q = bicyclo[3.1.0]hexane

(A) fused bicyclic compound = P bridged bicyclic compound = P
(B) fused bicyclic compound = P bridged bicyclic compound = Q
(C) fused bicyclic compound = Q bridged bicyclic compound = P
(D) fused bicyclic compound = Q bridged bicyclic compound = Q

53. Which of the following compounds can contain a benzene ring?
(A) $C_{10}H_{16}$ (B) $C_8H_6Cl_2$ (C) C_5H_4 (D) $C_{10}H_{16}O$

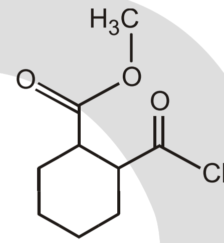
54. The given compound is called valproic acid used in treatment of epilepsy. What is the correct name of valproic acid?



Valproic acid

(A) Heptane-4-carboxylic acid
(B) 2-propylpentanoic acid
(C) 2-propylpentanecarboxylic acid
(D) Heptane-4-oic acid

55. IUPAC name of the given compound is :

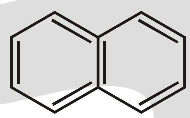


(A) Methyl 2-chloroformyl-1-cyclohexanecarboxylate
(B) 2-Methoxy carbonyl Cyclohexanecarbonyl chloride
(C) Methyl 2-chloroformyl-1-cyclohexanoate
(D) 2-Methoxy carbonyl Cyclohexanonyl chloride

56. The ratio of pi bonds to sigma bonds in benzene is :

(A) 1 : 2 (B) 1 : 3 (C) 1 : 4 (D) 4 : 1

57. Common name of the given compound is :



(A) Naphthalene (B) Anthracene (C) Phenanthracene (D) Phenanthrene

58. IUPAC name of the given compound is :

$$CH_3C(\quad O)N(CH_3)_2$$

(A) N-ethylethanamide (B) N-methylethanamide
(C) N,N-diethylethanamide (D) N,N-dimethylethanamide

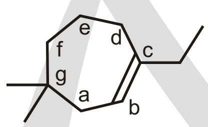
59. IUPAC name of Isoprene is :

(A) 2-methylbuta-1,3-diene (B) 2,3-dimethylbuta-1,3-diene
(C) Methylbuta-1,3-diene (D) 2-methylbut-1-ene

60. IUPAC name of the given compound is :

$$(CH_3)_2C \quad CHC(C_2H_5) \quad CH_2$$

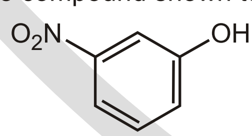
(A) 2-methyl-4-methylidenehex-2-ene (B) 4-ethyl-2-methylpenta-1,3-diene
(C) 2-ethyl-4-methylpenta-1,3-diene (D) 4-methyl-2-methylidenehex-2-ene

61. 

From the carbon indicated a-g, which will get the number 1?

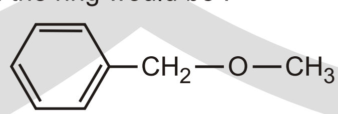
(A) a (B) b (C) c (D) g

62. An appropriate name for the compound shown to the right is :

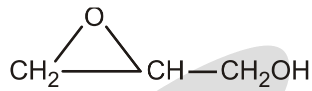


(A) p-nitrophenol (B) m-nitrophenol (C) o-nitrophenol (D) m-nitrophenyl

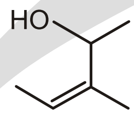
63. A correct name for the compound on the ring would be :



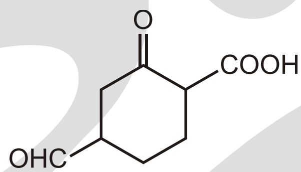
(A) methyl phenyl ether (B) benzyl methyl ether
(C) dimethyl phenyl ether (D) methoxybenzene

64. The IUPAC name of the compound  is :

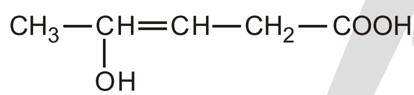
(A) 1, 2 - epoxy - 3 propanol
(B) 1, 2 - oxa - 3 - propanol
(C) 2, 3 - epoxy - 1 - propanol
(D) 2, 3 - epoxy allyl alcohol

65. The IUPAC name of the compound  is :

(A) 3, 4 - dimethyl but - 2 - en - 4 - ol
(B) 3 - methyl pent - 2 - en - 4 - ol
(C) 3 - methyl pent - 3 - en - 2 - ol
(D) 1, 2 - dimethyl but - 2 - en - 1 - ol

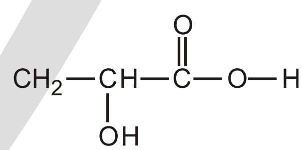
66. The correct IUPAC name of the compound  is

(A) 5 - carboxy - 3 - oxocyclohexane carboxaldehyde
(B) 2 - carboxy - 5 - formylcyclohexane
(C) 4 - formyl - 2 - oxocyclohexane carboxylic acid
(D) 4- carboxy - 3 - oxocyclohexanal

67. The IUPAC name of the compound  is :

(A) hydroxypentenoic acid
(B) 4 - hydroxy - 3 - pentenoic acid
(C) 4 - hydroxy - 4 - pentenoic acid
(D) 4 - hydroxy - 4 - methyl - 3 - ene pentenoic acid

68. What is the IUPAC name of the following compound?



(A) 2-hydroxypropanoic acid
(B) 2-methyl-2-hydroxyethanoic acid
(C) Propanoic acid
(D) 2-carboxyethanol

MATCH THE COLUMN

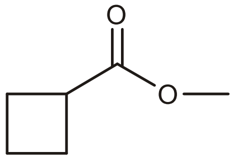
1. Which of the following pairs is correctly matched?

	Column I		Column II
P.	Paraffins	**1.**	Alkanes
Q.	Olefins	**2.**	Alkenes
R.	Di-olefins	**3.**	Alkynes
S.	Di-paraffins	**4.**	Buta-1,3-diene
T.	Poly-olefins	**5.**	Hexa-1,3,5-triene
U.	Poly-paraffins		

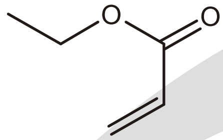
(A) P-1 Q-2 R-4 T-5
(B) P-1 Q-2 R-4 T-5
(C) P-2 Q-1 S-4 U-5
(D) P-2 Q-1 S-4 U-5

UNSOLVED EXAMPLE

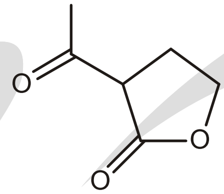
1. How many compounds shown below can be classified as an ester as well as a ketone?



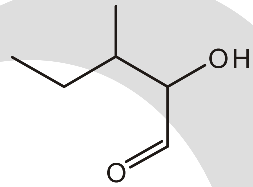
(1)



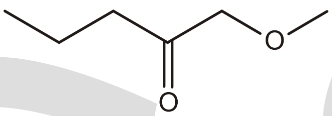
(2)



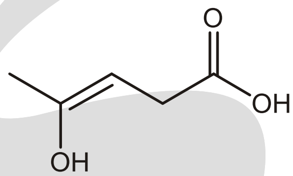
(3)



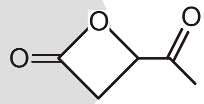
(4)



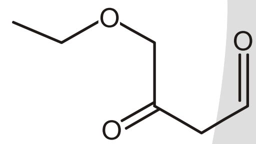
(5)



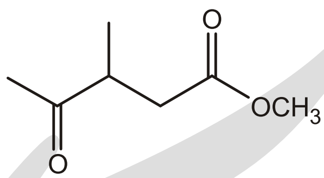
(6)



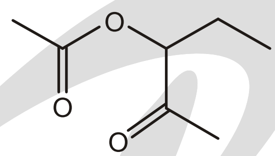
(7)



(8)



(9)

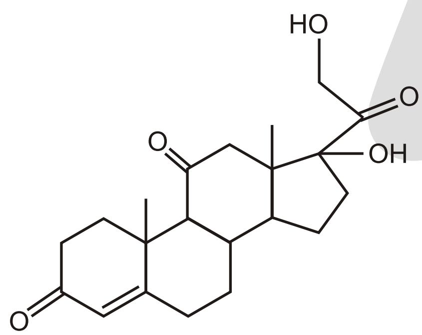


(10)

2. *X* Number of alcohols (structurally different) possible for $C_4H_{10}O$

Y Number of ketones (structurally different) possible for $C_5H_{10}O$

Z Number of different functional groups present in the below compound :



Find the value of $\frac{X \quad Y \quad Z}{2}$?

3. Draw the structures of

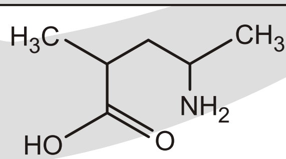
(A) 1, 6-hexanedioic acid

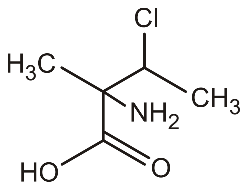
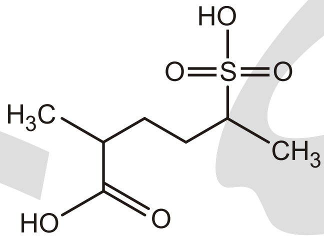
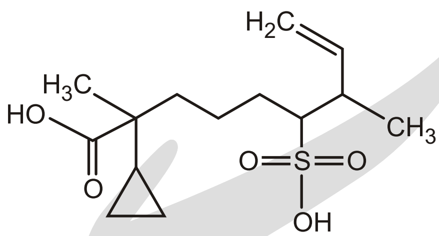
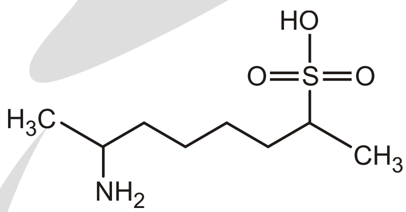
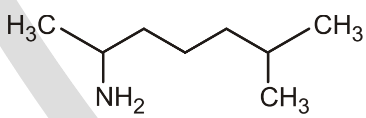
(B) ethyl 2-ethyl 2-hydroxybutanoate

(C) 2-amino-3-cyclohexyl-1-propanol

(D) 2,2 diethyl cyclobutane carboxylic acid

WORK SHEET

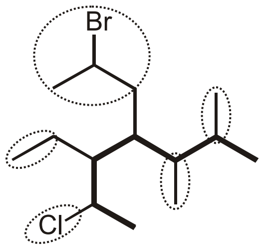
S.No.	Compounds	Write IUPAC - Name
1.		

2.		
3.		
4.		
5.		
6.		

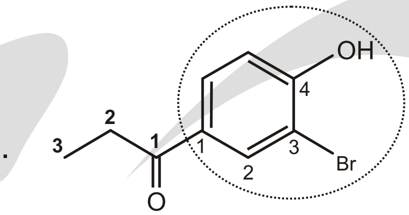
Answers

Single Choice Questions

1. (C)	**2.** (C)	**3.** (C)	**4.** (C)	**5.** (C)	**6.** (B)	**7.** (C)	**8.** (B)
9. (A)	**10.** (B)	**11.** (B)	**12.** (C)	**13.** (B)	**14.** (B)	**15.** (D)	**16.** (A)
17. (B)	**18.** (D)	**19.** (A)	**20.** (C)	**21.** (C)	**22.** (C)	**23.** (B)	**24.** (C)
25. (D)	**26.** (C)	**27.** (D)	**28.** (C)	**29.** (A)	**30.** (B)	**31.** (A)	**32.** (C)
33. (A)	**34.** (C)	**35.** (D)	**36.** (D)	**37.** (D)	**38.** (B)	**39.** (A)	**40.** (D)
41. (B)	**42.** (C)	**43.** (A)	**44.** (B)	**45.** (A)	**46.** (C)	**47.** (C)	**48.** (B)
49. (D)	**50.** (C)	**51.** (D)	**52.** (C)	**53.** (B)	**54.** (B)	**55.** (A)	**56.** (C)
57. (A)	**58.** (D)	**59.** (A)	**60.** (C)	**61.** (C)	**62.** (B)	**63.** (B)	**64.** (C)
65. (C)	**66.** (C)	**67.** (B)	**68.** (A)				

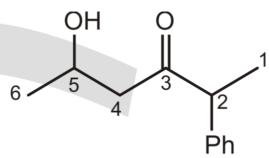
20. 

Total 5 substituents

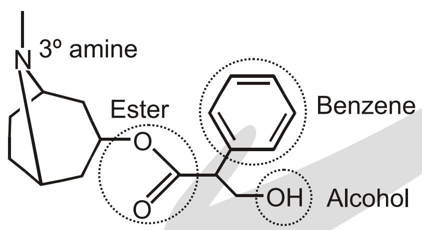
21. 

3-Bromo-4-hydroxy phenyl propan-1-one

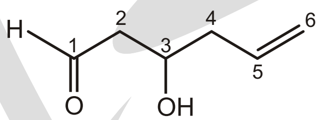
Ketone is main functional group.

22. 

5-hydroxy-2-phenyl hexan-3-one

23. 

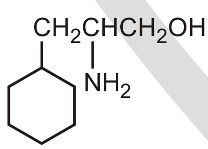
Not having phenol.

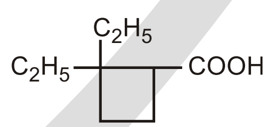
24. 

Principal functional group is –CHO (according to priority table).

69. (A) $HO\overset{O}{\overset{\|}{C}}(CH_2)_4\overset{O}{\overset{\|}{C}}OH$

(B) $CH_3CH_2\underset{CH_2CH_3}{\underset{|}{C}}(OH)-\overset{O}{\overset{\|}{C}}OCH_2CH_3$

(C) 

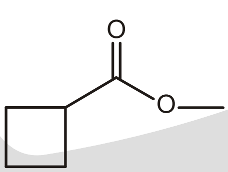
(D) 

Match the Column

1. (A)

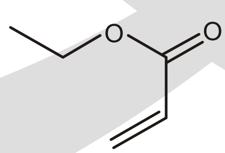
Unsolved Example

1. (4)



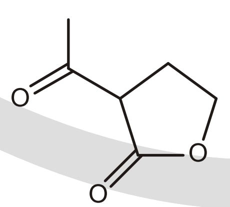
Ester

(1)



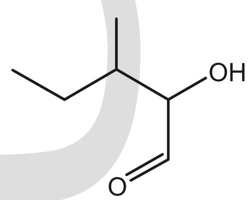
Ester, alkene

(2)



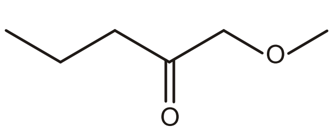
Ester, ketone

(3)



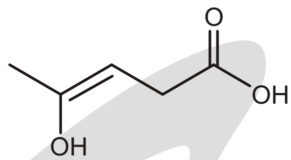
Aldehyde, alcohol

(4)



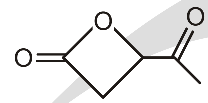
Ketone, ether

(5)



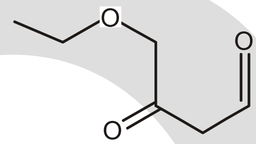
Alkene, alcohol, carboxylic acid

(6)



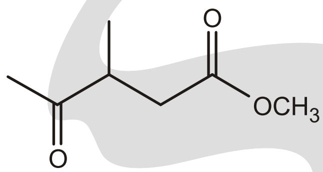
Ester, Ketone

(7)



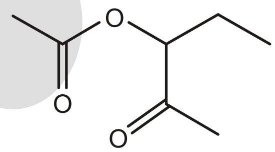
Ester, Ketone, aldehyde

(8)



Ketone ester

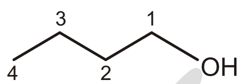
(9)



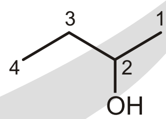
Ketone, ester

(10)

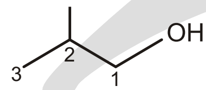
2. (5)



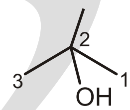
butan-1-ol



butan-2-ol

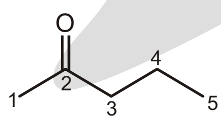


2-methyl propan-1-ol

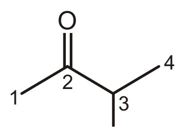


2-methyl propan-2-ol

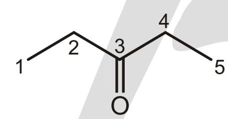
; X 4



pentan-2-one

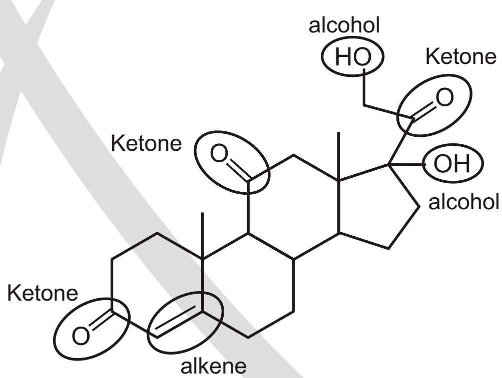


3-methyl butan-2-one alcohol



pentan-3-one

; Y 3



Only Ketone, Alkene, Alcohol are present ; Z 3

thus $\frac{X \quad Y \quad Z}{2}$ 5

Work Sheet

1. 4-amino-2-methylpentanoic acid

2. 2-amino-3-chloro-2-methylbutanoic acid

3. 2-methyl-5-sulfohexanoic acid

4. 2-cyclopropyl-2,7-dimethyl-6-sulfo non-8-enoic acid

5. 7-aminooctane-2-sulfonic acid

6. 6-methylheptan-2-amine

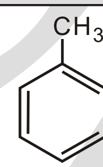
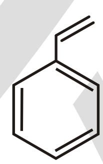
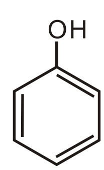
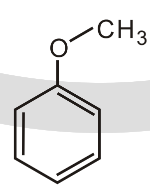
❑❑❑

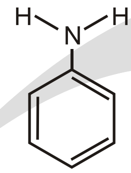
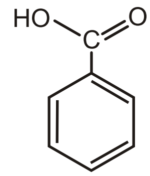
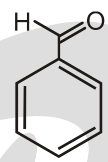
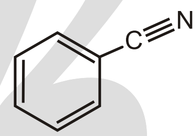
CHAPTER

12

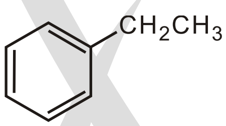
Nomenclature of Aromatic Compounds

COMMON SUBSTITUTED BENZENES

Common name	Substituted benzene	Formula
Toluene	Methylbenzene	 $C_6H_5CH_3$
Styrene	Ethenylbenzene	 $C_6H_5CH{=}CH_2$
Phenol		 C_6H_5OH
Anisole	Methoxybenzene	 $C_6H_5OCH_3$

Aniline	Aminobenzene	 $C_6H_5NH_2$
Benzoic acid		 C_6H_5COOH
Benzaldehyde		 C_6H_5CHO
Benzonitrile	Cyanobenzene	 C_6H_5CN

Solved Example

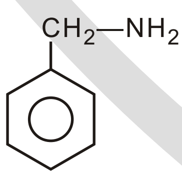


$C_6H_6CH_2CH_3$

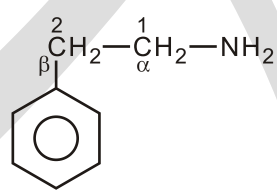
The substituent chain is C2 therefore **= ethylbenzene**

☞ When a benzene ring is attached to an aliphatic chain having a functional group, it is named as phenyl derivative of that aliphatic compound.

☞ **Aralkylamines :**

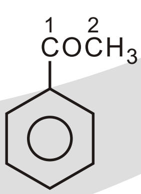


Phenylmethanamine (Benzylamine)

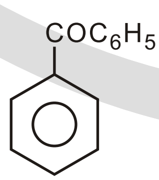


2-phenylethanamine (β-phenylethylamine)

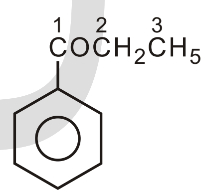
☞ **Ketones :**



1-phenylethan-1-one
(Accetophenone or methyl phenyl ketone)



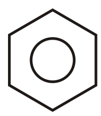
diphenylmethanone
(Benzophenone or Diphenyl ketone)



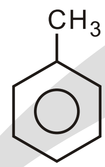
1-phenylpropan-1-one
(Propiophenone)

☞ **Aromatic hydrocarbons (Arenes) :**

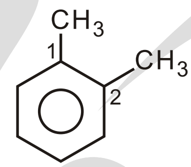
Hydrocarbons which contain both alphatic and aromatic units are called arenes.



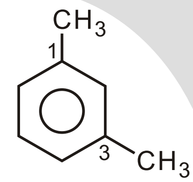
benzene



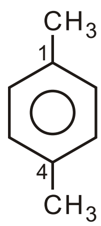
methylbenzene
(toluene)



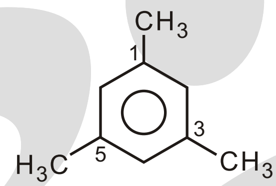
1,2-dimethylbenzene
(o-xylene)



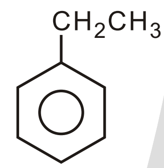
1,3-dimethylbenzene
(m-xylene)



1,4-dimethylbenzene (p-xylene)

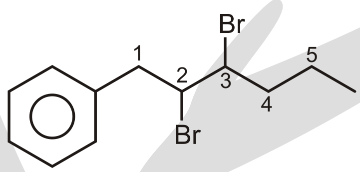


1,3,5-trimethylbenzene (mesitylene)

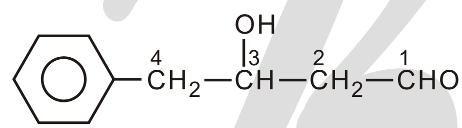


Ethylbenzene

Solved Example

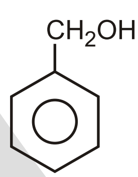


2, 3-Dibromo-1-phenylpentane

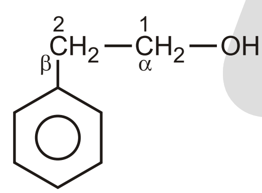


3-hydroxy-4-Phenylbutanal

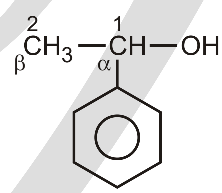
☞ **Aromatic alcohols :**



phenylmethanol
(benzyl alcohol)

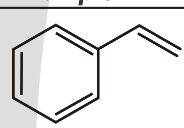


2-phenylethan-1-ol
(β-phenylethyl alcohol)



1-phenylethan-1-ol
(α-phenylethyl alcohol)

Solved Example

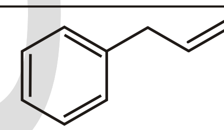


$C_6H_5CH{=}CH_2$ **Phenylethene**

Solved Example

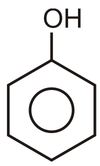
- Principal functional group is an alkene therefore suffix **= -ene**
- The longest continuous chain is C2 therefore root **= eth**
- The benzene ring is a substituent therefore **= phenyl**
- Numbering from the right as drawn to give the alkene the lowest locant **= 1**
- Phenyl locant **= 3**

3-phenylpropene

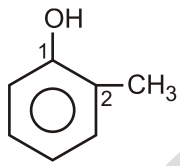


$C_6H_5{-}CH_2{-}CH{=}CH_2$

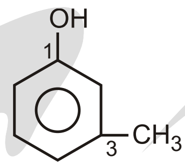
☞ **Phenols :**



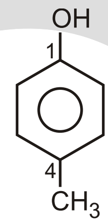
phenol



2-methylphenol (o-cresol)

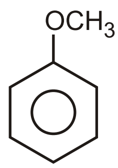


3-methylphenol (m-cresol)

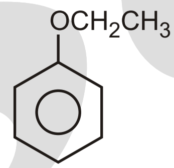


4-methylphenol (p-cresol)

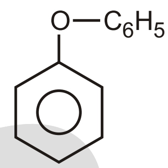
☞ **Aromatic ethers :**



methoxybenzene
(anisole or methyl phenyl ether)

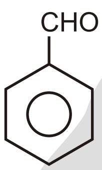


Ethoxybenzene
(phenetole or Ethyl phenyl ether)

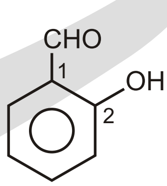


phenoxybenzene
(diphenyl ether)

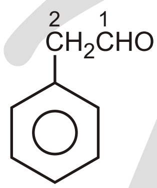
☞ **Aldehydes :**



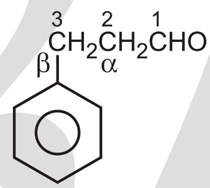
benzaldehyde



2-hydroxybenzaldehyde
(salicyladehyde)

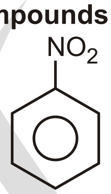


2-phenylethanal
(phenylacetaldehyde)

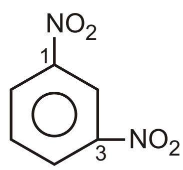


3-phenylpropanal
(β-phenylpropionaldehyde)

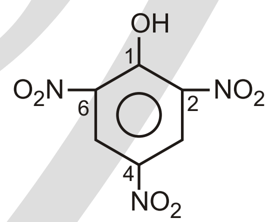
☞ **Nitro compounds :**



Nitrobenzene

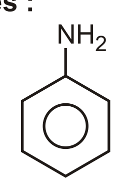


1,3-dinitrobenzene or m-dinitrobenzene

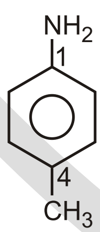


2,4,6-trinitrophenol (Picric acid)

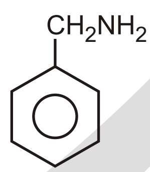
☞ **Amines :**



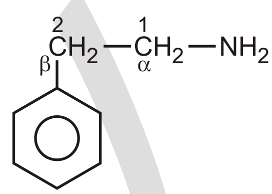
Benzenamine
(Aniline)



4-methylbenzenamine
(p-toluidine)

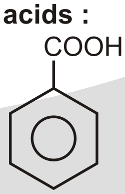


Phenylmethanamine
(Benzylamine)

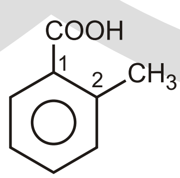


2-phenylethanamine
(β-phenylethylamine)

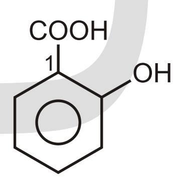
☞ **Carboxylic acids :**



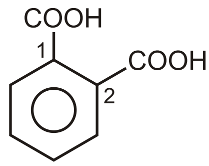
Benzoic acid
or benzenecarboxylic acid



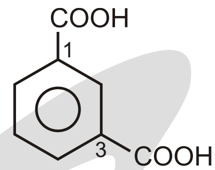
2-methylbenzoic acid
(o-Toluic acid)



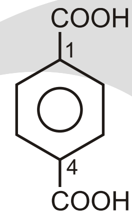
2-hydroxybenzoic acid
(Salicylic acid)



1,2-benzenedicarboxylic acid
(Phthalic acid)

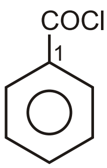


1,3-benzenedicarboxylic acid
(Isophthalic acdi)

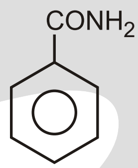


1,4-benzenedicarboxylic acid
(Terephthalic acid)

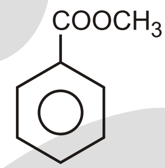
☞ **Acid derivatives :**



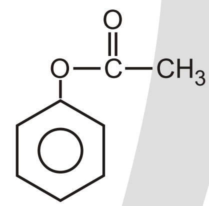
benzoyl chloride
or benzenecarbonylchloride



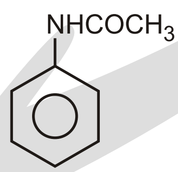
benzamide
or benzenecarboxamide



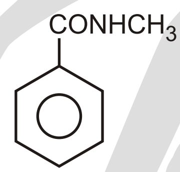
methyl benzoate
or methyl benzenecarboxylate



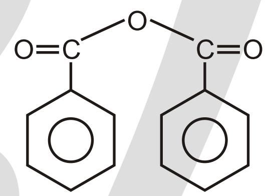
phenyl ethanoate
(Phenyl acetate)



N-phenylethanamide (N-phenylacetamide or Acetanilide)

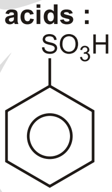


N-methylbenzamide

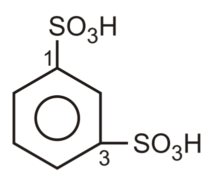


benzoic anhydride

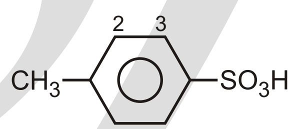
☞ **Sulphonic acids :**



benzenesulphonic acid

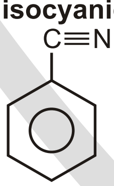


1,3-benzenedisulphonic acid

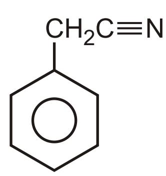


4-toluenesulphonic acid
(p-Toluenesulphonic acid)

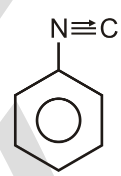
☞ **Cyanides and isocyanides :**



benzenenitrile
(Benzonitrile or Phenyl cyanide)



phenylethanenitrile
(Benzyl cyanide or Phenyl acetonitrile)

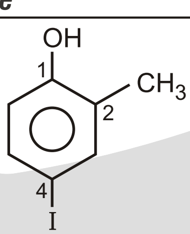


phenylisocyanide or
phenylcarbylamine

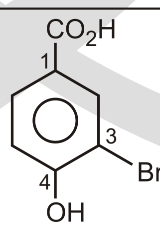
☞ When an aromatic compound contains two or more functional groups, it is named as a derivative of the compound with the principal functional group at position 1.

Solved Example

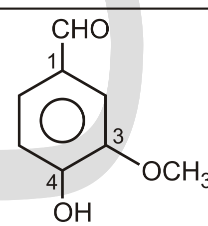
▸



4-Iodo-2-methylphenol
(OH is the principal functional group)



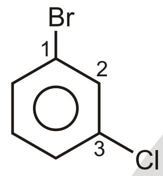
3-Bromo-4-hydroxybenzoic acid
(–COOH is the principal functional group)



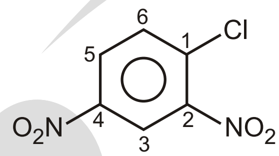
4-Hydroxy-3-methoxybenzaldehyde
(–CHO is the principal functional group)

☞ **Alphabatic Order Rule :** If lowest locant set rule is failed, give preference alphabetically.

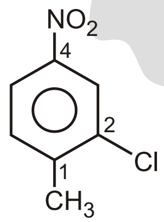
Solved Example



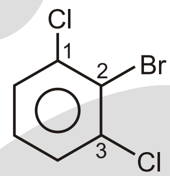
1-Bromo-3-Chlorobenzene
(not 1-chloro-3-bromobenzene)



1-Chloro-2, 4-dinitrobenzene
(and nor 4-chloro-1, 3-dinitrobenzene)



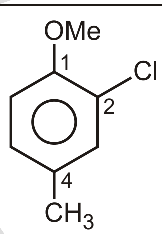
2-Chloro-1-methyl-4-nitrobenzene
(not 3-chloro-4-methylnitrobenzene)



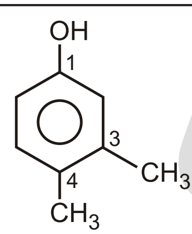
2-Bromo-1, 3-dichlorobenzene
(not 1-bromo-2, 6-dichlorobenzene)

☞ When a substituent is such which when taken together with the benzene ring gives a special name to the molecule, then it is named as a derivative of that molecule with the substituent at position 1.

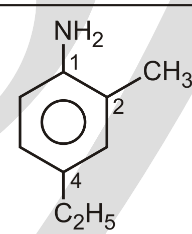
Solved Example



2-Chloro-4-methyl**anisole**

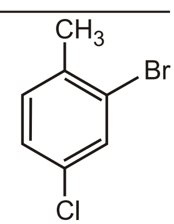


3, 4-Dimethyl**phenol**



4-Ethyl-2-methyl**aniline**

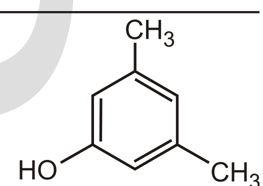
Solved Example



- Principal functional group is the methylbenzene therefore root **= toluene**
- There is a bromine substituent therefore **bromo**
- There is a chlorine substituent therefore **chloro**
- Numbering from the $-CH_3$ (priority group at C1) gives the substituents the locants **= 2** and **4**

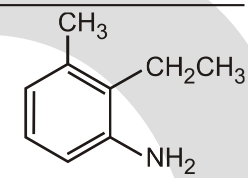
  2-bromo-4-chlorotoluene

Solved Example



- Principal functional group is the aromatic alcohol therefore **phenol**
- There are two C1 substituents therefore dimethyl
- Numbering from the -OH (priority group at C1) gives the substituent the locant **= 3, 5**
- **3,5-dimethylphenol**

Solved Example



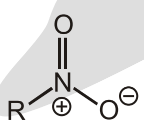
- Principle functional group is the aromatic amine therefore **= aniline**
- There is a C1 substituent therefore methyl
- There is a C2 substituent therefore ethyl
- Numbering from the -NH_2 (priority group at C1) gives the substituents the locants = 2 and 3

  2-ethyl-3-methylaniline

☑ SPECIAL TOPIC

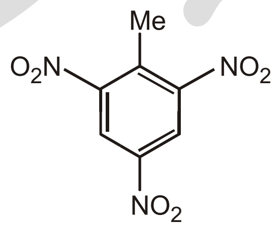
NITRO COMPOUNDS (R–NO_2) CONTAIN THE NITRO GROUP (NO_2)

The nitro group (NO_2) is often incorrectly drawn with five bonds to nitrogen which you will see in Chapter 4, is impossible. Make sure you draw it correctly when you need to draw it out in detail. If you write just NO_2 you are all right!. Several nitro groups in one molecule can make it quite unstable and even explosive. Three nitro groups give the most famous explosive of all.

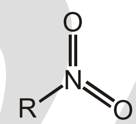


the nitro group

nitrogen cannot have five bonds!



TNT (trinitrotoluene)

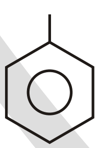


incorrect structure for the nitro group

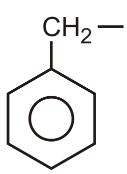
However, functional groups refuse to be stereotyped. Nitrazepam also contains a nitro group, but this compound is marketed as Mogadon®, the sleeping pill.

☞ **Aryl groups :**

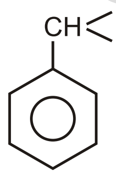
C_6H_5—



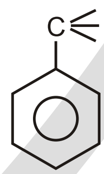
phenyl



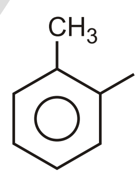
benzyl



benzal

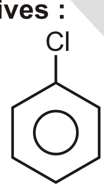


benzo

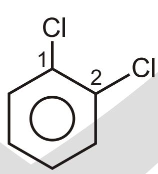


o–Tolyl

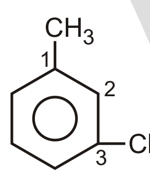
☞ **Halogen dervatives :**



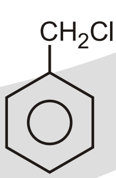
chlorobenzene



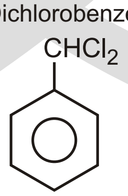
1,2-dichlorobenzene or
o-Dichlorobenzene



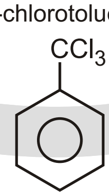
3-chlorotoluene or
m-chlorotoluene



phenylchloromethane
(benzyl chloride)

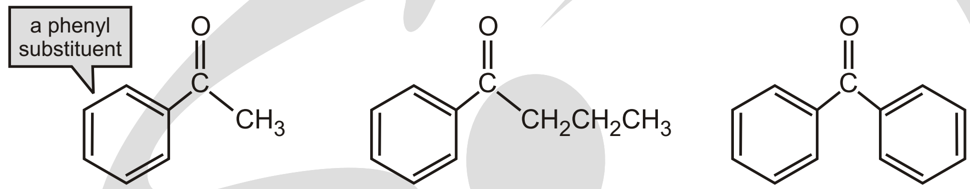


phenyldichloromethane
(benzal dichloride)



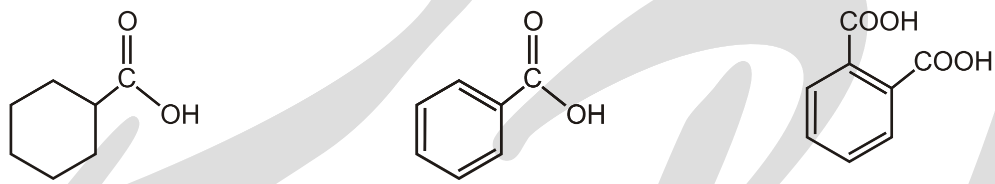
phenyltrichloromethane
(benzotrichloride)

Only a few ketones have common names. The smallest ketone, propanone, is usually referred to by its common name, acetone. Acetone is a widely used laboratory solvent. Common names are also used for some phenyl-substituted ketones; the number of carbons (other than those of the phenyl group) is indicated by the common name of the corresponding carboxylic acid, substituting "-ophenone" for "-ic acid."



common name:	acetophenone	butyrophenone	benzophenone
derived name:	methyl phenyl ketone	phenyl propyl ketone	diphenyl ketone

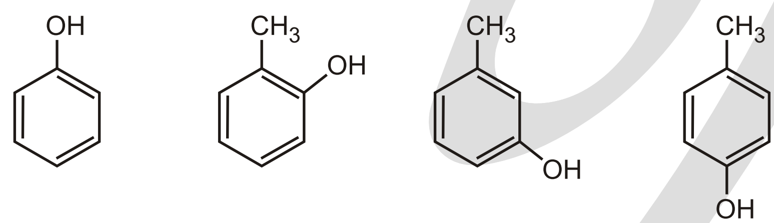
Carboxylic acids in which a carboxyl group is attached to a ring are named by adding "carboxylic acid" to the name of the cyclic compound.



cyclohexanecarboxylic acid — benzenecarboxylic acid, benzoic acid — 1,2-benzenedicarboxylic acid

☞ **Phenols :** The simplest hydroxy derivative of benzene is phenol. It is its common name and also an accepted IUPAC name.

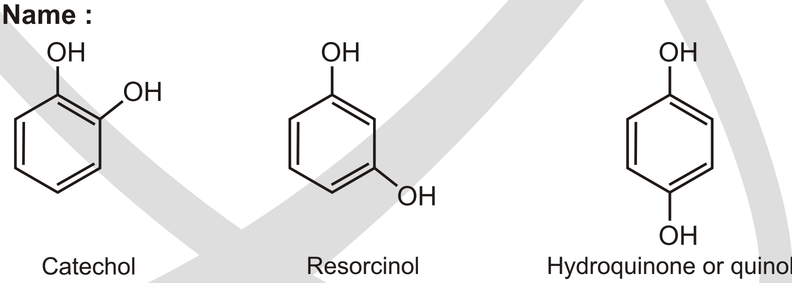
Common name IUPAC Name :



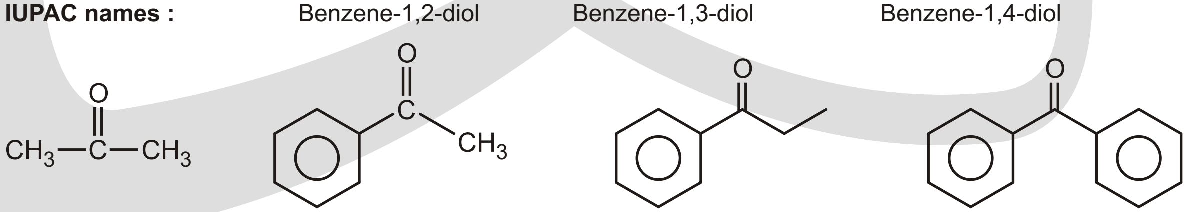
Common names :	Phenol	o-Cresol	m-Cresol	p-Cresol
IUPAC names :	Phenol	2-Methylphenol	3-Methylphenol	4-Methylphenol

Dihydroxy derivatives of benzene are known as 1,2-, 1, 3-and 1, 4-benzenediol.

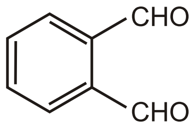
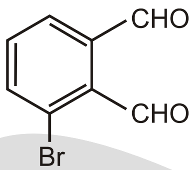
Common name IUPAC Name :



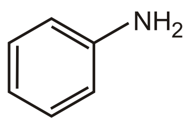
Common names :	Catechol	Resorcinol	Hydroquinone or quinol
IUPAC names :	Benzene-1,2-diol	Benzene-1,3-diol	Benzene-1,4-diol

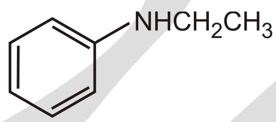


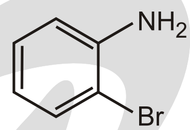
acetone — acetophenone — propiophenone — benzophenone

Structure	Common name	IUPAC name
	Phthaldehyde	Benzene-1,2-dicarbaldehyde
	*m*-Bromophthaldehyde	3-Bromobenzene-1,2-dicarbaldehyde

☞ **Aromatic Amines :**
Aromatic amines are named as derivatives of aniline.


aniline

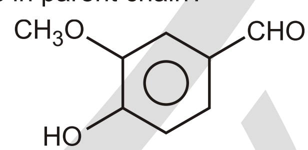

N-ethylaniline


o-Bromoaniline

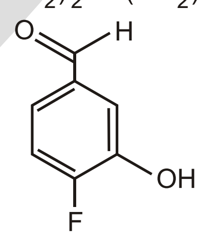
EXERCISE

SINGLE CHOICE QUESTIONS

1. Which of the following represent incorrectly named IUPAC compound?
(A) $CH_3COC_6H_5$ 1-Phenylethanone
(B) $CH_3CH_2COCH(CH_3)_2$ 3-Hexanone
(C) $(CH_3)_2CHCOCH(CH_3)_2$ 2, 4-Dimethyl 3-Pentanone
(D) $C_6H_5COC_6H_5$ Diphenyl methanone

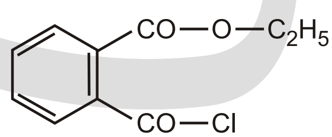
2. Which of the following represent incorrect number of carbons in parent chain?
(A) $CH_3(CH_2)_3CO(CH_2)_3CH_3$ 9
(B)  1
(C) $C_6H_5CHCH_2CHO$ 3
(D) $CH_3(CH_2)_2CO(CH_2)_2CH_3$ 3

3. IUPAC name of the given compound is :



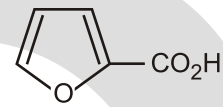
(A) 2-Fluoro-5-formylbenzenol
(B) 4-Fluoro-3-hydroxybenzenecarbaldehyde
(C) 1-Fluoro-4-formyl-2-hydroxybenzene
(D) 4-Fluoro-5-hydroxybenzenecarbaldehyde

4. Write the IUPAC name of the following compound :



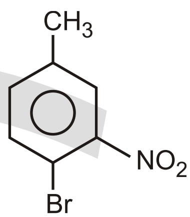
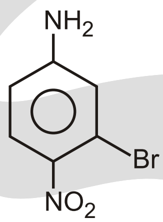
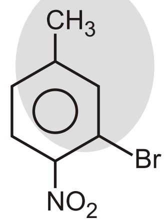
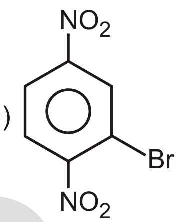
(A) ethyl-2-(chlorocarbonyl) benzoate
(B) ethyl-2-(chlorocarbonyl) hexanoate
(C) 2-(ethoxycarbonyl) benzoyl chloride
(D) None of these

5. The correct IUPAC name for the molecule is :

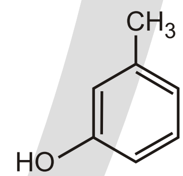


(A) 1-furoic acid
(B) furanyl carboxylic acid
(C) 2-furoic acid
(D) 3-furoic acid

6. Which of the following is 3-bromo-4-nitro toluene?

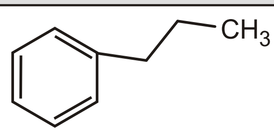
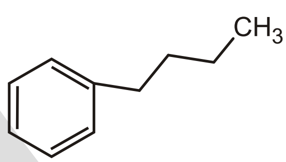
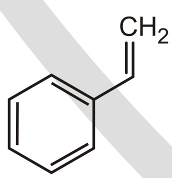
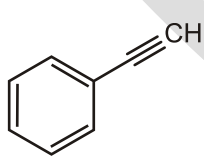
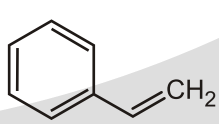
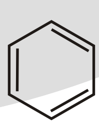
(A) 
(B) 
(C) 
(D) 

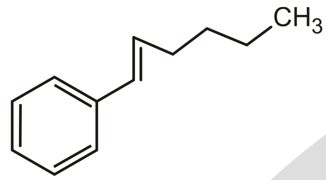
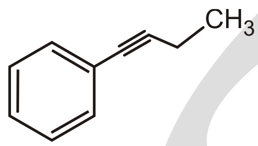
7. What would be the best name for the following compound?



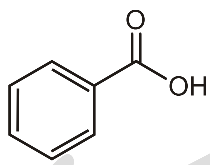
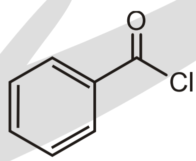
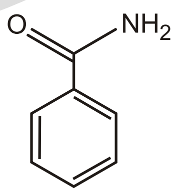
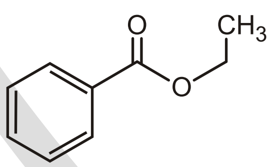
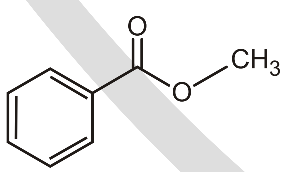
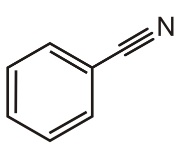
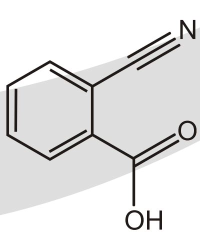
(A) 3-methylhydroxybenzene
(B) 3-methylcyclohexa-13,5-trien-1-ol
(C) 3-methylphenol
(D) 2-hydroxytoluene

WORK SHEET - 1

S.No.	Compounds	Write IUPAC - Name
1.		
2.		
3.		
4.		
5.		
6.		

7.		
8.		

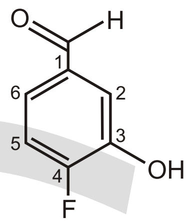
WORK SHEET - 2

S.No.	Compounds	Write IUPAC - Name
1.		
2.		
3.		
4.		
5.		
6.		
7.		

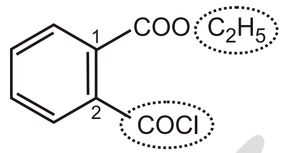
Answers

Single Choice Questions

1. (B) **2.** (D) **3.** (B) **4.** (A) **5.** (B) **6.** (C) **7.** (C) **8.** (B)

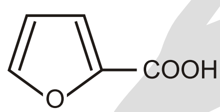
3. 

Additional suffix carbaldehyde is used for –CHO group.

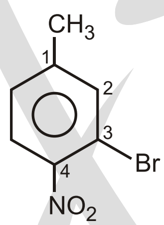
4. 

ethyl-2-(chlorocarbonyl) benzoate

chlorocarbonyl

5. 

= furanyl carboxylic acid.

6. 

3-bromo-4-nitro toluene

Work Sheet - 1

1. propylbenzene
2. butyl benzene
3. 1-phenyl ethene
4. phenyl ethyne
5. 1-phenyl ethene
6. benzene
7. 1-phenyl hex-1-ene
8. 1-phenyl but-1-yne

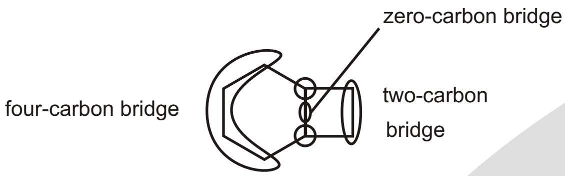
Work sheet - 2

1. benzenecarboxylic acid or benzoic acid
2. benzenecarbonyl chloride or benzoyl chloride
3. benzenecarboamide or benzamide
4. ethyl benzene carboxylate or ethyl benzoate
5. methyl benzene carboxylate or methyl benzoate
6. benzenecarbonitrile
7. 2-cyano benzenecarboxylic acid or 2-cyanobenzoic acid

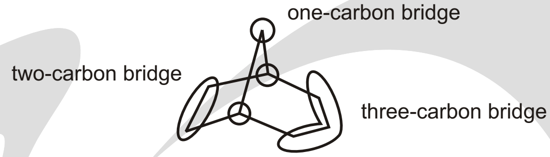
☑ SPECIAL TOPIC

BICYCLO COMPOUNDS

All fused and bridged bicyclic systems have three bridges connecting the two bridgehead atoms where the rings connect. The numbers in the brackets give the number of carbon atoms in each of the three bridges connecting the bridgehead carbons, in order of decreasing size.



bicyclo[4,2,0]octane

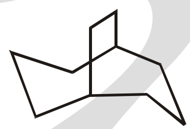


bicyclo[3,2,1]octane

EXERCISE

SINGLE CHOICE QUESTIONS

1. Which of the following best describes the compound given below?



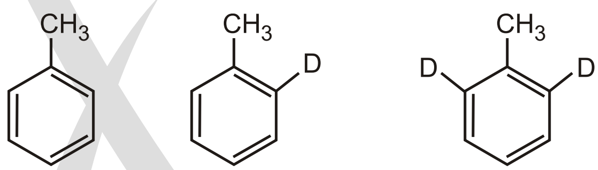
(A) bridged bicyclic
(B) fused bicyclic
(C) spiro bicyclic
(D) bridged tricyclic

SUBJECTIVE TYPE QUESTIONS

1. Which isomer of xylene can give three different monochloroderivatives ?

(a) *o*-xylene
(b) *m*-xylene
(c) *p*-xylene
(d) xylene cannot give a monochloro derivative

2.



(I) (II) (III)

The rate of o-nitration of the above compounds, (I) toluene, (II) 2-D-toluene and (III) 2, 6-D_2-toluene is in the following order

(a) I > II > III
(b) II > I > III
(c) III > I > II
(d) The rate is the same for all the three compounds

Answers

Single Choice Questions

1. (A)

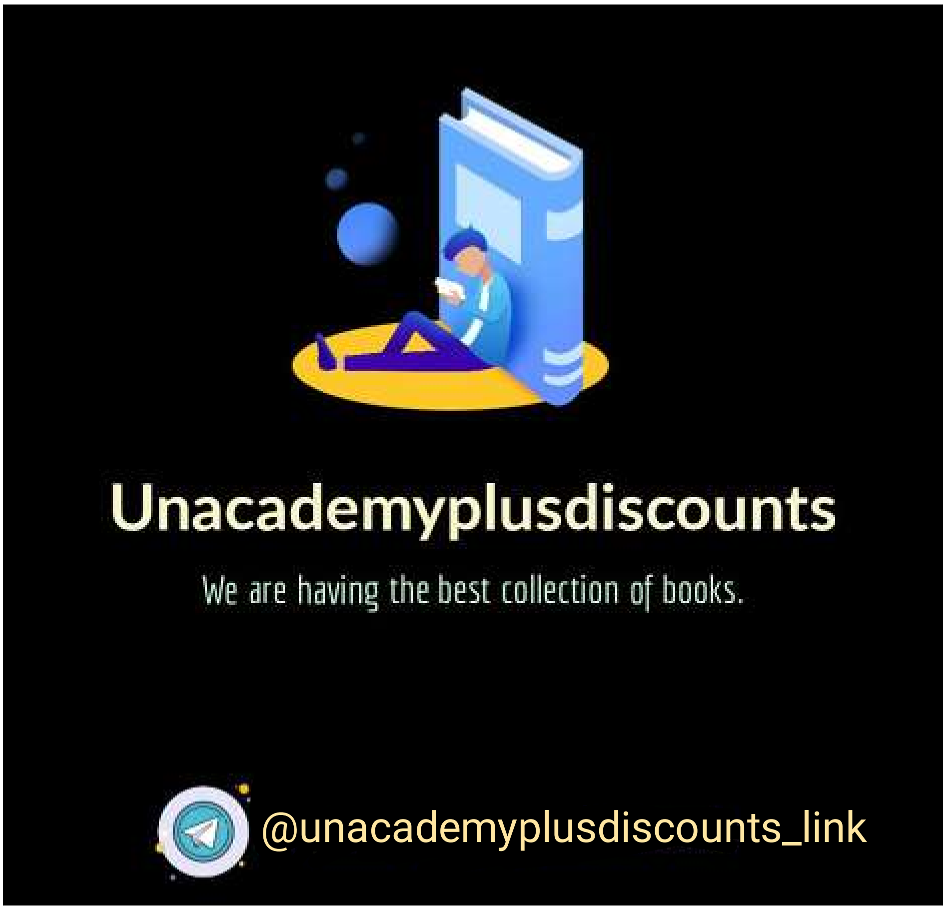
Subjective Type Questions

1. (b)

2. (d) I = II = III

In the rds step C — D bond cleavage is not involved.

❑❑❑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CHAPTER

13

Inductive Effect

ELECTRON DISPLACEMENT EFFECTS IN COVALENT BONDS

- Electron shifting in a covalent bond is known as electron displacement effects.
- Electron displacement effects are of mainly 3-types :
  - (i) Inductive effect
  - (ii) Resonance
  - (iii) Hyperconjugation

INDUCTIVE EFFECT

Inductive effect is the polarization of shared pair (covalently bonded pair) of electrons towards more electronegative atoms. Let us consider a molecule.

$$\overset{\delta\delta\delta+}{C} \rightarrow \overset{\delta\delta+}{C} \rightarrow \overset{\delta+}{C} \rightarrow \overset{\delta-}{C}$$

- A permanent effect
- The electrons never leave their original atomic orbital.
- Operates through s bonds
- Polarisation of electrons is always in single direction.
- Its magnitude (*i.e.,* electron withdrawing or donating power) decreases with increase in distance.
- There occurs partial movement of shared pair of electrons.
- This is a very weak effect.
- This effect is additive in nature, *i.e.* two electronegative atoms exert greater induction effect than one atom of greater electronegativity from the same position.

Inductive effect showing groups can be divided into 2 types.

Inductive effect

- +I (electron donating group)
- −I (electron withdrawing group)

INDUCTIVE EFFECTS

Inductive Effect (I Effect of Groups)

Electron donating groups show positive inductive effect (+I). For example,

$-T > -D > -H$

$-\bar{N}-R > -\bar{O}$

$-\bar{Se} > -\bar{S}$

$-C(CH_3)_3 > -CH(CH_3)_2 > -CH_2-CH_3 > -CH_3$

$-\bar{O} > -\overset{\overset{O}{\|}}{C}-\bar{O}$

Electron withdrawing groups show negative inductive effect (–I). For example,

$-\overset{+}{O}R_2 > -\overset{+}{N}R_3$

$-\overset{+}{N}CH_3(CH_3) > -\overset{+}{P}(CH_3)-CH_3 > -\overset{+}{As}(CH_3)-CH_3 > -\overset{+}{Sb}(CH_3)-CH_3$

$-\overset{+}{O}{<}^{R}_{R} > -\overset{+}{S}R_2 > -\overset{+}{Se}R_2 > -\overset{+}{Te}R_2$

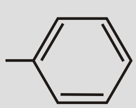
$-\overset{+}{O}R_2 > -OR$

$-\overset{+}{S}R_2 > -SR$

$-C\equiv N > -CH{=}NH_2 > -CH_2-NH_2$

$-CH{=}O > -CH{=}NH > -CH{=}CH_2$

$-C\equiv CN > -CH{=}CH_2$

 $> -CH{=}CH_2$

$-F > -OR > -NR_2 > -CR_3$

$-\overset{\overset{O}{\|}}{\underset{\underset{O}{\|}}{S}}-R > -\overset{\overset{O}{\|}}{S}-R > -S-R$

$-SiR_3 > -GeR_3$

$-C\equiv\overset{+}{N}H > -CH{=}\overset{+}{N}H_2 > -CH_2-\overset{+}{N}H_3$

$-\overset{+}{N}H_3 > -NH_2$

$-OCH_3 > -S-CH_3 > -Se-CH_3$

$-F > -Cl-Br > -I$

$-CH{=}O > -CH_2-O-CH_3$

APPLICATION OF INDUCTIVE EFFECT

Carbocations, carbanions and carbon radicals :

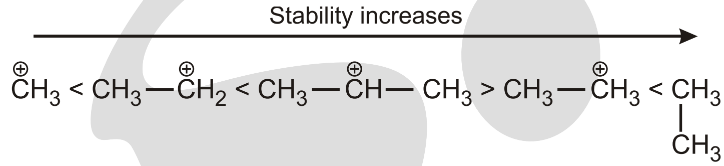
Carbocations, which include carbenium and carbonium ions, contain a positive charge on carbon. Carbenium ions have three bonds to the positively charged carbon (*e.g.* Me_3C).

☞ Carbenium ions (R_3C) are generally planar and contain an empty *p* orbital. They are stabilised by electron-donating groups (R is I and/or +M), which delocalise the positive charge; +M groups are generally more effective than I groups.

(a) Stability of Alkyl Carbocation :

Stability of alkyl carbocation I effect

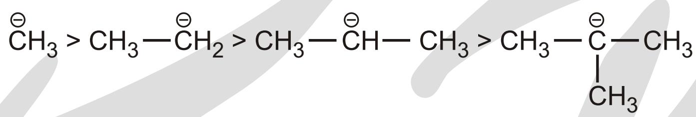
If the I effect will increase then magnitude of positive charge on carbon will decrease so it will stabilised carbocation.



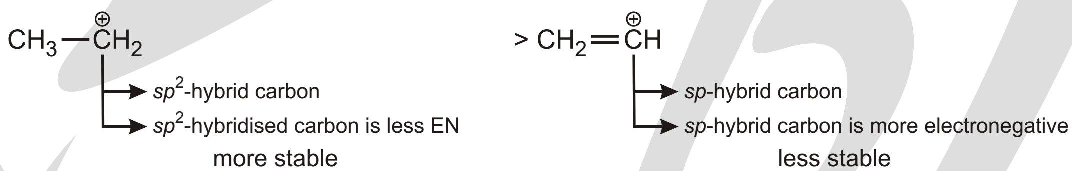
(b) Stability of Alkyl Carbanion :

Stability of alkyl carbanion I effect

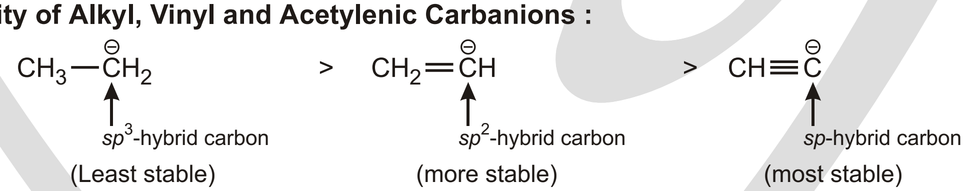
For examples :



(c) Stability of Alkyl and Vinyl Carbocation :



(d) Stability of Alkyl, Vinyl and Acetylenic Carbanions :



ORDER OF STABILITY

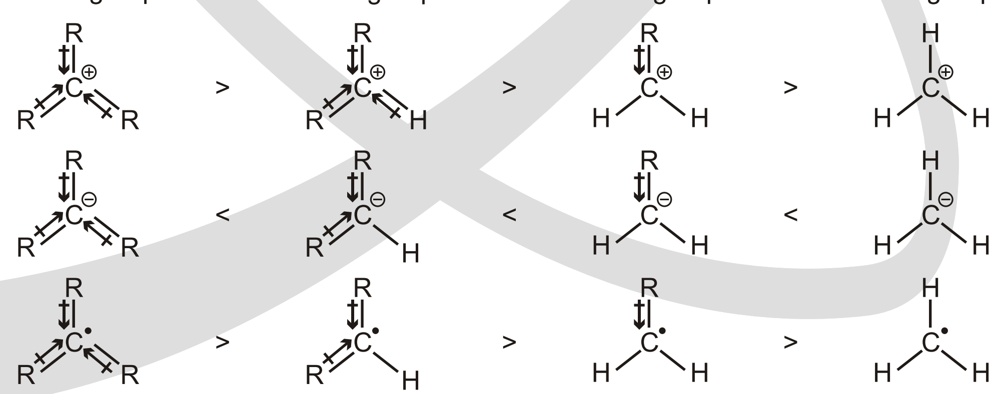
The order of stability of carbocations, carbanions and carbon radicals bearing electron donating (I) alkyl groups, R, is as follows. (Free radicals will be formed by homolytic fission)

	Tertiary three +I groups	**Secondary** two +I groups	**Primary** one +I groups	**Methyl** no +I groups

Cation Stability

Anion Stability

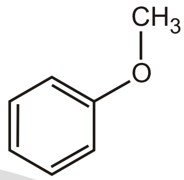
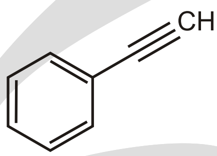
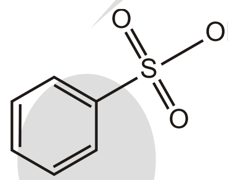
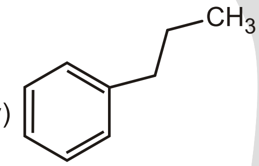
Radical Stability

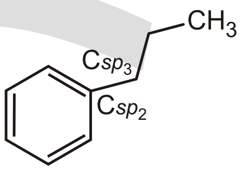


Carbanions can be stabilised by electron-withdrawing groups (I, –M groups), whereas carbocations can be stabilised by electron-donating groups (I, +M groups).

Solved Example

- Which of the side chain attached to benzene ring have I effect

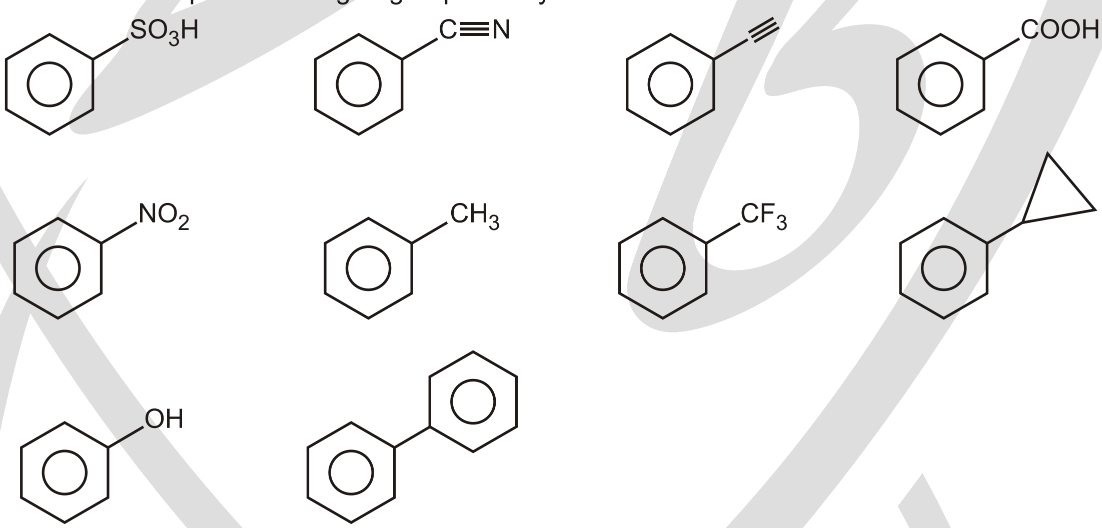
(i)  (ii)  (iii)  (iv) 

Sol.  I effect Due to E.N. difference.

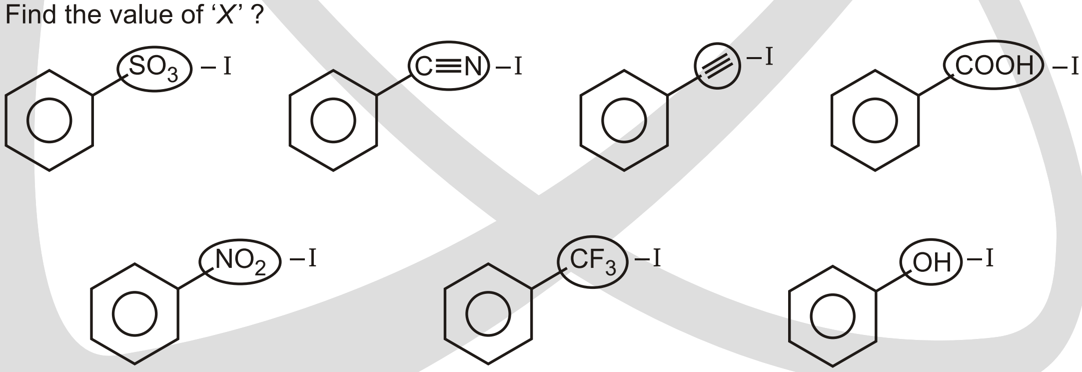
E.N. order : c_{sp} c_{sp_2} c_{sp_3} due to % s-character.

Solved Example

- '*X*' = Number of compound having I group directly attached to benzene.



Find the value of '*X*' ?

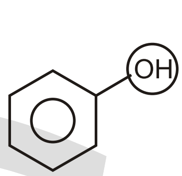
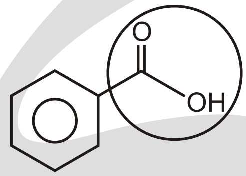
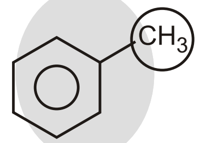
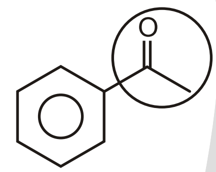
Sol. 

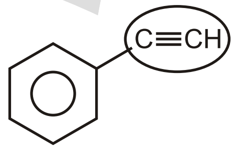
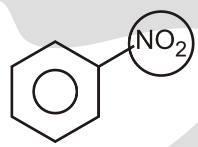
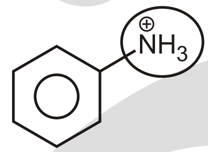
So *x* 7

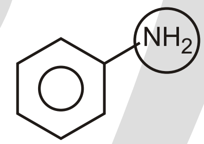
EXERCISE

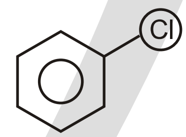
WORK SHEET

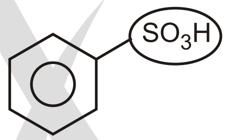
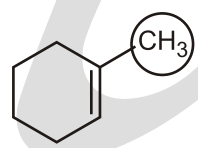
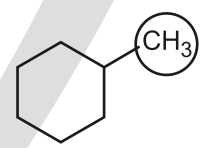
1. Identify I or I effect which is present by circled group ?

(a)  (b)  (c)  (d) 

(e)  (f)  (g)  (h) $CH_3 - CH{=}CH_2$

(i) $CH_3 - C{\equiv}CH$ (j) $H_2C{=}CH - C{\equiv}CH$ (k) $CD_3 - CH{=}CH_2$ (l) 

(m) $CH_3 - \overset{\oplus}{C}H_2$ (n) $NO_2 - CH_2 - \overset{\ominus}{C}H_2$ (o) $Cl - CH_2 - \overset{\ominus}{C}H_2$ (p) 

(q)  (r) $CH_3 - CH_3$ (s)  (t) 

Answers

Work Sheet

1. (a) I	(b I	(c) I	(d) I
(e) –I	(f) –I	(g) I	(h) I
(i) I	(j) I	(k) I	(l) I
(m) I	(n) I	(o) I	(p) I
(q) I	(r) None	(s) I	(t) None

(Note : Inductive effect does not operate in alkanes)

❑❑❑

CHAPTER

14

Resonance

RESONANCE

It is delocalisation of   electrons in a conjugated system.

Important Terminology used in Resonance :

(a) Curved arrows

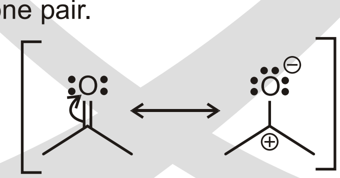
(b) Conjugation system

(c) Localized & delocalized electron

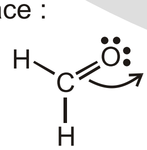
CURVED ARROWS : THE TOOLS FOR DRAWING RESONANCE STRUCTURES

Every curved arrow has a head and a tail. It is essential that the head and tail of every arrow be drawn in precisely the proper place. The tail shows where the electrons are coming from, and the head shows where the electrons are going (remember that the electrons aren't really going anywhere, but we treat them as if they were so we can make sure to draw all resonance structures) :

Or it must point to an atom to form a lone pair.



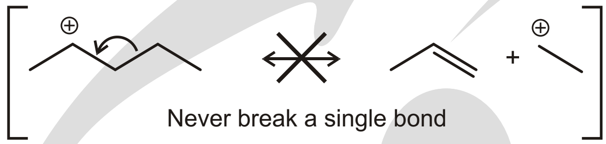
Never draw the head of an arrow going off into space :



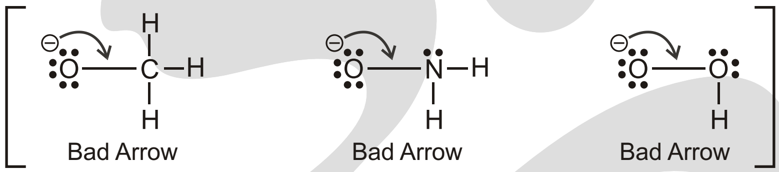
Bad Arrow

Let's focus on one at a time :

C-1. **Never break a single bond when drawing resonance structures.** By definition, resonance structures must have all the same atoms connected in the same order. Otherwise, they would be different compounds.



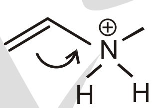
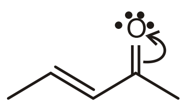
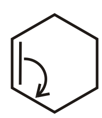
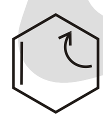
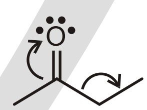
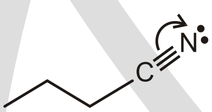
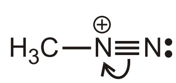
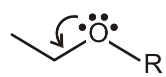
C-2. **Never violate the octet rule.** Let's review the octet rule. Atoms in the second period (C, N, O, F) have only four orbitals in their valence shell.



In each of these drawings, the central atom cannot form another bond because it does not have a fifth orbital that can be used. This is impossible. Don't ever do this.

Solved Examples

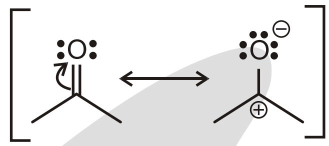
- For each of the problems below, determine which arrows violate either one of the two commandments C-1 or C-2, and explain why. (Don't forget to count all hydrogen atoms and all lone pairs. You must do this to solve these problems.)

1. 
2. 
3. 
4. 
5. 
6. 
7. 
8. 
9. 
10. 
11. 

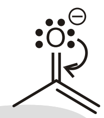
Ans.
1. C-2 ('N' can't accomodate 10*e* *s*)
2. No violation
3. C-2 ('C' can't accomodate 10*e* *s*)
4. C-2('O' can't accomodate 10*e* *s*)
5. C-2 ('C' can't accomodate 10*e* *s*)
6. C-2 ('C' can't accomodate 10*e* *s*)
7. C-1 (Breaking of single bond)
8. C-1(Breaking of single bond)
9. No violation
10. No violation
11. C-2 ('C' can't accomodate 10*e* *s*)

DRAWING GOOD ARROWS

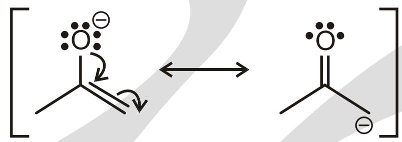
Now, we need to know where to put the head of the arrow. We look for any lone pairs or double bonds that are appearing. We see that there is a new lone pair appearing on the oxygen. So now we know where to put the head of the arrow :



For below example, we need two arrows. Let's start at the top. Loose a lone pair from the oxygen and form a C O. Let's draw that arrow :

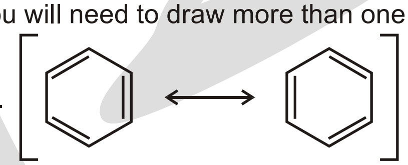


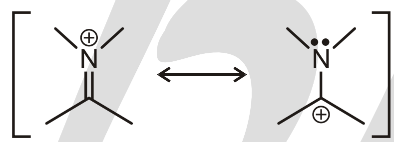
Notice that if we stopped here, we would be violating the second commandment C-2. The central carbon atom is getting five bonds. To avoid this problem, we must immediately draw the second arrow. The C C disappears (which solves our octet problem) and becomes a lone pair on the carbon. Now we can draw both arrows :

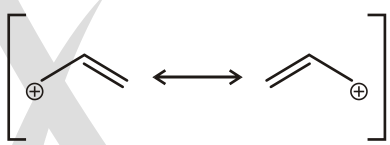


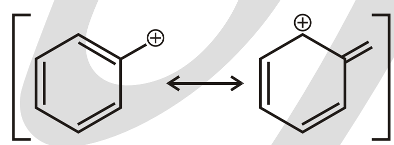
Solved Examples

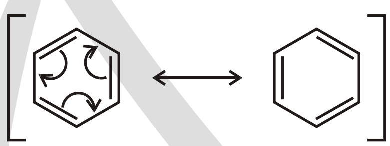
- For each drawing, try to draw' the curved arrow's that get you from one drawing to the next. In many cases you will need to draw more than one arrow.

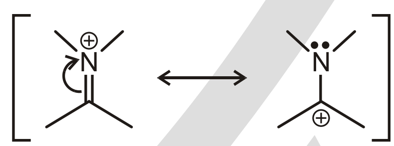
1. 

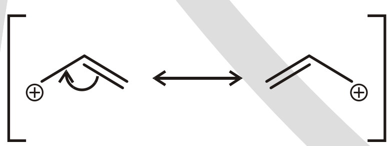
2. 

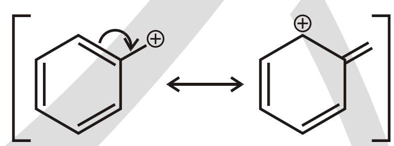
3. 

4. 

Ans. 1. 

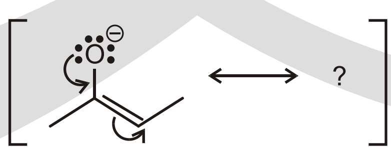
2. 

3. 

4. 

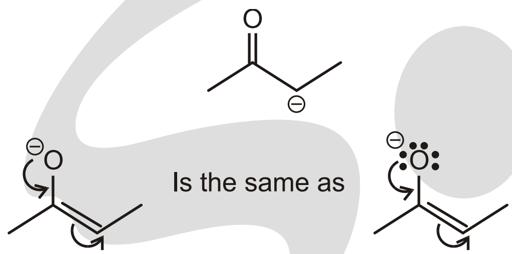
Solved Example

- Draw the resonance structure that you get when you push the arrows shown below. Be sure to include formal charges.



Ans. We read the arrows to see what is happening. One of the lone pairs on the oxygen is coming down to form a bond, and the C C double bond is being pushed to form a lone pair on a carbon atom. This is very similar to the example we just saw.

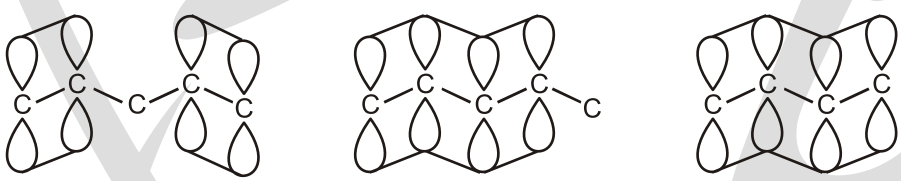
We just get rid of one lone pair on the oxygen, place a double bond between the carbon and oxygen, get rid of the carbon-carbon double bond, and place a lone pair on the carbon. Finally, we must put in any formal charges :



CONJUGATED SYSTEMS

In the dictionary, 'conjugated' is defined, as joined together, especially in pairs' and 'acting or operating as if joined'. This does indeed fit very well with the behaviour of such conjugated double bonds since the properties of a conjugated system are often different from those of the components parts. We are using *conjugation* to describe bonds and *delocalization* to describe electrons.

We shall use conjugation and delocalization : conjugation focuses on the sequence of alternating double and single bonds while delocalization focuses on the molecular orbitals covering the whole system. Electrons are delocalized over the whole of a conjugated system.

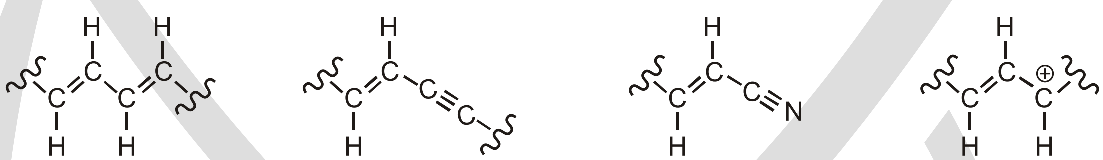


--------- delocalised overlap of *p* oribtals

for clarity, all hydrogen atoms have been omitted

1,4-pentadiene 1,3-pentadiene 1,3-butadiene

Figure : Overlap of *p* orbitals in dienes

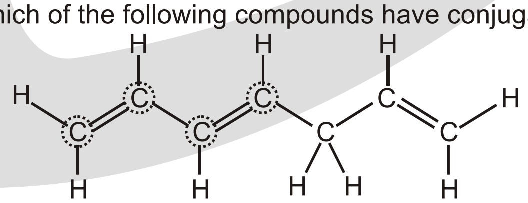


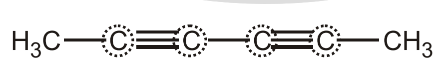
conjugated diene | conjugated ene-yne | conjugated nitrile | conjugated carbocation

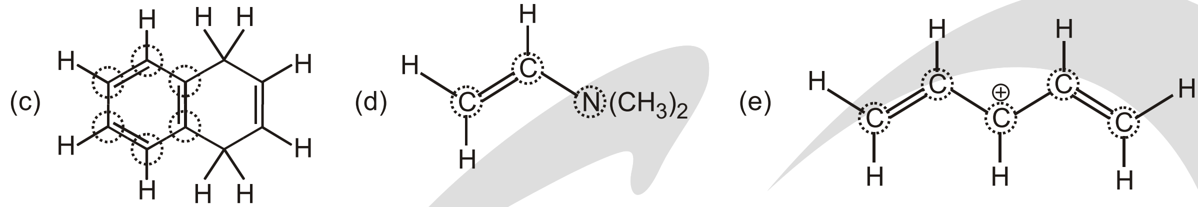
conjugated carbonyl | conjugated imine | conjugated radical | conjugated carbanion

Solved Examples

- Which of the following compounds have conjugated systems :

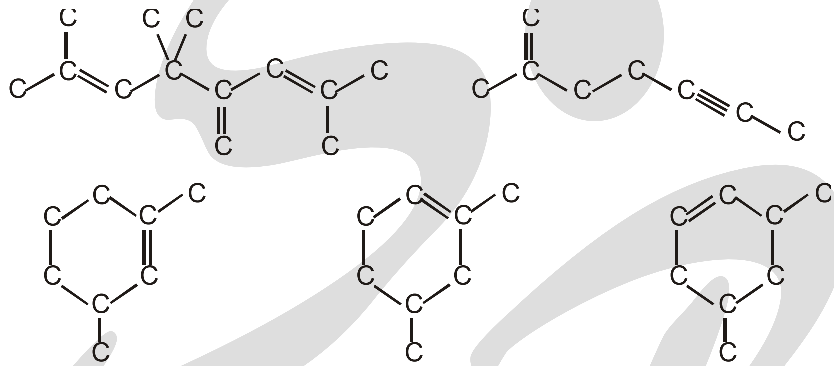
(a) 

(b) 



Ans. All have conjugated system.

- Are these molecules conjugated? Explain your answer in any reasonable way.



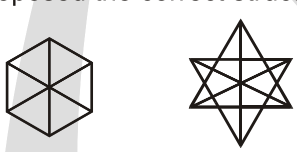
Ans. Only 1 of the above is conjugated.

LOCALIZED & DELOCALIZED ELECTRON

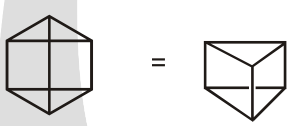
Electrons that are restricted to a particular region are called localized electrons. Localized electrons either belong to a single atom or are confined to a bond between two atoms.

Many organic compounds contain delocalized electrons. Delocalized electrons neither belong to a single atom nor are confined to a bond between two atoms-they are electrons that are shared by more than two atoms.

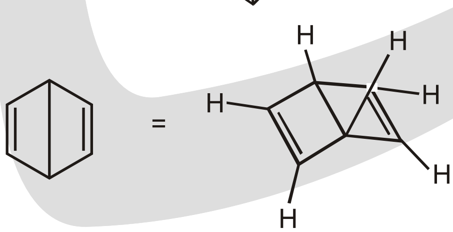
The rest of this chapter concerns molecules with more than one C — C double bond and what happens to the orbitals when they interact. To start, we shall take a bit of a jump and look at the structure of benzene. Benzene has been the subject of considerable controversy since its discovery in 1825. It was soon worked out that the formula was C_6H_6, but how were these atoms arranged? Some strange structures were suggested until Kekulé proposed the correct structure in 1865.



these diagrams represent old structures for benzene. They do not represent compounds that could ever be made

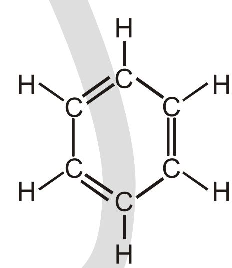


prismane synthesized 1973



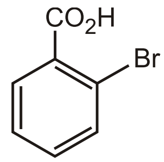
Dewar benzene synthesized 1963

these early suggestions for the structure of benzene have now been made. they are certainly not benzene, but entirely different compounds

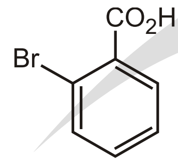


Kekule's structure for benzene

With benzene itself, these two forms are equivalent but,



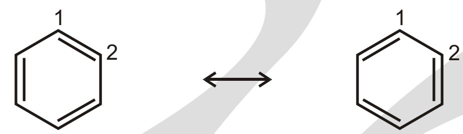
2-bromobenzoic acid



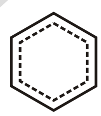
'6'-bromobenzoic acid

If the double bonds were localized then these two compound would be chemically different.
(the double bonds are drawn shorter than the single bonds to emphasize the difference)

If we had a 1,2- or a 1,3-disubstituted benzene compound, these two forms would be different. A synthesis was designed for these two compounds but it was found that both compounds were identical. This posed a bit of a problem to Kekulé—his structure didn't seem to work after all. His solution was that benzene rapidly equilibrates, or 'resonates' between the two forms to give an averaged structure in between the two.

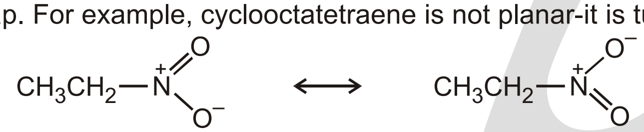


resonance contributor   resonance contributor

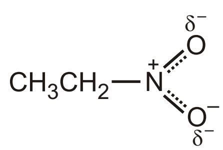


resonance hybrid

☞ Only if all the atoms sharing the delocalized electrons lie in or close to the same plane so their *p* orbitals can effectively overlap. For example, cyclooctatetraene is not planar-it is tub-shaped.

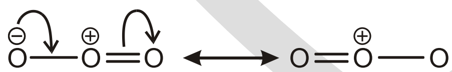


resonance contributor   resonance contributor

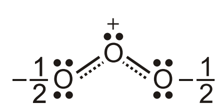


resonance hybrid

Alternatively, an average of Lewis structures is sometimes drawn using a dashed line to represent a "partial" bond. In the dashed-line notation the central oxygen is linked to the other two by bonds that are halfway between a single bond and a double bond, and the terminal oxygens each bear one half of a unit negative charge.



is equivalent to



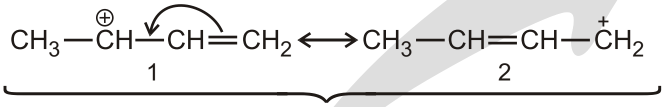
Curved arrow notation

Dashed - line notation

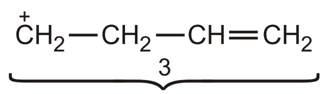
☞ **Rules for Writing Resonance Structures :**

1. **Resonance structures exist only on paper :** Although they have no real existence of their own, resonance structures are useful because they allow us to describe molecules, radicals, and ions for which a single Lewis structure is inadequate. We write two or more Lewis structures, calling them resonance structures or resonance contributors. We connect these structures by double-headed arrows ( ), and we say that the hybrid of all of them represents the real molecule, radical, or ion.

2. **In writing resonance structures, we are only allowed to move electrons :** The positions of the nuclei of the atoms must remain the same in all of the structures. For example, Structure 3 is not a resonance structure for the allylic cation, because in order to form it we would have to move a hydrogen atom and this is not permitted :



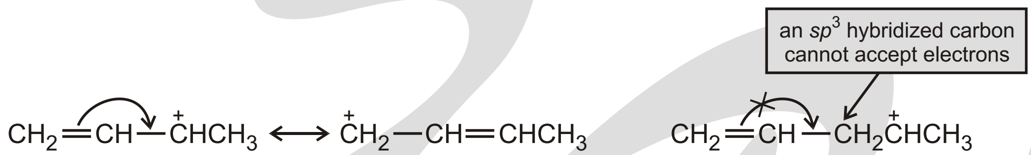
These are resonance structures for the allylic cation formed when 1, 3-butadiene accepts a proton.



This is not a proper resonance structure for the allylic cation because a hydrogen atom has been moved.

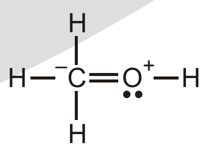
Generally speaking, when we move electrons we move only those of bond (as in the example above) and those of lone pairs.

(a) Move electrons toward a positive charge or toward a bond.

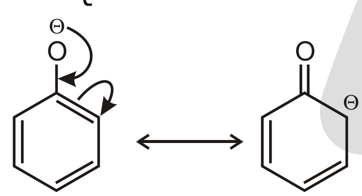


delocalized electrons — localized electrons

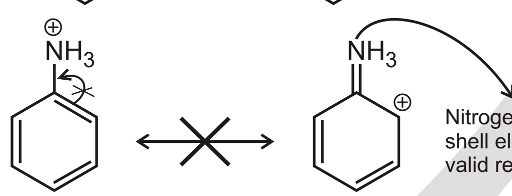
3. **All of the structures must be proper Lewis structures :** We should not write structures in which carbon has five bonds, for example :



This is not a proper resonance structure for methanol because carbon has five bonds. Elements of the first major row of the periodic table cannot have more than eight electrons in their valence shell.



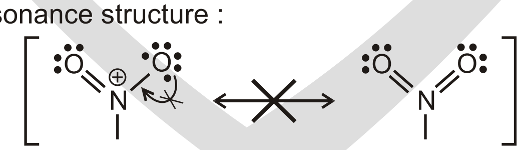
All atoms follow octet rule



(I) (II)

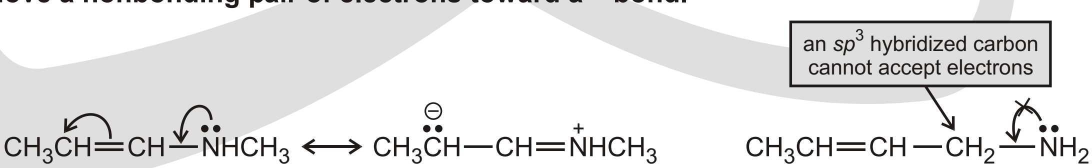
Nitrogen have 10 valance shell electron so it is not a valid resonating structure

We cannot draw this resonance structure :

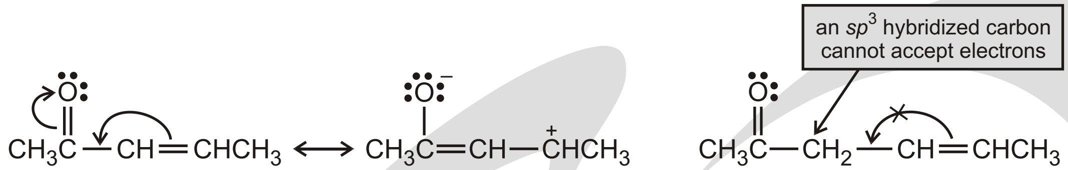


The nitrogen atom would have five bonds, which would violate the octet rule.

(b) Move a nonbonding pair of electrons toward a bond.

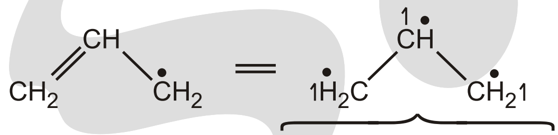


delocalized electrons — localized electrons



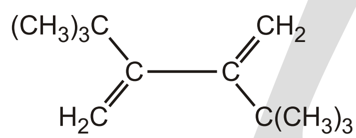
delocalized electrons localized electrons

4. **All resonance structures must have the same number of unpaired electrons :** The following structure is not a resonance structure for the allyl radical because it contains three unpaired electrons and the allyl radical contains only one :



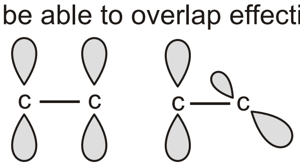
This is not a proper resonance structure for the allyl radical because it does not contain the same number of unpaired electrons as $CH_2=CHCH_2$.

5. **All atoms that are a part of the delocalized -electron system must lie in a plane or be nearly planar :** For example, 2,3-di-tert-butyl-1,3-butadiene behaves like a non-conjugated diene because the large tert-butyl groups twist the structure and prevent the double bonds from lying in the same plane. Because they are not in the same plane, the p orbitals at C_2 and C_3 do not overlap and delocalization (and therefore resonance) is prevented :



2, 3-Di-tert-butyl-1,3-butadiene

The bond contains two electrons and, since we fill up the energy level diagram from the lowestenergy orbital upwards, both these electrons go into the bonding molecular orbital. In order to have a strong bond, the two atomic *p* orbitals must be able to overlap effectively. This means they must be parallel.



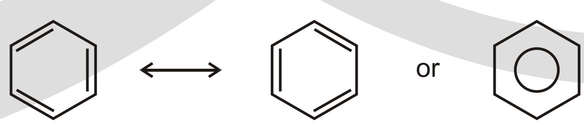
good overlap poor overlap

the two *p* orbitals can only overlap if they are parallel

6. **The energy of the actual molecule is lower than the energy that might be estimated for any contributing structure :** The actual allyl cation, for example, is more stable than either resonance structure **4** or **5** taken separately would indicate. Structures **4** and **5** resemble primary carbocations and yet the allyl cation is more stable (has lower energy) than a secondary carbocation.

$$\underset{\mathbf{4}}{CH_2=CH-\overset{+}{C}H_2} \longleftrightarrow \underset{\mathbf{5}}{\overset{+}{C}H_2-CH=CH_2}$$

We shall find that benzene is highly resonance stabilized because it is a hybrid of the two equivalent forms that follow :

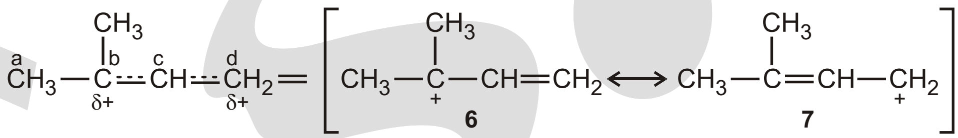


Resonance structures for benzene

Representation of hybrid

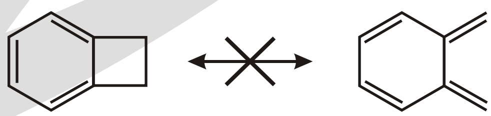
7. **Equivalent resonance structures make equal contributions to the hybrid, and a system describe by them has a large resonance stabilization :** Structures **4** and **5** make equal contributions to the allylic cation because they are equivalent. They also make a large stabilizing contribution and account for allylic cations being unusually stable. The same can be said about the contributions made by the equivalent structures for benzene.

8. **The more stable a structure is (when taken by itself), the greater is its contribution to the hybrid :** Structures that are not equivalent do not make equal contributions. For example, the following cation is a hybrid of structures **6** and **7**. Structure 6 makes a greater contribution than **7** because structure **6** is a more stable tertiary carbocation while structure 7 is a primary cation :



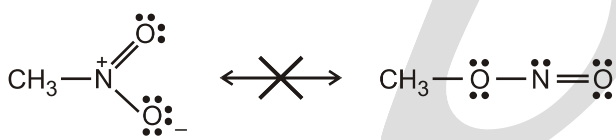
That **6** makes a larger contribution means that the partial positive charge on carbon **b** of the hybrid will be larger than the partial positive charge on carbon **d**. It also means that the bond between carbon atoms **c** and **d** will be more like a double bond than the bond between carbon atoms b and c.

9. Resonance forms do not differ in the position of nuclei. The two structures given below are not resonance forms because the position of the carbon and hydrogen atoms outside the ring are different in the two forms.



not resonance structures

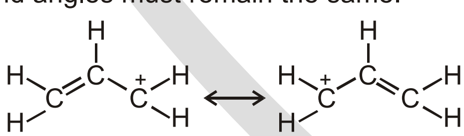
Another Example is of the structural formulas



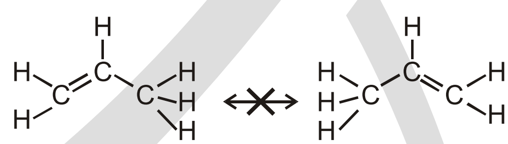
A B

which represent different compounds, not different resonance forms of the same compound. A is a Lewis structure for nitromethane ; B is methyl nitrite.

10. **Only the placement of the electrons may be shifted from one structure to another.** (Electrons in double bonds and lone pairs are the ones that are most commonly shifted.) Nuclei cannot be moved, and the bond angles must remain the same.

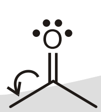


resonance forms

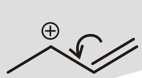


NOT resonance

11. If we try to push the pi bonds to form other pi bonds, we find



No: This violates the octet rule.



Yes : Does not violate the octet rule.

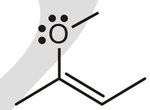
DRAWING RESONANCE STRUCTURES— BY RECOGNIZING PATTERNS

There are five patterns that you should learn to recognize to become proficient at drawing resonance structures. First we list them, and then we will go through each pattern in detail, with examples and exercises. Here they are :

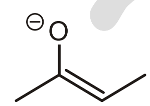
1. A lone pair next to a pi bond.
2. A lone pair next to a positive charge.
3. A pi bond next to a positive charge.
4. A pi bond between two atoms, where one of those atoms is electronegative.
5. Pi bonds going all the way around a ring.

A LONE PAIR NEXT TO A PI BOND

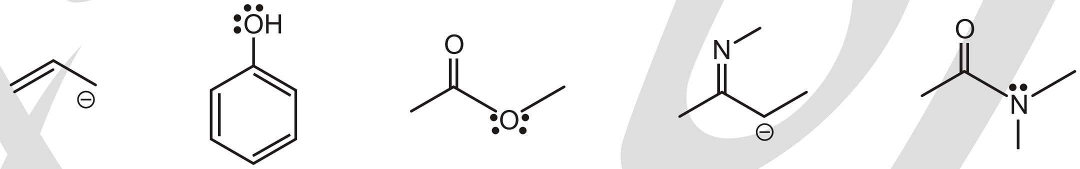
Let's see an example before going into the details :



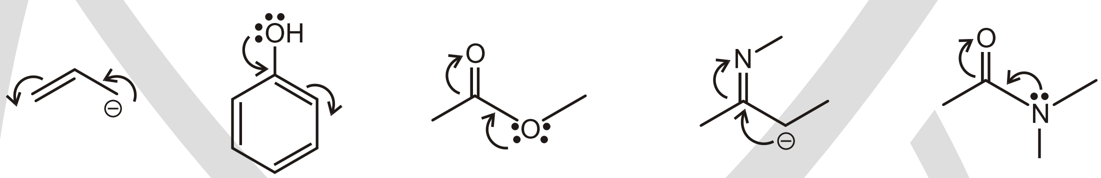
The atom with the lone pair can have no formal charge (as above), or it can have a negative formal charge as below :



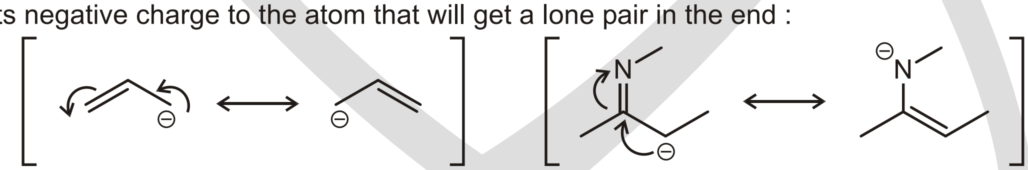
The important part is having a lone pair "next to" the pi bond. "Next to" means that the lone pair is separated from the double bond by exactly one single bond—no more and no less. You can see this in all of the examples below:



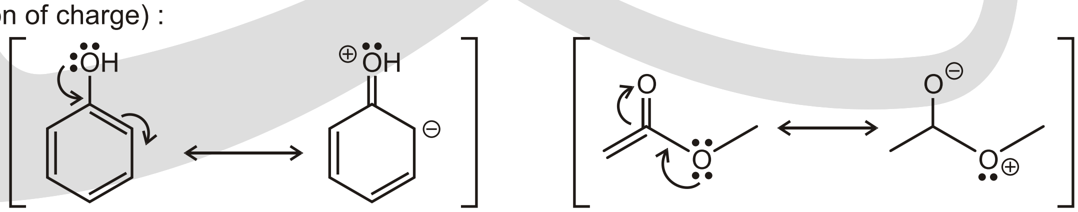
In each of these cases, you can bring down the lone pair to form a pi bond, and kick up the pi bond to form a lone pair



Notice what happens with the formal charges. When the atom with the lone pair has a negative charge, then it transfers its negative charge to the atom that will get a lone pair in the end :



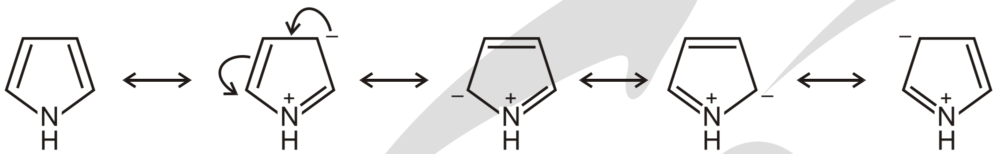
When the atom with the lone pair does not have a negative charge to begin with, then it will end up with a positive charge in the end, while a negative charge will go on the atom getting the lone pair in the end (remember conservation of charge) :



Solved Examples

- Pyrrole.

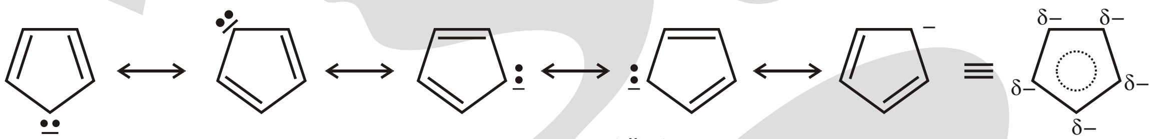
Sol.



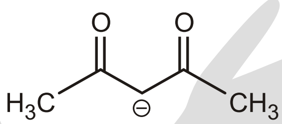
resonance contributors of pyrrole

- Cyclopentadienyl anion

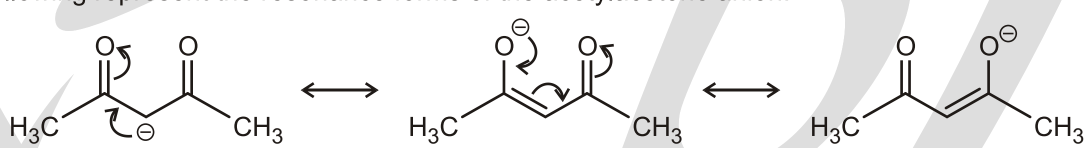
Sol.

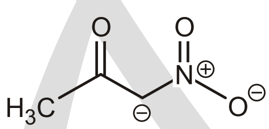


resonance contributors

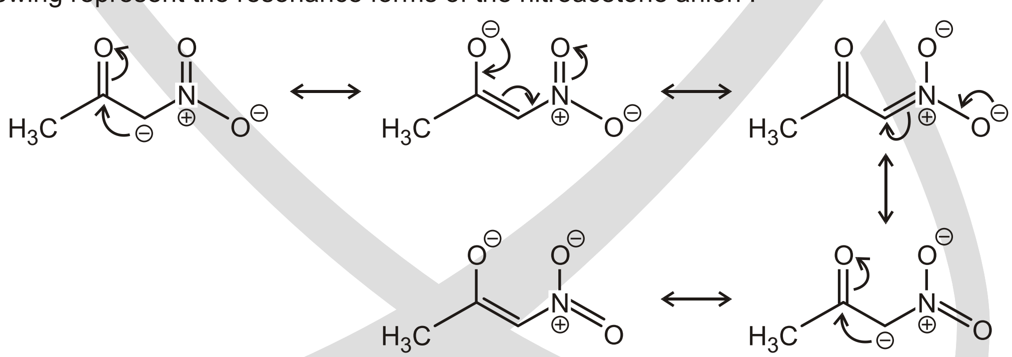
- 

Sol. The following represent the resonance forms of the acetylacetone anion:



- 

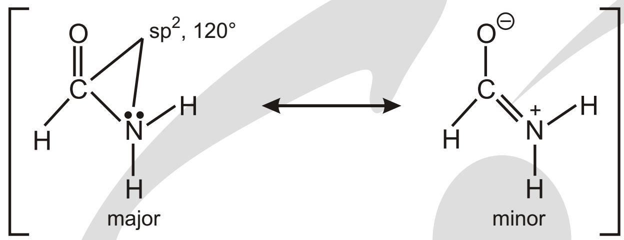
Sol. The following represent the resonance forms of the nitroacetone anion :



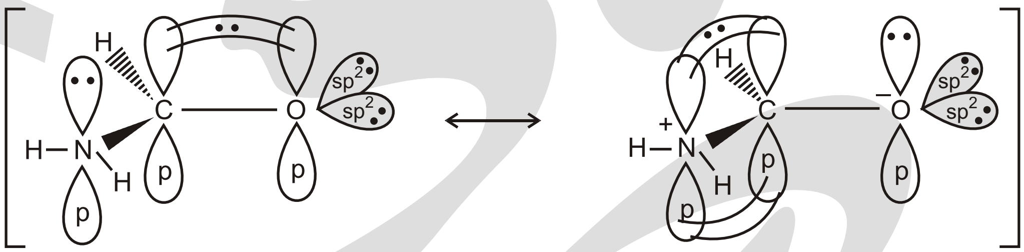
Please note that while nitro groups are so electron withdrawing that delocalization of their associated positive charge plays a minimal role in any family of resonance structures, this delocalization is technically possible. Try to identify additional resonance structures where the positive charge is delocalized.

- Use a three-dimensional drawing to show where the electrons are pictured to be in each resonance form.

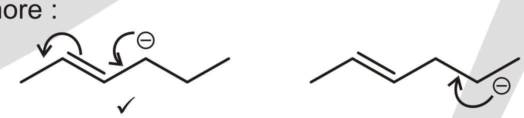
  (a) $HCONH_2$



Sol.



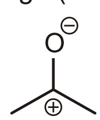
☞ Notice that the lone pair needs to be directly next to the pi bond. If we move the lone pair one atom away, this does not work anymore :



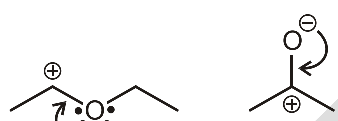
A LONE PAIR NEXT TO A POSITIVE CHARGE

Let's see an example :

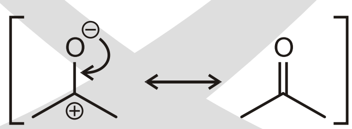
The atom with the lone pair can have no formal charge (as above) or it can have negative formal charge :



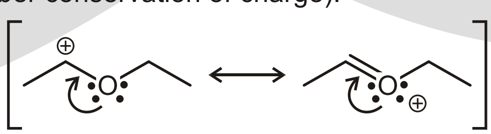
The important part is having a lone pair next to a positive charge. In each of the below cases, we can bring down the lone pair to form a pi bond:



Notice what happens with the formal charges. When the atom with the lone pair has a negative charge, then the charges end up canceling each other:

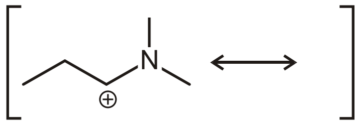


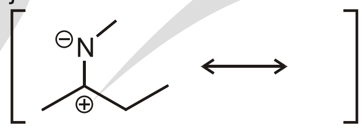
When the atom with the lone pair does not have a negative charge to begin with, then it will end up with the positive charge in the end (remember conservation of charge):

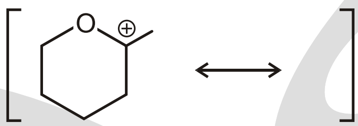


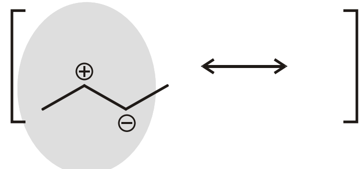
Solved Examples

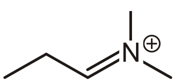
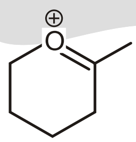
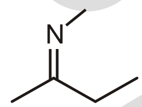
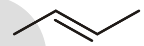
- For each of the compounds below, locate the pattern we just learned and draw the resonance structure.

1. 

3. 

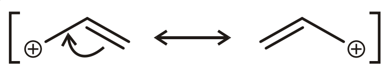
2. 

4. 

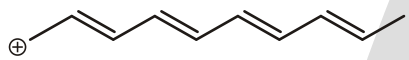
Ans. 1.  2.  3.  4. 

A PI BOND NEXT TO A POSITIVE CHARGE

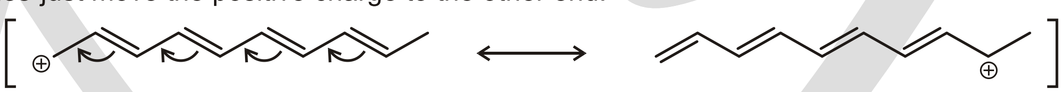
Notice what happens to the formal charge in the process. It gets moved to the other end:



It is possible to have many double bonds in conjugation (this means that we have many double bonds that are each separated by only one single bond) next to a positive charge:

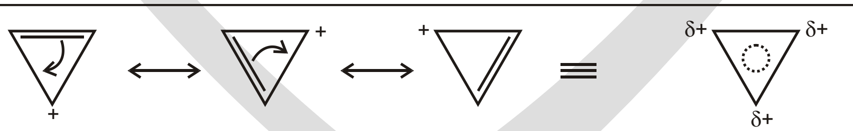


When this happens, we can push all of the double bonds over, and we don't need to worry about calculating formal charges-just move the positive charge to the other end:



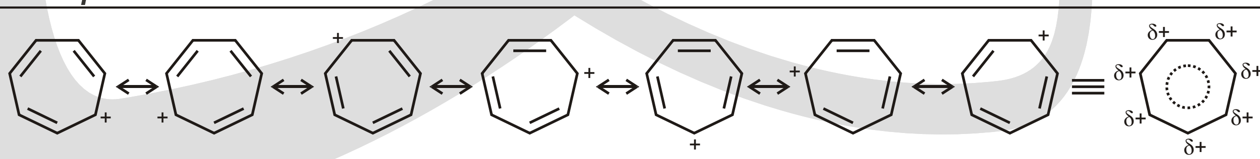
Of course, we should push one arrow at a time so that we can draw all of the resonance structures. But it is nice to know how the formal charges will end up so that we don't have to calculate them every time we push an arrow.

Solved Examples

- 

resonance contributors for the cyclopropenyl cation   resonance hybrid

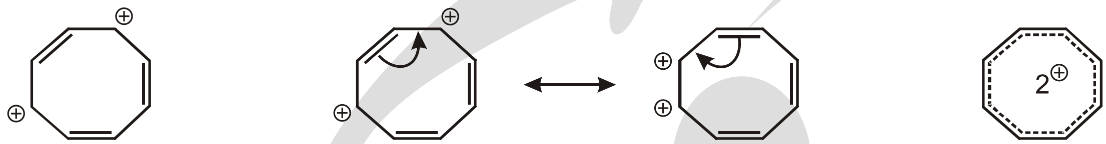
Solved Examples

- 

resonance contributors   R.H.

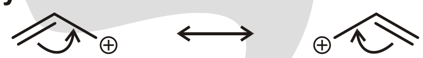
DRAWING THE DICATION

The dication still has the same number of atoms as the neutral species with only fewer electrons. Where have the electrons been taken from? The system now has two electrons less. We could draw a structure showing two localized positive charges but this would not be ideal since the charge is spread over the whole ring system.



one structure with localized charges — the charges can be delocalized all round the ring — structure to show equivalence of all the carbon atoms

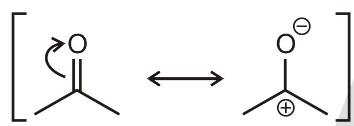
☞ **Representations of the allyl cation**



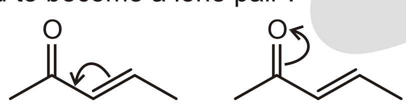
curly arrows show the positive charge is shared over both the end atoms

Do not confuse this delocalization arrow with the equilibrium sign. A diagram like this would be wrong :

A PI BOND BETWEEN TWO ATOMS, WHERE ONE OF THOSE ATOMS IS ELECTRONEGATIVE (N, O, ETC.)



As another example, consider the structure below. We cannot move the C C bond to become another bond unless we also move the C O bond to become a lone pair :

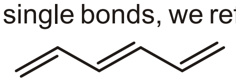


No Yes

In this way, we truly are "pushing" the electrons around.

PI BONDS BETWEEN SIMILAR ATOMS

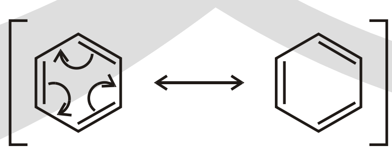
Whenever we have alternating double and single bonds, we refer to the alternating bond system as conjugated:



Conjugated double bonds

PI BONDS GOING ALL THE WAY AROUND A RING

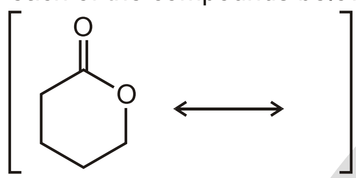
When we have a conjugated system that wraps around in a circle, then we can always move the electrons around in a circle:

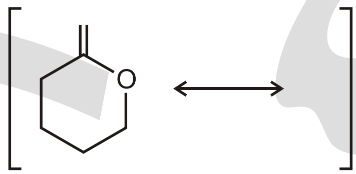


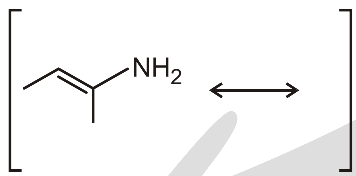
It does not matter whether we push our arrow's clockwise or counterclockwise (either way gives us the same result, and remember that the electrons are not really moving anyway).

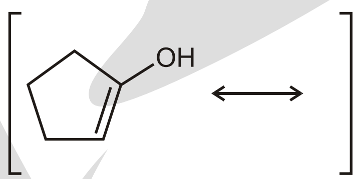
Solved Examples

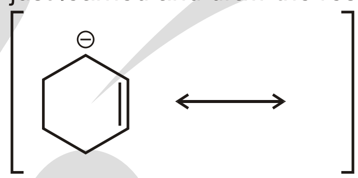
- For each of the compounds below, locate the pattern we just learned and draw the resonance structure.

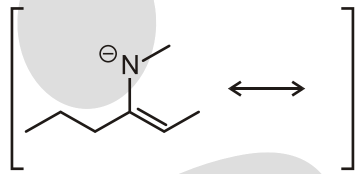
1. 

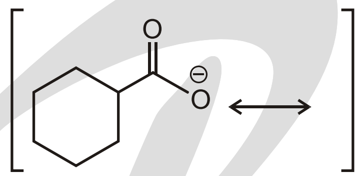
2. 

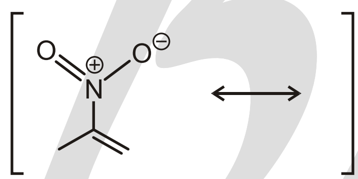
3. 

4. 

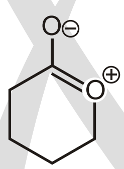
5. 

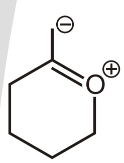
6. 

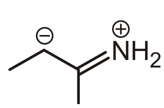
7. 

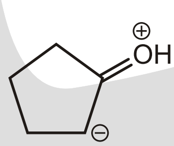
8. 

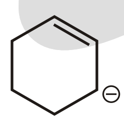
Ans.

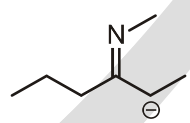
1. 

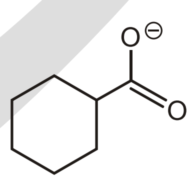
2. 

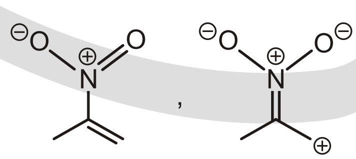
3. 

4. 

5. 

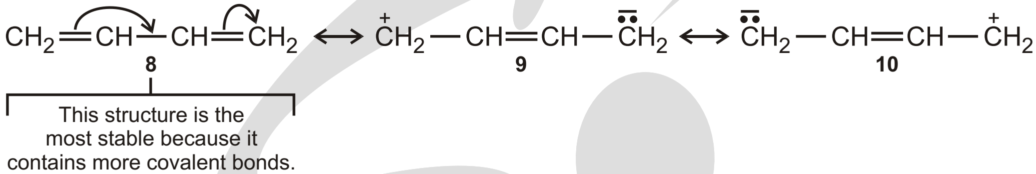
6. 

7. 

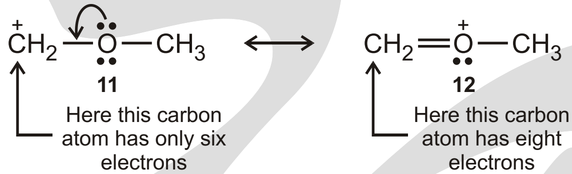
8. 

ESTIMATING THE RELATIVE STABILITY OF RESONANCE STRUTURES

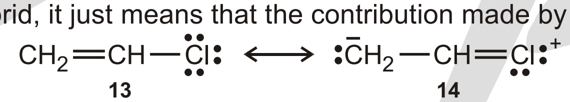
(a) The more covalent bonds a structure has, the more stable it is. We know that forming a covalent bond lowers the energy of atoms. 8 is by far the most stable and makes by far the largest contribution because it contains one more bond.



(b) Structures in which all of the atoms have a complete valence shell of electrons are especially stable and make large contributions to the hybrid.



(c) Charge separation decreases stability Separating opposite charges requires energy. Therefore, structures in which opposite charges are separated have greater energy (lower stability) than the those that have no charge separation. This means that of the following two structures for vinyl chloride, structure 13 makes a larger contribution because it does not have separated charge. (This does not mean that structure 14 does not contribute to the hybrid, it just means that the contribution made by 14 is smaller.



(d) Resonance contributors with negative charge on highly electronegative atoms are more stable than ones with negative charge on less or nonelectronegative atoms. Conversely, resonance contributors with positive charge on highly electronegative atoms are less stable than ones with positive charge on nonelectronegative atoms.

(e) To predict the energies of the resonance structures, we consider the energy of resonance hybrid structure. Resonance hybrid is the weighted average of the resonance contributors. The following structures are considered relatively stable:

(i) Structures having filled octets for second row elements (C, N, O, F) are stable.

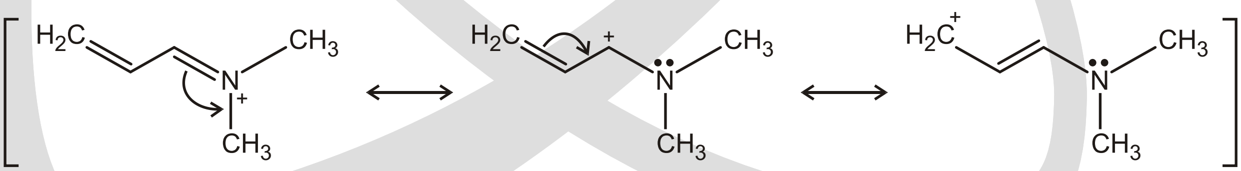
(ii) Structures having minimum number of formal charges and maximum number of bonds.

(iii) Structure in which negative charge is on the most electronegative atom (C<N<O).

(iv) Structure in which there is minimal charge separation while keeping the formal charges closer together.

Solved Examples

- Which of the following structures is relatively stable?



(A) (B) (C)

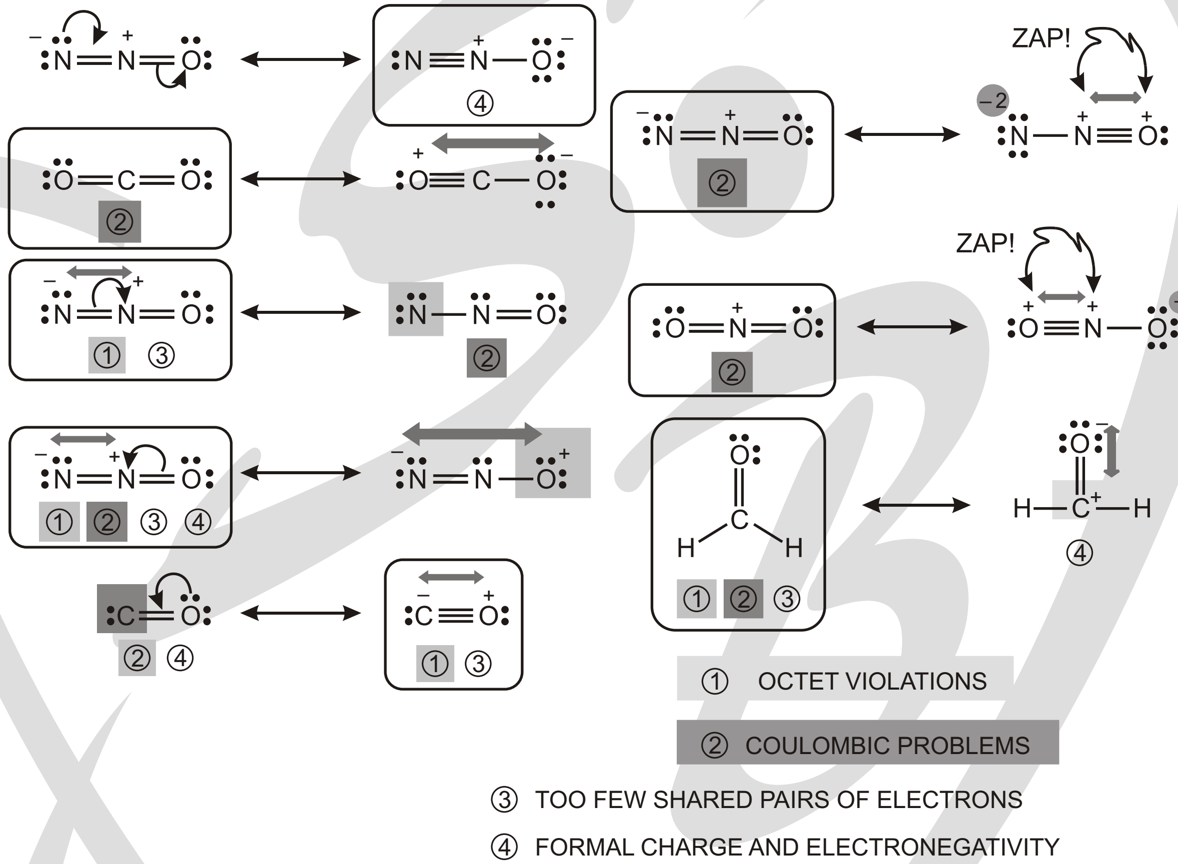
Ans. Structure (A) is most stable as it follows all guidelines as per points (a) to (e).

Structure (B) is more stable as it violates point (a).

Structure (C) is least stable for violating point (a) and primary carbocation. Stability order is A > B > C

Solved Examples

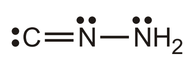
- The carbon monoxide molecule is unusual and interesting in many ways. A collection of examples of relative stability of resonance forms is found in the diagram below. You should go through them very carefully and see if you can understand them and make the same predictions that you see here.

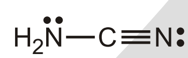


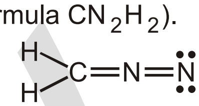
Solved Examples

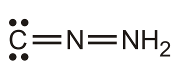
- The following Lewis/Kekule structures (A) – (L) are isomeric (with molecular formula CN_2H_2).


(A)


(B)


(C)

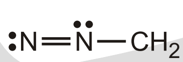

(D)

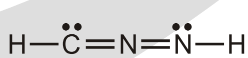

(E)

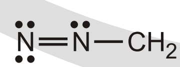

(F)


(G)


(H)


(I)


(J)


(K)

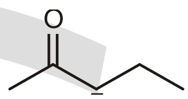
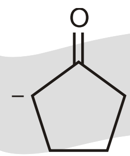
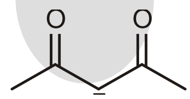
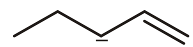
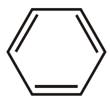
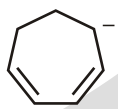
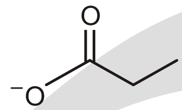
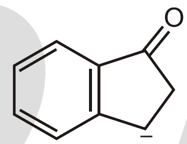
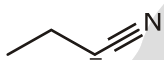
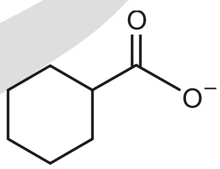
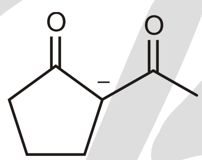

(L)

Which of these structures match the following properties ? Indicate with letters (A) to (L). If no structure fit the property write the letter X.

1. Which of the structures have no atoms with formal charge?
2. Which of the structures have at least one nitrogen atom with a(+) formal charge
3. Which of the structure have at least one nitrogen atom with a(–) formal charge?
4. Which of the structures have at least one carbon atom with a(+) formal charge?
5. Which of the structures have at least one carbon atom with a(–) formal charge?
6. Which of the structures have electron deficient heavy atoms (N or C)?

Ans. 1. (B, C, G), 2. (A, D, E, F, H, J), 3. (A, D, K, L), 4. (K, L), 5 (F, E, H, J), 6. (B, I, K, L).

- Write the possible resonance structures for the following molecules. Show the direction of the movement of electrons with the help of arrows.

1. 
2. 
3. 
4. 
5. 
6. 
7. 
8. 
9. 
10. 
11. 
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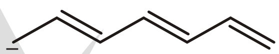
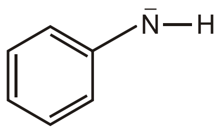
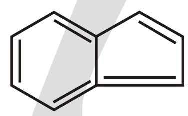
13. 
14. 
15. 
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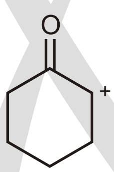
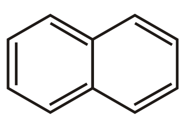
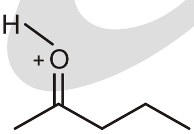
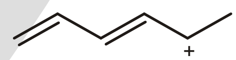
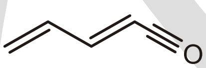
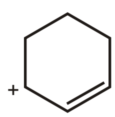
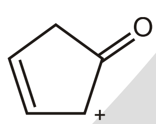
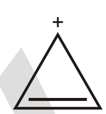
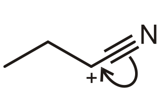
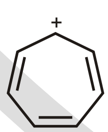
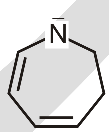
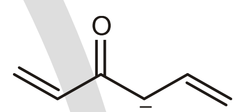
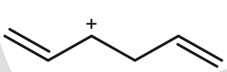
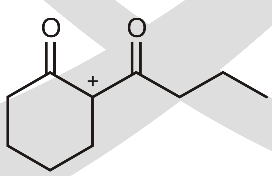
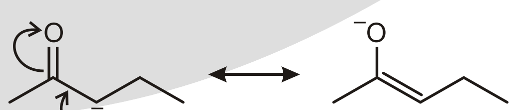
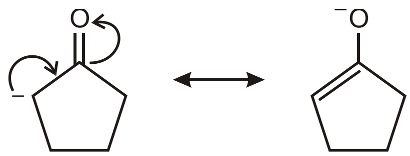
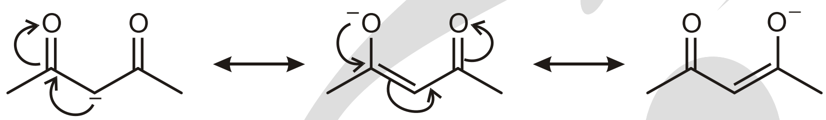
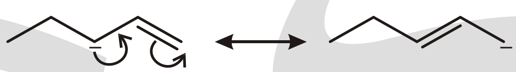
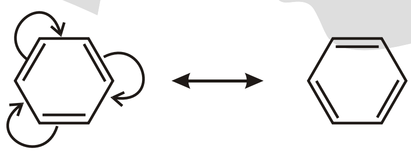
17. 
18. 
19. 
20. 
21. 
22. 
23. 
24. 
25. 
26. 
27. 
28. 
29. 
30. 

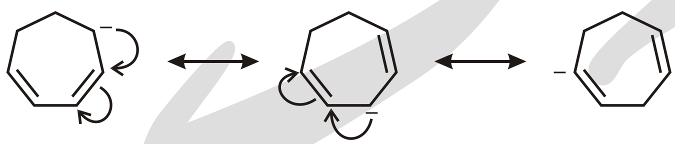
Ans. 1. 

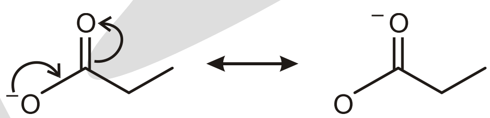
2. 

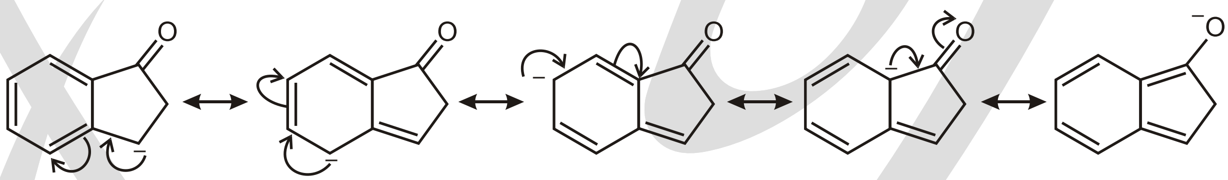
3. 

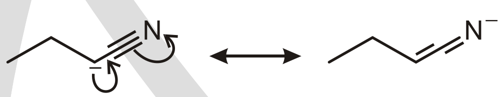
4. 

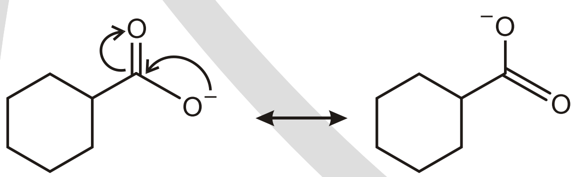
5. 

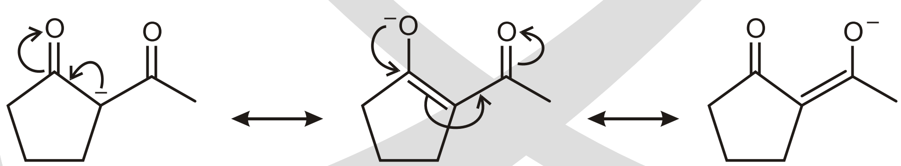
6. 

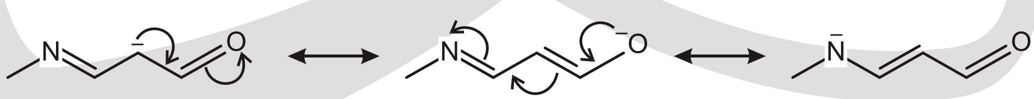
7. 

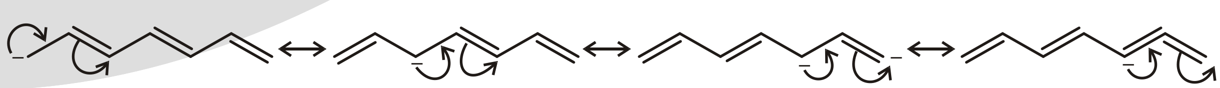
8. 

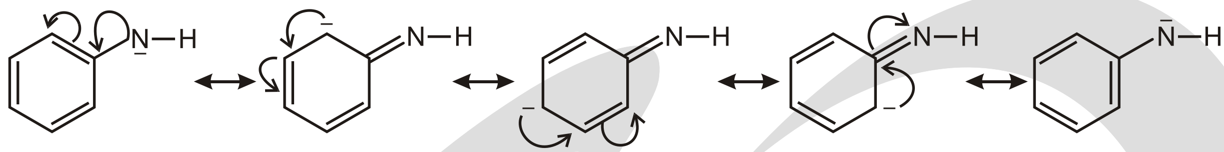
9. 

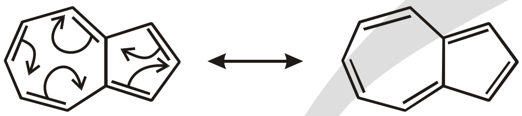
10. 

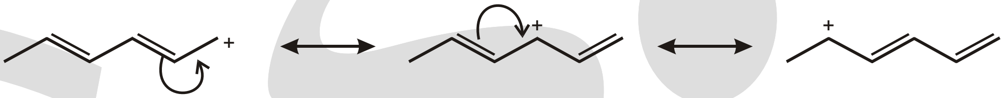
11. 

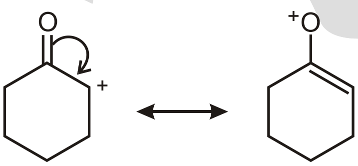
12. 

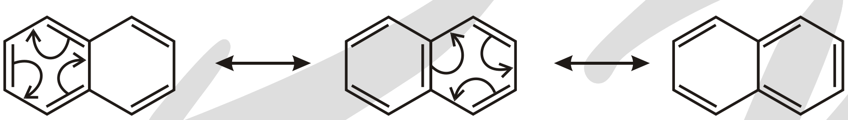
13. 

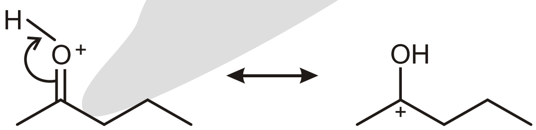
14. 

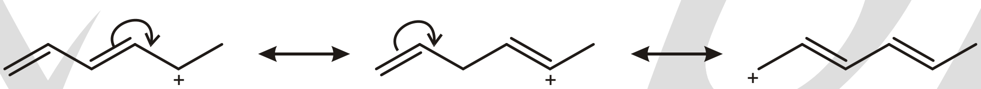
15. 

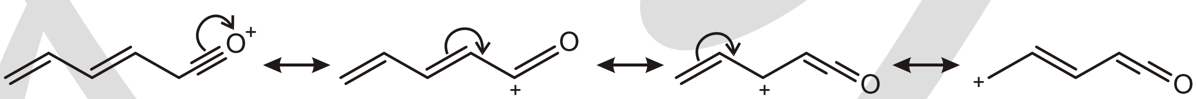
16. 

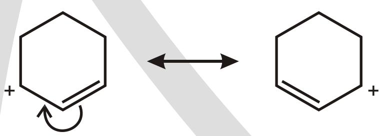
17. 

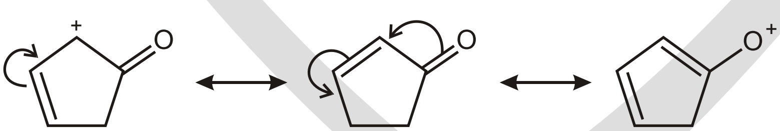
18. 

19. 

20. 

21. 

22. 

23. 

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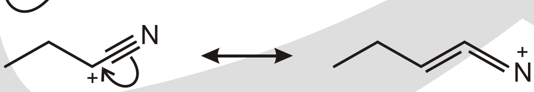
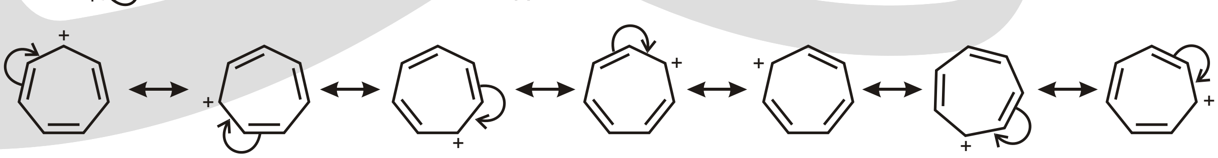
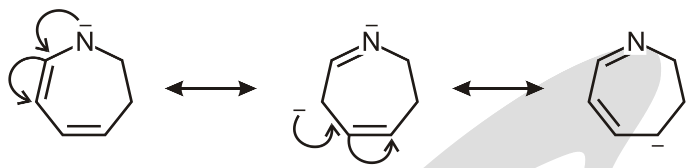
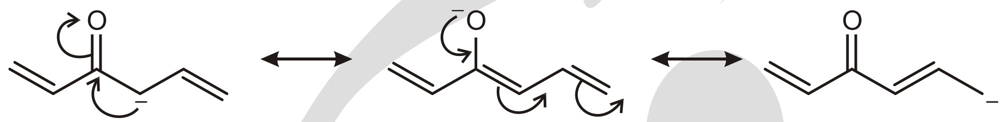
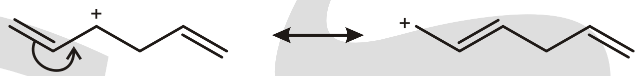
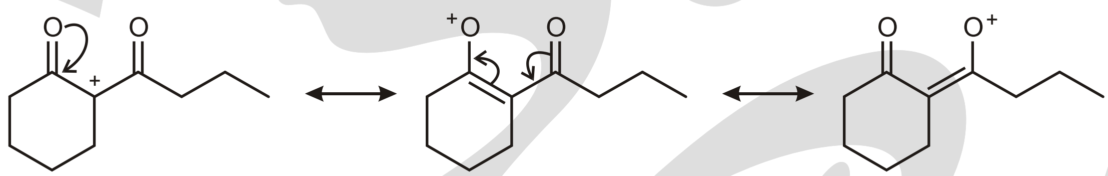
25. 

26. 

27. 

28. 

29. 

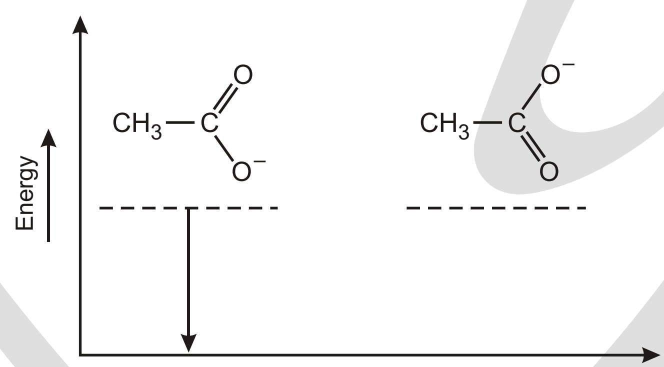
30. 

SCHEMATIC ENERGY LEVEL DIAGRAMS OF SOME MESOMERIC MOLECULES/IONS

Energy diagrams of some resonating structures are given as follows. (Note: The downwards arrow marks the conventional mesomeric energies and indicates energy level).

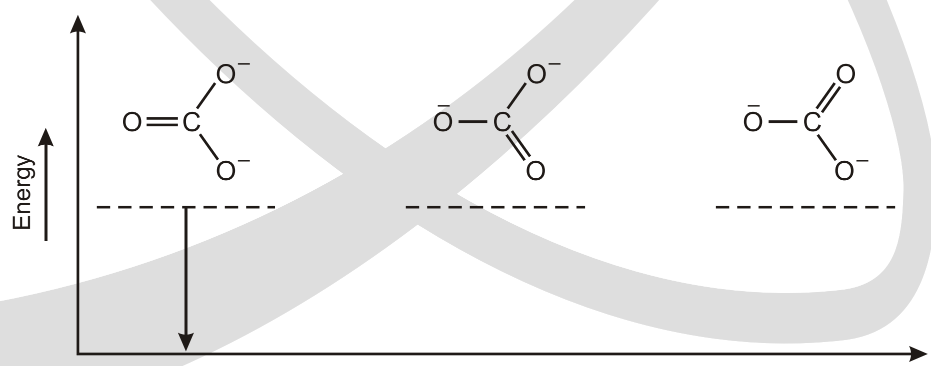
Solved Examples

- **Carboxylate ion**



Interpretation : The two equivalent resonating structures have the same energy.

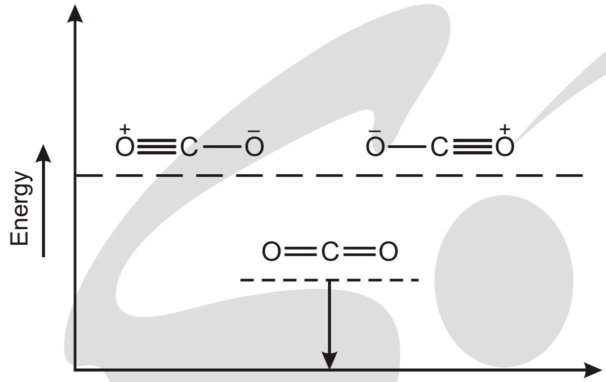
- **Carbonate ion**



Carbonate ion

Interpretation : The three equivalent resonating structures have the same energy.

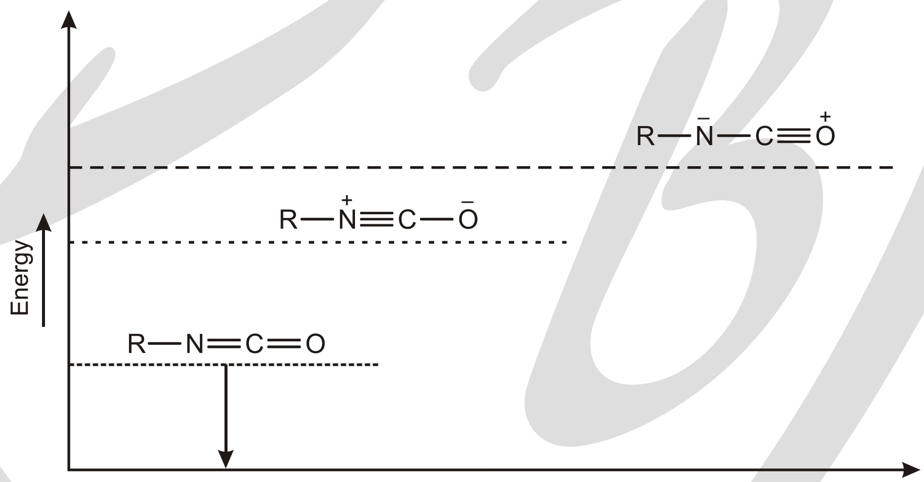
- **Carbon dioxide**



Carbon dioxide

Interpretation : Polarized structures have higher energy because separation of opposite charges requires energy.

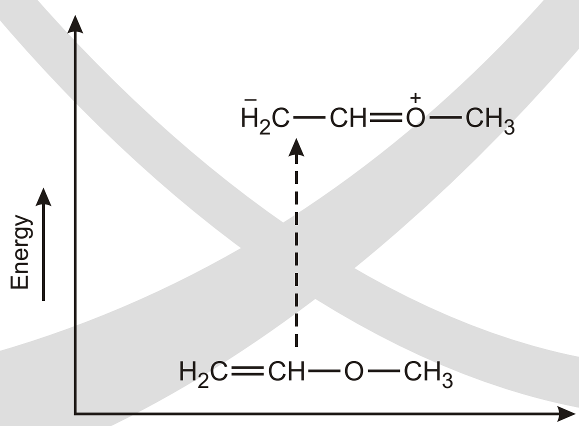
- **Alkylcyanate**



Alkyl cyanate

Interpretation : Negative charge on the more electronegative element (oxygen) makes the structure more stable.

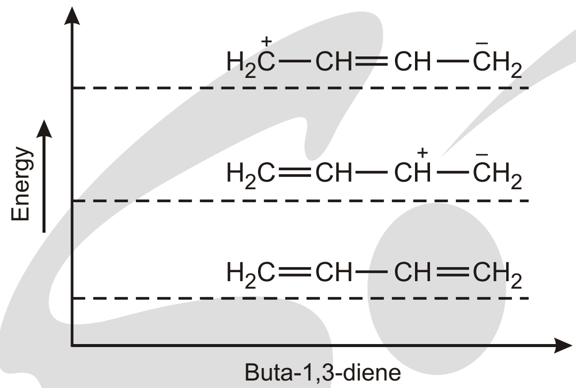
- **Vinyl ether**



Vinyl ether

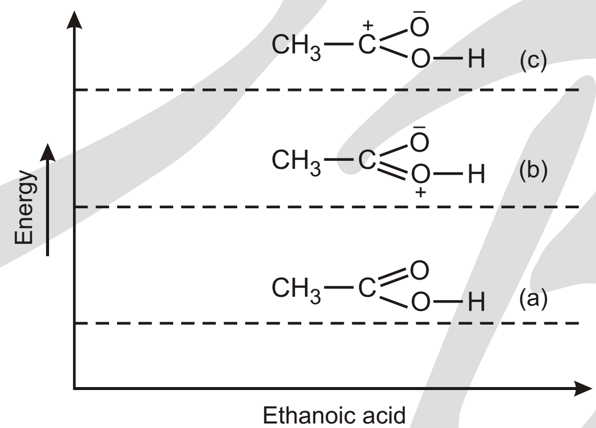
Interpretation : Separation of opposite charges requires energy.

- **But-1,3-diene**



Interpretation : Unlike charges must be closer to each other for the structure to be more stable.

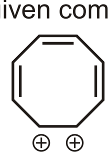
- **Ethanoic acid**



EXERCISE

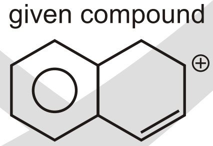
SINGLE CHOICE QUESTIONS

1. Number of Resonating structures of the given compound are :



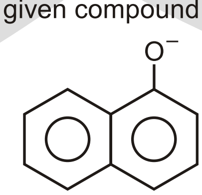
(A) 0 (B) 14 (C) 16 (D) 18

2. Number of Resonating structures of the given compound are :



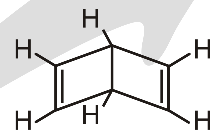
(A) 7 (B) 8 (C) 9 (D) 10

3. Number of Resonating structures of the given compound are :



(A) 7 (B) 8 (C) 9 (D) 10

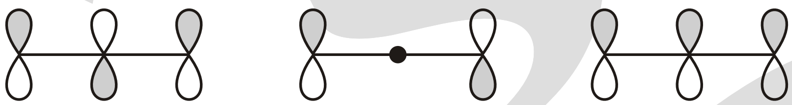
4. The compound shown to the right is known as Dewar Benzene. What relationship does this compound bear to benzene?



(A) It is a resonance structure of benzene (B) It is an enantiomer of benzene

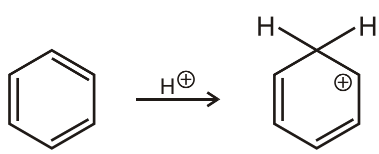
(C) It bears no relationship to benzene (D) It is a structural isomer of benzene

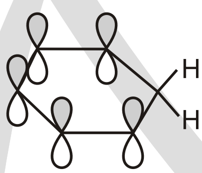
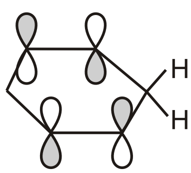
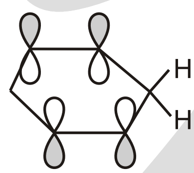
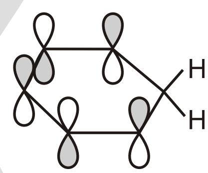
5. Choose the answer that has the molecular orbitals for the allyl anion (CH_2 CH — CH_2) correctly identified.

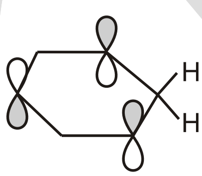


(A)	LUMO	HOMO	bonding
(B)	LUMO	bonding	HOMO
(C)	bonding	LUMO	HOMO
(D)	bonding	HOMO	LUMO
(E)	HOMO	bonding	LUMO

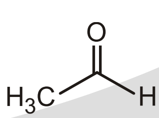
6. When benzene is protonated the resulting ion is the cyclohexadienyl cation. Which of the following MOs is the best representation of the LUMO of this cation?

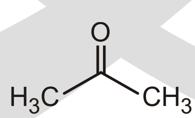

cyclohexadienyl cation

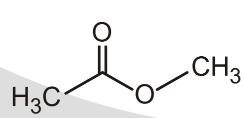
(A)  (B)  (C)  (D) 

(E) 

7. Choose the order that has the followingC = O groups correctly arranged with respect to increasing resonance stabilization.


i


ii

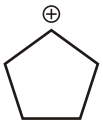

iii

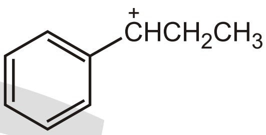
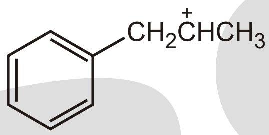
(A) i < ii < iii (B) i < iii < ii (C) ii < i < iii (D) ii < iii < i

(E) iii < ii < i

8. Which of the following pairs are resonance contributors?

(A) CH_3 — CH   CH — CH — CH CH_2 and CH_3 — CHCH   CHCH CH_2

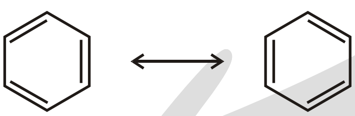
(B)  and 

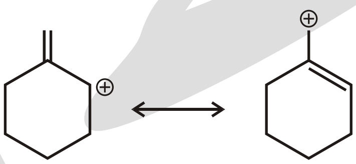
(C)  and 

(D) All of the above

9. Which of the following are not resonating structures of each other?

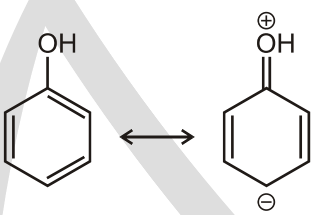
(A) $CH_3-\overset{O}{\overset{\|}{C}}-CH_2CH_3$ and $CH_3-\overset{OH}{\overset{|}{C}}=CHCH_3$

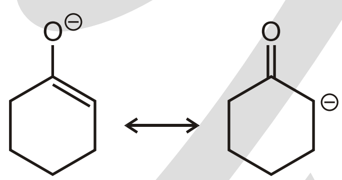
(B) 

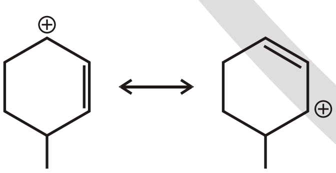
(C) 

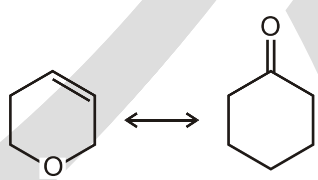
(D) CH_3CH CH — CH — CH CH_2 and CH_3 CH — CH   CH — CH CH_2

10. Which of the following is not resonating structure of each other?

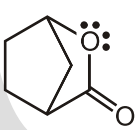
(A) 

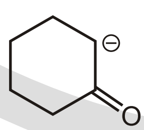
(B) 

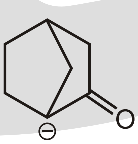
(C) 

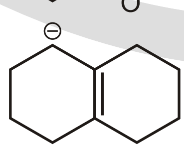
(D) 

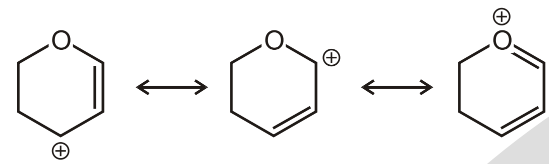
11. Which of the following compound is not resonance stabilized?

(A) 

(B) 

(C) 

(D) 

12. 

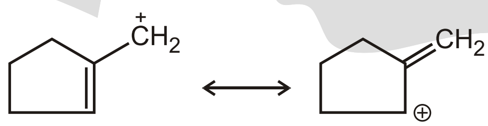
(I) (II) (III)

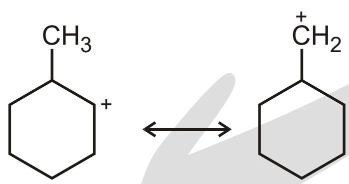
The most stable canonical structure among the given structure is :

(A) I (B) II (C) III (D) all are equally stable

13. Which is <u>not</u> an example of resonance?

(I) CH_2 $CH—CH_2$ $CH_2—CH$ CH_2

(II) 

(III) 

(IV) CH_2 $CH—CH—CH_2CCl_3$ $CH_2—CH$ $CH—CH_2CCl_3$

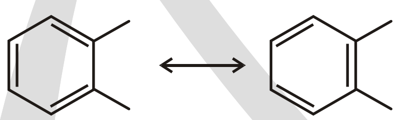
(A) I (B) II (C) III (D) IV

14. Which is <u>not</u> a proper resonance structure for 1, 3-butadiene?

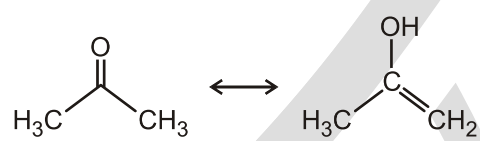
(A) CH_2 $CH—CH$ CH_2 (B) $CH_2—CH$ $CH—CH_2$

(C) $\bar{C}H_2—CH$ $CH—CH_2$ (D) $CH_2—CH$ $CH—C\bar{H}_2$

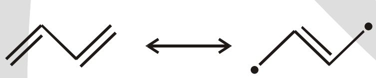
15. Which of the following pairs are resonance structures of each other?



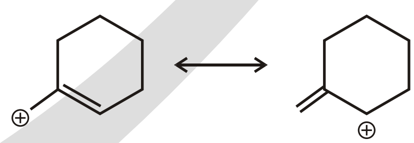
(I)



(II)



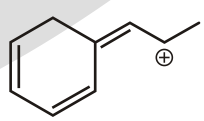
(III)



(IV)

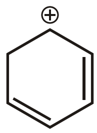
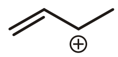
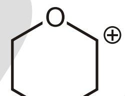
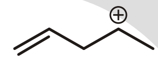
(A) I, II, III (B) I, IV (C) II, III (D) I, III, IV

16. Total number of resonating structure is :

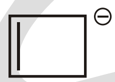
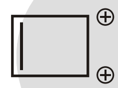
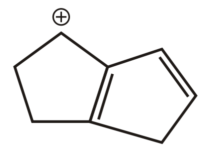


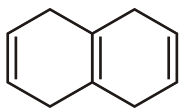
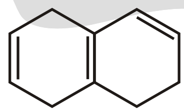
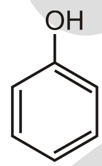
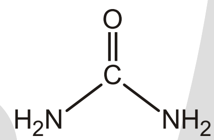
(A) 2 (B) 3 (C) 4 (D) 5

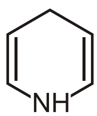
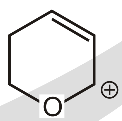
17. Which of the carbocation is not resonance stabilized?

(A)  (B)  (C)  (D) 

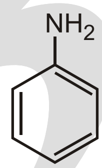
18. How many number of compounds which are Resonance stabilised ?

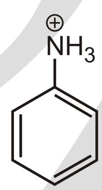
(1)  (2)  (3)  (4) 

(5)  (6)  (7)  (8) 

(9)  (10) 

(A) 7 (B) 8 (C) 9 (D) 10

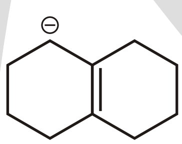
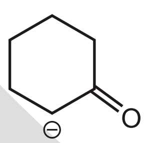
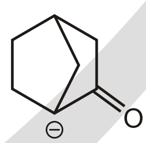
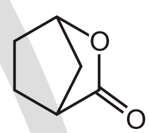
19. X number of Resonance Structures contributed in Resonance Hybrid of 

Y number of Resonance Structures contributed in Resonance Hybrid of 

Find the sum of X Y

(A) 5 (B) 6 (C) 7 (D) 8

20. Which of the following compound is not resonance stabilized?

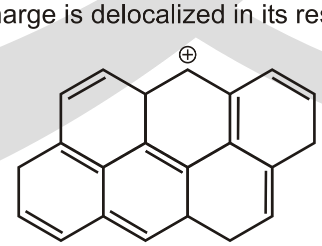
(A)  (B)  (C)  (D) 

21. Which compound below does not contain any conjugated multiple bonds?

(A) 1,2,4-pentatriene (B) 1,3-cyclobutadiene

(C) 1,5-hexadiene (D) 3-methyl-2,4-hexadiene

22. Number of C-atoms where charge is delocalized in its resonance hybrid.



(A) 5 (B) 6 (C) 7 (D) 8

23. Number of o/p-directing directing groups present on benzene ring :

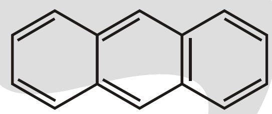
(a) $-\ddot{N}=O$ (b) $-NO_2$ (c) $-CH_3$ (d) $-SO_3H$

(e) $-\underset{\underset{R}{|}}{S}=O$ (f) $-COOH$ (g) $-Cl$ (h) $-OH$

(i) $-COO^{\ominus}$

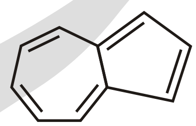
(A) 5 (B) 6 (C) 7 (D) 8

24. How many resonance structures are there for anthracene?



(A) 6 (B) 5 (C) 4 (D) 2

25. How many uncharged resonance structures are there for azulene?

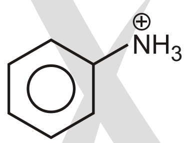


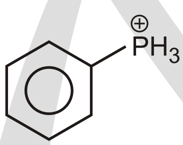
(A) 1 (B) 2 (C) 3 (D) 4

26. Which of the following is the correct resonance hybrid of buta-1,3-diene?

(A) $CH_2\overset{\cdots}{-}CH\overset{\cdots}{-}CH-CH_2$ (B) $CH_2\overset{\cdots}{-}CH\overset{\cdots}{-}CH\overset{\cdots}{-}CH_2$

(C) $CH_2\overset{\cdots}{-}CH-CH\overset{\cdots}{-}CH_2$ (D) None of these

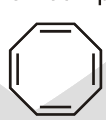
27.  $\longrightarrow$ *X* Resonating structures

 $\longrightarrow$ *Y* Resonating structures

Sum of *X* *Y* is :

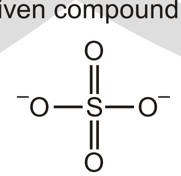
(A) 5 (B) 6 (C) 7 (D) 8

28. Number of Resonating structures of the given compound are :



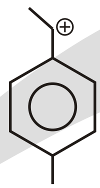
(A) 0 (B) 2 (C) 4 (D) 8

29. Number of Resonating structures of the given compound are :

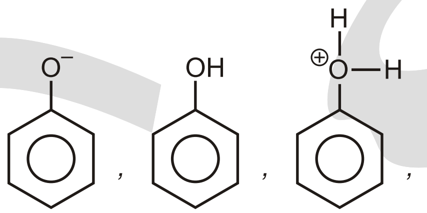


(A) 7 (B) 6 (C) 9 (D) 8

30. Number of Resonating structures having 2° carbocation



(A) 2 (B) 3 (C) 4 (D) 5

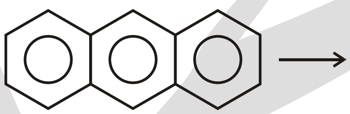
31. 

x y z

x, y, and z denote the number of resonating structures of the given compunds.

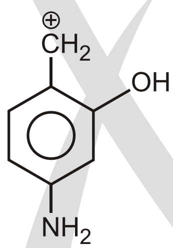
What is the value of x y z?

(A) 12 (B) 13 (C) 10 (D) 11

32. 

Number of resonating structures of the given compound are?

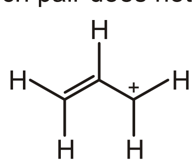
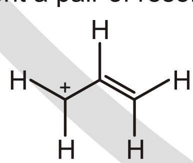
(A) 3 (B) 4 (C) 5 (D) 6

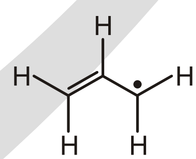
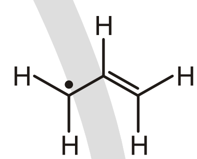
33. 

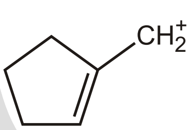
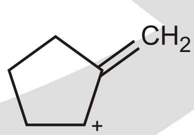
Number of resonating structures of the given compound are?

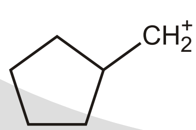
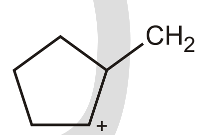
(A) 5 (B) 6 (C) 7 (D) 8

34. Which pair does not represent a pair of resonance structures?

(I)  and 

(II)  and 

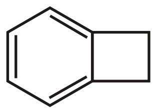
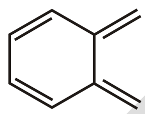
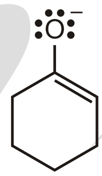
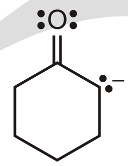
(III)  and 

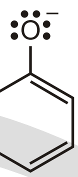
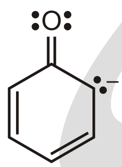
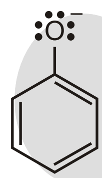
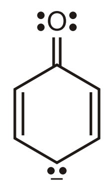
(IV)  and 

(A) I (B) II (C) III (D) IV

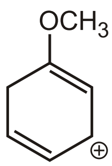
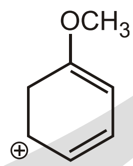
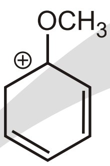
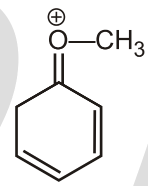
(E) All of these represent pairs of resonance structures.

35. Which of the following pairs of structures do not represent resonance forms?

(A)  and  (B)  and 

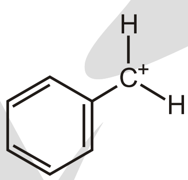
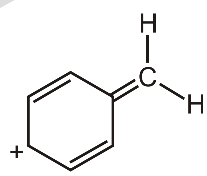
(C)  and  (D)  and 

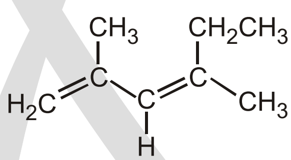
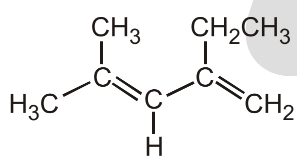
36. Which of the following resonance structure contributes the most to the resonance hybrid?

(A)  (B)  (C)  (D) 

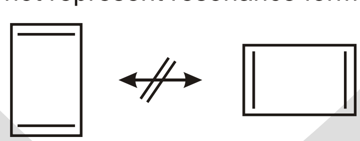
UNSOLVED EXAMPLES

1. Which of the following pairs of structures represent resonance forms, and which do not? Explain.

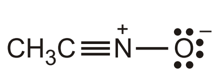
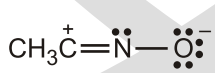
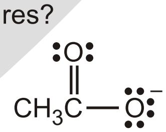
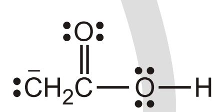
(a)  and 

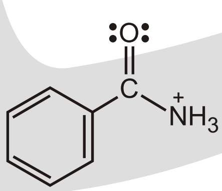
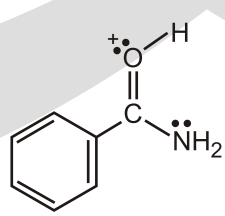
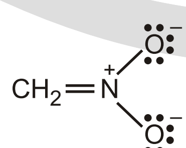
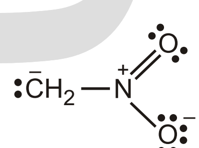
(b)  and 

2. 1,3-Cyclobutadiene is a rectangular molecule with two shorter double bonds and two longer single bonds. Why do the following structures not represent resonance forms?



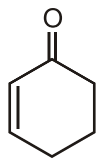
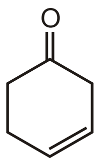
3. Which of the following pairs represent resonance Structures?

(a)  and  (b)  and 

(c)  and  (d)  and 

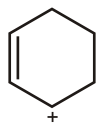
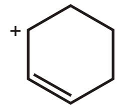
4. Naphthalene, has three resonance forms. Draw them.

5. Are the following pairs of structures resonance contributors or different compounds?

(a)  and 

(b) CH_3CH CHCHCH CH_2 and CH_3CHCH CHCH CH_2

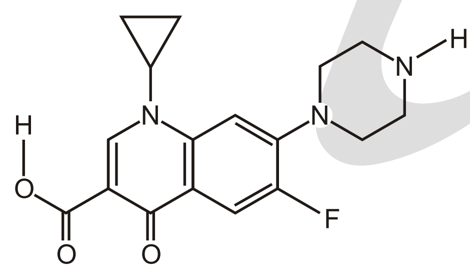
(c) $CH_3\overset{O}{\overset{\|}{C}}CH_2CH_3$ and $CH_3\overset{OH}{\overset{|}{C}}{=}CHCH_3$

(d)  and 

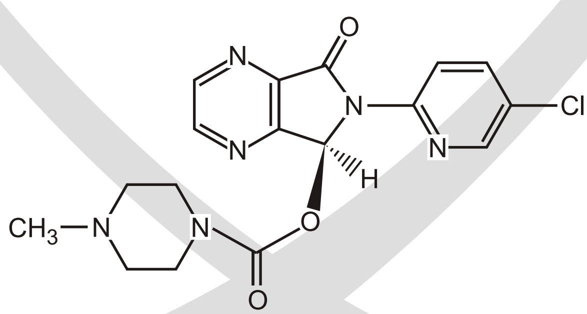
(e) CH_3CHCH $CHCH_3$ and CH_3CH $CHCH_2CH_2$

6. Write the contributing resonance structures and the delocalized hybrid for (a) BCl_3, (b) H_2CN_2 (diazomethane).

7. Circle the conjugated atoms of ciprofloxacin.
One circle per atom



8. How many conjugated atoms (marked with)



9. Consider structural formulas A, B, C and D :

(A) (B) (C) (D)

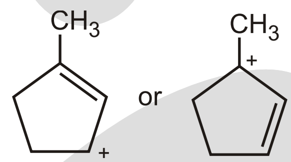
(a) Which structures contain a positively charged carbon?

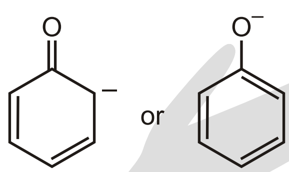
(b) Which structures contain a positively charged nitrogen?

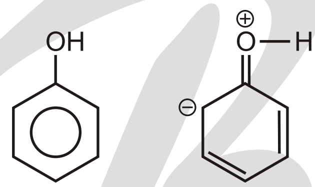
(c) Which structures contain a positively charged oxygen?
(d) Which structures contain a negatively charged carbon?
(e) Which structures contain a negatively charged nitrogen?
(f) Which structures contain a negatively charged oxygen?
(g) Which structures are electrically neutral (contain equal numbers of positive and negative charges)? Are any of them cations? Anions?
(h) Which structure is the most stable?
(i) Which structure is the least stable?

10. Which resonance contributor makes the greater contribution to the resonance hybrid?

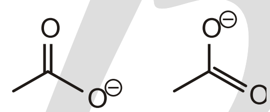
(a) $CH_3CHCH \quad CH_2$ or $CH_3CH \quad CHCH_2$

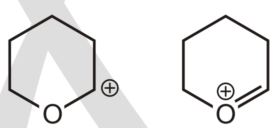
(b) 

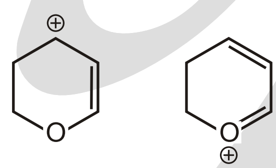
(c) 

(d) 

(e) $O{=}CH{-}\overset{\ominus}{C}H_2 \longleftrightarrow \overset{\ominus}{O}{-}CH{=}CH_2$

(f) 

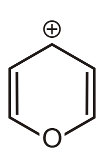
(g) 

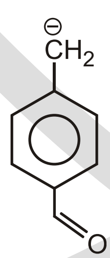
(h) 

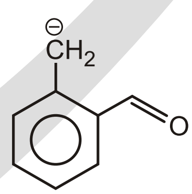
(i) $CH_3{-}NH{-}\overset{\oplus}{C}H_2 \longleftrightarrow CH_3{-}\overset{\oplus}{N}H{=}CH_2$

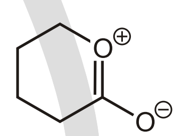
(j) 

11. Draw most stable resonating structure :

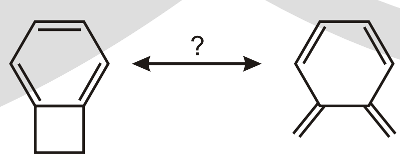
(a) 

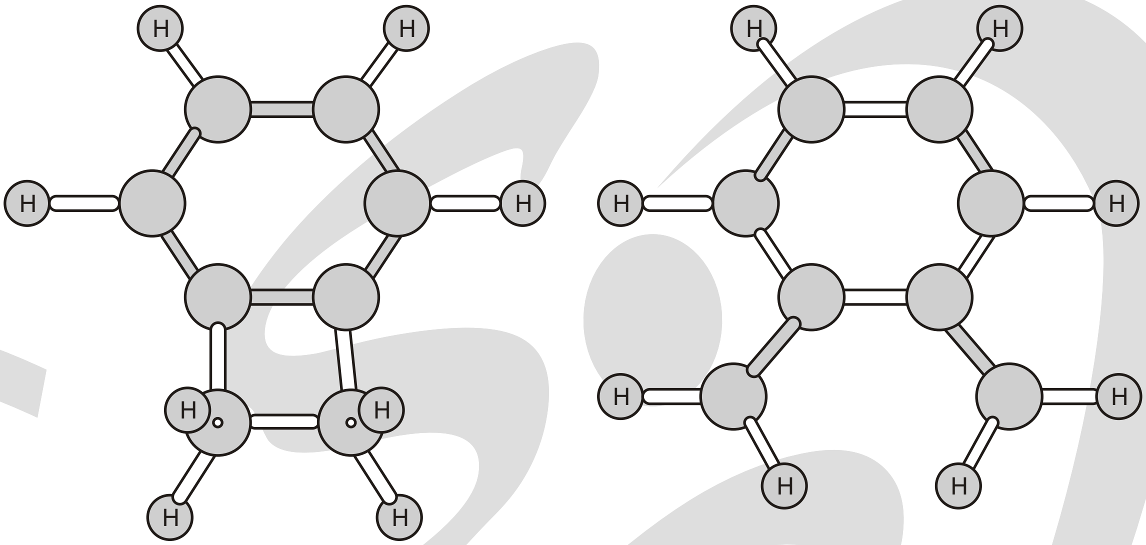
(b) 

(c) 

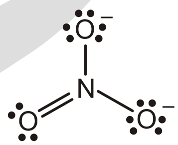
(d) 

12. Are the following two structures resonance forms of one another?



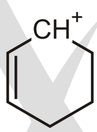


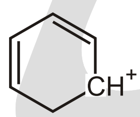
13. The following is one way of writing the structure of the nitrate ion. Draw others.

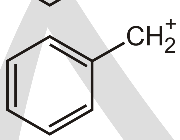


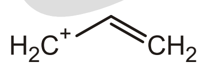
WORK SHEET-1

1. Draw Resonance hybrid of following?

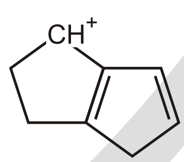
1. 

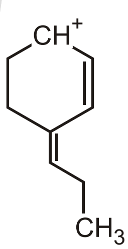
2. 

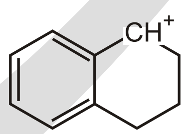
3. 

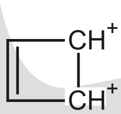
4. 

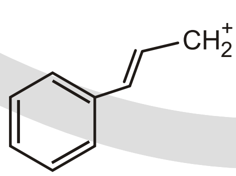
5. 

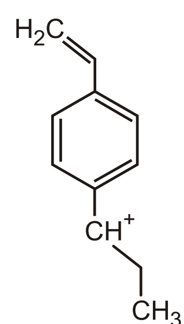
6. 

7. 

8. 

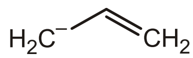
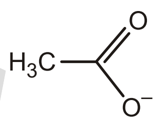
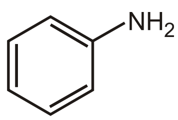
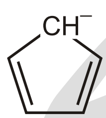
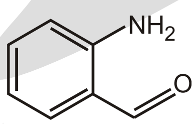
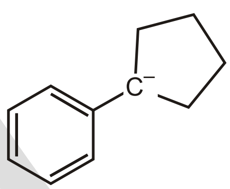
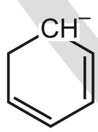
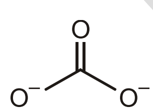
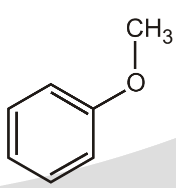
9. 

10. 

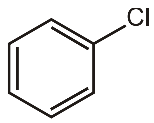
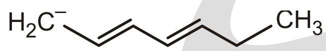
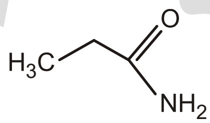
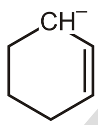
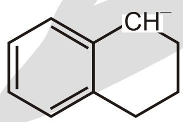
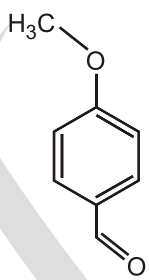
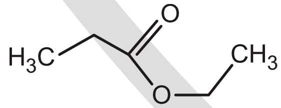
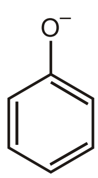
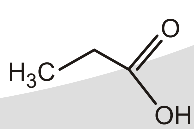
11. 

WORK SHEET-2

1. Draw the Resonating structure and Resonance Hybrid of following species?

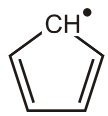
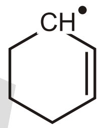
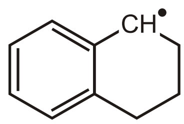
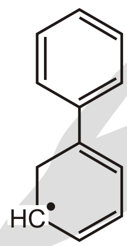
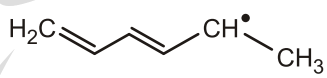
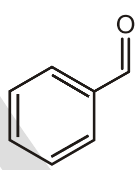
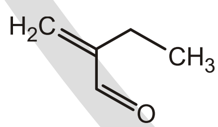
S.No.	Compound	Resonating Structure
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		

WORK SHEET-3

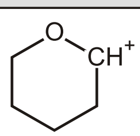
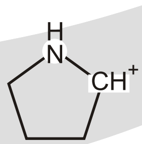
S.No.	Compound	Resonating Structure
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		
9.		

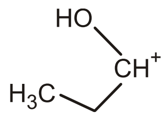
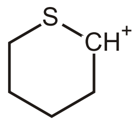
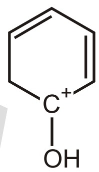
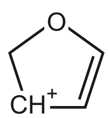
WORK SHEET-4

1. Draw the Resonating structure and Resonance Hybrid of following species?

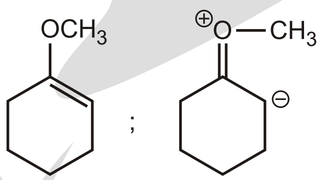
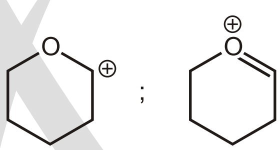
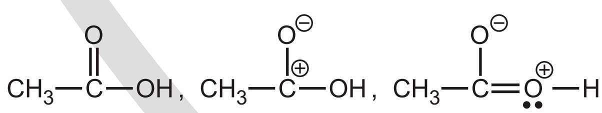
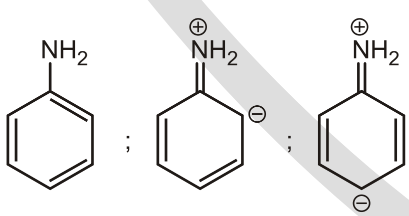
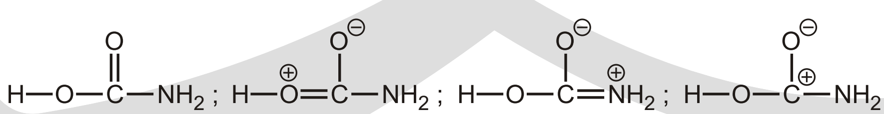
S.No.	Compound	Resonating Structure
1.		
2.		
3.		
4.		
5.		
6.		
7.		
8.		

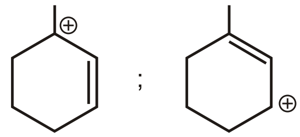
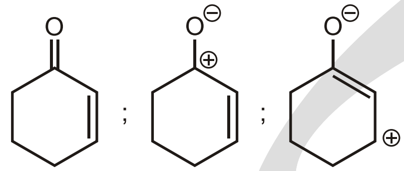
WORK SHEET-5

S.No.	Compound	Resonating Structure
1.		
2.		

3.		
4.		
5.		
6.		

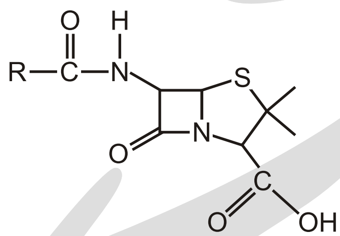
WORK SHEET-6

S.No.	Resonting Structure	Relative stability
1.		
2.		
3.		
4.		
5.	$O{=}C{=}O$; $^{\oplus}O{\equiv}C{-}O^{\ominus}$; $O{=}\overset{\oplus}{C}{-}O^{\ominus}$; $O^{\ominus}{-}\overset{+2}{C}{-}O^{\ominus}$	
6.		

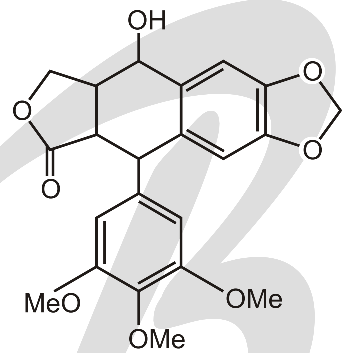
7.		
8.		

SUBJECTIVE TYPE QUESTIONS

1. How extensive are the conjugated systems in these molecules?



a β-lactam Autibiotic



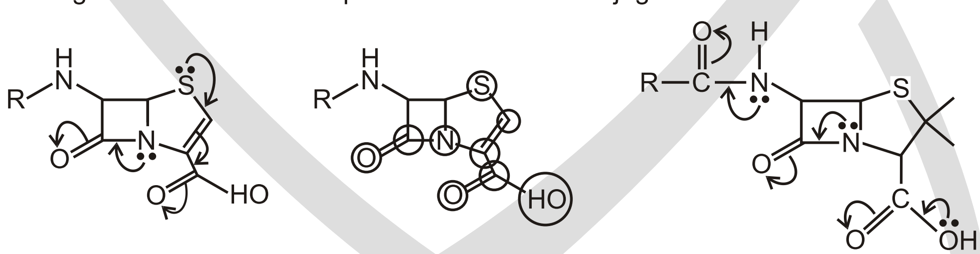
the anticancer compound podophyllotoxin

Purpose of the problem

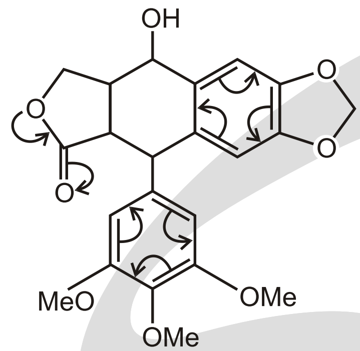
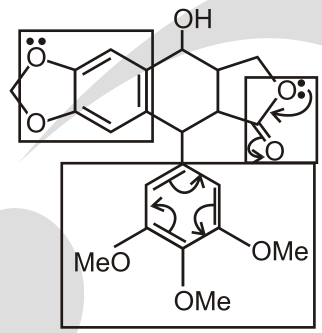
A chance to develop more deeply into what is meant by conjugation.

Suggested solution

The -lactam has two clearly defined conjugating systems : the amide and the more extended unsaturated acid going right through to the sulfur atom. These are shown by curly arrows on the first diagram. These systems are joined by a single bonds so they really are one system : all the *p* orbitals on the ringed atoms in the second diagram are more or less parallel and all are conjugated.



Podophyllotoxin has the obvious two benzene rings and cyclic ester conjugation, shown by curly arrows on the first diagram. Each benzene ring has substituents with lone pairs shown on the second diagram so this molecule has no less than six lone pairs of electrons involved in extended conjugation. There are three separate conjugated systems shown in boxes on the second diagram.

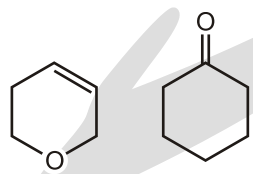



Answers

Single Choice Questions

1. (C) **2.** (A) **3.** (D) **4.** (D) **5.** (A) **6.** (E) **7.** (A) **8.** (A)

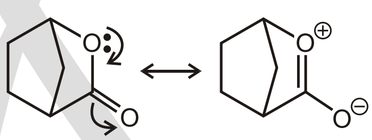
9. (A) **10.** (D)

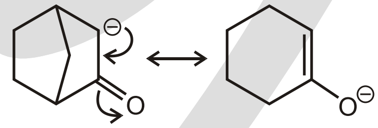
Sol. (D) 

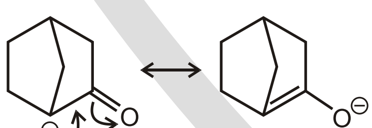
Not R.S.
(position of atom is changed)

11. (C)

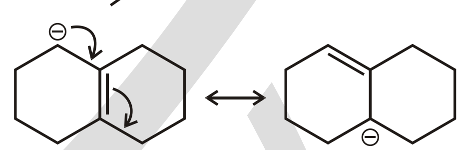
Ans. (C)

Sol. (A) 

(B) 

(C) 

Bredt's rule
violate

(D) 

12.

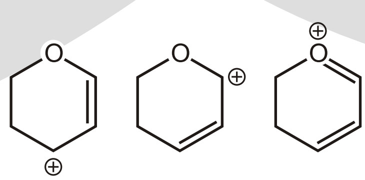
Ans. (C)

Sol. Strategy Two most dominating factors

(1) Number of bonds more Stability of R.S. more

(2) Atoms having complete octet Stability of R.S. more

Factors R.S.



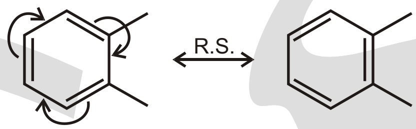
1.	No. of bonds (visible)	7	7	8
2.	Atoms having octet			✓
3.	Positive charge on highly e.n. atom			✓

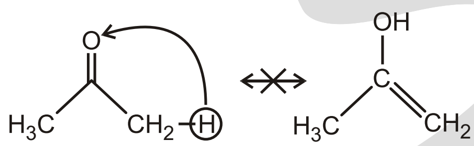
Stability order of R.S. III > I = II

13. (C)

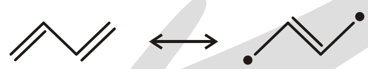
14. (B)

15. (B)

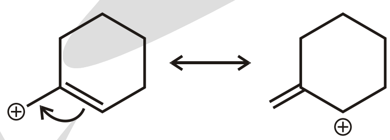
Sol. (I) 

(II) 

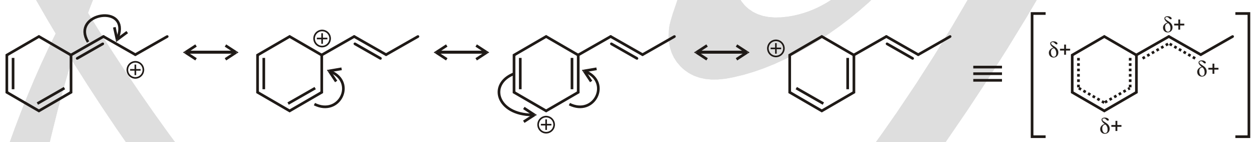
H-atom migrating

(III)  Number of paired-unpaired e s are not same

2 unpaired electrons

(IV) 

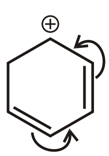
16. (C)

Sol. 

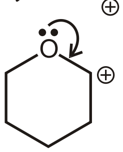
(I) (II) (III) (IV)

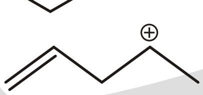
4-Resonance structures

17. (D)

Sol. (A)  Carbocation is resonance stabilised

(B)  Resonance stabilised

(C)  Resonance stabilised

(D)  Not resonance stabilised due to not conjugated with double bond.

18. (C)

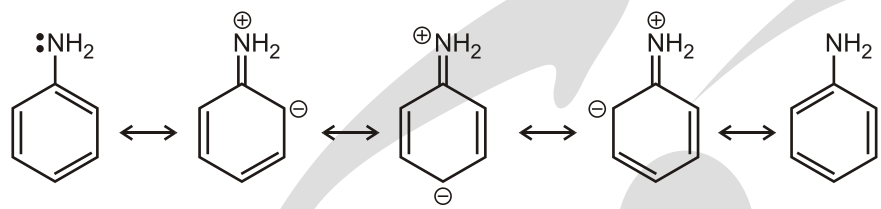
Ans. 9

Sol. Compounds 5 is not resonance stabilised.

19. (C)

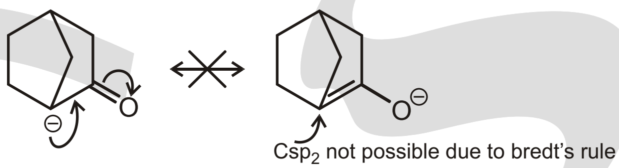
Ans. 7

Sol. X =



20. (C)

Sol.



Thus, this compound is not resonance stabilised.

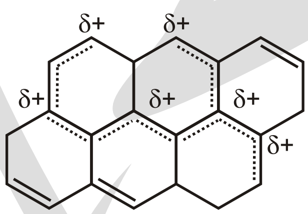
21. (C)

Sol.  not conjugated

22. (B)

Ans. 6

Sol.



23. (B)

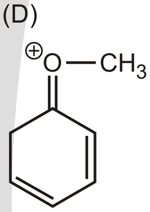
Ans. 6

Sol. (a, c, e, g, h, i) = 6

24. (C)	**25.** (B)	**26.** (B)	**27.** (C)	**28.** (A)	**29.** (B)	**30.** (C)	**31.** (A)
32. (B)	**33.** (C)	**34.** (D)	**35.** (A)				

36. (D)

Sol.



Maximum -bonds, thus, more stable R.S.

UNSOLVED EXAMPLES

1.

Ans. (a) only

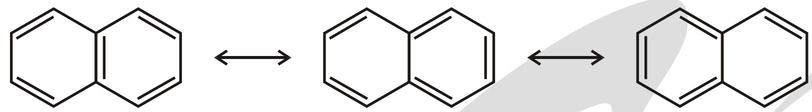
2.

Ans. Because of unstability due to delocalisation.

3.

Ans. (a) and (d)

4.

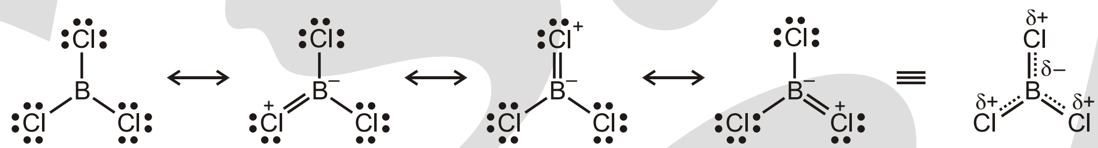
Ans. 

5.

Ans. (a) Different (b) Resonance (c) Different (d) Resonance (e) Different

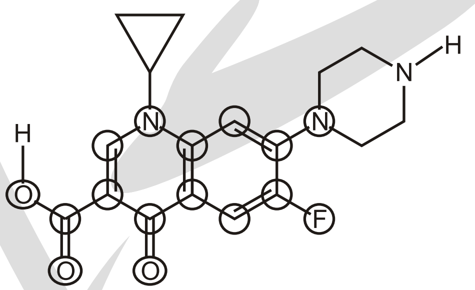
6.

Ans. (a) Boron has six electrons in its outer shell in BCl_3 and can accommodate eithe electrons by having a B–Cl bond assume some double-bond character.



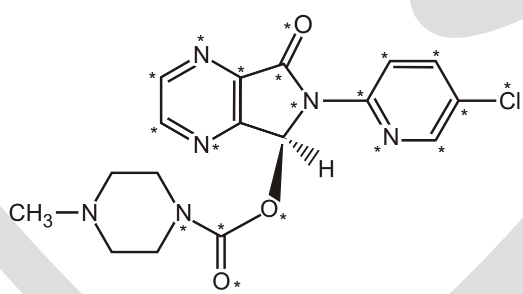
(b) $H_2C^{\ominus}-\overset{\oplus}{N}\equiv N \longleftrightarrow H_2C^{\ominus}-\ddot{N}=\overset{\oplus}{N} \longleftrightarrow H_2C=N^{\oplus}=N^{\ominus}$

7.

Ans. 

8.

Sol. 20 atoms are conjugated.



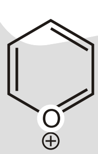
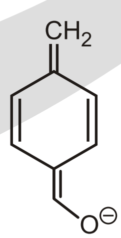
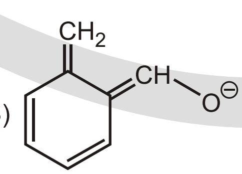
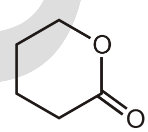
9.

Ans. (a - D), (b - A, B), (c-None), (d-A), (e - None), (f-B, D), (g-A, B, D), (h-B), (i-C),

10.

Ans. (a - i), (b-ii), (c-ii), (d-i), (e-ii), (f - same), (g-ii), (h-ii), (i-ii), (j-i)

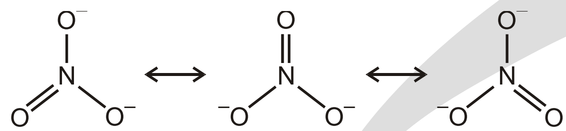
11.

Ans. (1)  (2)  (3)  (4) 

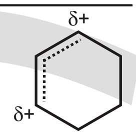
12.

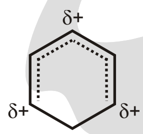
Ans. No they are different structures

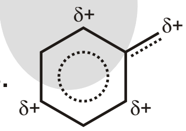
13.

Ans.  (3 equivalent resonating structures)

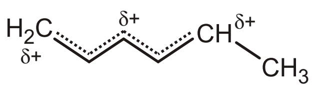
WORK SHEET-1

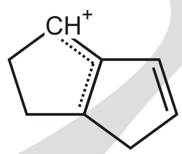
1. 

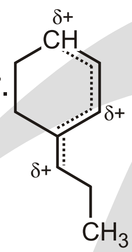
2. 

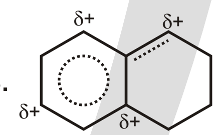
3. 

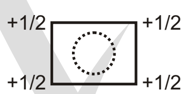
4. 

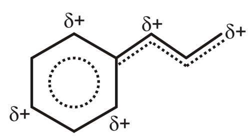
5. 

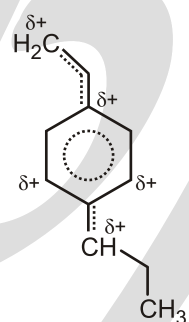
6. 

7. 

8. 

9. 

10. 

11. 

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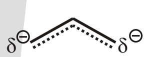
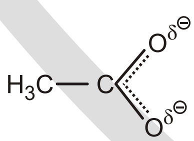
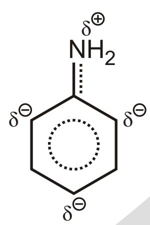
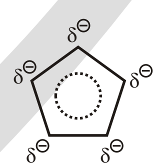
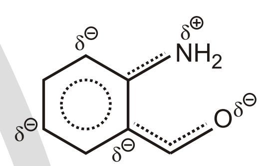
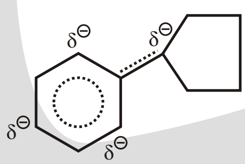
1. 

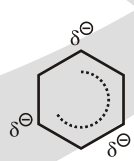
2. 

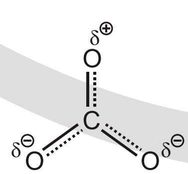
3. 

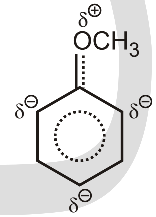
4. 

5. 

6. 

7. 

8. 

9. 

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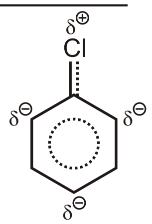
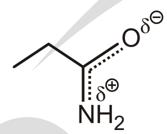
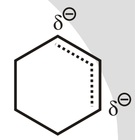
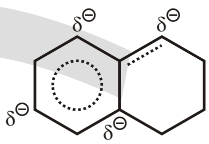
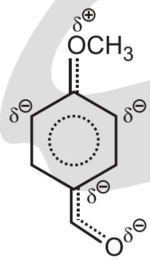
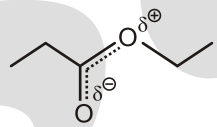
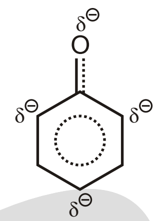
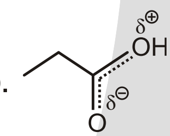
1.  **2.**  **3.**  **4.** 

5.  **6.**  **7.**  **8.**  **9.** 

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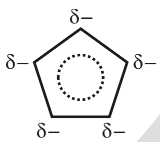
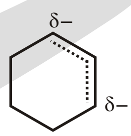
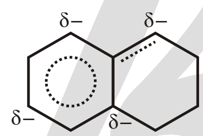
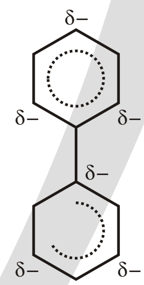
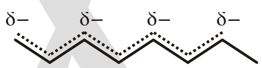
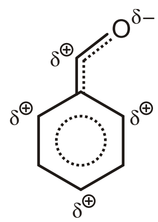
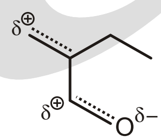
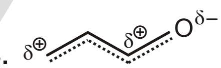
1.  **2.**  **3.**  **4.** 

5.  **6.**  **7.**  **8.** 

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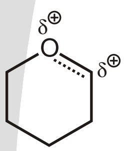
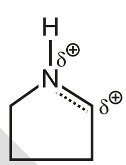
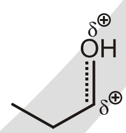
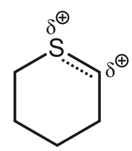
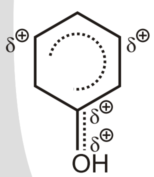
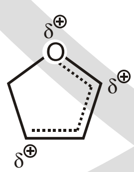
1.  **2.**  **3.**  **4.** 

5.  **6.** 

WORK SHEET-6

1. I II **2.** II I **3.** I III II **4.** I II III

5. I II III IV **6.** I III II IV **7.** I II **8.** I II III

CHAPTER

15

Mesomeric Effect

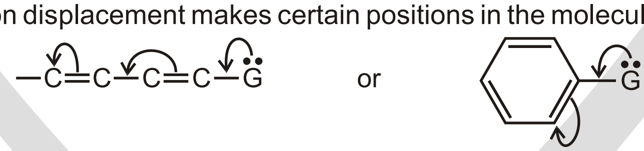
MESOMERIC OR RESONANCE EFFECT

The resonance effect is defined as "the polarity produced in the molecule by the interaction of two- -bonds or between a -bond and lone pair of electrons present on an adjacent atom". The effect is transmitted through the chain. There are two types of resonance or mesomeric effect designated as R or M effect.

It can be two types

(i) Positive Resonance Effect (+R or +M effect)

In this effect, the transfer of electrons is away from an atom or substituent group attached to the conjugated system. This electron displacement makes certain positions in the molecule of high electron densities.



When flow of *e* pair (movement) starts from the group (G). It takes place when G has a lone pair or an extra electron (in ion).

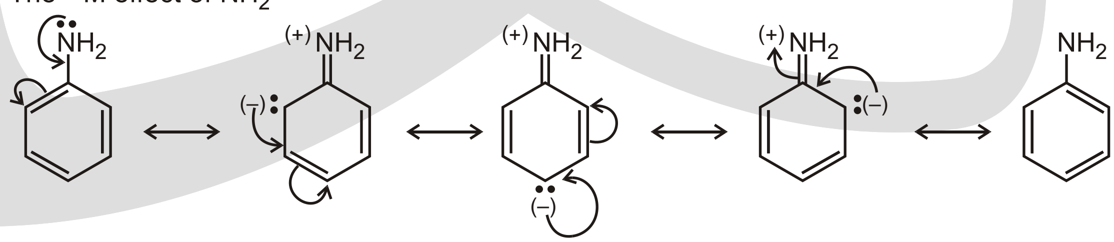
e.g. $-NH_2$, $-OH$, $-Cl$, $-OR$ etc.

★ **+M groups** generally contain a lone pair of electrons or a -bond (s) :

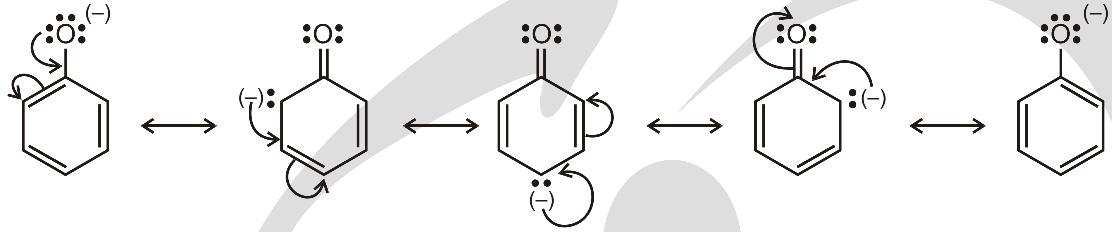
$\ddot{\text{Cl}}$, $\ddot{\text{Br}}$, $\ddot{\text{O}}\text{H}$, $\ddot{\text{O}}\text{R}$, $\ddot{\text{S}}\text{H}$, $\ddot{\text{S}}\text{R}$, $\ddot{\text{N}}\text{H}_2$, $\ddot{\text{N}}\text{HR}$, $\ddot{\text{N}}\text{R}_2$, aromatics, alkenes.

★ Aromatic (or aryl) groups and alkenes can be both +M and –M.

★ The +M effect of $\ddot{\text{N}}\text{H}_2$

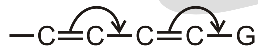


- ★ The lone pair of nitrogen in aniline gets delocalised, and the basic nature of aniline is less than that of ammonia. Similarly, the phenoxide ion gets stabilised by resonance.

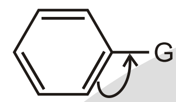


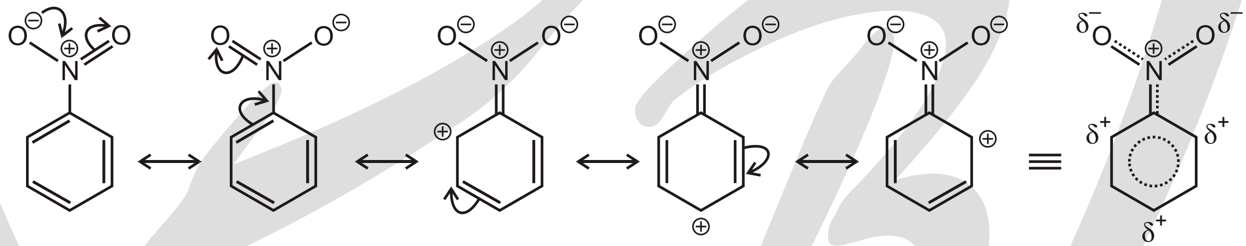
(ii) Negative Resonance Effect (– M or – R effect)

This effect is observed when the transfer of electrons is towards the atom or substituent group attached to the conjugated system. For example in nitrobenzene this electron displacement can be depicted as :



or





- ★ When *e*   pair movement takes place towards G from the molecule.
  e.g., $—NO_2$, —C   O, —C   N, $—SO_3H$ etc.

☑ **SPECIAL TOPIC**

DIFFERENCE BETWEEN MESOMERIC AND RESONANCE EFFECT

Strictly speaking, the term resonance effect (R) is not the same as the mesomeric effect (M). The mesomeric effect is a permanent polarisation, and the mechanism of electron transfer is the same as that in the electromeric effect, *i.e.,* the mesomeric effect is a permanent displacement of electron pairs which occurs in a system of the type Z — C   C; *e.g.,* Z R_2N, Cl :

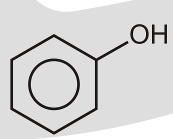
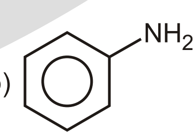
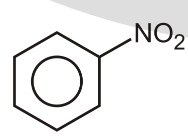
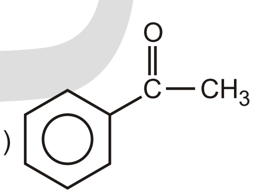
Thus the essential requirement for mesomerism is the presence of multiple bond in the molecule. On the other hand, the resonance effect embraces all permanent electron displacement in the molecule in the ground state. *e.g.,* the hydrogen chloride molecule is a resonance hybrid of two resonating structures :

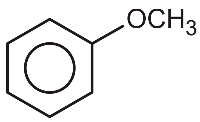
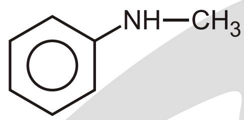
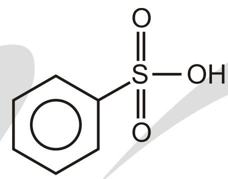
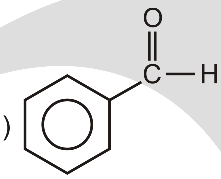
H — Cl   H Cl

Since there is no multiple bond in this molecule, the mesomeric effect is not possible.

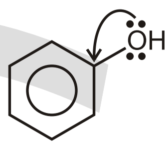
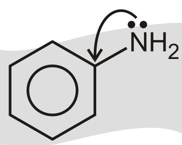
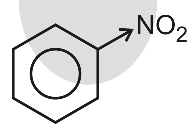
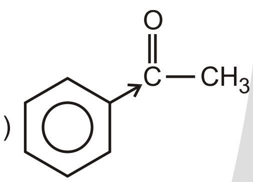
Solved Example

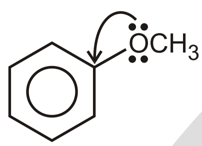
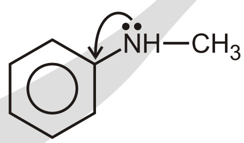
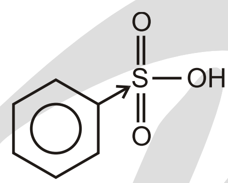
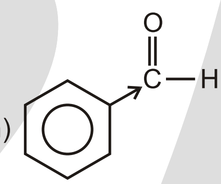
- *x*   number of (+M) group attached with phenyl ring, so the value of *x* is.

(a) 
(b) 
(c) 
(d) 

(e)  (f)  (g)  (h) 

Ans. 4

Sol. (a)  +M (b)  +M (c)  –M (d)  –M

(e)  +M (f)  +M (g)  –M (h)  –M

CHAPTER 16

Hyperconjugation

HYPERCONJUGATION

Hyperconjugation involves delocalisation of electrons through overlapping of '*p*' orbitals of a double bond and orbital of the adjacent single bond (conjugation). The structures are called ***hyperconjugative structures***. Since there is no bond between C and H, it is also called ***no bond resonance***. The free proton here is quite firmly bound to the electron cloud and is not free to move.

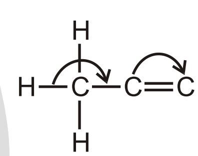
It is also evident that hyperconjugation occurs through H — atoms present on the carbon next to the double bond, that is, -hydrogen atoms. Naturally, the larger the number of such hydrogen, the more the hyperconjugative structures and greater the hyperconjugation effect. Thus, the order of this effect is

$$CH_3— \quad CH_3—CH_2— \quad (CH_3)_2CH— \quad (CH_3)_3C—$$

When (C — H) sigma electrons are in conjugation to pi bond, this conjugation is known as (C — H), conjugation, No bond resonance or -bond resonance or hyperconjugatioon.

BAKER-NATHAN EFFECT (1935)

As we have seen, the general inductive effect of alkyl groups is Me_3C Me_2CH $MeCH_2$ Me. This inductive order has been used satisfactorily to explain various physical data, etc. In some reactions, however, the inductive order is reversed, the -electrons of the H— C bond become less localised by entering into partial conjugation with the attached unsaturated system. *i.e.,* , -conjugation :



Structural requirement for hyperconjugation :

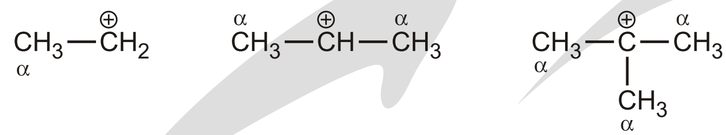
(A) Compound should have at least one sp^2-hybrid carbon of either alkene, alkyl carbocation or alkyl free radical.

(B) -carbon with respect to sp^2 hybrid carbon should have at least one hydrogen.

If both these conditions are fulfilled then hyperconjugation will take place in the molecule.

Hyperconjugation is of three types :

(i) (C — H), positive charge conjugation : This type of conjugation occurs in alkyl carbocation.



Sigma Bonding + Empty *p*-orbital

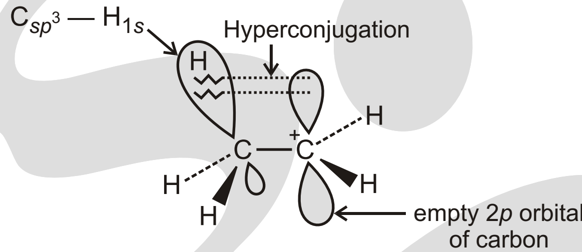
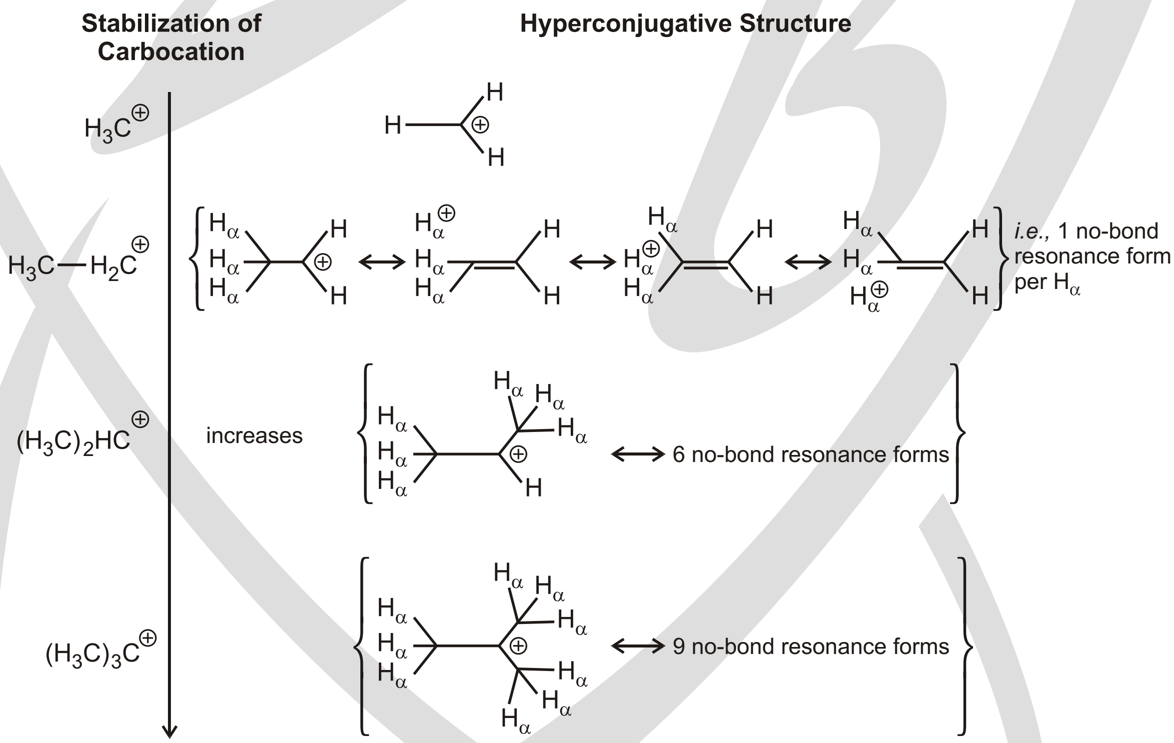


Figure : Orbital diagram showing hyper conjugation in ethyl cation

Table : Stabilization of Trivalent Carbenium Ion Centers by Methyl Substituents : Experimental Findings and Their Explanation by Means of no-bond Resonance Theory



(ii) (C — H), odd electron conjugation : This type of conjugation occurs in alkyl free radicals -

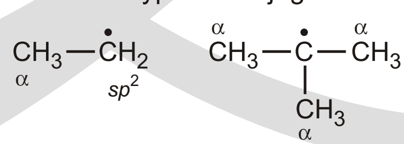
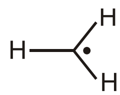
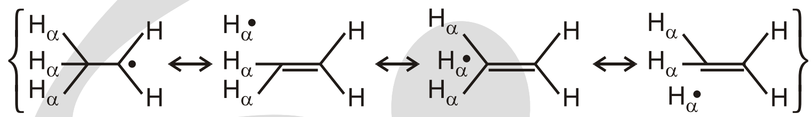
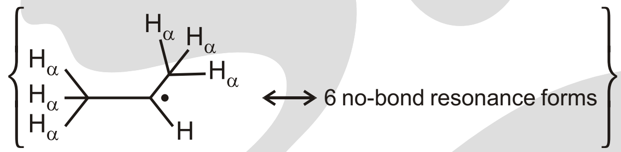
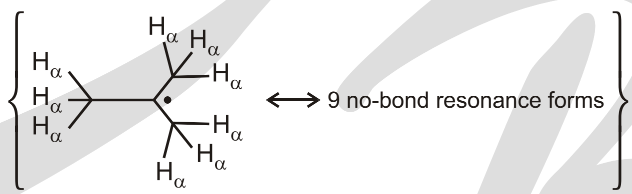


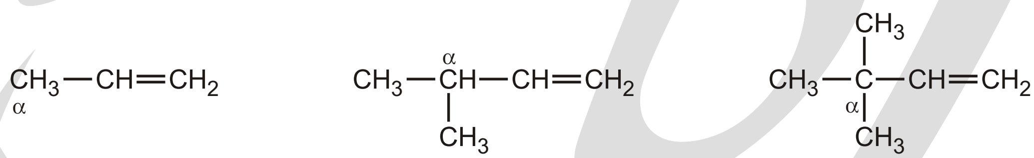
Figure : Some substrantes and products of radical substitution reactions

A primary radical is 6 kcal/mol more stable, a secondary radical is 9 kcal/mol more stable, and a tertiary radical is 12 kcal/mol more stable than the methyl radical.

Table : Stabilization of Radicals by Alkyl Substituents

	DE kcal/mol	Hyperconjugative structure formulation of the radical R
H_3C-H	104	
H_3C-H_2C-H	98	 *i.e.,* 1 no-bond resonance form per H
$(H_3C)_2HC-H$	95	
$(H_3C)_3C-H$	92	

(iii) (C — H), conjugation : This type of conjugation occurs in alkenes.



Resonating structures due to hyperconjugation may be written involving "no bond" between the alpha carbon and hydrogen atoms.

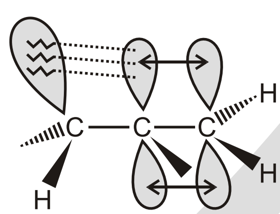
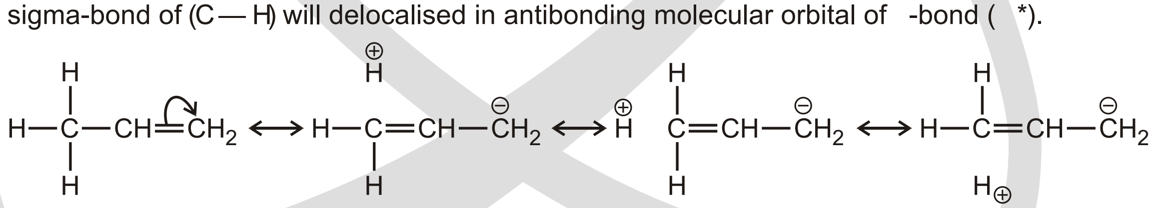


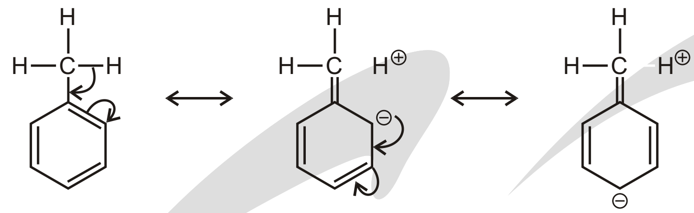
Figure : Orbital diagram showing hyper conjugation in propene

sigma-bond of (C — H) will delocalised in antibonding molecular orbital of -bond (*).



In the above resonating structures there is no covalent bond between carbon and hydrogen. From this point of view, hyperconjugation may be regarded as **"no bond resonance"**.

(C) Electron releasing (or donating) power of R in alkyl benzene : CH_3 — (or alkyl group) is +H group, ortho-para directing group and activating group for electrophilic aromatic substitution reaction because of the hyperconjugation.



★ The electron donating power of alkyl group will depends on the number of resonating structures, this depends on the number of hydrogens present on -carbon. The electron releasing power of some groups are as follows -

$$CH_3 > CH_3-CH_2 > \begin{matrix} CH_3 \\ CH_3 \end{matrix}\!\!> CH > CH_3-\underset{CH_3}{\overset{CH_3}{C}}-$$

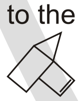
$$\longrightarrow$$

Electron donating power in decreasing order due to the hyperconjugation

★ Thus there is conjugation between electrons of single and those of multiple bonds. This type of conjugation is known as hyperconjugation, and is a permanent effect (this name was given by Mulliken, 1941).

BREDT'S RULE

"Bredt's Rule is an empirical observation in organic chemistry that states that a double bond cannot be placed at the bridgehead of a bridged ring system, unless the rings are large enough. The rule in named after Julius Bredt." The German chemist J. Bredt proposed in 1935 that bicycloalkenes such as 1-norbormene, which have a double bond to the bridgehead carbon, are too strained to exist. (Making a molecular model will be helpful.)

e.g.,  No hyperconjugate due to Bredt's Rule.

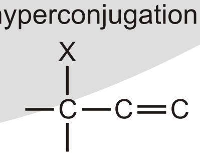
(A) Carbon-carbon double bond length in Alkenes : As we know that the more is the number of resonating structures, the more will be single bond character in carbon-carbon double bond. Thus, bond length between carbon-carbon double bond number of resonating structures.

Examples are :

Structures	Number of -hydrogens zero	Number of hyperconjugation structures	Carbon-carbon double bond length in Å
$CH_2 \quad CH_2$	zero	zero	1.34 Å
$CH_3-CH \quad CH_2$	3	3	1.39 Å
$CH_3-CH_2-CH \quad CH_2$	2	2	1.37 Å
$CH_3-\underset{CH_3}{CH}-CH=CH_2$	1	1	1.35 Å
$CH_3-\underset{CH_3}{\overset{CH_3}{C}}-CH=CH_2$	zero	zero	1.34 Å

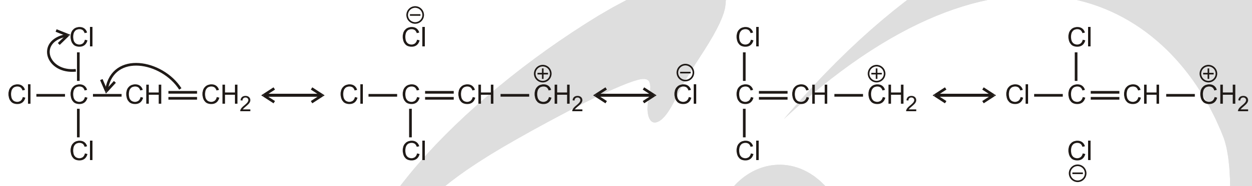
REVERSE HYPERCONJUGATION

The phenomenon of hyperconjugation is also observed in the system given below :

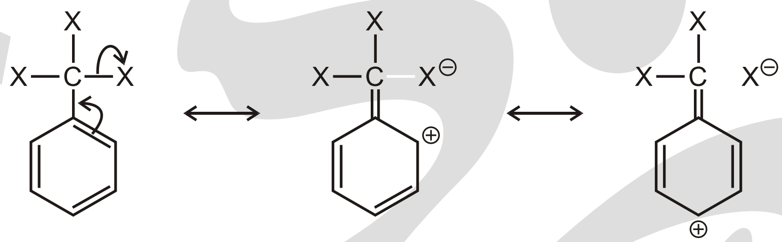


where X - halogen

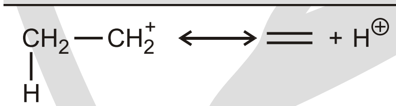
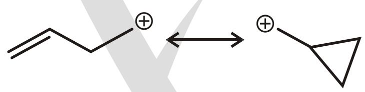
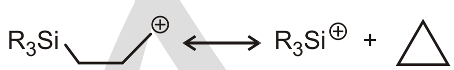
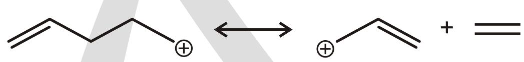
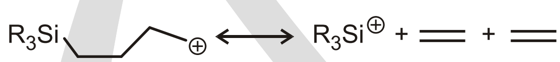
In such system the effect operates in the reverse direction. Hence the hyperconjugation in such system is known as reverse hyperconjugation.



The meta directing influence and dectivating effect of CX_3 group for electrophilic aromatic substitution reaction can be explained by this effect.



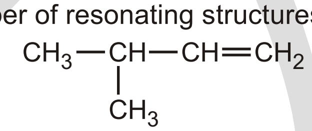
Types of hyperconjugation :

Valence Structures	Abbreviation	Name
		Sacrificial hyperconjugation
		Hyperconjugation
		Homoconjugation
		Homohyperconjugation
	/	Hyperconjugation/conjugation
	/	Double hyperconjugation

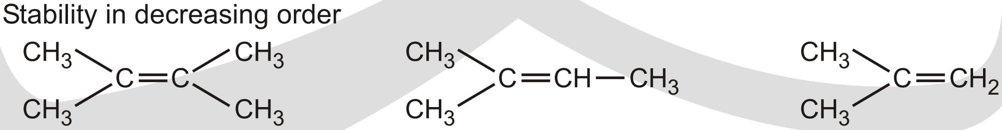
APPLICATION OF HYPERCONJUGATION

(A) Stability of Alkenes : Hyperconjugation explains the stability of certain alkenes over other alkenes :

(i) Stability of alkenes   Number of alpha hydrogens   Number of resonating structures

$CH_3-CH=CH_2$ $CH_3-CH_2-CH=CH_2$ 

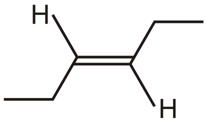
Stability in decreasing order



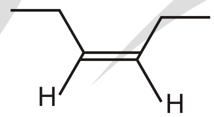
Number of alpha hydrogens in decreasing order stability of alkenes in decreasing order

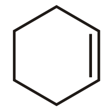
Solved Example

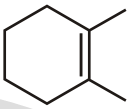
- Count those compounds having odd number of   -hydrogen.

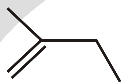
(a) 

(b) 

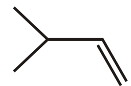
(c) 

(d) 

(e) 

(f) 

(g) 

(h) 

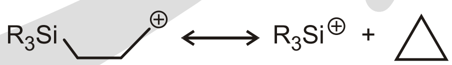
Bridt's rule

Ans. 3

EXERCISE

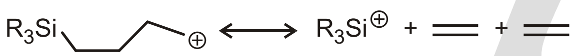
SINGLE CHOICE QUESTIONS

1. What is the type of conjugation for the given reaction?



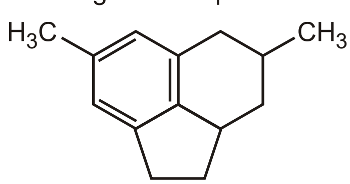
(A) Hyperconjugation
(B) Simple Conjugation
(C) Homoconjugation
(D) Homohyperconjugation

2. What is the type of conjugation for the given reaction?



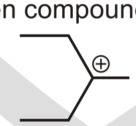
(A) Hyperconjugation
(B) Simple Conjugation
(C) Homoconjugation
(D) Double hyperconjugation

3. Number of Benzylic carbons in the given compound are :



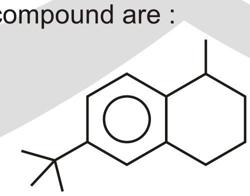
(A) 4 (B) 3 (C) 1 (D) 2

4. Number of   -   conjugation in the given compound is :



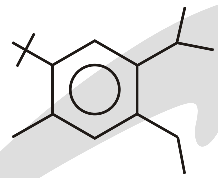
(A) 4 (B) 6 (C) 7 (D) 8

5. Number of   -hydrogen in given compound are :



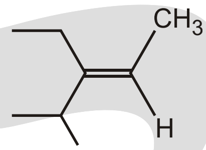
(A) 2 (B) 3 (C) 4 (D) 5

6. Number of -hydrogen in given compound are :



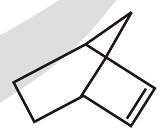
(A) 5 (B) 6 (C) 7 (D) 8

7. Number of -hydrogen in given compound are :



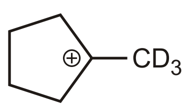
(A) 5 (B) 6 (C) 7 (D) 8

8. Number of hyperconjugating structure in given compound are :

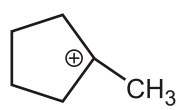


(A) 0 (B) 1 (C) 2 (D) 3

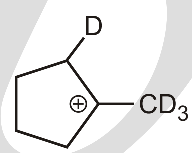
9. Rank the following according to stability (most stable to least stable).



(a)



(b)



(c)

(A) a > b > c (B) a > c > b (C) c > b > a (D) b > a > c

10. The type of delocalisation involving sigma bond orbitals is called :

(A) inductive effect (B) hyperconjugation (C) electromeric effect (D) mesomeric effect

11. The allyl radical has how many bonding -molecular orbitals?

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

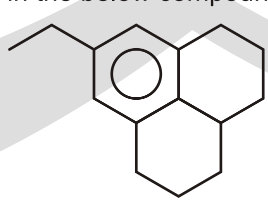
MULTIPLE CHOICE QUESTIONS

1. Which of the following alkene is more stable than 1-butene :

(A) cis-2-butene (B) trans-2-butene (C) Iso-butene (D) ethene

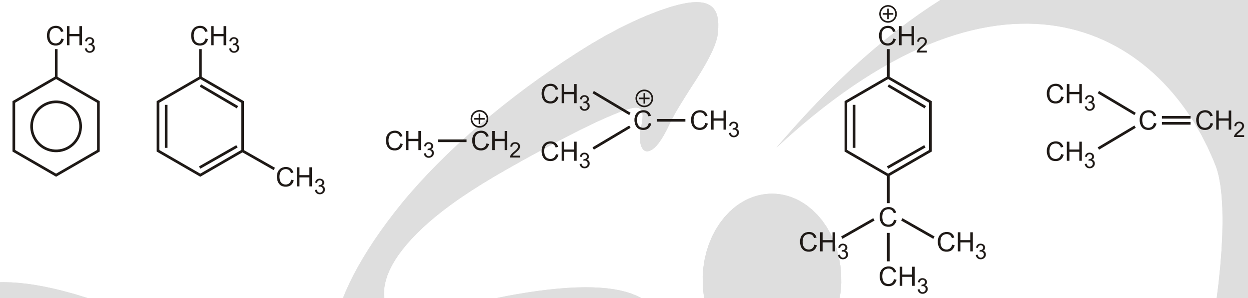
INTEGER TYPE QUESTIONS

1. *X* Total number of -hydrogen in the below compound?

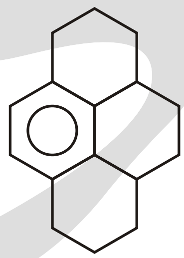


Find value of *X*?

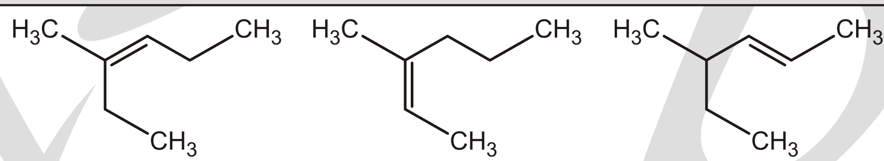
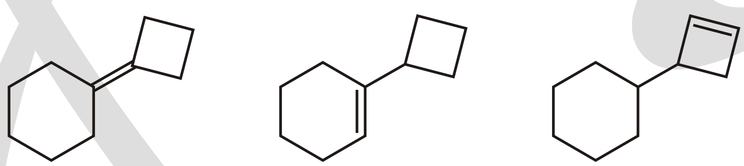
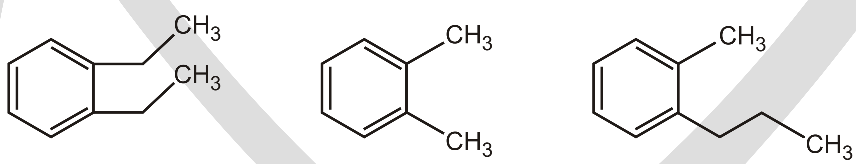
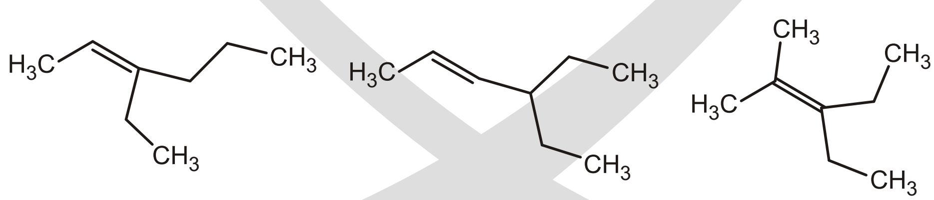
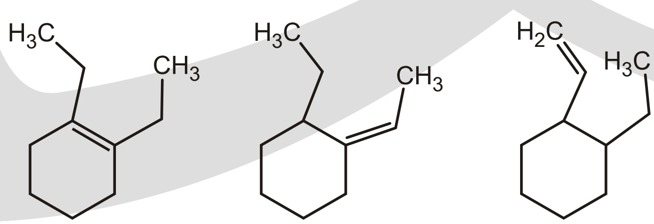
2. How many compounds have -hyperconjugation?

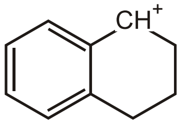
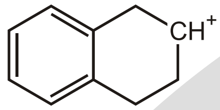
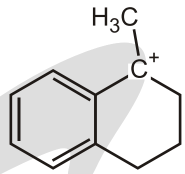
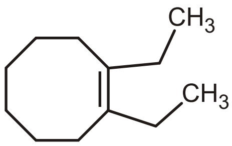
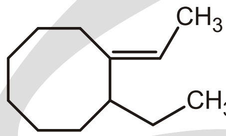
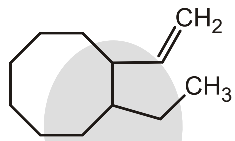
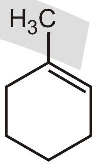
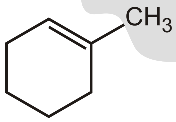
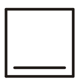


3. Total number of -hydrogen in the given compound is :



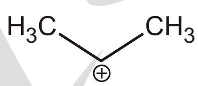
WORK SHEET - 1

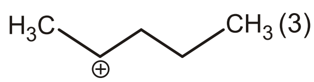
S.No.	Compound	Relative Stability
1.		
2.		
3.		
4.		
5.		

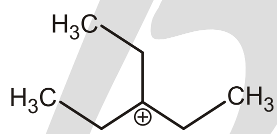
6.	  	
7.	  	
8.	 	
9.	 	

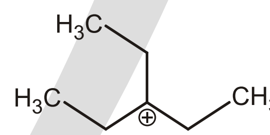
WORK SHEET - 2

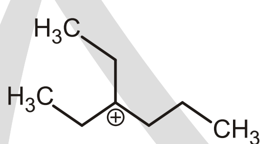
1. Write Number of -hydrogen in given carbocation in bracket :

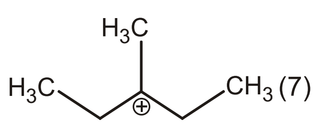
(1) 

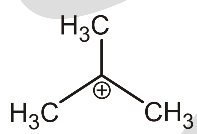
(2) 

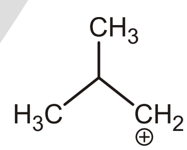
(3) 

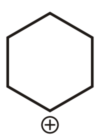
(4) 

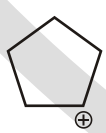
(5) 

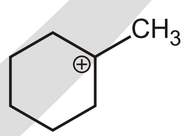
(6) 

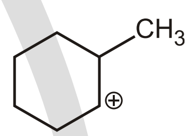
(7) 

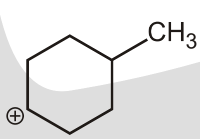
(8) 

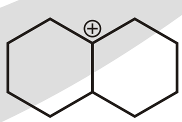
(9) 

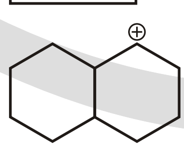
(10) 

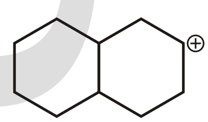
(11) 

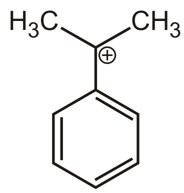
(12) 

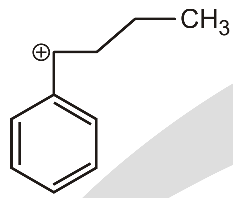
(13) 

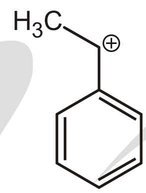
(14) 

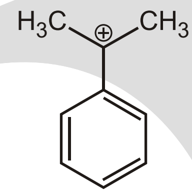
(15) 

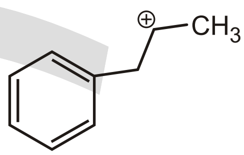
(16) 

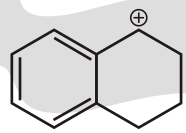
(17) 

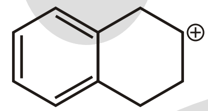
(18) 

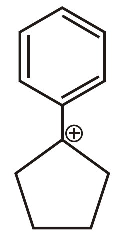
(19) 

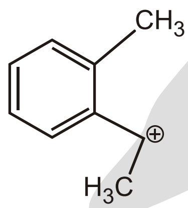
(20) 

(21) 

(22) 

(23) 

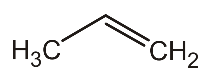
(24) 

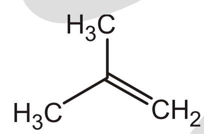
(25) 

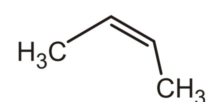
WORK SHEET - 3

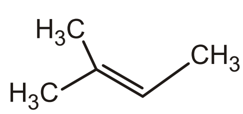
1. Write number of -Hydrogen in given compounds in the bracket :

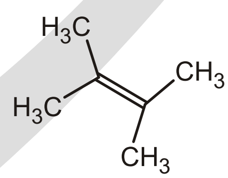
(1) H_2C CH_2

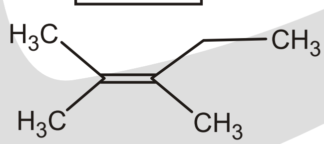
(2) 

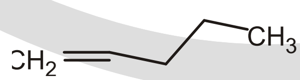
(3) 

(4) 

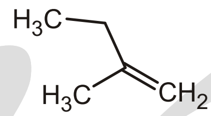
(5) 

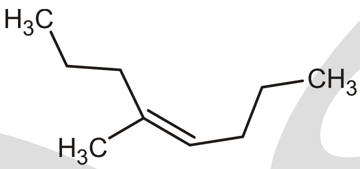
(6) 

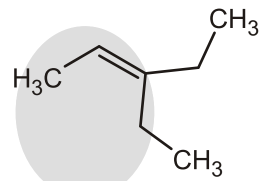
(7) 

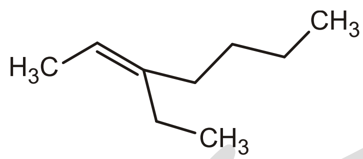
(8) 

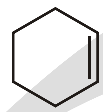
(9) 

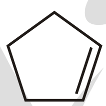
(10) 

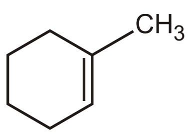
(11) 

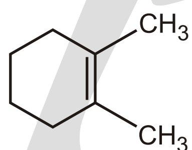
(12) 

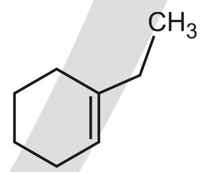
(13) 

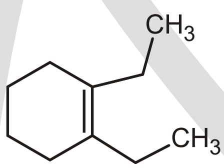
(14) 

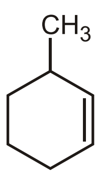
(15) 

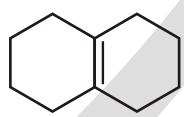
(16) 

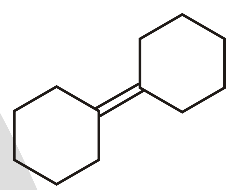
(17) 

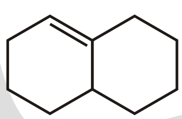
(18) 

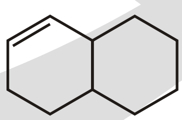
(19) 

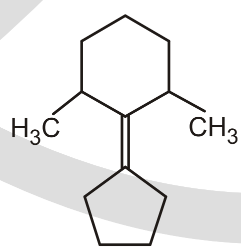
(20) 

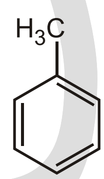
(21) 

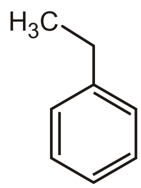
(22) 

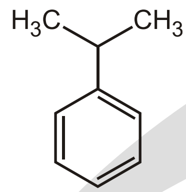
(23) 

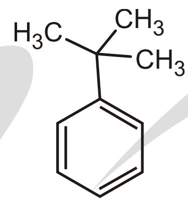
(24) 

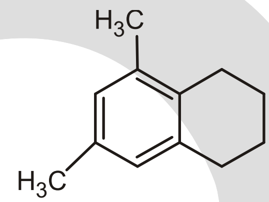
(25) 

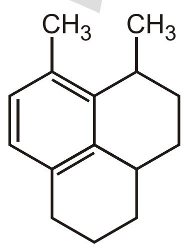
(26) 

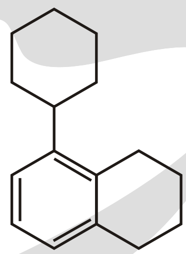
(27) 

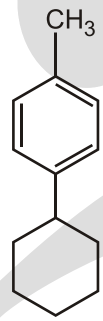
(28) 

(29) 

(30) 

(31) 

(32) 

(33) 

Answers

Single Choice Questions

1. (D)	**2.** (D)	**3.** (A)	**4.** (C)	**5.** (B)	**6.** (B)	**7.** (B)	**8.** (A)
9. (D)	**10.** (B)	**11.** (A)					

Multiple Choice Questions

1. (A, B, C)

Integer Type Questions

1. 7	**2.** 2	**3.** 6

Work Sheet - 1

1. ii > i > iii	**2.** i > ii > iii	**3.** ii > iii > i	**4.** iii > i > ii	**5.** i > ii > iii	**6.** iii > i > ii	**7.** i > ii > iii	**8.** i = ii	**9.** i > ii

Work Sheet - 2

1. 6	**2.** 5	**3.** 6	**4.** 6	**5.** 6	**6.** 7	**7.** 9	**8.** 1
9. 4	**10.** 4	**11.** 7	**12.** 3	**13.** 4	**14.** 5	**15.** 3	**16.** 4
17. 6	**18.** 2	**19.** 3	**20.** 6	**21.** 5	**22.** 2	**23.** 4	**24.** 4
25. 3							

Work Sheet - 3

1. 0	**2.** 3	**3.** 6	**4.** 6	**5.** 9	**6.** 12	**7.** 11	**8.** 2
9. 5	**10.** 5	**11.** 7	**12.** 7	**13.** 7	**14.** 4	**15.** 4	**16.** 7
17. 10	**18.** 6	**19.** 8	**20.** 3	**21.** 8	**22.** 8	**23.** 5	**24.** 3
25. 6	**26.** 3	**27.** 2	**28.** 1	**29.** 0	**30.** 10	**31.** 7	**32.** 5
33. 4							

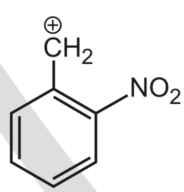
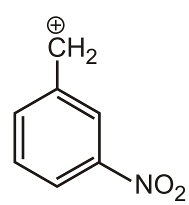
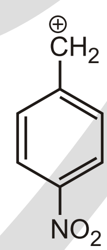
CHAPTER 17

Application of Resonance, Hyperconjugation & Inductive Effect

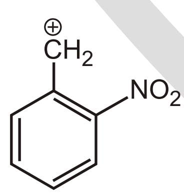
ADDITIONAL PROBLEMS BASED ON RESONANCE, HYPERCONJUGATION AND INDUCTIVE EFFECT

Solved Example

- What is correct order of Stability of given Carbocation :

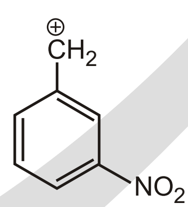




Sol.



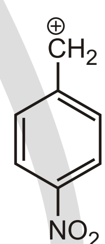
(I)

(i) Increase in the magnitude of positive charge by – I and – R effect
(ii) – I and – R power is maximum



(II)

Increase in positive charge only by – I effect



(III)

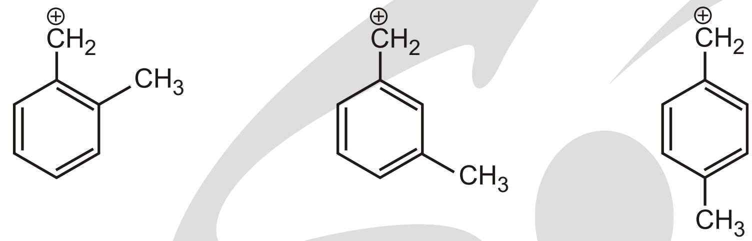
(i) Increase in positive charge by – I – R effect
(ii) – I and – R power is minimum

Hence (II) is more stable than (III) which is more stable than (I).

Thus meta derivative is more stable than p-derivative which is more stable than o-derivative.

Solved Example

- What is correct order of Stability of given Carbocation :



Sol.

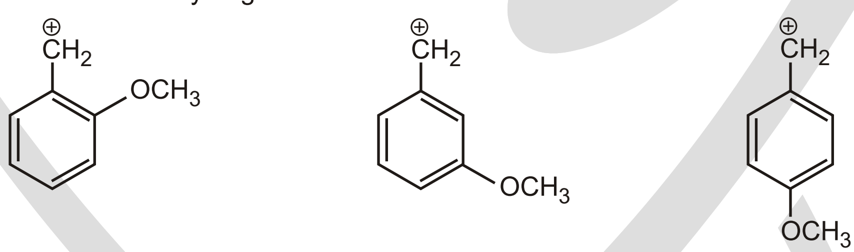
(I)	(II)	(III)
Positive charge is decreased by +I and +H group or stabilised by +I and +H group and +I and +H power is maximum	Stabilised by +I group only	Stabilised by +I and +H effect and +I and +H power is minimum

Hence (I) is more stable than (III) which is more stable than (II).

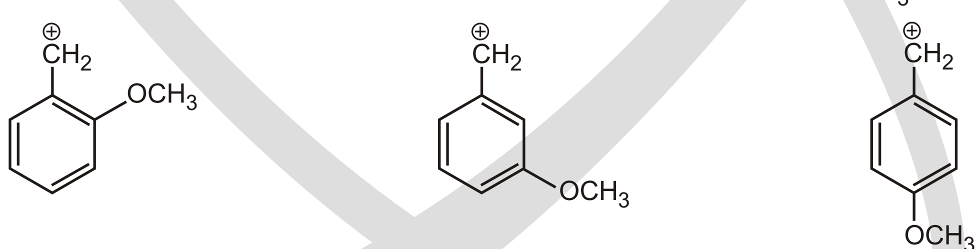
Thus o-derivative is more stable than p-derivative which is more stable than m-derivative.

Solved Example

- What is correct order of Stability of given Carbocation :



Sol.

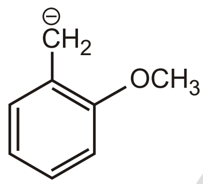
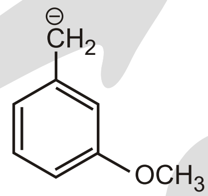
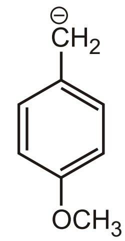


(I)	(II)	(III)
Stabilised by +R effect destabilised by −I effect −I power is maximum (due to distance)	Destabilised by −I effect	Stabilised by +R effect destabilised by −I effect −I power is minimum (due to distance)

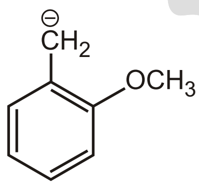
Hence III is more stable than I which is more stable than II.

Solved Example

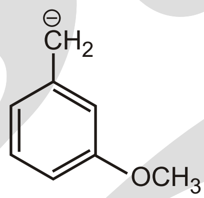
- What is correct order of Stability of given Carbanion :

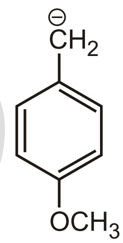
Sol.



(I)
Destabilised by + R effect and
stabilised by – I effect
– I power is maximum



(II)
Stabilised by – I effect



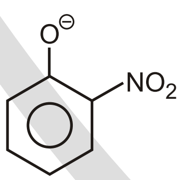
(III)
Destabilised by + R effect
stabilised by – I effect and
– I power is minimum

Thus m-derivative is more stable o-derivative which is more stable than p-derivative.

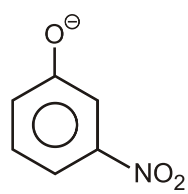
Solved Example

- What is the correct order of acidic strength of orthonitro phenol, metanitro phenol and paranitro phenol.

Sol. **Acidity of Substituted Phenols :** Acidity of substituted phenols depends on the stability of the phenoxide ion because acidity is the function of the stability of acid anion.



(I)
Phenoxide ion is stabilised by
– R and – I effect and
– I power is maximum
– R power is maximum



(II)
Stabilised by
– I effect only

(III)
Stabilised by – R and – I effect
and – I power is minimum
– R power is minimum

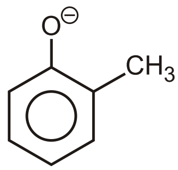
Thus according to stability of anions o-dervative will be more acidic than p-derivative which will be more acidic than m-derivative. But result is as follows in case of nitrophenols p-derivative is more acidic than o-derivative which is more acidic than m-derivative. In o-derivative, there is hydrogen bonding which decreases acidity. Thus order of acidity is as follows :

paranitro phenol > orthonitro phenol > metanitro phenol > phenol

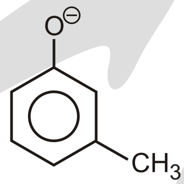
Solved Example

- What is the correct order of acidic strength of orthocresol, metacresol and paracresol.

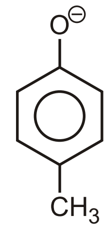
Sol.



(I)
Destabilised by +H and +I effect and +I power is maximum



(II)
Destabilised by +I effect



(III)
Destabilised by +H and +I effect and +I power is minimum

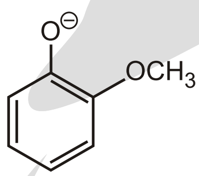
Thus, m-derivative is more acidic than p-derivative which will be more acidic than o-derivative.

$$\xrightarrow[\text{Acidity in decreasing order}]{\text{Phenol > m-derivative > p-derivative > o-derivative}}$$

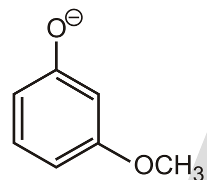
Solved Example

- What is the correct order of acidic strength of ortho, meta and para methoxy phenol.

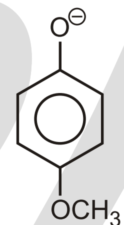
Sol.



(I)
Destabilised by +R effect stabilised by –I effect –I power is maximum



(II)
Stabilised by –I effect



(III)
Destabilised by +R effect stabilised by –I effect and –I power is minimum

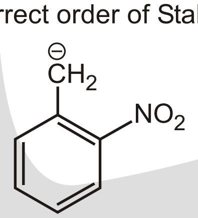
Therefore (II) is more stable than (I) which is more stable than (III).
Thus, m-derivative is more acidic than o-derivative which is more acidic than p-derivative.

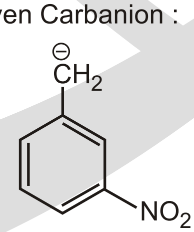
EXERCISE

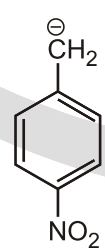
UNSOLVED EXAMPLE

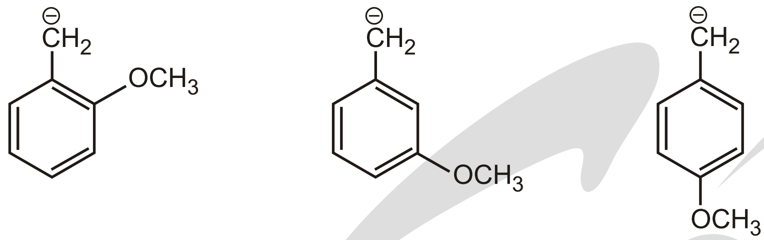
1. Correct order of Stability of given Carbanion :

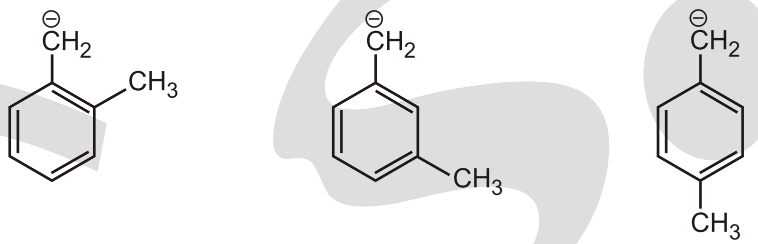
(i)





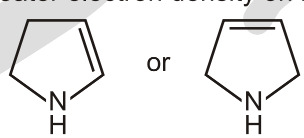


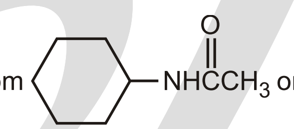
(ii) 

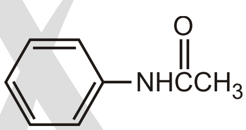
(iii) 

2. (a) Which oxygen atom has the greater electron density $CH_3\overset{\overset{\displaystyle O}{\|}}{C}OCH_3$

(b) Which compound has the greater electron density on its nitrogen atom



(c) Which compound has the greater electron density on its oxygen atom  or



3. In each of the following pairs, which species is more stable?

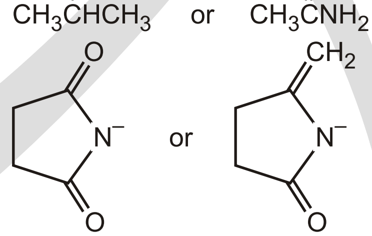
(a) $CH_3CH_2O^-$ or $CH_3\overset{\overset{\displaystyle O}{\|}}{C}O^-$

(b) $CH_3\overset{\overset{\displaystyle O}{\|}}{C}\bar{C}HCH_2\overset{\overset{\displaystyle O}{\|}}{C}H$ or $CH_3\overset{\overset{\displaystyle O}{\|}}{C}\bar{C}H\overset{\overset{\displaystyle O}{\|}}{C}CH_3$

(c) $CH_3\bar{C}HCH_2\overset{\overset{\displaystyle O}{\|}}{C}CH_3$ or $CH_3CH_2\bar{C}H\overset{\overset{\displaystyle O}{\|}}{C}CH_3$

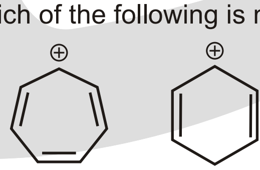
(d) $CH_3\overset{\overset{\displaystyle NH_2}{|}}{C}HCH_3$ or $CH_3\overset{\overset{\displaystyle NH}{\|}}{C}NH_2$

(e) $CH_3\underset{\underset{\displaystyle CH_3}{|}}{\overset{\ominus}{C}}-\overset{\overset{\displaystyle O}{\|}}{C}H$ or $CH_3\underset{\underset{\displaystyle CH_3}{|}}{\bar{C}}-\overset{\overset{\displaystyle CH_2}{\|}}{C}H$

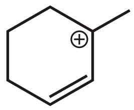
(f) 

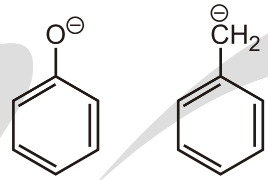
WORK SHEET - 1

1. Which of the following is more stable, Write in the box?

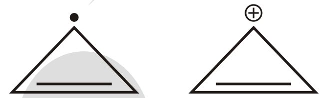
1. 

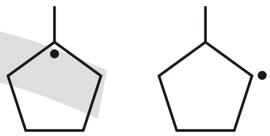
2. 

3. 

4. 

5. $HC\equiv C^{\oplus}$ $H_2C=\overset{\oplus}{C}H$

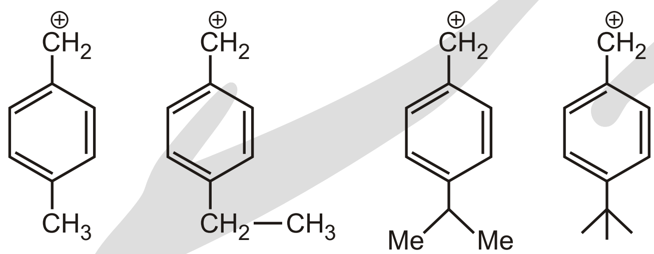
6. 

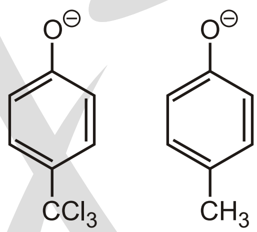
7. 

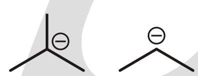
8. $F-\overset{\ominus}{C}H_2$ $Cl-\overset{\ominus}{C}H_2$

9. $Cl-CH_2-\overset{\ominus}{C}H_2$ $F-CH_2-\overset{\ominus}{C}H_2$

10. $\overset{\ominus}{C}H_2-O-CH_3$ $H_2\overset{\ominus}{C}-S-CH_3$

11. 

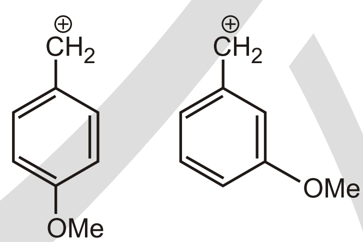
12. 

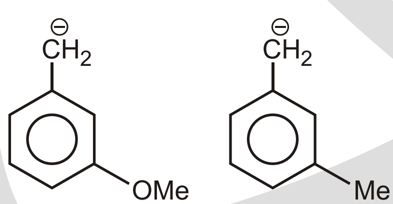
13. 

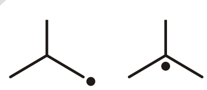
14. 

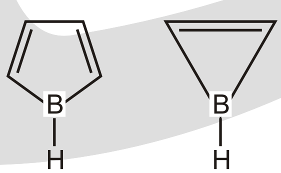
15. $D_3C-\overset{\oplus}{C}H_2$ $H_3C-\overset{\oplus}{C}H_2$

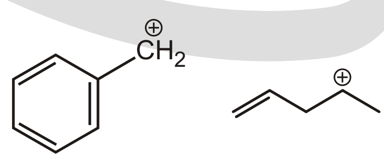
16. $DH_2C-\overset{\oplus}{C}H_2$ $D_2HC-\overset{\oplus}{C}H_2$

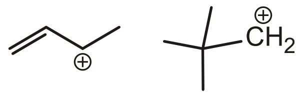
17. 

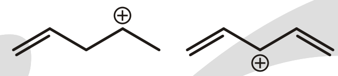
18. 

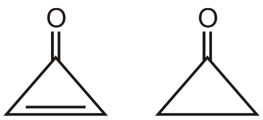
19. 

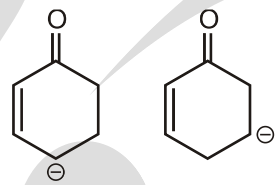
20. 

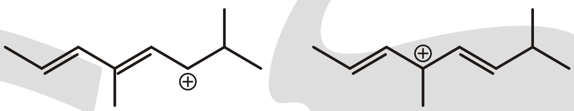
21. 

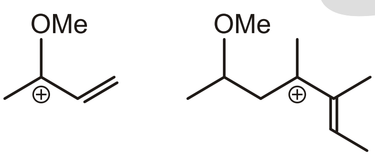
22. 

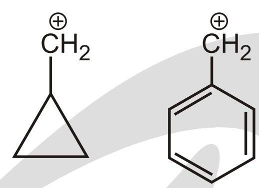
23. 

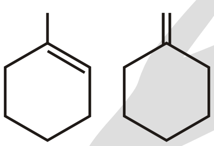
24. 

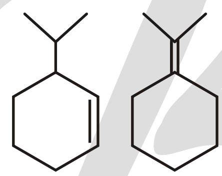
25. 

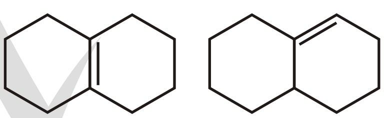
26. 

27. 

28. 

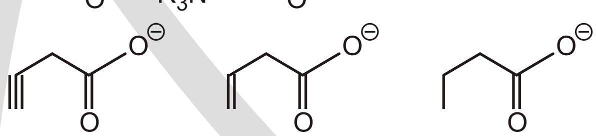
29. 

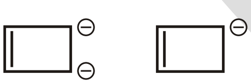
30. 

31. 

32. $H-C\equiv C^{\ominus}$ $H_2C=CH^{\ominus}$ $H_3C-CH_2^{\ominus}$

33. 

34. 

35. 

Answers

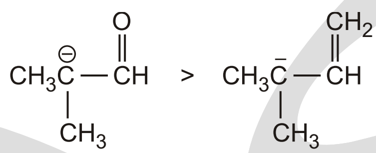
Unsolved Example

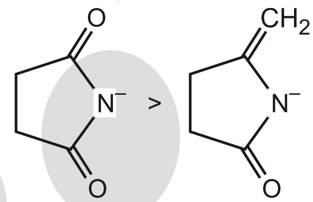
1. **(i) Stability order :** a > c > b
   (ii) Stability order : b > a > c
   (iii) Stability order : b > c > a

2. (a) $CH_3\overset{\overset{O}{\|}}{C}OCH_3$ (b) a < b (c) a > b

3. (a) $CH_3CH_2O^- \;<\; CH_3\overset{\overset{O}{\|}}{C}O^-$ (b) $CH_3\overset{\overset{O}{\|}}{C}\bar{C}HCH_2\overset{\overset{O}{\|}}{C}H \;<\; CH_3\overset{\overset{O}{\|}}{C}\bar{C}H\overset{\overset{O}{\|}}{C}CH_3$

(c) $CH_3\bar{C}HCH_2\overset{\overset{O}{\|}}{C}CH_3 \;<\; CH_3CH_2\bar{C}H\overset{\overset{O}{\|}}{C}CH_3$ (d) $CH_3\underset{}{\overset{\overset{NH_2}{|}}{C}}HCH_3 \;<\; CH_3\overset{\overset{NH}{\|}}{C}NH_2$

(e) 

(f) 

Work sheet - 1

1. a > b	**2.** a > b	**3.** a > b	**4.** a > b	**5.** a < b	**6.** a < b	**7.** a > b	**8.** a < b
9. b > a	**10.** a < b	**11.** a > b > c > d	**12.** a > b	**13.** a < b	**14.** a > b	**15.** a < b	**16.** a > b
17. a > b	**17.** a > b	**18.** a > b	**19.** a < b	**20.** a < b	**21.** a > b	**22.** a > b	**23.** a < b
24. a > b	**25.** a > b	**26.** a > b	**27.** a > b	**28.** a > b	**29.** a > b	**30.** a < b	**31.** a > b
32. a > b > c	**33.** a < b	**34.** a > b > c	**35.** a > b				

❑❑❑

CHAPTER

18

Bond Energy and Bond Length

APPLICATION OF RESONANCE

(i) Bond length : The compound in which resonance occurs, its double bond is slightly longer while its single bond is slightly shorter. Due to delocalization, the bond will acquire the character of partial double bond.

Ex. $H_2C{=}CH \overset{a}{—} \ddot{N}H_2$

$CH_3—CH_2 \overset{b}{—} \ddot{N}H_2$

If we compare bond length of C — N bond of above compounds, the bond length order is b > a.

Reason : Due to delocalization, the **a** bond will acquire the partial double bond character, thus bond length decreases.

Double Bond Character (DBC) $\dfrac{1}{\text{bond length}}$

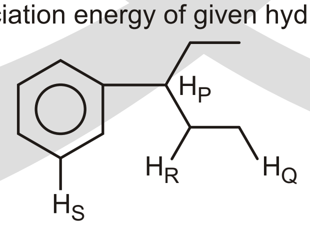
Single Bond Character (SBC) bond length

(ii) Bond dissociation energy : More stable free radical and less is the energy required to form a free radical

$$\text{Bond dissociation energy} \quad \frac{1}{\text{stability of free radical}}$$

Solved Example

- Compare order of bond dissociation energy of given hydrogens in given following compound :



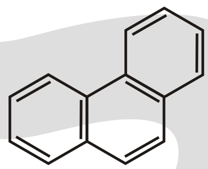
Sol. Bond dissociation energy $\dfrac{1}{\text{stabality of free radical}}$

Order of stability of free radical P > R > Q > S

So bond dissociation energy S > Q > R > P

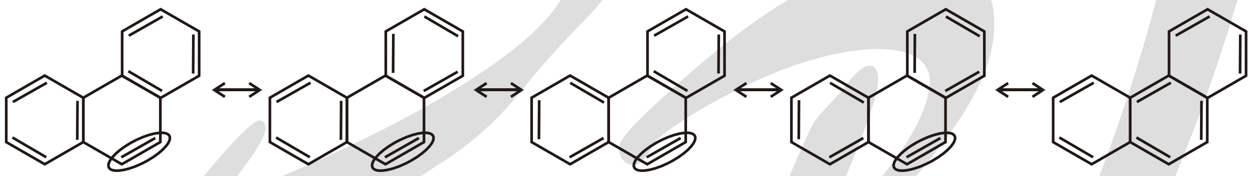
Solved Example

- Phenanthrene has five resonance structures, one of which is shown. Draw the other four and Look at the five resonance structures for phenanthrene and predict which of its carbon-carbon bonds is shortest.



Phenanthrene

Sol.

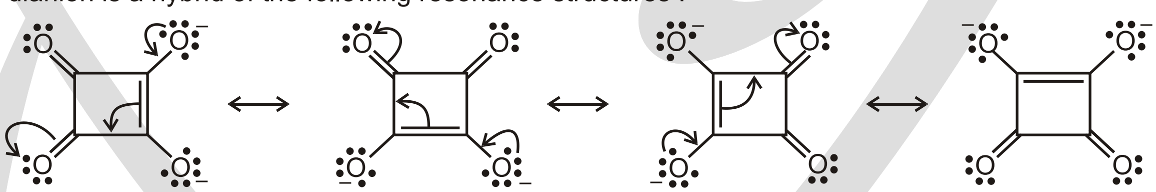


The circled bond is represented as a double bond in four of the five resonance forms of phenanthrene. This bond has more double-bond character and thus is shorter than the other carbon-carbon bonds of phenanthrene.

Solved Example

- Draw four Resonating Structures of dianion of squaric acid.

Sol. The dianion is a hybrid of the following resonance structures :

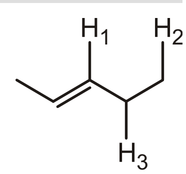


❑ **NOTE :** It is more stable than acetate ion.

EXERCISE

Single Choice Questions

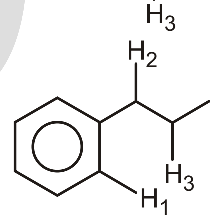
1. Compare the bond strength of the indicated bonds in the given compound :



(A) 1 > 2 > 3 (B) 3 > 1 > 2

(C) 2 > 1 > 3 (D) 2 > 3 > 1

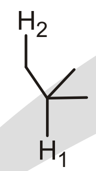
2. Compare the bond strength of the indicated bonds in the given compound :



(A) 1 > 3 > 2 (B) 3 > 1 > 2

(C) 2 > 1 > 3 (D) 2 > 3 > 1

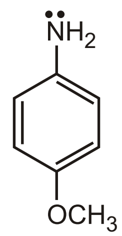
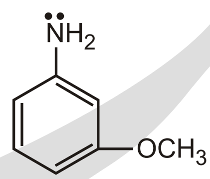
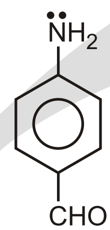
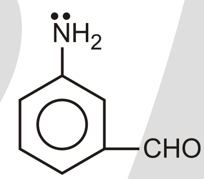
3. Compare the bond strength of the indicated bonds in the given compound :



(A) 1 > 2 (B) 1 = 2
(C) 2 > 1 (D) can not be predicted

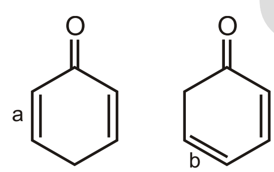
4. Which of the following sequences regarding ease of abstraction of hydrogen atom is correct?
(A) 3° > 2° > 1° (B) 3° < 2° < 1°
(C) 3° < 2° > 1° (D) 3° > 2° < 1°

5. Which C — N bond having more bond strength.

(A) 
(B) 
(C) 
(D) 

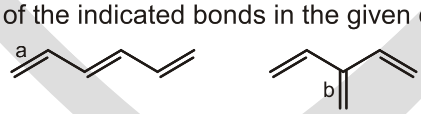
6. Which one of the following statement is not correct?
(A) Amines are stronger bases than water.
(B) Basic strength of amines decreases in the following order : $R_3N > R_2NH > RNH_2$ (In gaseous state).
(C) Carbon-nitrogen bond length in aniline is shorter than that of C — N bond length in hydrogen cyanide.
(D) Aromatic compound has $(4n \quad 2)$ electrons in the loop.

7. Compare the bond lengths of the indicated bonds in the given compound :



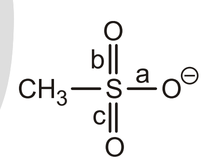
(A) a > b (B) a < b
(C) a = b (D) Cannot be predicted

8. Compare the bond lengths of the indicated bonds in the given compound :



(A) a > b (B) a < b
(C) a = b (D) Cannot be predicted

9. Compare the bond lengths of the indicated bonds in the given compound :

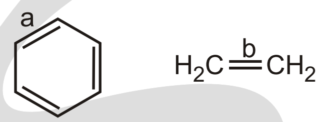


(A) a < b = c
(B) a > b = c
(C) a = b = c
(D) cannot be compared

10. Compare the bond lengths of the indicated bonds in the given compound :

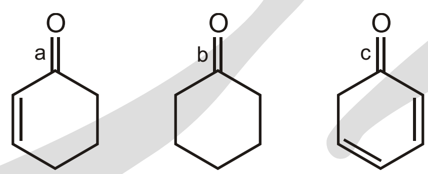
(A) a > b
(B) a < b
(C) a = b
(D) Cannot be predicted

11. Compare the bond lengths of the indicated bonds in the given compound :



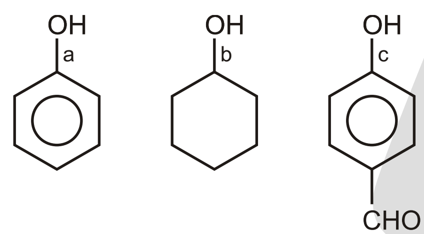
(A) a > b
(B) a < b
(C) a = b
(D) Cannot be predicted

12. Compare the bond lengths of the indicated bonds in the given compound :



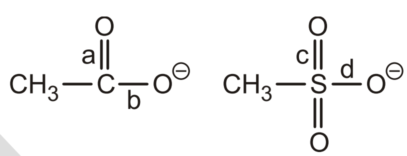
(A) a > c > b
(B) c > a > b
(C) b > c > a
(D) b > a > c

13. Compare the bond lengths of the indicated bonds in the given compound :



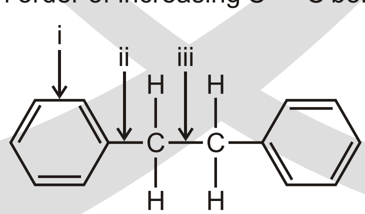
(A) a > c > b
(B) c > a > b
(C) b > c > a
(D) b > a > c

14. Compare the bond order of the indicated bonds in the given compound :



(A) a = b = c = d
(B) a = b > c = d
(C) a = b < c = d
(D) cannot be compared

15. Arrange the following bonds in order of increasing C — C bond strength.



(A) i < ii < iii
(B) i < iii < ii
(C) ii < iii < i
(D) iii < ii < i
(E) iii < i < ii

Answers

Single Choice Questions

1. (A)	**2.** (A)	**3.** (C)	**4.** (A)	**5.** (C)	**6.** (C)	**7.** (B)	**8.** (A)
9. (C)	**10.** (B)	**11.** (A)	**12.** (B)	**13.** (D)	**14.** (B)	**15.** (D)	

1. Stablity of free radical 2 > 3 > 1,
so bond strength 1 > 3 > 2

2. Stablity of free radical 2 > 3 > 1,
so bond strength 1 > 3 > 2

3. Stablity of free radical 1 > 2
so bond strength 2 > 1

4. Stablity of free radical 3º > 1º > 1º

5. M of – OCH_3 is not operate at meta position

6. Bond length = single bond > partial double bond > double bond > triple bond

7. Bond b have more resonance then bond a so its bond length is more.

8. Bond a have extended conjugation but bond b have cross conjugation.

9. Due to equivalent RS all bonds are same.

10. Bond b have more double bond character due to — M — CH O CH CH_2.

11. Bond length = single bond > partial double bond > double bond > triple bond

12. Bond c have more resonance so it have more single bond character thus bond length is longest.

13. Bond c have more resonance so it have more double bond character thus bond length is shortest.

❑❑❑

CHAPTER

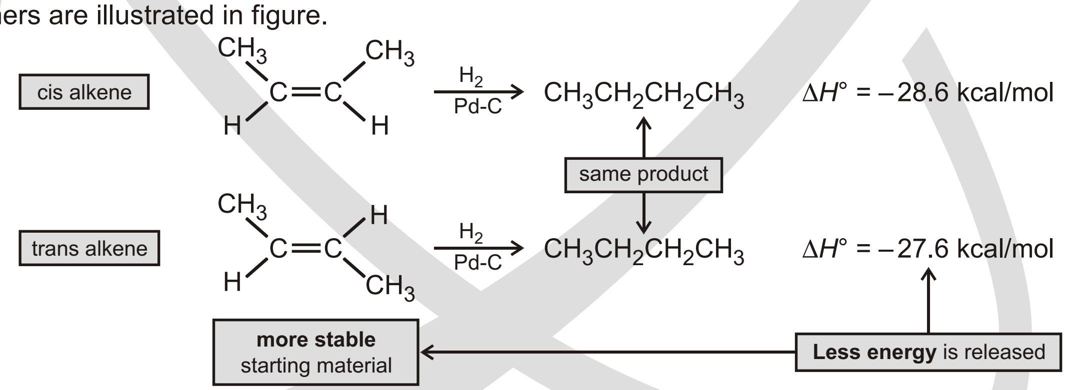
19

Heat of Hydrogenation and Heat of Combustion

HEAT OF HYDROGENATION

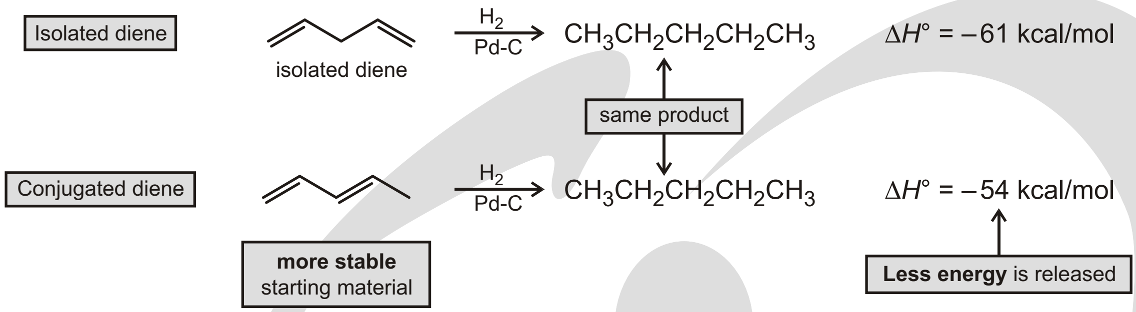
Hydrogenation reactions are exothermic because the bonds in the product are stronger than the bonds in the starting materials, making them similar to other alkene addition reactions. The H for hydrogenation, called the **heat of hydrogenation**, can be used as a measure of the relative stability of two different alkenes that are hydrogenated to the same alkane.

☞ For example, both cis- and trans-2-butene are hydrogenated to butane, and the heat of hydrogenation for the trans isomer is less than that for the cis isomer. Because less energy is released in converting the trans alkene to butane, it must be lower in energy (more stable) to begin with. The relative energies of the butene isomers are illustrated in figure.

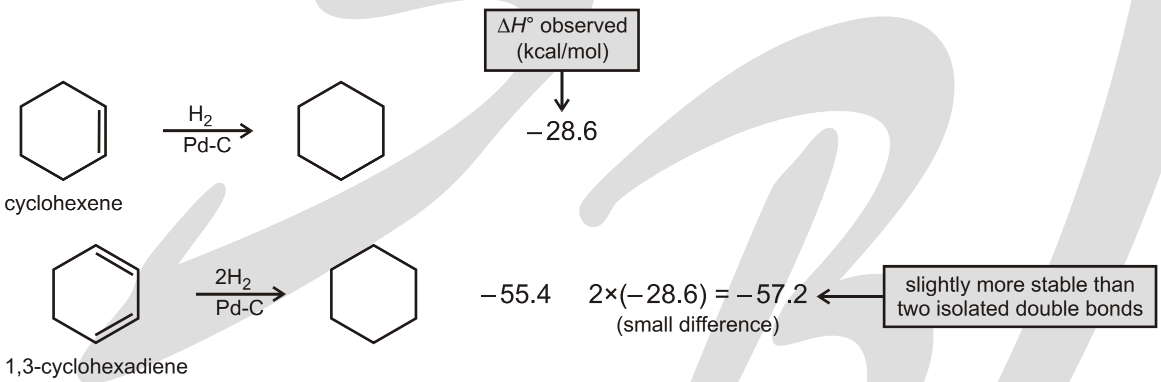


☞ When hydrogenation of two alkenes gives the same alkane, the more stable alkene has the smaller heat of hydrogenation.

Heat of hydrogenation $\frac{\mathbf{1}}{\textbf{Stability}}$

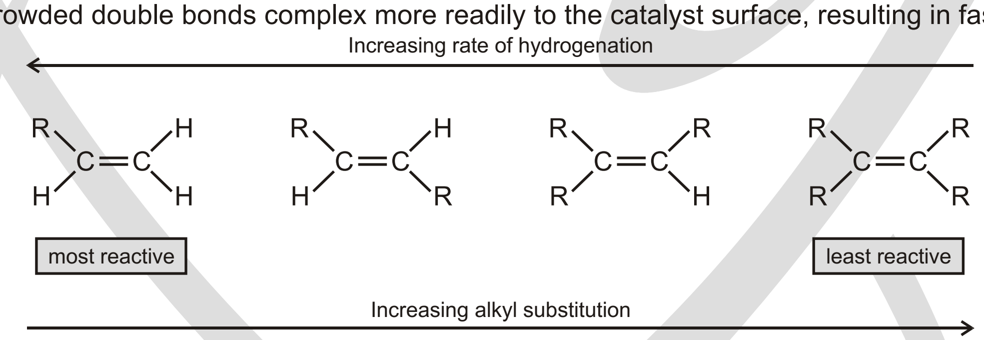


☞ When one compound have more -bonds than other, its heat of hydrogenation is also more.
Heat of hydrogenation number of -bonds present in compound.
For Example :



☞ Rapid, sequential addition of H_2 occurs from the side of the alkene complexed to the metal surface, resulting in syn addition. (syn addition means same side addition)

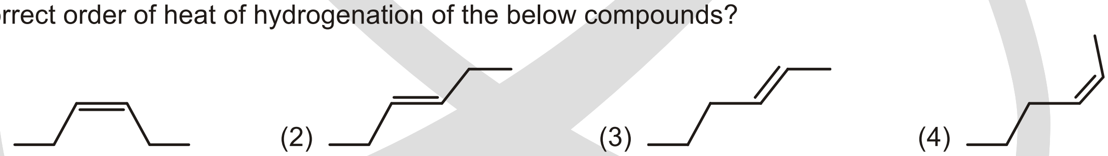
☞ Less crowded double bonds complex more readily to the catalyst surface, resulting in faster reaction.



Solved Example

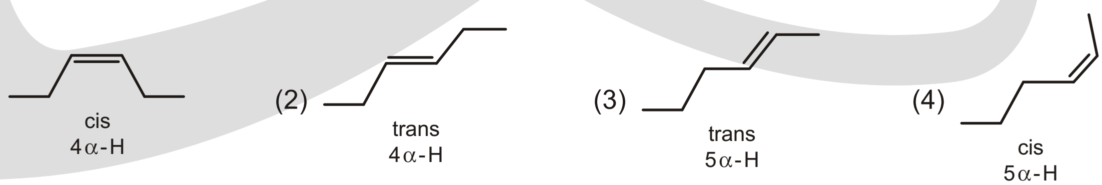
▸ Correct order of heat of hydrogenation of the below compounds?

(1) (2) (3) (4)



Ans. 1 > 2 > 4 > 3

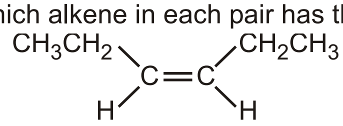
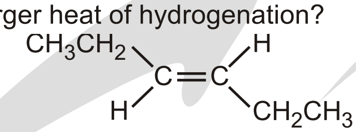
Sol. (1) (2) (3) (4)

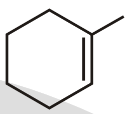
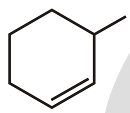


Stability order 3 > 4 > 2 > 1

Heat of hydrogenation 1 > 2 > 4 > 3

- Which alkene in each pair has the larger heat of hydrogenation?

(I)  or 

(II)  or 

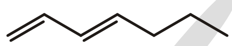
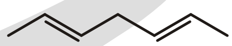
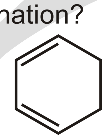
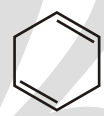
Sol. (I) a > b (II) b > a

In pairs (I) stability b(trans) > a(cis)

Number of H a(7) > b(3)

Heat of hydrogenation $\frac{1}{\text{Stability}}$

- Which diene in each pair has the larger heat of hydrogenation?

(a)  or  (b)  or 

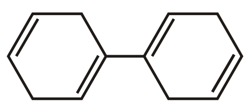
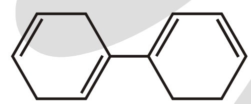
Sol. (a) II > I (b) II > I

Heat of hydrogenation $\frac{1}{\text{Stability}}$

In pair (a) compound I is resonance stabilized

In pair (b) compound I is resonance stabilized

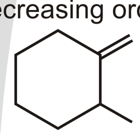
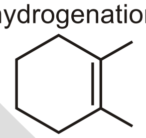
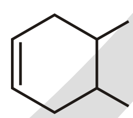
- Rank the following compounds in order of increasing stability.

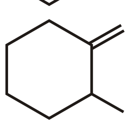
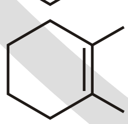
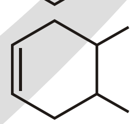
(a)  (b)  (c) 

Sol. c > b > a

In compound (c) more number of -bonds (3) are in resonance thats why it is most stable.

- Decreasing order of heat of hydrogenation?

(I)  (II)  (III) 

Sol. (I)  (II)  (III) 

$\alpha H = 3$ $\alpha H = 10$ $\alpha H = 4$

Stab. order II > III > I

HOH order II < III < I

HEAT OF COMBUSTION(HOC)

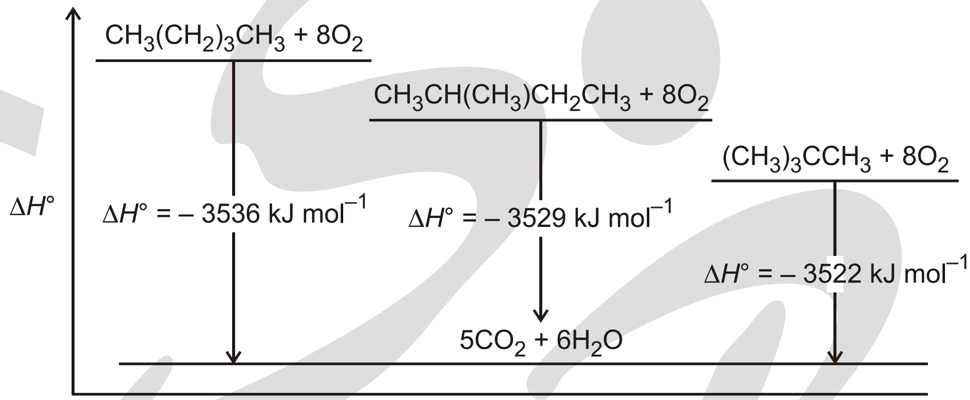
The heat of combustion is the total energy released as heat when a substance undergoes. Complete combustion with oxygen under standard conditions. The chemical reaction is typically a hydrocarbon or other compound with oxygen to form carbon dioxide and water and release heat.

Ex. : C_3H_8 $5O_2$ $3CO_2$ $4H_2O$ H 2202 KJ mol ^{1}

Heat of combustion is more of that isomer which is less stable since it has more potential energy.

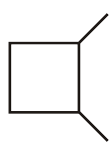
HOC $\frac{\mathbf{1}}{\mathbf{Stability}}$

Ex. : $(CH_3)_3CCH_3$ is the most stable isomer of pentane (*i.e.*, it is the isomer with the lowest potential energy) because it evolves the least amount of heat on a molar basis when subjected to complete combstion.

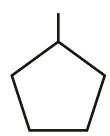


Solved Examples

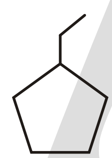
- Heat of combustion of following compounds is :



I



II



III

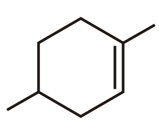
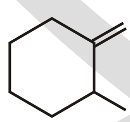
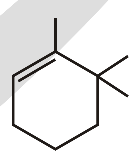
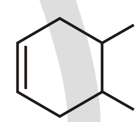
Ans. III > I > II

Sol. Heat of combustion increases with the number of carbons is the compound which is maximum is case of (III) In case number of carbon is same, heat released is inversely related to stability.

EXERCISE

SINGLE CHOICE QUESTIONS

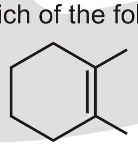
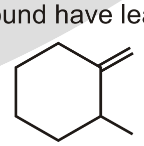
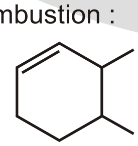
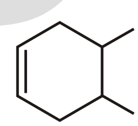
1. Correct order for the heat of combustion of the following is :

(I)  (II)  (III)  (IV) 

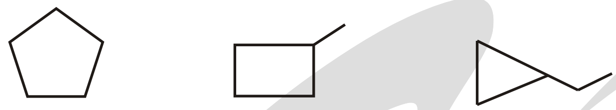
(A) I > II > IV > III (B) IV > I > II > III

(C) III > I > IV > II (D) III > II > IV > I

2. Which of the following compound have least Heat of Cumbustion :

(A)  (B)  (C)  (D) 

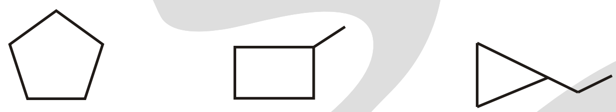
3. Which of the following will have largest heat of combustion?



(I) (II) (III)

(A) I
(B) II
(C) III
(D) All will have same heat of combustion because DBE = 1 for all.

4. Which of the given is most stable?



(I) (II) (III)

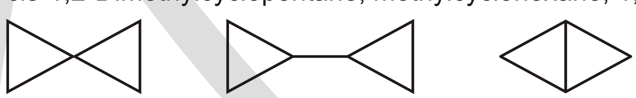
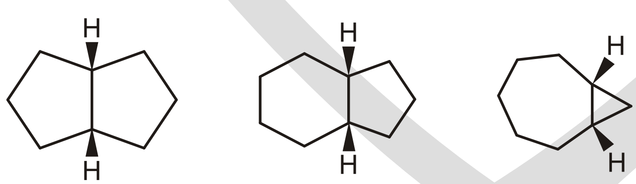
(A) I
(B) II
(C) III
(D) All are equally stable because DBE = 1 for all.

5. Which of the following alkene have maximum heat of combustion?

(A) 2, 4-Dimethyl-1-pentene
(B) 1-Heptene
(C) 2, 4-Dimethyl-2-pentene
(D) 4, 4-dimethyl-2-pentene

WORK SHEET

1. In each of the following groups of compounds, identify the one with the largest heat of combustion and the one with the smallest. In which cases can a comparison of heats of combustion be used to assess relative stability?

(a) Cyclopropane, cyclobutane, cyclopentane
(b) cis-1,2-Dimethylcyclopentane, methylcyclohexane, 1,1,2,2-tetramethylcyclopropane
(c) 
(d) 

2. In each of the following groups of compounds, identify the one with the largest heat of combustion and the one with the smallest. (Try to do this problem without consulting Table).

(a) Hexane, heptane, octane
(b) Isobutane, pentane, isopentane
(c) Isopentane, 2-methylpentane, neopentane
(d) Pentane, 3-methylpentane, 3,3-dimethylpentane
(e) Ethylcyclopentane, ethylcyclohexane, ethylcycloheptane

Answers

Single Choice Questions

1. (D) **2.** (A) **3.** (C) **4.** (A) **5.** (B)

1. Stability of compound I > IV > II
So heat of combustion III > II > IV > I
compound III have more number of carbon atoms so its HOC is maximum

2. More the stability least is HOC

3. Compound C is least stable due to ring strain so HOC is maximum.

4. Compound C is least stable due to ring strain.

5. H.O.C $\frac{1}{\text{Stability}}$, if number of 'C' are same.

Work Sheet

1. (a) c > b > a (b) c > a > b (c) b > a > c (d) b > c > a

2. (a) c > b > a (b) b > c > a (c) b > a > c (d) c > b > a (e) c > b > a

❑❑❑

CHAPTER

20

Aromaticity

☑ SPECIAL TOPIC

THE SWEET BOUQUET OF AROMATIC COMPOUNDS

The word aromatic usually gets people's minds moving. Some people may associate the word with the beautiful scent of roses, while others may think instead of a freshly cut lawn on a warm spring morning. People with a somewhat darker mindset may think of things with a less pleasant smell such as garbage or sweaty socks.

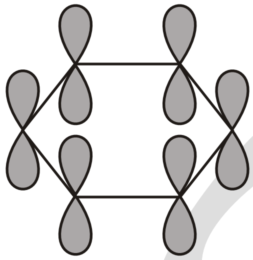
AROMATICITY

The compound we know as benzene was first isolated in 1825 by Michael Faraday, who extracted the compound from a liquid residue obtained after heating whale oil under pressure to produce a gas used to illuminate buildings in London. Because of its origin, chemists suggested that it should be called ''pheno'' from the Greek word phaine in (''to shine'').

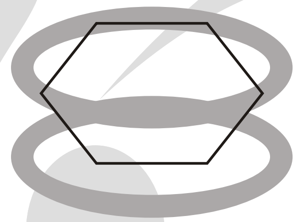
In 1834, Eihardt Mitscherlich correctly determined benzene's molecular formula (C_6H_6) and decided to call it benzin because of its relationship to benzoic acid, a known substituted form of the compound. Later its name was changed to benzene.

Compounds like benzene, which have relatively few hydrogens in relation to the number of carbons, are typically found in oils produced by trees and other plants. Early chemists called such compounds aromatic compounds because of their pleasing fragrances. In this way, they were distinguished from aliphatic compounds, with higher hydrogen-to-carbon ratios, that were obtained from the chemical degradation of fats. The chemical meaning of the word ''aromatic'' now signifies certain kinds of chemical structures. We will now examine the criteria that a compound must satisfy to be classified as aromatic.

At this point, we can define an aromatic compound to be a cyclic compound containing some number of conjugated double bonds and having an unusually large resonance energy. Using benzene as the example, we will consider how aromatic compounds differ from aliphatic compounds. Then we will discuss why an aromatic structure confers extra stability and how we can predict aromaticity in some interesting and unusual compounds.



(a)
Each carbon of Benzene has a *p* orbital.



(b)
The overlap of the *p* orbitals forms a cloud of π electrons above and below the plane of the Benzene ring.

☑ SPECIAL TOPIC

CHEMISTRIVIA

2,4,6-trinitrotoluene is better known as the explosive TNT. Before this explosive property was discovered, TNT was used as a yellow dye. Because TNT dyed their hair green and skin yellow, the women who filled explosive artillery shells during World War I were nicknamed “canary girls.”

Benzene is a planar, cyclic compound with a cyclic cloud of delocalized electrons above and below the plane of the ring. Because its electrons are delocalized, all the C — C bonds have the same length-pathway between the length of a typical single and a typical double bond. We also saw that Benzene is a particularly stable compound because it has an unusually large resonance energy (36kcal/mol). Most compounds with delocalized electrons have much smaller resonance energies. Compounds such as benzene with unusually large resonance energies are called aromatic compounds.

HOW TO DETERMINE AROMATIC, ANTIAROMATIC AND NONAROMATIC COMPOUNDS

How can we tell whether a compound is aromatic by looking at its structure?

In other words, what structural features do aromatic compounds have in common?

To be classfied as aromatic, a compound must follow the criteria :

AROMATIC

1. The structure must be cyclic, containing some number of conjugated pi bonds.
2. Each atom in the ring must have an unhybridized *p* orbital. (The ring atoms are usually sp^2 hybridized or occasionally *sp* hybridized.)
3. The unhybridized *p* orbitals must overlap to form a continuous ring of parallel orbitals. In most cases, the structure must be planar (or nearly planar) for effective overlap to occur.
4. Delocalization of the pi electrons over the ring must *lower* the electronic energy.
5. Huckel rule :

   It must have (4*n*   2) -electrons where *n*   0, 1, 2, 3 .... *n* represent the whole number.

   n   0   2 -electron
   n   1   6 -electron
   n   2   10 -electron
   n   3   14 -electron

ANTI-AROMATIC

1. The structure must be cyclic, containing some number of conjugated pi bonds.
2. Each atom in the ring must have an unhybridized *p* orbital. (The ring atoms are usually sp^2 hybridized or occasionally *sp* hybridized.)
3. The unhybridized *p* orbitals must overlap to form a continuous ring of parallel orbitals. In most cases, the structure must be planar (or nearly planar) for effective overlap to occur.
4. Delocalization of the electrons over the ring must *increase* the electronic energy.
5. It must have $(4n)$ -electrons where n 1, 2, 3 .... n always represent the natural number.

   n 1   4 -electron
   n 2   8 -electron
   n 3   12 -electron

Non-aromatic

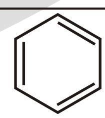
The compound which is neither aromatic nor anti-aromatic

RELATIVE STABILITIES

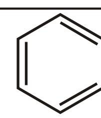
Aromatic compound > cyclic compound with localized electrons > antiaromatic compound

increasing stability ←

Solved Example



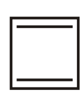
more stable (aromatic)



less stable

Sol. Aromatic structures are more stable than their open-chain counterparts. Hence, Benzene is more stable than hexa-1,3,5-triene.

Solved Example



less stable (antiaromatic)

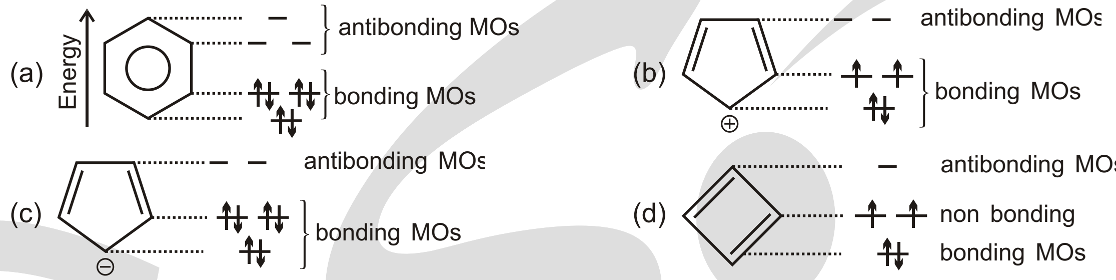
more stable

Sol. An anti aromatic compound is one that meets the first three criteria, but delocalization of the pi electrons over the ring increases the electronic energy. Cyclobutadiene meets the first three criteria for a continuous ring of overlapping p orbitals, but delocalization of the pi electrons increases the electronic energy. Cyclobutadiene is less stable than its open-chain counterpart (buta-1,3-diene), and it is antiaromatic.

MOLECULAR ORBITAL THEORY (M.O.T.) DESCRIPTION OF AROMATICITY AND ANTIAROMATICITY

Why are planar molecules with uninterrupted cyclic electron clouds very stable (aromatic) if they have an odd number of pairs of electrons and very unstable (anti-aromatic) if they have an even number of pairs of electrons? To answer this question, we must turn to molecular orbital theory.

The relative energies of the molecular orbitals of planar molecules with uninterrupted cyclic electron clouds can be determined, without having to use any math, by first drawing the cyclic compound with one of its vertices pointed down. The relative energies of the molecular orbitals correspond to the relative levels of the vertices (Figure). Molecular orbitals below the midpoint of the cyclic structure are bonding molecular orbitals, those above the midpoint are antibonding molecular orbitals, and any at the midpoint are nonbonding molecular orbitals. Notice that the number of molecular orbitals is equal to the number of atoms in the ring since each ring atom contributes a orbital.



The distribution of electrons in the molecular orbitals of
(a) Benzene (b) Cyclopentadienyl cation
(c) Cyclopentadienyl anion, and (d) Cyclobutadiene.

The relative energies of the molecular orbitals in a cyclic compound correspond to the relative levels of the vertices. Molecular orbitals below the midpoint of the cyclic structure are bonding, those above the midpoint are antibonding, and those at the midpoint are non-bonding.

THE POLYGON RULE

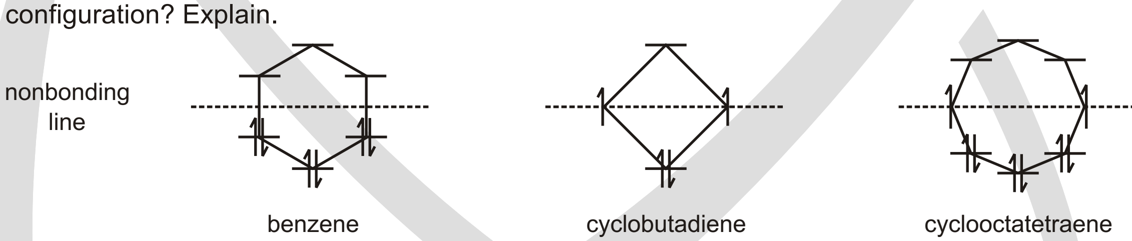
The patterns of molecular orbitals in benzene and in cyclobutadiene are similar to the patterns in other annulenes : The lowestlying MO is the unique one with no nodes; thereafter, the molecular orbitals occur in degenerate (equal-energy) pairs until only one highest-lying MO remains. In benzene, the energy diagram looks like the hexagon of a benzene ring. In cyclobutadiene, the pattern looks like the diamond of the cyclobutadiene ring.

The **polygon rule** states that the molecular orbital energy diagram of a regular, completely conjugated cyclic system has the same polygonal shape as the compound, with one vertex (the all-bonding MO) at the bottom. The nonbonding line cuts horizontally through the center of the polygon. Following solved example shows how the polygon rule predicts the MO energy diagrams for benzene, cyclobutadiene, and cyclooctatetraene. The pi electrons are filled into the orbitals in accordance with the aufbau principle (lowest energy orbitals are filled first) and Hund's rule.

Solved Example

- Does the MO energy diagram of cyclooctatetraene (see figure) appear to be a particularly stable or unstable configuration? Explain.

Ans.



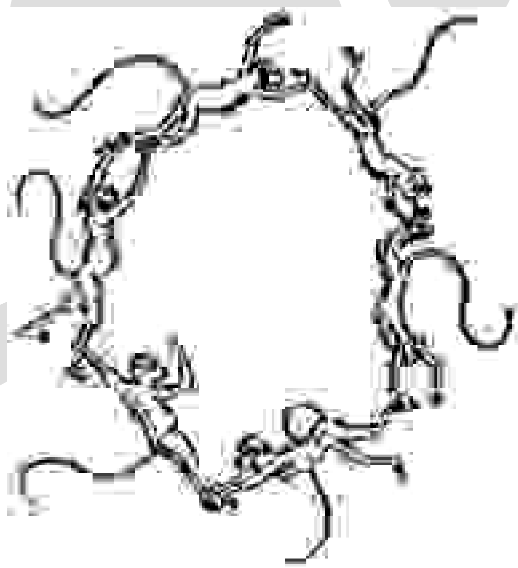
The polygon rule predicts that the MO energy diagrams for these annulenes will resemble the polygonal shapes of the annulenes.

☑ SPECIAL TOPIC

SPECIAL TOPIC : KEKULE'S DREAM

Friedrich August Kekulé von Stradonitz (1829–1896) was born in Germany. He entered the University of Giessen to study architecture but switched to chemistry after taking a course in the subject. He was a professor of chemistry at the University of Heidelberg, at the University of Ghent in Belgium, and then at the University of Bonn. In 1890, he gave an extemporaneous speech at the twenty-fifth-anniversary celebration of his first paper on the cyclic structure of benzene. In this speech, he claimed that he had arrived at the structures as a result of dozing off in front of a fire while working on a textbook. He dreamed of chains of carbon atoms twisting and turning in a snakelike motion, when suddenly the head of one snake seized hold of its own tail and formed a spinning ring.

Recently, the veracity of Kekulé's snake story has been questioned by those who point out that there is no written record of the dream from the time he experienced it in 1861 until the time he related it in 1890. Others counter that dreams are not the kind of evidence one publishes in scientific papers, and it is not uncommon for scientists to experience creative ideas emerging from their subconscious at moments when they were not thinking about science.

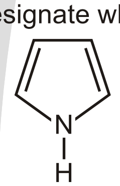
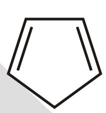
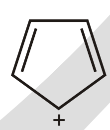
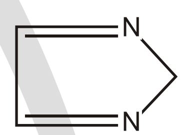
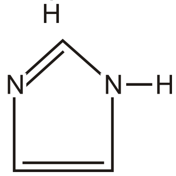
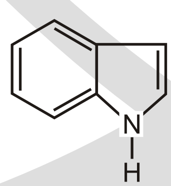
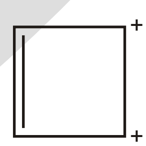
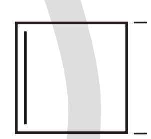
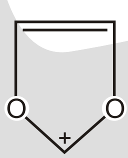


Kekule's snake dream inspired this figure that appeared in a spoof edition of the German chemical Journal, Berichte der Deutschen Chemischen Geselischaft in 1886.

HETEROCYCLIC AROMATIC COMPOUNDS

Solved Problem

- Designate whether each of the following compounds is aromatic, non-aromatic or anti-aromatic.

(a) 
(b) 
(c) 
(d) 
(e) 
(f) 
(g) 
(h) 
(i) 

Ans. (a) 6 e. All carbons are sp^2 hybridized. The compound is aromatic.

(b) 6 e. All carbons are sp^2 hybridized. The compound is aromatic.

(c) 4 e. All carbons are sp^2 hybridized. The compound is anti-aromatic.

(d) The compound is non-aromatic.

(e) 6 e. The compound is aromatic.

(f) 10 e. The compound is aromatic.

(g) 2 e. The compound is aromatic.

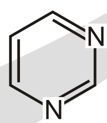
(h) 6 e. The compound is aromatic.

(i) 6 e. The compound is aromatic.

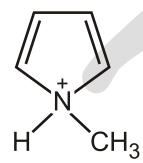
Solved Example

- Describe the following as aromatic, anti-aromatic or non-aromatic (neither aromatic nor anti-aromatic). Assume each is planar.

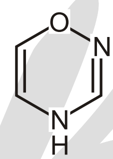

Anti-Aromatic



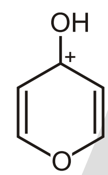
Aromatic



Anti-Aromatic



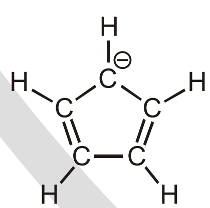
Anti-Aromatic

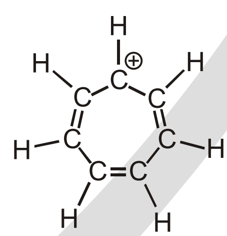


Aromatic

AROMATIC IONS

According to the Huckel criteria for aromaticity, a molecule must be cyclic, conjugated (nearly planar with a *p* orbital on each atom), and have $4n$ 2 electrons. Nothing in this definition says that the number of *p* electrons must be the same as the number of atoms in the ring or that the substance must be neutral. In fact, the numbers can be different and the substance can be an ion. Thus, both the cyclopentadienyl anion and the cycloheptatrienyl cation are aromatic even though both are ions and neither contains a six-membered ring.


Cyclopentadienyl anion

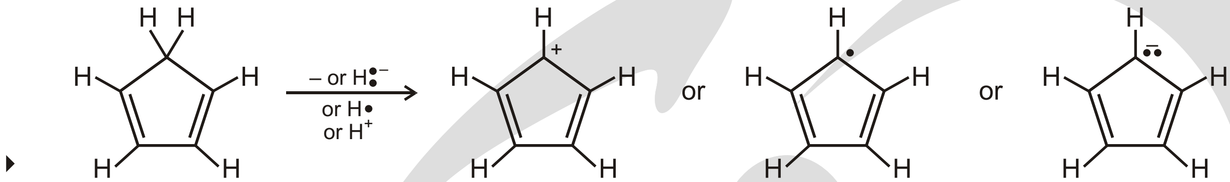

Cycloheptatrienyl cation

Six π electrons; aromatic ions

To see why the cyclopentadienyl anion and the cycloheptatrienyl cation are aromatic, imagine starting from the related neutral hydrocarbons, 1,3-cyclopentadiene and 1,3,5-cycloheptatriene, and removing one hydrogen from the saturated CH_2 carbon in each. If that carbon then rehybridizes from sp^3 to sp^2, the resultant products would be fully conjugated, with a *p* orbital on every carbon. There are three ways in which the hydrogen might be removed.

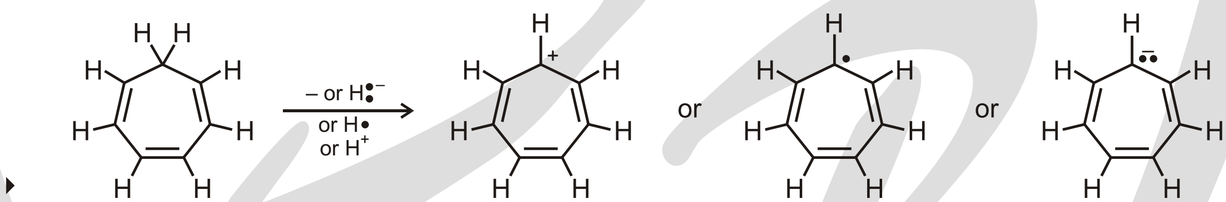
- The hydrogen can be removed with both electrons ($H:^-$) from the C — H bond, leaving a carbocation as product.
- The hydrogen can be removed with one electron (H• from the C — H bond, leaving a carbon radical as product.
- The hydrogen can be removed with no electrons (H+) from the C — H bond, leaving a carbanion as product.

Solved Example

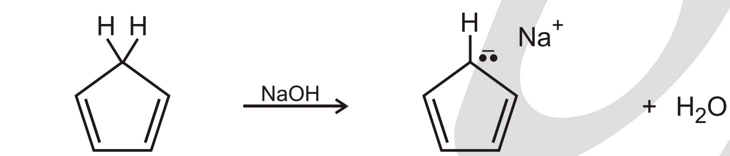


1,3-Cycloentadiene | **Cyclopentadienyl cation** (four π electrons) | **Cyclopentadienyl radical** (five π electrons) | **Cyclopentadienyl anion** (six electrons) (Aromatic)

Solved Example

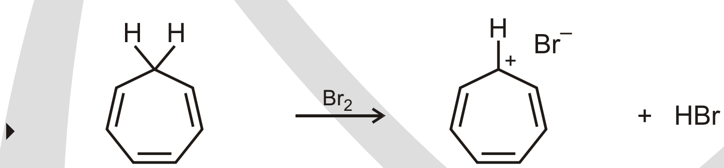


1,3,5-Cycloheptatriene | **Cycloheptatrienyl cation** (six π electrons) | **Cycloheptatrienyl radical** (seven π electrons) | **Cycloheptatrienyl anion** (eight π electrons) (not Aromatic)



1,3,-Cyclopentadiene | Cyclopentadienyl anion (Aromatic)

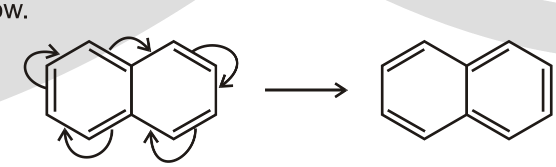
Solved Example



1,3,5-Cycloheptatriene | Cycloheptatrienyl cation

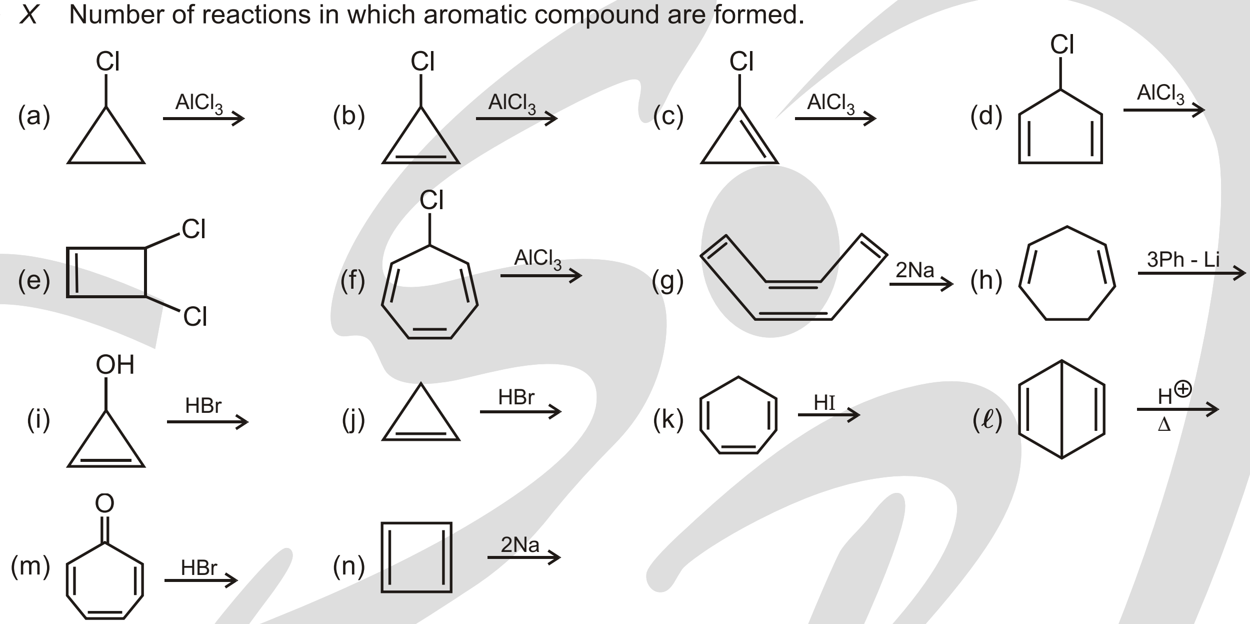
PERIPHERAL OR PERIMETER RESONANCE

Although Huckel's rule strictly applies only to monocyclic compounds, it does appear to have application to certain bicyclic compounds, provided the important resonance structures involve only the perimeter double bonds, as in naphthalene below.



Solved Example

- X Number of reactions in which aromatic compound are formed.



Find the value of $(x \quad 4) \quad ?$

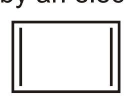
Ans. 7

Sol. Reactions in which aromatic compound are formed are : (b), (e), (f), (g), (h), (i), (j), (k), (ℓ), (m), (n)

$X \quad 11; \quad (X \quad 4) \quad 7$

CYCLOBUTADIENE

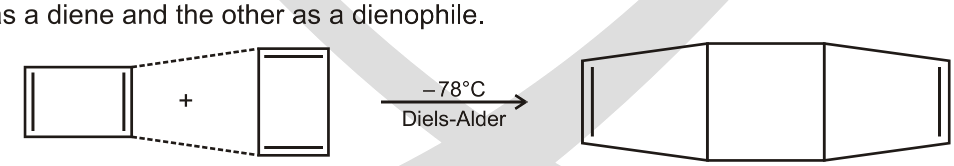
Cyclobutadiene has four *p* electrons and is antiaromatic. The electrons are localized in two double bonds rather than delocalized around the ring, as indicated by an electrostatic potential map.



Cyclobutadiene

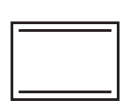
Two double bonds; four electrons

Cyclobutadiene is highly reactive and shows none of the properties associated with aromaticity. In fact, it was not even prepared until 1965, when Rowland Pettit of the University of Texas was able to make it at low temperature. Even at 78 C, however, cyclobutadiene is so reactive that it dimerizes by a Diels-Alder reaction. One molecule behaves as a diene and the other as a dienophile.

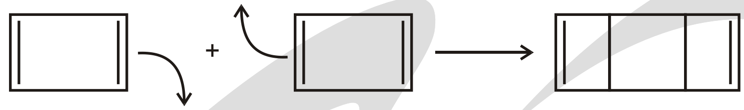


Solved Problem

- Explain why the given compound undergoes dimerization at room temperature and readily reacts with active metals

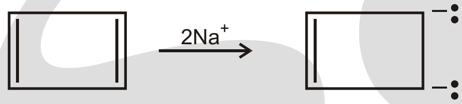


Ans. The given compound dimerizes as follows :



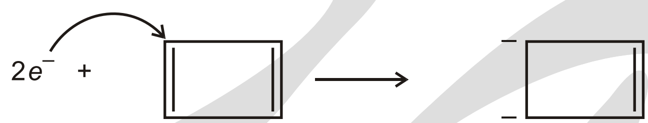
Here, the substrate is anti-aromatic and therefore unstable. It readily undergoes dimerization to give a product which is non-aromatic.

It reacts with an active metal as follows :



The mechanism of the above reaction is believed to proceed as follows :

2Na 2Na 2e

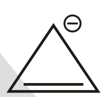
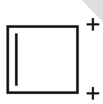
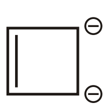


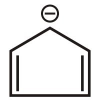
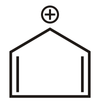
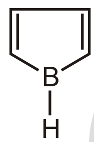
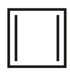
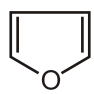
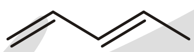
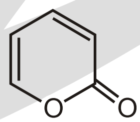
6πe, cyclic and planar systme, (aromatic)

Obviously, the reaction with an a active metal converts the unstable anti-aromatic substrate into a stable aromatic system and proceeds readily.

Solved Example

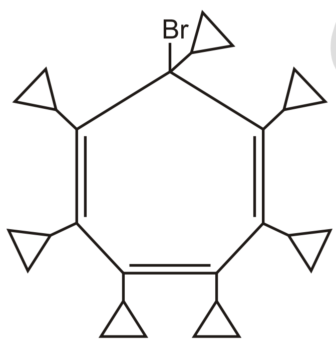
- Use the Huckel rule to indicate whether the following species are aromatic or antiaromatic or non aromatic ?

S.No.	Compound	Number of electron	Nature of compound
1.		2	Non aromatic
2.		4	Anti aromatic
3.		2	Aromatic
4.		2	Aromatic
5.		6	Aromatic
6.		2	Non aromatic
7.		4	Non aromatic

8.		6	Aromatic
9.		4	Anti aromatic
10.		4	Anti aromatic
11.		4	Anti aromatic
12.		6	Aromatic
13.		4	Non aromatic
14.		6	Aromatic

Solved Problem

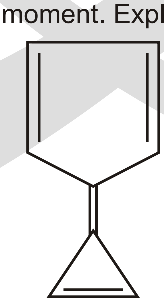
- In the given compound, will Br ionize in the form of (a) Br or (b) Br .



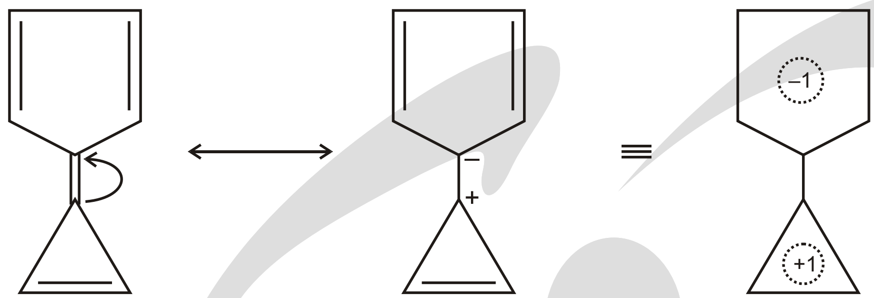
Ans. In this compound, Br will ionize in the form of Br so as to attain aromatic character for the molecule.

Solved Problem

- The given compound has high dipole moment. Explain.



Ans.

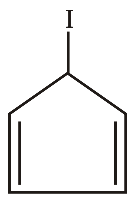


It is apparent that the given compound tends to acquire a resonance stabilized dipolar structure to gain aromaticity in each and so would have high dipole moment.

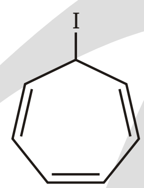
Solved Problem

- Compare the rate of reaction of the following compounds with $AgNO_3$.

(a)

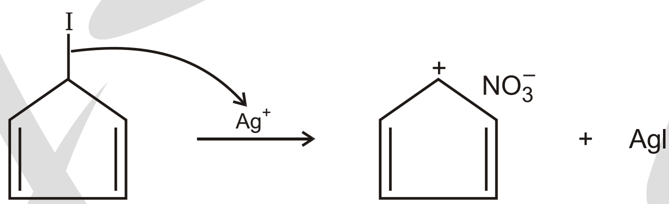


(b)



Ans. More the stability of the resultant carbocation, greater will be the rate of reaction.

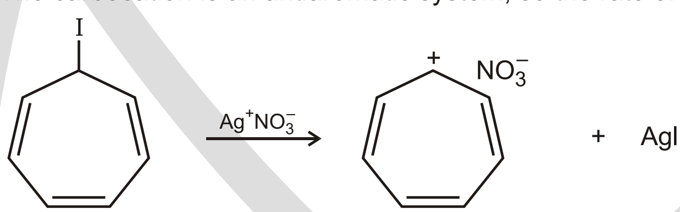
(a)



Anti-aromatic system

The carbocation is an antiaromatic system, so the rate of reaction is slow.

(b)

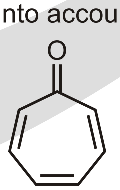


Aromatic system

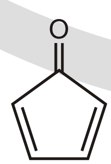
The carbocation in this case is stable aromatic system and so the reaction proceeds faster.

Solved Example

- Cycloheptatrienone is stable, but cyclopentadienone is so reactive that it can't be isolated. Explain, taking the polarity of the carbonyl group into account.



Cyclopropanone



Cyclopentadienone

Sol. 

A B C D

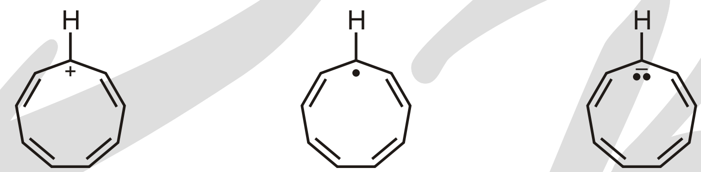
Cycloheptatrienone Cyclopentadienone

As in the previous problem, we can draw resonance forms in which both carbonyl n electrons are located on oxygen. The cycloheptatrienone ring in **B** contains six *n* electrons and is aromatic according to Huckel's rule. The cyclopentadienone ring in **D** contains four electrons and is antiaromatic.

Solved Example

- Which would you expect to be most stable, cyclononatetraenyl radical, cation, or anion?

Sol. Check the number of electrons in the system of each compound. The species with a Huckel (4*n* 2) number of electrons is the most stable.



Cation Radical anion

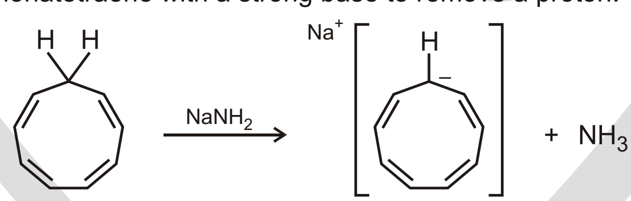
8π electrons 9π electrons 10π electrons

The 10 electron anion is the most stable.

Solved Example

- How might you convert 1,3,5,7-cyclononatetraene to an aromatic substance?

Sol. Treat 1,3,5,7-cyclononatetraene with a strong base to remove a proton.

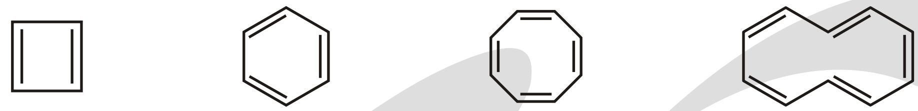


FAILURES OF THE RESONANCE

Large-Ring Annulenes

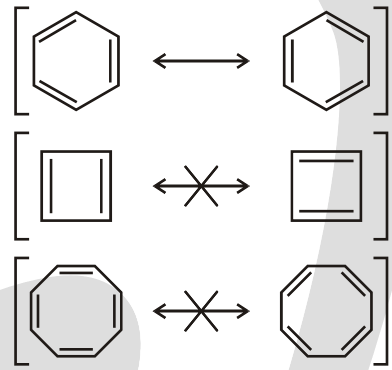
Like cyclooctatetraene, larger annulenes with (4*N*) systems do not show antiaromaticity because they have the flexibility to adopt nonplanar conformations. Even though [12] annulene, [16] annulene and [20] annulene are (4*N*) systems (with *N* 3, 4 and 5 respectively), they all react as partially conjugated polyenes.

For many years, chemists assumed that benzene's large resonance energy resulted from having two identical, stable resonance structures. They thought that other hydrocarbons with analogous conjugated systems of alternating single and double bonds would show similar stability. These cyclic hydrocarbons with alternating single and double bonds are called annulenes. For example, benzene is the six-membered annulene, so it can be named [6] annulene. Cyclobutadiene is [4] annulene, cyclooctatetraene is [8] annulene, and larger annulenes are named similarly.



cyclobutadiene [4] annulene — benzene [6] annulene — cyclooctatetraene [8] annulene — cyclodecapentaene [10] annulene

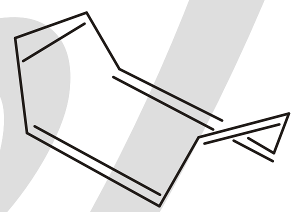
For the double bonds to be completely conjugated, the annulene must be planar so the *p* orbitals of the pi bonds can overlap. As long as an annulene is assumed to be planar, we can draw two Kekulé-like structures that seem to show a benzene-like resonance. The given figure shows proposed benzene-like resonance forms for cyclobutadiene and cyclooctatetraene. Although these resonance structures suggest that the [4] and [8] annulenes should be unusually stable (like benzene), experiments have shown that cyclobutadiene and cyclooctatetraene are not unusually stable. These results imply that the simple resonance picture is incorrect.



Cyclobutadiene and cyclooctatetraene have alternating single and double bonds similar to those of benzene. These compounds were mistakenly expected to be aromatic.

Cyclobutadiene has never been isolated and purified. It undergoes an extremely fast Diels–Alder dimerization. To avoid the Diels–Alder reaction, cyclobutadiene has been prepared at low concentrations in the gas phase and as individual molecules trapped in frozen argon at low temperatures. This is not the behavior we expect from a molecule with exceptional stability!

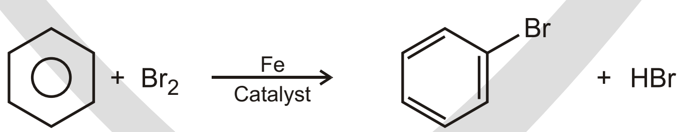
In 1911, Richard Willstäer synthesized cyclooctatetraene and found that it reacts like a normal polyene. Bromine adds readily to cyclooctatetraene, and permanganate oxidizes its double bonds. This evidence shows that cyclooctatetraene is much less stable than benzene. In fact, structural studies have shown that cyclooctatetraene is not planar. It is most stable in a ''tub'' conformation, with poor overlap between adjacent pi bonds.



"tub" conformation of cyclooctatetraene

RESONANCE ENERGY

Benzene (C_6H_6) has six fewer hydrogens than the corresponding six-carbon cycloalkane (C_6H_{12}) and is clearly unsaturated, usually being represented as a six-membered ring with alternating double and single bonds. Yet it has been known since the mid-1800s that benzene is much less reactive than typical alkenes and fails to undergo typical alkene addition reactions. Cyclohexene, for instance, reacts rapidly with Br_2 and gives the addition product 1,2-dibromocyclohexane, but benzene reacts only slowly with Br_2 and gives the substitution product C_6H_5Br.



Bromobenzene
(substitution product)

We can get a quantitative idea of benzene's stability by measuring heats of hydrogenation. Cyclohexene, an isolated alkene, has H_{hydrog} 118 kJ/mol (–28.2 kcal/mol), and 1,3-cyclohexadiene, a conjugated diene, has H_{hydrog} 230 kJ/mol (–55.0 kcal/mol). As noted before, this value for 1,3-cyclohexadiene is a bit less than twice that for cyclohexene because conjugated dienes are more stable than isolated dienes.

Carrying the process one step further, we might expect H hydrog for ''cyclo-hexatriene'' (benzene) to be a bit less than – 356 kJ/mol, or three times the cyclohexene value. The actual value, however, is – 206 kJ/mol, some 150 kJ/mol (36 kcal/mol) less than expected. Since 150 kJ/mol less heat than expected is released during hydrogenation of benzene, benzene must have 150 kJ/mol less energy to begin with. In other words, benzene is more stable than expected by 150 kJ/mol (Figure).

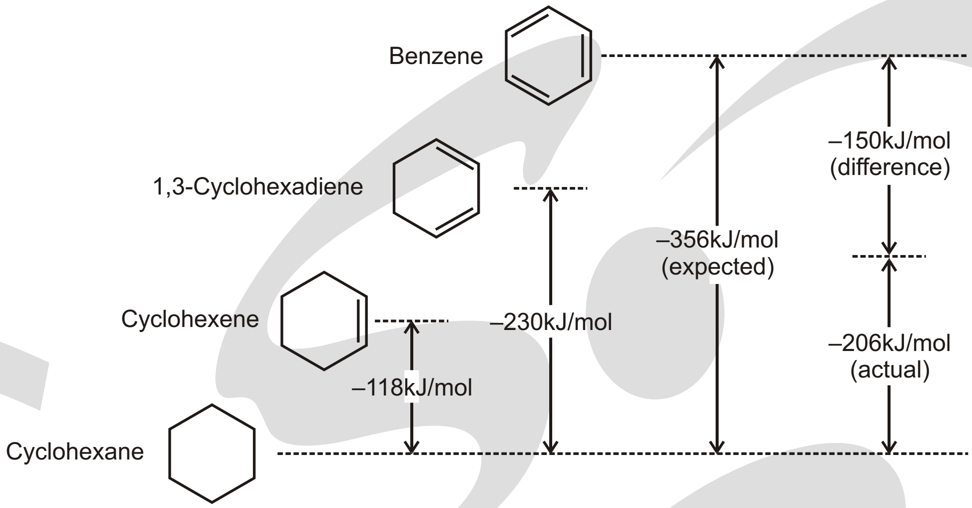


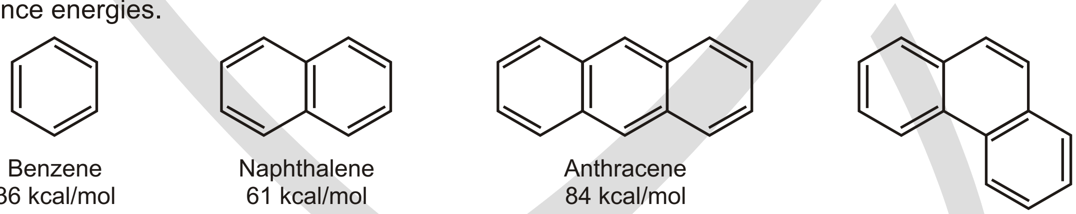
Figure : A comparison of the heats of hydrogenation for cyclohexene, 1,3-cyclohexadiene and benzene. Benzene is 150 kJ/mol (36 kcal/mol) more stable than might be expected for "cyclohexatriene."

Further evidence for the unusual nature of benzene is that all its carbon-carbon bonds have the same length—139 pm—intermediate between typical single (154 pm) and double (134 pm) bonds. In addition. Thus, benzene is a planar molecule with the shape of a regular hexagon. All C — C — C bond angles are 120°, all six carbon atoms are *sp*^{2}-hybridized, and each carbon has a *p* orbital perpendicular to the plane of the six-membered ring.

Because all six carbon atoms and all six *p* orbitals in benzene are equivalent, it's impossible to define three localized bonds in which a given *p* orbital overlaps only one neighboring *p* orbital. Rather, each *p* orbital overlaps equally well with both neighboring *p* orbitals, leading to a picture of benzene in which all six electrons are free to move about the entire ring. In resonance terms, benzene is a hybrid of two equivalent forms. Neither form is correct by itself; the true structure of benzene is somewhere in between the two resonance forms but is impossible to draw with our usual conventions. Because of this resonance, benzene is more stable and less reactive than a typical alkene.

Solved Example

- Shown below are the structures of the first four benzene-based aromatic hydrocarbons and their associated resonance energies.



Benzene
36 kcal/mol

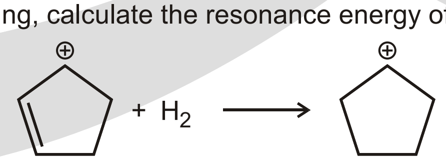
Naphthalene
61 kcal/mol

Anthracene
84 kcal/mol

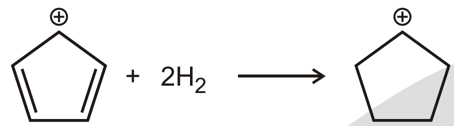
Phenanthrene
92 kcal/mol

Solved Example

- From the following, calculate the resonance energy of the cyclopentadienyl cation.



ΔH = – 14.58 kcal/mol

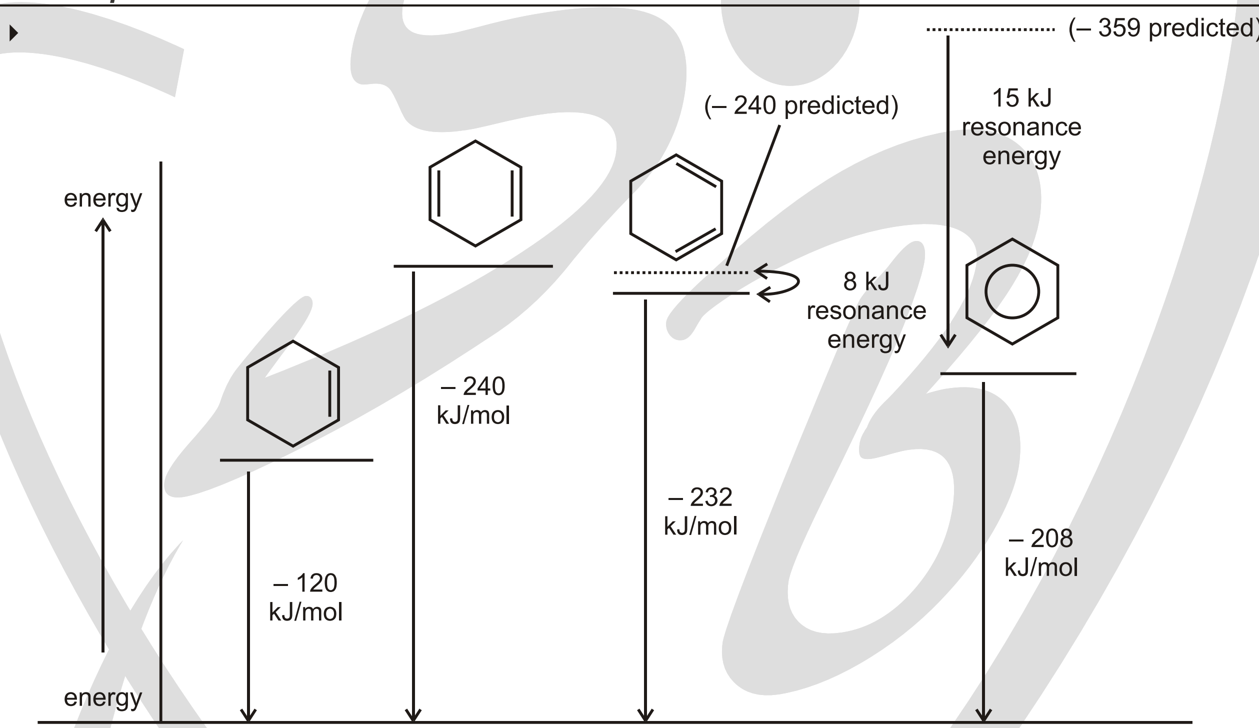


ΔH = – 72.91 kcal/mol

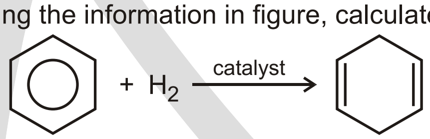
Sol. One hydrogeneration of the allyl cation liberates – 14.58 kcal/mol, so two should liberate – 29.16 kcal/mol. Hydrogenation actually gains – 72.91 kcal/mol making the cation HIGHER in energy by 43.75 kcal/mol. The resonance energy is NEGATIVE : – 43.75 kcal/mol.

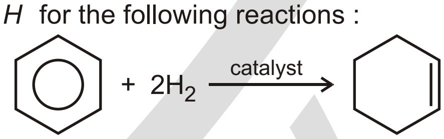
Solved Example

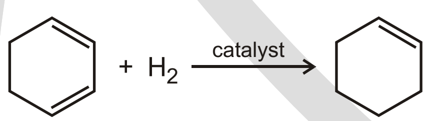
▸



Using the information in figure, calculate the values of H for the following reactions :

(a) 

(b) 

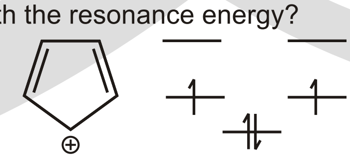
(c) 

Ans. (a) + 32 kJ/mol (b) – 88 kJ/mol (c) – 112 kJ/mol

Solved Example

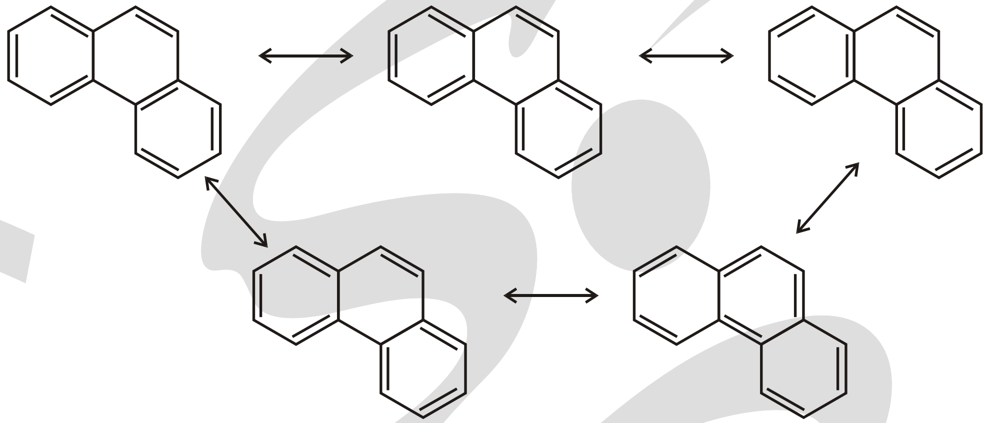
▸ Sketch the MO diagram for this species and use it to determine whether this is an aromatic or antiaromatic molecule. Is this consistent with the resonance energy?

Sol.

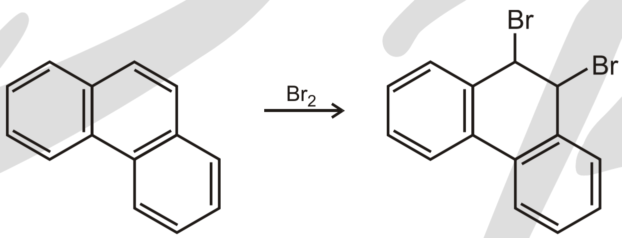


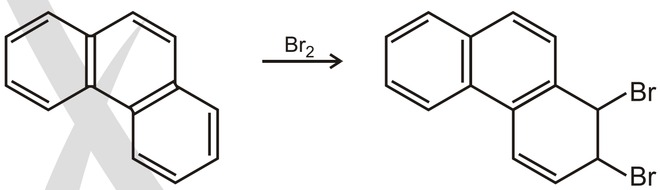
Solved Example

- Phenanthrene has five Kekule resonance structures. One of the bonds of phenanthrene reacts with bromine to give an **addition reaction** just like an alkene. Which bonds? Explain.



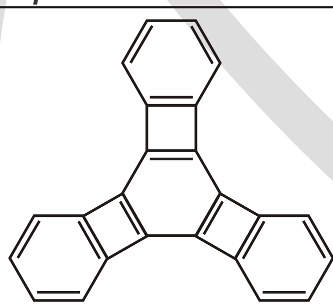
Reaction at this bond costs 92-72 (two benzenes) = 20 kcal/mol in resonance.

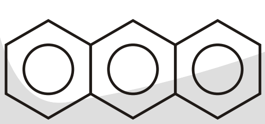


Sol. 

Reaction in the other ring costs 92-61 or 31 Kcal/mol

Solved Examples

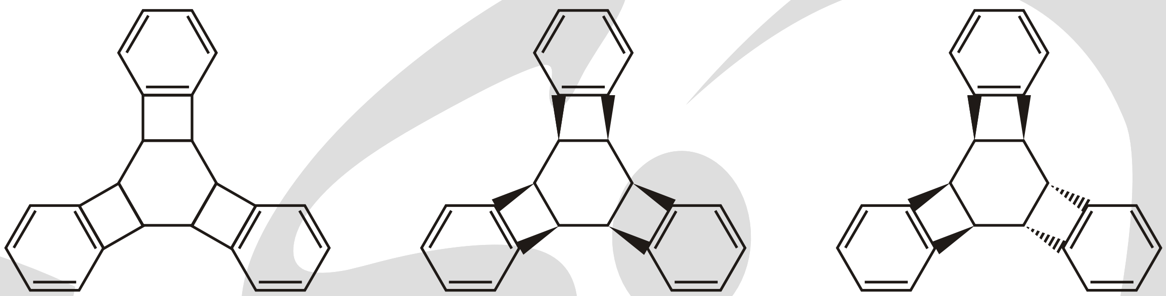
- (i)  $\xrightarrow{3H_2/Pd}$ (x) number of product including stereo isomers

(ii) 

When two hydrogen of anthracene is replaced by bromine then number of meso isomer (y) obtained is so,
x y ?

Ans. 2 0 2

Sol. (i) After hydrogenation we get here we get 3 aromatic 4 non aromatic rings in all other cases we get at least 1 anti aromatic ring which is very unstable. Possible isomers.



All three on the same side

Any two on same side

Hence x 2.

(ii) For a meso product, we need at least 2 chiral centres.

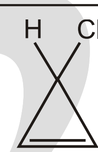
After dibromination we don't get any such products.

Hence y 0

x y 2

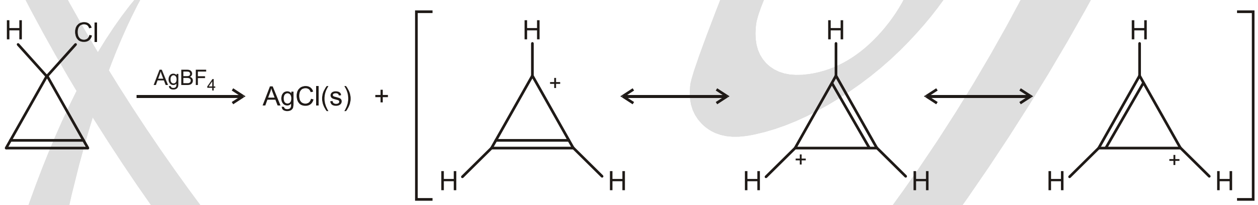
Solved Example

- 3-Chlorocyclopropene, on treatment with $AgBF_4$, gives a precipitate of AgCl and a stable solution of a product. What is a likely structure for the product, and what is its relation to Huckel's rule?



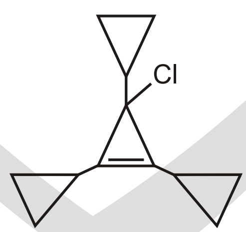
3-Chlorocyclopropene

Sol.

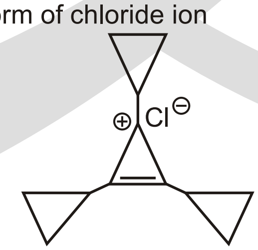


Solved Example

- When this compound ionizes, is Cl or Cl formed?

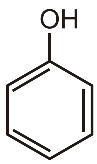


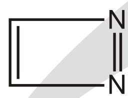
Sol. Chlorine atom will ionised in the form of chloride ion

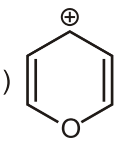


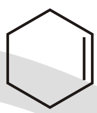
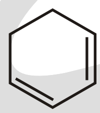
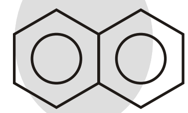
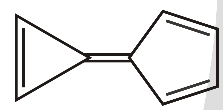
Solved Example

- Number of aromatic compounds in the following

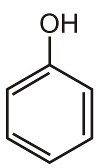
(a) 

(b) 

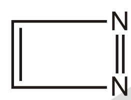
(c) 
(d) 

(e) 
(f) 
(g) 
(h) 

Ans. 4

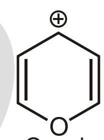
Sol. (a) 

Aromatic

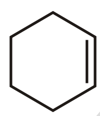
(b) 

Anti-aromatic

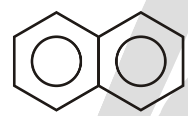
(c) 
Non-aromatic

(d) 

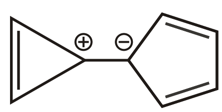
Quasi-
Aromatic

(e) 
Non-aromatic

(f) 
Non-aromatic

(g) 
Aromatic

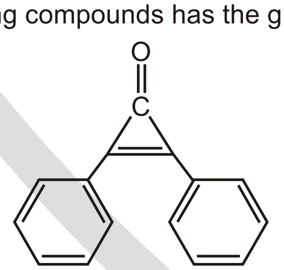
(h) 
Aromatic

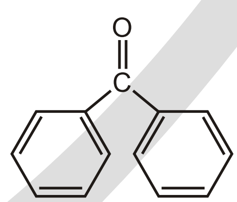


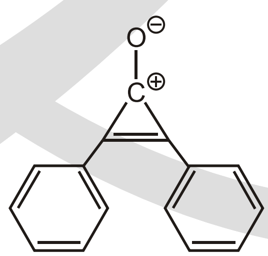
Quasi-aromatic

Solved Example

- Which of the following compounds has the greater dipole moment?



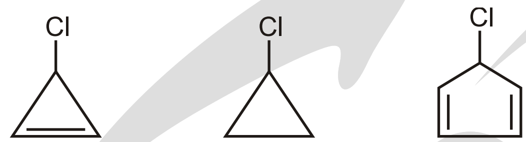


Sol. Ist compound has the greater dipole moment 

quasi aromatic compound
(highly stable)

Solved Example

- Order of rate of reaction with $AgNO_3$

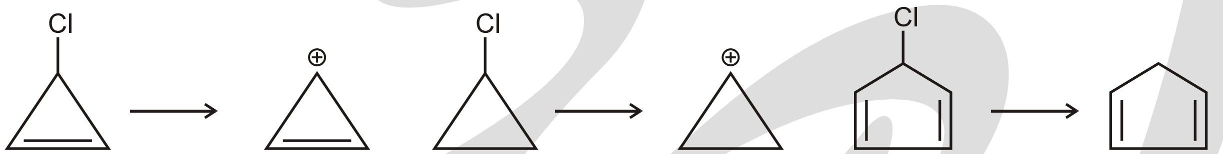


(I) (II) (III)

(A) I > III > II (B) II > III > I
(C) I > II > III (D) III > I > II

Ans. (C)

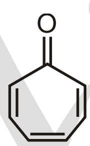
Sol.

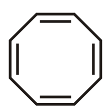


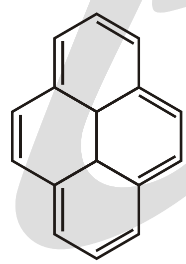
(I) Quasi-aromatic (II) Non-aromatic (III) Anti-aromatic

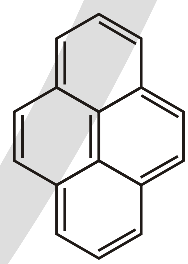
Solved Example

- Comment on the nature of given compounds

1. 

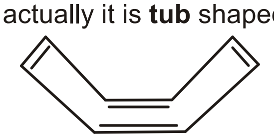
2. 

3. 

4. 

Sol. (1) 6- -electron are present in the compound (-bond outside the ring will not counted in number of -electrons so compound is **aromatic** in nature.

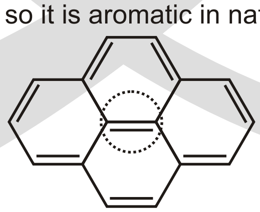
(2) This compound is not planar, actually it is **tub** shaped so it's **non-aromatic** in nature.



Tub shaped due to ring strain

(3) 14- -electron's are present in the compound (so the value of n is 3) so it is aromatic in nature.

(4) The circled -bond will not be counted in number of -electron. 14- -electron's are present in the compound (so the value of n is 3) so it is aromatic in nature.



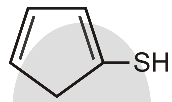
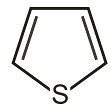
EXERCISE

SINGLE CHOICE QUESTIONS

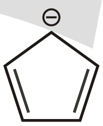
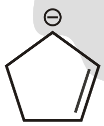
1. In which of the following pair Ist compound has more resonance energy than IInd?

(A)  

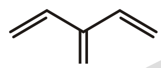
(I) (II)

(B)  

(I) (II)

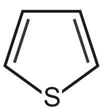
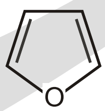
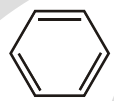
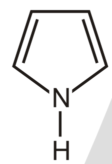
(C)  

(I) (II)

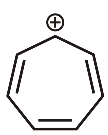
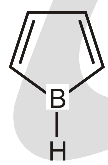
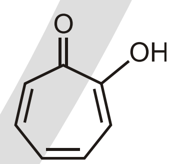
(D)  

(I) (II)

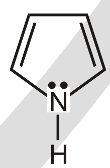
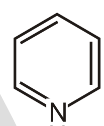
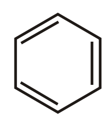
2. Which of the following aromatic compounds is the most stable?

(A) 
(B) 
(C) 
(D) 

3. Which of the following is unstable at room temperature?

(A) 
(B) 
(C) 
(D) 

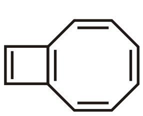
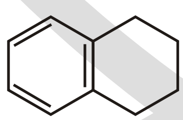
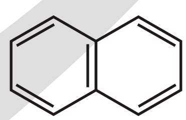
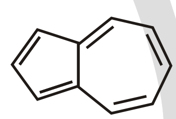
4. Which of these would you expect to have significant resonance energy?

(I) (II) (III)

(A) I
(B) II
(C) III
(D) All of the above
(E) None of the above

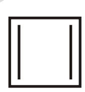
5. Which of these is the single best representation for naphthalene?

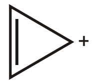
   

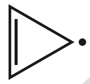
(I) (II) (III) (IV)

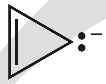
(A) I (B) II (C) III (D) IV

6. Which cyclization(s) should occur with a decrease in pi-electron energy?

(I) CH_2 CH — CH CH_2  H_2

(II) CH_2 CH — CH_2  H_2

(III) CH_2 CH — CH_2  H_2

(IV) CH_2 CH — CH_2  H_2

(A) I (B) II (C) III (D) IV

(E) All of the above

7. Which of the following statements about cyclooctatetraene is NOT true?

(A) The compound rapidly decolorizes Br_2/CCl_4 solutions. (test for unsaturated hydrocarbon)

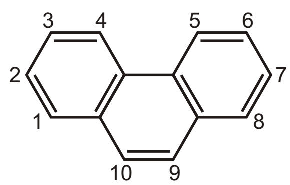
(B) The compound rapidly decolorizes aqueous solutions of $KMnO_4$. (test for unsaturated hydrocarbon)

(C) The compound readily adds hydrogen.

(D) The compound is nonplanar.

(E) The compound is comparable to benzene in stability.

8. Recalling that benzene has a resonance energy of 152 kJ mol ^{1} and naphthalene has a resonance energy of 255 kJ mol ^{1}, predict the positions which would be occupied by bromine when phenanthrene (below) undergoes addition of Br_2.



(A) 1, 2 (B) 1, 4 (C) 3, 4 (D) 7, 8

(E) 9, 10

9. Which of the following are not aromatic :

(A) Benzene (B) Cyclo-octatetrarenyl dianion

(C) Tropyllium cation (D) Cyclopentadienyl cation

10. Which of the following statements is correct for the given molecule?

Cyclobutadiene

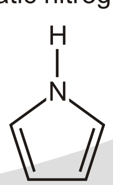
(A) Shape of the given molecule is square

(B) Shape of the given molecule is rectangular

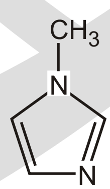
(C) Given molecule is Non-aromatic

(D) Given molecule is more stable than Non-aromatic molecules

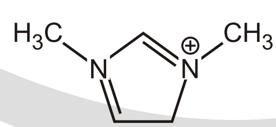
11. Choose the aromatic nitrogen heterocycles.



(i)



(ii)

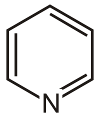


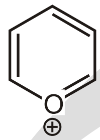
(iii)

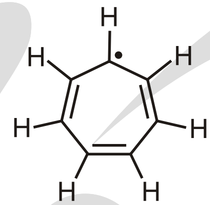
(A) i (B) ii (C) iii (D) i and ii

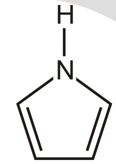
(E) i, ii and iii

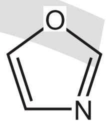
12. Which of the following compounds is not aromatic?

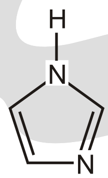
(A) 

(B) 

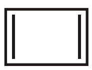
(C) 
Cycloheptatrienyl radical

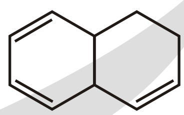
(D) 

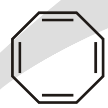
(E) 

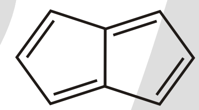
(F) 

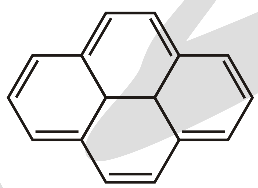
13. Which one of the following compounds is aromatic?

(A) 

(B) 

(C) 

(D) 

(E) 

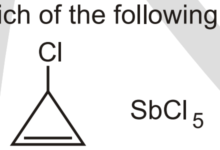
14. Aromatic properties of benzene are proved by :

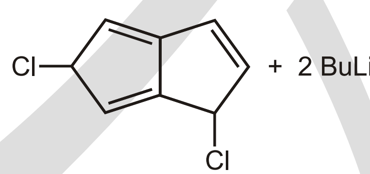
(A) Aromatic sextet theory
(B) Resonance theory
(C) Molecular orbital theory
(D) All of these

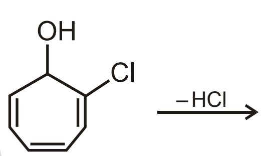
15. An aromatic compouns among other things should have a p-electron cloud containing (4n   2)   electrons where n can't be (huckel rule) :

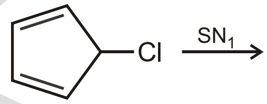
(A) 1/2
(B) 3
(C) 2
(D) 1

16. Which of the following reactions will NOT proceed?

(A) 

(B) 

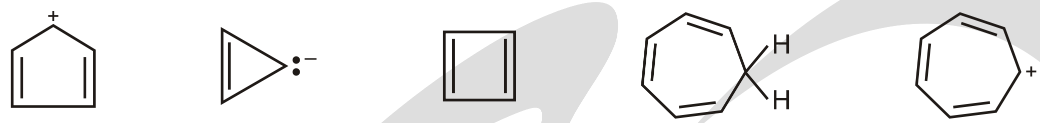
(C) 

(D) 

17. Which of the following statements regarding the cyclopentadienyl radical is correct?

(A) It is aromatic.
(B) It is not aromatic.
(C) It obeys Huckel's rule.
(D) It undergoes reactions characteristic of benzene.
(E) It has a closed shell of 6 pi-electrons.

18. Which of the following would you expect to be aromatic?



(I) (II) (III) (IV) (V)

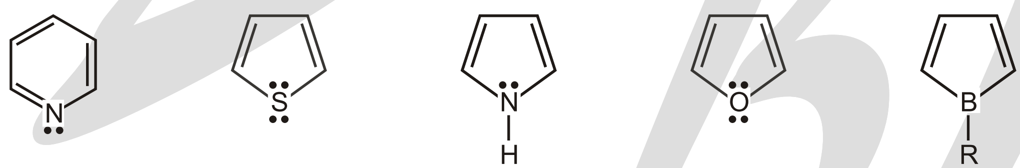
(A) I (B) II (C) III (D) IV
(E) V

19. Which of the following would you expect to be aromatic?

(I) (II) (III) (IV)

(A) I (B) II (C) III (D) IV
(E) None of these

20. Which compound would you NOT expect to be aromatic?



(I) (II) (III) (IV) (V)

(A) I (B) II (C) III (D) IV
(E) V

21. Which annulene would you NOT expect to be aromatic?

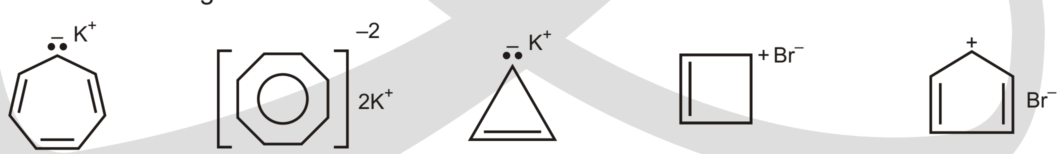
(A) [6]-Annulene (B) [14]-Annulene (C) [16]-Annulene (D) [18]-Annulene
(E) [22]-Annulene

22. Which of the following would you expect to be aromatic?

(I) (II) (III) (IV)

(A) I (B) II (C) III (D) IV
(E) None of these

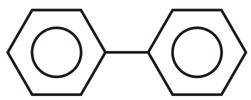
23. Which of the following structures would be aromatic?

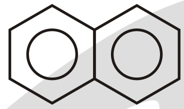


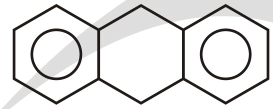
(I) (II) (III) (IV) (V)

(A) I (B) II (C) III (D) IV
(E) V

24. Compare resonance Energy of following


(a)


(b)


(c)

(A) c > a > b (B) b > a > c (C) a > b > c (D) a > c > b

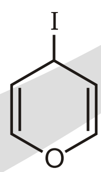
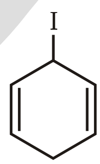
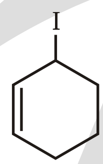
25. In the molecular orbital model of benzene, how many pi-electrons are delocalized about the ring?

(A) 2 (B) 3 (C) 4 (D) 5
(E) 6

26. In the molecular orbital model of benzene, how many pi-electrons are in bonding molecular orbitals?

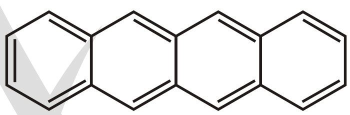
(A) 6 (B) 5 (C) 4 (D) 3
(E) 2

27. Compare the stability of carbocation which are formed when react with $AgNO_3$.

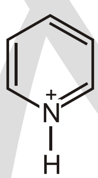




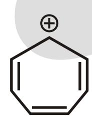
(A) a > b > c (B) a > c > b (C) b > c > a (D) b > a > c

28. How many compounds are aromatic :

(A) 

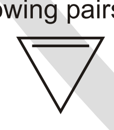
(B) 

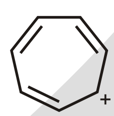
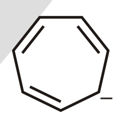
(C) 

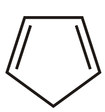
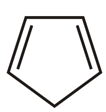
(D) 

(E) All of above

29. Which of the following pairs indicates correct order of stability?

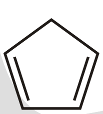
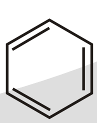
(A)  > 

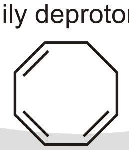
(B)  < 

(C)  > 

(D) All are correct

30. Which member of each of the following pairs of compounds is more readily deprotonated?

(A)  > 

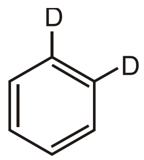
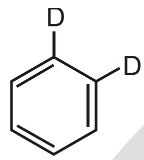
(B)  < 

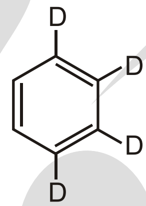
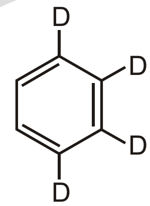
(C)  > 

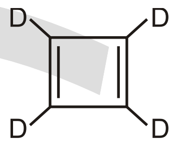
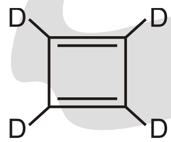
(D) All are correct

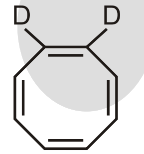
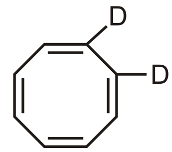
MULTIPLE CHOICE QUESTIONS

1. Which of the following represents same molecules.

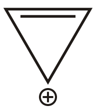
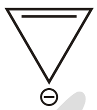
(A)  and 

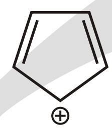
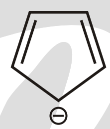
(B)  and 

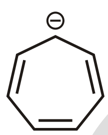
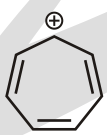
(C)  and 

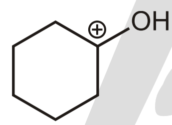
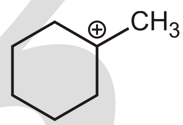
(D)  and 

2. In which pair second ion is more stable than first ?

(A)  

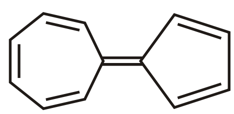
(B)  

(C)  

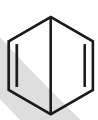
(D)  

UNSOLVED EXAMPLE

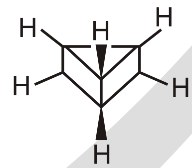
1. The following hydro carbon has an unusually high large dipole moment, explain why



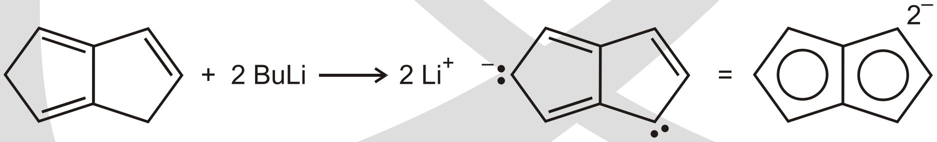
2. Between 1865 and 1890, other possible structures were proposed for benzene such as those shown here. Considering what nineteenth-century chemists knew about benzene, which is a better proposal for benzene's structure, Dewar benzene or Ladenburg benzene? Why?



Dewar benzene

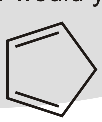


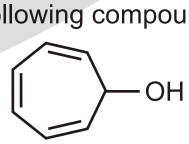
Ladenburg benzene

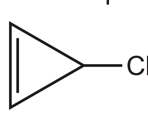
3. 

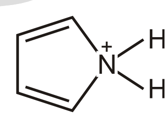
pentalene dianion

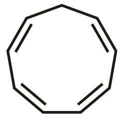
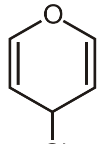
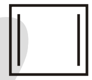
4. How would you convert the following compounds to aromatic compounds?

(a) 

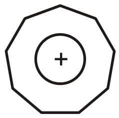
(b) 

(c) 

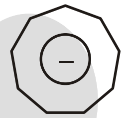
(d) 

(e)  (f)  (g) 

5. Is either of the following ions aromatic?

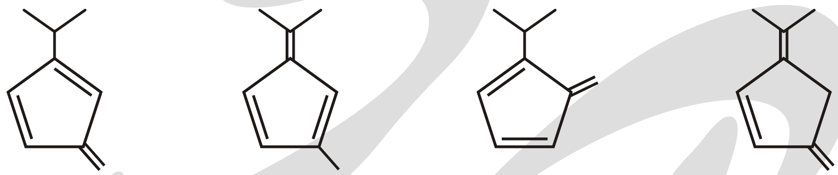
(a) 

Cyclononatetraenyl cation

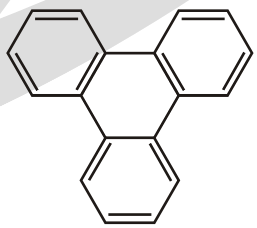
(b) 

Cyclononatetraenide anion

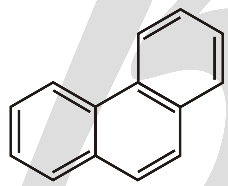
6. One of the following hydrocarbon is much more acidic than the others. Explain why?



7. Explain why compound **A** is much more stable than compound **B**.



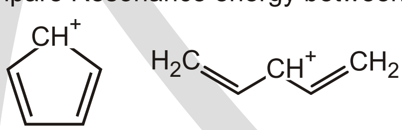
triphenylene
(A)

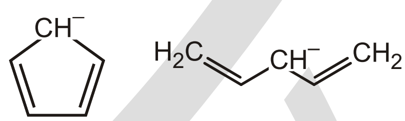


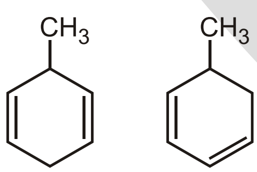
phenanthrene
(B)

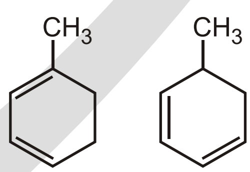
WORK SHEET - 1

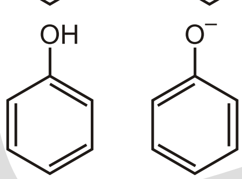
1. Compare Resonance energy between the given compounds:

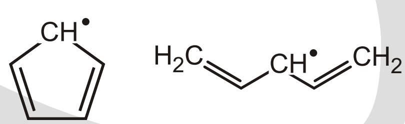
(a) 

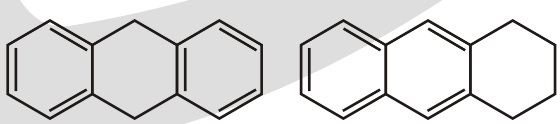
(b) 

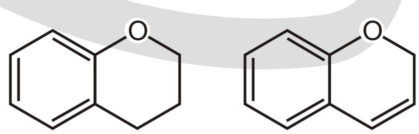
(c) 

(d) 

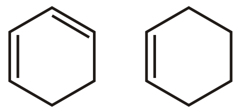
(e) 

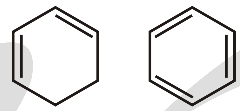
(f) 

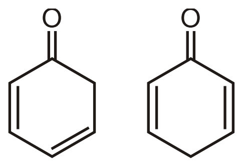
(g) 

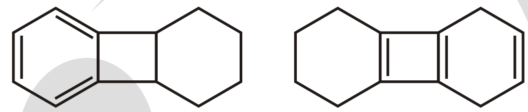
(h) 

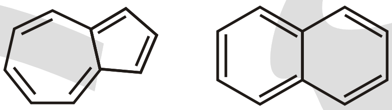
Chroman 2H-Chromene

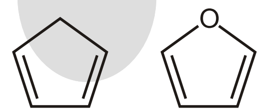
(i) 

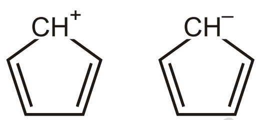
(j) 

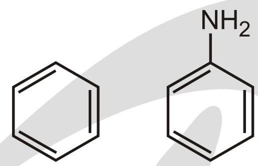
(k) 

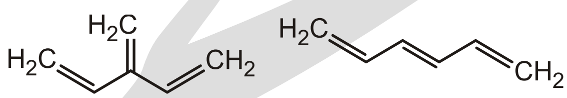
(l) 

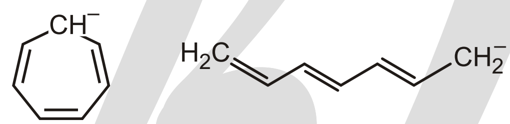
(m) 

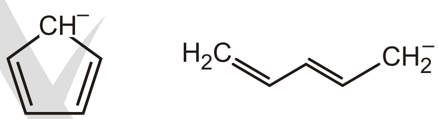
(n) 

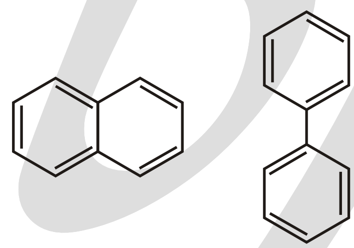
(o) 

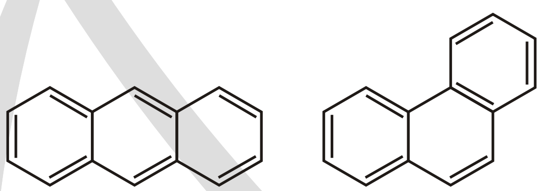
(p) 

(q) 

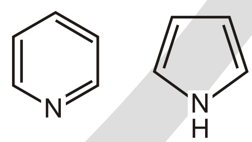
(r) 

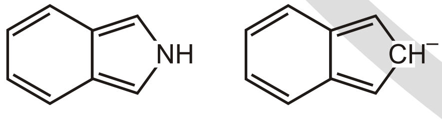
(s) 

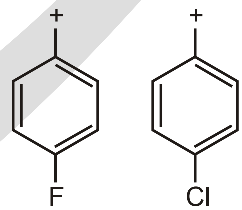
(t) 

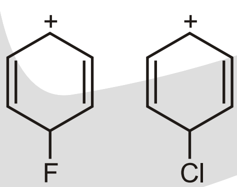
(u) 

Anthracene Phenanthrene

(v) 

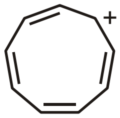
(w) 

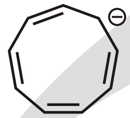
(x) 

(y) 

WORK SHEET - 2

1. Classify each as aromatic or antiaromatic or non-aromatic :

(1) 
thecyclonona-tetraenecation

(2) 
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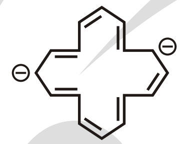
(3) 
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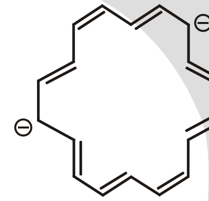
(4) 
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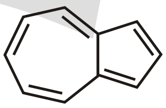
(5) 
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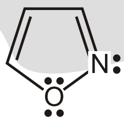
(6) 
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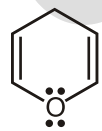
(7) 
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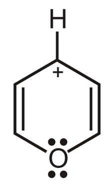
(8) 
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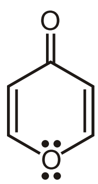
(9) 
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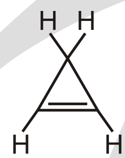
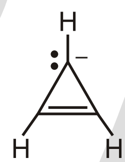
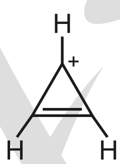
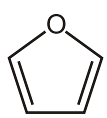
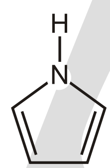
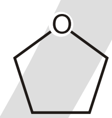
(11) 

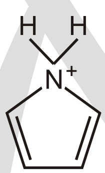
(12) 

(13) 

(14) 

(15) 

(16) 

(17) 

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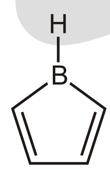
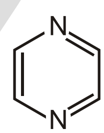
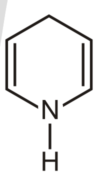
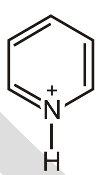
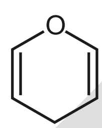
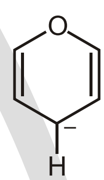
(19) 

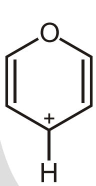
(20) 

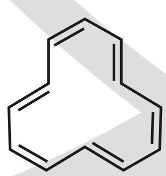
(21) 

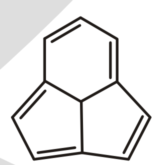
(22) 

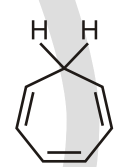
(23) 

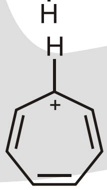
(24) 

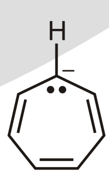
(25) 

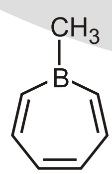
(26) 

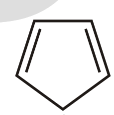
(27) 

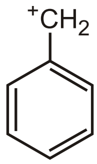
(28) 

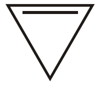
(29) 

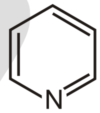
(30) 

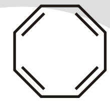
(31) 

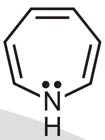
(32) 

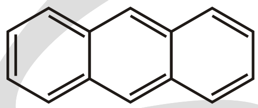
(33) 

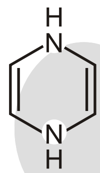
(34) 

(35) 

(36) 

(37) 

(38) 

(39) 

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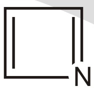
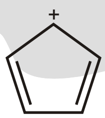
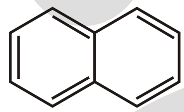
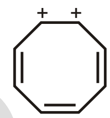
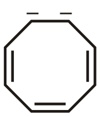
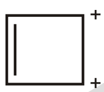
(41) 

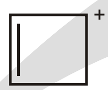
(42) 

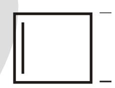
(43) 

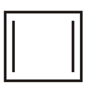
(44) 

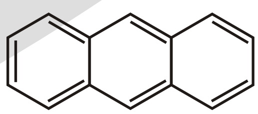
(45) 

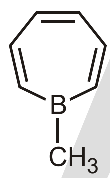
(46) 

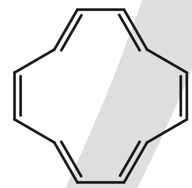
(47) 

(48) 

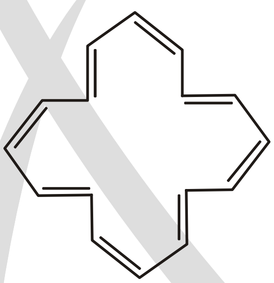
(49) 

(50) 

(51) 

(52) 

[12] annulene

(53) 

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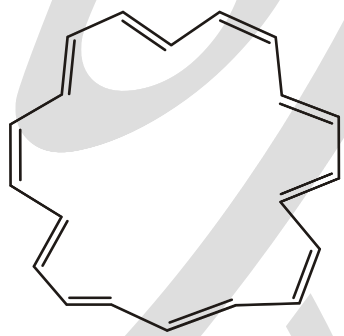
(54) 

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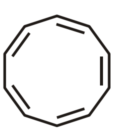
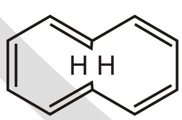
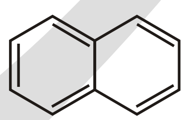
(55) 

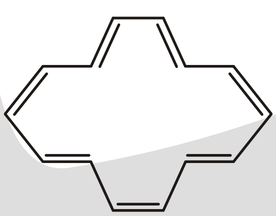
all-cis

(56) 

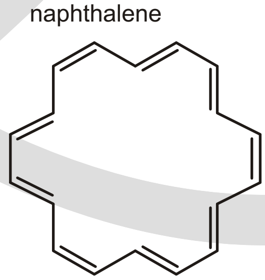
naphthalene

(57) 

naphthalene

(58) 

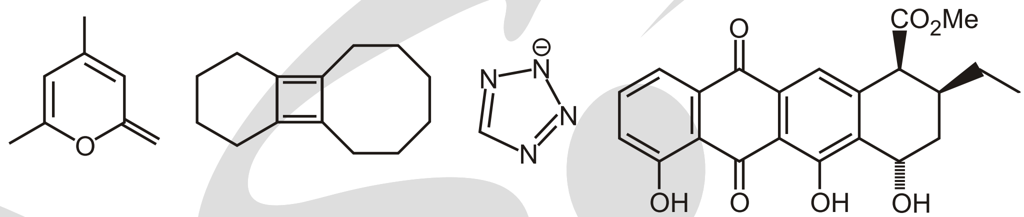
[14] annulene

(59) 

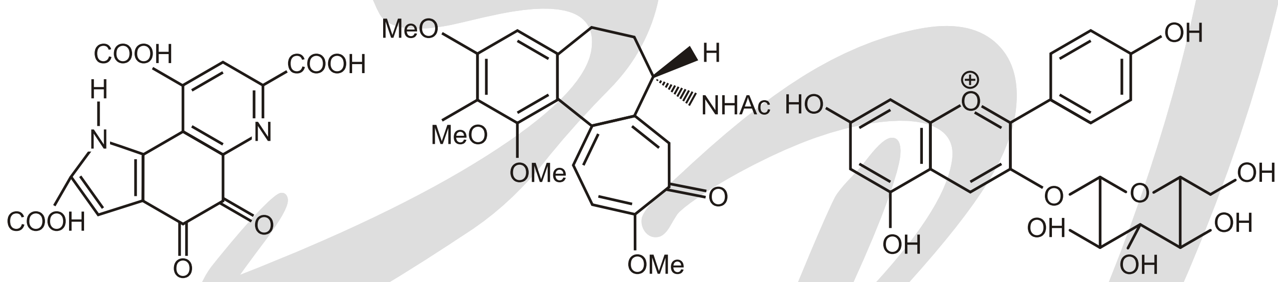
[18] annulene

SUBJECTIVE TYPE QUESTIONS

1. Which of these compounds are aromatic? Justify your answer with some electron counting. You are welcome to treat each ring separately or two or more rings together, whichever you prefer,



alkavinone : tetracycline antibotic



methoxatin coenzyme from bacteria living on methane

colchicine : compound from Autumn crocus used to treat gout

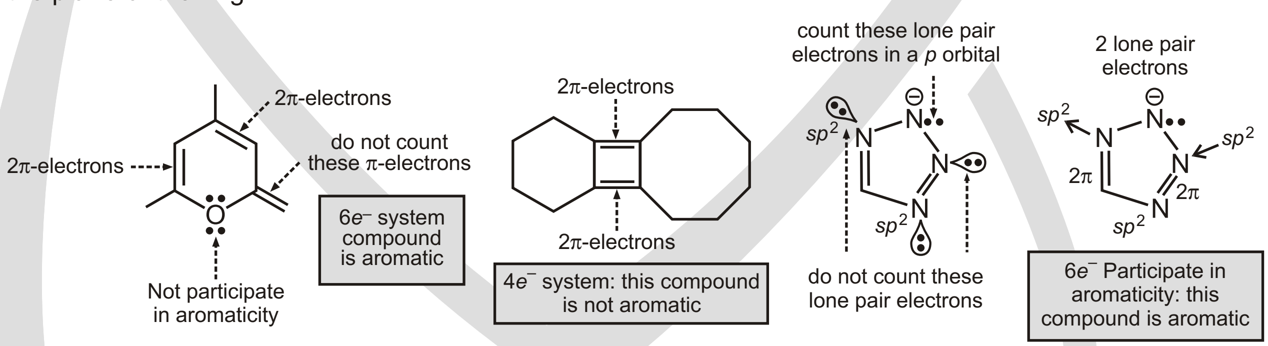
callistephin; natural red flower pigment

Purpose of the problem

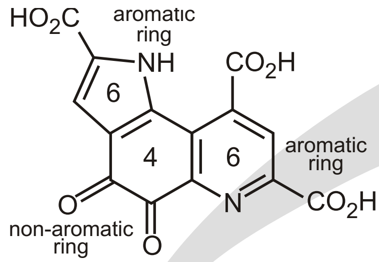
A simple exploration of the idea of aromaticity; can you count to six?

Suggested solution

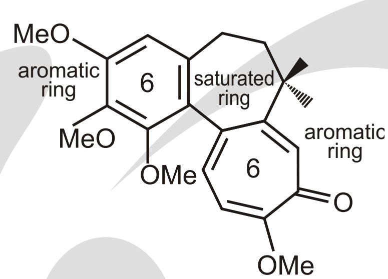
The first three compound are straight forward providing you count lone pair electron on atomic in the ring and do not count electrons outside the ring such as those in the carbonyl bond in the first compound. Nor should you count the lone pair represented by the negative charge in the third compound. They are in an sp^2 orbital in the plane of the ring.

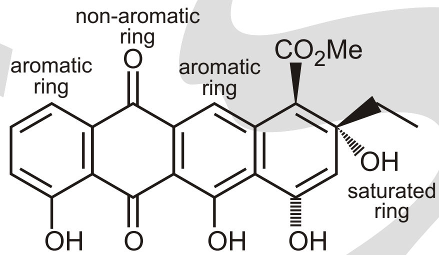


The rest offer variations on the benzene ring and each ring must be considered separately. Methoxatin has five-and six-membered ring with nitrogen in them. Count the lone pair in a *p*-orbital on the nitrogen atom in the five-membered ring but not those in an sp^2 orbital in the six-membered ring (pyridine). Both are aromatic. Colchicine has an aromatic seven-membered ring with six electrons (don't count the C O group) while callistephin has an interesting positively charged aromatic ring with three double bonds. We summarize these answer briefly by giving the number of electrons in each conjugated ring.

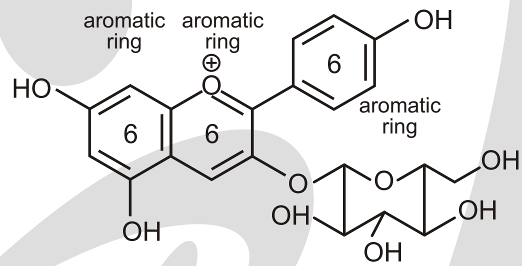


methoxatin: coenzyme from bacteria living on methane



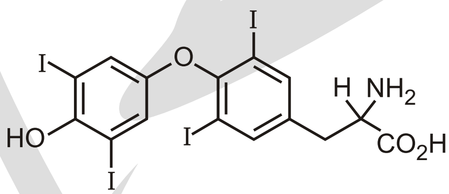


aklavinone: a tetracycline antibiotic



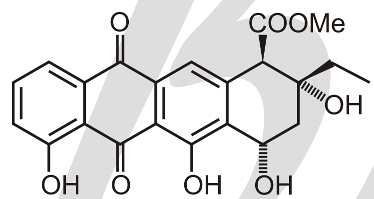
callistephin: natural red flower pigment

2. All you have to do is to spot the aromatic rings in these compounds. It may not be as easy as you think and you should state some reasons for your choice.

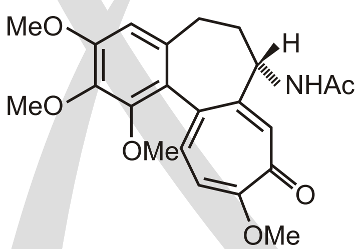


Thyroxine-T_4

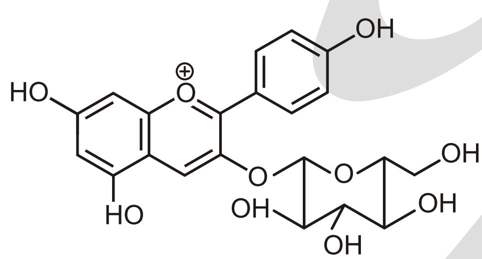
thyroxine (human hormone) Iodine carrier in thyroid gland



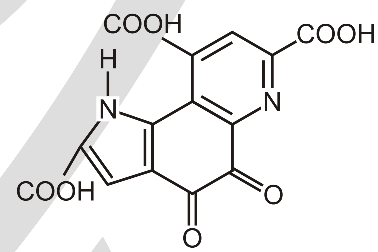
aklavinone tetracycline antibiotic [why tetracycline ?]



colchiche compound from autumn treat gout



calistephin natural red flower pigment



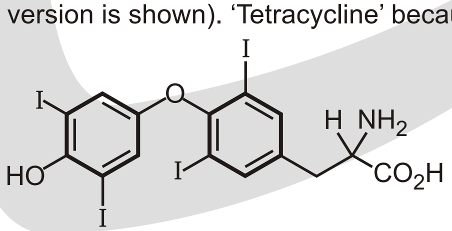
methoxatin coenzyme from bacteria living on methane

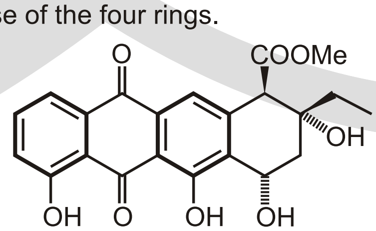
Purpose of the problem

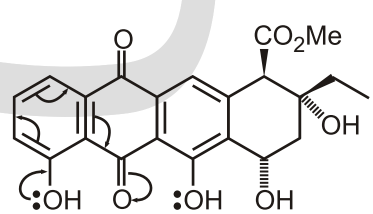
Simple exercise in counting electrons with a few hidden tricks.

Suggested solution

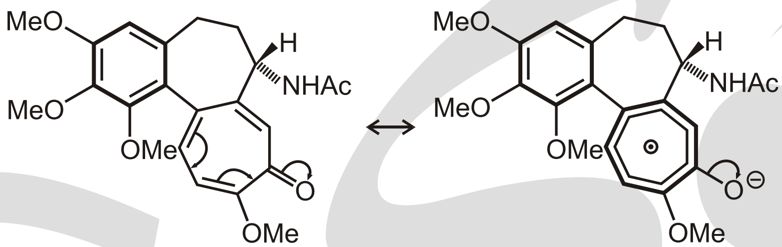
Truly aromatic rings are marked with thick lines. Thyroxine has two benzene rings, which are aromatic and c that is that. Aklavinone has again two aromatic rings, one definitely nonaromatic ring (D) and one (B) that we might argue about. However, try as you may you can't get six electrons into ring B (one extreme delocalized version is shown). 'Tetracycline' because of the four rings.

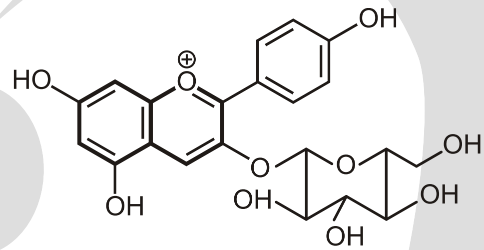






Colchicine has one benzene ring and one aromatic seven-membered ring with six electrons (don't count the electrons in the C O bond) as the delocalized structure makes clear. Callistephin has a benzene ring and a two-ring oxygen-based cation, which is like a naphthalene. You can count it as one ten electron system or as two fused six-electron systems sharing one C C bond, whichever you perfer.



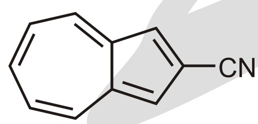


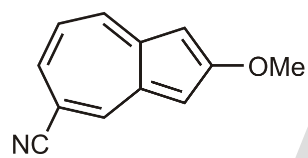
3. Azulene has an unexpectedly high dipole moment. Explain.

4. Compare the dipole moments of compounds in each of the following pairs.

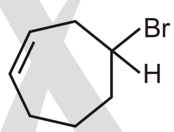
(a) *p*-Toluidine and *p*-Anisidine
(b) Vinyl bromide and Ethyl bromide
(c) 2,3-Diphenylcyclopropenone and Acetophenone
(d) MeCl and MeF
(e) *p*-chlorophenol and *p*-Fluorophenol
(f) Tropolone and 2-Hydroxytropolone

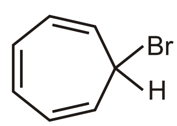
5. Compare the dipole moments of the following compounds with reasons.

(a) 

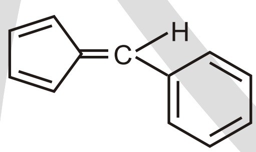
and 

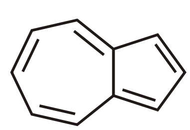
(i) (ii)

(b) 

and 

(iii) (iv)

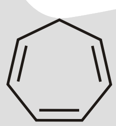
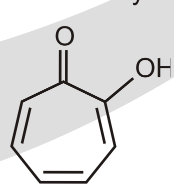
(c) 

and 

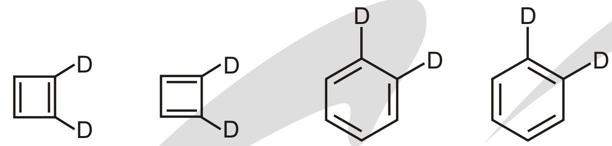
(v) (vi)

6. Predict which member of each of the following pairs of compounds has higher resonance energy and justify our choice :

(a) Anthracene or phenanthrene
(b) Ammonium acetate or acetamide
(c) Cyclooctatetraene or styrene
(d) Benzene or hexamethylbenzene
(e) *p*-benzoquinone or benzaldehyde
(f) furan or thiophene

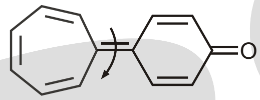
(g) 
or 

7. Using the theory of aromaticity, explain the fact that *A* and *B* are different compounds, but *C* and *D* are identical?

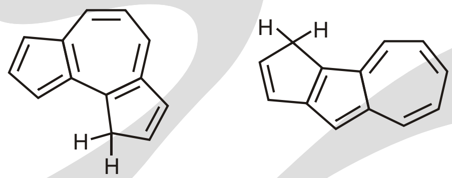


8. Account for the following observations.

(a) The barrier for rotation about the marked bond in the following compound is only about 14 kcal/mol.



(b) Hydrocarbon *A* (pK – 14) is much more acidic than *B* (pK-22)

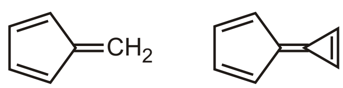


(A) (B)

(c) Cyclopentadienone is a kinetically unstable molecule.

9. (a) In what direction is the dipole moment in fulvene? Explain.

(b) In what direction is the dipole moment in calicene? Explain.



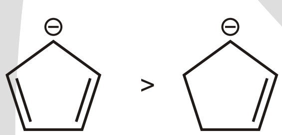
fulvene calicene

10. Why is [18] annulene more stable than [14] annulene ?

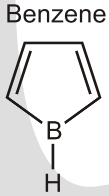
Answers

Single Choice Questions

1. (C)	**2.** (C)	**3.** (C)	**4.** (D)	**5.** (C)	**6.** (B)	**7.** (E)	**8.** (E)
9. (B)	**10.** (B)	**11.** (D)	**12.** (C)	**13.** (E)	**14.** (D)	**15.** (A)	**16.** (D)
17. (B)	**18.** (E)	**19.** (B)	**20.** (E)	**21.** (C)	**22.** (A)	**23.** (B)	**24.** (D)
25. (E)	**26.** (A)	**27.** (A)	**28.** (E)	**29.** (A)	**30.** (C)		

1. 

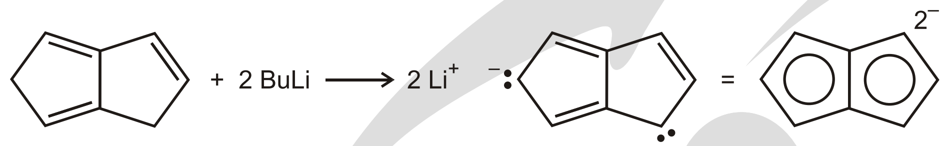
2. Benzene has highest resonance energy.

3.  Contain 4 electrons is the loop undergoing resonance and boron contains vacant p orbital to undergo complete resonance.

Multiple Choice Questions

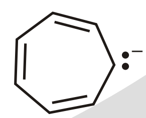
1. (A, B, C) **2.** (B, C)

Unsolved Example

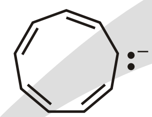
1. It exist in zewterion form
2. Dewar
3. 

Pentalene dianion

4. (a) NaH (b) HI (c) $AlCl_3$ (d) NaOH
   (e) NaH (f) $AlCl_3$ (g) 2Na
5. (a) The cycloheptatrienyl anion has 8 electrons and does not obey Huckel's rule; the cyclononatetraenyl anion with 10 *p* electrons obeys Huckel's rule.

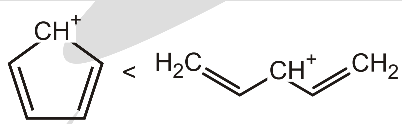


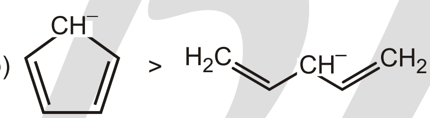
Cucloheptatrienyl anion
8π-electrons
Antiaromatic

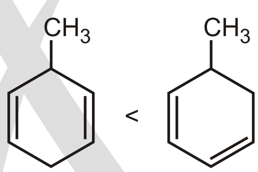


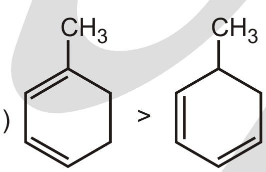
Cyclononatetraenyl anion
10π-electrons
Aromatic

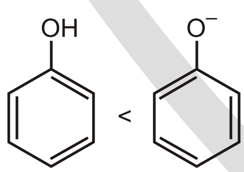
Work Sheet - 1

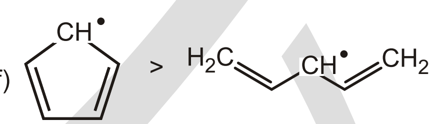
1. (a) 

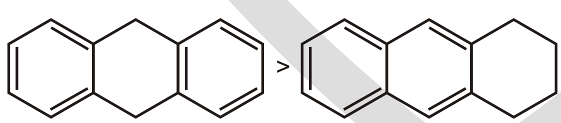
(b) 

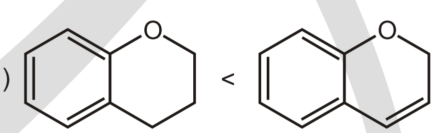
(c) 

(d) 

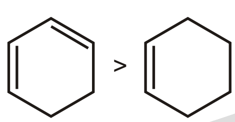
(e) 

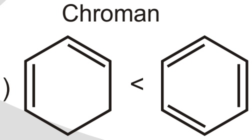
(f) 

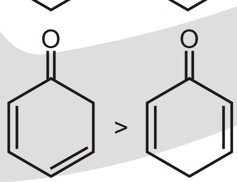
(g) 

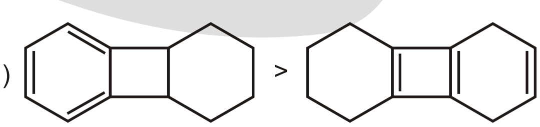
(h) 

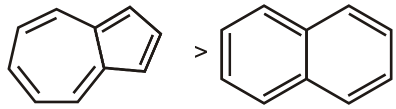
Chroman 2H-Chromene

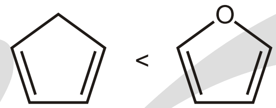
(i) 

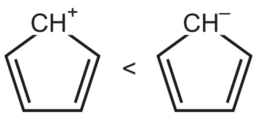
(j) 

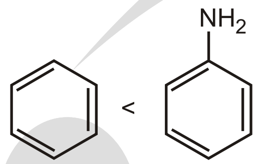
(k) 

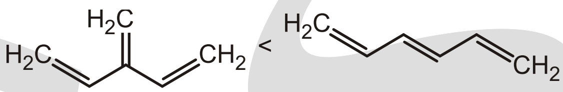
(l) 

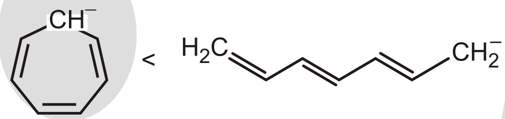
(m) 

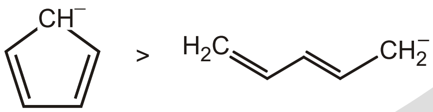
(n) 

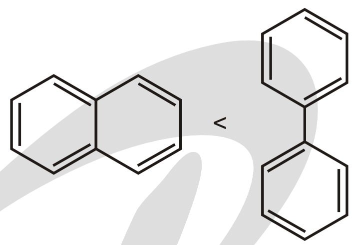
(o) 

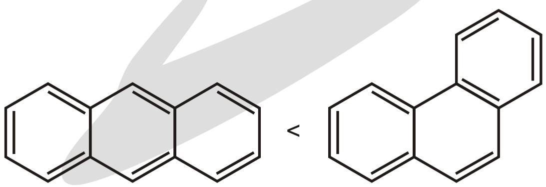
(p) 

(q) 

(r) 

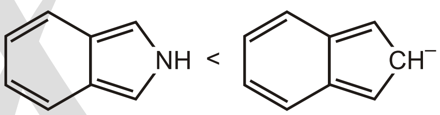
(s) 

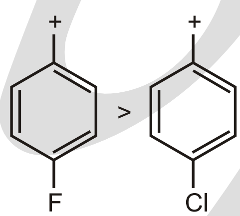
(t) 

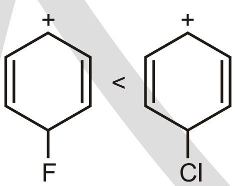
(u) 

Anthracene Phenanthrene

(v) 

(w) 

(x) 

(y) 

Work Sheet - 2

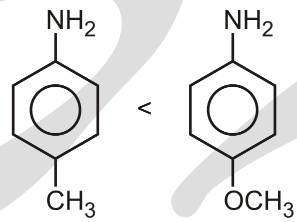
1. (1) Anti Aromatic (2) Aromatic (3) Aromatic (4) Anti Aromatic
(5) Aromatic (6) Aromatic (7) Non Aromatic (8) Aromatic
(9) Aromatic (10) Non Aromatic (11) Aromatic (12) Anti Aromatic
(13) Aromatic (14) Aromatic (15) Aromatic (16) Non Aromatic
(17) Non Aromatic (18) Non Aromatic (19) Anti Aromatic (20) Aromatic
(21) Non Aromatic (22) Aromatic (23) Non Aromatic (24) Anti Aromatic
(25) Aromatic (26) Anti Aromatic (27) Aromatic (28) Non Aromatic
(29) Aromatic (30) Anti Aromatic (31) Aromatic (32) Anti Aromatic
(33) Aromatic (34) Non Aromatic (35) Aromatic (36) Anti Aromatic
(37) Anti Aromatic (38) Aromatic (39) Anti Aromatic (40) Anti Aromatic
(41) Anti Aromatic (42) Anti Aromatic (43) Aromatic (44) Aromatic

(45) Aromatic	(46) Aromatic	(47) Non Aromatic	(48) Aromatic
(49) Anti Aromatic	(50) Aromatic	(51) Aromatic	(52) Anti Aromatic
(53) Anti Aromatic	(54) Non Aromatic	(55) Non Aromatic	(56) Non Aromatic
(57) Aromatic	(58) Aromatic	(59) Aromatic	

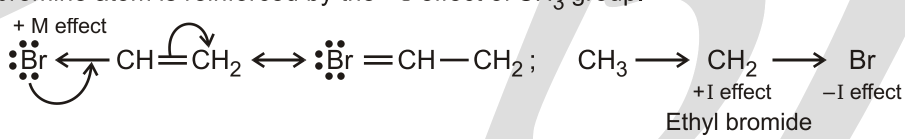
Subjective Type Questions

3. Azulene is a bycyclic compound with 10 -electrons. Distribution of these electrons between the two rings generates an aromatic system comprising a cycloheptatrienyl cation (tropylium ion) and a cyclopentadienyl ion. Since this aromatic system is quite stable, azulene remains mostly in bipolar form and shows high dipole moment value.

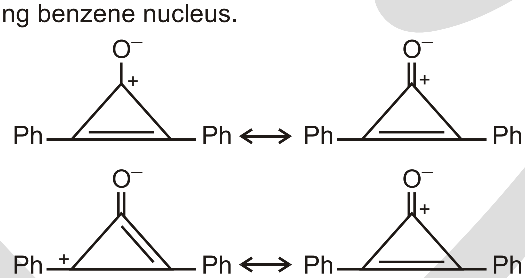
4. (a) *p*-Anisidine has greater dipole moment than *p*-Toluidine because in case of *p*-anisidine. Delocalization (*M*) effect is more pronounced due to the presence of an —NH_2 group having a lone pair of electrons on the nitrogen atom.



(b) Vinyl bromide has less dipole moment than ethyl bromide. In the case of vinyl bormide the *I* effect of bromine atom is opposed by the *M* effect due to pi-electrons delocalization. In case of ethyl bromide, *I* effect of the bromine atom is reinforced by the *I* effect of CH_3 group.

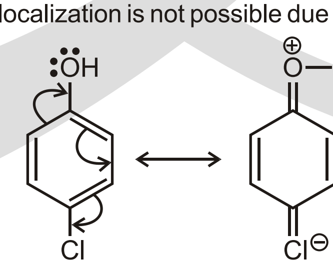


(c) 2,3-Diphenylcyclopropenone has higher dipole moment than acetophenone. Cyclopropenone system can assume aromatic stability due to delocalization. The generated cyclopropenium ion is further stabilized by the resonance involving benzene nucleus.

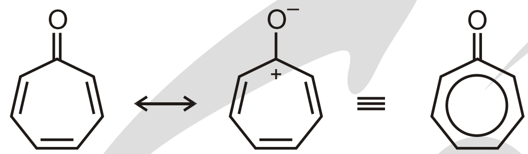


(d) CH_3Cl has higher dipole moment than CH_3F, because the C — Cl bond distance is much larger than the C — F bond distance in CH_3F, although CH_3F is more polar due to greater electronegativity of F atom. We know that dipole moment charge distance [*q* *d*].

(e) *p*-Chlorophenol has higher dipole moment than *p*-fluorophenol. The reason is that in case of *p*-chlorophenol, extended resonance is possible due to the available vacant *d*-orbital with the chlorine atom. In case of *p*-fluorophenol, extended delocalization is not possible due to the absence of the orbital with the fluorine atom.

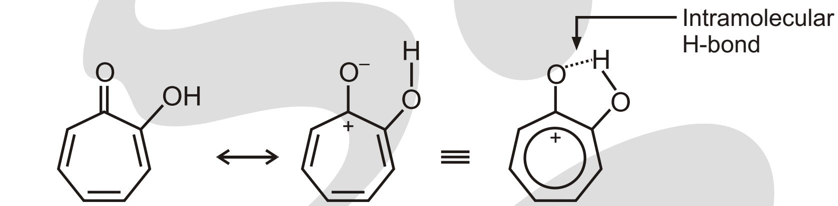


(f) Tropolone exhitibs aromatic character in its ionic form and hence has dipole moment. 2-Hydroxytropolone also exhibits greater dipole moment because of its aromatic ionic form but that ionic form further stabilized by intramolecular hydrogen bonding.



Tropolone

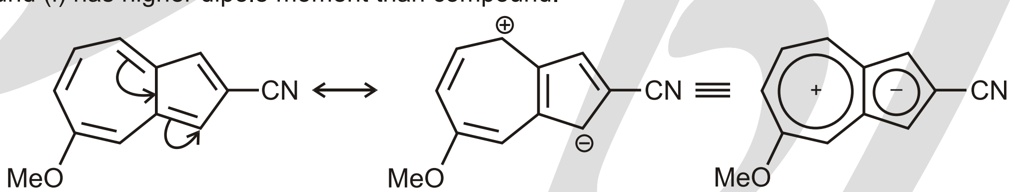
6π electron
atomatic system



2-Hydroxy tropolone

6π electron
atomatic system

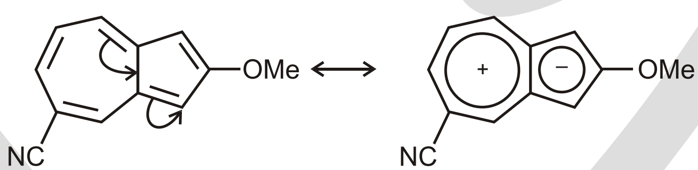
5. (a) These two are substituted azulenes. Due to redistribution of pi-electrons, the seven-membered ring becomes electron deficient and the five-membered ring becomes electron rich. In this form, azulene exhibits aromatic stability. Therefore –OMe group can more stabilize the electron deficient ring. Therefore, compound (i) has higher dipole moment than compound.



(i) (i) (i)

Stabilized molecule

(ii)



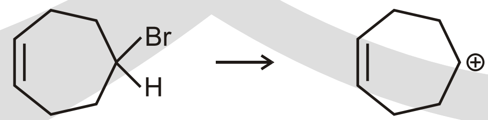
(ii) (ii)

Less stable

(b) Between the compounds (iii) and (iv), (iv) is more polar and stable. Therefore, it has greater dipole moment.

(iv)

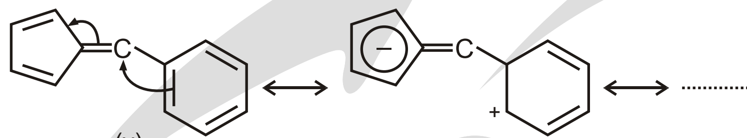
Stablize by
aromatic nature



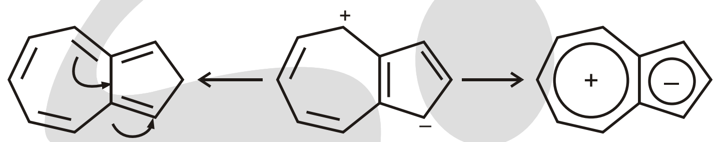
(iii)
Less ionic

Non aromatic unstable

(c) In this case, the compound (v) has higher dipole moment. In compound (v), extended resonance is possible along with the development of aromatic character to the five membered-ring. The compound (vi) is azulene and is also ionic due to delocalization of -electrons however charge separation is less.



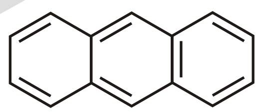
(v)



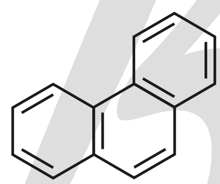
(vi)

6. It is known that the more stable a reasonating structure of the more would be its resonance energy and vice versa. Therefore, we will have to find out the more stable structure in each case of the compound in the above mentioned pairs of compounds.

(a) Between anthracene and phenanthrene, phenantherene has higher resonance energy. It has got the maximum number of benzenoid rings and can have greater number of resonating structures. Structures of anthracene and phenanthrene are given here.



Anthracene



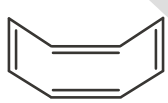
Phenanthrene

(b) Between ammonium acetate and acetamide, resonance is more effective in case of acetamide. In case of ammonium acetate, the negative charge on the oxygen atom of the acetate ion is not sufficiently free because of ammonium ion and therefore, resonance is inhibited. Therefore, acetamide has higher resonance energy.

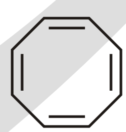
$$CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-O-\overset{\oplus}{N}H_4 \qquad CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2$$

Ammonium acetate　　Acetamide

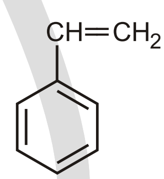
(c) Cyclooctatetraene almost always remains in non-planar tub-like structure with little scope for resonance. Planar structure of cyclooctatetraene is an unstable anti-aromatic system. Styrene is a planar molecule with an olefinic double conjugated to a phenyl ring and give a large number of resonating structures. Therefore, styrene has higher resonance energy,



Cyclooctatetraene
(Tub-like non-planar structure)



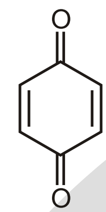
Cyclooctatetraene
(Planar, anti-aromatic structure)



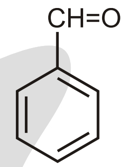
Styrene (Planar)

(d) Between beznene and hexamethylbenzene, hexamethylbenzene is more stable due to hyperconjugation. Therefore, it has higher resonance energy.

(e) *p*-benzoquinone is a crossed-conjugated system and has no aromatic stability. Benzaldehyde is an aromatic compound with an extended conjugation. Therefore, benzaldehyde is a very stable molecule having higher resonance energy.



p-Benzoquinone

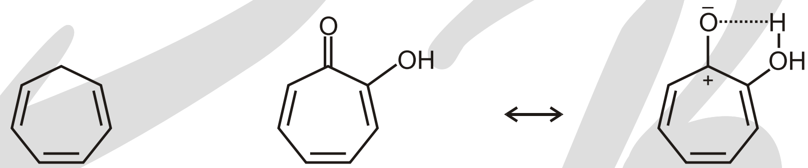


Benzaldehyde

(f) Between furan and thiophene, thiophene has higher aromatic character due to greater polarizability of the electron pair on the sulphur atom. Therefore, thiophene is more stable and has higher resonance energy.

Furan (Less effective resonance) Thiophene (More effective resonance)

(g) Between cyclohepta-1, 3, 5-triene and 2-hydroxycyclohepta-2, 4, 6-trienone, the latter is more stable, because it can form a stable aromatic system by the ionization of the ketonic group followed by intramolecular hydrogen bonding. Therefore, 2-hydroxycyclohepta-2, 4, 6-trienone has higher resonance energy.

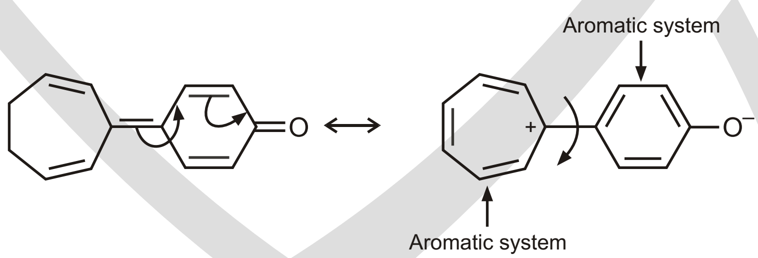


Cyclohepta-1, 3, 5-triene 2-Hydroxycyclohepta-2, 4,6-trienone Cycloheptatrienyl cation from (Very stable due to the formation of aromatic system)

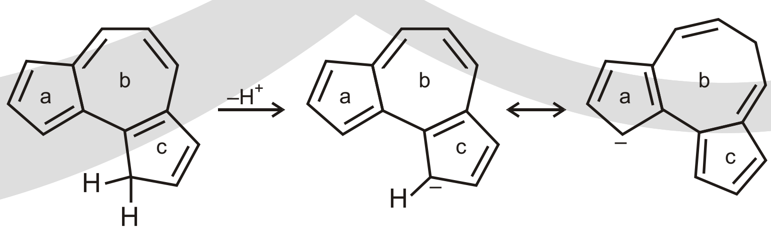
7. According to the condition of aromaticity, the contributing structures should differ in -framework. Now in case of compounds *A* and *B*. They differ in framework and antiaromatic system and one cannot be transformed into another by delocalization of -bonds. Two deuterium atoms are differently attached to the cyclobutadiene systems. Both *A* and *B* represent antiaromatic system but are not the same compounds.

The compounds *C* and *D* represent aromatic systems and are interconvertible by delocalization. Therefore they represent the same compound.

8. In the case of (a) delocalization of -electrons gives a stable aromatic system and concomitantly the double bond joining the two rings becomes a single bond appreciably. This is why rotational barrier between the two rings is only 14 kcal/mol.

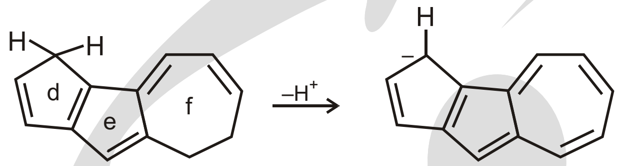


In problem (b), the acidity of compound depends on the stabilities of the conjugate base (carbanion after the loss of a proton.



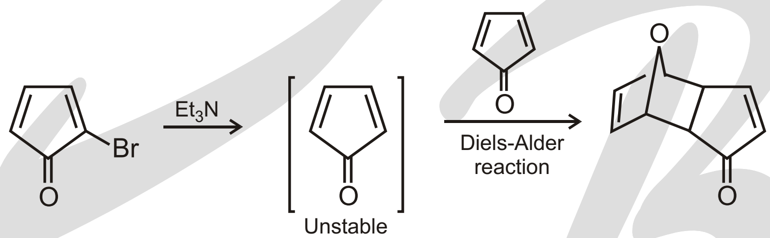
In compound (I), the terminal five-membered rings (a, c) can assume stable cyclopentadienyl anion (aromatic anion) by delocalization, after deprotonation from the sp^3 carbon.

II compound (II), only one terminal ring (d) can have aromatic stability after deprotonation, being converted into cyclopentadienide ion. further delocalization makes the cyclohetptatrienyl ring antiaromatic.

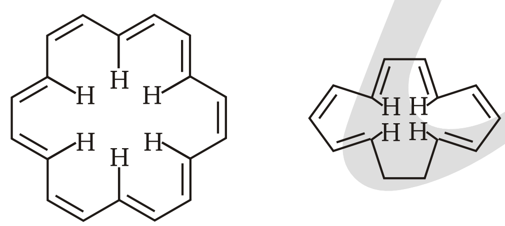


This is why, (I) is more acidic than (II).

(c) Cyclopenta-2, 3-dienone is a very unstable compound because by electromerization of the C O group, the ring becomes an antiaromatic system and consequently its reactivity is enhanced requiring less activation energy (kinetic instability). In fact, attempt to prepare cyclopentadienone by treatment of 5-bromocyclopentadienone with Et_3N : gave virtually a quantitative yield of Diels-Alder adduct showing extreme instability of the cyclopentadienone compound.



10. The structure of [18] annulene and [14] annulene are given here.



[18] Annulene [14] Annulene

Both [18] annulene and [14] annulene are aromatic in the sense that both of them satisfy the so called Huckel's rule for exhibiting aromatic character. however, because of a greater number of carbon atoms in the [18] annulene ring system, interannular space is more and as a result of which interannular steric greater aromatic character. In case of [14] annulene, interannular steric interactions of hydrogen atoms is large alnd the system assumes some non-planarity. This inhibits delocalization of the -electron system and reduce aromatic character.

❑❑❑

CHAPTER

21

Acidic and Basic Strength

ACID-BASE REACTIONS

To appreciate the reason for teaching acid-base chemistry early in the course, we need to first have a very simple understanding of what acid-base chemistry is all about. Let's summarize with a simple equation:

$$HA \rightleftharpoons H^+ + A^-$$

In the equation above, we see an acid (HA) on the left side of the equilibrium and the conjugate base (A) on the right side. HA is an acid by virtue of the fact that it has a proton (H) to give. A is a base by virtue of the fact that it wants to take its proton back (acids give protons and bases take protons). Since A is the base that we get when we deprotonate HA, we call A the conjugate base of HA.

► **Acid and conjugate base strength**
- The stronger the acid HA, the weaker its conjugate base, A^-
- The stronger the base A^-, the weaker its conjugate acid AH

Acids

Regarding **anionic stability,** there are many relevant factors. Among these are external influences such as **solvent effects.** Specifically, a **polar solvent** has the ability to stabili ze ionic species through charge – charge interactions or charge – heteroatom interactions. Conversely, a nonpolar solvent generally inhibits formation of charged species because it cannot interact with the ions.

$$HA \rightleftharpoons H^{\oplus} + A^{\ominus}$$

General representation of acid dissociation

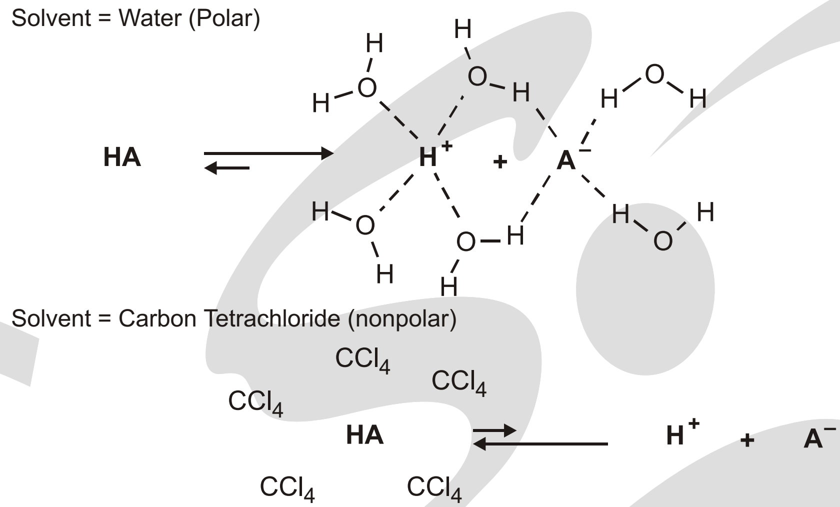
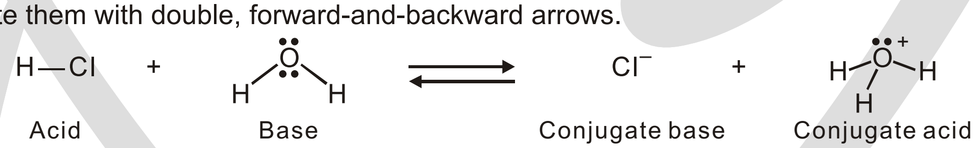


Figure 21.1 : Solvent effects on acid dissociation.

ACIDS AND BASES : THE BRONSTED-LOWRY DEFINITION

Perhaps the most important of all concepts related to electronegativity and polarity is that of acidity and basicity. We'll soon see, in fact, that the acid–base behavior of organic molecules explains much of their chemistry. You may recall from a course in general chemistry that two definitions of acidity are frequently used : the Bronsted–Lowry definition and the Lewis definition. We'll look at the Brønsted–Lowry definition in this and the following three sections and then discuss the Lewis definition.

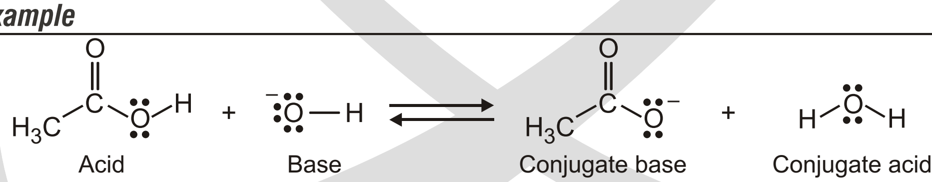
A **Brønsted–Lowry acid** is a substance that donates a hydrogen ion, H and a Brønsted–Lowry base is a substance that accepts a hydrogen ion. When gaseous hydrogen chloride dissolves in water, for example, a polar HCl molecule acts as an acid and donates a proton, while a water molecule acts as a base and accepts the proton, yielding chloride ion (Cl) and hydronium ion (H_3O). This and other acid–base reactions are reversible, so we'll write them with double, forward-and-backward arrows.

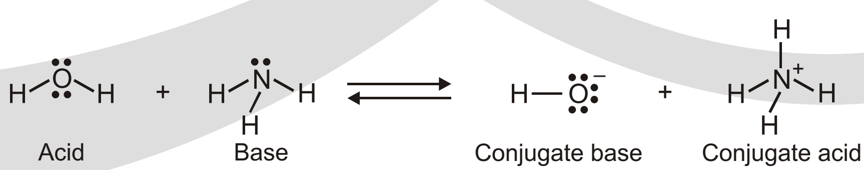


Chloride ion, the product that results when the acid HCl loses a proton, is called the **conjugate base** of the acid, and hydronium ion, the product that results when the base H_2O gains a proton, is called the conjugate acid of the base. Other common mineral acids such as H_2SO_4 and HNO_3 behave similarly, as do organic acids such as acetic acid, CH_3CO_2H. In a general sense,

$$\underset{\text{Acid}}{H{-}A} + \underset{\text{Base}}{:B} \rightleftharpoons \underset{\text{Conjugate base}}{:A^-} + \underset{\text{Conjugate acid}}{H{-}B^+}$$

Solved Example





► **For any acid and any base**

$$AH + B \rightleftharpoons BH^+ + A^-$$

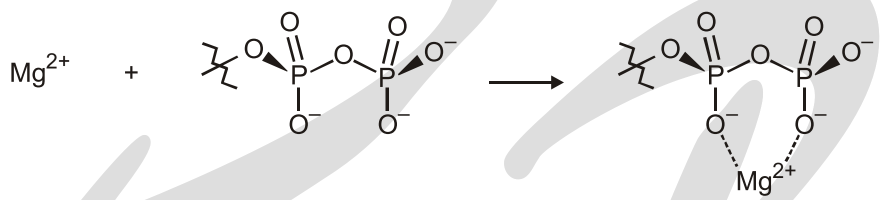
where AH is an acid and A^- is its conjugate base and B is a base and BH^+ is its conjugate acid, that is, *every acid has a conjugate base associated with it and every base has a conjugate acid associated with it.*

For example, with ammonia and acetic acid

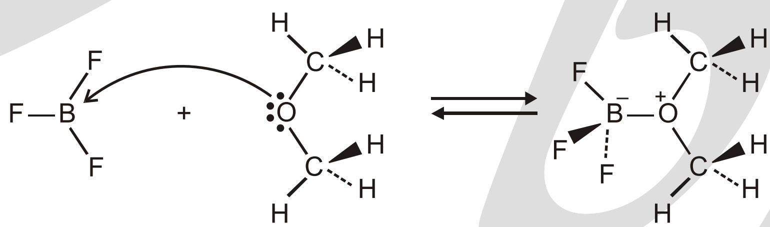
$$CH_3COOH \quad NH_3 \rightleftharpoons NH_4 \quad CH_3COO$$

ACIDS AND BASES : THE LEWIS DEFINITION

The Lewis definition of acids and bases is broader and more encompassing than the Brønsted–Lowry definition because it's not limited to substances that donate or accept just protons. A **Lewis acid** is a substance that accepts an electron pair and a **Lewis base** is a substance that donates an electron pair. The donated electron pair is shared between the acid and the base in a covalent bond.



Lewis acid — Lewis base (an organodiphosphate ion) — Acid–base complex



Boron trifluoride (Lewis acid) — Dimethyl ether (Lewis base) — Acid–base complex

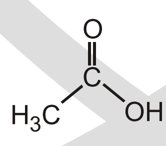
Dimethyl ether is the lewis base in the above written reaction which donates the electron pair to the valence orbital of the boron atom in BF_3, a lewis acid. Curved arrow shows the electron flow.

SOME FURTHER EXAMPLES OF LEWIS ACIDS FOLLOW

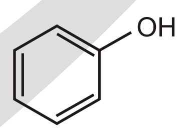
Some Lewis acids

Some neutral proton donors:

H_2O HCl   HBr HNO_3 H_2SO_4



A carboxylic acid



A phenol

CH_3CH_2OH

An alcohol

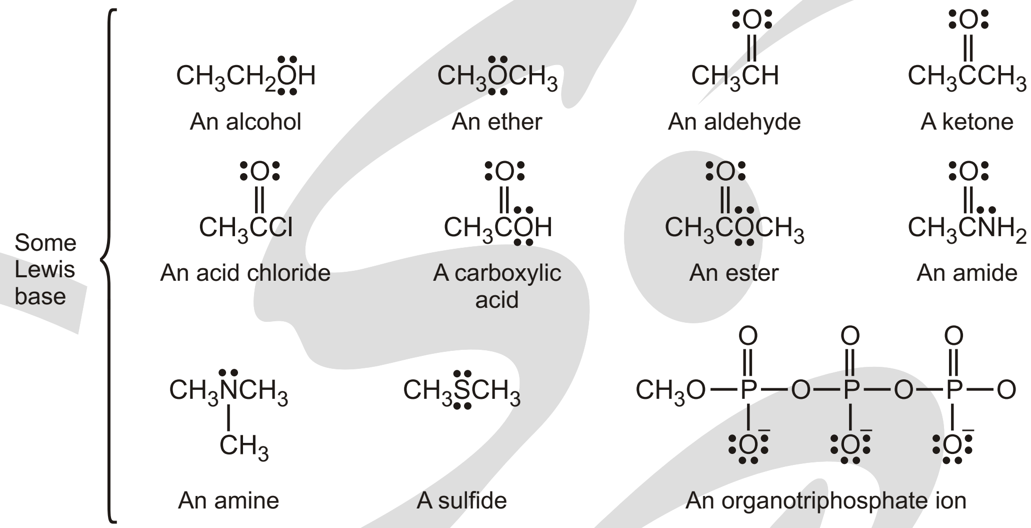
Some cations :

Li^+ Mg^{2+}

Some metal compounds :

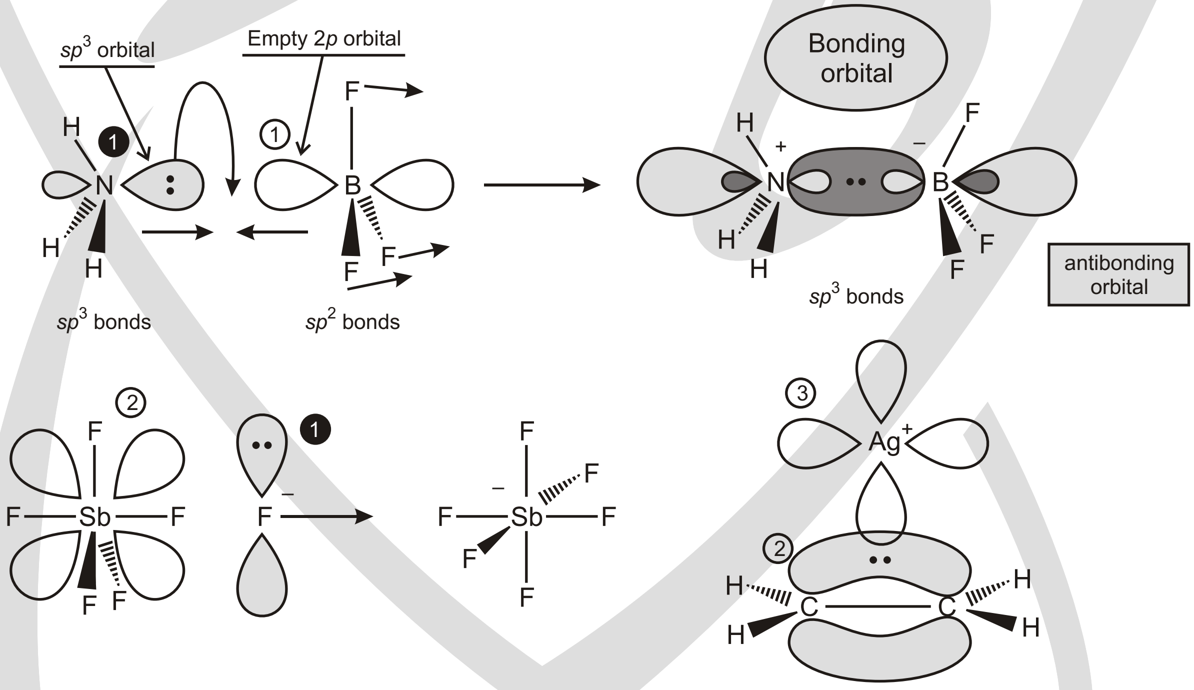
$AlCl_3$ $TiCl_4$ $FeCl_3$ $ZnCl_2$

☞ Compounds with open sextets (Empty *p* orbitals) [BF_3, BH_3] and with expandable valence shells [SbF_5, $FeBr_3$, $AlCl_3$] and Cation with vacant d orbitals [Ag , Zn^{2}].



☞ Compounds with atoms carrying unshared pair electron for example NH_3, H_2O and compound with multiple bonds CH_2 CH_2, CH CH are lewis base.

Lewis Acids and Lewis Bases



Lewis Bases

❶ Unshared pairs of electrons [NH_3, H_2O, F^{1}]

❷ orbitals [CH_2 CH_2]

Lewis Acid

① Open sextets (Empty *p* orbitals) [BF_3, BH_3]

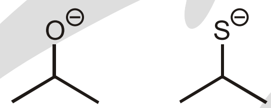
② Expandable valence shells (low-lying empty *d* orbitals) [SbF_5, $FeBr_3$, $AlCl_3$]

③ Positive ions with low-lying empty *d* orbitals [Ag , Zn^{2}, but not NH_4, Na]

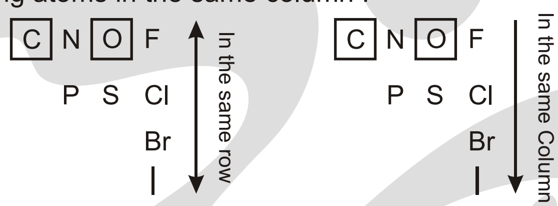
FACTOR AFFECTING THE ACIDIC STRENGTH

Factor-1 : Atom Carrying the Charge

The most important factor for determining charge stability is to ask what atom the charge is on. For example, consider the two charged compounds below :

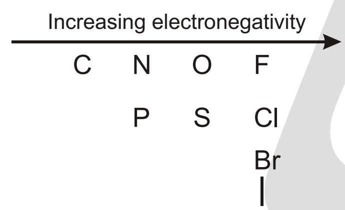


The one on the left has a negative charge on oxygen, and the one on the right has the charge on sulfur. How do we compare these? We look at the periodic table, and we need to consider two trends: comparing atoms in the same row and comparing atoms in the same column :



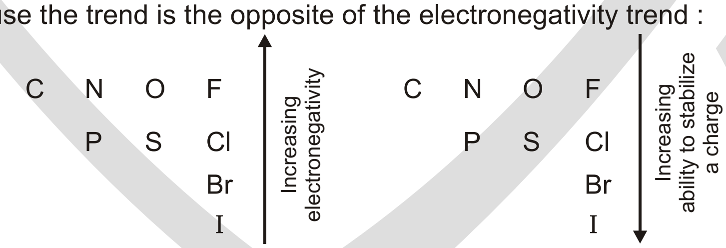
Let's start with comparing atoms in the same row. For example, let's compare carbon and oxygen :

The compound on the left has the charge on carbon, and the compound on the right has the charge on oxygen. Which one is more stable? Recall that electronegativity increases as we move to the right on the periodic table :

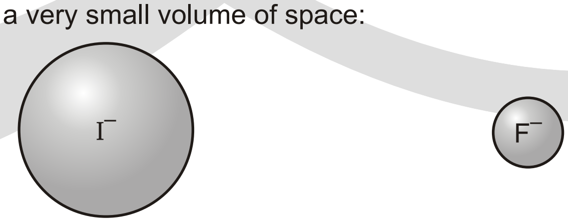


Since electronegativity is the measure of an element's affinity for electrons (how willing the atom will be to accept a new electron), we can say that a negative charge on oxygen will be more stable than a negative charge on carbon.

Now let's compare atoms in the same column, for example, iodide (I^-) and fluoride (F). Here is where it gets a little bit tricky, because the trend is the opposite of the electronegativity trend :



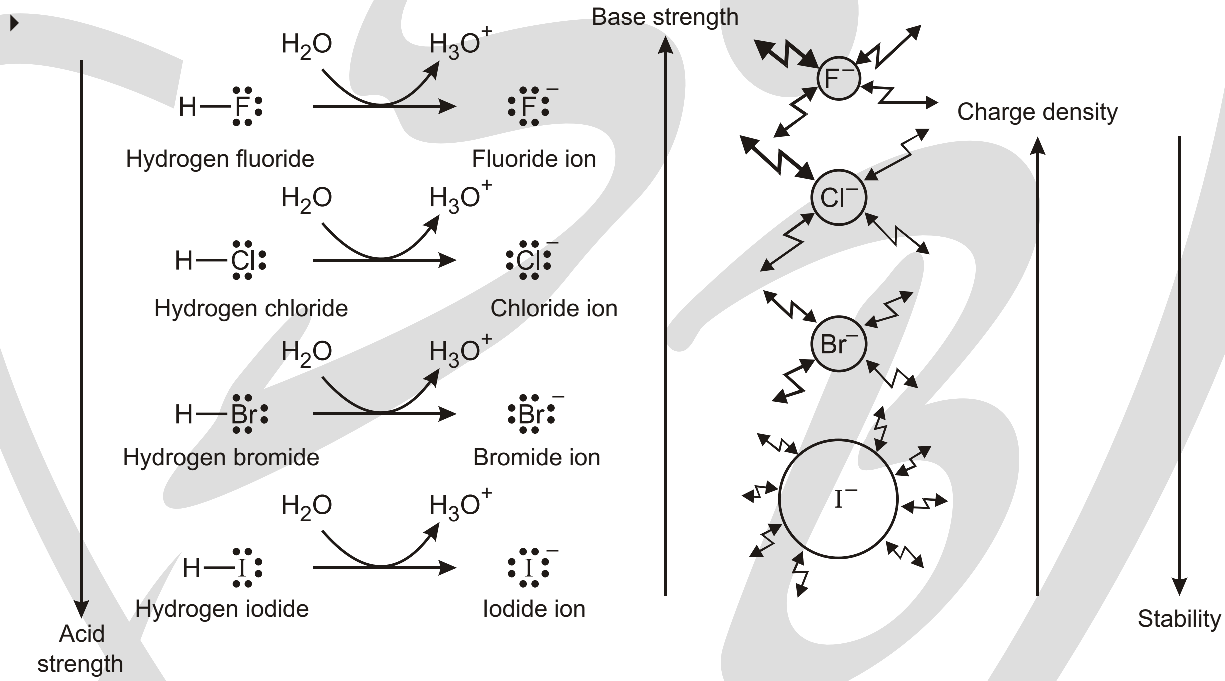
It is true that fluorine is more electronegative than iodine, but there is another more important trend when comparing atoms in the same column : the size of the atom. Iodine is huge compared to fluorine. So when a charge is placed on iodine, the charge is spread out over a very large volume. When a charge is placed on fluorine, the charge is stuck in a very small volume of space:



Even though fluorine is more electronegative than iodine, nevertheless, iodine can better stabilize a negative charge. If I^- is more stable than F , then HI must be a stronger acid than HF, because HI will be more willing to give up its proton than HF.

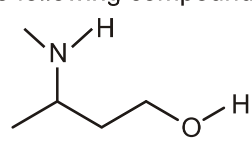
☞ To summarize, there are two important trends : electronegativity (for comparing atoms in the same row) and size (for comparing atoms in the same column). The first factor (comparing atoms in the same row) is a much stronger effect. In other words, the difference in stability between C and F is much greater than the difference in stability between I^- and F .

Solved Example



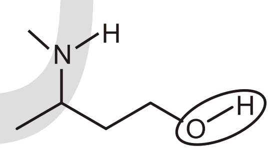
Solved Example

- Compare the two protons in the following compound. Which one is more acidic?



Ans. We begin by pulling off one proton and drawing the conjugate base that we get. Then, we do the same thing for the other proton :

Now we need to compare these conjugate bases and ask which one is more stable. In other words, which negative charge is more stable? We are comparing a negative charge on nitrogen with a negative charge on oxygen. So we are comparing two atoms in the same row of the periodic table, and the important trend is electronegativity. Oxygen can better stabilize the negative charge, because oxygen is more electronegative than nitrogen. The proton on the oxygen will be more willing to come off, so it is more acidic:



Relative acid strengths :

NH_3	<	H_2O	<	HF
pKa = 36		pKa = 15.7		pKa = 3.2

Relative elecronegativities of carbon atoms :

$$sp > sp^2 > sp^3$$

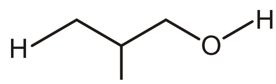
increasing electronegativity ←

Relative electronegativities :

N < O < F

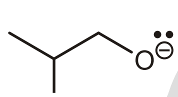
Solved Example

- Compare the two protons clearly shown in the following compound. (There are more protons in the compound, but only two are shown.) Which of these two protons is more acidic? Remember to begin by drawing the two conjugate bases, and then compare them.

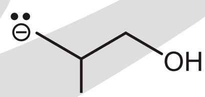


____________ Conjugate base 1   ____________ Conjugate base 2

Ans.



Conjugate base 1

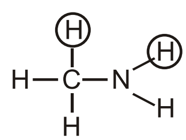


Conjugate base 2

; 1 is more stable than 2

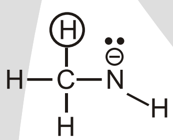
Solved Example

- Compare the two protons clearly shown in the following compound. Which of these two protons is more acidic?

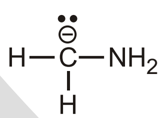


____________ Conjugate base 1   ____________ Conjugate base 2

Ans.



Conjugate base 1

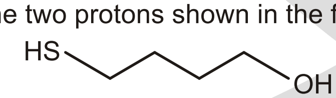


Conjugate base 2

; 1 is more stable than 2

Solved Example

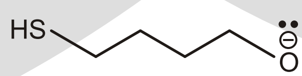
- Compare the two protons shown in the following compound and identify which proton is more acidic:



____________ Conjugate base 1   ____________ Conjugate base 2

Ans.

Conjugate base 1

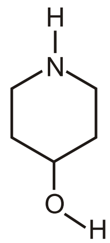


Conjugate base 2

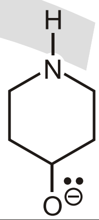
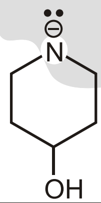
; 1 is more stable than 2

Solved Example

- Compare the two protons shown in the following compound, and identify which proton is more acidic:



Conjugate base 1   Conjugate base 2

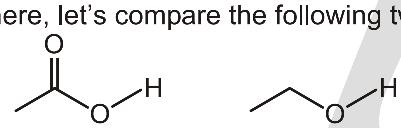
Ans.   ; 1 is more stable than 2

Conjugate base 1   Conjugate base 2

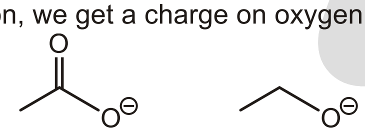
Factor-2 (Resonance) :

The last chapter was devoted solely to drawing resonance structures. If you have not yet completed that chapter, do so before you begin this section. We said in the last chapter that resonance would find its way into every single topic in organic chemistry. And here it is in acid-base chemistry.

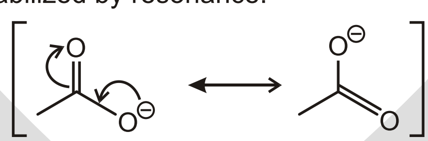
To see how resonance plays a role here, let's compare the following two compounds:



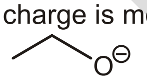
In both cases, if we pull off the proton, we get a charge on oxygen:



So we cannot use factor 1 (what atom is the charge on) to determine which proton is more acidic. In both cases, we are dealing with a negative charge on oxygen. But there is a critical difference between these two negative charges. The one on the left is stabilized by resonance:



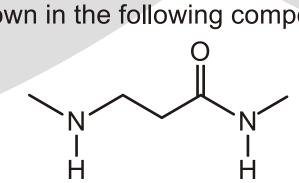
A delocalized negative charge is more stable than a localized negative charge (stuck on one atom) :

 Charge is stuck on one atom ("localized")

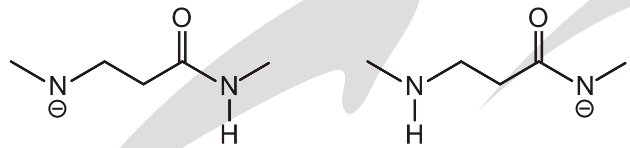
we have one molecule losing its proton for every 10,000 molecules that do not give up their proton. In the world of acidity, this is not very acidic, but everything is relative.

Solved Example

- Compare the two protons shown in the following compound. Which one is more acidic?



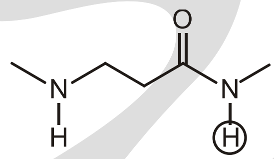
Ans. We begin by pulling off one proton and drawing the conjugate base that we get. Then we do the same thing for the other proton:



Now we need to compare these conjugate bases and ask which one is more stable. In the compound on the left, we are looking at a charge that is localized on a nitrogen atom. For the compound on the right, the negative charge is delocalized over a nitrogen atom and an oxygen atom (draw resonance structures).

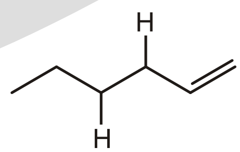
It is more stable for the charge to be delocalized, so the second compound is more stable.

The more acidic proton is that one that leaves to give the more stable conjugate base.



Solved Example

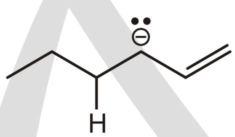
- Compare the two protons identified below. There are more protons in the compound, but only two of them are shown.



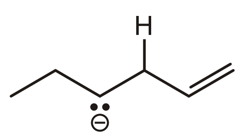
Conjugate base 1   Conjugate base 2

Identify which of these protons is more acidic, and explain why by comparing the stability of the conjugate bases.

Ans.



Conjugate base 1

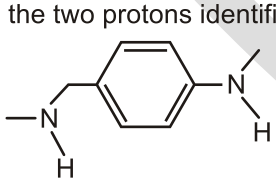


Conjugate base 2

; 1 is more stable than 2

Solved Example

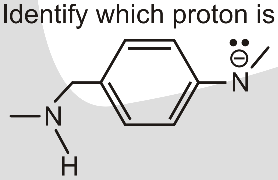
- Compare the two protons identified below :



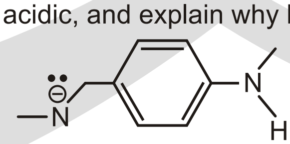
Conjugate base 1   Conjugate base 2

Identify which proton is more acidic, and explain why by comparing the stability of the conjugate bases.

Ans.



Conjugate base 1

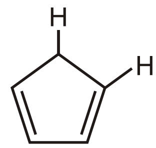


Conjugate base 2

; 1 is more stable than 2

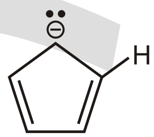
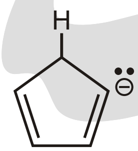
Solved Example

- Compare the two protons identified below : H



Conjugate base 1 Conjugate base 2

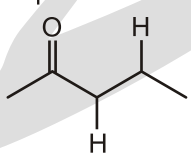
Identify which proton is more acidic, and explain why by comparing the stability of the conjugate bases.

Ans.   ; 1 is more stable than 2

Conjugate base 1 Conjugate base 2

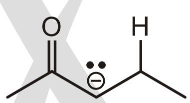
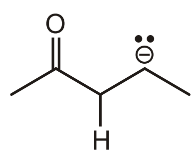
Solved Example

- Compare the two protons identified below :



Conjugate base 1 Conjugate base 2

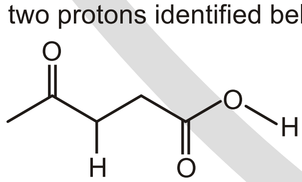
Identify which proton is more acidic, and explain why by comparing the stability of the conjugate bases.

Ans.   ; 1 is more stable than 2

Conjugate base 1 Conjugate base 2

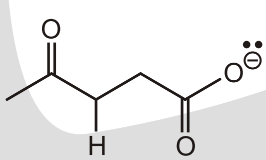
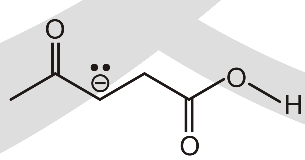
Solved Example

- Compare the two protons identified below :



Conjugate base 1 Conjugate base 2

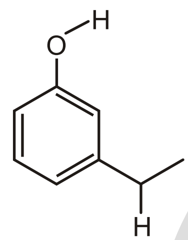
Identify which proton is more acidic, and explain why by comparing the stability of the conjugate bases.

Ans.   ; 1 is more stable than 2

Conjugate base 1 Conjugate base 2

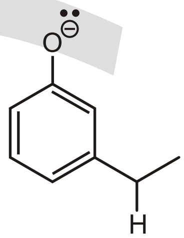
Solved Example

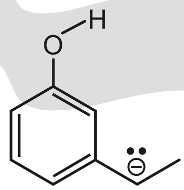
- Compare the two protons identified below :



Conjugate base 1   Conjugate base 2

Ans.





; 1 is more stable than 2

Conjugate base 1   Conjugate base 2

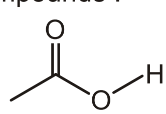
► **Get a feel of *p*K_a*s*!**

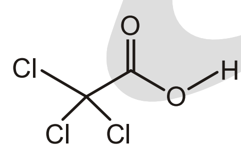
Notice that these oxygen acids have *p*K_a*s* that conveniently fall in units of 5 (approximately).

Acid	RSO_2OH	RCO_2H	PhOH	ROH
Approx. *p*K_a	0	5	10	15

Identify which proton is more acidic, and explain why by comparing the stability of the conjugate bases.

Let's compare the following compounds :

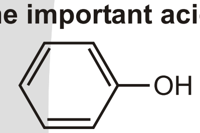
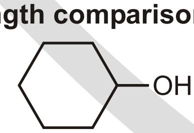




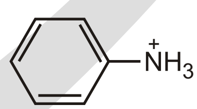
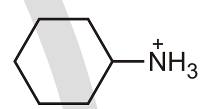
Which compound is more acidic? The only way to answer that question is to pull off the protons and draw the conjugate bases :

Let's go through the factors we learned so far. Factor 1 does not answer the problem: in both cases, the negative charge is on oxygen. Factor 2 also does not answer the problem: in both cases, there is resonance that delocalizes the charge over two oxygen atoms. **NOW WE NEED FACTOR 3.**

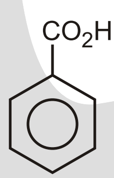
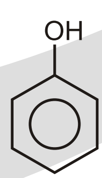
Some important acidic strength comparisons :

1.  > 

phenol *p*K_a = 10   cyclohexanol *p*K_a = 16

2.  > 

protonated aniline *p*K_a = 4.60   protonated cyclohexylamine *p*K_a = 11.2

3.  > 

Benzoic acid *p*K_a = 4.2   Phenol *p*K_a = 10

☑ SPECIAL TOPIC

Buffer Solutions

A solution of a weak acid (HA) and its conjugate base (A) is called a buffer solution.

A buffer solution will maintain nearly constant pH when small amounts of acid or base are added to it, because the weak acid can give a proton to any HO added to the solution, and its conjugate base can accept any H that is added to the solution.

can give an H^+ to HO^-

$$HA + HO^- \longrightarrow A^- + H_2O$$

$$\bar{A} + H_3O^+ \longrightarrow HA + H_2O$$

can accept an H^+ from H_3O^+

acid's pK_a is 3.75, the majority of the buffer will be in the basic form at pH = 4.2. Acetic acid with pK_a 4.76, will have more buffer in the acidic form than in the basic form. Thus, it would be better to use acetic acid/acetate buffer for your reaction.

Blood : A buffered solution

Blood is the fluid that transports oxygen to all the cells of the human body. The normal pH of human blood is ~7.4. Death will result if this pH decreases to less than ~6.8 or increases to greater than ~8.0 for even a few seconds.

Oxygen is carried to cells by a protein in the blood called haemoglobin (HbH). When haemoglobin binds O_2, hemoglobin loses a proton, which would make the blood more acidic if it did not contain a buffer to maintain its pH.

$$HbH \quad O_2 \rightleftharpoons HbO_2 \quad H$$

A carbonic acid/bicarbonate (H_2CO_3 / HCO_3) buffer controls the pH of blood. An important feature of this buffer is that carbonic acid decomposes to CO_2 and H_2O, as shown below :

$$\underset{\text{bicarbonate}}{HCO_3} \quad H \rightleftharpoons \underset{\text{carbonic acid}}{H_2CO_3} \rightleftharpoons CO_2 \quad H_2O$$

During exercise our metabolism speeds up, producing large amounts of CO_2. The increased concentration of CO_2 shifts the equilibrium between carbonic acid and bicarbonate to the left, which increases the concentration of H . Significant amounts of lactic acid are also produced during exercise, which further increases the concentration of H . Receptors in the brain respond to the increased concentration of H by triggering a reflex that increases the rate of breathing. Haemoglobin then relases more oxygen to the cells and more CO_2 is eliminated by exhalation. Both processes decrease the concentration of H in the blood by shifting the equilibrium of the top reaction to the left and the equilibrium of the bottom reaction to the right.

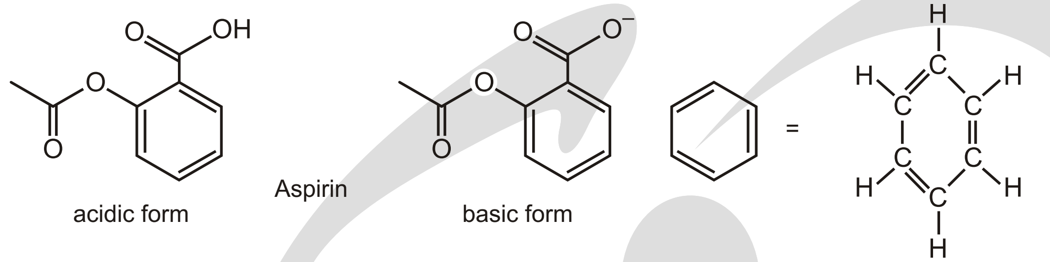
Thus, any disorder that decreases the rate and depth of ventilation, such as emphysema, will decrease the pH of the blood–a condition called acidosis. In contrast, any excessive increase in the rate and depth of ventilation, as with hyperventilation due to anxiety, will increase the pH of blood – a condition called alkalosis.

☑ SPECIAL TOPIC

Aspirin Must be in its basic form to Be physiologically active.

Aspirin has been used to treat fever, mild pain, and inflammation since it first became commerically available in 1899. It was the first drug to be tested clinically before it was marketed. Currently one of the most widely used drugs in the world, aspirin is one of a group of over-the-counter drugs known as NSAIDs (non steroidal anti-inflammatory drugs).

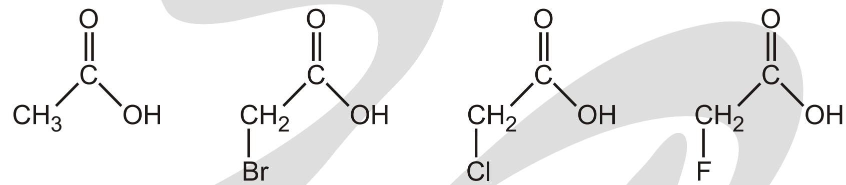
Aspirin is a carboxylic acid. The carboxylic acid group must be in its basic form to be physiologically active.



acidic form — Aspirin — basic form

Factor-3 : How Substituents Affect the Strength of an Acid

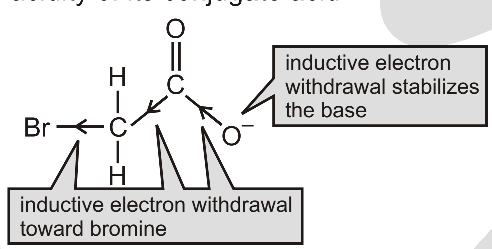
Although the acidic proton of each of the following carboxylic acids is attached to the same atom (an oxygen), the four compounds have different pK_a values :



Weakest acid — $pK_a = 4.76$ — $pK_a = 2.86$ — $pK_a = 2.81$ — $pK_a = 2.66$ — Strongest acid

The different pK_a values indicate that there must be another factor that affects acidity other than the nature of the atom to which the hydrogen is bonded. From the pK_a values of the four carboxylic acids, we see that replacing one of the hydrogens of the CH_3 group with a halogen increases the acidity of the compound. (The term for replacing an atom in a compound is substitution , and the new atom is called a substituent.) The halogen is more electronegative than the hydrogen it has replaced, so the halogen pulls the bonding electrons toward itself more than a hydrogen would. Pulling electrons through sigma (s) bonds is called inductive electron withdrawal.

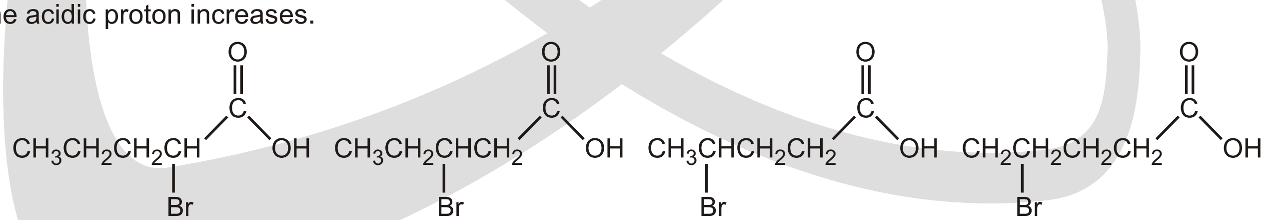
If we look at the conjugate base of a carboxylic acid, we see that inductive electron withdrawal decreases the electron density about the oxygen that bears the negative charge, thereby stabilizing it. And we know that stabilizing a base increases the acidity of its conjugate acid.



The pK_a values of the four carboxylic acids shown above (become more acidic) as the electron-withdrawing ability (electronegativity) of the halogen increases.

Thus, the fluoro-substituted compound is the strongest acid because its conjugate base is the most stabilized (is the weakest).

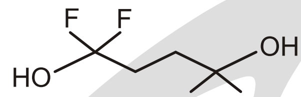
The effect a substituent has on the acidity of a compound decreases as the distance between the substituent and the acidic proton increases.



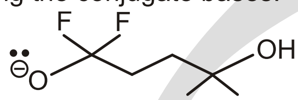
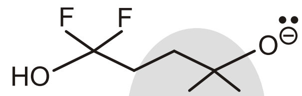
Strongest acid — $pK_a = 2.97$ — $pK_a = 4.01$ — $pK_a = 4.59$ — $pK_a = 4.71$ — Weakest acid

Solved Example

- Compare two protons shown in the following compound. Which proton is more acidic?

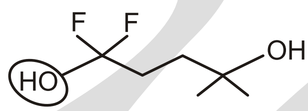


Ans. Begin by drawing the conjugate bases:

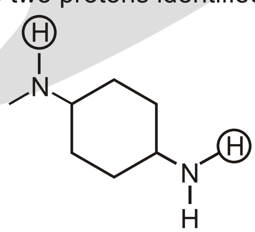
In the compound on the left, the charge is somewhat stabilized by the inductive effects of the two neighboring fluorine atoms. In contrast, the compound on the right is destabilized by the presence of two carbon atoms (methyl groups) that donate electron density. Therefore, the compound on the left is more stable.

The more acidic proton is the one that will leave to give the more stable negative charge. So the circled proton is more acidic:



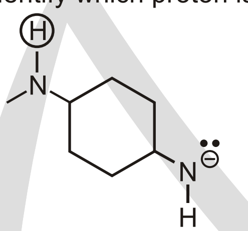
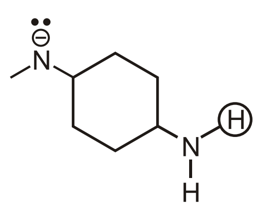
Solved Example

- Compare the two protons identified below :



Conjugate base 1 | Conjugate base 2

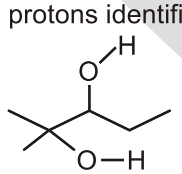
Identify which proton is more acidic, and explain why by comparing the stability of the conjugate bases.

Ans.   ; 1 is more stable than 2

Conjugate base 1 | Conjugate base 2

Solved Example

- Compare the two protons identified below :



Conjugate base 1 | Conjugate base 2

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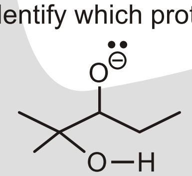
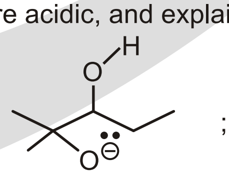
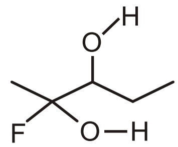
Ans.   ; 1 is more stable than 2

Conjugate base 1 | Conjugate base 2

Solved Example

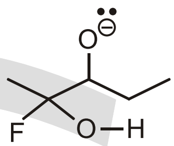
- Compare the two protons identified below :



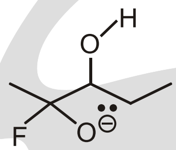
________________ Conjugate base 1 ________________ Conjugate base 2

Identify which proton is more acidic, and explain why by comparing the stability of the conjugate bases.

Ans.



Conjugate base 1



Conjugate base 2

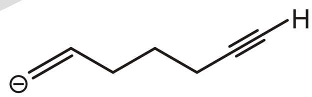
; 2 is more stable than 1

FACTOR-4 (Orbitals)

The three factors we have learned so far will not explain the difference in acidity between the two identified protons in the compound below :

If we pull off the protons and look at the conjugate bases to compare them, we see this:

So a negative charge on an sp hybridized carbon is more stable than a negative charge on an sp^3 or sp^2 hybridized carbon:

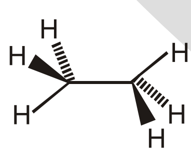


More stable

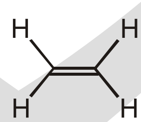
Determining which carbon atoms are sp, sp^2 or sp^3 is very simple: a carbon with a triple bond is sp, a carbon with a double bond is sp^2, and a carbon with all single bonds is sp^3. For more on this topic, turn to the next chapter (covering geometry).

Hybridization can also affect the pK_a

The hybridization of the orbital from which the proton is removed also affects the pK_a. Since *s* orbitals are held closer to the nucleus than are *p* orbitals, the electrons in them are lower in energy, that is, more stable. Consequently, the more s character an orbital has, the more tightly held are the electrons in it. This means that electrons in an *sp* orbital (50% s character) are lower in energy than those in an sp^2 orbital (33% s character), which are, in turn, lower in energy than those in an sp^3 orbital (25% s character). Hence the anions derived from ethane, ethene, and ethyne increase in stability in this order and this is reflected in their pK_as. Cyanide ion, –CN, with an electronegative element as well as an *sp* hybridized anion, is even more stable and HCN has a pK_a of about 10.

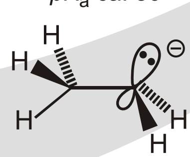


pK_a ca. 50

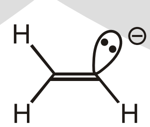


pK_a ca. 44

pK_a ca. 26



lone pair of CHCH
in sp orbital



lone pair of CHCH
in sp orbital

lone pair of
in sp orbital

SUMMARY OF ACIDIC STRENGTH

1. What atom is the charge on? (Remember the difference between comparing atoms in the same row and comparing atoms in the same column.)
2. Are there any resonance effects making one conjugate base more stable than the others?
3. Are there any inductive effects (electronegative atoms or alkyl groups) that stabilize or destabilize any of the conjugate bases?
4. In what orbital do we find the negative charge for each conjugate base that we are comparing?

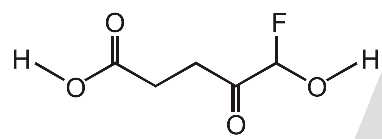
   There is an important exception to this order. Compare the two compounds below :

   If we wanted to know which compound was more acidic, we would pull off the protons and compare the conjugate bases :

   When comparing these two negative charges, we find two competing factors : the first factor (what atom is the charge on?) and the fourth factor (what orbital is the charge in?). The first factor says that a negative charge on nitrogen is more stable than a negative charge on carbon. However, the fourth factor says that a negative charge in an sp orbital is more stable than a negative charge in an sp^3 orbital (the negative charge on the nitrogen is an sp^3 orbital).

Solved Example

- Compare the two protons identified below :



Identify which proton is more acidic, and explain why.

Now we can compare them and ask which negative charge is more stable, using our four factors :

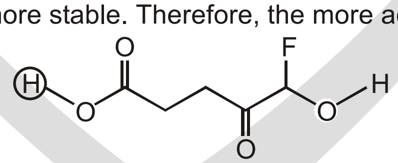
(i) Atom In both cases, the charge is on an oxygen, so this doesn't help us.

(ii) Resonance The compound on the left has resonance stabilization and the compound on the right does not. Based on this factor alone, we would say the compound on the left is more stable.

(iii) Induction The compound on the right has an inductive effect that stabilizes the charge, but the compound on the left does not have this effect. Based on this factor alone, we would say the compound on the right is more stable.

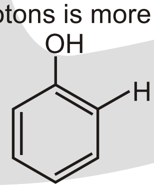
(iv) ***Orbital*** This does not help us.

So, we have a competition of two factors. In general, resonance will beat induction, so we can say that the negative charge on the left is more stable. Therefore, the more acidic proton is the one circled here :

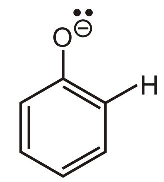
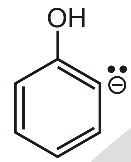


Solved Example

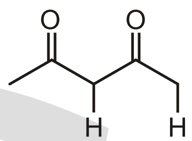
- For each compound below, two protons have been identified. In each case, determine which of the two protons is more acidic.

(i) 

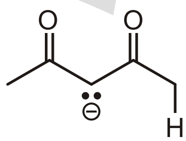
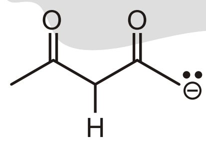
Conjugate base 1   Conjugate base 2

Ans.   ; 1 is more stable than 2

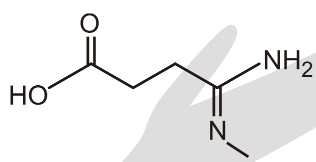
Conjugate base 1 — Conjugate base 2

(ii) 

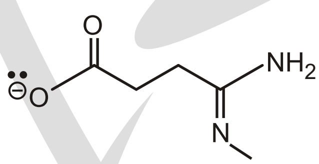
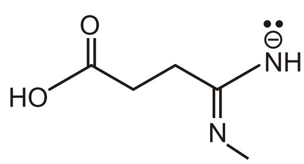
______ Conjugate base 1 ______ Conjugate base 2

Ans.   ; 1 is more stable than 2

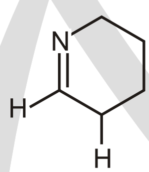
Conjugate base 1 — Conjugate base 2

(iii) 

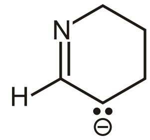
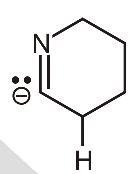
______ Conjugate base 1 ______ Conjugate base 2

Ans.   ; 1 is more stable than 2

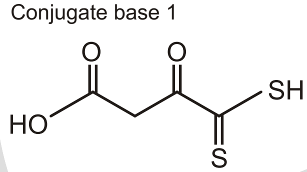
Conjugate base 1 — Conjugate base 2

(iv) 

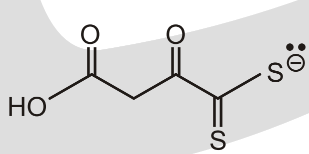
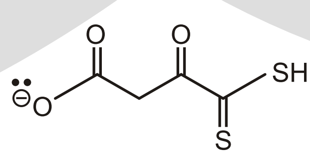
______ Conjugate base 1 ______ Conjugate base 2

Ans.   ; 1 is more stable than 2

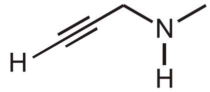
Conjugate base 1 — Conjugate base 2

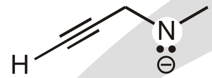
(v) 

______ Conjugate base 1 ______ Conjugate base 2

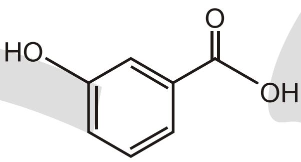
Ans.   ; 1 is more stable than 2

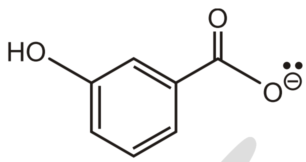
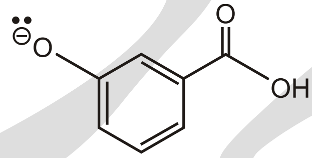
Conjugate base 1 — Conjugate base 2

(vi)  ______ Conjugate base 1 ______ Conjugate base 2

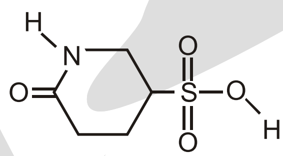
Ans.   ; 1 is more stable than 2

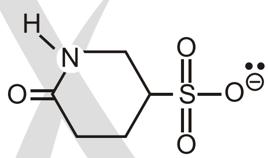
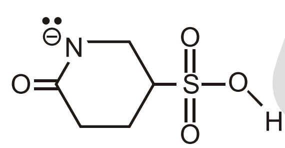
Conjugate base 1 — Conjugate base 2

(vii)  ______ Conjugate base 1 ______ Conjugate base 2

Ans.   ; 1 is more stable than 2

Conjugate base 1 — Conjugate base 2

(viii)  ______ Conjugate base 1 ______ Conjugate base 2

Ans.   ; 1 is more stable than 2

Conjugate base 1 — Conjugate base 2

Solved Example

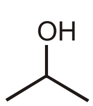
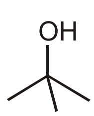
- For each pair of compounds below, predict which will be more acidic.

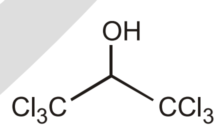
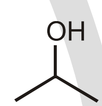
(i) HCl ; HBr

(ii) H_2O; H_2S

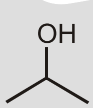
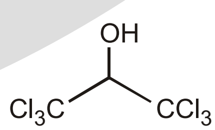
(iii) NH_3; CH_4

(iv)  ; 

(v)  ; 

(vi)  ; 

Ans. (i) HBr (ii) H_2S (iii) NH_3 (iv) 

(v)  (vi) 

☑ SPECIAL TOPIC

Derivation of the Henderson-Hasselbalch equation

The Henderson-Hasselbalch equation can be derived from the expression that defines the acid dissociation constant:

$$K_a \quad \frac{[H_3O\ \ ][A\ \ ]}{[HA]}$$

Take the logarithms of both sides of the equation and remember that when expressions are multiplied, their logs are added. Thus, we obtain

$$\log K_a \quad \log[H_3O\ \ ] \quad \log\frac{[A\ \ ]}{[HA]}$$

Multiplying both sides of the equation by 1 gives us

$$\log K_a \quad\quad \log[H_3O\ \ ] \quad \log\frac{[A\ \ ]}{[HA]}$$

Substituting pKa for $\log K_a$, pH for $\log[H_3O\ \ ]$, and inverting the fraction (which means the sign of its log changes), we get

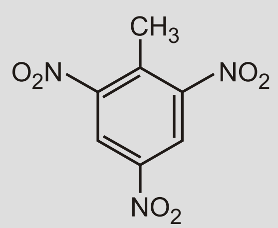
$$pK_a \quad pH \quad \log\frac{[HA]}{[A\ \ ]} \quad \text{or} \quad pH \quad pK_a \quad \log\frac{[HA]}{[A\ \ ]}$$

☑ SPECIAL TOPIC

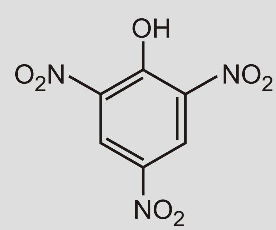
► **Picric acid is a very acidic phenol**

Electron-withdrawing effects on aromatic rings will be covered in more detail in Chapter 22 but for the time being note that electron-withdrawing groups can considerably lower the *pK*as of substituted phenols and carboxylic acids, asillustrated by picric acid.

2, 4, 6-Trinitrophenol's more common name, picric acid, reflects the strong acidity of this compound (pK_a 0.7 compared to phenol's 10.0). Picric acid used to be used in the dyeing industry but is little used now because it is also a powerful explosive (compare its structure with that of TNT!).



trinitrotoluene, TNT

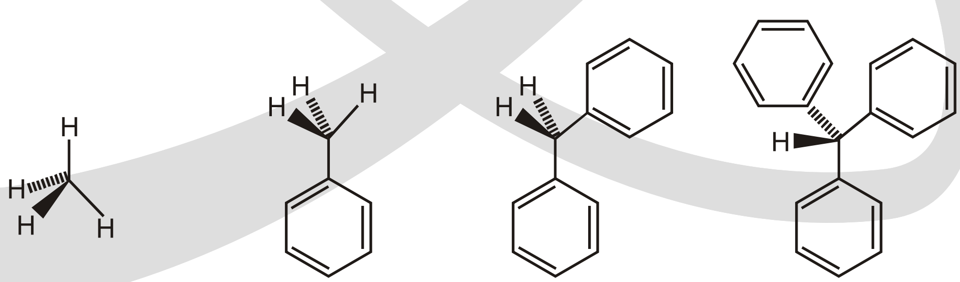


picric acid

FACTOR-5 (Steric Factor)

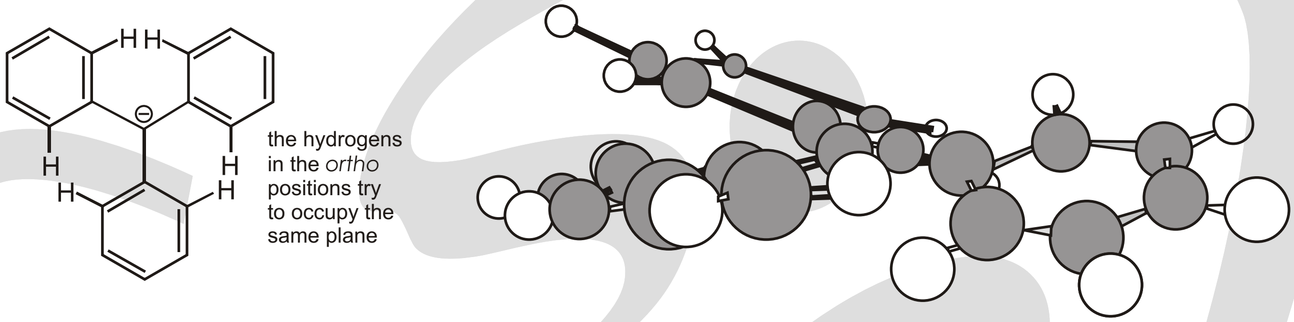
Highly conjugated carbon acids

If we can delocalize the negative charge of a conjugate anion on to oxygen, the anion is more stable and consequently the acid is stronger. Even delocalization on to carbon alone is good if there is enough of it, which is why some highly delocalized hydrocarbons have remarkably low pKas for hydrocarbons. Look at this series.



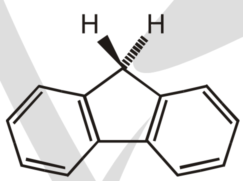
pK_a ca. 48 pK_a ca. 40 pK_a ca. 33 pK_a ca. 32

Increasing the number of phenyl groups decreases the pK_a—this is what we expect, since we can delocalize the charge over all the rings. Notice, however, that each successive phenyl ring has less effect on the pK_a: the first ring lowers the pK_a by 8 units, the second by 7, and the third by only 1 unit. In order to have effective delocalization, the system must be planar. Three phenyl rings cannot arrange themselves in a plane around one carbon atom because the ortho-hydrogens clash with each other (they want to occupy the same space) and the compound actually adopts a propeller shape where each phenyl ring is slightly twisted relative to the next.

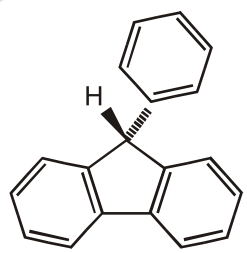


each phenyl ring is staggered relative to the next

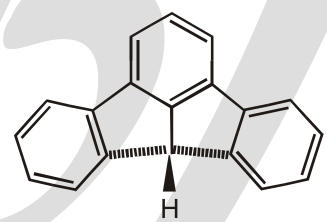
Even though complete delocalization is not possible, each phenyl ring does lower the pK_a because the sp^2 carbon on the ring is electron-withdrawing. If we force the system to be planar, as in the compounds below, the pK_a is lowered considerably.



fluorene, pK_a 22.8
in the anion, the whole
system is planar



9-phenylfluorene, pK_a 18.5
in the anion, only the two
fused fings can be planar

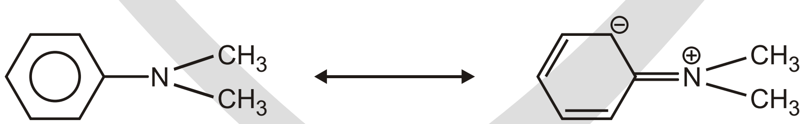


fluoradene pK_a 11
in the anion, the whole
system is planar

☑ SPECIAL TOPIC

Steric Inhibition of Resonance (S.I.R effect) :

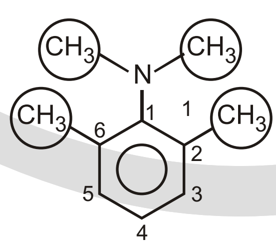
Restriction of resonance due to steric hindrance is known as S.I.R effect



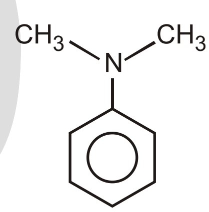
(X) (Y)

For instance in the above compound carbon atom of the Phenylring and Nitrogen are in same plane. If substituents are present in 2, 6 position, this coplanarity is inhibited and lone pair of Nitrogen is not involved in Resonance.

In compound (*P*) due to steric hindrance across Nitrogen. Bond rotation across C and Nitrogen will take place and lonepair of Nitrogen is perpendiculars to ring thereafter it will not involve in resonance with Phenylring, this is known as S.I.R. effect.



(*P*)

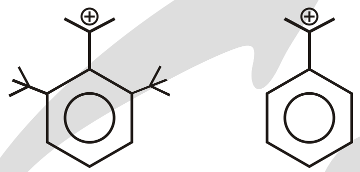


(*Q*)

P is more basic than Q, because in (*P*) lonepair of Nitrogen is not involved in resonance.

Solved Example

- Compare carbocation stability in given pairs,

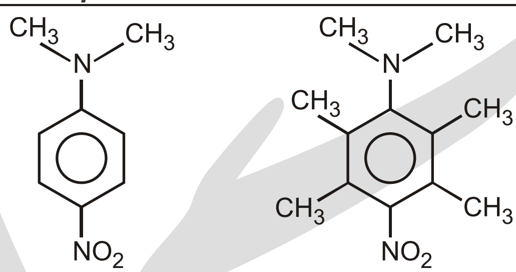


(*A*) (*B*)

Sol. *B* is more stable than (*A*) because carbocation (*B*) is resonance stabilizer. In compound (*A*) resonance does not take place because of S.I.R. effect.

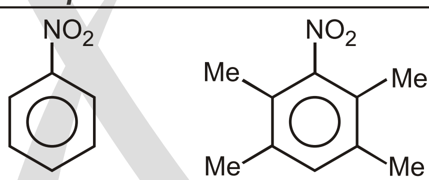
S.I.R. effect can also affect dipole of the compound for example.

Solved Example

- 

μ = 6.87D μ = 4.11D

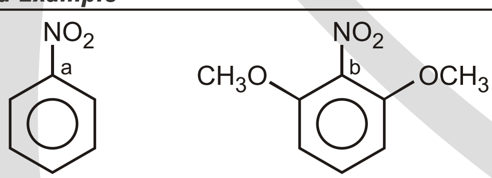
Solved Example

- 

μ = 3.95D μ = 3.39D

S.I.R. effect can also affect bond length

Solved Example

- 

(*X*) (*Y*)

Bond length of Carbon-Nitrogen are **b > a.**

Due to S.I.R. effect in (*Y*), NO_2 will not participate in resonance with phenyl ring, therefore has single bond character (b).

While in compound (*X*). Nitrogen and phenyl ring are co-planar, therefore resonance will take place.

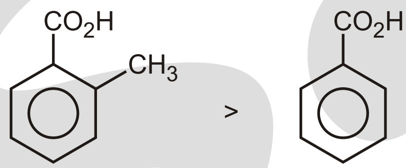
☑ SPECIAL TOPIC

Ortho and Para effect :

Classic concept of resonance, steric effect and steric inhibition of resonance have been widely used when interpreting reactivites of crowded conjugated molecules.

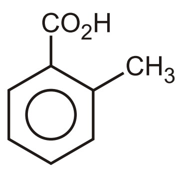
Ortho effect is used in following case

(1) Benzoic acid (ortho - substituted benzoic acid is stronger acidic than benzoic acid)

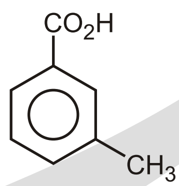


Compare acidic strength

(1)



(a)



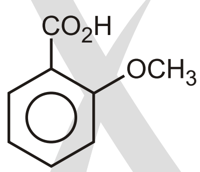
(b)

(c)

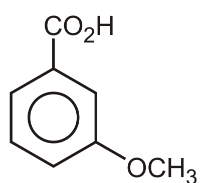
Ans. **a** > **b** > **c**

ortho effect — I effect — + H effect

(2)



(a)



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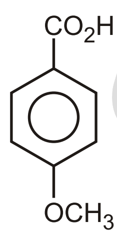
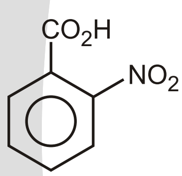


(c)

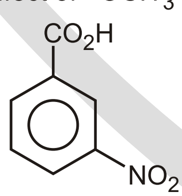
Ans. **a** > **b** > **c**

ortho effect — $-\mathrm{I}$ effect of $-OCH_3$ — + m effect of ($-OCH_3$)

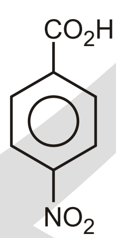
(3)



(a)



(b)



(c)

Ans. **a** > **c** > **b**

ortho effect — Due to –m effect of $-NO_2$, para is more acidic — $-\mathrm{I}$ effect

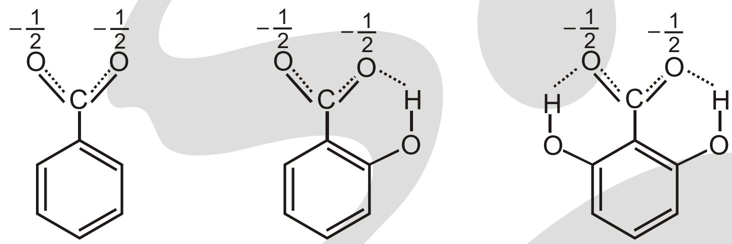
FACTOR-6 (Hydrogen Bonding) :

o-Hydroxybenzoic acid (salicylic acid) is far stronger than the corresponding m- and p-isomers.

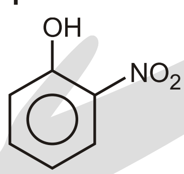
The explanation offered is hydrogen bonding; H of the o-OH group can form a hydrogen bond with the carboxyl group. The carboxylate ions of o-hydroxybenzoic acids are stabilised by intramolecular hydrogen bonding and support for this is given by the following order of acidic strength :

2, 6-di-OH benzoic acid (pK_a 2.30) > 2-OH benzoic acid(pK_a 2.98) > benzoic acid (pK_a 4.17)

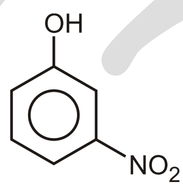
It can be seen that two hydrogen bonds would be expected to bring about more stabilisation than one hydrogen bond.



Acidic strength comparison of nitrophenol :



(a)
ortho-nitrophenol

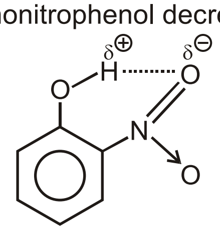


(b)
meta-nitrophenol

(c)
para-nitrophenol

Acidic strength : c > a > b

Reason : Intramolecular H-bonding in orthonitrophenol decreases it's acidic strength a bit.

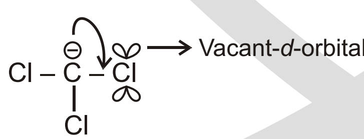


FACTOR-7 (*d*-orbital Resonance) :

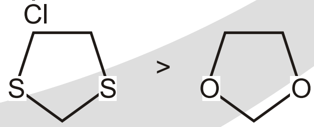
Due to *d*-orbital resonance the stability of conjugate base increases which results in the increase of acidic strength.

(1) Acidic strength : $CHCl_3 > CHF_3$

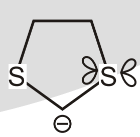
Explanation :



(2) Acidic strength :



Explanation :



is *d*-orbital resonance stabilised anion.

☑ SPECIAL TOPIC

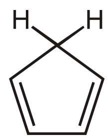
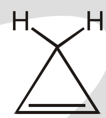
► **Acids as preservatives**

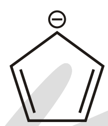
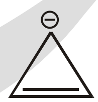
Acetic acid is used as a preservative in many foods, for example, pickles, mayonnaise, bread and fish products, because it prevents bacteria and fungi growing. However, its fungicidal nature is not due to any lowering of the pH of the foodstuff. In fact, it is the *undissociated* acid that acts as a bactericide and a fungicide in concentrations as low as 0.1–0.3%. Besides, such a low concentration has little effect on the pH of the foodstuff anyway.

Although acetic acid can be added directly to a foodstuff (disguised as E260), it is more common to add vinegar which contains between 10 and 15% acetic acid. This makes the product more 'natural' since it avoids the nasty 'E numbers'. Actually, vinegar has also replaced other acids used as preservatives, such as propionic (propanoic) acid (E280) and its salts (E281, E282 and E283).

FACTOR-8 (Aromatic and Non-aromatic) :

Aromatic nature of conjugate base increases it's stability which results in the increase acidic strength.

(1) Acidic strength :  > 

Explanation :  > 

Aromatic anion   Anti-Aromatic

FACTOR-9 (Combined effect of resonance, hyperconjugation and inductive effect) :

Remember the 8 factors, and what order they come in :

1. Atom
2. Resonance
3. Induction
4. Orbital
5. Steric effects
6. Hydrogen bond
7. d-orbital resonance
8. Aromaticity

☞ If you have trouble remembering the order, try remembering this acronym: **ARIOSHDA.**

Acid Base Equilibrium : An acids pKa depends on the stability of its conjugate base

$$HCl + H_2O \rightleftharpoons H_3O^{\oplus} + Cl^{\ominus} \qquad Ka \quad 10^{7}$$

$$CH_3COOH + H_2O \rightleftharpoons H_3O^{\oplus} + CH_3COO^{\ominus} \qquad Ka \quad 1.74 \quad 10^{\ 5}$$

The stronger the acid HA, the weaker its conjugate base $A^{\ominus}$

The stronger the base A , the weaker its conjugate acid HA

$$\underset{\boxed{\text{stronger acid}}}{RC\equiv CH} + \underset{\boxed{\text{stronger base}}}{\underset{\text{amide anion}}{^{-}NH_2}} \rightleftharpoons \underset{\boxed{\text{weaker base}}}{\underset{\text{acetylide anion}}{RC\equiv C^{-}}} + \underset{\boxed{\text{weaker acid}}}{NH_3}$$

$$\underset{\boxed{\text{weaker acid}}}{RC\equiv CH} + \underset{\boxed{\text{weaker base}}}{\underset{\text{hydroxide anion}}{HO^{-}}} \rightleftharpoons \underset{\boxed{\text{stronger base}}}{\underset{\text{acetylide anion}}{RC\equiv C^{-}}} + \underset{\boxed{\text{stronger acid}}}{H_2O}$$

- **Reactions with metals and alkalies:**

$$2R-COOH + 2Na \qquad \underset{\text{sodium carboxylate}}{2RCOONa} \quad H_2$$

$$R-COOH + NaOH \qquad R-COONa + H_2O$$

$$R-COOH + NaHCO_3 \qquad RCOONa + H_2O + CO_2$$

Solved Example

- Predict the direction of the equillibrium of the following :

(a) $NH_3 + {}^{\ominus}OH \rightleftharpoons {}^{\ominus}NH_2 + H_2O$ Keq – $10^{17.26}$ (left side favoured)
pK_a 33 ; pK_a 15.74

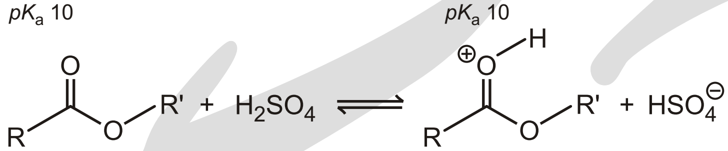
(b) $CH_3CO_2H + CO_3^{\ominus} \rightleftharpoons CH_3CO_2^{\ominus} + HCO_3^{\ominus}$ Keq ~ $10^{5.33}$ (right side favoured)
pK_a ~ 5 ; pK_a 10.33

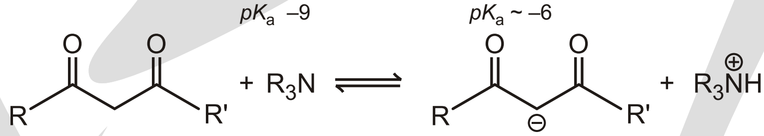
(c) $RCO_2H + R_2NH \rightleftharpoons RCO_2^{\ominus} + R_2NH_2^{\oplus}$ Keq ~ 10^6 (right side favoured)
pK_a ~ 5 ; pK_a 11

(d)  Keq ~ 10 (right side favoured)
pK_a ~ 18 ; pK_a ~ 19

(e) $NaF + HCl \rightleftharpoons NaCl + HF$ Keq ~ $10^{9.22}$ (right side favoured)
pK_a ~ 6 ; pK_a 3.22

(f) $ArOH + RNH_2 \rightleftharpoons Ar\overset{\ominus}{O} + R\overset{\oplus}{N}H_3$ Keq ~ 1 (niether side favoured)
pK_a 10 ; pK_a 10

(g)  Keq ~ 10^3 (right side favoured)
pK_a –9 ; pK_a ~ –6

(h)  Keq ~ 10^3 (right side favoured)
pK_a 9 ; pK_a ~ 12

(i) $R-C\equiv C^{\ominus} + RCH_2OH \rightleftharpoons R-C\equiv C-H + RCH_2O^{\ominus}$ Keq ~ 10^9 (right side favoured)
pK_a 16 ; pK_a 25

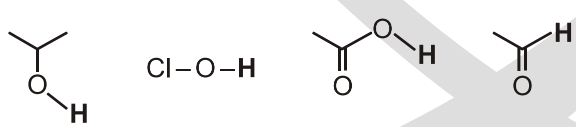
(j) $CH_3Li + (CH_3)_2C{=}CH_2 \rightleftharpoons CH_4 + (CH_3)_2C{=}\overset{\ominus}{C}H\ Li^{\oplus}$ Keq ~ 10^6 (right side favoured)
pK_a 44 ; pK_a 50

Solved Example

- Compare the acidic strength of the following :

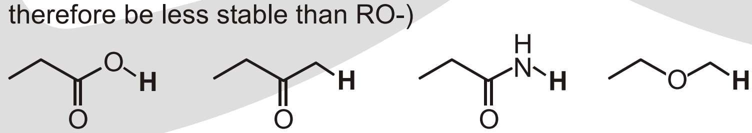
(a) $CH_3CH_2-\mathbf{H}$ (1) $CH_3O-\mathbf{H}$ (3) $\mathbf{H}-F$ (4) $CH_3CH_2NH-\mathbf{H}$ (2) (1 = least acidic, 4 = most acidic)

(b) $\mathbf{H}-Cl$ (2) $\mathbf{H}-F$ (1) $\mathbf{H}-Br$ (3) $\mathbf{H}-I$ (4) (1 = least acidic, 4 = most acidic)

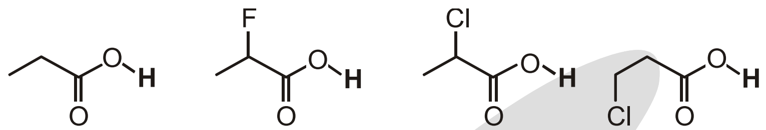
(c)  (1 = least acidic, 4 = most acidic)

3 2 4 1

(ClOH less acidic than the alcohol since the conjugate base Cl-O- will have neighbouring lone pairs and therefore be less stable than RO-)

(d)  (1 = least acidic, 4 = most acidic)

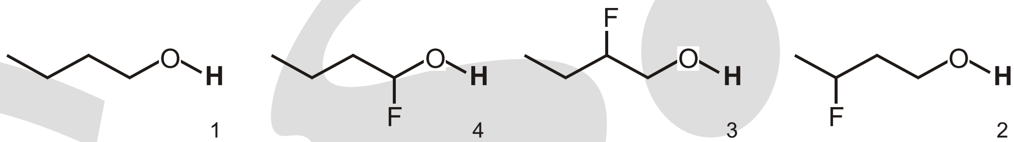
4 2 3 1

(e)  (1 = least acidic, 4 = most acidic)

1 4 3 2

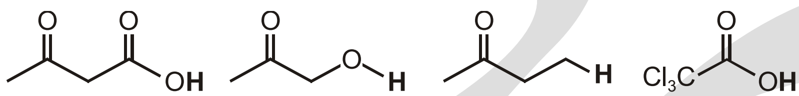
(f)  (1 = least acidic, 4 = most acidic)

4 2 3 1

(g) 

1 4 3 2

(1 = least acidic, 4 = most acidic)

(h)  (1 = least acidic, **4** = most acidic)

3 2 1 4

Table 1 : Inductive effects of substituent groups of some substituted acetic acids

Acid	pK_a	I effect of the substituent
$\boxed{H}-CH_2-CO_2H$	4.75	H reference point
$\boxed{CH_3}-CH_2-CO_2H$	4.90	+I
$\boxed{CH_3-O}-CH_2-CO_2H$	4.30	−I
$\boxed{I}-CH_2-CO_2H$	3.00	−I
$\boxed{Br}-CH_2-CO_2H$	2.90	−I
$\boxed{Cl}-CH_2-CO_2H$	2.80	−I
$\boxed{(CH_3)_3N^+}-CH_2-CO_2H$	1.80	−I

Here, —I effect Acid strength

Table 2 : Position of the substituent and inductive effect

Acid	pK_a	I effect of the substituent
$CH_3-CH_2-\underset{\boxed{Cl}}{\underset{\vert}{CH}}-CO_2H$	2.84	−I
$CH_3-\underset{\boxed{Cl}}{\underset{\vert}{CH}}-CH_2-CO_2H$	4.06	−I
$\underset{\boxed{Cl}}{\underset{\vert}{CH_2}}-CH_2-CH_2-CO_2H$	4.52	−I

Substituent at the position exerts the maximum effect.

Table 3 : Distance between the groups and the extent of inductive effect.

Acid	pK_a	I effect of the substituent
$\boxed{HO_2C}-CO_2H$	1.23	–I
$\boxed{HO_2C}-CH_2-CO_2H$	2.88	–I
$\boxed{HO_2C}-CH_2-CH_2-CO_2H$	4.19	–I

Inductive effect is distance dependent; as distance increases, inductive effect decreases.

Table 4 : Additive nature of inductive effect

Acid	pK_a	I effect of the substituent
$\boxed{H}-CH_2-CO_2H$	4.75	H = reference point
$\boxed{Cl}-CH_2-CO_2H$	2.86	One (–I) effect
$\boxed{Cl}-CH(\boxed{Cl})-CO_2H$	1.25	Two (–I) effect
$\boxed{Cl}-C(\boxed{Cl})(\boxed{Cl})-CO_2H$	0.65	Three (–I) effect

Inductive effect is an additive effect.

Table 5 : Aromatic acids and inductive effect

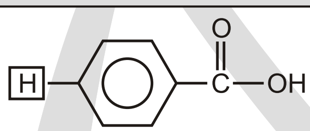
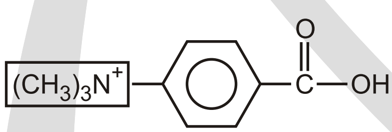
Acid	pK_a	I effect of the substituent
	4.2	H = reference point
	3.9	–I effect

Table 6 : Isotopic substituents and inductive effect

Acid	pK_a	I effect of the substituent
$\boxed{CH_3}-C(=O)-OH$	4.75	+I
$\boxed{CD_3}-C(=O)-OH$	4.6	+I

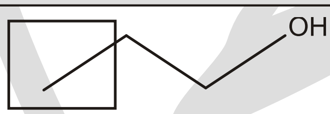
+I effect of deuterium is stronger than that of hydrogen. +I($-CD_3 \quad -CH_3$)

Table 7 : Electron withdrawing groups and inductive effects

Acid	pK_a	I effect of the substitutent
$\boxed{CH_3}-OH$	15.5	+I
$\boxed{F_3C} \leftarrow CH_2-OH$	12.4	–I
$\boxed{F_3C} \leftarrow CH_1-OH$ with $\downarrow \boxed{CF_3}$	9.3	–I
$\boxed{CF_3} \leftarrow C-OH$ with $\uparrow \boxed{CF_3}$ and $\downarrow \boxed{CF_3}$	5.4	–I

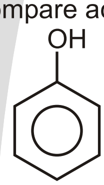
Electron withdrawing groups lower the pK_a of alcohols.

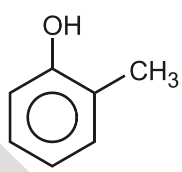
Table 8 : Hybridization state of the carbon atom in the substituent and inductive effect

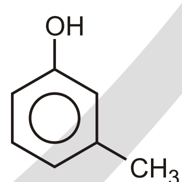
Acid	pK_a	I effect of the substitutent
	16	+I
	15.5	–I effect
	13.5	–I effect

Solved Example

- Compare acidic strength :

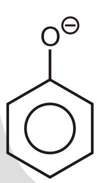
(a) 

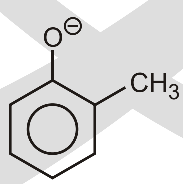
(b) 

(c) 

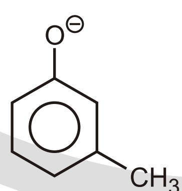
(d) 

Ans. a > c > d > b

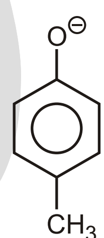
Exp. (a) 

(b) 

De-stabilized by +H and +I and (+I) is maximum

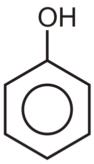
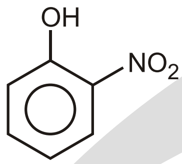
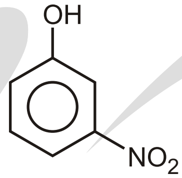
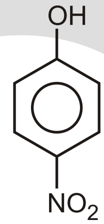
(c) 

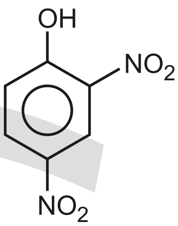
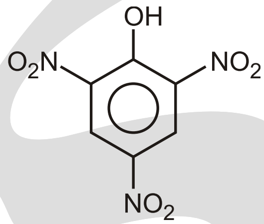
De-stabilized by +I only

(d) 

De-stabilized by +H and +I and (+I) is minimum

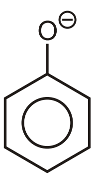
- Compare the acidic strength :

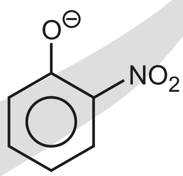
(a)  (b)  (c)  (d) 

(e)  (f) 

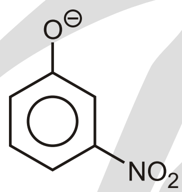
Sol. f > e > d > b > c > a

Due to intramolecular hydrogen bonding ortho isomer (b) is less acidic than Para isomer(d).

Exp. (a) 

(b) 

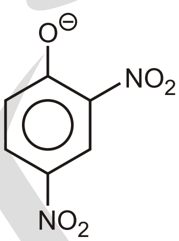
Stabilised by –M and –I effect of NO_2

(c) 

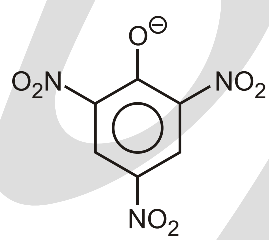
Stabilised by –I effect only

(d) 

Stabilised by –M and –I effect of NO_2

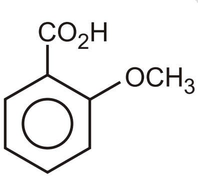
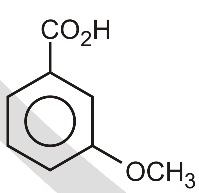
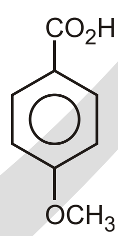
(e) 

Stabilised by –M and –I effect of two NO_2 groups

(f) 

Stabilised by –M and –I effect of three NO_2 groups

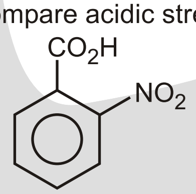
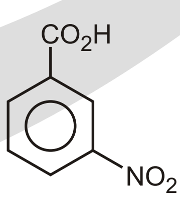
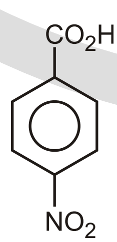
- Compare acidic strength :

(a)  (b)  (c) 

Ans. a > b > c

ortho effect – I effect of $–OCH_3$ + m effect of $(–OCH_3)$

- Compare acidic strength :

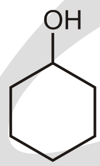
(a)  (b)  (c) 

Ans. a > c > b

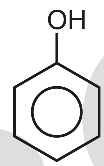
ortho effect | Due to –m effect of $-NO_2$, para is more acidic | I effect

- Correct order of acidic strength of given compounds will be?

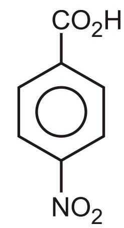
(I)



(II)



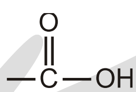
(III)



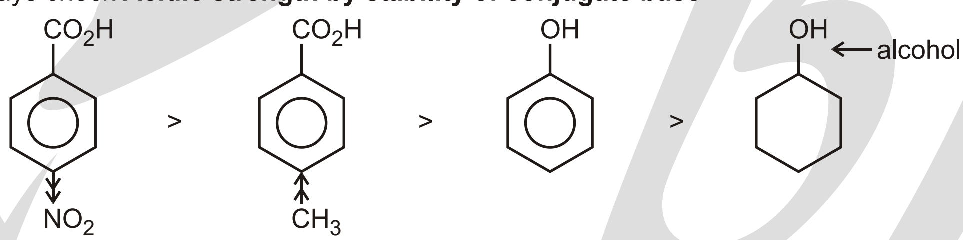
(IV)

(A) IV II I III
(B) IV III I II
(C) IV I II III
(D) IV I III II

Ans. (D)

Sol. **Strategy** (I)  group is always more acidic than alcohol and phenol

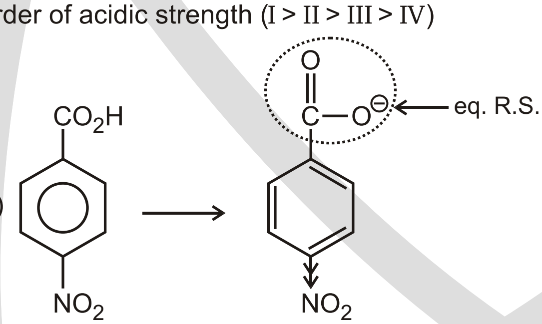
(II) Always check **Acidic strength by stability of conjugate base**

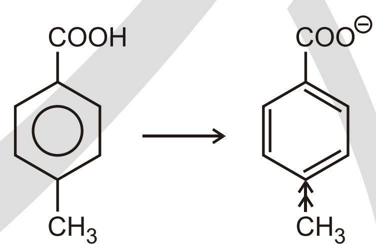


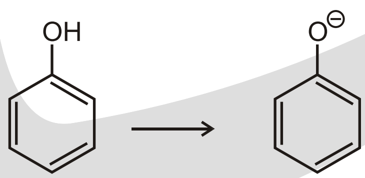
–M, –I (I) > +I, +H (II) > (III) > (IV)

Always check Acidic strength by stability of conjugate base

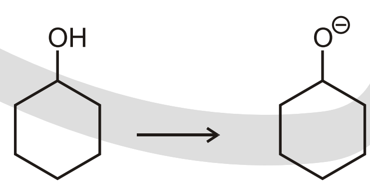
Order of acidic strength (I > II > III > IV)

(I) 

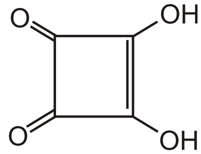
(II) 

(III) 

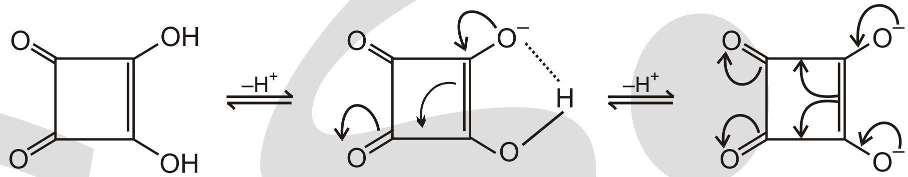
(Reso. stabalised)

(IV) 

(Not Reso. stabalised)

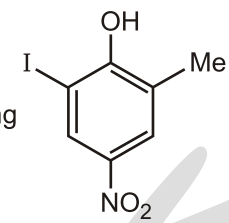
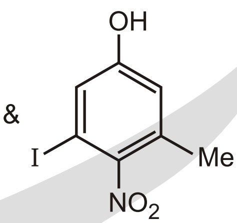
- Squaric acid  is a diprotic acid with both protons being more acidic than acetic acid. In the di-anion after the loss of both protons all of the C-C bonds are the same length as well as all of the C-O bonds. Provide a explanation for these observations.

Sol.

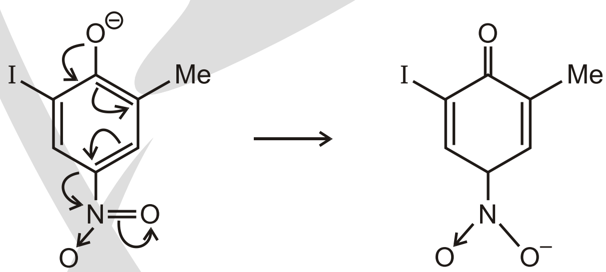


R-stablized & intra H-bonding

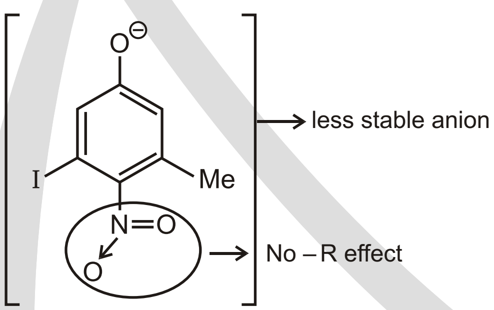
4 Eq. R.S.

- Among  &  which is more acidic & why? Explain through canonical forms.

Sol.

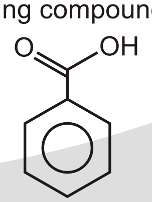


stable anion

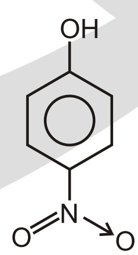


Out of plane

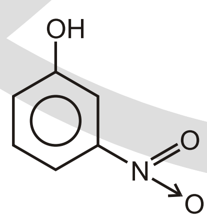
- In the following compound



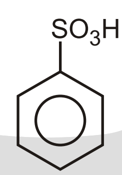
(I)



(II)



(III)



(IV)

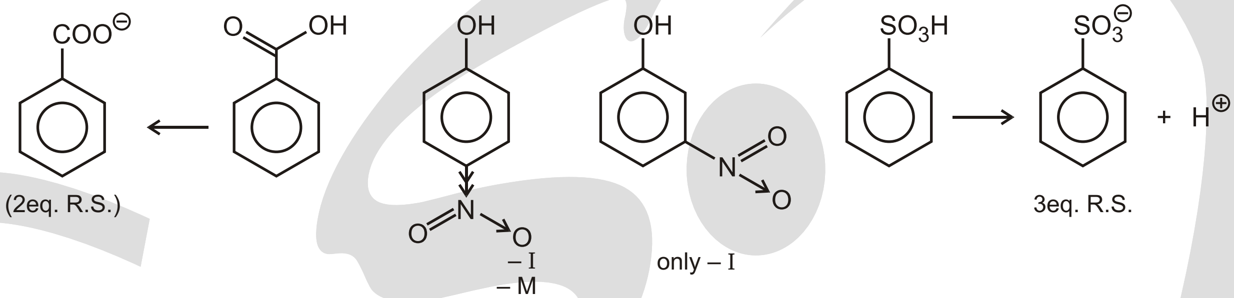
the decreasing order of acidic strength :

(A) I II IV III (B) II I IV III

(C) II IV I III (D) IV I II III

Ans. (D)

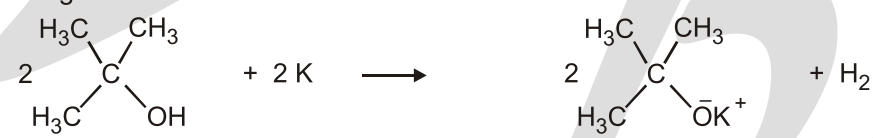
Sol.



(I) (II) (III) (IV)

REACTION OF ACTIVE METAL

Because alcohols are weak acids, they don't react with weak bases, such as amines or bicarbonate ion, and they react to only a limited extent with metal hydroxides, such as NaOH. Alcohols do, however, react with alkali metals and with strong bases such as sodium hydride (NaH), sodium amide ($NaNH_2$), and Grignard reagents (RMgX). Alkoxides are themselves bases that are frequently used as reagents in organic chemistry. They are named systematically by adding the -ate suffix to the name of the alcohol. Methanol becomes methanolate, for instance.



tert-Butyl alcohol (2-methyl-2-propanol)   Potassium tert-butoxide (Potassium 2-methyl-2-propanlate)

CH_3OH NaH CH_3O Na H_2

Methanol   Sodium methoxide (sodium methanolate)

CH_3CH_2OH $NaNH_2$ CH_3CH_2O Na NH_3

Ethanol   Sodium ethoxide (sodium ethanolate)

Solved Example

- $Ph-\overset{O}{\overset{\|}{C}}-OH$ $\;^{NaNH_2}$ (*A*) gas

  $Ph-\overset{O}{\overset{\|}{C}}-OH$ $\;^{NaH}$ (*B*) gas

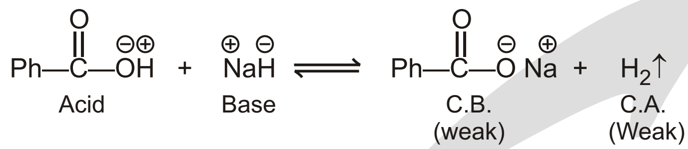
  (*A*) & (*B*) are molecular mass of gas so the value of *A* *B* 10

Ans. 9

Sol. **Strategy** Acid - Base reaction always **favour** the formation of **weaker acid and weaker base**

$$\underset{\text{Strong Acid}}{Ph-\overset{O}{\overset{\|}{C}}-OH} + \underset{\text{Strong Base}}{\overset{\oplus\ \ominus}{NaNH_2}} \rightleftharpoons \underset{\text{C.B.(weak)}}{Ph-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{O}\overset{\oplus}{Na}} + \underset{\text{C.A. (Weak)}}{NH_3\uparrow}$$

(A) NH_3 gas

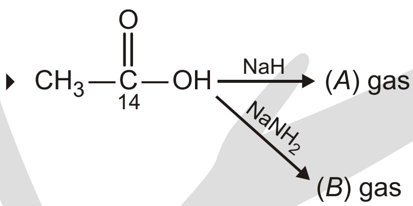


(B) H_2 gas

		Mol. wt.
(A)	NH_3	17
(B)	H_2	2

A B 19 $\quad\quad$ A B 10 9

Solved Example



Sum of molar mass of (A B) is suppose (X) so the value of (X 15) is :

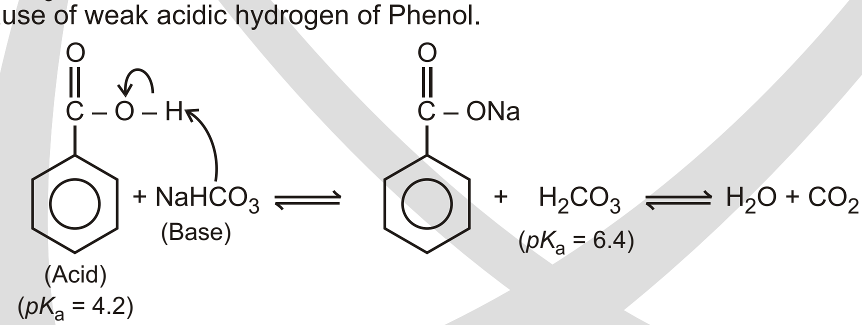
Ans. A H_2 ; B NH_3 ; $\quad x = 2 + 17 - 15 = 4$

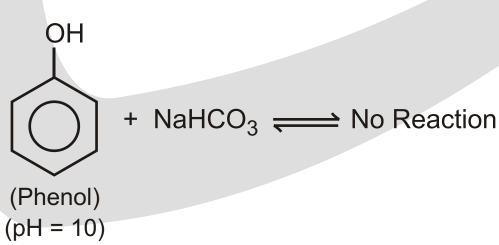
Sol. $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-O^{\ominus}\,Na^{\oplus} + H_2\uparrow$ molar mass = 2 ; $CH_3-\underset{14}{\overset{\overset{\displaystyle O}{\|}}{C}}-O^{\ominus}\,Na^{\oplus} + NH_3\uparrow$ molar mass = 17

A B 19; $\quad$ 19 – 15 = 4

Separation of binary mixture

$NaHCO_3$ is used to separate a binary Mixture of Phenol and benzoic acid. Phenol will not react with $NaHCO_3$ because of weak acidic hydrogen of Phenol.

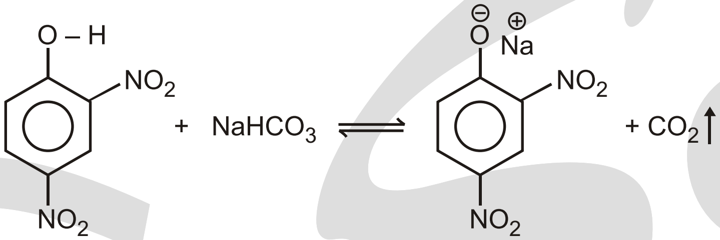
(1) 

(2) 

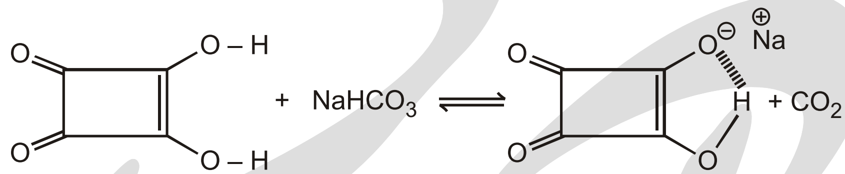
(3) $$Ph-\overset{O}{\overset{\|}{C}}-O-H + NaH\overset{\circledast}{C}O_3 \rightleftharpoons Ph-\overset{O}{\overset{\|}{C}}-\overset{\ominus}{O}\,\overset{\oplus}{Na} + \overset{\circledast}{C}O_2$$

($\overset{\circledast}{C} = {}^{14}C$)

(4)



(5) Squaric acid will also react with $NaHCO_3$ because both acidic hydrogens are more acidic than carboxylic acid.

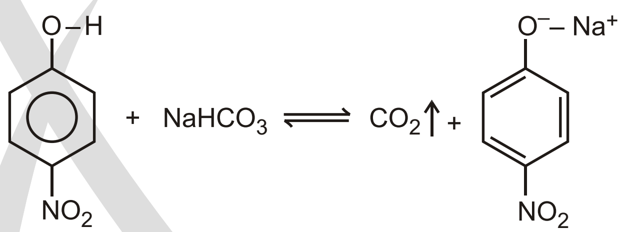


Solved Example

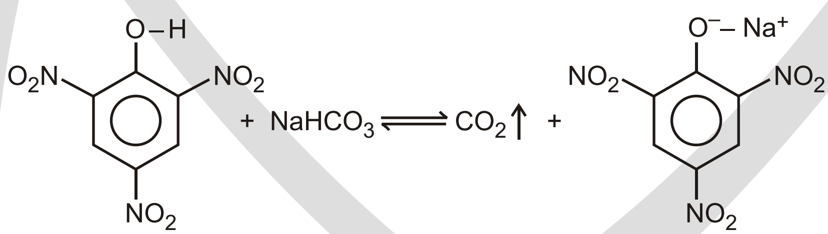
- Which gas is evolved in each reactions?

(1) $$Ph-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-O-H + NaHCO_3 \rightleftharpoons CO_2\uparrow + Ph-\underset{\underset{O}{\|}}{\overset{\overset{O}{\|}}{S}}-O^-Na^+$$

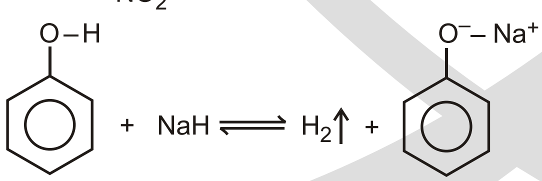
(2)



(3)



(4)

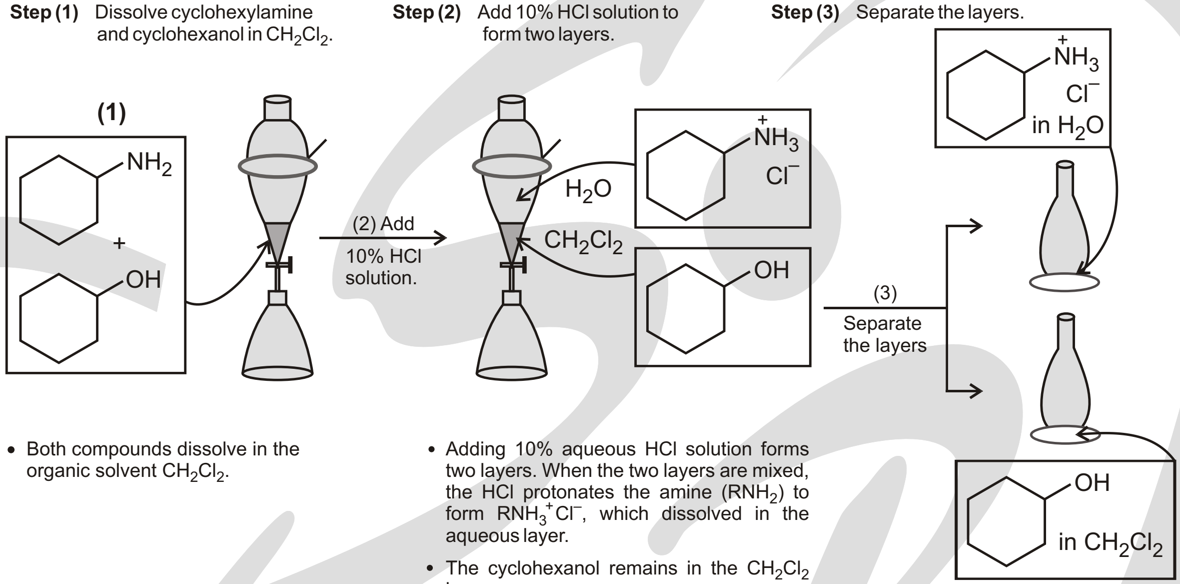


Separation of cyclohexanamine and cyclohexanol by an extraction procedure :

Because amines are protonated by aqueous acid, they can be separated from other organic compounds by extraction using a separatory funnel. **Extraction separates compounds based on solubility differences.** When an amine is protonated by aqueous acid, its solubility properties change.

For example, when cyclohexylamine is treated with aqueous HCl, it is protonated, forming an ammonium salt. Because the ammonium salt is ionic, it is soluble in water, but insoluble in organic solvents. A similar acid-base reaction does not occur with other organic compounds like alcohols, which are much less basic.

Step (1) Dissolve cyclohexylamine and cyclohexanol in CH_2Cl_2. **Step (2)** Add 10% HCl solution to form two layers. **Step (3)** Separate the layers.



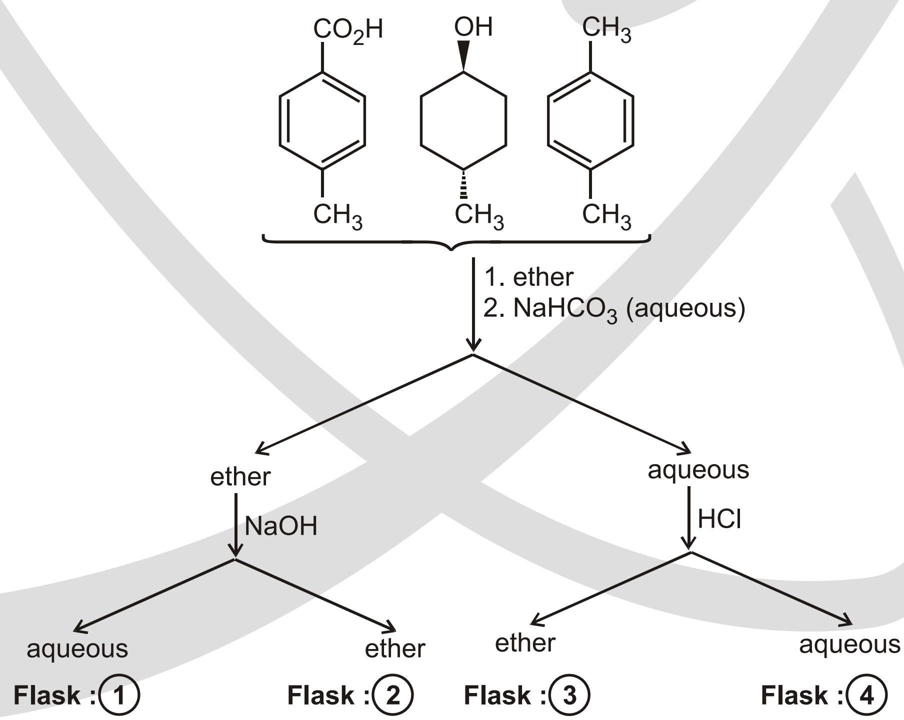
- Both compounds dissolve in the organic solvent CH_2Cl_2.

- Adding 10% aqueous HCl solution forms two layers. When the two layers are mixed, the HCl protonates the amine (RNH_2) to form $RNH_3^+Cl^-$, which dissolved in the aqueous layer.
- The cyclohexanol remains in the CH_2Cl_2 layer.

- Draining the lower layer out the bottom stopcock separates the two layers and the separation process is complete.
- Cyclohexanol (dissolved in CH_2Cl_2) is in one flask. The ammonium salt, $RNH_3^+Cl^-$ (dissolved in water), is in another flask.

Solved Example

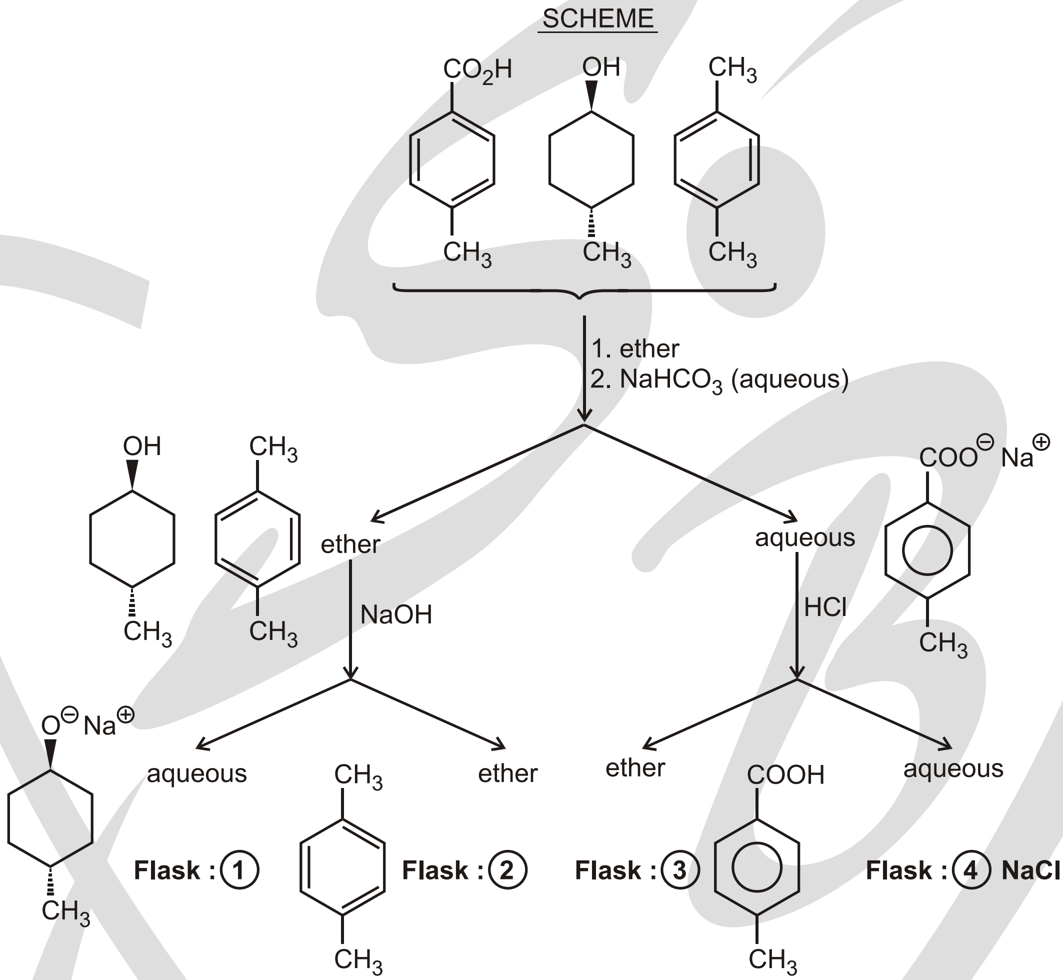
SCHEME/PATTERN



Which flask contains the 4-methyl benzoic acid?

(A) 1 (B) 2 (C) 3 (D) 4

Sol.



BASIC STRENGTH COMPARISON

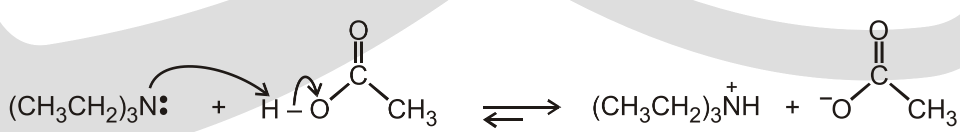
Amines as Bases : Amines react as Bases with a variety of organic and inorganic acids.

A bronsted-Lowry acid-base reaction

$$\underset{\text{base}}{R-NH_2} + \underset{\text{acid}}{H-A} \rightleftharpoons \underset{\substack{\text{conjugate acid} \\ pK_a \approx 10-11}}{R-\overset{+}{N}H_3} + :A^-$$

To favor the products, the **pK_a of HA must be <10.**

$$CH_3CH_2-\ddot{N}H_2 + \underset{pK_a = -7}{H-Cl} \rightleftharpoons \underset{pK_a = 10.8}{CH_3CH_2-\overset{+}{N}H_3} + Cl^-$$



$pK_a = 4.8$ $pK_a = 11.0$

Equilibrium favours the products

Table 1 : State of hybridization and base strength

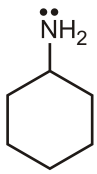
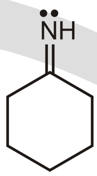
	Base	Base strength	Comment
(a)		Decreases ↓	Lone pair on sp^3 hybridized atom is more protonating
(b)			

Table 2 : Carbon chain length and base strength

	Base	Base strength	Comment
(a)	$CH_3 - NH_2$	increase ↓	I effect increases with the length alkyl chain and so does base strength.
(b)	$CH_3 - CH_2 - CH_2 - NH_2$		

Table 3 : Center atom and base strength

	Base	Base strength	Comment
(a)	$CH_3-\underset{CH_3}{\overset{CH_3}{\underset{\vert}{\overset{\vert}{C}}}}-CH_2-NH_2$	increase ↓	Here, the central atom silicon being more electro-positive than carbon, pushes the electron density more.
(b)	$CH_3-\underset{CH_3}{\overset{CH_3}{\underset{\vert}{\overset{\vert}{Si}}}}-CH_2-NH_2$		

Table 4 : Extent of inductive effect and base strength

	Base	Base strength	Comment
(a)	$Cl_3C - CH_2 - NH_2$	increase ↓	I effect (1/basic strength) I effect of Cl_3C group in less effective in (b) than in (a).
(b)	$Cl_3C - CH_2 - CH_2 - NH_2$		

Table 5 : Inductive effect and base strength

	Base	Base strength	Comment
(a)	$F_3C - CH_2 - NH_2$	increase ↓	I effect of $-CF_3$ $-CCl_3$
(b)	$Cl_3C - CH_2 - NH_2$		

Table 6. Delocalization of lone pair and base strength

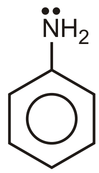
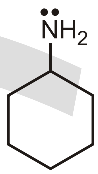
Base		Base strength	Comment
(a)	 Aniline	Increase ↓	Lone pair of electrons on nitrogen in aniline is delocalized
(b)	 Cyclohexylamine		

Table 7 : Delocalization of negative charge and base strength

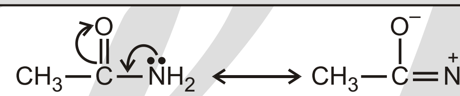
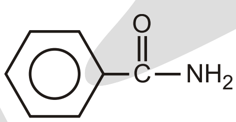
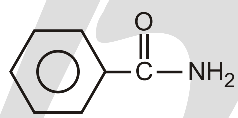
Base		Base strength	Comment
(a)	$CH_3-\overset{O}{\overset{\Vert}{C}}-NH_2$ Acetamide	Increase ↓	 Effective delocalization of negative charge on oxygen.
(b)	 Benzamide		 –I effect decreases basic strength

Table 8 : Basic strength of pyridine and pyrrole

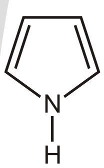
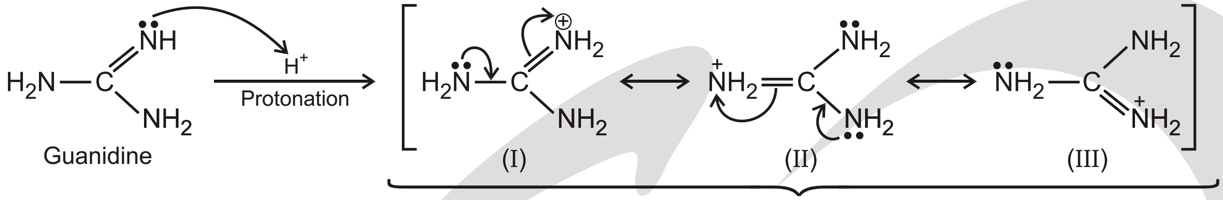
Base		Base strength	Comment
(a)	 Pyridine	Decreases ↓	Lone pair of nitrogen is delocalized in pyrrole because it is conjugated to the *p* bond. Therefore, it is not available for protonation.
(b)	 Pyrrole		

Table 9 : Basic strength of guanidine and imine

Base		Base strength	Comment
(a)	$NH_2-C(=NH)-NH_2$ Guanidine	Decreases ↓	Guanidine is the strongest organic nitrogen base. On protonation, the positive charge can be delocalized over three nitrogen atoms to give a very stable cation.
(b)	$(CH_3)_2C=NH$ Isopropyl		



All three have equal contribution in resonance hybrid. This makes Guanidine the strongest base.

Table 10 : Amine inversion and base strength

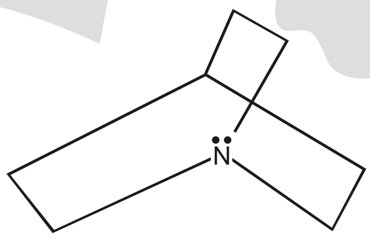
	Base	Base strength	Comment
(a)	 Quinuclidene	Decreases ↓	Quinuclidene will not undergo amine inversion.
(b)	$Et_3\ddot{N}$ Triethylamine		

Table 11 : Bredt's rule and base strength

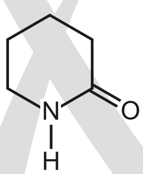
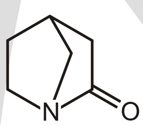
	Base	Base strength	Comment
(a)		Increase ↓	Lone pair is not delocalized in (b) because it violates Bredt's rule.
(b)			

Table 12 : Mesomeric effect and base strength

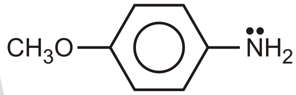
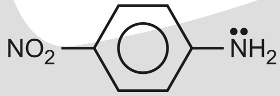
	Base	Base strength	Comment
(a)		Decreases ↓	M basic strength M $\frac{1}{\text{basic strength}}$
(b)			

Table 13 : Delocalization of lone pair and base strength

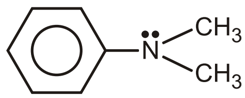
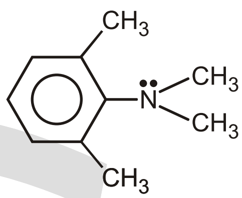
<table>
<tr><th colspan="2">Base</th><th>Base strength</th><th>Comment</th></tr>
<tr><td>(a)</td><td></td><td rowspan="2">Increase</td><td rowspan="2">Lone pair of electrons is not delocalized in (b) due to SIR and so it is more basic.</td></tr>
<tr><td>(b)</td><td></td></tr>
</table>

Table 14 : Ortho effect and base strength

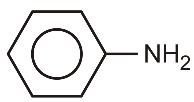
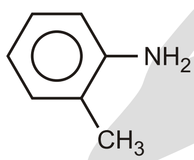
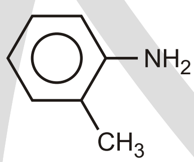
<table>
<tr><th colspan="2">Base</th><th>Base strength</th><th>Comment</th></tr>
<tr><td>(a)</td><td></td><td rowspan="2">Decreases</td><td rowspan="2">Base strength decreases due to ortho effect.</td></tr>
<tr><td>(b)</td><td></td></tr>
</table>

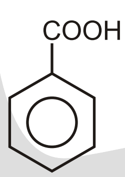
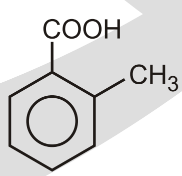
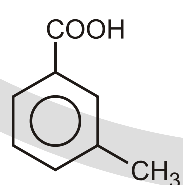
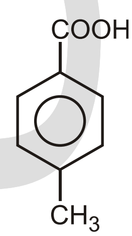
Table 15 : Para effect and base strength

<table>
<tr><th colspan="2">Base</th><th>Base strength</th><th>Comment</th></tr>
<tr><td>(a)</td><td></td><td rowspan="2">Decreases</td><td rowspan="2">Base strength decreases due to para effect.</td></tr>
<tr><td>(b)</td><td></td></tr>
</table>

EXERCISE

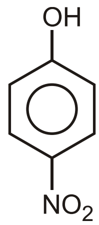
SINGLE CHOICE QUESTIONS

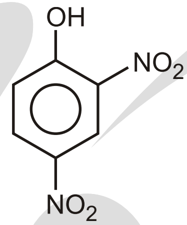
1. Which of the following is most acidic in nature.

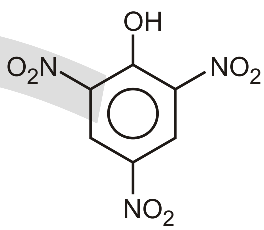
(I) 
(II) 
(III) 
(IV) 

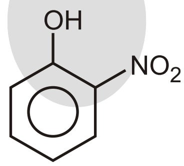
(A) I (B) II (C) III (D) IV

2. Which of the following substituted phenol is most acidic in nature?

(I) 

(II) 

(III) 

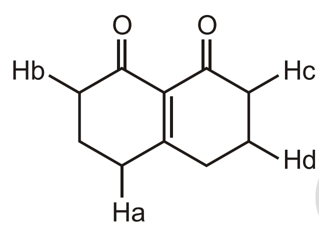
(IV) 

(A) I (B) II (C) III (D) IV

3. Which of the following substituted carboxylic acid has the highest Ka value?

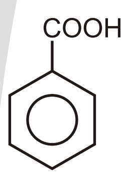
(A) -chlorobutyric acid (B) -chlorobutyric acid
(C) -fluorobutyric acid (D) -fluorobutyric acid

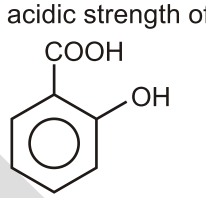
4. Which of the following indicated H-atom is most acidic in nature.

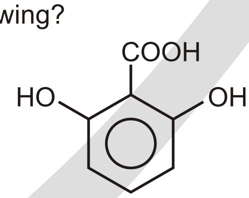


(A) Ha (B) Hb
(C) Hc (D) Hd

5. Write the correct order for the acidic strength of the following?

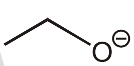
(a) 

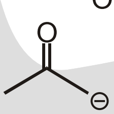
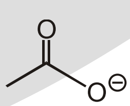
(b) 

(c) 

(A) a > b > c (B) b > c > a
(C) c > b > a (D) None of these

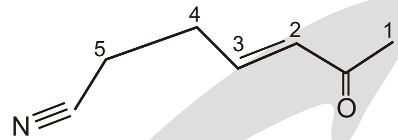
6. Which bases are strong enough to deprotonate H_2O?

(a)  (b)  (c) $Cl^{\ominus}$ (d) $\overset{\ominus}{N}H_2$

(e)  (f) 

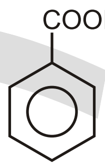
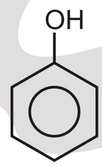
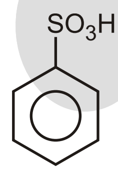
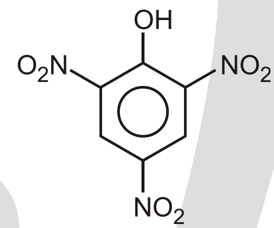
(A) a,b,c,d,e (B) b,c,d,e,f
(C) a,b,d,e (D) b,c,d,e

7. Which of the following carbon will be deprotonated first on treatment with base.

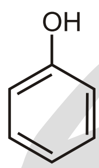


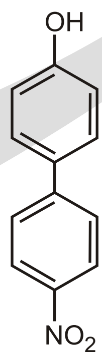
(A) 4 (B) 5 (C) 2 (D) 3

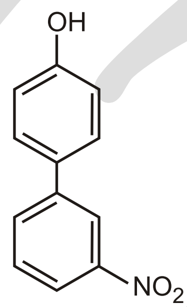
8. Which of the following compound do not react with $NaHCO_3$ to evolve CO_2 gas?

(A) 
(B) 
(C) 
(D) 

9. Correct order of Acidic strength among the following compounds :


(I)

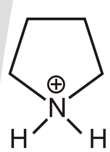
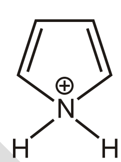
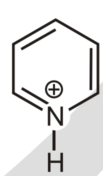
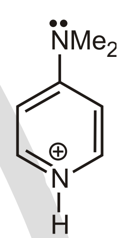

(II)


(III)

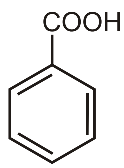

(IV)

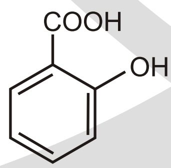
(A) I II III IV (B) II III IV I
(C) II III I IV (D) III II IV I

10. The Most acidic species is :

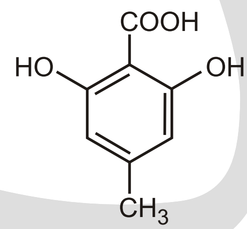
(A) 
(B) 
(C) 
(D) 

11. Among the following given compounds


(I)


(II)


(III)


(IV)

The decreasing order of their acidity is :

(A) I II III IV (B) I III II IV (C) IV II III I (D) IV II I III

12. What is the correct order of acidic strength?

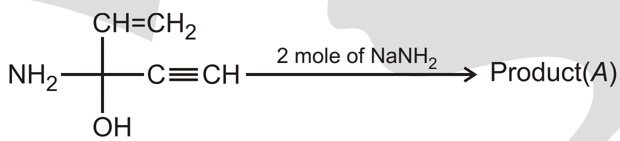
I II III IV

(A) I II III IV (B) IV I II III (C) III IV I II (D) I III II IV

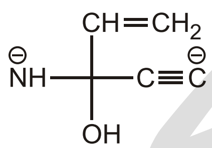
13. Which compounds release CO_2 gas on reaction with $NaHCO_3$(aq)?

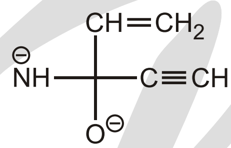
(A) Phenol (carbolic acid) (B) Cyclohexanol

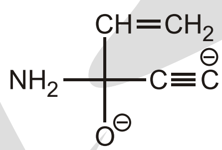
(C) Benzoic acid (D) Benzamide

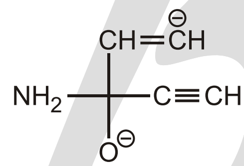
14. 

The product (*A*) will be :

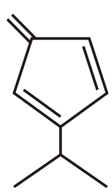
(A) 

(B) 

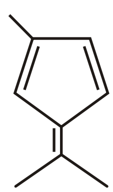
(C) 

(D) 

15. Which of the given compounds is the strongest acid?



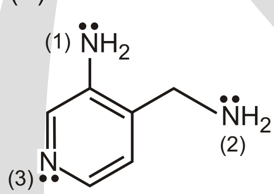
1



2

3

(A) 1 (B) 2 (C) 3 (D) Both 1 & 2

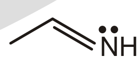
16. 

Find the basic strength order among nitrogen atoms :

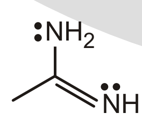
(A) 2 > 1 > 3 (B) 2 > 3 > 1 (C) 3 > 2 > 1 (D) 3 > 1 > 2

17. Correct order of Basic Strength (K_b order) is :

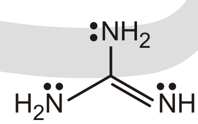
1



2



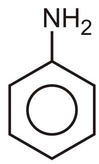
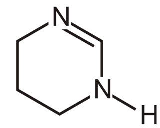
3



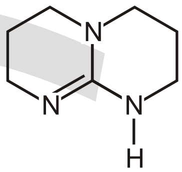
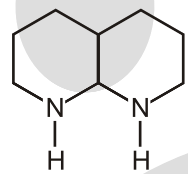
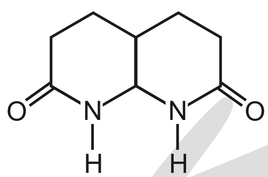
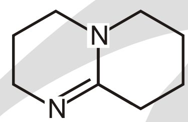
4

(A) 4 > 3 > 2 > 1 (B) 1 > 2 > 3 > 4 (C) 4 > 3 > 1 > 2 (D) 1 > 4 > 3 > 2

18. Which of the following base is weaker than $CH_3 — CH_2 — NH_2$ (ethyl amine) :

(A) 
(B) $CH_3 - C(=NH)-NH_2$
(C) $NH_2 - C(=NH)-NH_2$
(D) 

19. Compare the Basic strength :

(1) 
(2) 
(3) 
(4) 

(A) 1 > 4 > 2 > 3 (B) 2 > 1 > 4 > 3 (C) 2 > 4 > 1 > 3 (D) 1 > 2 > 4 > 3

20. The most basic compound among the following is :

(A) Benzylamine (B) Aniline
(C) Acetaniline (D) *p*-nitro aniline

21. The correct decreasing order of basic strength of the following species is ________.

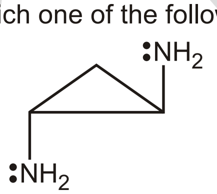
$H_2O, NH_3, OH\ \ , NH_2$

(A) $NH_2\quad OH\quad NH_3\quad H_2O$ (B) $OH\quad NH_2\quad H_2O\quad NH_3$
(C) $NH_3\quad H_2O\quad NH_2\quad OH$ (D) $H_2O\quad NH_3\quad OH\quad NH_2$

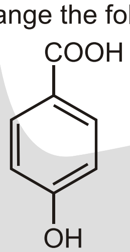
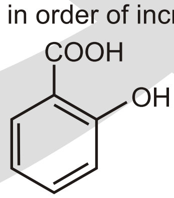
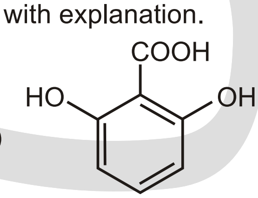
SUBJECTIVE TYPE QUESTIONS

1. -Aminoacids have greater dipole moments that expected. Explain.

2. Which one of the following two compounds is more basic and why?

(a) 
(b) 

3. Arrange the following compounds in order of increasing acid strength with explanation.

(a) 
(b) 
(c) 

4. What is squaric acid. Why is it a very strong acid?

Answers

Single Choice Questions

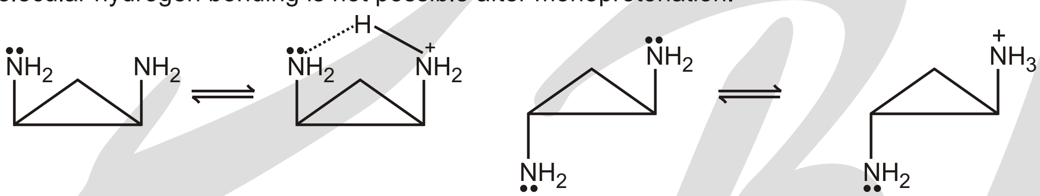
1. (B)	**2.** (C)	**3.** (C)	**4.** (A)	**5.** (C)	**6.** (C)	**7.** (A)	**8.** (B)
9. (B)	**10.** (B)	**11.** (D)	**12.** (D)	**13.** (C)	**14.** (C)	**15.** (B)	**16.** (B)
17. (C)	**18.** (A)	**19.** (A)	**20.** (A)	**21.** (A)			

Subjective Type Questions

1. An -amino acid remains as a dipolar zwitter ion and that enhances polarity. That is why amino acids have higher dipole moments than expected.

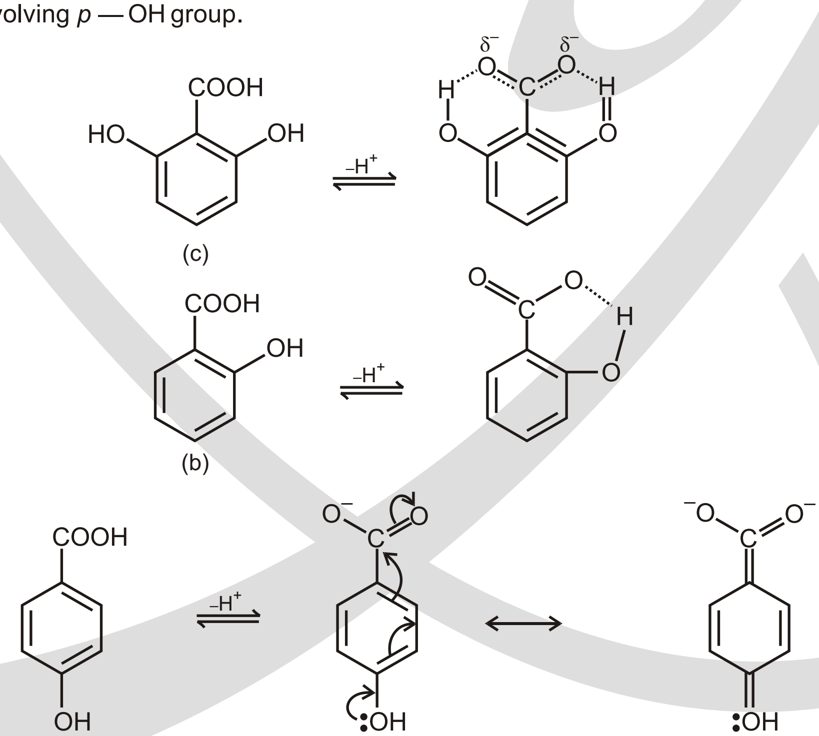
$$\underset{\text{Glycene}}{H_2\ddot{N}-CH_2-COOH} \rightleftharpoons \underset{\text{Zwitter ion (More polar)}}{H_2\overset{+}{N}-CH_2-COO^-}$$

2. The above mentioned compounds are rigid molecules. The compound (b) is more basic, because monoprotonated form gets stabilized by intramolecular hydrogen bonding. In the case of compound (a). such intramolecular hydrogen bonding is not possible after monoprotonation.



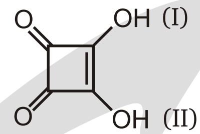
Not stabilized by Intramolecular hydrogen bonding

3. The decreasing aciding order is (c) > (b) > (a). In case of (c), the corresponding carboxylate anion is stabilized by intramolecular hydrogen bonding from the two ortho —OH groups. In case of (b), the stabilization comes from intramolecular hydrogen bonding from one ortho —OH group. In case of (a), carboxylate ion is destabilized by resonance involving *p* —OH group.

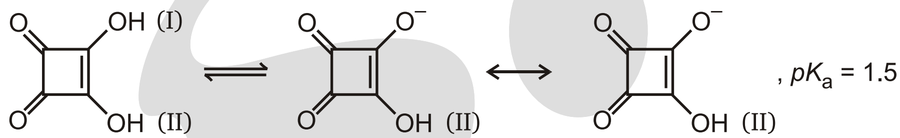


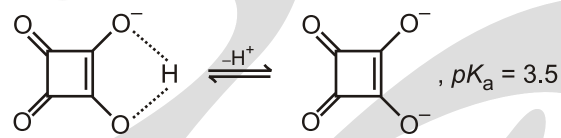
Electron density increases on the carboxylate group making the acid weaker

4. The structure of squaric acid is given here.



Structurally it is symmetrical. Its two acidic hydrogen atoms are associated with two OH groups identified as (I) and (II). The first ionization is found to give a more acidic hydrogen ion, because of vinylogous effect. Second ionization becomes slightly difficult because of the formation of a weak intramolecular hydrogen bonding.





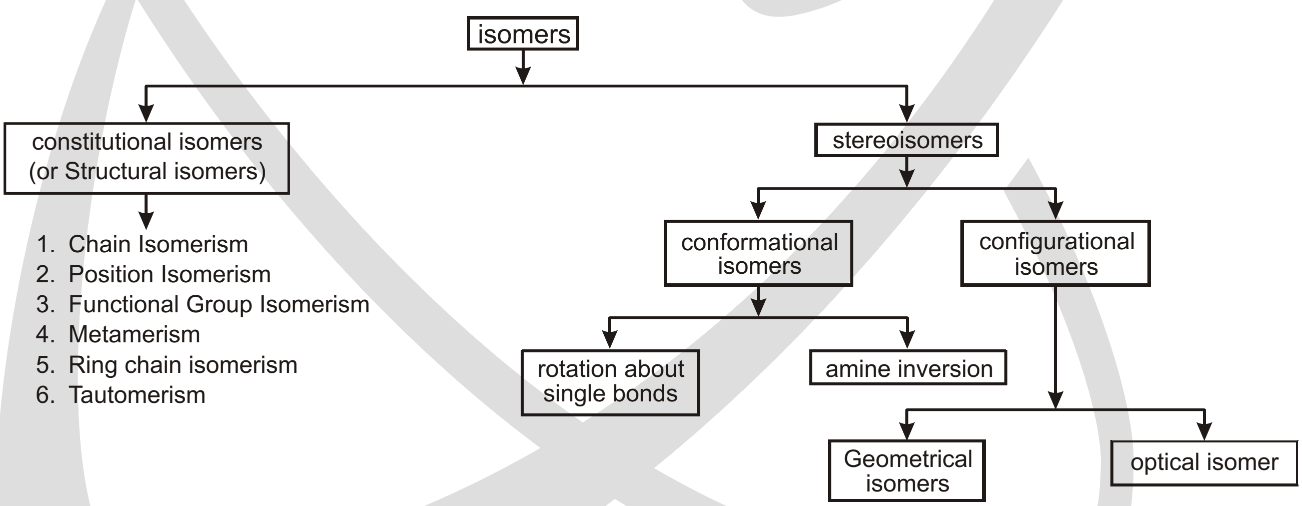
❑❑❑

Isomerism

Different compounds that have the same molecular formula are called isomers (Greek : isos, equal; meros = part).

They contain the same numbers of the same kinds of atoms, but the atoms are attached to one another in different ways. Isomers are different compounds because they have different molecular structures.

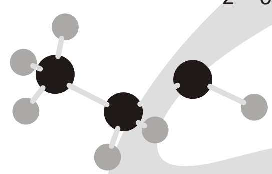
CLASSIFICATION OF ISOMERISM



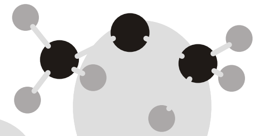
Analysis of dimethyl ether shows that it contains carbon, hydrogen, and oxygen in the same proportion as ethyl alcohol, 2C : 6H : 1O. It has the same molecular weight as ethyl alcohol, 46. We conclude that it has the same molecular formula C_2H_6O.

The compound dimethyl ether is a gas with a boiling point of –24ºC. It is clearly a different substance from ethyl alcohol, differing not only in its physical properties but also in its chemical properties. It does not react at all with sodium metal.

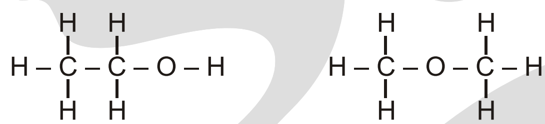
The compound ethyl alcohol is a liquid boiling at 78ºC. It contains carbon, hydrogen, and oxyen in the proportions 2C : 6 H : 1O. It has a molecular weight of 46. The molecular formula of ethyl alcohol must therefore be C_2H_6O. Ethyl alcohol is a quite reactive compound. For example, if a piece of sodium metal is dropped into a test tube containing ethyl alcohol, there is a vigorous bubbling and the sodium metal is consumed; hydrogen gas is evolved and there is left behind a compound of formula C_2H_5ONa. Ethyl alcohol reacts with hydriodic acid to form water and a compound of formula C_2H_5I.



Ethyl alcohol



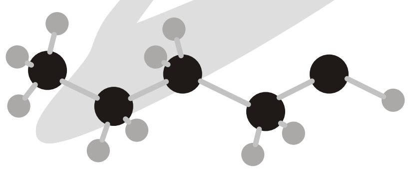
Dimethyl ether



I
Ethyl alcohol

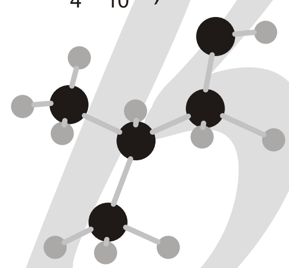
II
Dimethyl ether

e.g. n-butyl alcohol and isobutyl alcohol (same molecular formula $C_4H_{10}O$) are isomers :



n-butyl alcohol

$CH_3CH_2CH_2CH_2OH$



isobutyl alcohol

$CH_3—CH(CH_3)—CH_2—OH$

Broadly speaking, isomerism is of two types :

(i) Structural isomerism

(ii) Stereoisomerism

Structural isomerism : Structural isomers possess the same molecular formula but different connectivity of atoms. The term constitutional isomerism is a more modern term for structural isomerism.It arises because of the difference in the sequence of covalently bonded atoms in the molecule without reference to space.

It is further classified into following types.

Chain Isomerism : The different arrangement of carbon atoms give rise to chain isomerism. Chain isomers possess different lengths of carbon chains (straight or branched). Such isomerism is shown by each and every family of organic compounds.

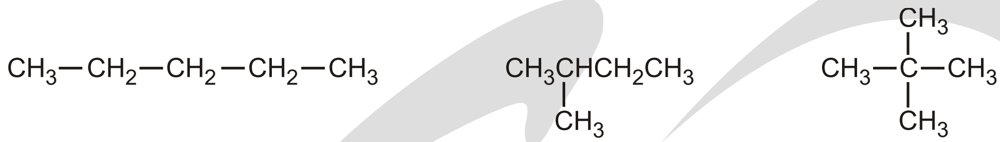
(i) Butane : C_4H_{10}

$CH_3—CH_2—CH_2—CH_3$ (*n*-butane)

$CH_3—CH(CH_3)—CH_3$ (iso butane)

n-butane has the chain of four carbon while isobutane has three. Hence they are chain isomers.

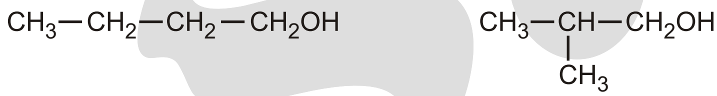
(ii) Pentane : C_5H_{12}



n-Pentane   isopentane   Neopentane

n-Pentane, isopentane and neopentane possesss the chain of five, four and three carbons, respectively. hence they are chain isomers.

(iii) Butyl alcohol : C_4H_9OH



n-Butyl alcohol   isobutyl alcohol

Position Isomerism : Position isomerism is shown by the compounds in which there is difference in the position of attachment of functional group, multiple bond or substituent along the same chain length of carbon atoms. Its characterestics are :

(i) The same molecular formula

(ii) The same length of carbon chain

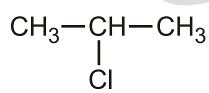
(iii) The same functional group.

e.g.,

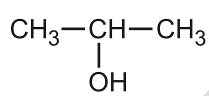
(i) Molecular formula : C_3H_7X (X = halogen, NH_2OH or OR)

(i)   (ii)

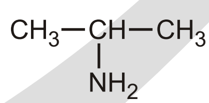
In these structure three carbon atoms form a chain, and X is joined at the end in (i), while at the middle carbon in (ii). Let us look at the following examples along with type of isomerism shown by them.

(a) $CH_3—CH_2—CH_2Cl$ and      :   Position isomers

1-Chloropropane   2-Chloropropane

(b) $CH_3—CH_2—CH_2OH$ and      :   Position isomers

1-Propanol   2-Propanol

(c) $CH_3—CH_2—CH_2—NH_2$ and      :   Position isomers

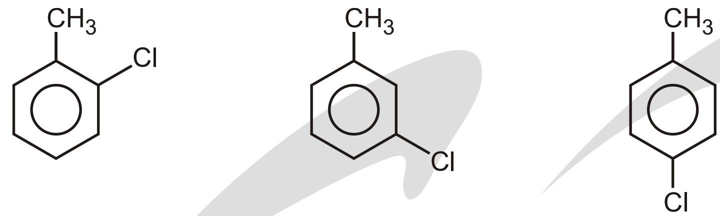
n-Propylamine   Isopropylamine

(ii) Similarly, Molecular formula : C_4H_8

$CH_3 — CH_2 — CH \quad CH_2$ and $CH_3 — CH \quad CH — CH_2$:   Position isomers

1-Butene   2-Butene

In the disubstituted benzene derivatives position isomerism also exists because of the relative position occupied by the substituents on the benzene ring. Thus, Chlorotoluene, $C_6H_4(CH_3)Cl$ exists in three isomeric forms–ortho, meta and para.



O-Chorotoluene *m*-Chlorotoluene *p*-Chlorotoluene

❑ **NOTE : Carboxylic acid, nitrile, aldehyde will not show positional isomerism**

Functional Group Isomerism :

These isomers possess same molecular formula but different functional groups. Such compounds are called functional group isomers.

(i) Molecular formula : C_2H_6O

CH_3-CH_2-OH (Ethyl alcohol (Alcohol)) and CH_3-O-CH_3 (Dimethyl ether (Ether)) : Functional isomers

(ii) Molecular formula : C_3H_6O

$CH_3-CH_2-\overset{\overset{O}{\|}}{C}-H$ (Aldehyde (Alcohol)) and $CH_3-\overset{\overset{O}{\|}}{C}-CH_3$ (Propanone (Ketone)) : Functional isomers

(iii) Molecular formula : $C_3H_6O_2$

CH_3-CH_2-COOH (Propanoic acid (Acid)) and $CH_3-\overset{\overset{O}{\|}}{C}-O-CH_3$ (Methyl acetate (Ester)) : Functional isomers

(iv) Molecular formula : CH_3NO_2

$CH_3-\overset{\overset{O}{\|}}{N}\rightarrow O$ (Nitromethane) and $CH_3-O-N=O$ (Methyl nitrite) : Functional isomers

(v) Molecular formula : C_2H_5NO

$CH_3-CH=N-OH$ (Ethyl oxime) and $CH_3\overset{\overset{O}{\|}}{C}-NH_2$ (Ethanamide) : Functional isomers

Metamerism :

(1) This type of Isomerism is due to unequal distribution of substituents on either side of the functional group.

(2) Members belong to the same homologous series

***e.g.,*:** (i) Diethyl ether and methyl propyl ether

$CH_3CH_2OCH_2CH_3$ (Diethyl ether) $CH_3OCH_2CH_2CH_3$ (Methyl propyl ether)

(ii) Diethyl amine and methyl propylamine

$CH_3CH_2-NH-CH_2CH_3$ (Diethyl amine) $CH_3CH_2CH_2-NH-CH_3$ (Methyl propyl amine)

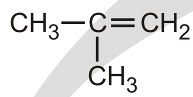
Ring chain isomerism :

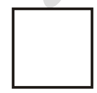
Such isomerism arises because of the difference of carbon–chain or ring.

For example :

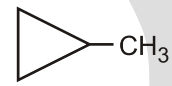
(i) Molecular formula : C_4H_6

$CH_3-CH_2-CH=CH_2$
(1-butene)


isobutylene
or 2-methyl propene


Cyclobutane

$CH_3-CH=CH-CH_3$


Methylcyclopropane

each of 1-butene and 2-butene is the chain isomer of 2-methylpropene while cyclobutane is the ring chain isomer of each of 1-butene, 2-butene and iso-butylene. Several; similar examples may be cited. Alkanes in no case exhibit ring-chain isomerism.

EXERCISE

SINGLE CHOICE QUESTIONS

1. How many distinct terminal alkynes exist with a molecular formula of C_5H_8?

(A) 1 (B) 2 (C) 3 (D) 4
(E) 5

2. How many distinct terminal alkynes exist with a molecular formula of C_6H_{10}?

(A) 1 (B) 2 (C) 3 (D) 4
(E) 5

3. Which of the following statements are correct?

(1) A pair of position isomers differs only in the position of the functional group(s).
(2) A pair of structural isomers has the same relative molecular mass.
(3) A pair of functional group isomers belongs to different homologous series.

(A) (1) and (2) only (B) (1) and (3) only
(C) (2) and (3) only (D) (1), (2) and (3)

4. Butanoic acid can be reduced to a primary alcohol. Which of the following compounds is the position isomer of the primary alcohol?

(A) $CH_3COCH_2CH_3$ (B) $CH_3CH_2CH_2CH_2OH$
(C) $CH_3CH_2CH_2CHO$ (D) $CH_2CH_2C(OH)HCH_3$

5. How many structural isomers does C_4H_8 have ?

(A) 3 (B) 4 (C) 5 (D) 6

6. Which of the following compounds are structural isomers of C_5H_{10}?

(1) 2-methylbut-2-ene (2) 3-methylbut-1-ene (3) Pent-1-ene

(A) (1) and (2) only (B) (1) and (3) only
(C) (2) and (3) only (D) (1), (2) and (3)

7. How many structural isomers does $C_3H_6Cl_2$ have?

(A) 2 (B) 3 (C) 4 (D) 5

8. Which of the following compounds are the structural isomers of $C_5H_{10}O$?

(1) 2-methylbutanal (2) Propyl ethanoate (3) Pentanal

(A) (1) and (2) only (B) (1) and (3) only

(C) (2) and (3) only (D) (1), (2) and (3)

9. How many different dibromophenols are possible?

(A) 8 (B) 7 (C) 6 (D) 5

(E) 4

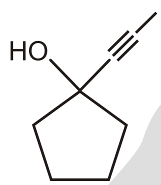
10. How many alcohols are possible for the molecular formula $C_5H_{12}O$ (consider only structural isomers) :

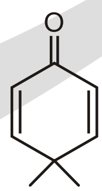
(A) 6 (B) 7 (C) 8 (D) 9

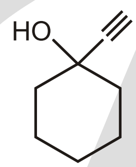
11. How many different structural isomers of molecular formula C_7H_{16} are possible, which contains five membered Parent Chain?

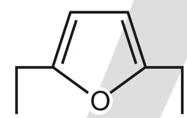
(A) 4 (B) 5 (C) 6 (D) 7

12. Among the following compounds, which is not a structural isomers of others?

(A) 

(B) 

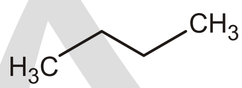
(C) 

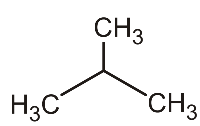
(D) 

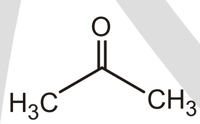
13. Number of structural isomer of C_6H_{14}?

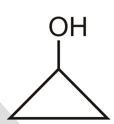
(A) 4 (B) 5 (C) 6 (D) 7

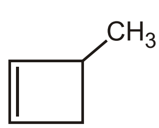
14. Which of the following pairs of molecules are NOT structural isomers?

(A) 

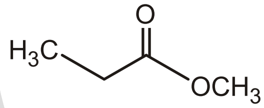


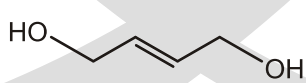
(B) 



(C) 

$H_3C-C\equiv C-CH_3$

(D) 



15. How many structurally different alkynes are formed having molecular formula C_5H_8.

(A) 2 (B) 3 (C) 4 (D) 5

16. How many possible structural isomers for $C_5H_{11}Cl$

(A) 6 (B) 7 (C) 8 (D) 9

17. What are the number of structural isomers possible in 1-butene and 1,3-butadiene one H is replaced by D.
(A) 2, 0 (B) 4, 2 (C) 2, 4 (D) 4, 4

18. Position isomers of 1-bromo-1-chloro cyclohexane are :
(A) 6 (B) 5 (C) 4 (D) 3

19. The number of strutural isomers of C_8H_{18} is '*a*' and the number of structural isomers of C_7H_{16} is '*b*.' What is the sum of *a* *b*?
(A) 15 (B) 24 (C) 27 (D) 20

20. Number of structural isomers of C_3H_6O are :
(A) 5 (B) 7 (C) 9 (D) 10

21. Number of structural isomers of $C_4H_8Br_2$ are :
(A) 7 (B) 8 (C) 9 (D) 10

22. Number of structural isomers of C_4H_7Cl are :
(A) 5 (B) 7 (C) 8 (D) 12

23. Number of structural isomers of C_3H_8O are :
(A) 2 (B) 3 (C) 5 (D) 6

24. Two isomeric forms of a saturated hydrocarbon :
(A) have the same structure
(B) have different compositions of elements
(C) have the same molecular formula
(D) All of these are correct

25. Number of structural isomers of C_4H_9Br are :
(A) 4 (B) 5 (C) 6 (D) 7

26. Number of structural isomers of C_4H_6 are :
(A) 4 (B) 5 (C) 6 (D) 9

27. Number of 6 membered aromatic compounds with molecular formula $C_2H_2N_4$ are :
(A) 3 (B) 4 (C) 5 (D) 6

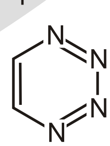
28. Number of cyclic isomers of C_6H_{12} are :
(A) 11 (B) 12 (C) 13 (D) 14

29. Number of structural isomers of C_3H_6O are :
(A) 8 (B) 9 (C) 5 (D) 6

30. Number of isomers of C_5H_{10} are :
(A) 10 (B) 11 (C) 9 (D) 8

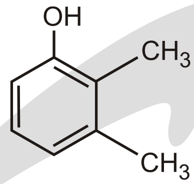
31. Number of non-cyclic isomers of C_4H_8O are :
(A) 6 (B) 7 (C) 8 (D) 15

32. Number of positional isomers of given compound with 6 membered aromatic ring are :



(A) 3 (B) 4 (C) 5 (D) 6

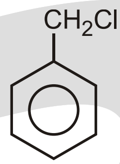
33. Number of positional isomers of given compound with 6 membered aromatic ring are :



(Xylenol)

(A) 3 (B) 4 (C) 5 (D) 6

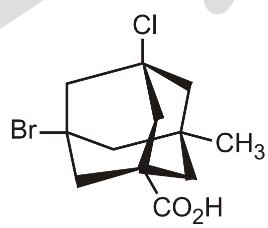
34. Number of positional isomers of given compound with 6 membered aromatic ring are :



(A) 3 (B) 4 (C) 5 (D) 6

SUBJECTIVE TYPE QUESTIONS

1. How many stereoisomers are theoretically possible for the following adamantane derivative?



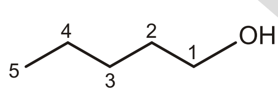
Answers

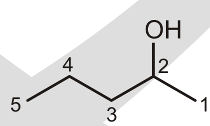
Single Choice Questions

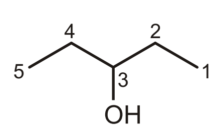
1.	(B)	**2.**	(C)	**3.**	(C)	**4.**	(D)	**5.**	(C)	**6.**	(D)	**7.**	(C)	**8.**	(D)
9.	(C)	**10.**	(C)	**11.**	(B)	**12.**	(B)	**13.**	(B)	**14.**	(C)	**15.**	(B)	**16.**	(C)
17.	(B)	**18.**	(D)	**19.**	(C)	**20.**	(C)	**21.**	(B)	**22.**	(D)	**23.**	(B)	**24.**	(C)
25.	(A)	**26.**	(D)	**27.**	(A)	**28.**	(B)	**29.**	(B)	**30.**	(A)	**31.**	(C)	**32.**	(A)
33.	(D)	**34.**	(B)												

10. $C_5H_{12}O$ D.B.E. value = 0 that means, saturated alcohol is been made.

Taking parent chain to be 5 membered :

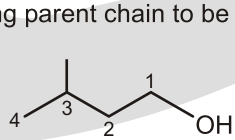


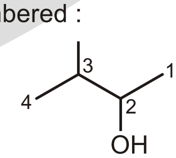


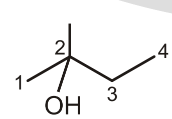


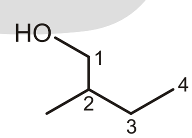
Above three are structurally different (means have different IUPAC names)

Taking parent chain to be 4 membered :



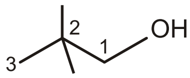






Above four are also structually different.

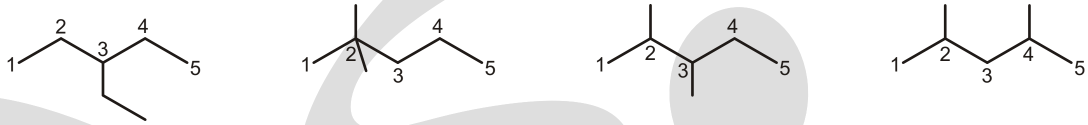
Now, taking parent chain to be 3 membered :

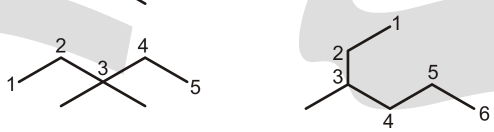


Only 1 possible alcohol

Thus, total 8 alcohols are possible for molecular formula $C_5H_{12}O$.

11.

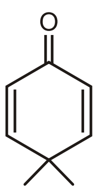




(Six carbon parent chain)

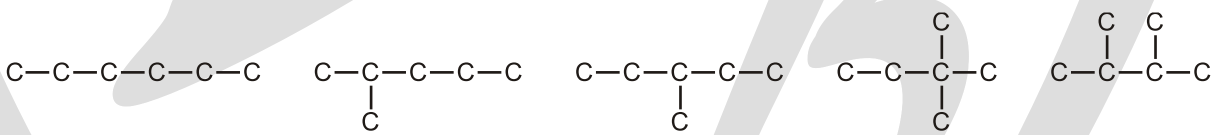
Thus, 5 structures are possible.

12.



have D.B.E. value = 4, rest all have DBE value = 3

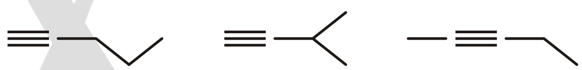
13.



Total structural isomers = 5

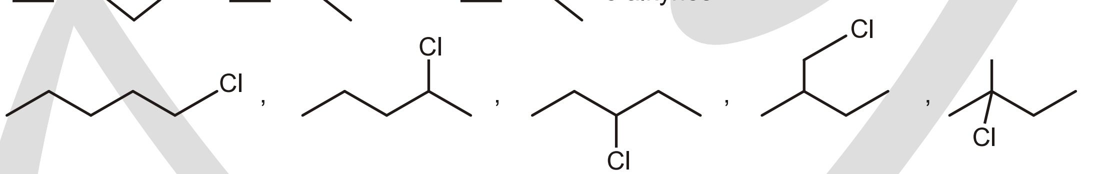
14. In (C) number of carbons are not same.

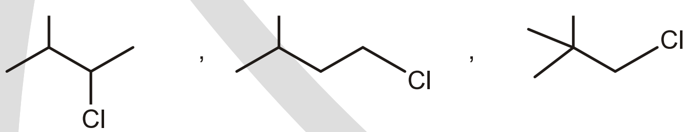
15.



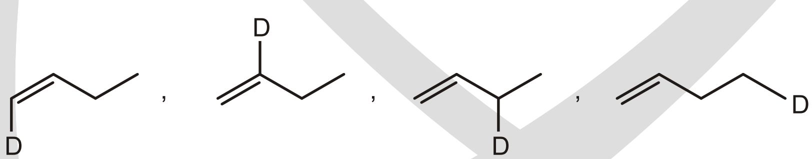
3 alkynes

16.

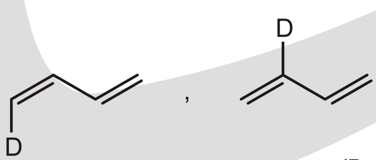




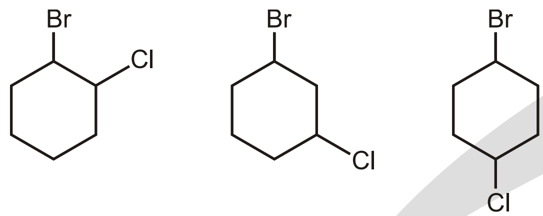
17.



(1-Butene)

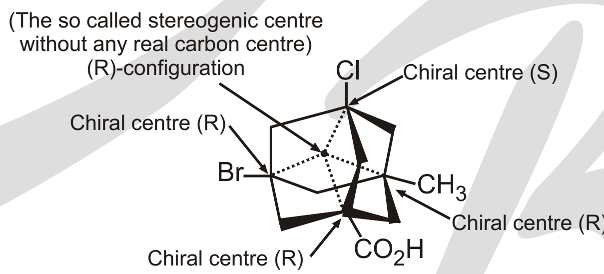


(Buta-1, 3-diene)

18. 

Subjective Type Questions

1. Although four asymmetric carbon atoms are present in this compound, it exists in the form of only two stereoisomers (enantiomers) rather than the sixteen predicted by the 2^n rule. There is, in fact, only one stereogenic centre, shown by the black dot at the centre of the molecule (in the following diagram), and the configuration shown here is (R). The asymmetric carbon atoms are not stereogenic centres, because any two substituents on any of these bridgehead carbons cannot be exchanged without destroying the constitutional integrity of this rigid, highly bridged molecule. The configurations of the asymmetric bridgehead carbon units in this structure are (R), (S), (R) and (R) respectively, for the Br, Cl, CO_2H and CH_3 substituents. Its enantiomer would, of course, have the opposite configurations.



❑❑❑

CHAPTER

23

Tautomerism

TAUTOMERISM

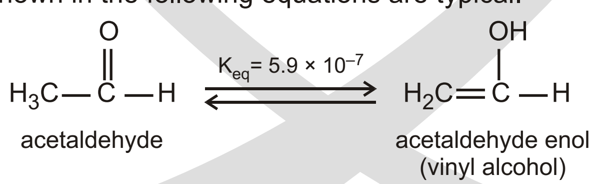
You may hear the word tautomers used to describe the relationship between enols and their corresponding carbonyl compounds. The term tautomers means "constitutional isomers that undergo such rapid interconversion that they cannot be independently isolated." Indeed under most common circumstances, carbonyl compounds and their corresponding enols are in rapid equilibrium. However, chemists now know that the interconversion of enols and their corresponding carbonyl compounds is catalyzed by acids and bases. This reaction can be very slow in dilute solution in the absence of acid or base catalysts, and indeed, enols have actually been isolated under very carefully controlled conditions.

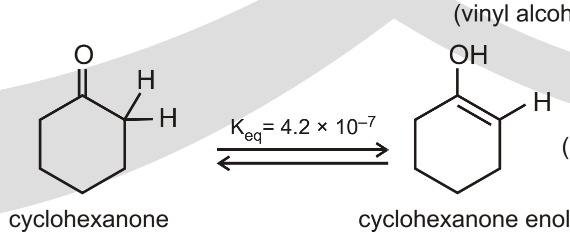
Tautomerism is a special type of functional group isomerism which arises due to the transfer of H–atom as proton from a polyvalent atom to other polyvalent atom.

☞ The other names of tautomerism are **'desmotroprism'** or **'prototropy'.**

ENOLIZATION OF CARBONYL COMPOUNDS

Carbonyl compounds with a-hydrogens are in equilibrium with small amounts of their enol isomers. The equilibrium constants shown in the following equations are typical.

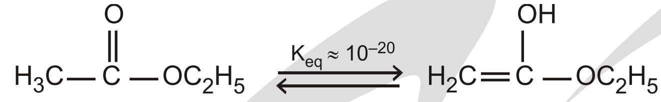




(Enol = 1.2% in liquid phase)

Unsymmetrical ketones are in equilibrium with more than one enol.

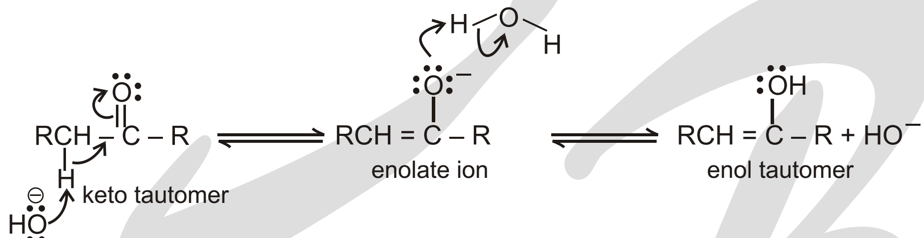
Ester contain even smaller amounts of enol isomers than aldehydes or keotnes.



Mechanism of the tautomerism : $R - CH_2 - \overset{O}{\overset{\|}{C}} - R \rightleftharpoons R - CH = \overset{OH}{\overset{|}{C}} - R$

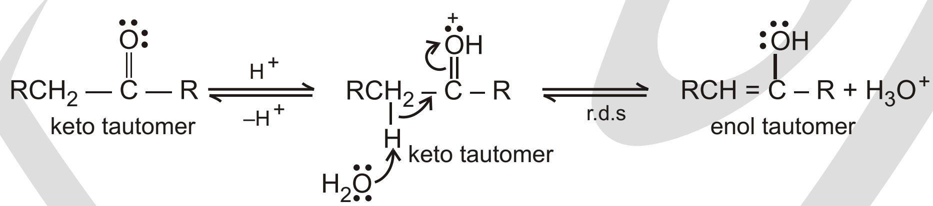
In a basic solution, hydroxide ion removes a proton from the -carbon of the keto tautomer, forming an enolate ion. Protonation on oxygen forms the enol tautomer, whereas protonation on the -carbon re-forms the keto tautomer.

BASE - CATALYZED KETO-ENOL INTERCONVERSION



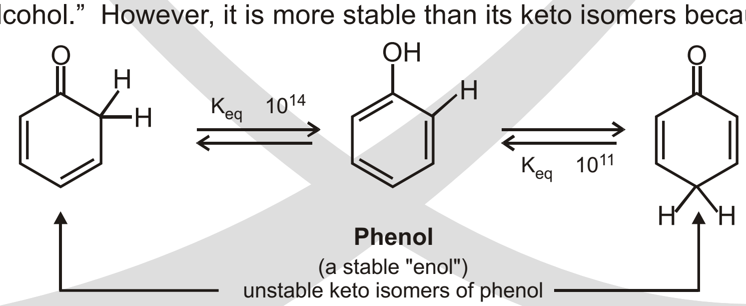
In an acidic solution, the carbonyl oxygen of the keto tautomer is protonated and water removes a proton from the -carbon, forming the enol.

ACID-CATALYZED KETO-ENOL INTERCONVERSION



The major reason for the instability of enols is that the C = O double bond of a carbonyl group is a stronger bond than the C = C double bond of a enol. With esters and acids, the additional instability of enols results from loss of the stabilizing resonance interaction between the carboxylate oxygen and the carbonyl electrons that is present in the carbonyl forms.

Some enols are more stable than their corresponding carbonyl compounds. Notice that phenol is conceptually an enol – a "vinylic alcohol." However, it is more stable than its keto isomers because phenol is aromatic.



DIFFERENCE BETWEEN TAUTOMERISM & RESONANCE

(a) In tautomerism, an atom changes place but resonance involves a change of position of pi-electrons or unshared electrons.

(b) Tautomers are different compounds and they can be separated by suitable methods but resonating structures cannot be separated as they are imaginary structures of the same compound.

(c) Two tautomers have different functional groups but there is same functional group in all canonical structures of a resonance hybrid.

(d) Two tautomers are in dynamic equilibrium but in resonance only one compound exists.

(e) Resonance in a molecule lowers the energy and thus stabilises a compound and decreases its reactivity but no such effects occur in tautomerism.

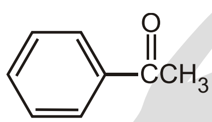
(f) In resonance, bond length of single bond decreases and that of double bond increases *e.g.*, all six C-C bonds in benzene are equal and length is in between the length of a single and a double bond.

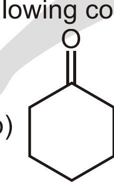
(g) Resonance occurs in planar molecule but atoms of tautomers may remain in different planes as well.

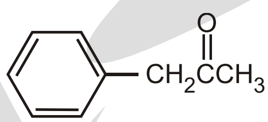
(h) Tautomers are indicated by double arrow $\rightleftharpoons$ in between the two isomers but double headed single arrow is put between the canonical (resonating) structures of a resonating molecule and effect of temperature and pressure is observed in tautomers.

Solved Example

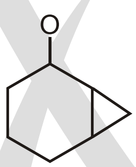
- Draw the most stable enol tautomers for each of the following compounds.

(a) 

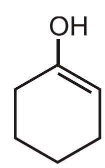
(b) 

(c) 

(d) $CH_3 — CH_2 — \overset{\overset{\displaystyle O}{||}}{C} — CH_2 — \overset{\overset{\displaystyle O}{||}}{C} — CH_2 — CH_3$

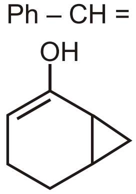
(e) 

Sol. (a) $Ph - \overset{\overset{\displaystyle OH}{|}}{C} = CH_2$

(b) 

(c) $Ph - CH = \overset{\overset{\displaystyle OH}{|}}{C} - CH_3$

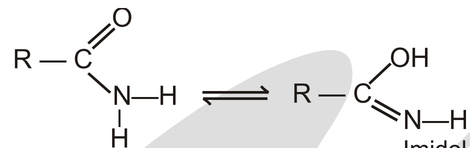
(d) $Et - \overset{\overset{\displaystyle OH}{|}}{C} = CH - \overset{\overset{\displaystyle O}{||}}{C} - Et$

(e) 

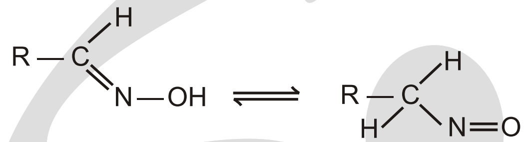
THREE ATOM TAUTOMERISM

$$\overset{H}{\diagdown}A — B = C \rightleftharpoons A = B — C\overset{H}{\diagup}$$

This type of Tautomerism is very prevalent, with one tautomer usually being much more stable than the other. For example, the summations of bond energies correctly predict the greater stabilities of amides, oximes, and aliphatic hydrazones overs the isomeric imidols, nitroso compounds, and aliphatic azo compounds, respectively.



Amide Imidol



Oxime Nitroso compound

$R_2C \quad N—NHR \rightleftharpoons R_2CH—N \quad N—R$

Hydrazone Azo compound

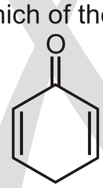
Another common pair of tautomers is the ketone - enol couple.

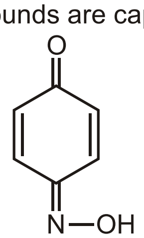
$$\overset{H}{}CH_2—CH=O \rightleftharpoons CH_2=CH—OH$$

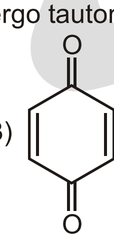
(i)	Amidine system :	$NH_2—CH \quad NH \rightleftharpoons HN \quad CH—NH_2$
(ii)	Nitroso-oximino system :	$CH_3—N \quad O \rightleftharpoons CH_2 \quad N—OH$
(iii)	Amido - imidol system :	$NH_2—CH \quad O \rightleftharpoons HN \quad CH—OH$
(iv)	Azo-hydrazone system :	$HN \quad N—CH_3 \rightleftharpoons HN—N \quad CH_2$
(v)	Diazo-nitrosamine system :	$Ar—N \quad N—OH \rightleftharpoons Ar—NH—N \quad O$
(vi)	Diazo-amino (triazen) system :	$HN \quad N—NH_2 \rightleftharpoons NH_2—N \quad NH$

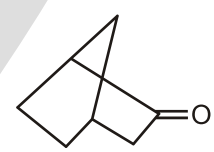
Solved Example

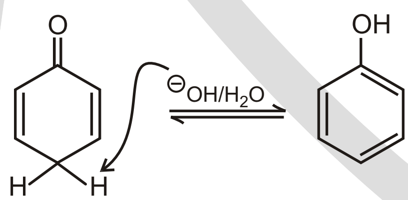
- Which of the following compounds are capable to undergo tautomerism in base catalysed medium?

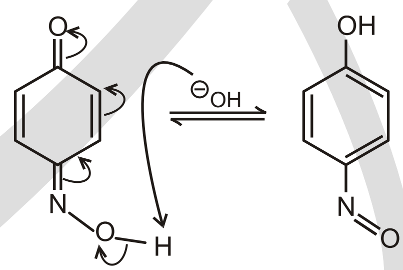
(1) 

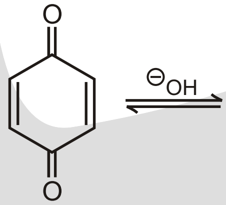
(2) 

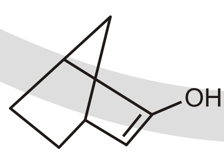
(3) 

(4) 

Sol. (1) 

(2) 

(3)  No tautomerism

(4) 

EXPERIMENTAL EVIDENCE

Acetoacetic ester $CH_3 — \overset{\overset{\large O}{||}}{C} — CH_2 — \overset{\overset{\large O}{||}}{C} — OEt$, was first discovered by Geuther (1863), who prepared it by the action of sodium on ethyl acetate, and suggested the formula $CH_3C(OH) = CHCO_2C_2H_5$ (-hydroxycrotonic ester). In 1865, Frankland and Duppa, who, independently of Geuther, also prepared acetoacetic ester by the action of sodium on ethyl acetate, proposed the formula $CH_3COCH_2CO_2C_2H_5$ (-ketobutyric ester). These two formula immediately gave rise to two schools of thought, one upholding the Geuther formula, and the other the Frankland-Duppa formula, each school bringing forward evidence to prove its own claim.

1. **Experimenal evidence in favour of the Geuther fomula :**
   (i) When acetoacetic ester is treated with an ethanolic solution of bromine, the colour of the latter is immediately discharged. This indicates the presence of an olefinic double bond $\left(>C=C< \right)$
   (ii) When acetoacetic ester is treated with sodium, hydrogen is evolved and the sodium derivative is formed. This indicates the presence of a hydroxyl group (– OH group).
   (iii) When acetoacetic ester is treated with ferric chloride, a reddish-violet colour is produced.
   This is characteristic of compounds containing the group $—C(OH)=C<$

2. **Experimenal evidence in favour of the Frankland – Duppa formula test for (enol).**
   (i) Acetoacetic ester forms a cyanohydrin with hydrogen cyanide.
   (ii) Acetoacetic ester forms a bisulphite compound with sodium hydrogen sulphite.
   Controversy continued until about 1910, when chemists were coming to the conclusion that both formulae were correct, and that the two compounds existed together in equilibrium in solution.

$$CH_3COCH_2CO_2C_2H_5 \rightleftharpoons CH_3\overset{\overset{\large OH}{|}}{C}=CHCO_2C_2H_5$$

DIRECTION OF EQUILIBRIUM IN KETO–ENOL TAUTOMERISM

$$G^{\ominus} \quad H^{\ominus} \quad T\ S^{\ominus} \quad RT \ln K$$

keto $\rightleftharpoons$ enol ; K [enol] / [keto]

A knowledge of the values of the enthalpy and entropy changes would enable us to calculate $G^{\ominus}$ and from this, K, the equilibrium constant of the reaction :

The major reason for the instability of enols is that the C O double bond of a carbonyl group is a stronger bond than the C C double bond of a enol.

If the equilibrium mixture contains 1 per cent enol, then

$$G^{\ominus} = –5.7 \log 1/99 = 11.42 \text{ kJ}$$

If the equilibrium mixture contains 99 per cent enol, then

$$G^{\ominus} = –5.7 \log 99 = –11.42 \text{ kJ}$$

Thus difference of 22.84 kJ mol ^{1} in G shifts the equilibrium from 1 per cent enol to 99 per cent enol. It can therefore be seen that small changes in G will have large effects on the position of the equilibrium. Hence any attempt to calculate K from estimated values of H and S is almost certainly doomed to failure. However, a semi-quantitative approach is instructive. Any factors that affect $H^{\ominus}$ and $S^{\ominus}$ will affect the value of $G^{\ominus}$.

The sum of the bond energies of the $–CH_2CO–$ group is 1870 and that of the $– CH = C(OH) –$ group is 1820 kJ (from Table 2.2) :

$$2C – H(828.4) + C – C(347.3) + C = O (694.5) = 1870 \text{ kJ}$$

$$C – H(414.2) + C = C(606.7) + C – O (334.7) + O – H (464.4) = 1820 \text{ kJ}$$

This means that the enthalpy of formation of the CH_2CO group is –1870 kJ and that of the CH = C(OH) group is –1820 kJ. Thus the keto form is more stable than the enol by 50 kJ mol ^{1} and hence there must be some driving formce to bring about enolisation.

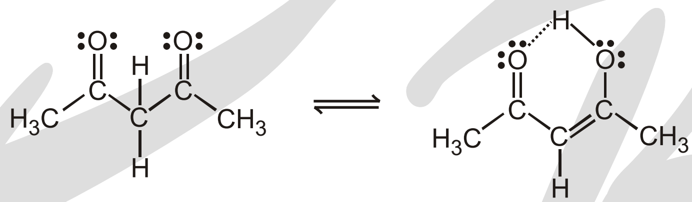
It's important that you understand that the keto form is way more stable than the enol form. (Typically, only 1 of every 100,000 to 10,000,000 molecules is in the enol form at any particular time.)

☑ SPECIAL TOPIC

GERO ENTROPY

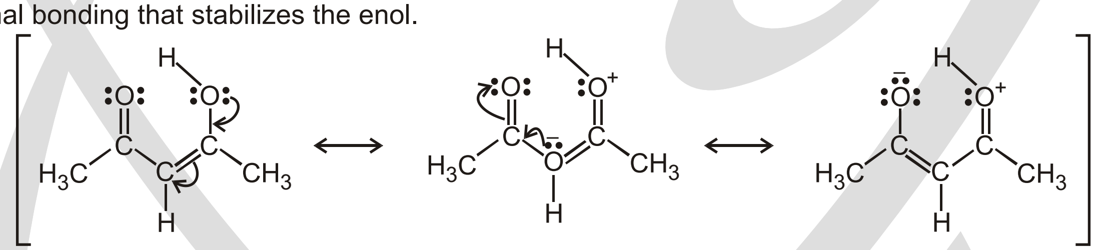
Gero (1962) has found that cyclic monoketones contain more enol than the corresponding acyclic 2-one. In the cyclic ketones, change from keto to enol involves a relatively small change in strain in the ring due to the introduction of a double bond. For acyclic ketones, the introduction of the double bond has a much greater effect on the freedom to take up different conformations.

The enols of -dicarbonyl compounds are also relatively stable. (-Dicarbonyl compounds have two carbonyl groups separated by one carbon).

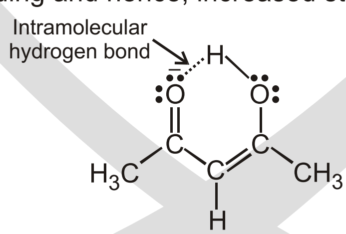


2,4-pentanedione (acetylacetone) (a b-dicarbonyl compound) — enol form 92% in hexane solution

There are two reasons for the stability of these enols. First, they are conjugated, but their parent carbonyl compounds are not. The resonance stabilization (-electron overlap) associated with conjugation provides additional bonding that stabilizes the enol.

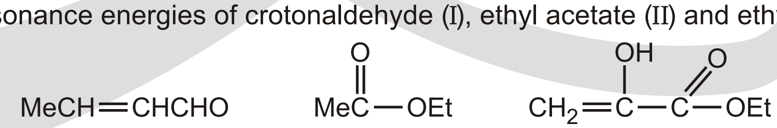


The second stabilizing effect is the intramolecular hydrogen bond present in each of these enols. This provides another source of increased bonding and hence, increased stabilization.



The enol form, in each case, is stabilised by intramolecular hydrogen bonding (say, 29.3 kJ mol ^{1}).

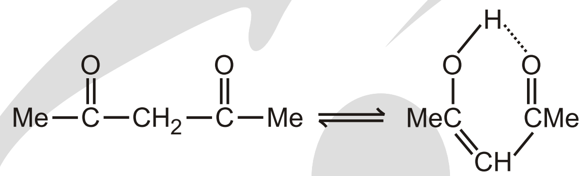
Now let us consider the resonance energies of crotonaldehyde (I), ethyl acetate (II) and ethyl

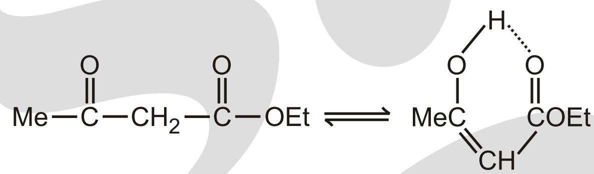


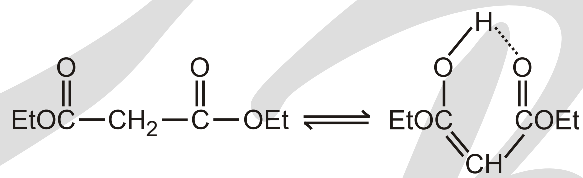
R.E. (kJ0 (I) 10.40 (II) 75.35 (III) 75.35

methacrylate (III). The important point here is that in (III) the ethylenic bond does not enter into resonance with the CO_2Et group. (III) contains crossed conjugation, and the contribution to resonance is only the path that leads to the greater R.E., which , in this case, is the carbethoxy group.

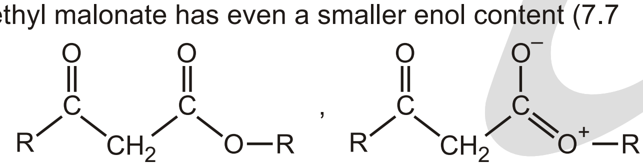
Consider the following equilibria :

Acetylacetone  76 %

Ethyl acetoacetate  8 %

Ethyl malonate  10 ^{3}%

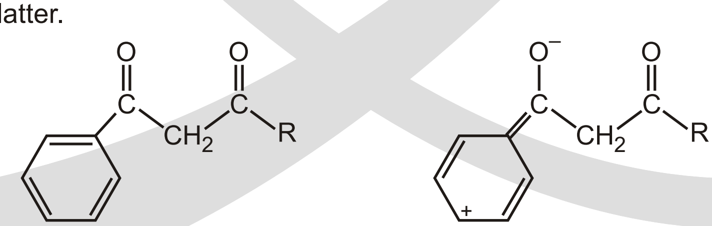
When an OR group is substituted for one of the R groups in, thereby producing an acetoacetic ester, there is a decrease in enol content. This can be attributed largely to a cross conjugation between the ester group and the enol structure. The resonance of the ester group occurs at the expense of the resonance of the enol ring. This destabilizes the enol with respect to the keto tautomer and shifts the equilibrium toward the ketone. As to be expected, the diester ethyl malonate has even a smaller enol content (7.7 10 ^{3} per cent).



	Enol content
$EtO — CO — CH_2 — CO — OEt$	< 1%
$CH_3 — CO — CH_2 — CO — OEt$	7.7
$— CO — CH_2 — CO — OEt$	21
$CH_3 — CO — CH_2 — CO — CH_3$	76
$— CO — CH_2 — CO — CH_3$	89
$— CO — CH_2 — CO —$	100

= Ph

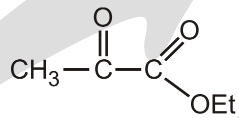
The substitution of a phenyl group increases the resonance energy of the keto as well as the enol tautomer, but much more so for the latter.



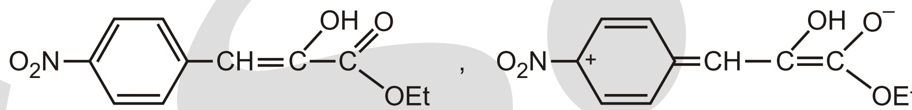
(a) (b)

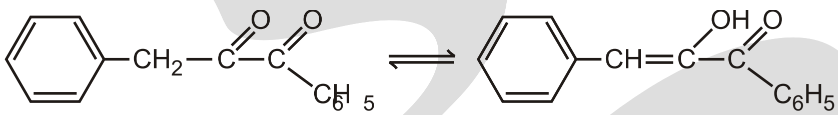
Consequently, diaroylmethanes, ($CO)_2CH_2$, are essentially 100 per cent enolic.

The effect of resonance in stabilizing the enols of -diketones is also demonstrable. For illustration, the enol content in ethyl pyruvate is too small to give a reaction with diazomethane. However, ethyl *p*-nitropyruvate is 42 per cent enolic in alcohol solution. Hence again the enol is stabilized by the “cinnamoyl” resonance energy. Other benzyl glyoxals have large enolic contents, too, and some range up to 100 per cent.

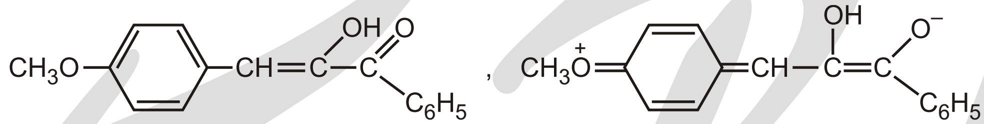


Ethyl pyruvate





30% enolic

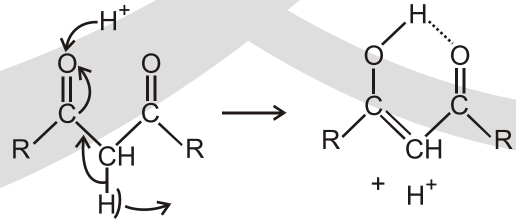


100% enolic

Table : Enol Content of some 1, 3-Diketones in various solvents

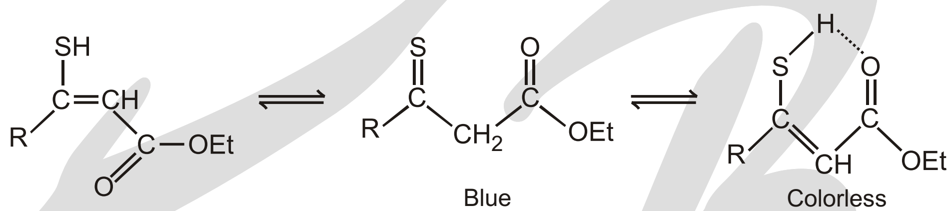
Diketone	Per cent enol	Solvent
C_6H_5 — CS — CH_2 — CO_2 — C_2H_5	95	Isooctane
	87	Ethanol
CH_3 — CO — CH_2 — $CO_2C_2H_5$	46.4	Hexane
	16.2	Benzene
	10.5	Ethanol
	10.1	Nitrobenzen
CH_3 — CO — CH_2 — CO — CH_3	92	Hexane
	15	Water
CH_3 — CO — CH — $CO_2C_2H_5$	67	Hexane
	31	Water

Enolization probably takes place by ionization of an hydrogen atom and short electron - pair shifts. Therefore, the more electronegative are the groups attached to the a carbon atom, the easier it is for an hydrogen atom to escape as a proton. This expectation is fully substantiated in the substituted acetoacetic esters CH_3CO — CHR — CO_2Et, in which the enol content increases with increasing electronegativity of the R group.



R in $CH_3COCHCO_2Et$ (R on CH)	Per cent enol (liquid)
C_2H_5	3
CH_3	4
H	7.5
C_6H_5	30
CO_2Et	44

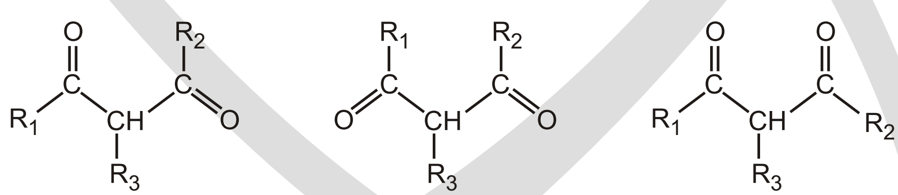
The inductive effect alone, however, is only a minor factor. For example, ethyl cyanoacetate has only a 0.25 per cent enol content. Another illustration is provided by the ketosulfones (RSO_2CH_2COR) and -disulfones ($RSO_2 — CH_2 — SO_2R$) which are hardly enolized at all and are weaker acids than the corresponding -diketones. The sulfones do not form chelates and the keto sulfones exhibit typical C = O infrared absorption and no O – H----O bands. On the other hand, thiobenzoylacetates are more enolic than the corresponding benzoylacetates.



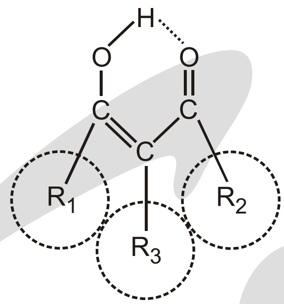
	Enol content (in alcohol)
$CH_3 — CO — CH_2 — CO_2Et$	11
$CH_3 — CS — CH_2 — CO_2Et$	41
$— CO — CH_2 — CO_2Et$	29(liq)
$— CS — CH_2 — CO_2Et$	87

The weakness of the C = S bond shifts the equilibrium in favor of the thioenol. Since the S – H --- O bond is not as strong as the O – H --- O bond, some of the thioenol exists in a nonchelated structure.

Steric and entropy effects : Finally, it can be shown that steric requirements and entropy changes can have a prolound effect upon the keto-enol equilibrium. A diketone $R_1 — CO — CHR_3 — CO — R_2$ may exist in several conformations.



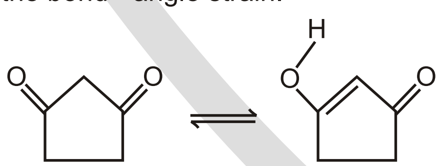
R in $R_1—CO—CH(R_2)—CO—R_3$	% Enol (liquid phase)
R_1 R_3 CH_3, R_2 H	76%
R_1 R_2 R_3 CH_3	44%
R_1 R_2 R_3 t B_4	0%



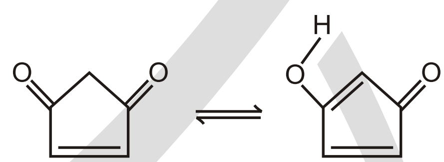
Compound	Per cent enol content	Compound	Per cent enol content
2-Butanone	0.12	3-Pentanone	0.07
2-Pentanone	0.01	3-Hexanone	0.05
2-Hexanone	0.11	3-Heptanone	0.17
2-Heptanone	0.10	3-Octanone	0.01
2-Octanone	0.92		

Ring size of cyclanone	Per cent enol content
4	0.55
5	0.09
6	1.18
7	0.56
8	9.3
9	4.0
10	6.1

It is interesting, in this connection, that cyclopentane-1,3-dione(LV) is completely enolic while cyclopenten-3, 5-dione (LVI) is completely ketonic. This further supports the notion that the cyclopentadienone is very unstable because of the bond - angle strain.



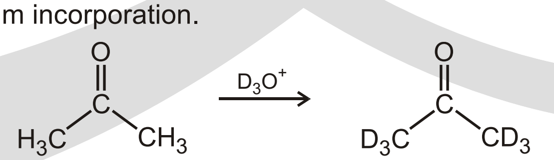
LV



LVI

REPLACEMENT OF HYDROGEN BY DEUTERIUM

Reaction of acetone with D_3O yields hexadeuterioacetone. That is, all the hydrogens in acetone are exchanged for deuterium. Review the mechanism of mercuric ion-catalyzed alkyne hydration and then propose a mechanism for this deuterium incorporation.



Acetone Hexadeuterioacetone

Acetone in deuterium oxide (D_2O) containing NaOD undergoes hydrogen-deuterium exchange and is transformed in successive steps into CD_3COCD_3 :

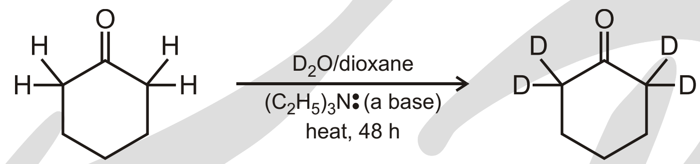
$$CH_3COCH_3 + DO^- \rightleftharpoons CH_3COCH_2\!:^- + HOD$$

$$CH_3COCH_2\!:^- + D_2O \rightleftharpoons CH_3COCH_2D + DO^-\!: \quad \text{etc.}$$

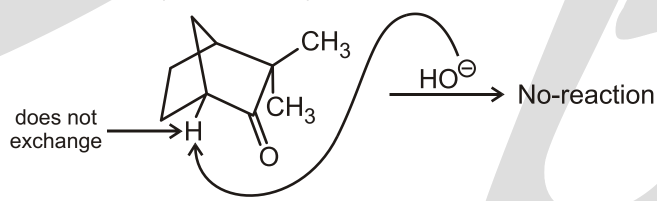
This exchange reaction is often of diagnostic value in determining the number of (replaceable) hydrogen atoms in an unknown carbonyl compound. Exchange of this kind, followed by analysis of the recovered compound for its deuterium content, shows the number of hydrogen atoms.

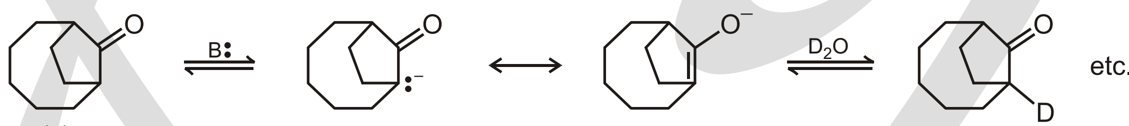
The acidities of aldehydes, ketones, and esters are particularly important because enolate ions are key reactive intermediates intermediates in many important reactions of carbonyl compounds. Let's consider the types of reactivity we can expect to observe with enolate ions.

First, enolate ions are Bronsted bases, and they react with Bronsted acids. The formation of enolate ions and their reactions with Bronsted acids have two simple but important consequences. First, the -hydrogens of an aldehyde or ketone – and no others – can be exchanged for deuterium by treating the carbonyl compound with a base in D_2O.



The following compound does not undergo base-catalyzed exchange in D_2O even though it has an -hydrogen because -hydrogen is less acidic - (Bredt's rule) is violated.



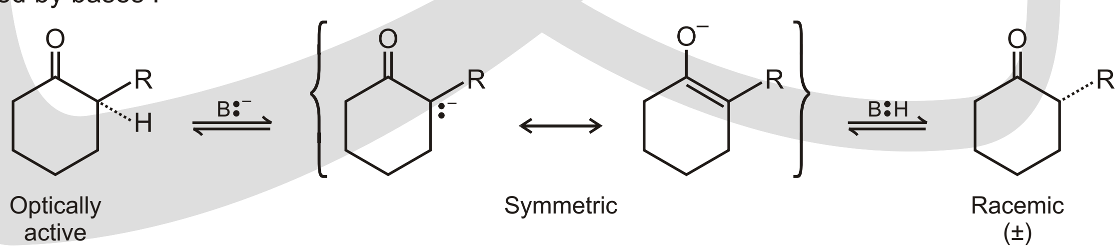


(a)

(b)

The non-exchangeability of the hydrogens in (b) is expressed in Bredt's rule, which states that a bicyclic compound cannot contain a double bond at a bridgehead. It is apparent that the rule does not apply when the ring system is large enough to accommodate the double bond, as in compound (a).

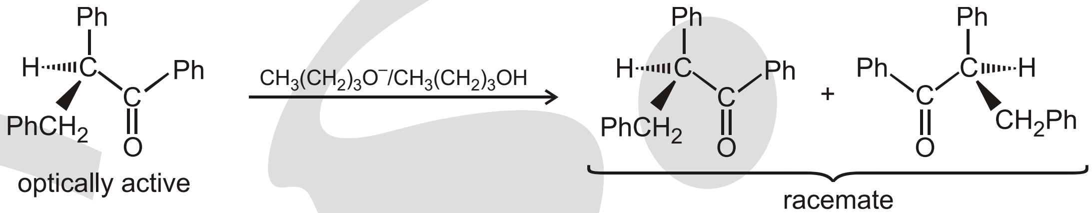
When the ketone (or aldehyde or ester) possesses an asymmetric carbon atom, the formation of the enolate anion destroys this asymmetry, with the result that optically active carbonyl compounds of this type are racemized by bases :



The symmetrical enolate anion can be reprotonated with equal probability to give either enantiomer.*

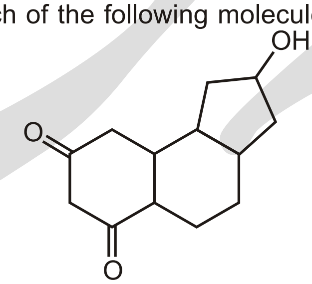
The equilibrium between the keto and enol forms is established by a deprotonation-protonation reaction that, when base catalyzed, can be regarded as proceeding via the enolate anion.

The second consequence of enolate - ion formation and protonation is that if an optically active aldehyde or ketone owes its chirality solely to an asymmetric -carbon, and if this carbon bears a hydrogen, the compound will be racemized by base.

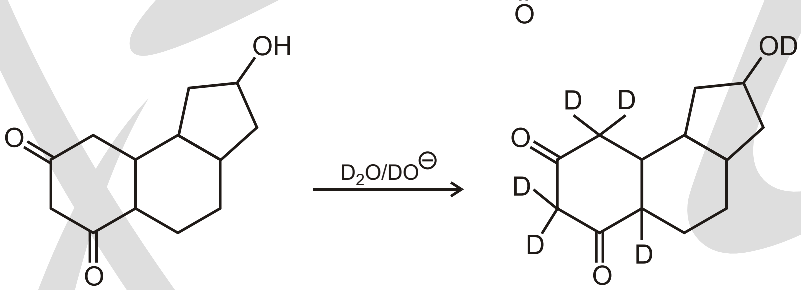


Solved Example

- Indicate which hydrogen(s) in each of the following molecules (if any) would be exchanged for deuterium following base treatement in D_2O:



Sol.



EXERCISE

SINGLE CHOICE QUESTIONS

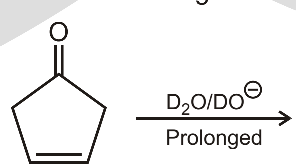
1. Tautomers are :

(A) Structural isomers
(B) Conformational isomers
(C) Configurational isomers
(D) Geometric isomers

2. Mixed ether are also known as :

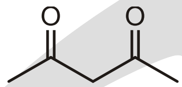
(A) Symmetric ethers
(B) Unsymmetric ethers
(C) Dialkyl ethers
(D) Oxane

3. Number of hydrogen replaced by Deuterium in the given reaction are :



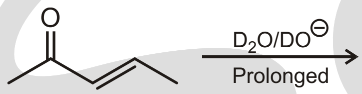
(A) 3 (B) 6 (C) 8 (D) 10

4. Total number of enol possible for the given compound are :



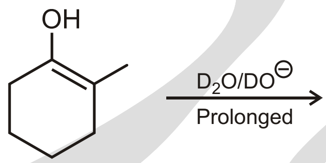
(A) 1 (B) 2 (C) 3 (D) 4

5. Number of hydrogen replaced by Deuterium in the given reaction are :



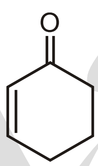
(A) 3 (B) 7 (C) 8 (D) 6

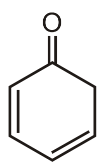
6. Number of hydrogen replaced by Deuterium in the given reaction are :

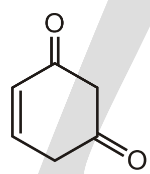


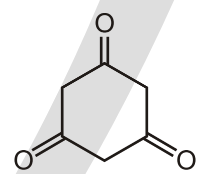
(A) 3 (B) 7 (C) 8 (D) 1

7. Which of the given compounds will have minimum enol content?

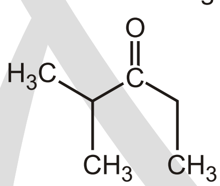
(A) 

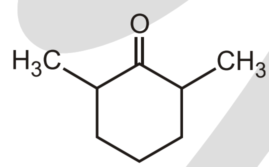
(B) 

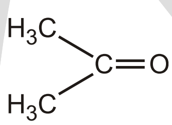
(C) 

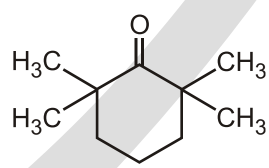
(D) 

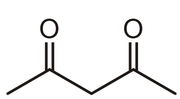
8. Which of the following compounds cannot exhibit keto-enol tautomerism?

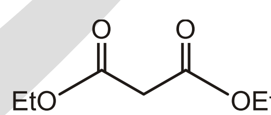
(A) 

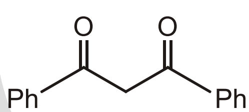
(B) 

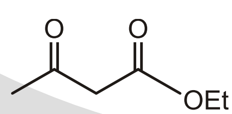
(C) 

(D) 

9. (I) 

(II) 

(III) 

(IV) 

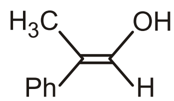
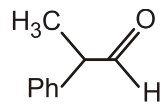
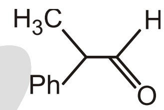
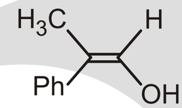
Correct order of enol content of given compound will be :

(A) III I IV II

(B) IV II I III

(C) II III I IV

(D) I III II IV

10. (I)  (II)  (III)  (IV) 

Correct relationship between above compounds :

(A) II and III are Geometrical isomers (B) I and II are Metamers

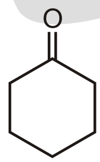
(C) III and IV are Tautomers (D) I and IV are Tautomers

11. Tautomerism is also known as :

(A) Protropy (B) Dynamic isomerism

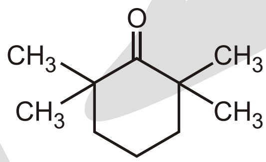
(C) Kryptomerism (D) all of above

12. Maximum enol content is observed in :

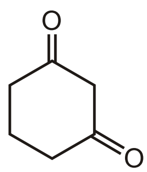
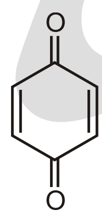
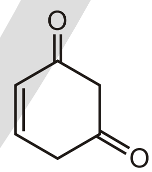
(A) $CH_3-\overset{\overset{O}{||}}{C}-CH_3$ (B)  (C) $CH_3COCH_2COCH_3$ (D) $CH_3CH_3\overset{\overset{O}{||}}{C}H$

13. Keto enol tautomerism is observed in :

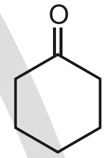
(A) C_6H_5CHO (B) $C_6H_5COC_6H_5$

(C)  (D) $C_6H_5COCH_2-COCH_3$

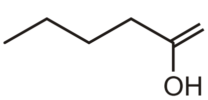
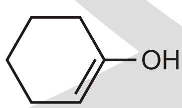
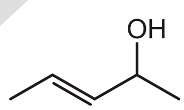
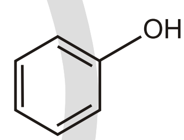
14. Keto enol tautomerism does not observed in :

(A)  (B)  (C)  (D) 

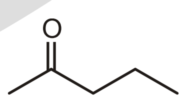
15. Tautomer of which of the following can show geometrical isomerism :

(A) CH_3-CHO (B) CH_3CH_2-CHO (C) $(CH_3)_2CH-CH$ O (D) 

16. Which of the following can't show Tautomerism?

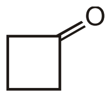
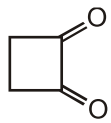
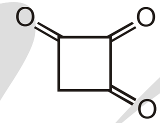
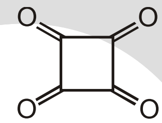
(A)  (B)  (C)  (D) 

17. Number of enols of the given compound are :

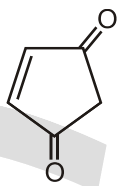


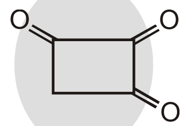
(A) 1 (B) 2 (C) 3 (D) 4

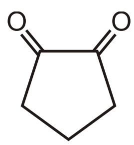
18. Maximum enol content is in which of the given compound :

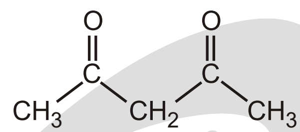
(A)  (B)  (C)  (D) 

19. Which of the following compounds has negligible enol?

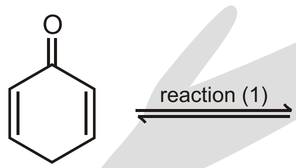
(A) 

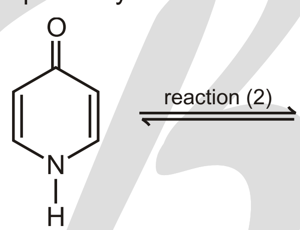
(B) 

(C) 

(D) 

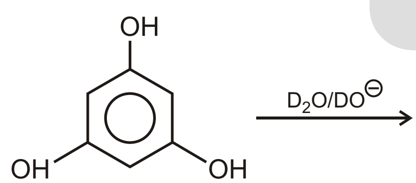
20. Direction of equilibrium for the given reactions 1 and 2, respectively are :

(1) 

(2) 

(A) forward, forward
(B) forward, backward
(C) backward, forward
(D) backward, backward

21. Number of hydrogen replaced by Deuterium in the given reaction are :



(A) 3
(B) 6
(C) 8
(D) 10

22. $\underset{(A)}{Ph-\underset{\substack{|\\ OH}}{CH}-CHO} \underset{H_2O}{\overset{HO^{\ominus}}{\rightleftharpoons}} (B)$

(*A*) & (*B*) are isomer and isomerization is effectively carried out by trace of base. Identify isomer (*B*).

(A) $Ph-CH_2-CO_2H$

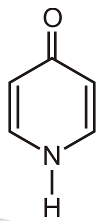
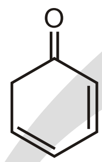
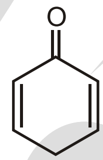
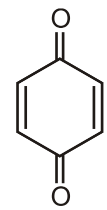
(B) $Ph-\overset{\overset{\displaystyle O}{||}}{C}-O-CH_3$

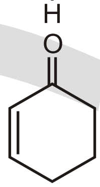
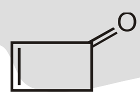
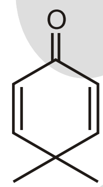
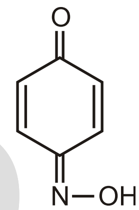
(C) $Ph-\overset{\overset{\displaystyle O}{||}}{C}-CH_2-OH$

(D) $H-\overset{\overset{\displaystyle O}{||}}{C}-CH_2-O-Ph$

UNSOLVED EXAMPLE

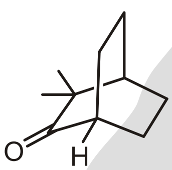
1. *X* Number of compounds which undergo Tautomerisation to form an Aromatic product.

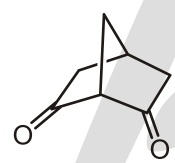
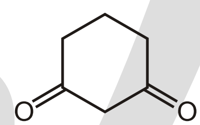
(a) 
(b) 
(c) 
(d) 

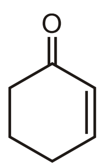
(e) 
(f) 
(g) 
(h) 

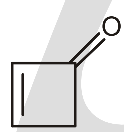
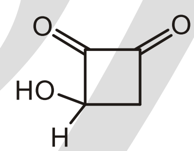
Find the value of *X*?

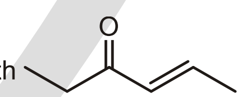
2. Which compound in each of the following pairs would be more extensively enolized?

(i)  and 

(ii)  and 

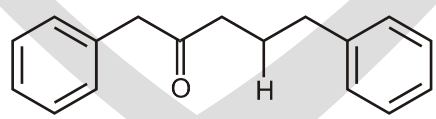
(iii)  and 

(iv)  and 

3. *Z* Number of deuterium exchanged on prolonged treatment of $\overset{\ominus}{O}D/D_2O$ with  . Find the value of *Z*?

SUBJECTIVE TYPE QUESTIONS

1. Treatment of this ketone with basic D_2O leads to rapid replacement of two hydrogen atoms by deuterium. Then, more slowly, all the other nonaromatic hydrogens except the one marked 'H' are replaced. How is this possible?

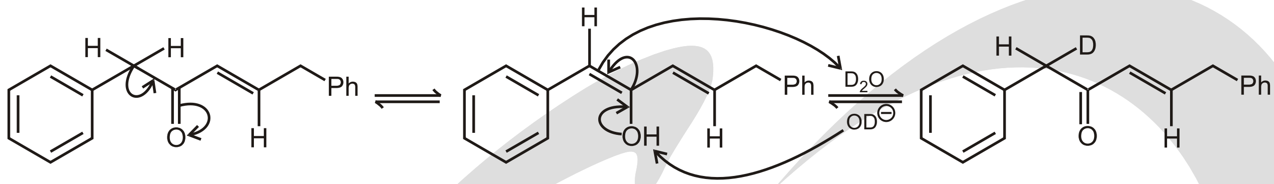


Purpose of the problem

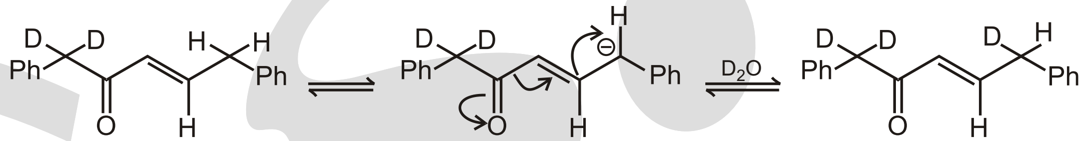
Working through the various ways in which enols and enolates can exchange hydrogen atoms.

Suggested solution

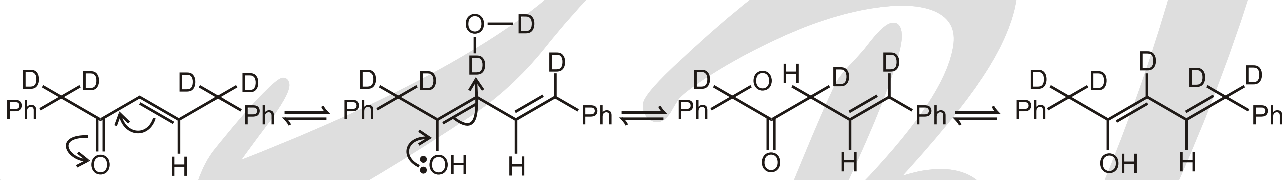
The protons on the group next to the ketone exchange by simple enolization and reversion to the ketone. Repetition of this process replaces both H atoms by D. Since D_2O is in large excess, the equilibrium favours D_2-ketone.



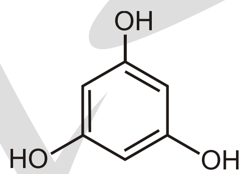
Next, enolization can occur at the other end of the molecule to form a dienol. This leads to replacement of the other CH_2 group with a CD_2 group after two exchanges.

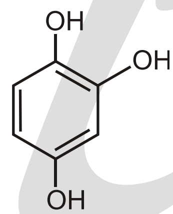


The same dienol can lead to exchange of the remaining proton by a more complicated series of reactions. Deuteration at the carbon atom next to the ketone and then loss of remaining proton back to the dienol is all that is needed.

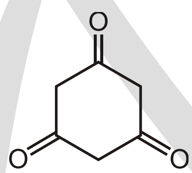


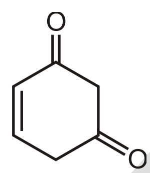
2. Which of the following two compounds will exhibit easy keto-enol tautomerism? Give explanation.

(a) 

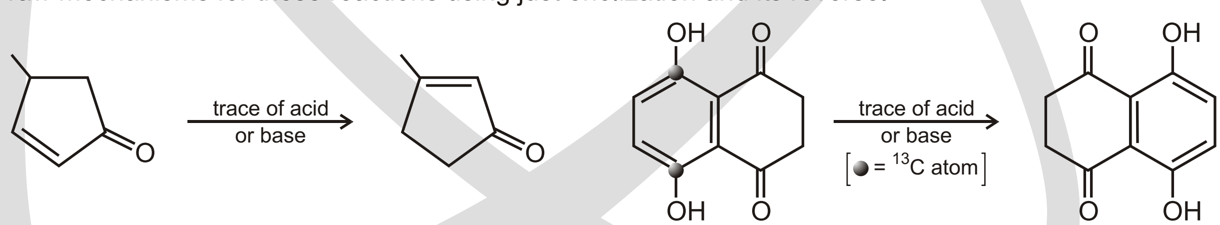
(b) 

3. Which one of the following has higher enol content? Give reasons for your answer.

(a) 

(b) 

4. Draw mechanisms for these reactions using just enolization and its reverse.

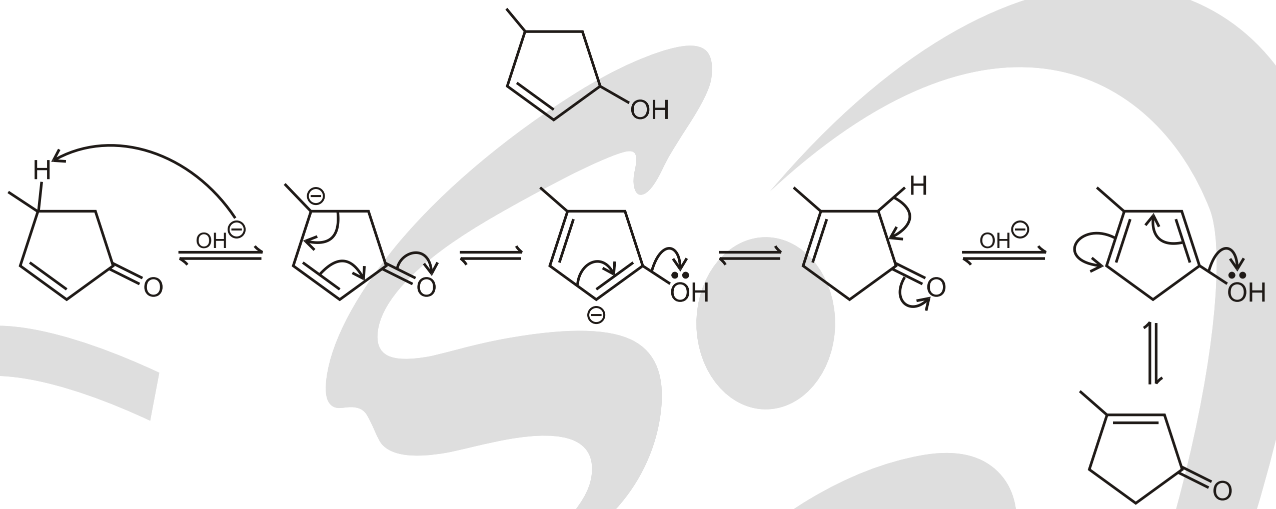


Purpose of the problem

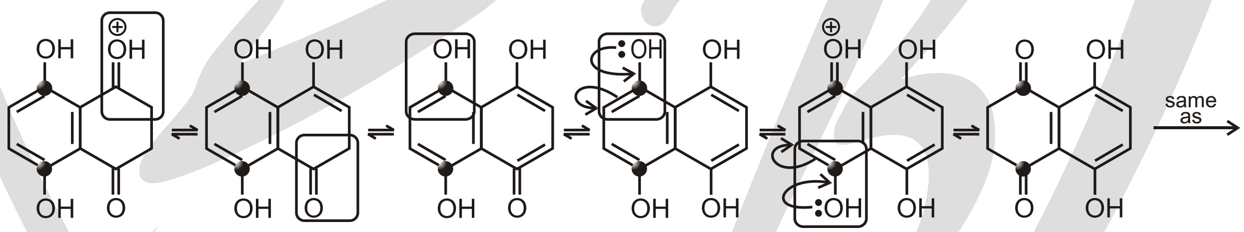
Exercise in using enolization of carry out simple reactions.

Suggested solution

Two enols are possible from the first compound: one (in the margin) leads back only to starting material but the other leads on to product. The whole system is in equilibrium favouring the enone with the more highly substituted alkene.



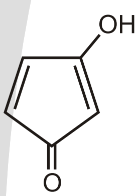
The second example is a bit of a trick. Of course, the ^{13}C label hasn't moved. The molecule just keeps enolizing and going back to a ketone until the functional groups is the two rings have changed places. Here we leave out the mechanism and just shown which ketones or enol are tautomerizing with a frame.

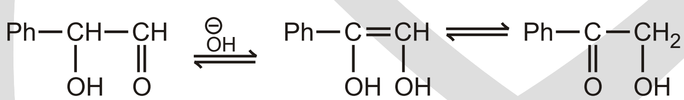


Answers

Single Choice Questions

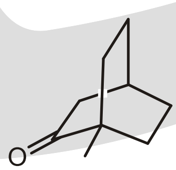
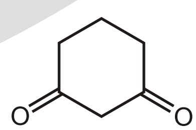
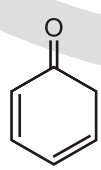
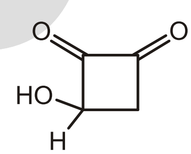
1. (A)	**2.** (B)	**3.** (B)	**4.** (C)	**5.** (B)	**6.** (A)	**7.** (A)	**8.** (D)
9. (A)	**10.** (C)	**11.** (A)	**12.** (C)	**13.** (D)	**14.** (C)	**15.** (B)	**16.** (C)
17. (C)	**18.** (C)	**19.** (A)	**20.** (B)	**21.** (B)	**22.** (C)		

19.  enol is anti-aromatic. No tautomerisation, 100% keto content.

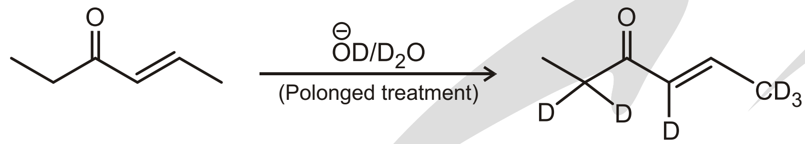
22. 

Unsolved Example

1. 4 (a,b,c,h)

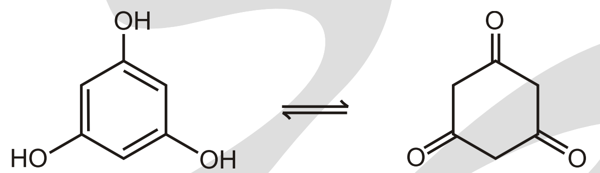
2. (i)  (ii)  (iii)  (iv) 

3. 6
Z 6

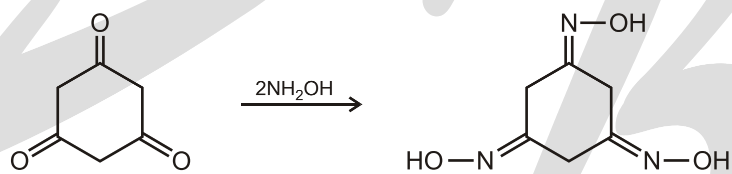


Subjective Type Questions

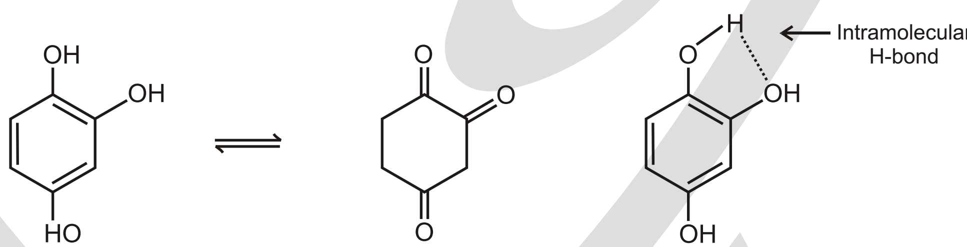
2. The compound 1, 3, 5-trihydroxybenzene (a) is trivially known as phloroglucinol. It exhibits easy keto-enol tautomerism. Its tri keto-form is known and reacts with hydroxylamine to give a trioxime. The loss of aromatic resonance energy of the benzene nucleus in phloroglucinol is counterbalanced by the bond energy of three C O groups, which is around 18 kcal/mol for each C O group.



Phloroglucinol   1, 3, 5-Cyclohexatrione



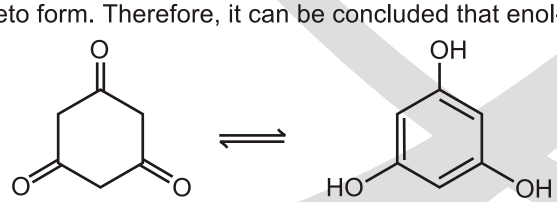
The compound (b) can also exhibit keto-enol tautomerism but the keto-form is very unstable due to the two C O groups on adjacent carbon atoms. The enol-form is stabilized by intramolecular hydrogen bonding. The structures are shown here.



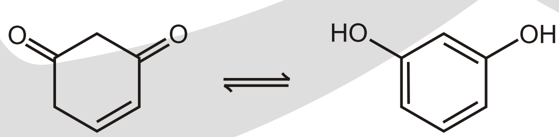
Hydroxyquinol
Enol-form (More stable)   (Keto-form less stable)

3. The compound (a) is the triketo-form of phloroglucinol which is the enolic form. The enolic form of compound (b) is resorcinol. Now, phloroglucinol can readily form its triketo form. However, resorcinol cannot readily form its diketo form. Therefore, it can be concluded that enol-content of the compound is higher than that of (a).

(a)



(b)



❑❑❑

CHAPTER

24

Conformers

CONFORMATIONS OF ALKANES : ROTATION ABOUT CARBON-CARBON BONDS

In a **sawhorse projection,** you are looking at the carbon-carbon bond from an oblique angle. In a **Newman projection,** you are looking down the length of a particular carbon-carbon bond. The carbon in front is represented by the point at which three bonds intersect, and the carbon in back is reprsented by a circle.

The electrons in a carbon-hydrogen bond will repel the electrons in another carbon-hydrogen bond if the bonds get too close to each other. The **staggered conformation,** therefore, is the most stable conformation because the carbon-hydrogen bonds are as far away from each other as possible. The **eclipsed conformation** is the least stable conformation because in no other conformation are the carbon hydrogen bonds as close to one another. The extra energy of the eclipsed conformtion is called torsional strain. **Torsional strain** is the name given to the repulsion felt by the bonding electrons of one substituent as they pass close to the bonding electrons of an other substituent.

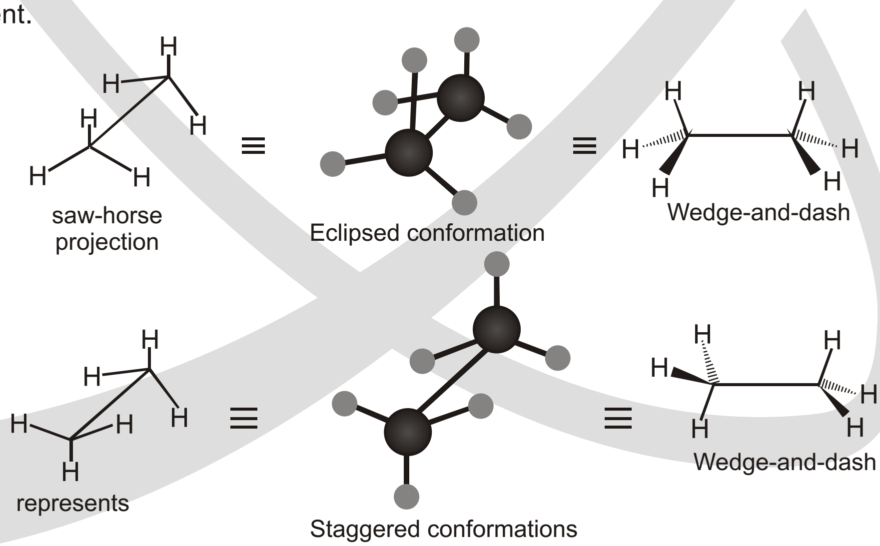


Fig. : Ethane in the eclipsed and staggered conformations.

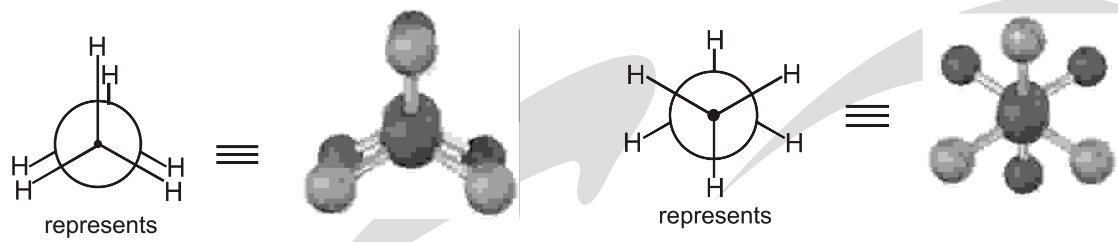
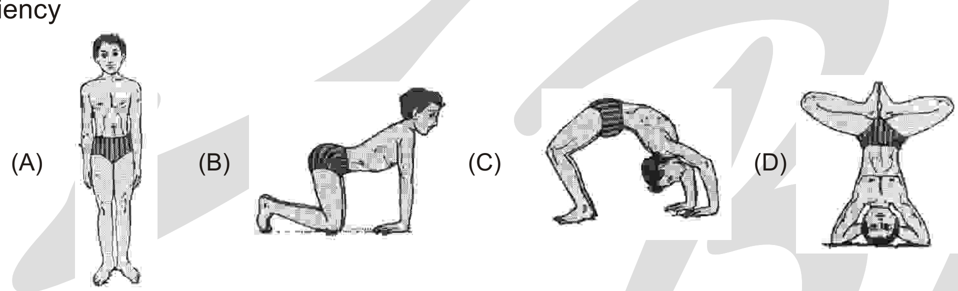


Figure : Newman projections of ethane in the eclipsed and staggered conformations.

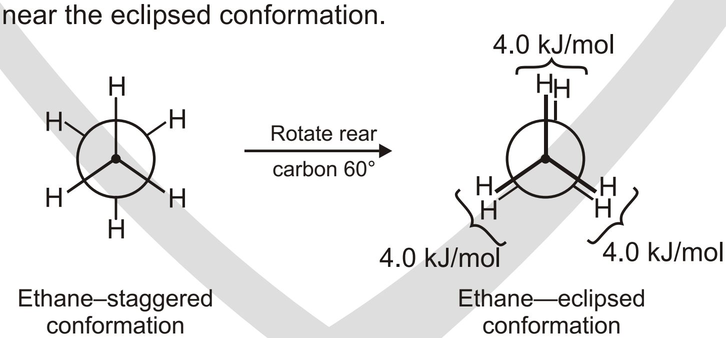
The picture is not yet complete. Certain physical properties show that rotation is not quite free : there is an energy barrier of about 3 kcal/mol.

(i) Most stable conformation of a person is *A* & *B*,*C*,*D* are different yoga pose.
Yoga pose *C* is Chakrasana and benefits are Strengthens back muscles, tones adrenals, helps kidneys, front part of the body is being stretch entirely, which is good for people who want to be more expressive as the openness in the heart may work on their heart chakra. Due to the stretch at the upper part of the abdomen muscles, it gives some pressure on the internal organs of the abdomen and therefore, increasing their efficiency



Energy

Despite what we've just said, we actually don't observe perfectly free rotation in ethane. Experiments show that there is a small (12 kJ/mol; 2.9 kcal/mol) barrier to rotation and that some conformations are more stable than others. The lowestenergy, most stable conformation is the one in which all six C — H bonds are as far away from one another as possible — staggered when viewed end-on in a Newman projection. The highest-energy, least stable conformation is the one in which the six C — H bonds are as close as possible — eclipsed in a Newman projection. At any given instant, about 99% of ethane molecules have an approximately staggered conformation and only about 1% are near the eclipsed conformation.



The extra 12 kJ/mol of energy present in the eclipsed conformation of ethane is called torsional strain. Its cause has been the subject of controversy, but the major factor is an interaction between C — H bonding orbitals on one carbon with antibonding orbitals on the adjacent carbon, which stabilizes the staggered conformation relative to the eclipsed one. Because the total strain of 12 kJ/mol arises from three equal hydrogen–hydrogen eclipsing interactions, we can assign a value of approximately 4.0 kJ/mol (1.0 kcal/mol) to each single interaction. The barrier to rotation that results can be represented on a graph of potential energy versus degree of rotation in which the angle between C — H bonds on front and back carbons as viewed end-on (the *dihedral angle*) goes full circle from 0 to 360°. Energy minima occur at staggered conformations, and energy maxima occur at eclipsed conformations, as shown.

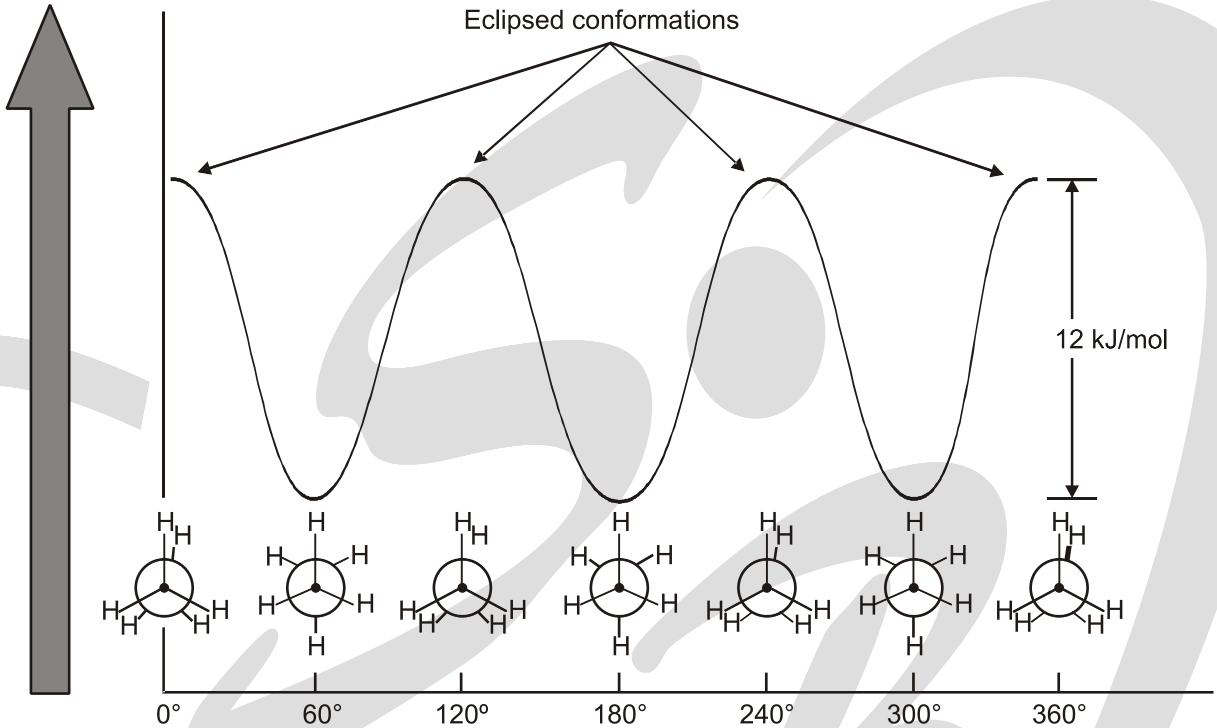


Fig. A graph of potential energy versus bond rotation in ethane.

The staggered conformations are 12 kJ/mol lower in energy than the eclipsed conformations.

CONFORMATION OF OTHER ALKANES

Propane, the next higher member in the alkane series, also has a torsional barrier that results in hindered rotation around the carbon–carbon bonds. The barrier is slightly higher in propane than in ethane — a total of 14 kJ/mol (3.4 kcal/mol) versus 12 kJ/mol.

The eclipsed conformation of propane has three interactions — two ethanetype hydrogen–hydrogen interactions and one additional hydrogen–methyl interaction. Since each eclipsing H H interaction is the same as that in ethane and thus has an energy "cost" of 4.0 kJ/mol, we can assign a value of 14 (2 4.0) 6.0k J/mol (1.4 kcal/mol) to the eclipsing H CH_3 interaction.

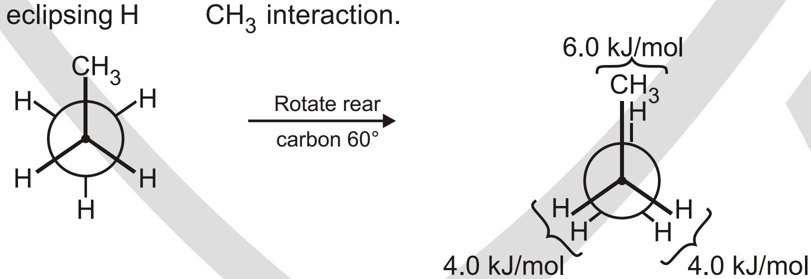
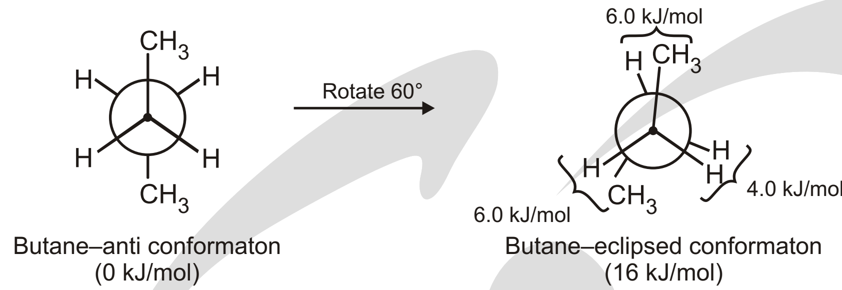
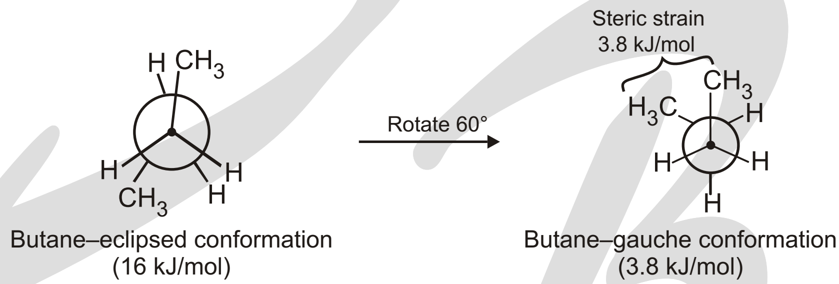


Fig. Newman projections of propane showing staggered and eclipsed conformations. The staggered conformer is lower in energy by 14 kJ/mol.

The conformational situation becomes more complex for larger alkanes because not all staggered conformations have the same energy and not all eclipsed conformations have the same energy. In butane, for instance, the lowest-energy arrangement, called the anti conformation, is the one in which the two methyl groups are as far apart as possible — 180° away from each other. As rotation around the C2 — C3 bond occurs, an eclipsed conformation is reached in which there are two CH_3 H interactions and one H H interaction. Using the energy values derived previously from ethane and propane, this eclipsed conformation is more strained than the anti conformation by 2 × 6.0 kJ/mol + 4.0 kJ/mol (two CH_3 H interactions plus one H H interaction), for a total of 16 kJ/mol (3.8 kcal/mol).



As bond rotation continues, an energy minimum is reached at the staggered conformation where the methyl groups are 60° apart. Called the **gauche conformation**, it lies 3.8 kJ/mol (0.9 kcal/mol) higher in energy than the anti conformation even though it has no eclipsing interactions. This energy difference occurs because the hydrogen atoms of the methyl groups are near one another in the gauche conformation, resulting in what is called *steric strain*. **Steric strain** is the repulsive interaction that occurs when atoms are forced closer together than their atomic radii allow. It's the result of trying to force two atoms to occupy the same space.



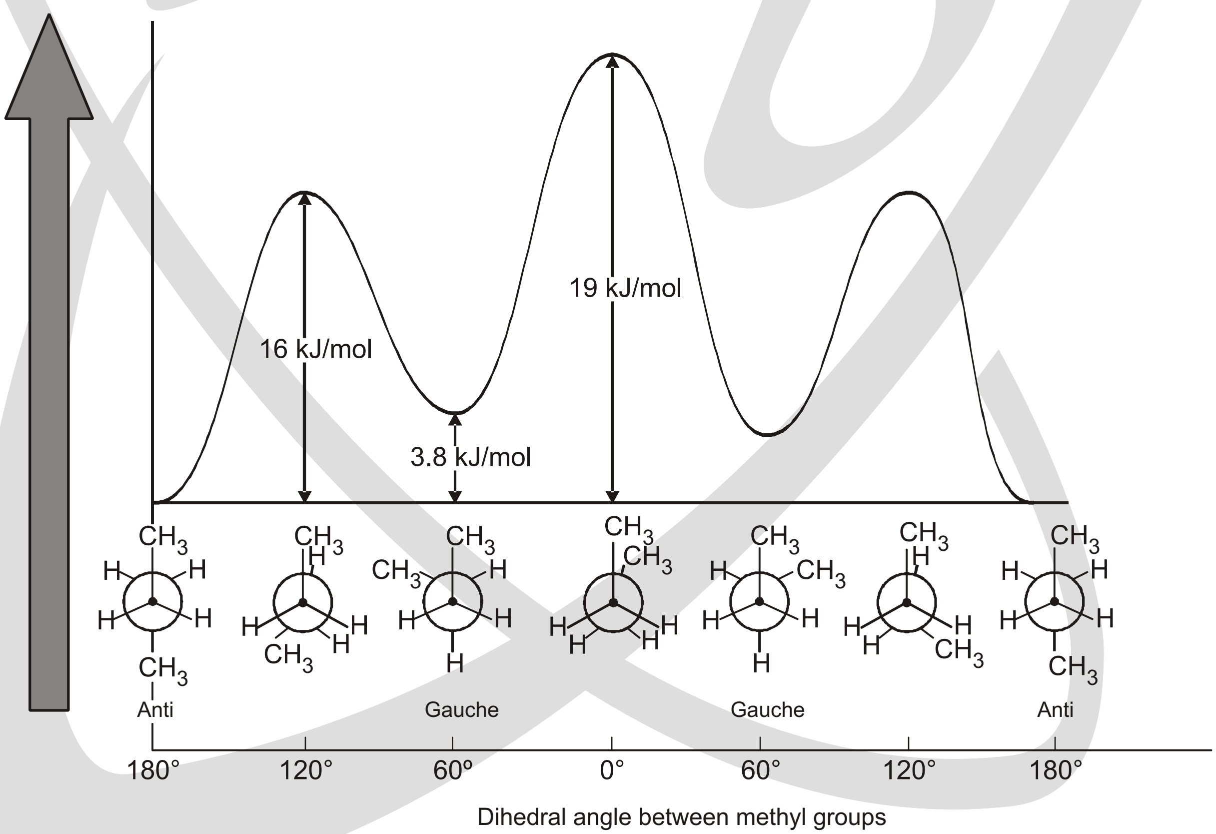


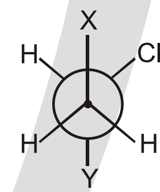
Fig. A plot of potential energy versus rotation for the C2—C3 bond in butane. The energy maximum occurs when the two methyl groups eclipse each other(fully eclipsed), and the energy minimum occurs when the two methyl groups are 180° apart (anti).

Table : Energy Costs for Interactions in Alkane Conformers

Interaction	Cause	Energy cost (kJ/mol)	Energy cost (kcal/mol)
H H eclipsed	Torsional strain	4.0	1.0
H CH_3 eclipsed	Mostly torsional strain	6.0	1.4
CH_3 CH_3 eclipsed	Torsional and steric strain	11	2.6
CH_3 CH_3 gauche	Steric strain	3.8	0.9

Solved Example

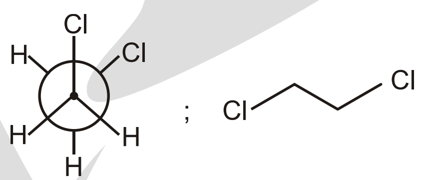
- In the Newman projection for 1, 2-dichloro propane *X* and *Y* can respectively be :

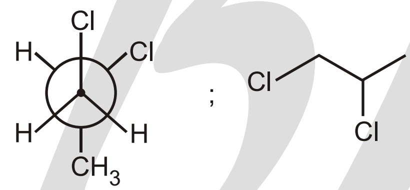


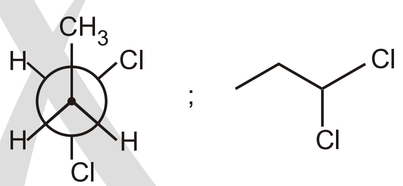
(A) Cl and H
(B) Cl and CH_3
(C) CH_3 and Cl
(D) H and CH_2Cl

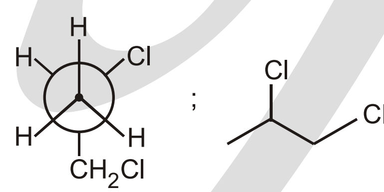
Ans. (B, D)

Put the value of *X* and *Y* in each structure

(A) 

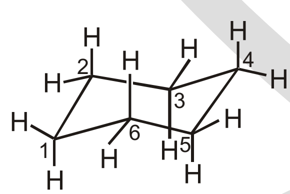
(B) 

(C) 

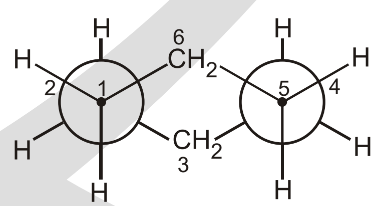
(D) 

CONFORMATION OF CYCLOHEXANE

The cyclic compounds most commonly found in nature contain six-membered rings because six-membered rings can exist in a conformation that is almost completely free of strain. The conformation is called the **chair conformation.**



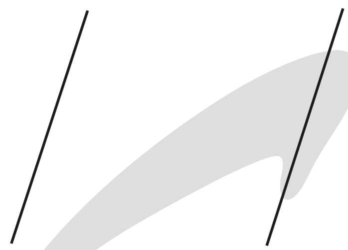
chair conformer of cyclohexane



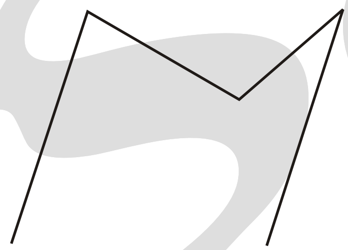
Newman projection of the chair conformer

HOW TO DRAW CHAIR CONFORMER OF CYCLO HEXANE AND IDENTIFYING AXIAL AND EQUATORIAL BONDS

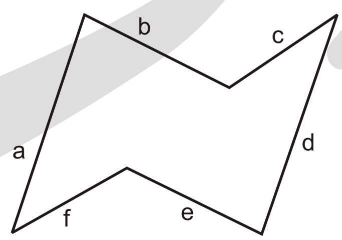
(1) Draw two parallel lines of same length, slanted upward and beginning at same level.



(2) Connect top-end by a V whose left hand side is greater than right hand side.

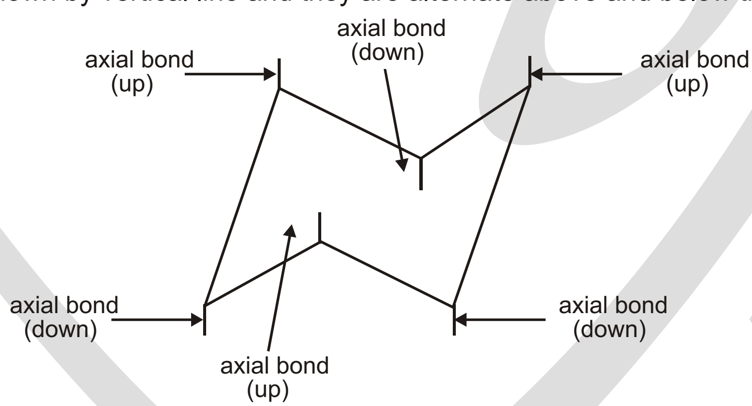


(3) Connect bottom end by an inverted V, whose L.H.S. is less than R.H.S.

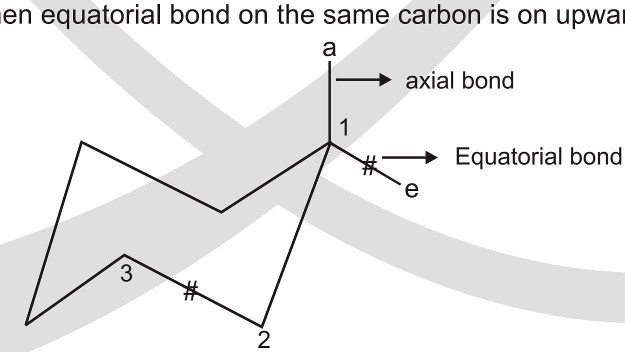


a||d (a is parallel to d)
b||e
c||f

(4) Axial bond are shown by vertical line and they are alternate above and below the ring.

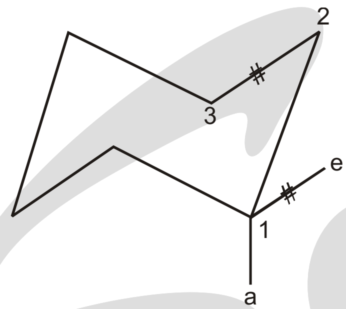


(5) If the axial bond is up then equatorial bond on the same carbon is on upward slant.

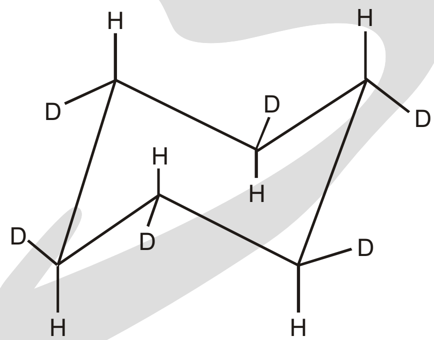


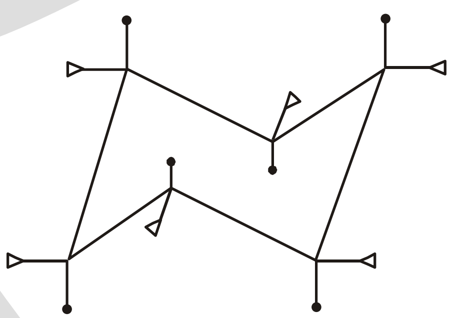
Equatorial bond is parallel to second carbon bond of the ring.

C_1 at equatorial is parallel to C_2 — C_3 bond of the ring.



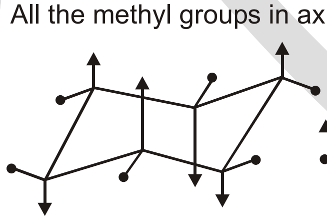
Solved Example





Solved Example

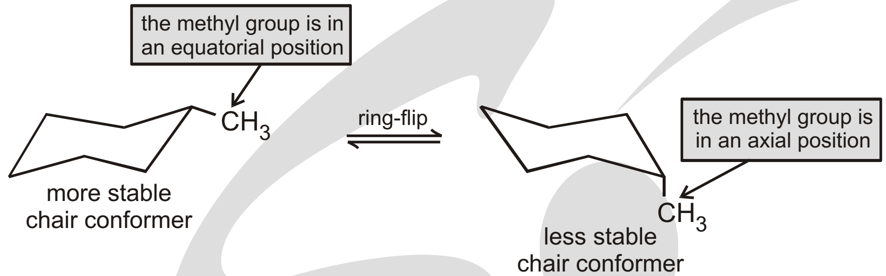
- Draw 1, 2, 3, 4, 5, 6-hexamethylcyclohexane with :
  (a) All the methyl groups in axial positions. (b) All the methyl groups in equatorial positions.

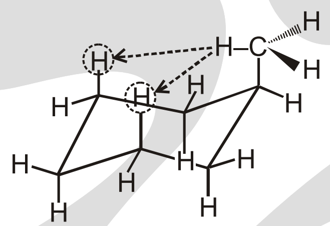
Ans. (a) 

(b) In above diagram of (a) ● = axial bond have $-CH_3$ groups
▲ = equatorial bond have –H-atom

CONFORMATIONS OF DISUBSTITUTED CYCLOHEXANES

A substituent is in the equatorial position in one chair conformer and in the axial position in the other chair conformer. The conformer with the substituent in the equatorial position is more stable, because in axial position group feels steric repulsion from hydrogen atom of the 3rd carbons of the ring it is known is **1,3-diaxial interactions (repulsion).**





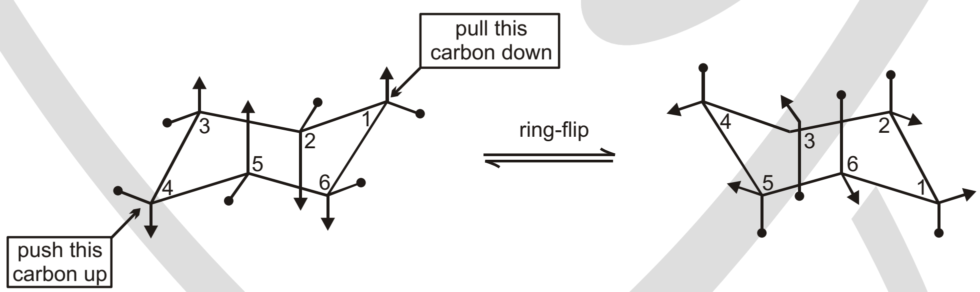
1,3-diaxial interactions

Ring Flip

In ring flip the bonds that are axial in one chair conformer are equatorial in the other chair conformer. The bonds that are equatorial in one chair conformer are axial in the other chair conformer.

☞ **During ring flip bonds that are equatorial in one chair conformer become axial in the other chair conformer.**

As a result of the ease of rotation about its carbon-carbon single bonds, cyclohexane rapidly interconverts between two stable chair conformations. This interconversion is known as **ring-flip.** When the two chair conformers interconvert, bonds that are equatorial in one chair conformer become axial in the other chair conformer.



Solved Example

- The equilibrium constant for the ring-flip of fluorocyclohexane is 1.5 at 25ºC. Calculate the percentage of the axial conformer at the temperature.

Ans.

$$K_{eq} \quad \frac{[eq]}{ax} \quad 1.5$$

$$\%\ \text{axial} \quad \frac{[ax]}{[eq]\quad[ax]} \quad 100\%$$

$$\frac{[1]}{[1.5]\quad[1]} \quad 100\%$$

$$40\%$$

CONFORMATIONS OF DISUBSTITUTED CYCLOHEXANES

Monosubstituted cyclohexanes are always more stable with their substituent in an equatorial position, but the situation in disubstituted cyclohexanes is more complex because the steric effects of both substituents must be taken into account. All steric interactions in both possible chair conformations must be analyzed before deciding which conformation is favoured.

Let's look at 1,2-dimethylcyclohexane as an example. There are two isomers, cis-1,2-dimethylcyclohexane and trans-1,2-dimethylcyclohexane, which must be considered separately. In the cis isomer, both methyl groups are on the same face of the ring and the compound can exist in either of the two chair conformations shown.

cis-1,2-Dimethylcyclohexane

One gauche
interaction (3.8 kJ/mol)
Two $CH_3 \longleftrightarrow H$ diaxial
interactions (7.6 kJ/mol)
Total strain: 3.8 + 7.6 = 11.4 kJ/mol

One gauche
interaction (3.8 kJ/mol)
Two $CH_3 \longleftrightarrow H$ diaxial
interactions (7.6 kJ/mol)
Total strain : 3.8 + 7.6 = 11.4 kJ/mol

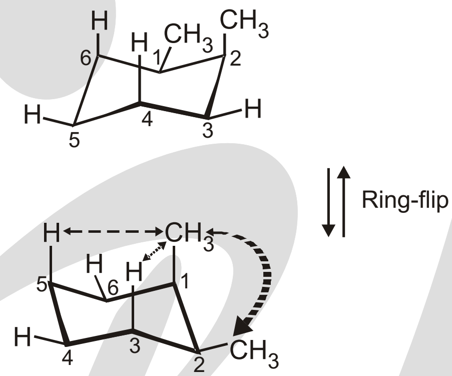


Figure 4.15 Conformations of cis-1,2-dimethylcyclohexane. The two chair conformations are equal in energy because each has one axial methyl group and one equatorial methyl group.

Both chair conformations of cis-1,2-dimethylcyclohexane have one axial methyl group and one equatorial methyl group. The top conformation in Figure 4.15 has an axial methyl group at C2, which has 1,3-diaxial interactions with hydrogens on C4 and C6. The ring-flipped conformation has an axial methyl group at C1, which has 1,3-diaxial interactions with hydrogens on C3 and C5. In addition, both conformations have gauche butane interactions between the two methyl groups. The two conformations are equal in energy, with a total steric strain of 3 3.8 kJ/mol = 11.4 kJ/mol (2.7 kcal/mol).

In trans-1, 2-dimethylcyclohexane, the two methyl groups are on opposite faces of the ring and the compound can exist in either of the two chair conformations. The situation here is quite different from that of the cis isomer. The top conformation in Figure 4.16 has both methyl groups equatorial and therefore has only a gauche butane interaction between them (3.8 kJ/mol) but no 1,3-diaxial interactions. The ring-flipped conformation, however, has both methyl groups axial. The axial methyl group at C1 interacts with axial hydrogens at C3 and C5, and the axial methyl group at C2 interacts with axial hydrogens at C4 and C6. These four 1,3-diaxial interactions produce a steric strain of 4 3.8 kJ/mol = 15.2 kJ/mol and make the diaxial conformation 15.2 – 3.8 = 11.4 kJ/mol less favourable than the diequatorial conformation. We therefore predict that trans-1,2-dimethylcyclohexane will exist almost exclusively in the diequatorial conformation.

trans-1,2-Dimethylcyclohexane

One gauche
interaction (3.8 kJ/mol)

Four $CH_3 \longleftrightarrow H$ diaxial
interactions (15.2 kJ/mol)

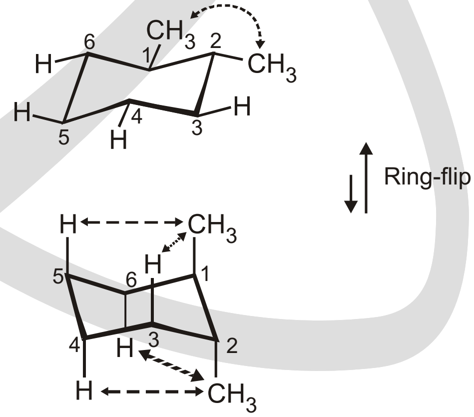


Figure 4.16 Conformations of trans-1,2-dimethylcyclohexane. The conformation with both methyl groups equatorial (top) is favored by 11.4 kJ/mol (2.7 kcal/mol) over the conformation with both methyl groups axial (bottom).

Solved Example

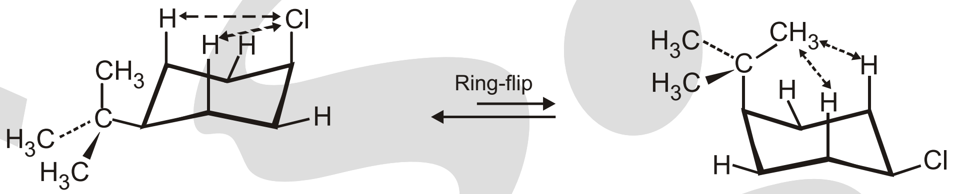
- **Drawing the Most Stable Conformation of a Substituted Cyclohexane**

  Draw the more stable chair conformation of cis-1-tert-butyl-4-chlorocyclohexane. By how much is it favored?

  Strategy

  Draw the two possible chair conformations, and calculate the strain energy in each. Remember that equatorial substituents cause less strain than axial substituents.

Sol. First draw the two chair conformations of the molecule :



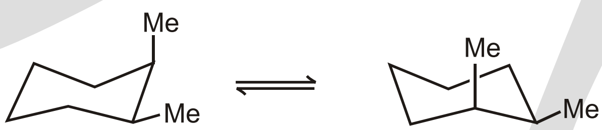
2 × 1.0 = 2.0 kJ/mol steric strain   2 × 11.4 = 22.8 kJ/mol steric strain

Disubstituted Cyclohexane

If 2 substituents are on cyclohexane the lowest energy conformation:

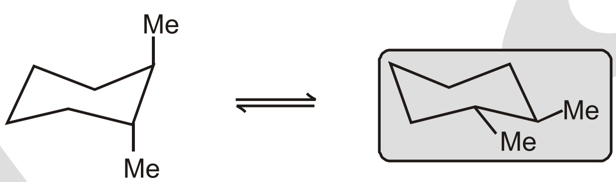
(a) Has both substituents equatorial (if possible)
(b) The group with the largest A value equatorial
(c) t-Bu is NEVER axial

1,2-cis



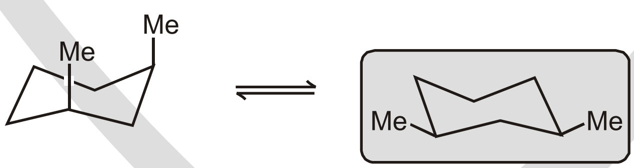
1 ax. 1 eq. versus 1ax. 1 eq. same energy

1,2-trans



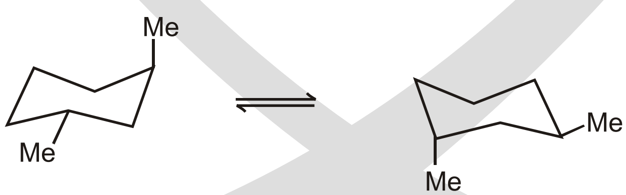
2 ax. versus 2 eq. all equatorial preferred

1,3-cis



2 ax. versus 2 eq. all equatorial preferred

1,3-trans

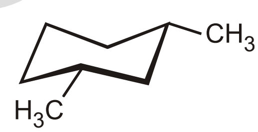


1 ax. 1 eq. versus 1ax. 1eq. same energy

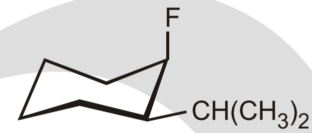
Solved Example

- Draw the ***lowest energy*** conformation adopted by each of the following molecules :

(a) ***cis***-1,3-Dimethylcyclohexane

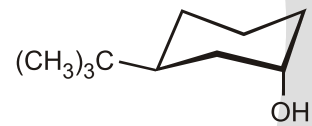


(b) ***cis***-1-Fluro-2-isopropylcyclohexane



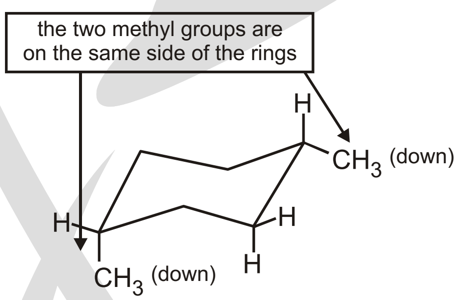
(c) ***trans***-1-Bromo-4-methylcyclohexane

(d) ***trans***-3-tert-butylcyclohexanol

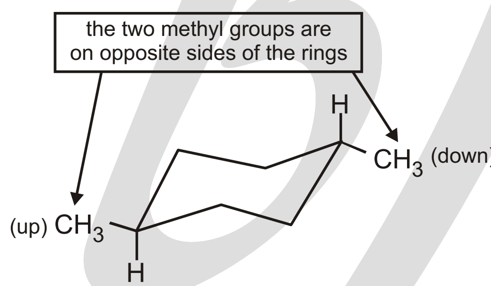


IDENTIFYING CIS & TRANS

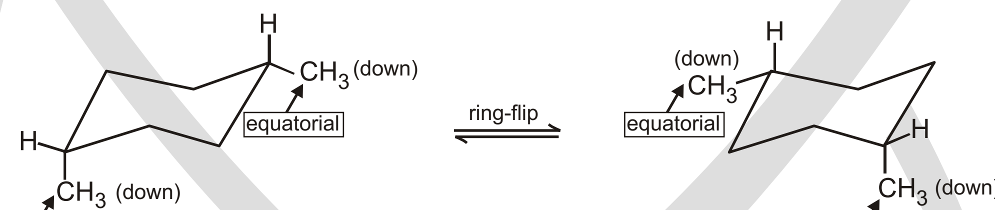
If there are two substituents on a cyclohexane ring, both substituents have to be taken into account when determining which of the two chair conformers is the more stable. Let's start by looking at 1,4-dimethylcyclohexane. First of all, there are two different dimethylcyclohexanes. One has both methyl substituents on the same side of the cyclohexane ring; it is the **cis isomer** (cis is Latin for ''on this side''). The other has the two methyl substituents on opposite sides of the ring; it is the **trans isomer** (trans is Latin for ''across''). cis-1,4-Dimethylcyclohexane and tran-1,4-dimethylcyclohexane are called **geometric isomers or cis-trans stereoisomers-**they have the same atoms, and the atoms are linked in the same order, but they differ in the spatial (stereo) arrangement of the atoms.



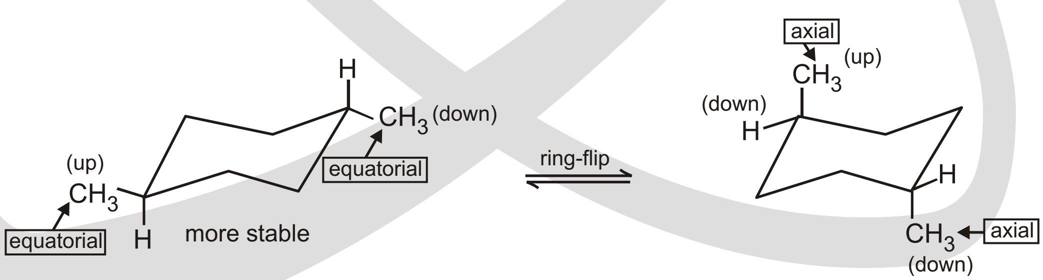
cis-1,4-dimethylcyclohexane



trans-1,4-dimethylcyclohexane



cis-1,4-dimethylcyclohexane
(equally stable)

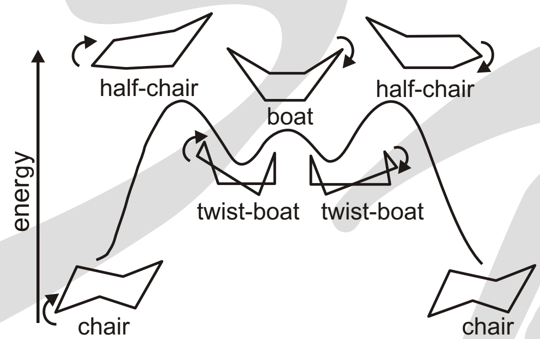


A summary of the various axial and equatorial relationships among substituent groups in the different possible cis and trans substitution patterns for disubstituted cyclohexanes is given in Table.

Table : Axial and Equatorial Relationships in Cis- and Trans-Disubstituted Cyclohexanes.

Cis/trans substitution pattern	Axial/equatorial relationships
1,2-Cis disubstituted	a, e or e, a
1,2-Trans disubstituted	a, a or e, e
1,3-Cis disubstituted	a, a or e, e
1,3-Trans disubstituted	a, e or e, a
1,4-Cis disubstituted	a, e or e, a
1,4-Trans disubstituted	a, e or e, e

Chair forms convert into various other forms(conformers *i.e.*, halfchair, twist boat and boat) during ringflip

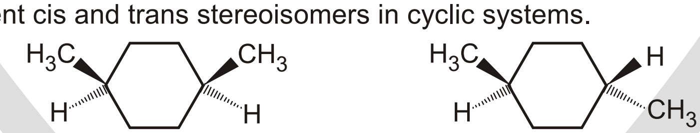


The boat conformer is further destabilized by the close proximity of the **flagpole hydrogens** which causes steric strain.

The twist-boat conformer is more stable than the boat conformer because the flagpole hydrogens have moved away from each other, thus reducing the steric strain somewhat.

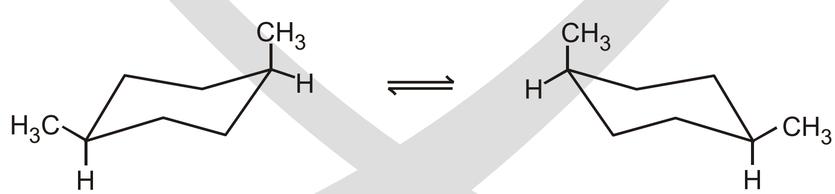
Conformation Analysis of Disubstituted Cyclohexanes

We'll begin with cis- and trans-1,4-dimethylcyclohexane. A conventional method uses wedge and dash descriptions to represent cis and trans stereoisomers in cyclic systems.



cis-1,4-Dimethylcyclohexane trans-1,4-Dimethylcyclohexane

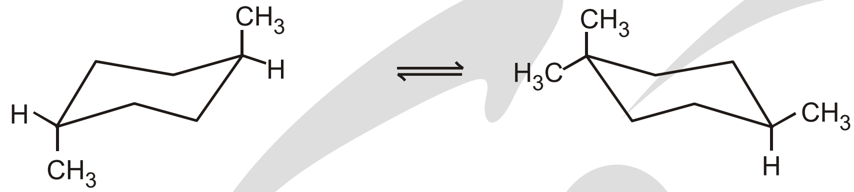
Each other by ring flipping. The equatorial methyl group becomes axial, and the axial methyl group becomes equatorial.



(One methyl group is axial, the other equatorial) (One methyl group is axial, the other equatorial)

(Both methyl groups are up)
cis-1,4-Dimethylcyclohexane

The methyl groups are described as cis because both are up relative to the hydrogen present at each carbon. If both methyl groups were down, they would still be cis to each other. Notice that ring flipping does not alter the cis relationship between the methyl groups. Nor does it alter their up-versus-down quality; substituents that are up in one conformaion remain up in the ring-flipped form.



(Both methyl groups are axial: less stable chair conformation)

(Both methyl groups are equatorial: more stable chair conformation)

(One methyl group is up, the other down)
trans-1,4-Dimethylcyclohexane

The more stable chair-the one with both methyl groups equatorial is adopted by most of the trans-1,4-dimethylcyclohexane molecules.

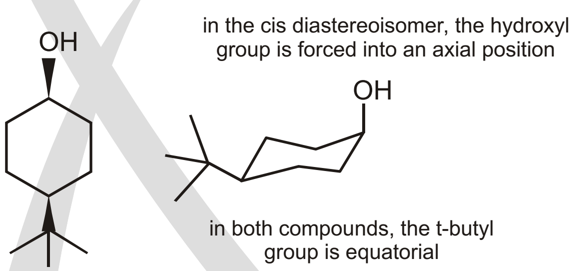
Trans-1,4-Dimethylcyclohexane is more stable than cis-1,4-dimethylcyclohexane because both of the methyl groups are equatorial in its most stable conformation. One methyl group must be axial in the cis stereoisomer.

☑ SPECIAL TOPIC

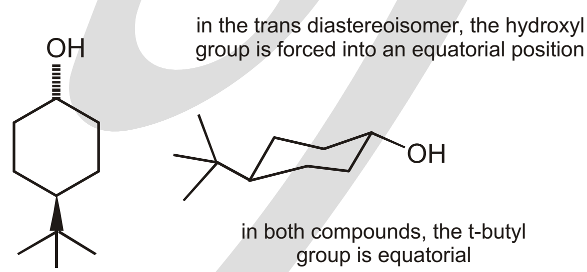
t-BUTYL GROUPS

We have already seen how a *t*-butyl group always prefers an equatorial position in a ring. This makes it very easy to decide which conformation the two different compounds below will adopt

cis-4-t-butylcyclohexanol

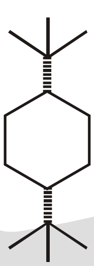


trans-4-t-butylcyclohexanol

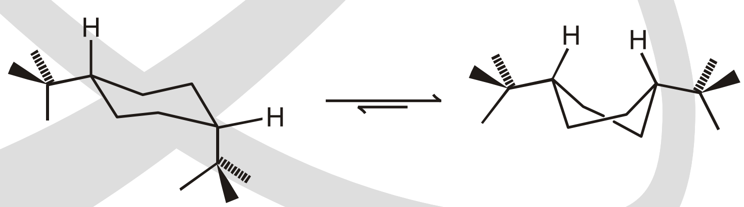


Cis-1,4-di-t-butylcyclohexane

An axial *t*-butyl group really is very unfavourable. In cis-1,4-di-*t*-butylcyclohexane, one *t*-butyl group would be forced axial if the compound existed in a chair conformation. To avoid this, the compound prefers to pucker into a twist boat so that the two large groups can both be in equatorial positions (or ‘pseudoequatorial’, since this is not a chair).



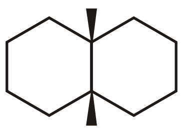
cis-1,4-di-t-butylcyclohexane



the twist-boat conformer (with both t-butyl groups in pseudoequatorial positions) is lower in energy than the chair conformer.

☑ SPECIAL TOPIC

1. Fused rings



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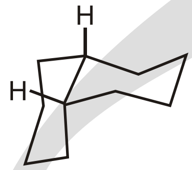
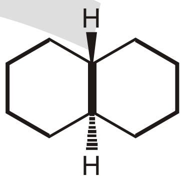


cis-decalin

can ring flip



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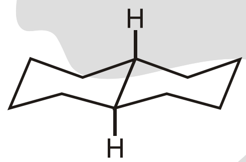
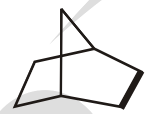


trans-decalin

cant' ring flip (locked)

bicyclo[4.4.0] decane

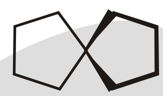
2. Bridged rings



conformationally locked

bicyclo[2.2.1] heptane

3. Spirocyclic rings (rare)



spiro[4.4] nonane

Conformation of cyclohexane

Cycloalkanes ring strain :

Compounds with three- and four-membered rings were not as stable as compounds with five or six-membered rings.

The German chemist Adolf von Baeyer first proposed in 1885 that the instability of three- and four-membered rings was due to angle strain. We know that an sp^3 hybridized carbon has ideal orbital angles of 109.5°. Baeyer suggested that the stability of a cycloalkane could be predicted by determining how close the bond angle of a planar cycloalkane is to the optimal tetrahedral bond angle of 109.5°. The angles in a regular triangle are 60°. The bond angles in cyclopropane, therefore, are compressed from the desired tetrahedral angle of 109.5° to 60°. The deviation of the bond angle from the desired bond angle causes the strain known as **angle strain.**

Cyclic compounds twist and bend in order to achieve a structure that minimizes the three different kinds of strain that can destabilize a cyclic compound.

(1) Angle strain is the strain induced in a molecule when the bond angles are different from the desired tetrahedral bond angle of 109.5°.

(2) Torsional strain is caused by repulsion of the bonding electrons of one substituent with the bonding electrons of a nearby substituent.

(3) Steric strain is caused by atoms or groups of atoms approaching each other too closely.

Solved Examples

- **Alkanes Stereochemistry**

(1) For the molecules below :

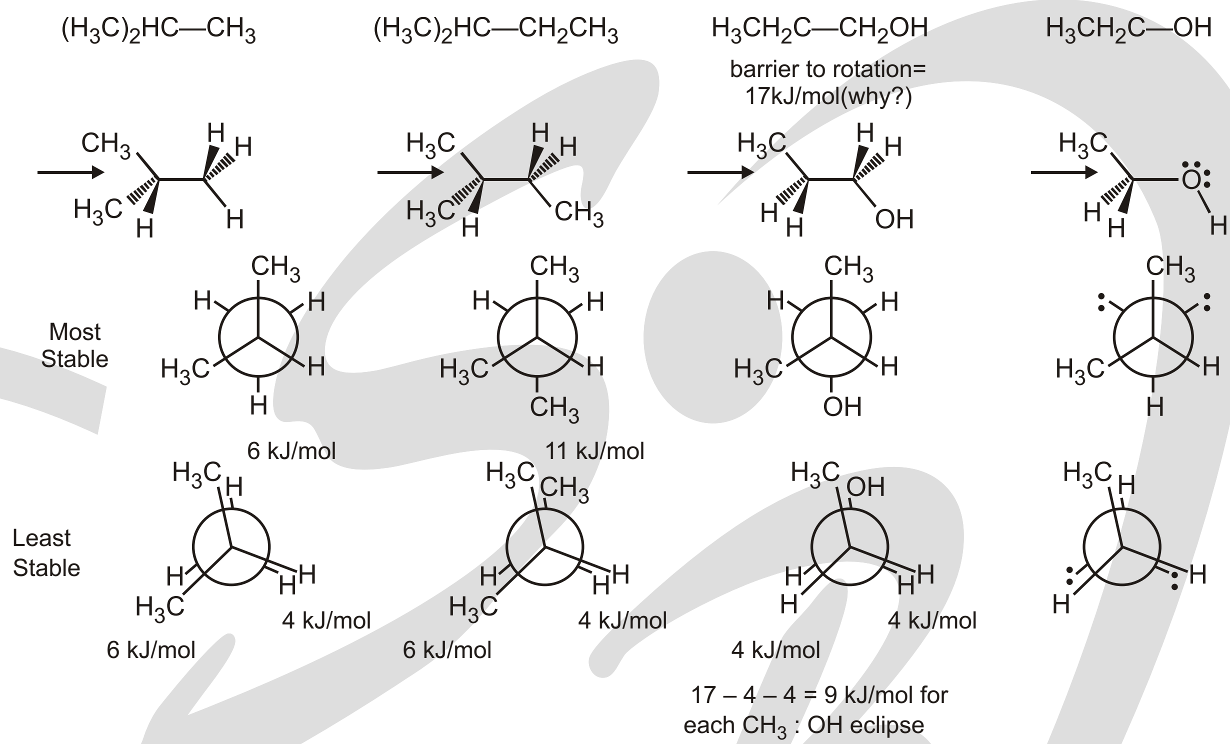
(a) Provide a 3 dimensional structure at the indicated atoms.

(b) Draw the Newman projection for your structure indicating the direction of sight with an arrow.

(c) Draw the Newman projection for the most stable conformation.

(d) Draw the Newman projection for the least stable conformation.

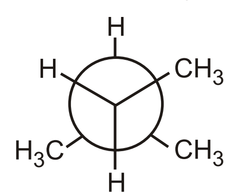
(e) If possible, calculate the energy difference between the most and least stable conformations.



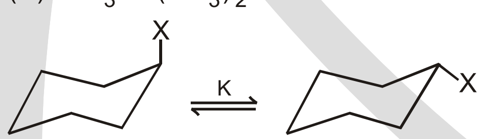
EXERCISE

SINGLE CHOICE QUESTIONS

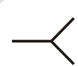
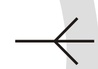
1. What compound is represented by the Newman projection shown?



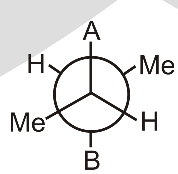
(A) $CH_3CH_2CH_3$
(B) $(CH_3)_2CHCH_2CH_3$
(C) $CH_3CH(CH_3)_2$
(D) $CH_3CH_2CH_2CH(CH_3)_2$

2. 

K Equilibrium constant is maximum for

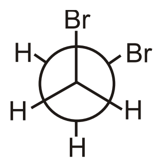
(A) –Me
(B) –Et
(C) 
(D) 

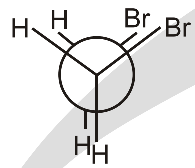
3. Identify (*A*) and (*B*) in Newman's projection of 2, 3-dimethyl pentane

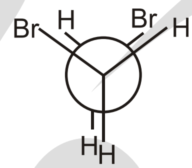


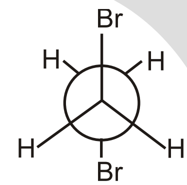
(A) $—CH_3, —CH_3$ (B) $—H, —C_2H_5$ (C) $—C_2H_5, —CH_3$ (D) $—C_2H_5, —C_2H_5$

4. Rank the following conformations in order of increasing energy

(I) 

(II) 

(III) 

(IV) 

(A) IV I III II (B) III II IV I
(C) II III I IV (D) IV III II I

5. Dipole moment of cis-1, 2-dichloro ethene is :

(A) 3 (B) $\sqrt{\ }$ (C) $\sqrt{3}$ (D) ^{2}

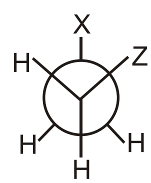
6. When methyl group is in axial position in methyl cyclohexane, the molecule has :

(A) One Gauche interaction (B) Two Gauche interaction
(C) No Gauche interection (D) Three Gauche interection

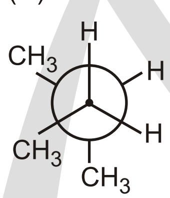
7. Bridge head hydrogen of the conformer of cis-decaline is positioned as :

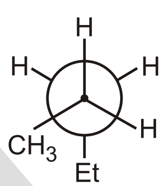
(A) a, a (B) e, e (C) a, e (D) pseudo-a, pseudo-e

8. Gauche conformer is stable when



(A) $X \quad Z \quad -CH_3$ (B) $X \quad Z \quad -OH$
(C) $X \quad Z \quad -Br$ (D) $X \quad Z \quad -C_2H_5$

9. 



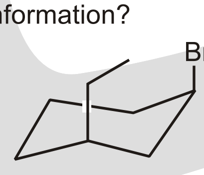
Type of isomerism shown by given pairs is :

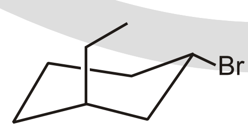
(A) conformatioinal (B) Positional
(C) Nuclear (D) Metamerism

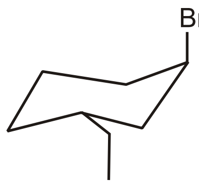
10. Isomers which can be interconverted through rotation around a single bond are :

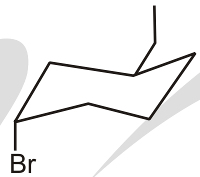
(A) conformers (B) diastereomers
(C) enantiomers (D) positional isomers.

11. Which one of the following drawings shows trans-1-bromo-3-ethylcyclohexane in its highest energy conformation?

(A) 

(B) 

(C) 

(D) 

12. Which of the following best explains the relative stabilities of the eclipsed and staggered forms of ethane? The .... form has the most .... strain.

(A) eclipsed; steric
(B) eclipsed; torsional
(C) staggered; steric
(D) staggered; torsional

13. Which of the following statements about the conformers that result, from rotation about the C2-C3 bond of butane is correct?

(A) The highest energy conformer is one in which methyl groups are eclipsed by hydrogens
(B) The gauche conformer is an eclipsed one
(C) Steric strain is absent in the eclipsed forms
(D) Torsional strain is absent in the eclipsed forms
(E) none of the above

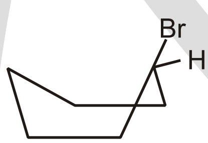
14. Which of the following correctly ranks the cycloalkanes in order of increasing ring strain per methylene?

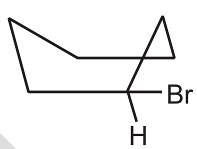
(A) cyclopropane < cyclobutane < cyclohexane < cycloheptane
(B) cyclohexane < cycloheptane < cyclobutane < cyclopropane
(C) cycloheptane < cyclobutane < cyclopentane < cyclopropane
(D) cyclopentane < cyclopropane < cyclobutane < cyclohexane
(E) cyclopropane < cyclopentane < cyclobutane < cyclohexane

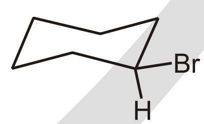
15. Which of the following correctly lists the conformations of cyclohexane in order of increasing energy ?

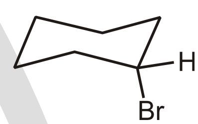
(A) chair < boat < twist-boat < half-chair
(B) half-chair < boat < twist-boat < chair
(C) chair < twist-boat < half-chair < boat
(D) chair < twist-boat < boat < half-chair
(E) half-chair < twist-boat < boat < chair

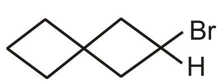
16. Which of the following is the most stable conformation of bromocyclohexane?

(I) 

(II) 

(III) 

(IV) 

(V) 

(A) I
(B) II
(C) III
(D) IV
(E) V

17. In the boat conformation of cyclohexane, the "flagpole" hydrogens are located :

(A) on the same carbon
(B) on adjacent carbons
(C) on C-1 and C-3
(D) on C-1 and C-4
(E) none of the above

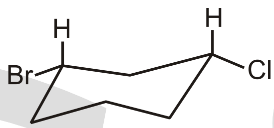
MULTIPLE CHOICE QUESTIONS

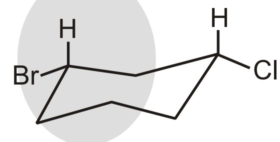
1. Twist boat conformer of cyclohexane is more stable than

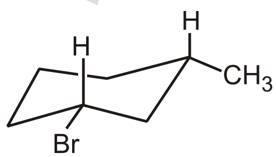
(A) boat (B) half-chair (C) chair (D) All of these

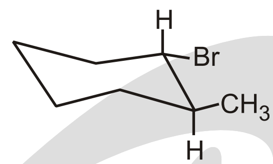
UNSOLVED EXAMPLE

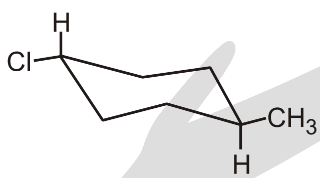
1. Determine whether each of the following compounds is a cis isomer or a trans isomer.

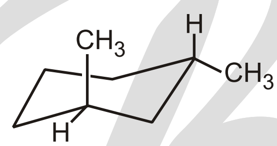
(a) 

(b) 

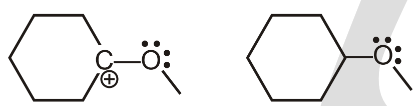
(c) 

(d) 

(e) 

(f) 

2. For the conformation of lowest energy estimate the atomic angles in the cation and the neutral molecule drawn below. Provide one number only for each question.



(a) C — O — C bond angle in cation.

(b) C — O — C bond angle in neutral molecule.

(c) C — O — C — C dihedral angle in the cation

(d) C — O — C — C dihedral angle in the neutral molecule.

3. (a) Draw a chair cyclohexane and put in all the axial bonds.

(b) On a second chair cyclohexane. put in all the equatorial bonds.

(c) Draw a cyclohexane with a bromine in an equtorial position.

(d) Draw a cyclohexane with a bromine in an axial position.

4. Draw both possible conformations for the indicated cyclohexane :

(a) trans -1-chloro-2-methylcyclohexane.

(b) cis-1-chloro-2-methylcyclohexane

(c) trans-1-chloro-3-methylcyclohexane

(g) a cyclohexane with 2 methyl groups, both axial.

(d) cis-1-chloro-3-methylcyclohexane

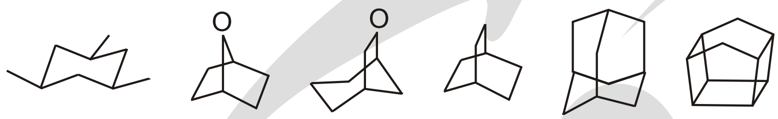
(e) trans-1-chloro-4-methylcyclohexane.

(f) cis-1-chloro-4-methylcyclohexane

(h) a cyclohexane with 2 methyl groups, both equatorial.

SUBJECTIVE TYPE QUESTIONS

1. Identify the chair or boat six-membered rings in the following structures and say why that particular shape is adopted.



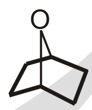
Purpose of the problem

Simple examples of chair and boat forms.

Suggested solution

The first three are relatively simple

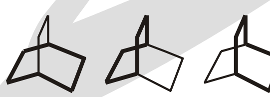
simple chair with all substituents equatorial



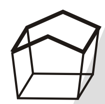
boat in cage molecule with no choice : chair impossible

chair in cage molecule with some choice : C* can move

The next two have several rings each, all boat in the first. We shall link that to the sixth molecule as it also has a boat and neither of these cage structures has any choice.

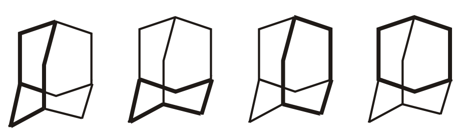


three boats cage compound has no choice



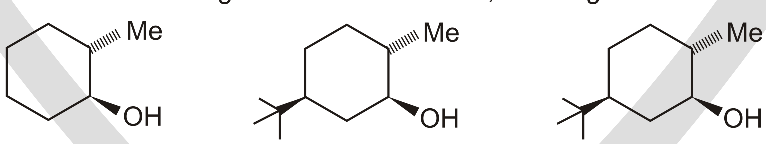
one boat cage compound has no choice

The rings are all chairs in the fifth molecule adamantane – a tiny fragment of a diamond molecule. The rings don't all look very chair-like in these diagrams – making a model of adamantane is the only way to appreciate this beautiful and symmetrical structure and to see all the chairs.



adamantane : four chairs cage compound has no choice

2. Draw clear conformational drawings for these molecules, labelling each substituent as axial or equatorial.

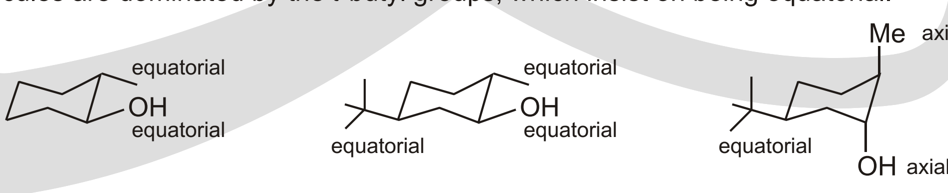


Purpose of the problem

Simple practice at drawing chair cyclohexanes with axial and equatorial substituents.

Suggested solution

Your drawings may look different from ours but make sure the rings have parallel sides and don't 'climb upstairs'. Make sure that the axial bonds are vertical and the equatorial bonds parallel to the next-ring-bond-but-one. The first molecule has a free choice so it puts both substituents equatorial. The last two molecules are dominated by the *t*-butyl groups, which insist on being equatorial.



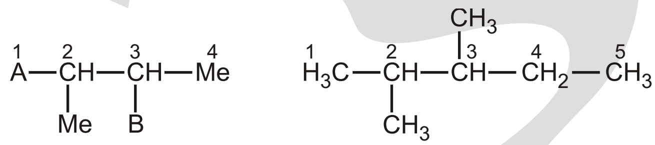
3. Contrary to $ClCH_2 — CH_2Cl$, in $FCH_2 — CH_2F$ gauche conformer is the more stable conformer. Explain this fact.

4. (a) What is 'gauche effect' ? Give examples.
   (b) Draw the eclipsed and bisecting conformations of propylene (propene)

Answers

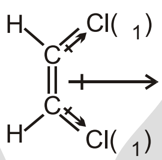
Single Choice Questions

1. (B)	2. (D)	3. (C)	4. (A)	5. (C)	6. (A)	7. (D)	8. (B)
9. (B)	10. (A)	11. (B)	12. (B)	13. (E)	14. (B)	15. (D)	16. (C)
17. (D)							

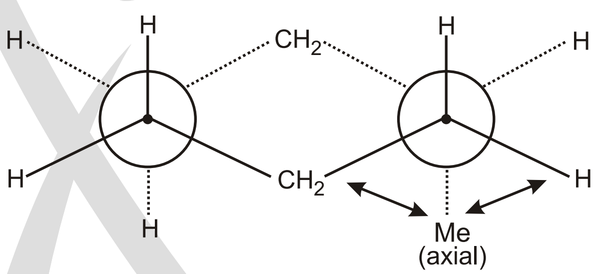
3. 

One of the combinations is (A) – C_2H_5, (B) – CH_3

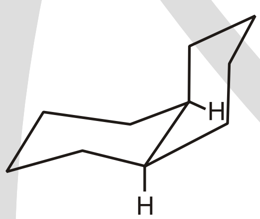
4. Order of increasing energy : Anti < Gauche < Partially eclipsed < Fully eclipsed.

5. 

$\sqrt{\ _1^2 \quad _1^2 \quad 2 \quad _1 \ _1 \cos 60} \quad \sqrt{2 \ _1^2 \quad 2 \quad _1 \ _1 \ \frac{1}{2}} \quad \sqrt{3 \ ^2} \quad \sqrt{3}$

6. 

one-n-butane gauch interaction

7. 

Bridge head hydrogen are pseudo a & pseudo e.
because this (H) are axial for 1 ring & equatorial for other thing.

Multiple Choice Questions

1. (A, B)

Stability order

Half chair < Boat < Twist boat < Chair

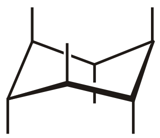
Unsolved Example

1. (a) = cis ; (b) = trans ; (c) = cis ; (d) = trans ;

(e) = trans ; (f) = trans.

2. (a) 120º (b) 108º - 110º (c) 0º or 180º (d) 60º

HINT : *The last two questions might be assisted by some Newman Projections.*

3. (a) 

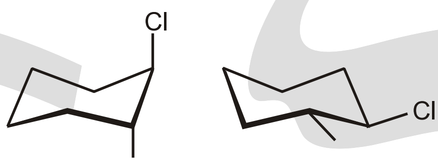
axial bonds

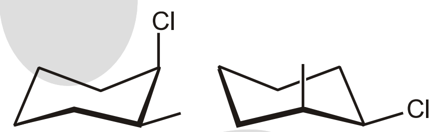
(b) 

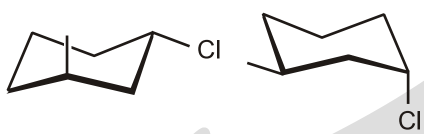
equatorial bonds

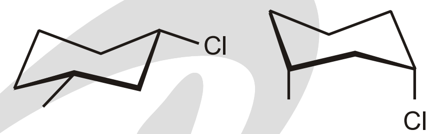
(c) 

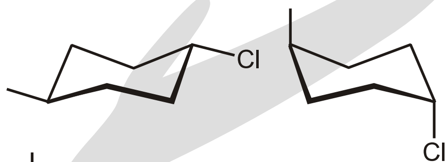
(d) 

4. (a) 

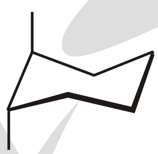
(b) 

(c) 

(d) 

(e) 

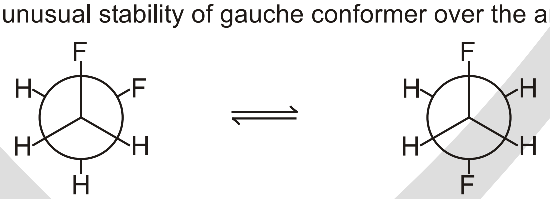
(f) 

(g)  one possible answer

(h)  one possible answer

Subjective Type Questions

3. The situation in FCH_2CH_2F is found to be quite different from either $ClCH_2CH_2Cl$ or $BrCH_2CH_2Br$ in the fact that gauche conformer is found to more stable than the anti conformer, even in gaseous state. Different explanations have been offered for this unusual stability of gauche conformer over the anti form

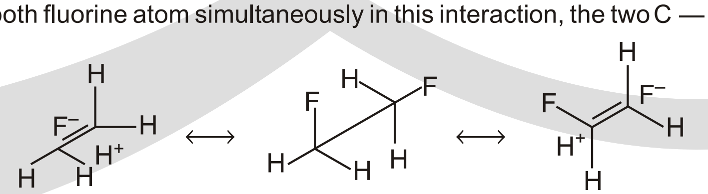


gauche-form (more stable) anti-form (Less stable)

Allinger *et al.* (1977) has explained the stability of gauche form on the basis of hyperconjugative interaction of the type, which is initiated by the high electronegativity of fluorine atom.

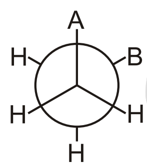
$F — CH_2 — CH_2 — F$ $\quad$ $F — CH \quad CH — FH$ $\quad$ $H\ F — CH \quad CH_2F$

In order to involve both fluorine atom simultaneously in this interaction, the two C — F bonds must be orthogonal.

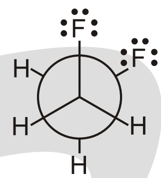


Small size of fluorine atom minimizes the van der Waals repulsive interaction in gauche conformer. Another explanation for preferred gauche conformation in the case of FCH_2CH_2F is the so called gauche effect a chain segment *A – B – C – D* will prefer gauche conformation when *A* and *D* are highly electronegative relative to *B* and *C* or *A* and *D* are themselves unshared electron pairs. In this case *A* and *D* are fluorine atom of high electronegativity.

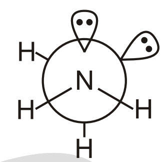
4. (a) In a conformational array, where *A* and *B* are second-row electronegative atoms such as *N*, *O*, or *F*, or unshared electron pair, the often observed preference for the gauche conformation (skew position) of *A* and *B* is called the 'gauche effect'. A few examples are given here.



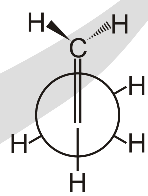
'gauche effect'



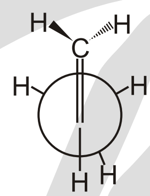
Preferred conformation of FCH_2CH_2F due to gauche effect



Preferred conformation of H_2NNH_2 due to gauche effect



Eclipsed conformation



Bisecting conformation

CHAPTER

25

Geometrical Isomerism

GEOMETRICAL ISOMERISM IS EXHIBITED BY A WIDE VARIETY OF COMPOUNDS

They may be classified into following groups :

(i) Compounds containing a double bond : C = C , C = N , N = N.

(ii) Compounds containing a cyclic structure – homocyclic, heterocyclic and fused ring systems.

(iii) Compounds which may exhibit geometrical isomerism due to restricted rotation about a single bond

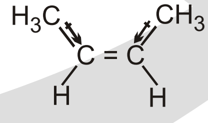
(iv) Bicyclo compound.

NOMENCLATURE OF GEOMETRICAL ISOMERS

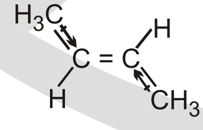
When geometrical isomerism is due to the presence of one double bond in a molecule, it is easy to name the geometrical isomers if two groups are identical, *e.g.*, in molecules (I) and (IV), (I) is the cis-isomer and (II) the trans ; similarly (III) is cis and (IV) is trans.

cis isomer (cis is Latin for "on this side"). trans isomer (trans is Latin for "across").

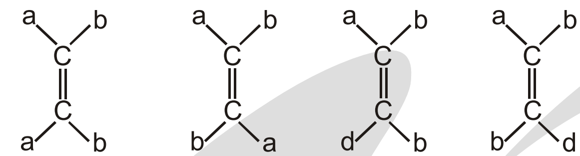
The *cis isomer has its substituents on the same side of the ring;the trans isomer has its substituents on opposite sides of the ring.* (A solid wedge represents a bond that points out of the plane of the paper toward the viewer and a hatched wedge represents a bond that points into the plane of the paper away from the viewer.)



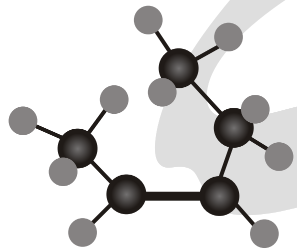
the cis isomer
bp 3.7 °C
= 0.33 D



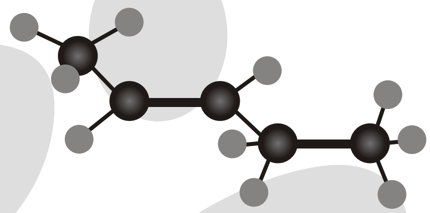
the trans isomer
bp = 0.9 °C
= 0 D



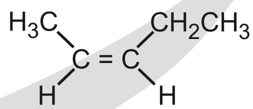
(I) cis (II) trans (III) cis (IV) trans



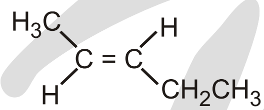
cis-2-pentene



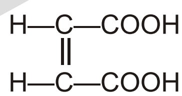
trans-2-pentene



cis-2-pentene

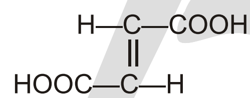


trans-2-pentene



Maleic acid (cis)

and



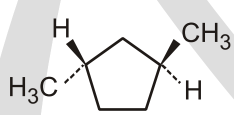
Fumaric acid (trans)

NAMING CYCLOALKANES

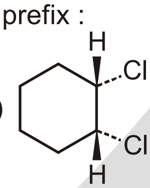
Solved Example

- Name the following substances, including the cis- or trans- prefix :

(a)



(b)

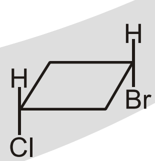


Strategy

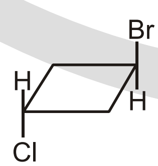
In these views, the ring is roughly in the plane of the page, a wedged bond protrudes out of the page, and a dashed bond recedes into the page. Two substituents are cis if they are both out of or both into the page, and they are trans if one is out of and one is into the page.

Ans. (a) *trans*-1,3-Dimethylcyclopentane (b) *cis*-1,2-Dichlorocyclohexane

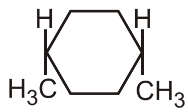
Solved Example



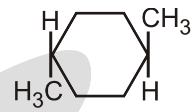
cis-1-bromo-3-chlorocyclobutane



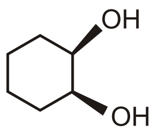
trans-1-bromo-3-chlorocyclobutane



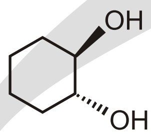
cis-1,4-dimethylcyclohexane



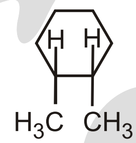
trans-1,4-dimethylcyclohexane



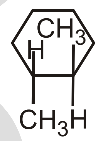
(cis)



(trans)



(cis 1, 2)

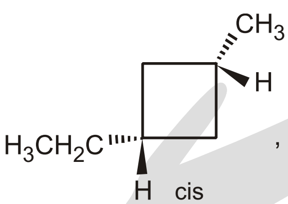


(trans 1, 2)

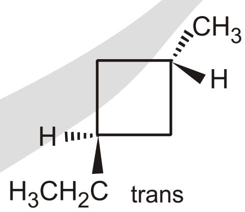
Solved Example

- Draw the cis and trans isomers for the following compounds :
  - (A) 1-ethyl-3-methylcyclobutane
  - (B) 3,4-dimethyl-3-heptene
  - (C) 1-bromo-4-chlorocyclohexane
  - (D) 1,3-dibromocyclobutane

Sol. (A)

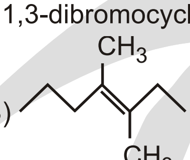


cis ,

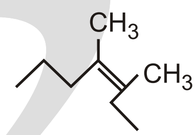


trans

(B)



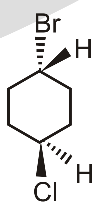
Ttrans



cis

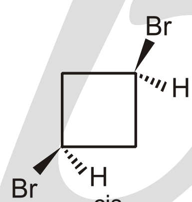
(C)

cis

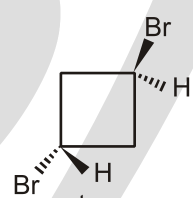


trans

(D)

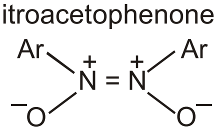


cis

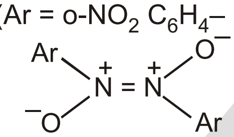


trans

Harley-Mason have offered evidence to show that they have isolated the three theoretically possible geometrical isomers of o-nitroacetophenone azine (Ar = o-NO_2 C_6H_4–) :

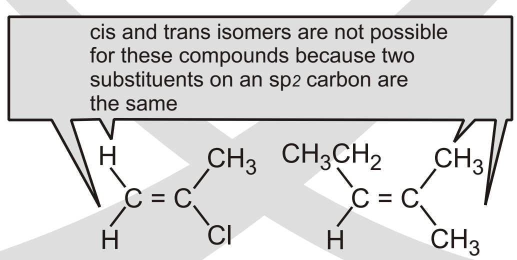


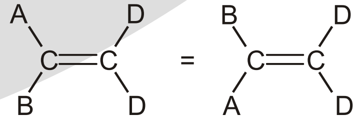
cis



trans

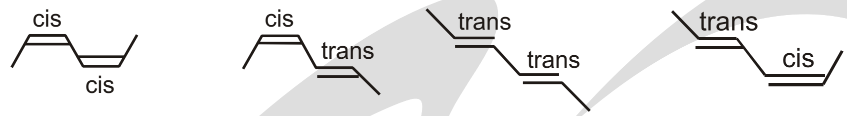
If one of the *sp*2 carbons is attached to two identical substituents, then the compound cannot have cis and trans isomers.



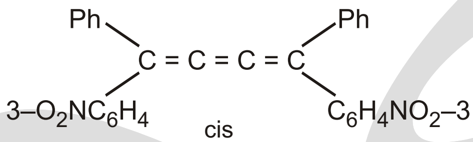


These two compounds are identical; they are not cis–trans isomers.

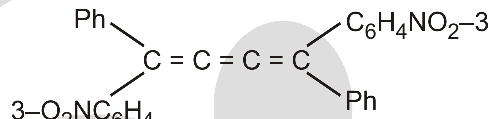
CIS & TRANS IN CONJUGATED DIENE



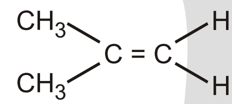
Alkenes like propene and 1-butene do not show geometrical isomerism



cis



trans



2-methylprop-1-ene

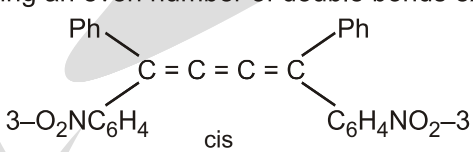
DIFFERENTIATING PROPERTIES OF CIS-TRANS ISOMERISM

(i) Dipole moment: Usually dipole moment of cis is larger than the trans-isomer.

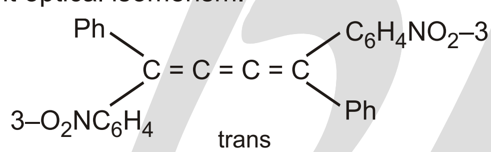
(ii) Melting point: The steric repulsion of the group (same) makes the cis isomer less stable than the trans isomers hence trans form has higher melting point than cis.

(iii) Different chemical properties: Syn-addition makes cis forms into meso and trans into d and l, anti addition makes cis into d - l and trans into meso.

Geometrical isomerism is also exhibited by cumulenes provided that the number of adjacent double bonds is odd, e.g., 1, 4-di-3nitrophenyl-1-4-diphenylbutatriene exists in two forms. On the other hand, cumulenes containing an even number of double bonds exhibit optical isomerism.

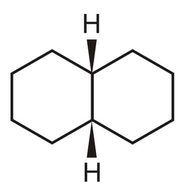


cis

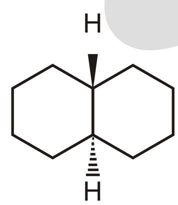


trans

GEOMETRICAL ISOMERS OF DECALINE

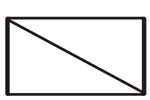


(I)
cis-

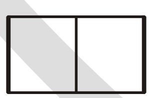


(II)
trans-

❑ **NOTE:**



(a)



(b)

(a) and (b) will not show geometrical isomerism

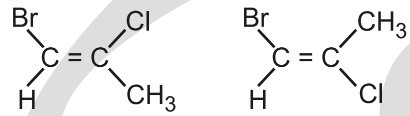
ALKENES STEREOCHEMISTRY AND THE E, Z DESIGNATION

If all the four groups/ atoms attached to C = C double bond are different, then Cis and trans nomenclature fails in such cases and a new nomenclature called E and Z system of nomenclature replace it.

The group / atom attached to carbon - carbon double bond is given to higher rank, whose atomic number is higher. If the two higher ranked group are across, it is called E form (E stands for the German word entgegen meaning thereby opposite) and the two higher ranked groups are on the same side, they are called Z-form (Z stands for German word Zusammen meaning thereby on the same side).

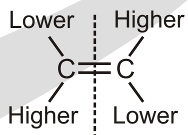
The cis–trans naming system used in the previous section works only with disubstituted alkenes—compounds that have two substituents other than hydrogen on the double bond. With trisubstituted and tetrasubstituted double bonds, a more general method is needed for describing double-bond geometry. (Trisubstituted means three substituents other than hydrogen on the double bond; tetrasubstituted means four substituents other than hydrogen.)

But how would you determine cis and trans isomers for a compound such as 1-bromo-2-chloropropene ?

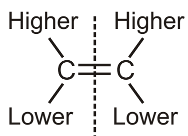


Which isomer is cis and which is trans ?

For a compound such as 1-bromo-2-chloropropene, the cis-trans system of nomenclature cannot be used because there are four different substituents on the two vinylic carbons. The E, Z system of nomenclature was devised for these kinds of situations. In order to name an isomer by the E, Z system, first determine the relative priorities of the two groups bonded to one of the sp2 carbons and then the relative priorities of the two groups bonded to the other sp2 carbon. (Rules for assigning relative priorities are explained next.) If the high-priority groups are on the same side of the double bond, the isomer is said to have the Z configuration (Z is for zusammen, German for "together"). If the high-priority groups are on opposite sides of the double bond, the isomer has the E configuration (E is for entgegen, German for "opposite").



E double bond
(Higher-ranked groups
are on opposite sides.)

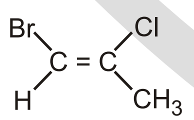


Z double bond
(Higher-ranked groups
are on *same* sides.)

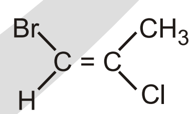
RULES TO DETERMINE THE PRIORITIES OF THE GROUPS BONDED TO THE SP^2 CARBONS

Rule 1.

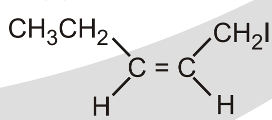
The relative priorities of the groups depended on the atomic numbers of the atoms bonded directly to a particular *sp*2 carbon. The greater the atomic number, the high the priority. For example, in 1-bromo-2-chloropropene, one of the sp2 carbons is bonded to a bromine and to a hydrogen. Bromine has a greater atomic number than hydrogen, so bromine has the higher priority. The other *sp*^{2} carbon is bonded to a chlorine and to a carbon. Chlorine has a greater atomic number than carbon, so chlorine has the higher priority. The isomer on the left has the high-priority groups (Br and Cl) on the same side of the double bond, so it is the Z isomer. The isomer on the right has the high-priority atoms on opposite sides of the double bond, so it is the E isomer.



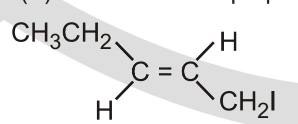
(Z)-1-bromo-2-chloropropene



(E)-1-brmo-2-chloropropene

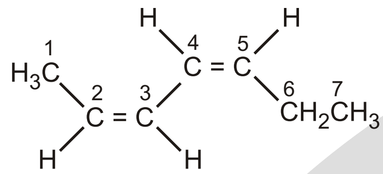


cis-or(Z)-1-Iodo-2-pentene

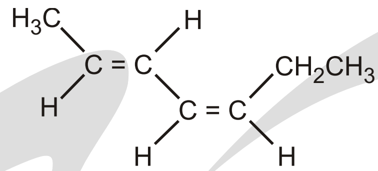


trans-or(E)-1-Iodo-2-pentene

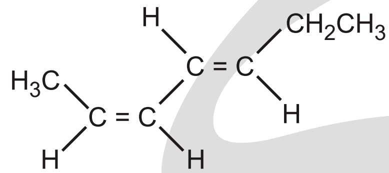
Both double bonds meet the conditions for geometric isomers and there are four diastereomers of 2,4-heptadiene.



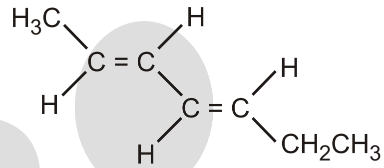
cis,cis or (Z, Z)



trans,cis or (E, Z)



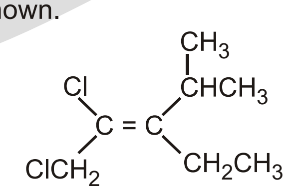
cis,trans or (Z, E)



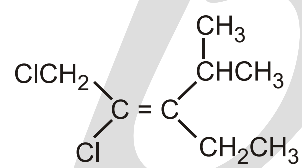
trans trans, or (E, E)

Rule 2.

If the two substituents on an sp^2 carbon have the same atomic number (there is a tie), the atomic numbers of the atoms that are attached to the "tied" atoms must be considered. In the following pair of isomers, both atoms bonded to one of the sp^2 carbons are carbons (from an ethyl group and an isopropyl group), so there is a tie at this point. One carbon is bonded to C, C, and H (the isopropyl group); the other is bonded to C, H, and H (the ethyl group). One C cancels in each of the two groups, leaving C and H in the isopropyl group and H and H in the ethyl group. Carbon has a greater atomic number than hydrogen, so the isopropyl group has a higher priority than the ethyl group. The other sp^2 carbon is bonded to a chlorine and to a chloromethyl group. Chlorine has a greater atomic number than carbon, so the chloro group has the higher priority. The E and Z isomers are as shown.



(Z)-1,2-dichloro-3-ethyl-
4-methyl-2-pentene



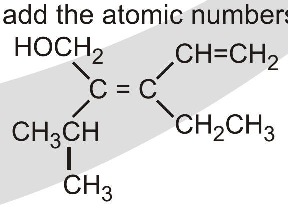
(E)-1,2-dichloro-3-ethyl-
4-methyl-2-pentene

Rule 3.

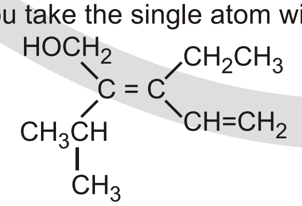
If an atom is doubly bonded to another atom, the priority system treats it as if it were singly bonded to two of those atoms. If an atom is triply bonded to another atom, the priority system treats it as if it were singly bonded to three of those atoms. One of the sp^2 carbons in the following pair of isomers is bonded to an ethyl group and to a vinyl group. Because the atoms immediately bonded to the sp^2 carbon are both carbons, there is a tie. The carbon of the ethyl group is bonded to C, H and H. The carbon of the vinyl group is bonded to an H and doubly bonded to a C. Therefore, the first carbon of the vinyl group is considered to be bonded to C, C, and H. Consequently, the vinyl group has a higher priority than the ethyl group. Both atoms that are bonded to the other sp^2 carbon are carbons, too, so there is a tie there as well. The carbon of the hydroxymetyl group is bonded to O, H, and H, and the carbon of the isopropyl group to C, C, and H. Of these six atoms, oxygen has the greatest atomic number, so the hydroxymethyl group has the higher priority.

- **Mistake to avoid**

Notice that you do not add the atomic numbers; you take the single atom with the greatest atomic number.



the Z isomer

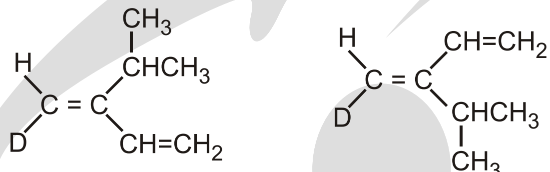


the E isomer

Rule 4.

In the case of isotopes (atoms with the same atomic number but different mass numbers), the mass number is used to determine their relative priorities.

–T > –D > –H

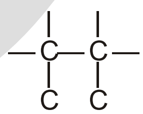
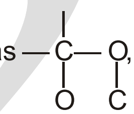


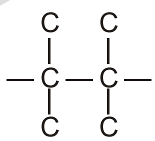
the Z isomer   the E isomer

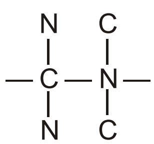
Notice that in all these examples you never count the atom bonded to the s bond from which you originate. In differentiating between the isopropyl and vinyl groups in the last example, you did count the atom bonded to the p bond from which you originated.

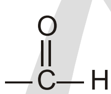
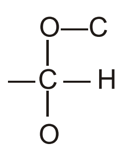
Rule 5.

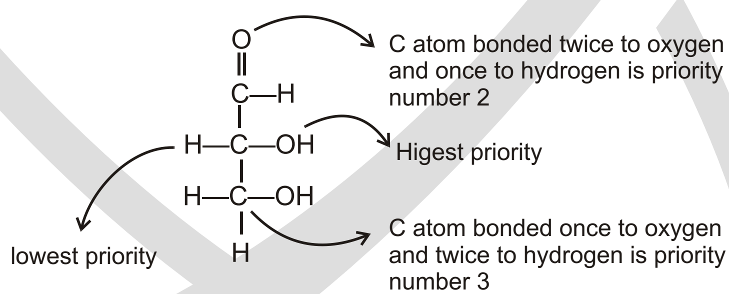
If the atom in consideration is further attached to an atom via a double bond, then it is treated as if it is attached to two such atoms.

For example, —C = C— may be treated as  and —C = O is treated as , that is, the atom at the end of the extreme end of the multiple bond is like as if it is equal to equivalent number of single bonds.

Thus —C C— is treated as 

—C N— is treated as 

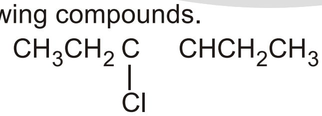
 is treated as 

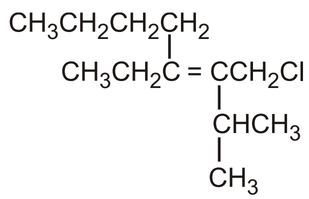


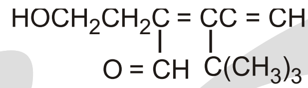
Solved Example

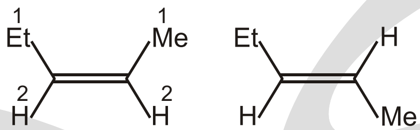
- Draw and label the E and Z isomers of each of the following compounds.

1. $CH_3CH_2CH = CHCH_3$

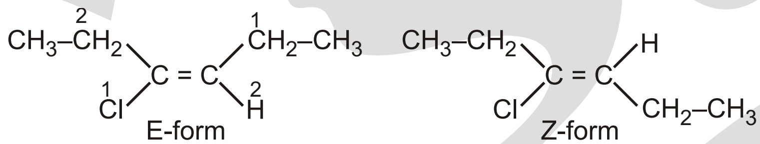
2. 

3. 

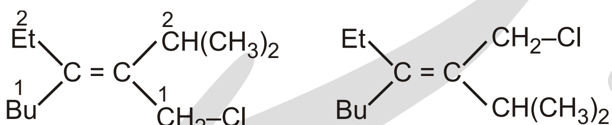
4. 

Ans. 1. 

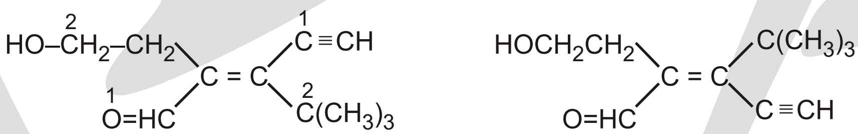
Z-form E-form

2. 

E-form Z-form

3. 

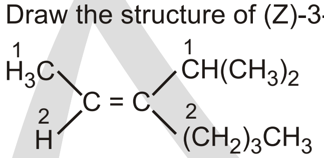
Z-form E-form

4. 

E-form Z-form

Solved Example

- Draw the structure of (Z)-3-isopropyl-2-heptene.

Sol. 

Z-form

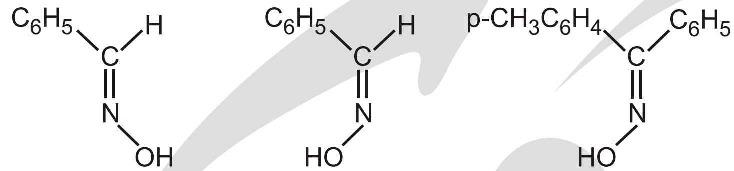
The priority order is followed as

I, Br, Cl, SO3H, SH, F, COOR, OR, OH, NO_2, NR_2, NHCOR, NHR, NH_2, CO_2R, COOH, $CONH_2$, CHO, CH_2OH, CN, CR_3, C_6H_5, CHR_2, CH_2R, CH_3, D, H.

NOMENCLATURE OF THE OXIMES

In oxidme chemistry the terms syn and anti are used instead of the terms cis and trans. When dealing with aldoximes, the syn-form is the one in which both the hydrogen atom and the hydroxyl group are on the same side; when these groups are on opposite sides, the configuration is anti. Thus (I) is syn-and (II) is anti-benzaldoxime. With ketoximes, the prefix indicates the spatial relationship between the first group named and the hydroxyl group. Thus III may be named as syn- p-tolyl phenyl ketoxime or anti-phenyl p-tolyl ketoxime.

Solved Example

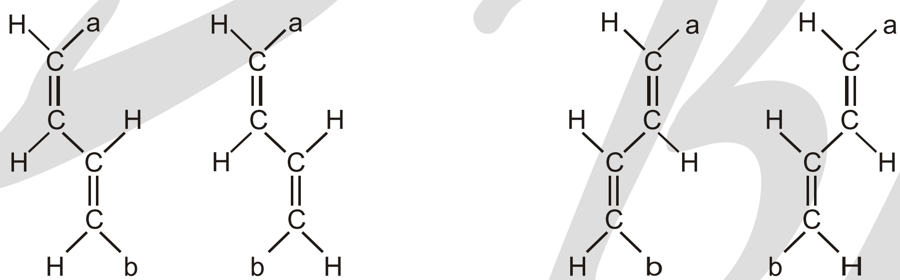


(I) syn (II) anti (III)

The E – Z system of nomenclature is also applied to oximes. Thus, the syn-oxime (I) is named benzaldehyde (E) - oxime or (E) - benzaldehyde oxime ; (II) is the corresponding (Z) oxime. The group with the greater priority (phenyl) is taken as being cis with respect to they hydroxyl group. Since p-tolyl has priority over phenyl, (III) is (Z) p-tolyl phenyl ketoxime.

TOTAL NUMBER OF GEOMETRICAL ISOMERS

If a compound has two double bonds, e.g., CHa = CH – CH = CHb, four geometrical isomers are possible :



cis-trans cis-cis trans-cis trans-trans

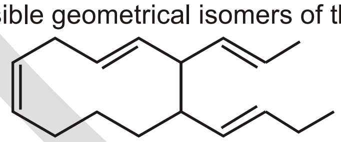
The number of geometrical isomers is 2^n, where n is the number of double bonds.

This formula applies only to molecules in which the ends are different

If the ends are identical, e.g., CHa = CH – CH = CHa, then the number of stereoisomers is $\mathbf{2^{n-1} + 2^{p-1}}$ **, where p = n/2 when n is even, and p = (n + 1)/2 when n is odd.**

Solved Example

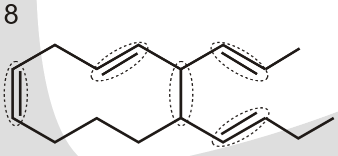
- **X** = Total number of possible geometrical isomers of the below compound.



The value of $\frac{X}{4}$ is :

Ans. 8

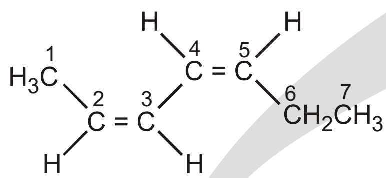
Sol.



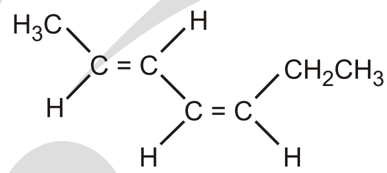
Bonds which are circled show geometrical isomers thus value of n = 5 ; Total G.I. = 2^5 = 32

The value of $\frac{X}{4}$ $\frac{32}{4}$ 8 = 8

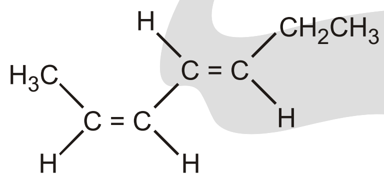
Solved Example



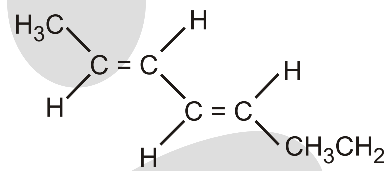
cis,cis or (Z, Z)



trans,cis or (E, Z)



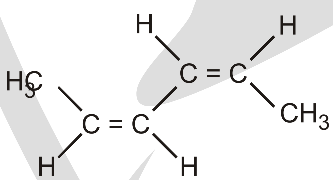
cis,trans or (Z, E)



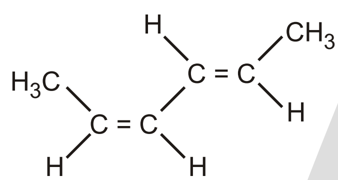
trans, trans or (E, E)

❑ **NOTE:** *That cis and trans and E and Z are listed in the same order as the bonds are numbered.*

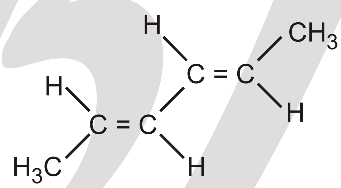
Solved Example



cis,cis-2,4-Hexadiene or
(Z,Z)-2,4-Hexadiene



cis,trans-2,4-Hexadiene or
(Z,E)-2,4-Hexadiene

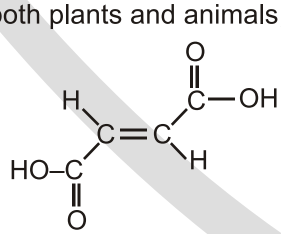


trans,trans-2,4-Hexadiene or
(E,E)-2,4-Hexadiene

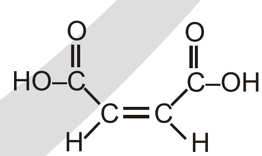
☑ SPECIAL TOPIC

MALEIC AND FUMARIC ACID

The cis isomer is called Maleic acid, and the trans isomer is called Fumaric acid. Fumaric acid is an essential metabolic intermediate in both plants and animals, but maleic acid is toxic and irritating to tissues.



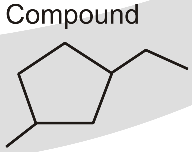
fumaric acid, mp 287ºC
essential metabolite

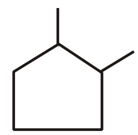


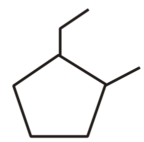
maleic acid, mp 138ºC
toxic irritant

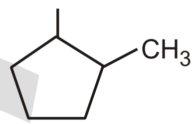
- Whether the following compound will show G.I. ?

Sol. Compound — Geometrical Isomerism

(A) 

(B) 

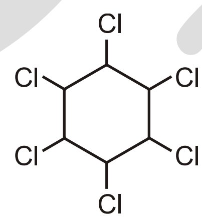
(C) 

(D) 

EXERCISE

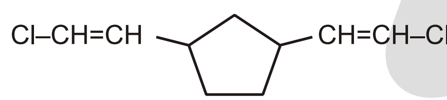
SINGLE CHOICE QUESTIONS

1. Number of geometrical isomers of the given compound are:



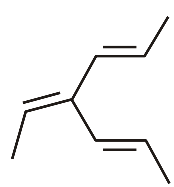
(A) 6 (B) 7 (C) 8 (D) 9

2. Number of geometrical isomers of the given compound are:



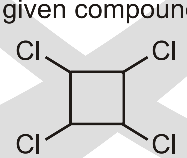
(A) 2 (B) 4 (C) 6 (D) 8

3. Number of geometrical isomers of the given compound are:



(A) 1 (B) 2 (C) 3 (D) 4

4. Number of geometrical isomers of the given compound are:



(A) 1 (B) 2 (C) 3 (D) 4

5. Number of geometrical isomers of the given compound are:

CH_3–CH = C=CH–CH=CH–Cl

(A) 1 (B) 2 (C) 3 (D) 4

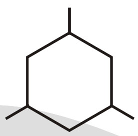
6. Total number of geometrical isomers possible for given compounds are:

Me-CH=CH-CH=C=CH-CH=CH-CH_3

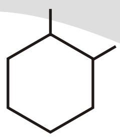
(A) 16 (B) 8 (C) 4 (D) 2

7. Find total number of Geometrical isomerism of following compounds.

CH_3–CH=CH–CH=N–OH has x geometrical isomers



has y geometrical isomers



has z geometrical isomers

What is the value of x+y+z?

(A) 16 (B) 8 (C) 4 (D) 10

8. Find total number of Geometrical isomerism of following compounds.

(A) 0 (B) 8 (C) 4 (D) 2

9. Which of the following compounds will exhibit geometrical isomerism :

(A) 1-Phenyl-2-butene (B) 3-Phenyl-1-butene

(C) 2-Phenyl-1-butene (D) 1,1-Diphenyl-1-propene.

10. The number of isomers for the compound with molecular formula $C_2BrClFI$ is :

(A) 2 (B) 3 (C) 5 (D) 6

11. Maleic acid and fumaric acid are :

(A) Position isomers (B) Geometric isomers

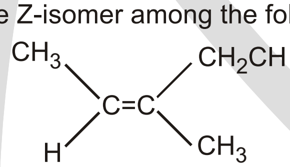
(C) Enantiomers (D) Functional isomers.

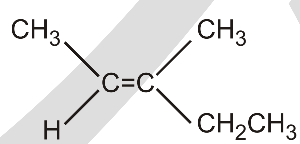
12. Which of the following compounds does not exhibit geometric isomerism :

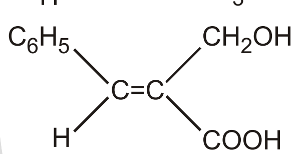
(A) 1,1-Dichloro-2-butene (B) 1,2-Dichloro-2-butene

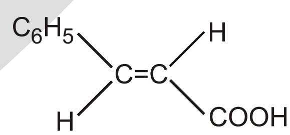
(C) 1,1-Dichloro-1-butene (D) 2,3-Dichloro-2-butene.

13. The Z-isomer among the following is

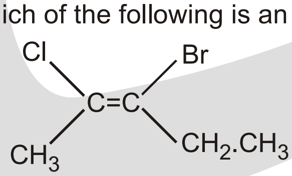
(A) 

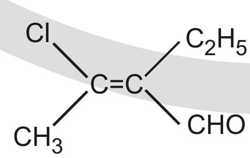
(B) 

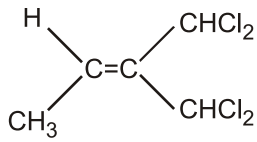
(C) 

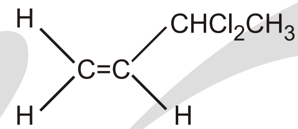
(D) 

14. Which of the following is an 'E' isomer?

(A) 

(B) 

(C) 

(D) 

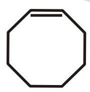
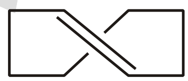
15. Which of the following statements about cis-trans isomerism is/are correct ?

(1) All alkenes exhibit cis-trans isomerism.

(2) But-2-ene exhibits, cis-trans isomerism.

(3) A pair of cis-trans isomers may be optically active

(A) (1) only
(B) (2) only
(C) (1) and (3) only
(D) (2) and (3) only

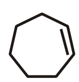
16.  

(A) (B)

Relationship between (A) and (B) is :

(A) Diastereomers
(B) Enantiomer
(C) Identical
(D) Structural isomer

17. Geometrical isomerism is possible in :

(A) 
(B) 
(C) 
(D) 

18. Which of the following compounds could exhibit geometrical isomerism?

(1) 3,4-dimethylhex-3-ene

(2) 2-methylpent-2-ene

(3) 1,6-dichlorohex-3-ene

(A) (1) and (2) only
(B) (1) and (3) only
(C) (2) and (3) only
(D) (1), (2) and (3)

19. Which of the following compounds has/have a pair of geometrical isomers ?

(1) $CH_3CH = CH_2$

(2) $CH_3OCCH = CHCOCH_3$

(3) $CH_2BrCH = CHCH_2Cl$

(A) (1) only
(B) (2) only
(C) (1) and (3) only
(D) (2) and (3) only

20. Which of the following statements concerning geometrical isomers is/are correct ?

(1) The cis isomer has a higher melting point than the trans isomer.

(2) A pair of geometrical isomers has the same functional group.

(3) Any organic compounds with a carbon-carbon double bondhave geometrical isomers.

(A) (1) only
(B) (2) only
(C) (1) and (3) only
(D) (2) and (3) only

21. Which of the following are the types of structural isomern?

(1) Geometrical isomerism

(2) Functional group isomerism

(3) Chain isomerism

(A) (1) and (2) only (B) (1) and (3) only

(C) (2) and (3) only (D) (1), (2) and (3)

22. Which of the following statements concerning a pair of geometrical isomers are correct ?

(1) They have different boiling points and melting points.

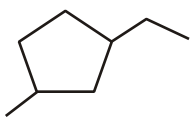
(2) They have the same relative molecular mass

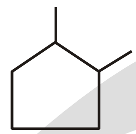
(3) Their atoms are joined in the same order.

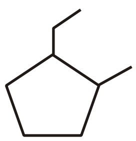
(A) (1) and (2) only (B) (1) and (3) only

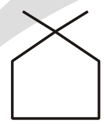
(C) (2) and (3) only (D) (1), (2) and (3)

23. Which of the cycloalkane is not cabable to show cis-trans isomerism ?

(A) 

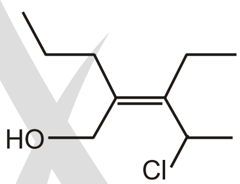
(B) 

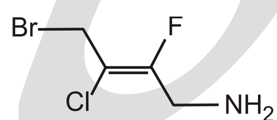
(C) 

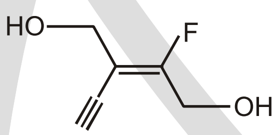
(D) 

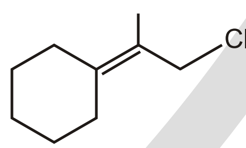
MULTIPLE CHOICE QUESTIONS

1. Which alkane has the Z configuration along '=' bond :

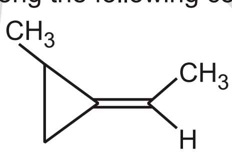
(A) 

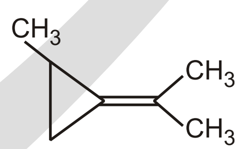
(B) 

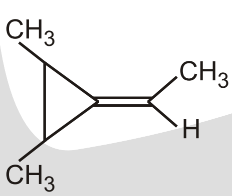
(C) 

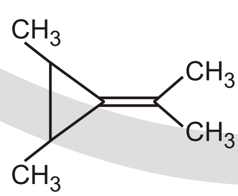
(D) 

2. Among the following compounds, which can show geometrical isomerism ?

(A) 

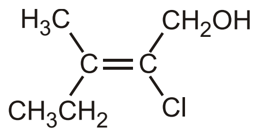
(B) 

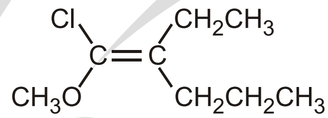
(C) 

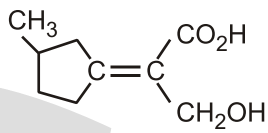
(D) 

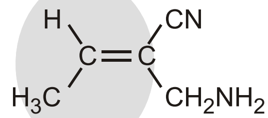
UNSOLVED EXAMPLE

1. Assign E or Z configuratoin to the following alkenes :

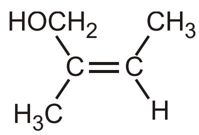
(a) 

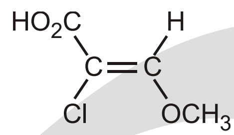
(b) 

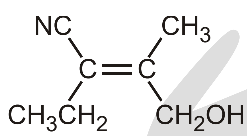
(c) 

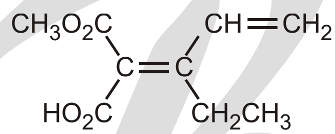
(d) 

2. Assign E or Z configuration to each of the following compounds :

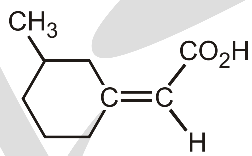
(a) 

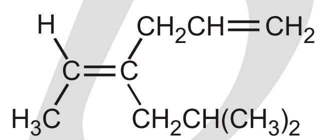
(b) 

(c) 

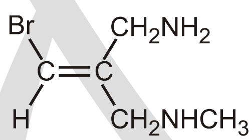
(d) 

3. Which of the following E, Z designations are correct, and which are incorrect ?

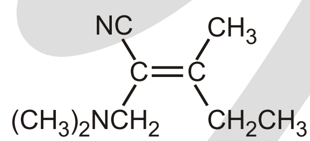
(a) 

(b) 

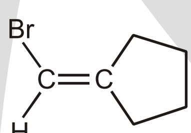
E

(c) 

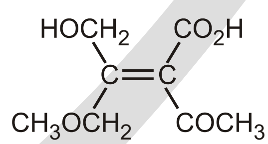
Z

(d) 

E

(e) 

Z

(f) 

E

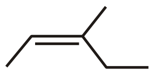
4. Which member in each of the following sets ranks higher?

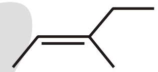
(a) –H or $–CH_3$
(b) –Cl or $–CH_2Cl$
(c) $–CH_2CH_2Br$ or $–CH=CH_2$
(d) $–NHCH_3$ or $–OCH_3$
(e) $–CH_2OH$ or –CH=O
(f) $–CH_2OCH_3$ or –CH=O

5. Rank the substituents in each of the following sets according to the sequence rules:

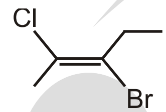
(a) $–CH_3$, –OH, –H, –Cl
(b) $–CH_3$, $–CH_2CH_3$, $–CH=CH_2$, $–CH_2OH$
(c) $–CO_2H$, $–CH_2OH$, –C N, $–CH_2NH_2$
(d) $–CH_2CH_3$, –C CH, –C N, $–CH_2OCH_3$

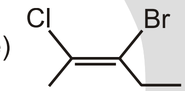
6. Write (E) and (Z) configuration of the following :

(a) 

(b) 

(c) 

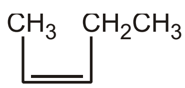
(d) 

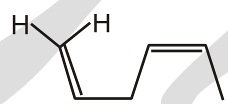
(e) 

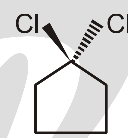
WORKSHEET-1

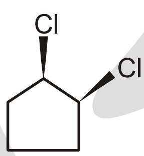
Q. (To find Geometrical isomerism)

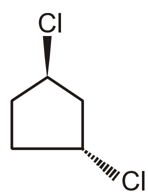
Identify which of the compound show geometrical isomerism?

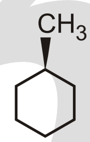
1. 

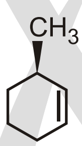
2. 

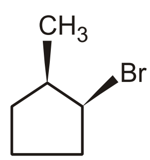
3. 

4. 

5. 

6. 

7. 

8. 

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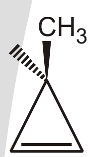
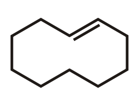
10. 

11. 

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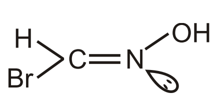
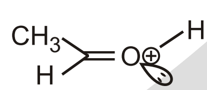
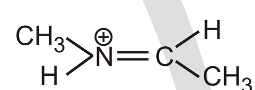
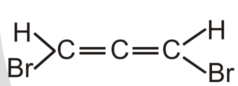
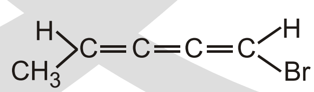
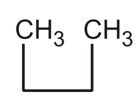
13. 

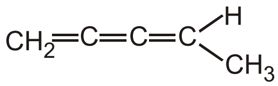
14. 

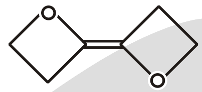
15. 

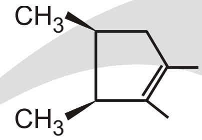
16. 

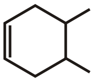
17. 

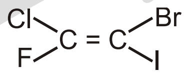
18. 

19. 

20. 

21. 

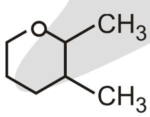
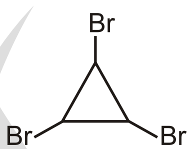
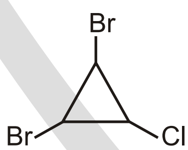
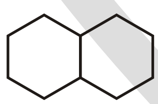
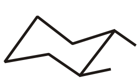
22. 

23. 

24. $CH_3 - C \quad C - CH_3$

WORKSHEET-2

(Total Geometrical Isomers)

S.N.	Compound	Total number of Geometrical Isomers possible.
1.	$CH_3 - CH = CH - CH_2CH_3$	
2.	$CH_3 - CH = CH - CH = CH_2$	
3.	$CH_3 - CH = CH - CH = CH - CH_3$	
4.	$CH_3 - CH = CH - CH = CH - Br$	
5.	$CH_3 - CH = CH - CH = CH - CH = CH_2$	
6.	$CH_3 \left(CH \quad CH_3 \right) Ph$	
7.		
8.		
10.		
11.		
12.		
13.	$CH_3 \quad CH \quad CH_2$	

Answers

Single Choice Questions

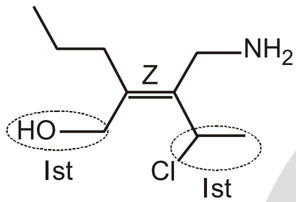
1. (C)	**2.** (C)	**3.** (D)	**4.** (D)	**5.** (B)	**6.** (C)	**7.** (B)	**8.** (D)
9. (A)	**10.** (D)	**11.** (B)	**12.** (C)	**13.** (A)	**14.** (B)	**15.** (D)	**16.** (A)
17. (D)	**18.** (B)	**19.** (D)	**20.** (B)	**21.** (C)	**22.** (D)	**23.** (D)	

Multiple Choice Questions

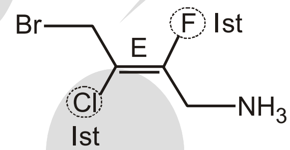
1. (A, C) **2.** (A, C, D)

1. According to CIP rule

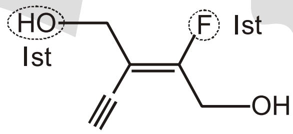
(A)



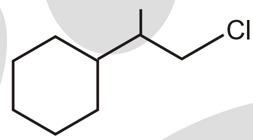
(B)



(C)



(D)



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❑❑❑

CHAPTER

26

Optical Isomerism

INTRODUCTION

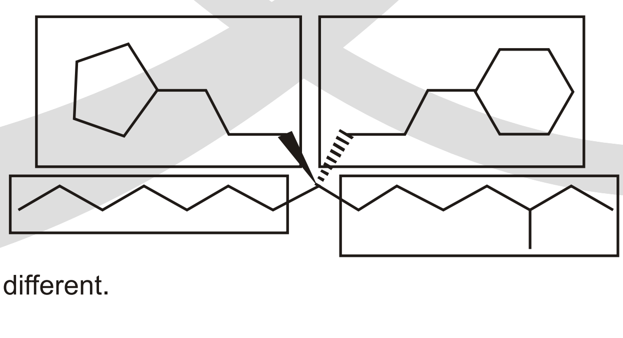
Plane polarised light and optical activity: Certain compounds rotate the plane polarised light (produced by passing ordinary light through Nicol prism) when it is passed through their solutions. Such compounds are called **optically active** compounds. The angle by which the plane polarised light is rotated is measured by an instrument called polarimeter. If the compound rotates the plane polarised light to the right, *i.e.,* clockwise direction, it is called dextrorotatory (Greek for right rotating) or the d-form and is indicated by placing a positive (+) sign before the degree of rotation. If the light is rotated towards left (anticlockwise direction), the compound is said to be laevorotatory or the l-form and a negative (–) sign is placed before the degree of rotation. Such (+) and (–) isomers of a compound are called optical isomers and the phenomenon is termed as **optical isomerism**.

CHIRAL CENTER

☞ Those centers which create unsymmetry (chirality) in the molecule are called chiral centers. Chiral center must be sp^3 with 4 different valencies.

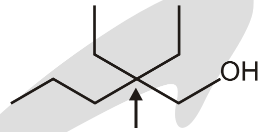
For purposes of this course, we will define a chiral center (stereocenter) as a carbon atom with four different groups on it. For example.

Whenever we look at the four groups connected to an atom, we are looking at the entire molecule, no matter how big those groups are. Consider the following example:



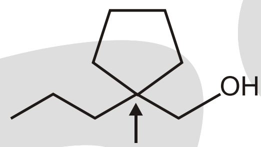
All four of these groups are different.

You must learn how to recognize when an atom has four different groups attached to it. To help you with this, let's begin by seeing the situations that are not chiral center (stereocenter):



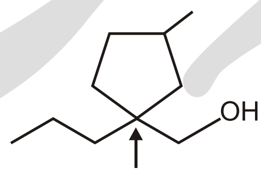
Not a stereocenter

The carbon atom indicated above is not a chiral center (stereocenter) because there are two groups that are the same (there are two ethyl groups). The same is true in the following case:



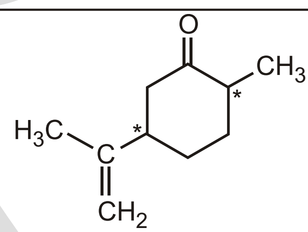
Not a stereocenter

Whether you go around the ring clockwise or counterclockwise, you see the same thing, so this is not a chiral center (stereocenter). If we wanted to make it a chiral center (stereocenter), we could do so by putting a group on the ring:

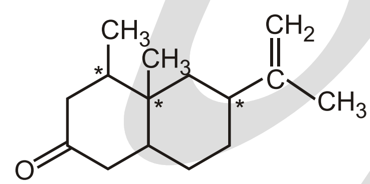


A stereocenter

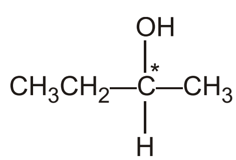
Solved Example



Carvone (spearmint oil)



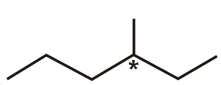
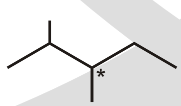
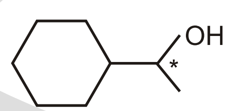
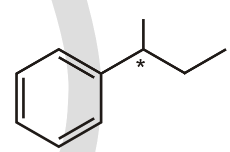
Nootkatone (grapefruit oil)

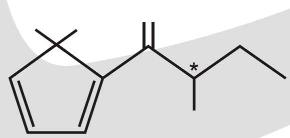
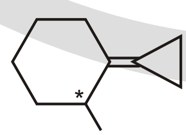


2-Butanol (chiral)

Solved Example

- In each of the compounds below', there is one stereocenter. Find it.

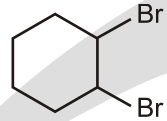
(a)  (b)  (c)  (d) 

(e)  (f) 

Stereocenters are marked with asterisck (*)

Solved Example

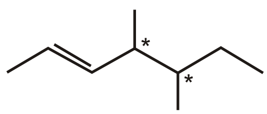
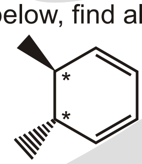
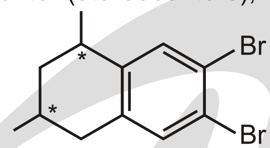
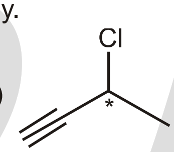
- In the following compound, find all of the chiral center (stereocenters), if any:

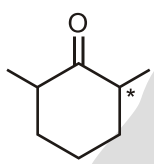
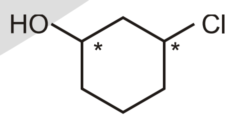
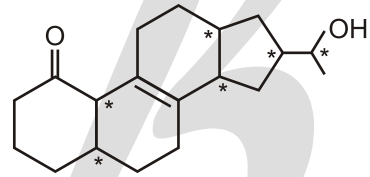


Ans. If we go around the ring, we find that there are only six carbon atoms in this compound. Four of them are CH_2 groups, so we know that they are not chiral center (stereocenters). If we look at the remaining two carbon atoms, we see that each of them is connected to four different groups. They are both chiral center (stereocenters).

Solved Example

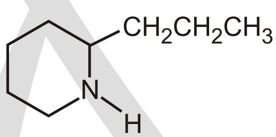
- For each of the compounds below, find all of the chiral center (stereocenters), if any.

(a)  (b)  (c)  (d) 

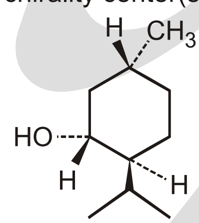
(e)  (f)  (g) 

Solved Example

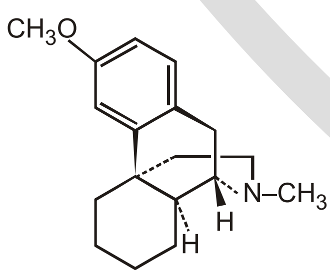
- Which of the following molecules are chiral? Identify the chirality center(s) in each.

(a) 

Coniine
(poison hemlock)

(b) 

Menthol
(flavoring agent)

(c) 

Dextromethorphan
(cough suppressant)

Ans. *a*, *b*, *c*, chiral centers in *a* = 2
b = 3
c = 4

Solved Example

▸ Which of the following compounds have chiral centers?

(a) $CH_3CH_2\underset{\underset{Cl}{|}}{C}HCH_3$ (b) $CH_3CH_2\underset{\underset{CH_3}{|}}{C}HCH_3$

(c) $CH_3CH_2\overset{\overset{CH_3}{|}}{\underset{\underset{Br}{|}}{C}}CH_2CH_2CH_3$ (d) CH_3CH_2OH

(e) $CH_3CH_2\underset{\underset{Br}{|}}{C}HCH_2CH_3$ (f) $CH_2 \quad CHC\underset{\underset{NH_2}{|}}{H}CH_3$

Ans. Only a, c and f has chirality centre (chiral carbon).

Solved Example

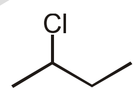
▸ Which of the following compound is chiral compound ?

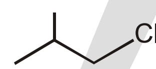
(A) 1-Chlorobutane (B) 2-Chlorobutane

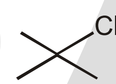
(C) 1-Chloro-2-methyl propane (D) 2-Chloro-2-methyl propane

Ans. (B)

(A) 

(B) 

(C) 

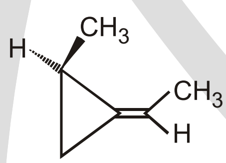
(D) 

STEREOCENTER

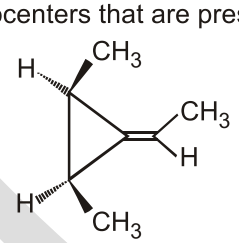
☞ Those centers in a molecule which can show optical or geometrical isomerism are called stereocenters. One Chiralcenter showing optical isomerism means one stereocenters and one bond showing geometrical isomerism means two stereocenters.

Solved Example

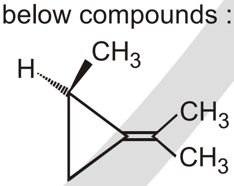
▸ Find the sum of all the stereocenters that are present in below compounds :



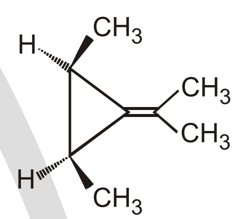
(I)



(II)



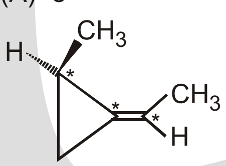
(III)



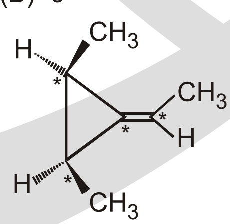
(IV)

(A) 8 (B) 9 (*C) 10 (D) 11

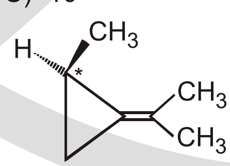
Sol.



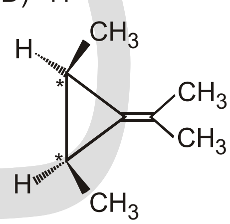
(I)



(II)



(III)

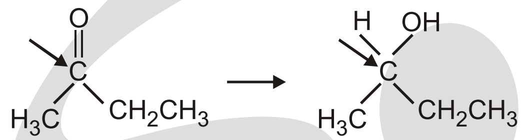


(IV)

Stereocenters are marked with asterisck (*)

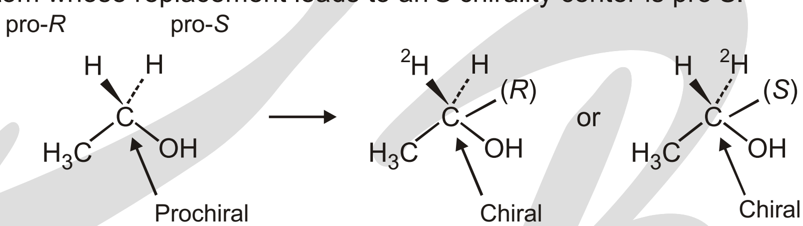
PROCHIRALITY

Closely related to the concept of chirality, and particularly important in biological chemistry, is the notion of prochirality. A molecule is said to be prochiral if it can be converted from achiral to chiral in a single chemical step. For instance, an unsymmetrical ketone like 2-butanone is prochiral because it can be converted to the chiral alcohol 2-butanol by addition of hydrogen.



2-Butanone (prochiral) 2-Butanol (chiral)

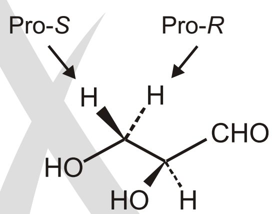
Of the two identical atoms in the original compound, that atom whose replacement leads to an *R* chirality center is said to be pro-*R* and that atom whose replacement leads to an *S* chirality center is pro-*S*.



Solved Example

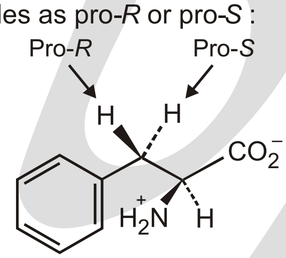
- Identify the indicated hydrogens in the following molecules as pro-*R* or pro-*S* :

(a)



(S)-Glyceraldehyde

(b)



Phenylalanine

R, S CONFIGURATION

Determining the Configuration of a Stereocenter

Now that we can find stereocenters, we must now learn how to determine whether a stereocenter is *R* or *S*. There are two steps involved in making the determination. First, we give each of the four groups a number (from 1 to 4). Then we use the orientation of these numbers to determine the configuration. So, how do we assign numbers to each of the groups?

We start by making a list of the four atoms attached to the stereocenter. Let's look at the following example:

The four atoms attached to the stereocenter are C, C, O, and H. We rank them from 1 to 4 based on atomic number. To do this, we must either consult a periodic table every time or commit to memory a small part of the periodic table — just those atoms that are most commonly used in organic chemistry:

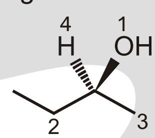
C	N	O	F
	P	S	Cl
			Br
			I

When comparing the four atoms in the example above, we see that oxygen has the highest atomic number, so we give it the first priority — we give it the number 1. Hydrogen is the smallest atom, so it will always get the number 4 (lowest priority) when a stereocenter has a hydrogen atom. We don't have to worry about what to do if there are two hydrogen atoms, because if there were, it would not be a stereocenter.

We compare the two lists and look for the first point of difference:

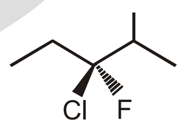
C	H
H	H
H	H

We see the first point of difference immediately: carbon beats hydrogen. So the left side of the stereocenter gets priority over the right side, and the numbering turns out like this:



Solved Example

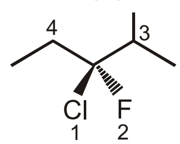
- In the compound below, find the stereocenter, and label the four groups from 1 to 4 using the system of priorities based on atomic number.



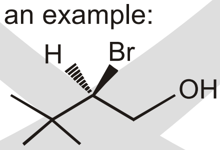
Ans. The four atoms attached to the stereocenter are C, C, Cl, and F. Of these, Cl has the highest atomic number, so its gets the first priority. Then comes F as number 2. We need to decide which carbon atom gets the number 2 and which carbon atom gets the number 3. We do this by listing the three atoms attached to each of them:

Left side	**Right side**
C	C
H	C
H	H

So the right side wins. Therefore, the numbering goes like this :



Also, you should know that we are looking for the first point of difference as we travel out, and we don't add the atomic number. This is best explained with an example:

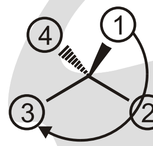


Left side	**Right side**
C	O
C	H
C	H

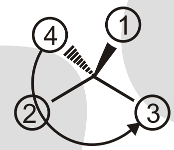
In this case, we do not add the atomic number and say that the left side wins. Rather we go down the list and compare each row. In the first row above, we have C versus O. That's it, end of story—the O wins. It doesn't matter what comes in the next two rows. Always look for the first point of difference. So the priorities so like this:

Now we need to learn how to use this numbering system to determine the configuration of a stereocenter. The idea is simple, but it is difficult to do if you have a hard time closing your eyes and rotating 3D objects in your mind. For those who cannot do this, don't worry. There is a trick. Let's first see how to do it without the trick.

If the number 4 group is pointing away from us (on the dash), then we ask whether 1, 2, and 3 are going clockwise or counterclockwise:

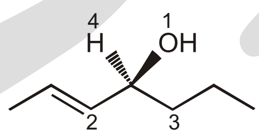


Clock wise
R



Counterclock wise
S

In the example on the left, we see that 1, 2, 3 go clockwise, which is called *R*. In the example on the right, we see that 1, 2, 3 go counterclockwise, which is called *S*. If the molecule is already drawn with the number 4 priority on the dash, then your life is very simple:

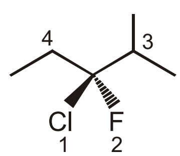


The 4 is already on the dash, so you just look at 1, 2 and 3. In this case, they go counterclockwise, so it is S.

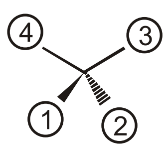
The solid wedges represent bonds that point out of the plane of the paper toward the viewer.

The hatched wedges (dash) represent bonds that point back from the plane of the paper away from the viewer.

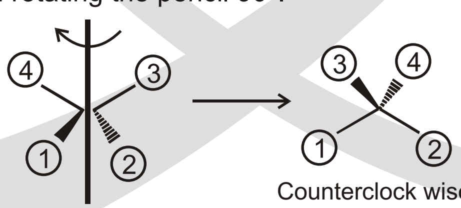
It gets a little more difficult when the number 4 is not on a dash, because then you must rotate the molecule in your mind. For example,



Let's redraw just the stereocenter showing the location of the four priorities:



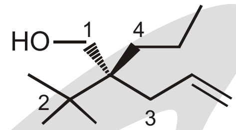
Now we need to rotate the molecule so that the fourth priority is on a dash. To do this, imagine spearing the molecule with a pencil and then rotating the pencil 90º.



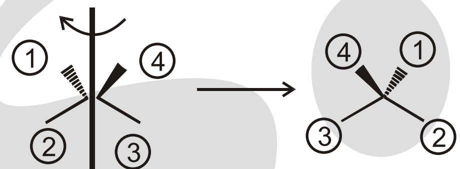
Counterclock wise
S

Now the 4 is on a dash, so we can look at 1, 2 and 3, and we see that they go counterclockwise. Therefore, the configuration is *S*.

Let’s see one more example:



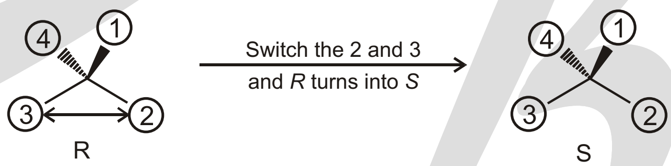
We redraw just the stereocenter showing the location of the four priorities, and then we spear the molecule with a pencil and rotate 180° to put the 4 on a dash:



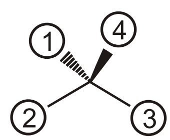
Clock wise

Now, the 4 is on a dash, so we can look at 1, 2 and 3, and we see that they go clockwise. Therefore, the configuration is *R*.

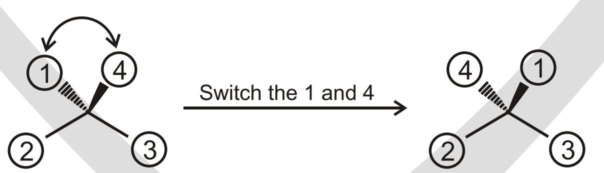
And now, for the trick. If you were able to see all of that, great! But if you had trouble seeing the molecules in 3*D*, there is a simple trick that will help you get the answer every time. To understand how the trick works, you need to realize that if you redraw the molecule so that any two of the four groups are switched, then you have switched the configuration (*R* turns into *S* and *S* turns into *R*):



You can switch any two groups and this will happen. You can use this idea to your advantage. Here is the trick: Switch the number 4 with whatever group is on the dash—then your answer is the opposite of what you see. Let’s do an example :



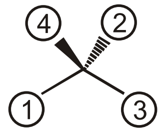
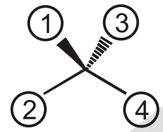
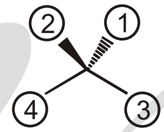
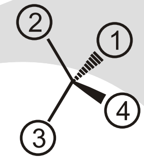
This looks tough because the 4 is on a wedge. But let’s do the trick: switch the 4 with whichever group is on the dash; in this case, we switch the 4 with the 1:

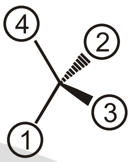
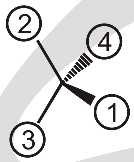
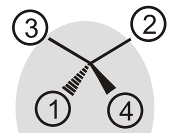
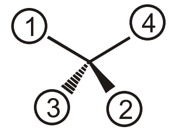


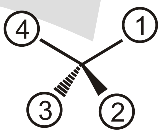
Counterclock wise
S

After doing the switch, the 4 is on a dash, and it becomes easy to figure out. It is counterclockwise, w'hich means *S*. We had to do one switch to make it easy to figure out, which means that we changed the configuration. So if it became *S* after the switch, then it must have been *R* before the switch. That's the trick. But be careful. This trick will work every time, but you must not forget that the answer you immediately get is the opposite of the real answer, because you did one switch.

Now, let's practice determining *R* or *S* when you are given the numbers, so that we can make sure you know how to do this step. You can either visualize the molecule in 3D, or you can use the trick—whatever works best for you.

(a) 
(b) 
(c) 
(d) 

(e) 
(f) 
(g) 
(h) 

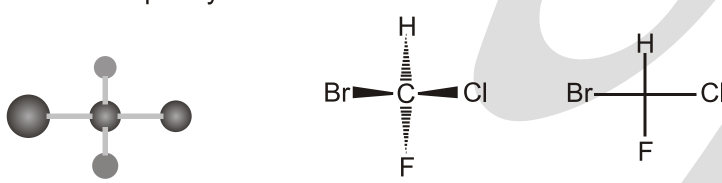
(i) 

Ans. (a) *S* (b) *R* (c) *S* (d) *R*
(e) *S* (f) *S* (g) *R* (h) *R*
(i) *S*

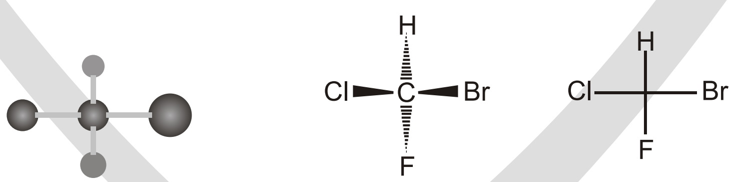
FISCHER PROJECTIONS

It is possible, however, to convey stereochemical information in an abbreviated form using a method devised by the German chemist Emil Fischer.

The molecule is oriented so that the vertical bonds at the chirality center are directed away from you and the horizontal bonds point toward you. A projection of the bonds onto the page is a cross. The chirality center lies at the center of the cross but is not explicitly shown.



(R)-Bromochlorofluoromethane



(S)-Bromochlorofluoromethane

☞ **How do we assign priority number (1   4) to four different groups attached to stereocenter?**

- ⋆ Higher atomic number precedes lower. *e.g.*, Br > Cl > S > O > N > C > H
- ⋆ For isotopes, higher atomic mass precedes lower. *e.g.*, T > D > H.
- ⋆ If atoms have the same priority, then secondary groups attached are considered. If necessary, the process is continued to the next atom in the chain.

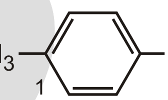
  e.g., $—CH_2 — CH_3 \quad —CH_2 — H$

  $-CH_2-CH(CH_3)_2 > -CH_2-CH_2-CH_2-CH_3$

First atom is carbon in both cases; consider the second atom: second atom is carbon in both cases; consider the next atom(s): carbon directly bonded to two further carbons has higher priority than carbon directly bonded to just one further carbon

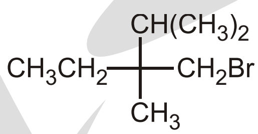
Solved Example

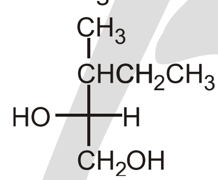
- Assign priority numbers to the following groups :
  (a) —CH_2OH — CH_3 — CH_2CH_2OH — H
  (b) —CH   O — OH — CH_3 — CH_2OH
  (c) —$CH(CH_3)_2$ — CH_2CH_2Br — Cl — $CH_2CH_2CH_2Br$
  (d) —CH CH_2 — CH_2CH_3 —  —CH_3

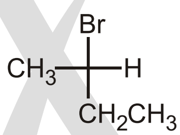
Ans. (a) —CH_2OH (1) —CH_3 (3) —CH_2CH_2OH (2) —H (4)
(b) —CH=O (2) —OH (1) —CH_3 (4) —CH_2OH (3)
(c) —$CH(CH_3)_2$ (2) —CH_2CH_2Br (3) —Cl (1) —$CH_2CH_2CH_2Br$ (4)
(d) —CH=CH_2 (2) —CH_2CH_3 (3) —  (1) —CH_3 (4)

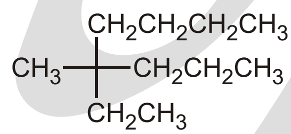
Solved Example

- Indicate whether each of the following structures has the *R* configuration or the *S* configuration.

(a) 

(b) 

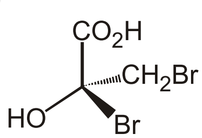
(c) 

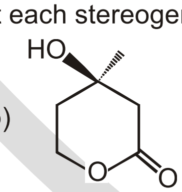
(d) 

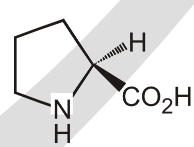
Ans. (a) *S* (b) *R* (c) *S* (d) *S*

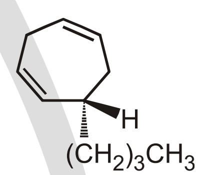
Solved Example

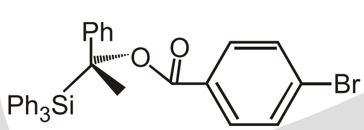
- Assign *R* / *S* configuration at each stereogenic centre in the following molecules :

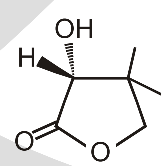
(a) 

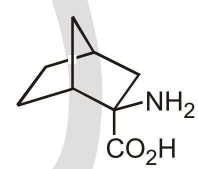
(b) 

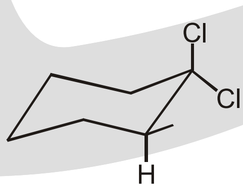
(c) 

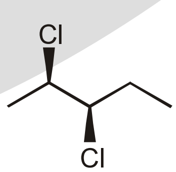
(d) 

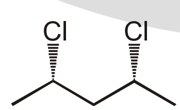
(e) 

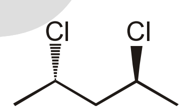
(f) 

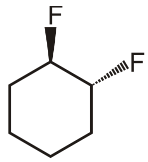
(g) 

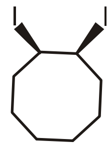
(h) 

(i) 

(j) 

(k) 

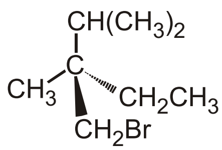
(l) 

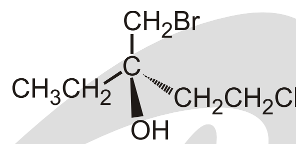
(m) 

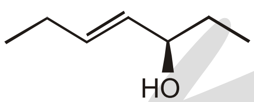
Ans. (a) *S* (b) *R* (c) *S* (d) *S*
(e) *R* (f) *R* (g) *R* (h) *S*
(i) 2*R*, 3*R* (j) 2*S*, 4*R* (k) 2*S*, 4*S* (l) 1*R*, 2*R*
(m) 1*R*, 2*S*

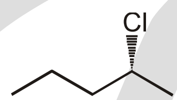
Solved Example

- Indicate whether each of the following structures has the R configuration or the S configuration:

(a) 

(b) 

(c) 

(d) 

Ans. (a) *R* (b) *R* (c) *R* (d) *R*

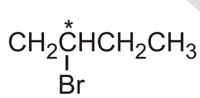
☑ SPECIAL TOPIC

ENANTIOMERS

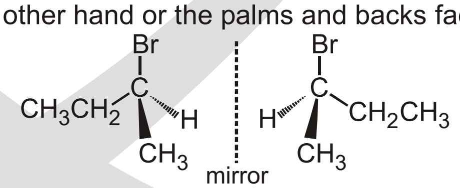
Drawing Enantiomers

Enantiomers are two compounds that are nonsuperimposable mirror images. Let's first clear up the term ''enantiomers," since students will often use this word incorrectly in a sentence. Let's compare it to people again. If two boys are born to the same parents, those boys are called brothers. Each one is the brother of the other. If you had to describe both of them, you say that they are brothers. Similarly, when you have two compounds that are non-super imposable mirror images, they are called enantiomers. Each one is the en-antiomer of the other. Together, they are a pair of enantiomers. But what do we mean by "nonsuperimposable mirror images"?

A compound with a chirality center, such as 2-bromobutane can exist as two different isomers. Because the two isomers are different, they cannot be superimposed. The two isomers are analogous to a left and a right hand. You cannot superimpose your left hand on your right hand. When you try to superimpose them, either the thumb of one hand lies on top of the little finger of the other hand or the palms and backs face opposite directions.

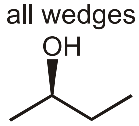


2-bromobutane

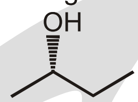


the two isomers of 2-bromobutane
a pair of enantiomers

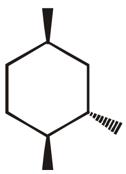
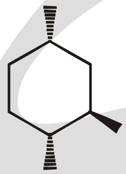
The simplest way to draw an enantiomer is to redraw the carbon skeleton, but invert all stereocenters. In other words, change all dashes into wedges and change all wedges into dashes. For example,



The compound above has a stereocenter (what is the configuration?). If we wanted to draw the enantiomer, we would redraw the compound, but we would turn the wedge into a dash:

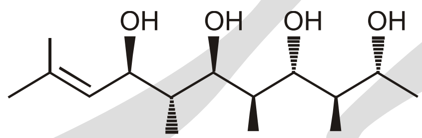


This is a pretty simple procedure for drawing enantiomers. It works for compounds with many stereocenters just as easily. For example,

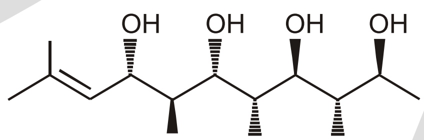
The enantiomer of  is 

Solved Example

- The enantiomer of the following compound :

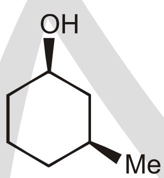


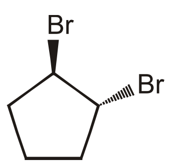
Sol. Redraw' the molecule, but invert every stereocenter. Convert all wedges into dashes, and convert all dashes into wedges :

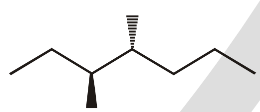


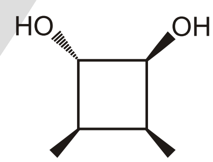
Solved Example

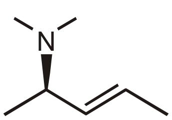
- Draw the enantiomer of each of the following compounds. OH

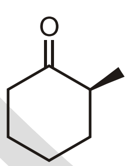
(a) 

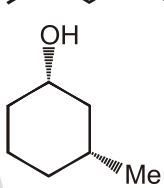
(b) 

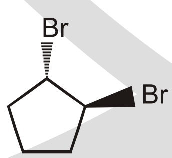
(c) 

(d) 

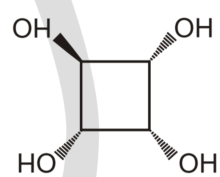
(e) 

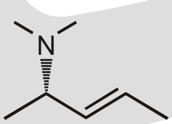
(f) 

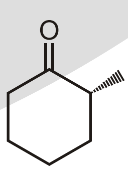
Ans. (a) 

(b) 

(c) 

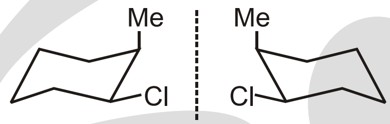
(d) 

(e) 

(f) 

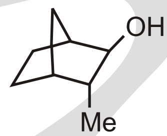
There is another way to draw enantiomers. In the previous method, we placed an imaginary mirror behind the compound, and we looked into that mirror to see the reflection. In the second method for drawing enantiomers, we place the imaginary mirror on the side of the compound, and we look into the mirror to see the reflection. Let's see an example:

But that is a lot of steps to go through when there is a simpler way to draw' the enantiomer—just put the imaginary mirror on the side (there is no need to actually draw the mirror), and draw' the enantiomer like this:

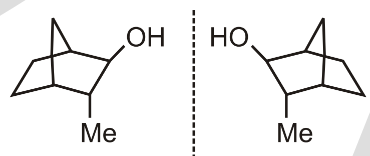


Solved Example

- Draw the enantiomer of the following compound :



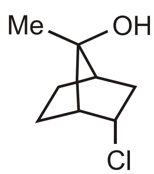
Ans. This is a rigid bicyclic system, and the dashes and wedges are not shown. Therefore, we will use the second method for drawing enantiomers. We will place the mirror on the side of the compound, and draw what would appear in the mirror:

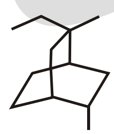


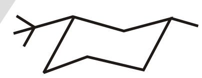
Solved Example

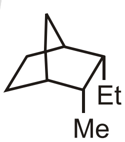
- Draw the enantiomer of each of the following compounds.

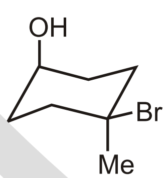
(a) 

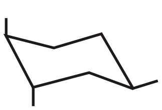
(b) 

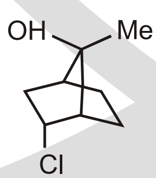
(c) 

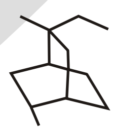
(d) 

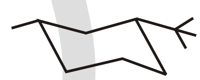
(e) 

(f) 

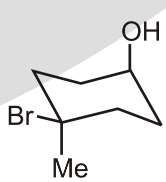
Ans. (a) 

(b) 

(c) 

(d) 

(e) 

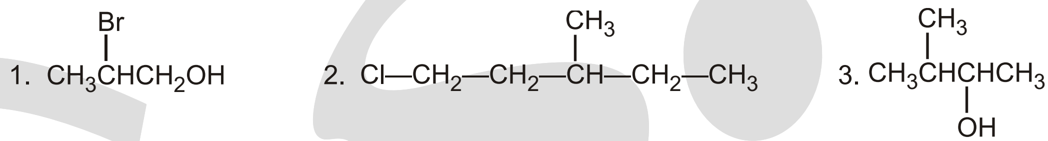
(f) 

ENANTIOMERS

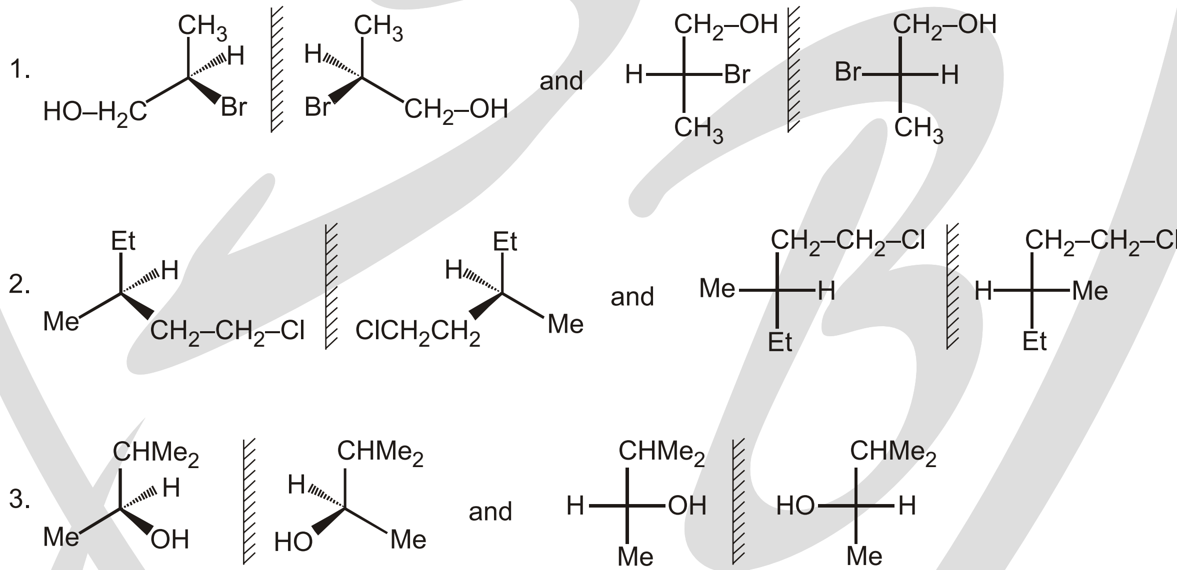
☞ Enantiomers have identical physical properties, except for the direction of rotation of the plane of polarized light.

Solved Example

▸ Draw enantiomers for each of the following compounds using :
(a) Perspective formulas (b) Fischer projections



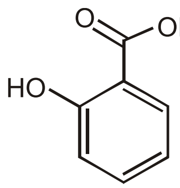
Ans.



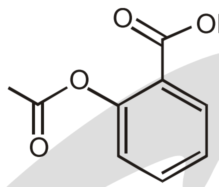
DRUGS BIND TO THEIR RECEPTORS

Many drugs exert their physiological effects by binding to specific sites, called receptors , on the surface of certain cells. A drug binds to a receptor using the same kinds of bonding interactions—van der Waals interactions, dipole–dipole interactions, hydrogen bonding—that molecules use to bind to each other. The most important factor in the interaction between a drug and its receptor is a snug fit. Therefore, drugs with similar shapes and properties, which causes them to bind to the same receptor, have similar physiological effects. For example, each of the compounds shown here has a nonpolar, planar, six-membered ring and substituents with similar polarities. They all have anti-inflammatory activity and are known as NSAIDs (non-steroidal anti-inflammatory agents).

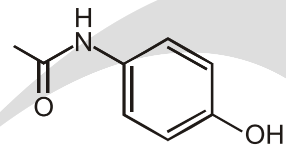
Salicylic acid has been used for the relief of fever and arthritic pain since 500 b.c. In 1897, acetylsalicylic acid (known by brand names such as Bayer Aspirin, Bufferin, Anacin, Ecotrin, and Ascriptin) was found to be a more potent anti-inflammatory agent and less irritating to the stomach; it became commercially available in 1899.



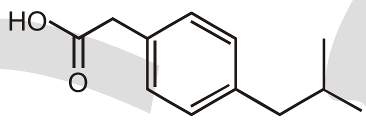
salicyclic acid



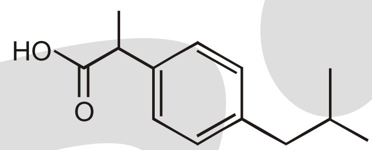
acetylsalicylic acid



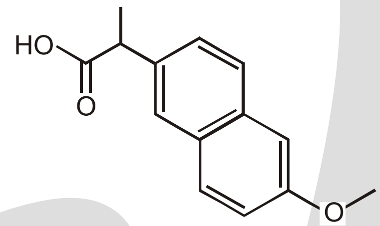
acetaminophen
(Tylenol)



ibufenac



ibuprofen
(Advil)

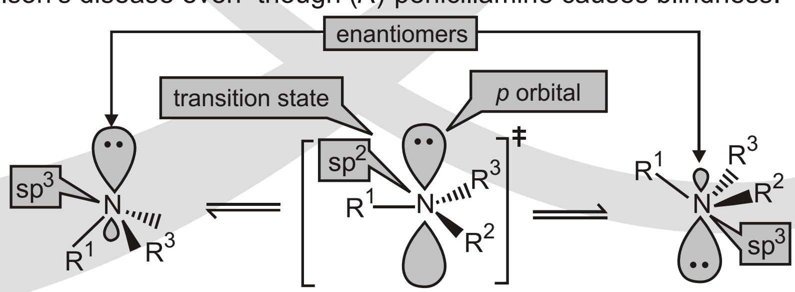


naproxen
(Aleve)

Changing the substituents and their relative positions on the ring produced acetaminophen (Tylenol), which was introduced in 1955. It became a widely used drug because it causes no gastric irritation. However, its effective dose is not far from its toxic dose. Subsequently, ibufenac emerged; adding a methyl group to ibufenac produced ibuprofen (Advil), which is a much safer drug. Naproxen (Aleve), which has twice the potency of ibuprofen, was introduced in 1976.

Chiral Drugs

Until relatively recently, most drugs with one or more asymmetric centers have been marketed as racemic mixtures because of the difficulty of synthesizing single enantiomers and the high cost of separating enantiomers. In 1992, however, the Food and Drug Administration (FDA) issued a policy statement encouraging drug companies to use recent advances in synthesis and separation techniques to develop single-enantiomer drugs. Now most new drugs sold are single enantiomers. Drug companies have been able to extend their patents by marketing a single enantiomer of a drug that was previously available only as a racemate (see page 303). If a drug is sold as a racemate, the FDA requires both enantiomers to be tested because drugs bind to receptors and, since receptors are chiral, the enantiomers of a drug can bind to different receptors (Section 6.18). Therefore, enantiomers can have similar or very different physiological properties. Examples are numerous. Testing has shown that (*S*)-(+)-ketamine is four times more potent an anesthetic than (*R*)-(–)-ketamine, and the disturbing side effects are apparently associated only with the (*R*)-(–)-enantiomer. Only the *S* isomer of the beta-blocker propranolol shows activity; the *R* isomer is inactive. The *R* isomer of Prozac, an antidepressant, is better at blocking serotonin but is used up faster than the R isomer. The activity of ibuprofen, the popular analgesic marketed as Advil, Nuprin, and Motrin, resides primarily in the (*S*)-(+)-enantiomer. Heroin addicts can be maintained with (–)- acetylmethadol for a 72-hour period compared to 24 hours with racemic methadone. This means less frequent visits to an outpatient clinic, because a single dose can keep an addict stable through an entire weekend . Prescribing a single enantiomer spares the patient from having to metabolize the less potent enantiomer and decreases the chance of unwanted drug interactions. Drugs that could not be given as racemates because of the toxicity of one of the enantiomers can now be used. For example, (*S*)-penicillamine can be used to treat Wilson's disease even though (*R*)-penicillamine causes blindness.



amine inversion

Why Are Drugs So Expensive?

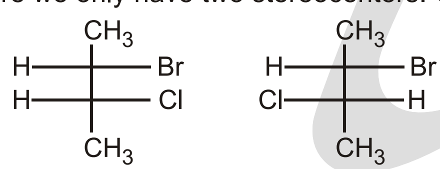
The average cost of launching a new drug is \$1.2 billion. The manufacturer has to recover this cost quickly because the patent has to be filed as soon as the drug is first discovered. Although a patent is good for 20 years, it takes an average of 12 years to bring a drug to market after its initial discovery, so the patent protects the discoverer of the drug for an average of 8 years. It is only during the eight years of patent protection that drug sales can provide the income needed to cover the initial costs as well as to pay for research on new drugs.

Why does it cost so much to develop a new drug? First of all, the Food and Drug Administration (FDA) has high standards that must be met before a drug is approved for a particular use. An important factor leading to the high price of many drugs is the low rate of success in progressing from the initial concept to an approved product. In fact, only 1 or 2 of every 100 compounds tested become lead compounds. A lead compound is a compound that shows promise of becoming a drug. Chemists modify the structure of a lead compound to see if doing so improves its likelihood of becoming a drug. For every 100 structural modifications of a lead compound, only one is worthy of further study. For every 10,000 compounds evaluated in animal studies, only 10 will get to clinical trials. Clinical trials consist of three phases. Phase I evaluates the effectiveness, safety, side effects, and dosage levels in up to 100 healthy volunteers; phase II investigates the effectiveness, safety, and side effects in 100 to 500 volunteers who have the condition the drug is meant to treat; and phase III establishes the effectiveness and appropriate dosage of the drug and monitors adverse reactions in several thousand volunteer patients. For every 10 compounds that enter clinical trials, only 1 satisfies the increasingly stringent requirements to become a marketable drug.

☑ SPECIAL TOPIC

DIASTEREOMERS

Let's start off with a simple case where we only have two stereocenters. Consider the two compounds below:



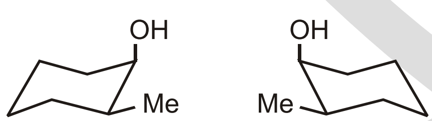
We can clearly see that they are not the same compound. In other words, they are nonsuperimposable. But, they are not mirror images of each other. The top stereocenter has the same configuration in both compounds. If they are not mirror images, then they are not enantiomers. So what is their relationship? They are called di-astereomers.

Diastereomers are any compounds that are nonsuperimposable stereoisomers that are not mirror images of each other.

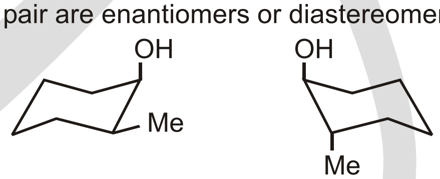
Solved Example

- For each pair compounds below, determine whether the pair are enantiomers or diastereomers.

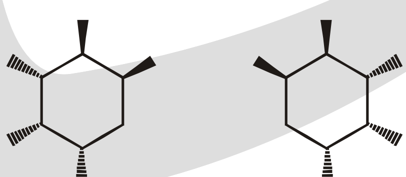
(a)



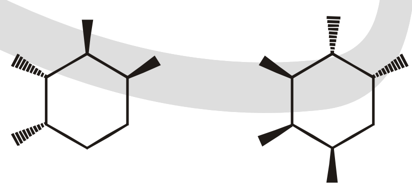
(b)

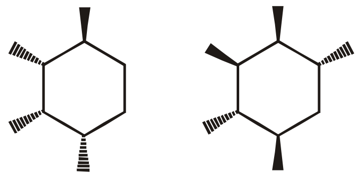


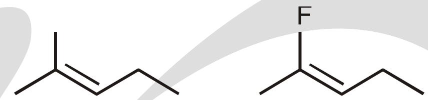
(c)



(d)

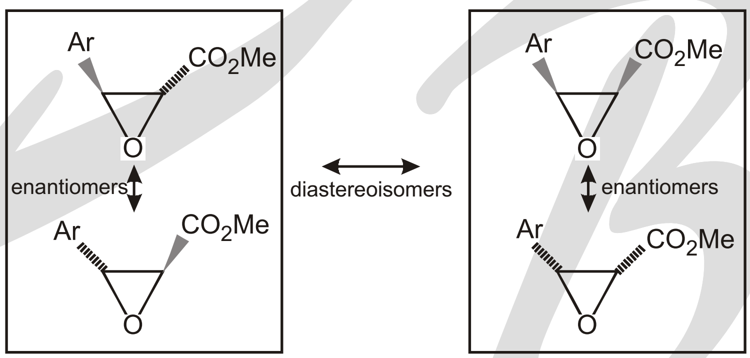


(e) 

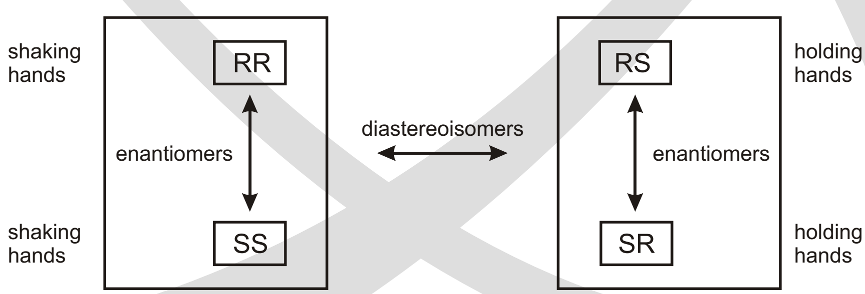
(f) 

Ans. (a) Enantiomers (b) Diastereomers (c) Enantiomers (d) Different compounds
(e) Different compound (f) Different compounds

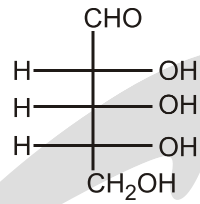
If a compound is chiral, it can exist as two enantiomers. We've just drawn the two enantiomers of each of the diastereoisomers of our epoxide. This set of four structures contains two diastereoisomers (stereoisomers that are not mirror images). These are the two different chemical compounds, the cis and trans epoxides, that have different properties. Each can exist as two enantiomers (stereoisomers that are mirror images) indistinguishable except for rotation. We have two pairs of diastereoisomers and two pairs of enantiomers. When you are considering the stereochemistry of a compound, always distinguish the diastereoisomers first and then split these into enantiomers if they are chiral.



We can illustrate the combination of two stereogenic centres in a compound by considering what happens when you shake hands with someone. Hand-shaking is successful only if you each use the same hand! By convention, this is your right hand, but it's equally possible to shake left hands. The overall pattern of interaction between two right hands and two left hands is the same: a right-handshake and a left-handshake are enantiomers of one another; they differ only in being mirror images. If, however, you misguidedly try to shake your right hand with someone else's left hand you end up holding hands. Held hands consist of one left and one right hand; a pair of held hands have totally different interactions from pair of shaking hands; we can say that holding hands is a diastereoisomer of shaking hands. We can summarize the situation when we have two hands, or two chiral centres, each one *R* or *S*.



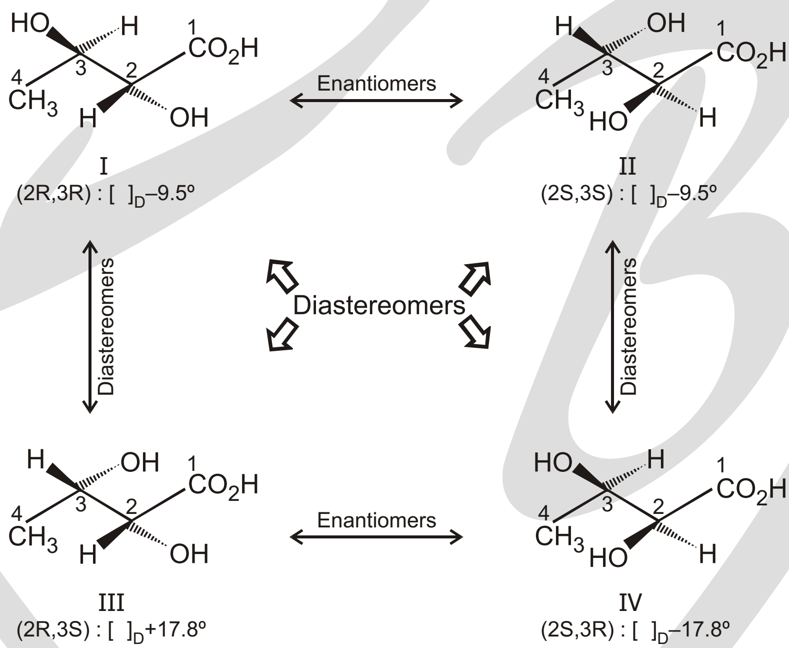
What about compounds with more than two stereogenic centres? The family of sugars provides lots of examples. Ribose is a 5-carbon sugar that contains three stereogenic centres. The enantiomer shown here is the one used in the metabolism of all living things and, by convention, is known as D-ribose. The three stereogenic centres of D-ribose have the *R* configuration. In theory we can work out how many 'stereoisomers' there are of a compound with three stereogenic centres simply by noting that there are 8 (2^3) ways of arranging *R* and *S*.



RRR	RRS	RSR	RSS
SSS	SSR	SRS	SRR

But this method blurs the all-important distinction between diastereoisomers and enantiomers. In each case, the combination in the top row and the combination directly below it are enantiomers (all three centres are inverted); the four columns are diastereoisomers. Three stereogenic centres therefore give four diastereoisomers, each a pair of two enantiomers.

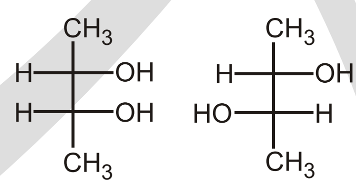
Chiral Molecules with Two Chirality Centers

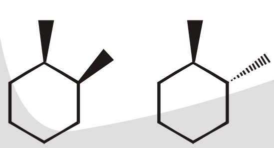


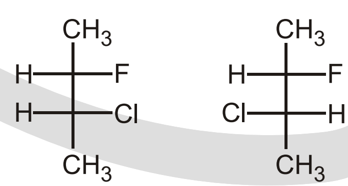
Solved Example

- Which of the following pair are diastereomers?

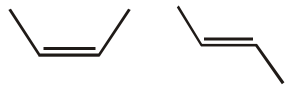
(A) cis-2-Butene,trans-2-Butene

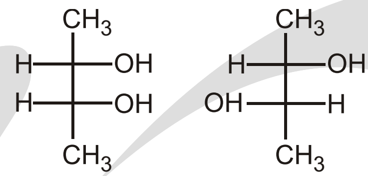
(B) 

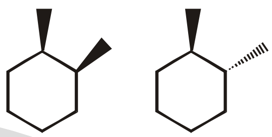
(C) 

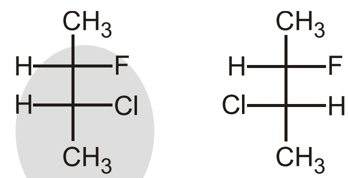
(D) 

Ans. (A, B, C, D)

Sol. (A) 

(B) 

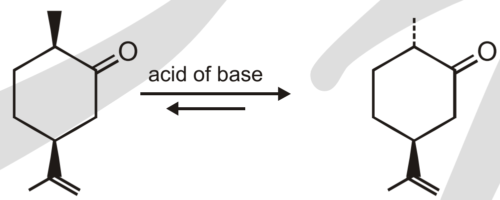
(C) 

(D) 

Diastereomers are not mirror image of each other

Interconversion of monoterpene stereoisomers through enolization

On heating with either acid or base, the monoterpene ketone **isodihydrocarvone** is largely converted into one product only, its stereoisomer **dihydrocarvone.**

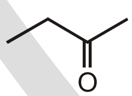


(–)-isodihydrocarvone (–)-dihydocarvone

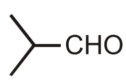
There are two chiral centres in isodihydrocarvone, but only one of these is adjacent to the carbonyl group and can participate in enolization. Under normal circumstances, we might expect to generate an equimolar mixture of two diastereoisomers. This is because two possible configurations could result from the chiral centre a to the carbonyl, whereas the other centre is going to stay unchanged.

Solved Example

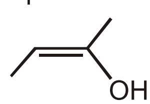
- Predict the correct relationship between given compounds :



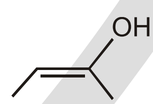
(I)



(II)



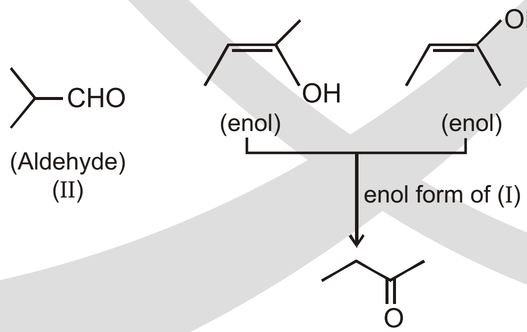
(III)



(IV)

(A) (I) and (II) are tautomers of each other.
(B) (III) and (IV) are enol form of compound (II)
(C) (II) and [(III), (IV)] are functional isomers
(D) (I) and (II) will give same enol forms.

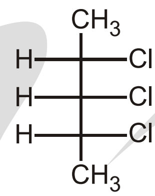
Sol. (C)



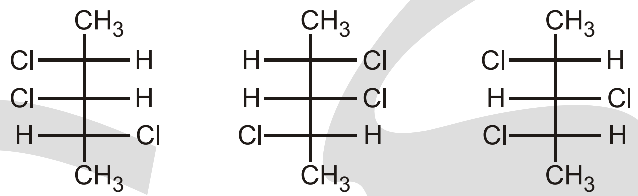
(I)

Solved Example

- Number of diastereomer of given compound :

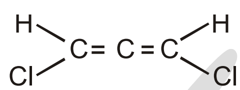
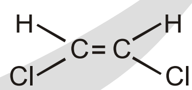
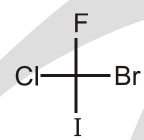
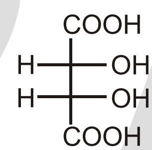


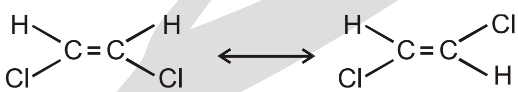
(A) 2
(B) 3
(C) 4
(D) 6

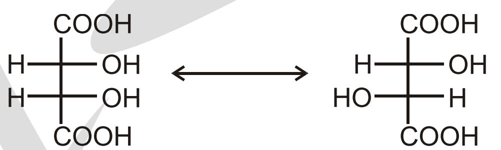
Sol.  3

Solved Example

- Which of the following can exist in diastereomeric form?

(A)  (B)  (C)  (D) 

Sol. (B)  ; diastereomers

(D)  ; diastereomers

(meso)-tartaric acid

☑ SPECIAL TOPIC

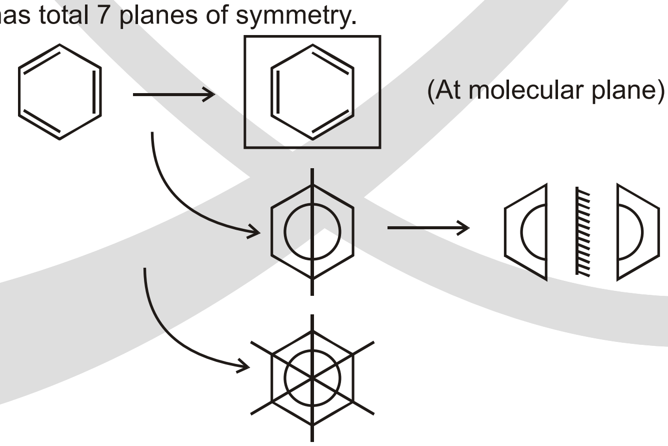
PLANE OF SYMMETRY

Plane of symmetry ( ) : An imaginary plane which bisects the molecule into two equal halves is called as plane of symmetry. It is also known as internal mirror plane.

Only one plane of symmetry in any conformer of compound is sufficient for optical inactivity.

Solved Example

- Benzene molecule has total 7 planes of symmetry.



Solved Example

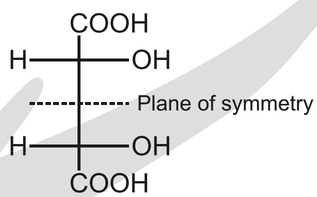
- Most of the alphabets from A to Z have a molecular plane (horizontal) and a vertical plane. The vertical plane is shown as follows :

A	B	C	D	E	F	G	H	I	J	K
✓	✓	✓	✓	✓	×	×	✓✓	✓✓	×	✓
L	M	N	O	P	Q	R	S	T	U	V
×	✓	×	✓✓	×	×	×	×	✓	✓	✓
W	X	Y	Z							
✓	✓✓	✓	×							

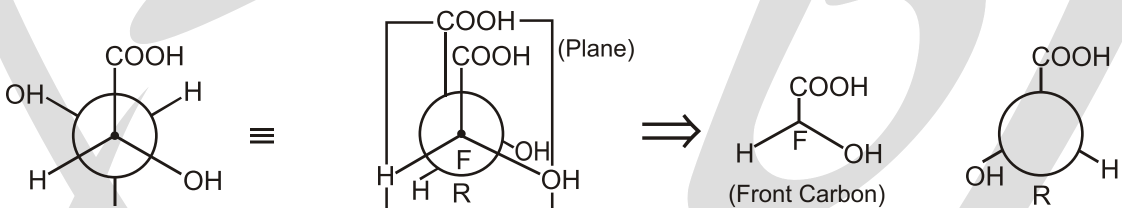
✓ One plane
✓✓ Two or more planes
× No plane

Solved Example

- Mesotartaric acid :



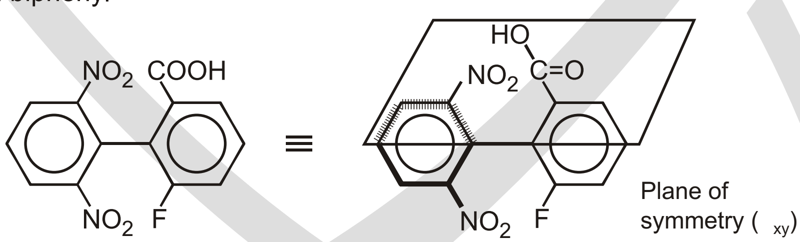
(Fischer Projection Formula)



(Newman Formula) (Front Carbon) (Rear Carbon)

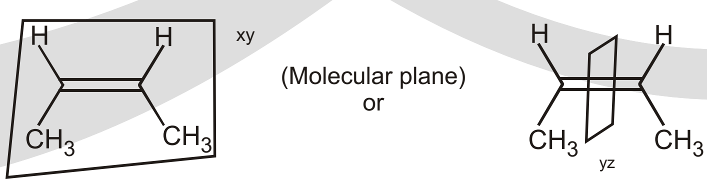
Solved Example

- A tetrasubstituted biphenyl



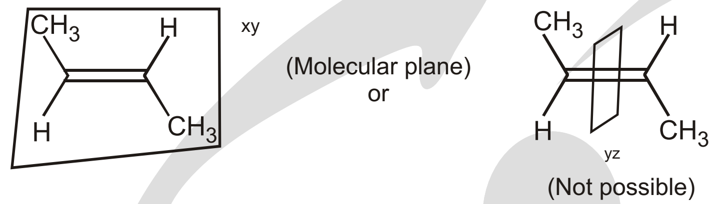
Solved Example

- Cis 2-butene



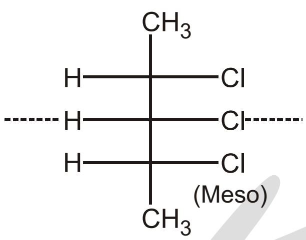
Solved Example

- Trans 2-butene



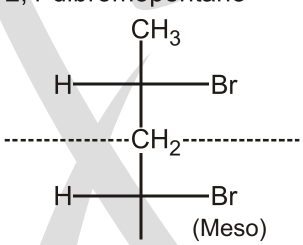
Solved Example

- 2,3,4-trichloropentane

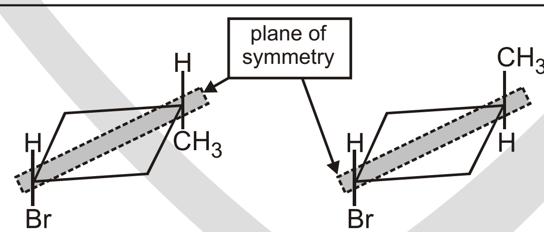


Solved Example

- 2,4-dibromopentane



Solved Example

- 

cis-1-bromo-3-methylcyclobutane   trans-1-bromo-3-methylcyclobutane

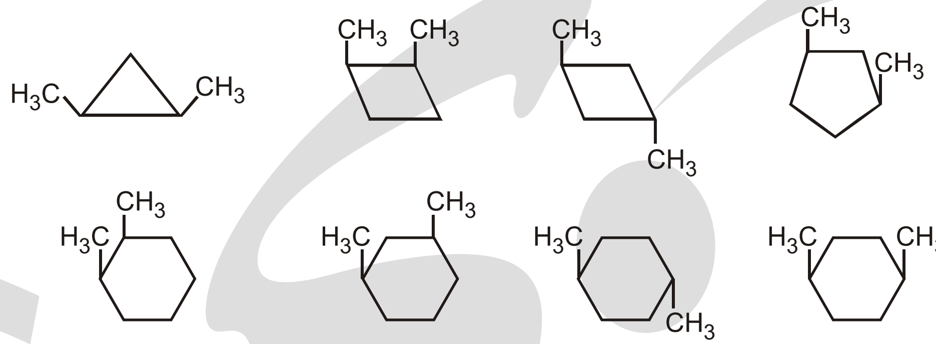
A molecule that has a nonidentical mirror image, such as either enantiomer of 2-bromobutane, is said to be **chiral** (ky-ral). A chiral molecule does not contain a plane of symmetry. A **plane of symmetry** is a plane that cuts a molecule into two halves, each of which is the mirror image of the other.

Chiral objects do not contain a plane of symmetry

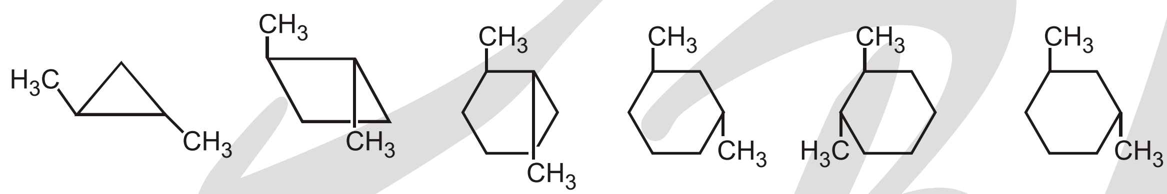
A molecule or object that does contain a plane of symmetry is said to be achiral (ayky-ral). If you cut the object in two halves along the plane of symmetry, the left half is the mirror image of the right half. A fork and a table each has a plane of symmetry, so they are achiral.

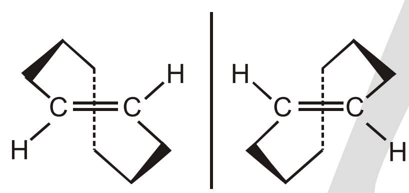
Solved Example

- Achiral structures :

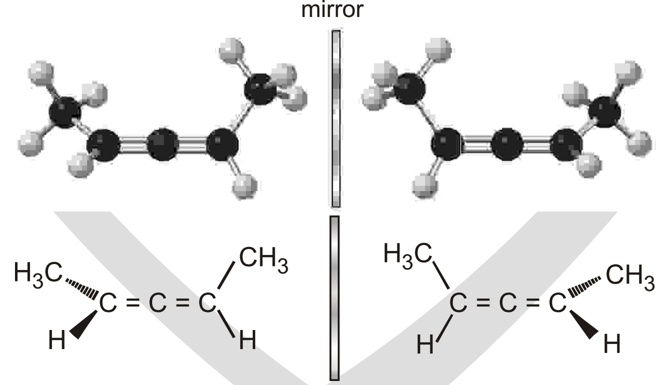


Chiral structures



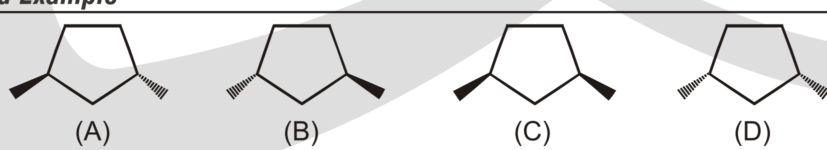


These conformations are nonsuperimposable mirror images, and they do not interconvert. They are enantiomers, and they can be separated and isolated. Each of them is optically active, and they have equal and opposite specific rotations.



enantiomers of penta-2,3-diene

Solved Example

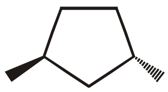
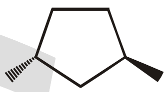
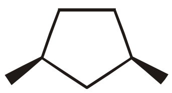
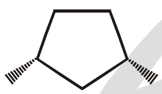


Which of the following will be optically active?

(A) Pure (A) (B) pure (C)
(C) an equal mixture of (A) and (B) (D) pure (D)

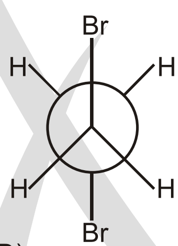
Ans. (A)

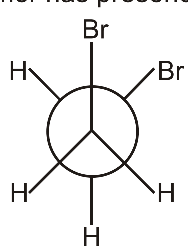
Sol.

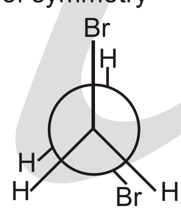
Compound	POS	optically active
 (A)	✗	✔
 (B)	✗	✔
 (C)	✔	✗
 (D)	✔	✗

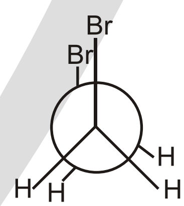
Solved Example

- Which of the following conformer has presence of plane of symmetry

(A) 

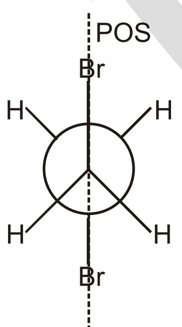
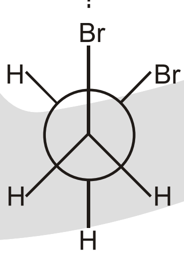
(B) 

(C) 

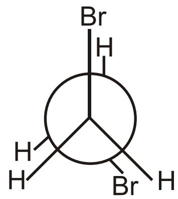
(D) 

Sol. (A,D)

Sol.

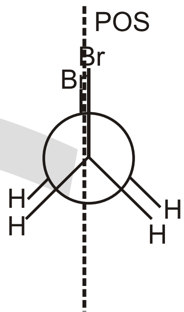
Compound	POS
(A) 	✔
(B) 	✗

(C)



✗

(D)



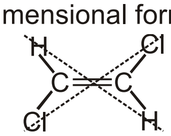
✗

☑ SPECIAL TOPIC

CENTER OF SYMMETRY (COS)

An imaginary center in a molecule through which if we draw two lines in opposite direction and they meet the same atom after the same distance (this rule should applicable for each atom in molecule). Then molecule is said to have center of symmetry it also called center of inversion (Ci).

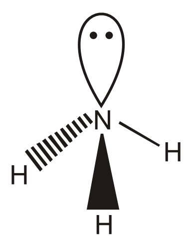
This operation is only applicable for three-dimensional formula and not for Fisher projection formula.



Point of symmetry

Solved Example

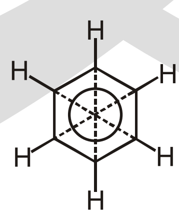
- Ammonia molecule



COS absent

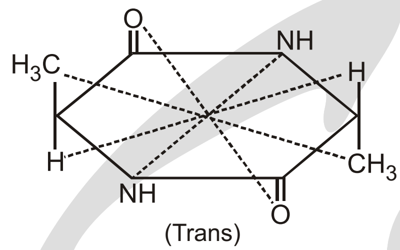
Solved Example

- Benzene molecule



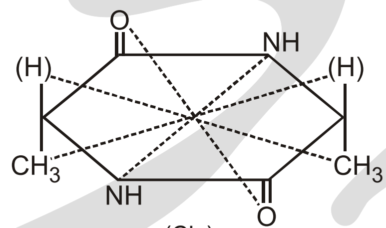
COS present

Solved Example



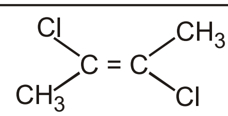
(Trans)
COS present

Solved Example



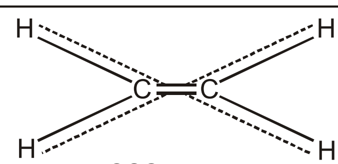
(Cis)
COS absent

Solved Example



COS present

Solved Example



COS present

☑ SPECIAL TOPIC

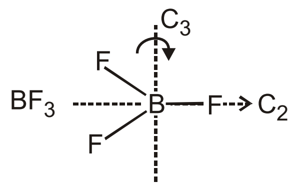
AXIS OF SYMMETRY (C_n)

An imaginary axis (passing through center) in a molecule through which if we rotate the molecule by a certain minimum angle ( ) and if molecule is again reappear than the molecule is said to have axis of symmetry

It is represented as C_n where $n \quad \frac{360}{\ }$.

AOS has nothing to do with optical activity. All objects of universe have (C_1) one AOS which is called natural axis of symmetry.

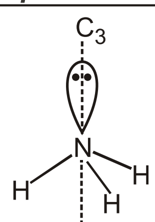
Solved Example



$n = 360/120 = 3$

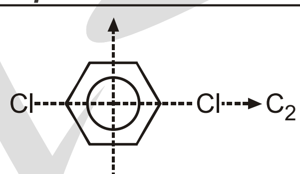
Therefore, C_n C_3 axis.

Solved Example



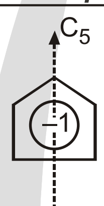
If there are two or more types of C_n axes, then the axis with higher value of n is the main axis.

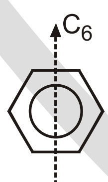
Solved Example

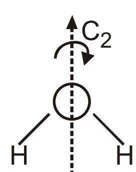


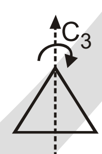
If there are two or more possible AOS with the same value of n, then preference is given to the axis that passes through more no. of atoms.

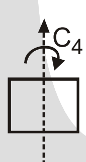
Solved Example

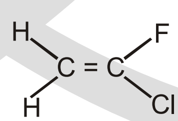




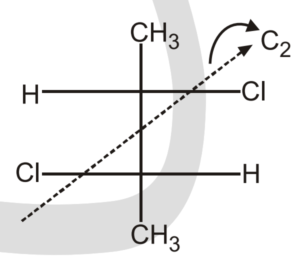




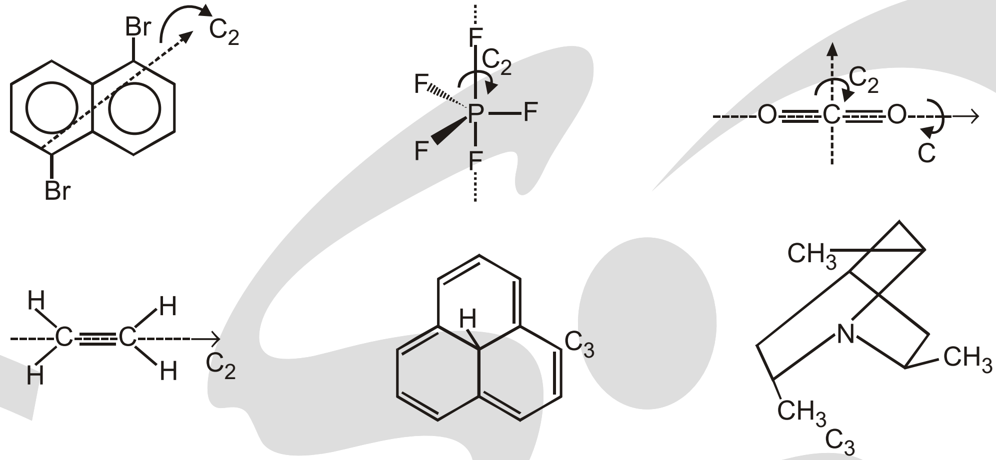




(AOS (absent)

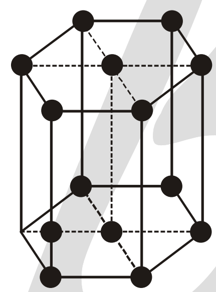
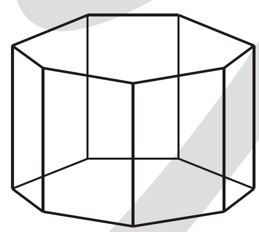
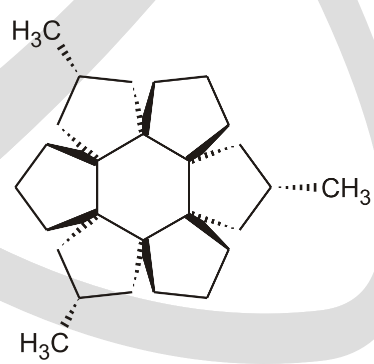


(In Fisher projection formula, AOS is visible.)



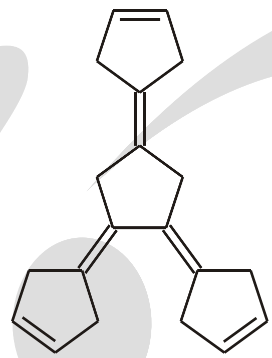
Solved Example

- **Match the Column**

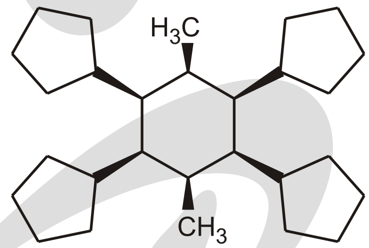
Column-I	Column-II
(A) C_2-axis of symmetry	(P) 
(B) C_3-axis of symmetry	(Q) 
(C) Plane of symmetry	(R) 

(D) Centre of symmetry

(S)

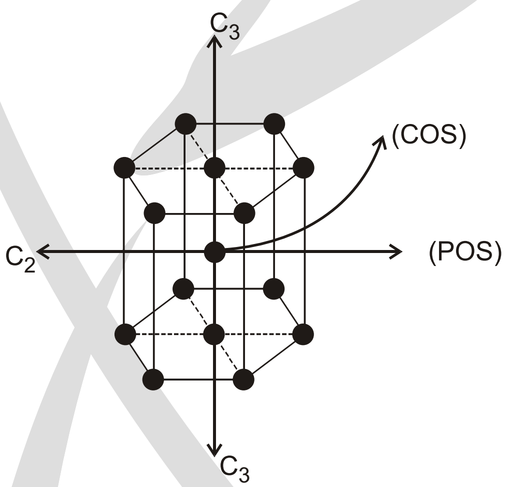


(T)



Ans. A P,Q,S,T; B PR; C PQRST; D P;

Sol.



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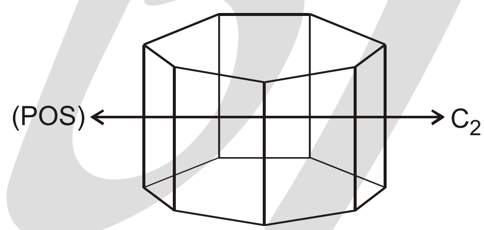
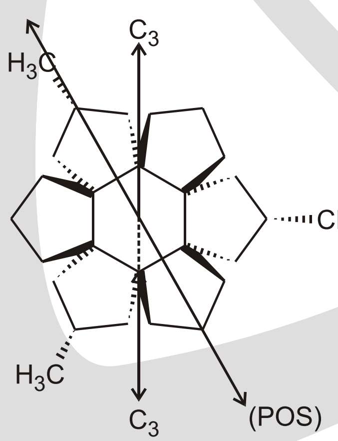
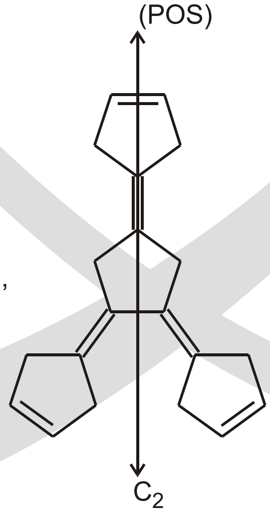
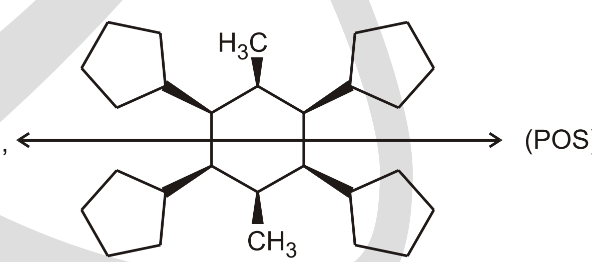




,

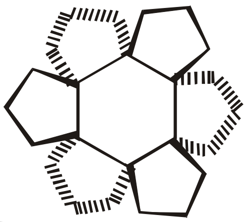
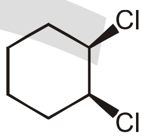
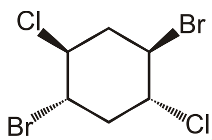
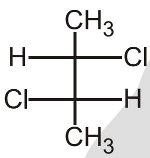


,

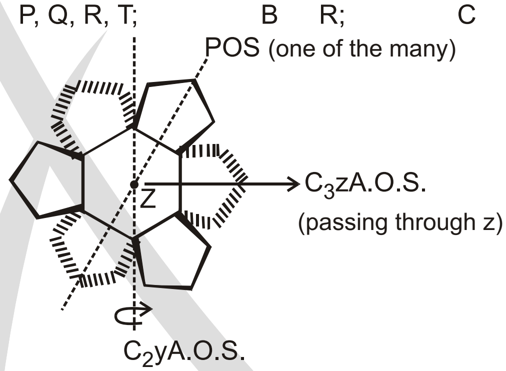


Solved Example

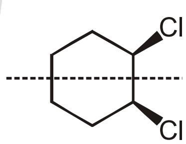
- **Match the Column**

Column-I	Column-II
(A) 	(P) Centre of symmetry
(B) 	(Q) C_2 axis of symmetry
(C) 	(R) Plane of symmetry
(D) 	(S) Optically active
	(T) C_3 axis of symmetry

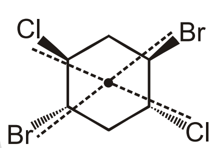
Ans. A P, Q, R, T; B R; C P; D Q, S

Sol. (A) 

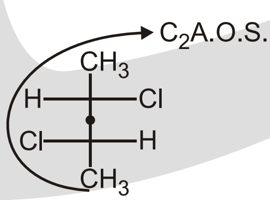
Molecule possess C.O.S. with centre z.

(B) 

P.O.S.

(C) 

C.O.S. present

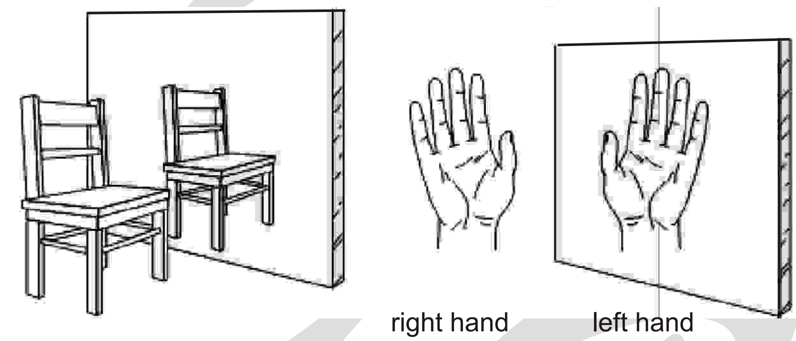
(D) 

Optically active

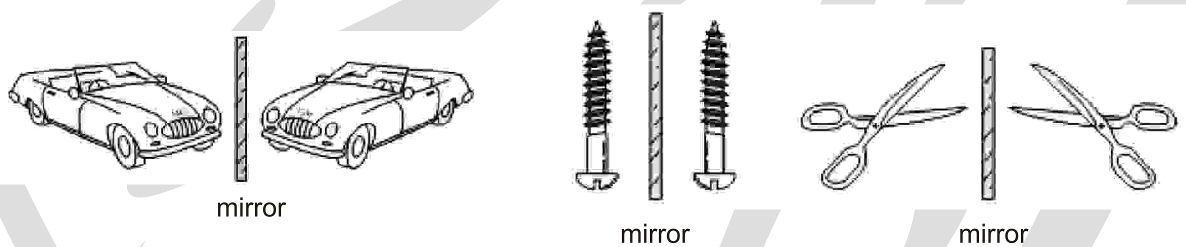
☑ SPECIAL TOPIC

CHIRALITY

The discovery of stereochemistry was one of the most important breakthroughs in the structural theory of organic chemistry. Stereochemistry explained why several types of isomers exist, and it forced scientists to propose the tetrahedral carbon atom.

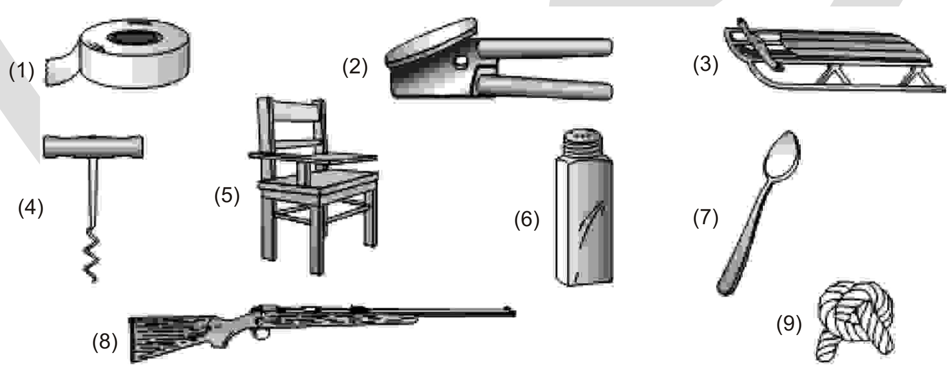


Use of a mirror to test for chirality. An object is chiral if its mirror image is different from the original object.



Common chiral objects. Many objects come in “left-handed” and “right-handed” versions.

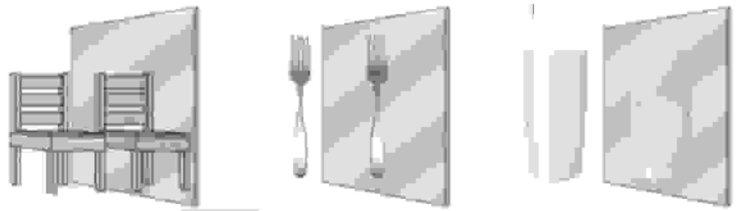
Determine whether the following objects are chiral or achiral.



(4, 6, 8 are chiral)

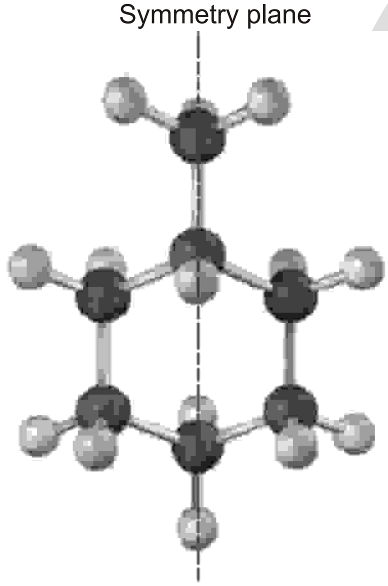
Why can’t you put your right shoe on your left foot? Why can’t you put your right glove on your left hand? It is because hands, feet, gloves, and shoes have right-handed and lefthanded forms. An object with a right-handed and a left-handed form is said to be chiral (ky-ral), a word derived from the Greek word cheir, which means “hand.” A chiral object has a nonsuperimposable mirror image. In other words, its mirror image is not the same as an image of the object itself. A hand is chiral because when you look at your right hand in a mirror, you see a left hand, not a right hand

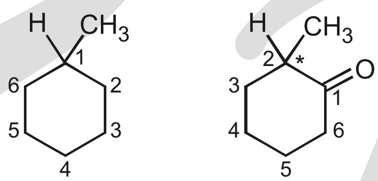
In contrast, a chair is not chiral; the reflection of the chair in the mirror looks the same as the chair itself. Objects that are not chiral are said to be achiral. An achiral object has a superimposable mirror image

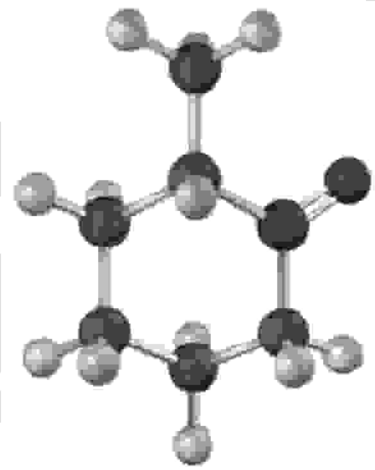


A chiral molecule has a nonsuperimposable mirror image.
An achiral molecule has a superimposable mirror image.

Symmetry plane







Methylcyclohexane (achiral) 2-Methylcyclohexanone (chiral)

► **Achiral structures are superimposable on their mirror images**

Solved Example

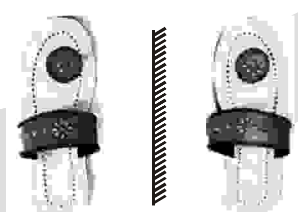
- Which of the following objects are chiral?
  (a) Soda can (b) Screwdriver (c) Screw (d) Shoe

Sol. (c, d)

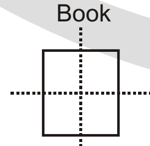
Solved Example

- Ignoring specific markings, which of the following objects are chiral ?
  (I) a shoe
  (II) a book
  (III) a pencil
  (IV) a pair of shoes (consider the pair as one object)
  (V) a pair of scissors
  (A) I only (B) I & V (C) I, IV, V (D) III, IV, V

Sol. (B)

(I) 

A shoe (chiral)

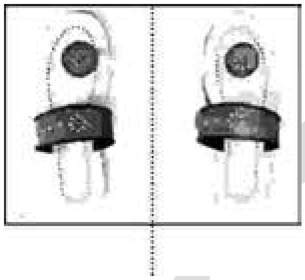
(II) 

(Achiral)

(III) 

A pencil
(Achiral)

Pair of shoe

(IV) 

Internal POS
(Achiral)

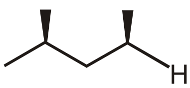
(V) 

(chiral)
(asymmetric)

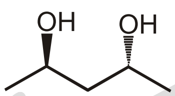
(a pair of scissors)

Solved Example

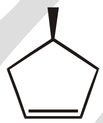
- How many compounds shown below are chiral?



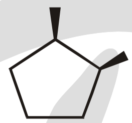
(I)



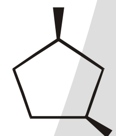
(II)



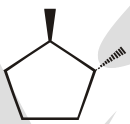
(III)



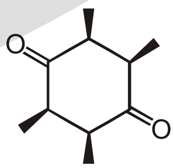
(IV)



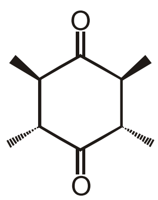
(V)



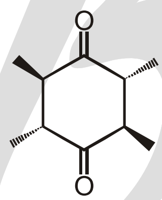
(VI)



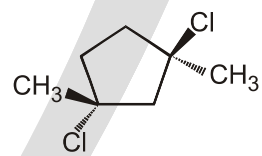
(VII)



(VIII)



(IX)

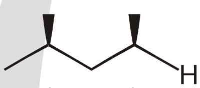


(X)

(A) 4 (B) 5 (C) 6 (D) 8

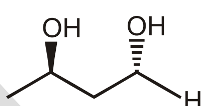
Ans. (A)

Sol. Compound which are chiral are (II), (VI), (IX), (X)



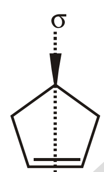
(achiral)

(I)

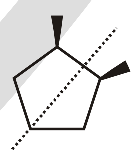


chiral
(no POS, COS)

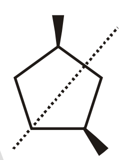
(II)



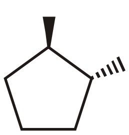
(III)



(IV)

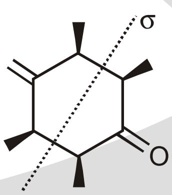


(V)

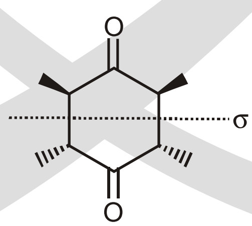


chiral
(no POS, COS)

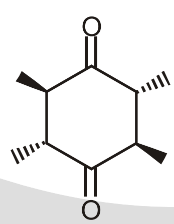
(VI)



(VII)

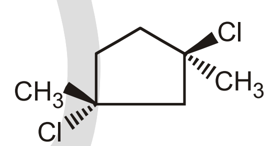


(VIII)



chiral
(no POS, COS)

(IX)

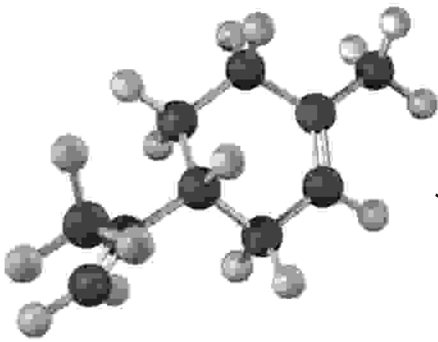


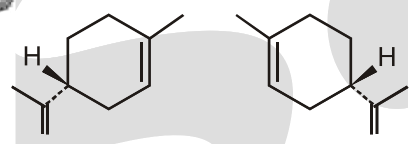
chiral
(no POS, COS)

(X)

Chirality in Nature and Chiral Environments

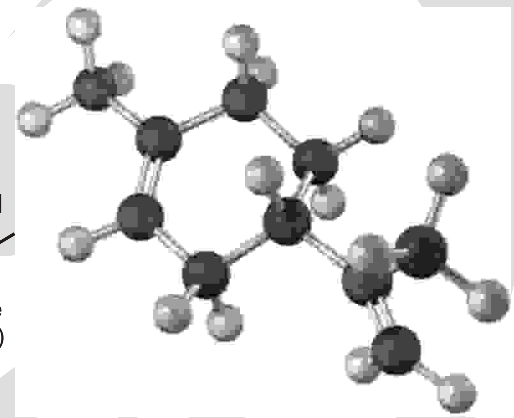
Although the different enantiomers of a chiral molecule have the same physical properties, they usually have different biological properties. For example, the (+) enantiomer of limonene has the odor of oranges and lemons, but the (–) enantiomer has the odor of pine trees.



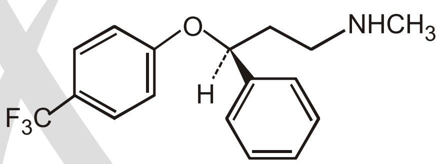


(+)-Limonene
(in citrus fruits)

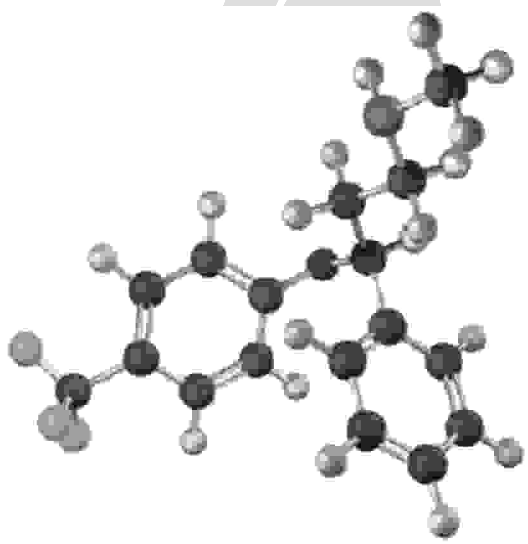
(–)-Limonene
(in pine trees)



More dramatic examples of how a change in chirality can affect the biological properties of a molecule are found in many drugs, such as fluoxetine, a heavily prescribed medication sold under the trade name Prozac. Racemic fluoxetine is an extraordinarily effective antidepressant but has no activity against migraine. The pure S enantiomer, however, works remarkably well in preventing migraine. Other examples of how chirality affects biological properties are given in A Deeper Look at the end of this chapter.



(S)-Fluoxetine
(prevents migraine)



☑ SPECIAL TOPIC

MESO COMPOUNDS

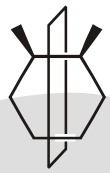
This is a topic that notoriously confuses students, so let's start off with analogy.

Now imagine that the parents, out of nowhere, have a one more child who is born without a twin—just a regular one-baby birth. When you look at this family, you would see a lot of sets of twins, and then one child who has no twin (and has tw'o moles—one on each side of his face). You might ask that child, where is your twin? Where is your mirror image? He would answer :

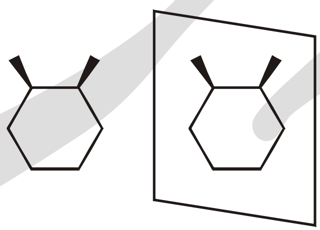
"I don't have a twin. I am the mirror image of myself. That's why the family has an odd number of children, instead of an even number."

The analogy goes like this: when you have a lot of stereocenters in a compound, there will be many stereoisomers (brothers and sisters). But, they will be paired up into sets of enantiomers (twins). Any one molecule will have many, many diastereomers (brothers and sisters), but it will have only one internal enantiomer (its mirror image twin). For example, consider the following compound :

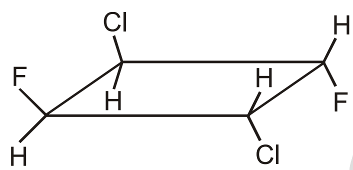
A meso compound has stereocenters, but the compound also has symmetry that allows it to be the mirror image of itself. Consider cis- l ,2-dimethylcyclohexane as an example. This molecule has a plane of symmetry cutting the molecule in half. Everything on the left side of the plane is mirrored by everything on the right side:



If a molecule has an internal plane of symmetry, then it is a meso compound. If you try to draw the enantiomer (using either one of the two methods we saw), you will find that you are drawing the same thing again. This molecule does not have a twin. It is its own mirror image :



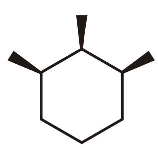
It can also happen when the compound has a center of inversion. For example,



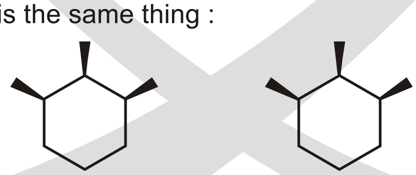
compound will be superimposable on its mirror image, and the compound is meso.

Solved Example

- Is the following a meso compound?

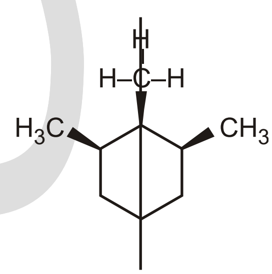


Ans. We need to try to draw the mirror image and see if it is just the same compound redrawn. If we use the second method for drawing enantiomers (placing the mirror on the side), then we will be able to see that the compound we would draw is the same thing :



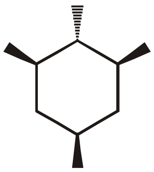
Therefore, it is a meso compound.

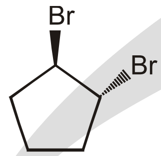
A simpler way to draw the conclusion would be to recognize that the molecule has an internal plane of symmetry that chops through the center of one of the methyl groups:



Solved Example

- Identify which of the following compounds is a meso compound.

(a) 

(b) 

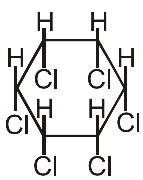
(c) 

Sol. (a, c)

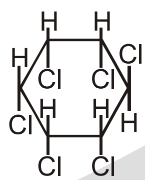
Solved Example

- Find the number of meso, enantiomer, diastereomer structures in gamexene ($C_6H_6Cl_6$)

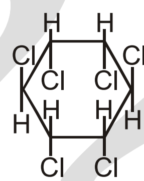
Sol.



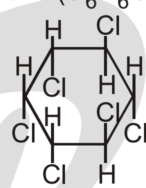
1
all cis
meso



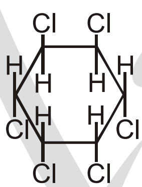
2
1-trans
meso



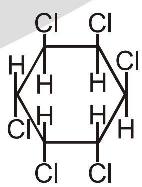
3
1,4-trans
meso



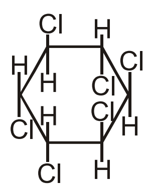
4
1,3-trans
meso



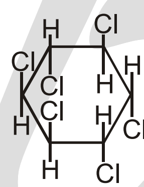
5
1,2-trans
meso



6
1,2,3-trans
meso

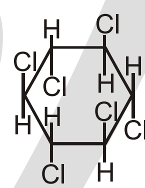


7



8

1,2,4-trans
Enantiomers

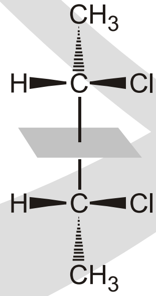


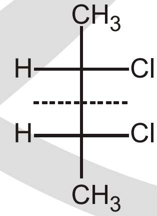
9
1,3,5-trans
meso

Ans. 7 meso, 2 enantiomer, 8 diastereomers

Final Definition of Meso Compound

A meso comound is one whose molecules are superimposable on their mirror images even though they contain chiral centers. A meso compound is optically inactive due to internal compensation (*i.e.,* cancellation) because half part of molecule rotate the PPL clockwise and other half part anticlockwise. The rotation caused by half part of molecule is cancelled by an equal and opposite rotation caused by another half part of molecule that is the mirror image of the first.

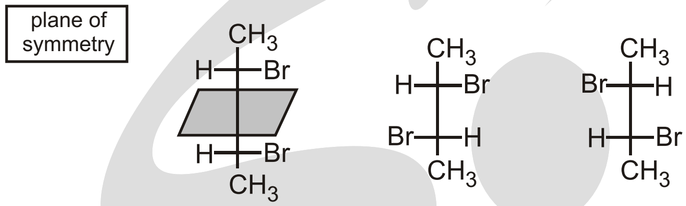




The molecule has a plane of symmetry, and cannot be chiral. (Caution : If we do not see a plane of symmetry, however, this does not necessarily mean that the molecule is chiral).

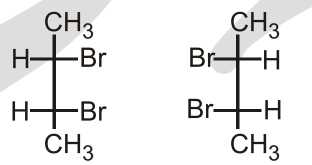
In the examples we have just seen, each compound with two chirality centers has four stereoisomers. However, some compounds with two chirality centers have only three stereoisomers. This is why we emphasized in that the maximum number of stereoisomers a compound with *n* chirality centers can have is 2^n, instead of stating that a compound with n chirality centers has 2^n stereoisomers.

An example of a compound with two chirality centers that has only three stereoisomers is 2,3-dibromobutane.



1 **2** **3**

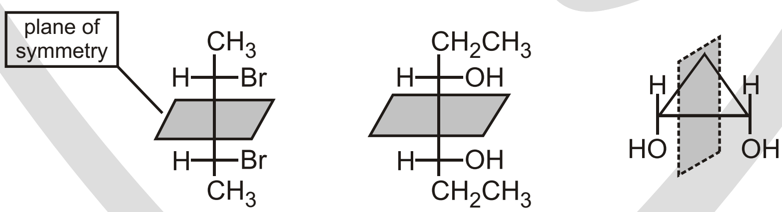
The ''missing'' stereoisomers is the mirror image of stereoisomer **1**. Stereoisomer **1** has a plane of symmetry, which means that it does not have a nonidentical mirror image. If we draw the mirror image of stereoisomer **1**, we find that it and stereoisomer **1** are identical. To convince yourself that the two structures are identical, rotate by 180° the mirror image of stereoisomer **1** that you just drew and you will see that it is identical to stereoisomer **1**. (Remember, you can move Fischer projections only by rotating them 180º in the plane of the paper.)



superimposable mirror images

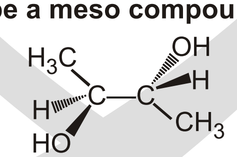
A meso compound is an achiral compound that has two or more chirality centers.

Stereoisomer **1** is called a meso compound. Even though a meso compound has chirality centers, it is an achiral molecule because it has a plane of symmetry. Mesos is the Greek word for ''middle.'' Because of the plane of symmetry, a meso compound does not rotate the plane of polarized light. It is optically inactive. A meso compound can be recognized by the fact that it has two or more chirality centers and a plane of symmetry. If a compound has a plane of symmetry, it will not be optically active even thugh it has chirality centers.

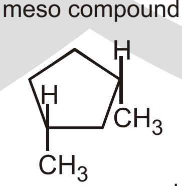


meso compounds

If a compound with two chirality centers has the same four groups bonded to each of the chirality centers, one of its stereoisomers will be a meso compound.



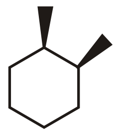
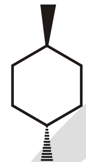
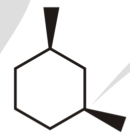
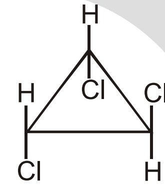
a meso compound



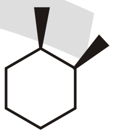
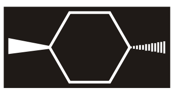
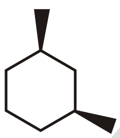
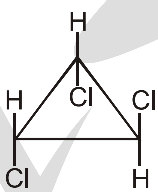
a meso compound

Solved Example

- Which of the following is a meso molecule

(A)  (B)  (C)  (D) 

Ans. (A,C,D)

	Compound	**Chiral Carbon**	**POS**
(A)		2	✔
(B)		0	✔
(C)		2	✔
(D)		–	✔

Solved Example

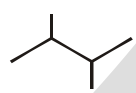
- How many compound has a stereoisomer that is a meso compound : suppose this value is x. So the value of x 7 is :

2-bromo-3-chlorobutane

(A)

2,2-dibromopropane

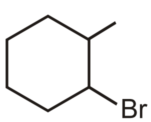
(B)



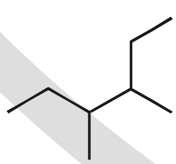
(C)

2, 3-dichlorobutane

(D)



(E)



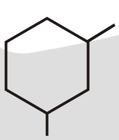
(F)

CO_2H
|
$(CH–OH)_2$
|
CO_2H
(Tartaric acid)

(G)

1, 2-dimethyl cyclohexane

(H)



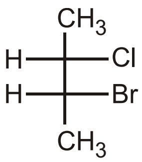
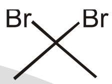
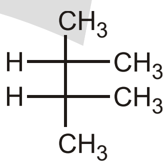
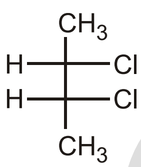
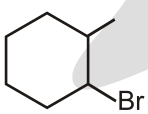
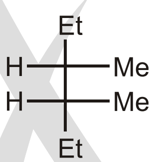
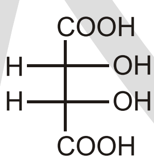
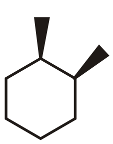
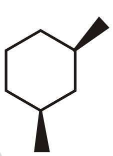
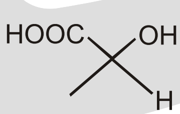
(I)

$CH_3 – \underset{\displaystyle OH}{\underset{|}{CH}} – CO_2H$
(Lactic acid)

(J)

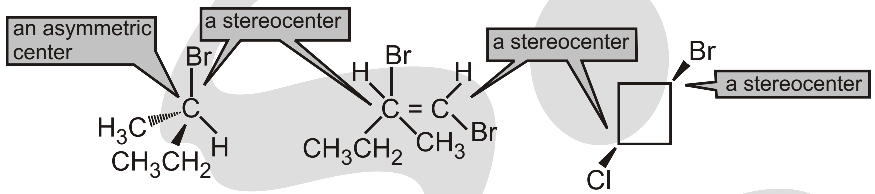
Ans. 12

Sol.

Compound		Chiral centre	POS	Meso
(A)		2	✗	✗
(B)		0	✔	✗
(C)		0	✔	✗
(D)		2	✔	✔
(E)		2	✗	✗
(F)		2	✔	✔
(G)		2	✔	✔
(H)		2	✔	✔
(I)		2	✔	✔
(J)		1	✗	✗

DIFFERENCE BETWEEN CHIRAL CENTER AND STEREOCENTER

An asymmetric center is also called a stereocenter (or a stereogenic center), but they do not mean quite the same thing. A stereocenter is an atom at which the interchange of two groups produces a stereoisomer. Thus, stereocenters include both (1) asymmetric centers, where the interchange of two groups produces an enantiomer, and (2) the sp^2 carbons of an alkene or the sp^3 carbons of a cyclic compound, where the interchange of two groups converts a cis isomer to a trans isomer or vice versa. This means that although all asymmetric centers are stereocenters , not all stereocenters are asymmetric centers.



Solved Example

- (a) How many asymmetric centers does the following compound have?
  (b) How many stereocenters does it have?

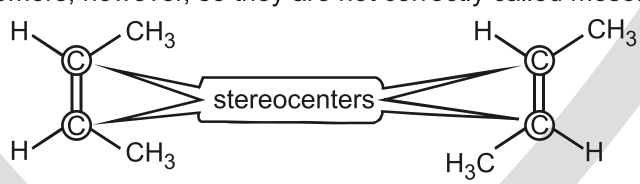
  $CH_3\underset{\substack{|\\Cl}}{C}HCH{=}CHCH_3$

Ans. (a) (1) (b) (3)

Solved Example

- One source defines a meso compound as "an achiral compound with stereocenters." Why is this a poor definition?

Sol. A stereocenter is an atom at which the interchange of two groups gives a stereoisomer. Stereocenters include both chirality centers and double-bonded carbons giving rise to cis-trans isomers. For example, the isomers of but-2-ene are achiral and they contain stereocenters (circled), so they would meet this definition. They have no chiral diastereomers, however, so they are not correctly called meso.



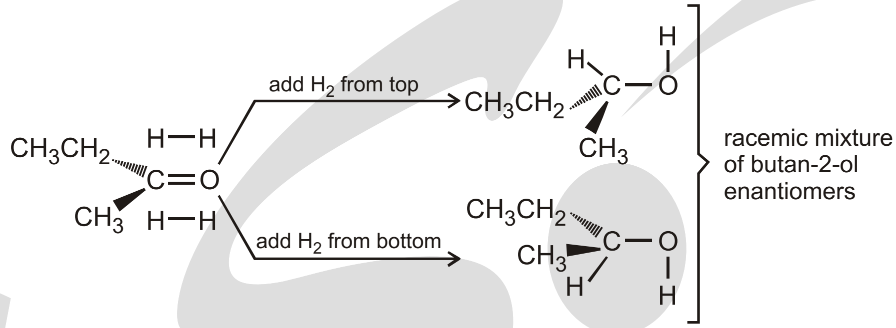
☑ SPECIAL TOPIC

RACEMIC MIXTURE

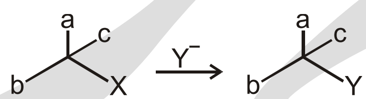
► **A racemic mixture is a mixture of two enantiomers in equal proportions. This principle is very important. Never forget that, if the starting materials of a reaction are achiral, and the products are chiral, they will be formed as a racemic mixture of two enantiomers.**

Racemic Mixture :

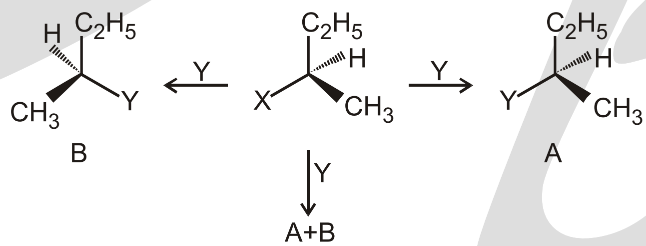
A mixture of equal amounts of a pair of enantiomers is called a racemic mixture, a racemic modification, or a racemate. Racemic mixtures do not rotate plane polarized light. They are optically inactive because for every molecule in a racemic mixture that rotates the plane of polarization in one direction, there is a mirror-image molecule that rotates the plane in the opposite direction. As a result, the light emerges from a racemic mixture with its plane of polarization unchanged.



(i) Retention: Retention of configuration is the preservation of integrity of the spatial arrangement of bonds to an asymmetric centre during a chemical reaction or transformation. It is also the configurational correlation when a chemical species XCabc is converted into the chemical species YCabc having the same relative configuration.



(ii) Inversion, retention and racemisation: There are three outcomes for a reaction at an asymmetric carbon atom. Consider the replacement of a group *X* by *Y* in the following reaction:



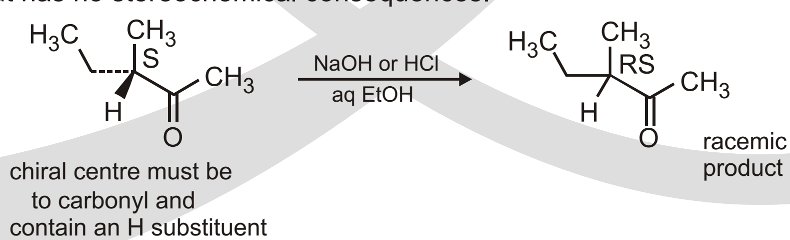
If (A) is the only compound obtained, the process is called retention of configuration.

If (B) is the only compound obtained, the process is called inversion of configuration.

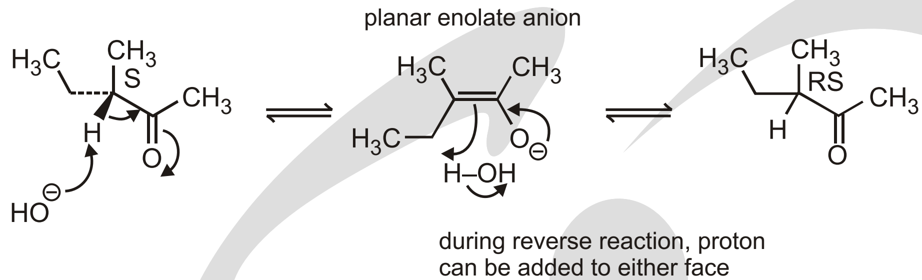
If a 50:50 mixture of the above two is obtained then the process is called recemisation and theproduct is optically inactive, as one isomer will rotate light in the direction opposite to another.

Racemization

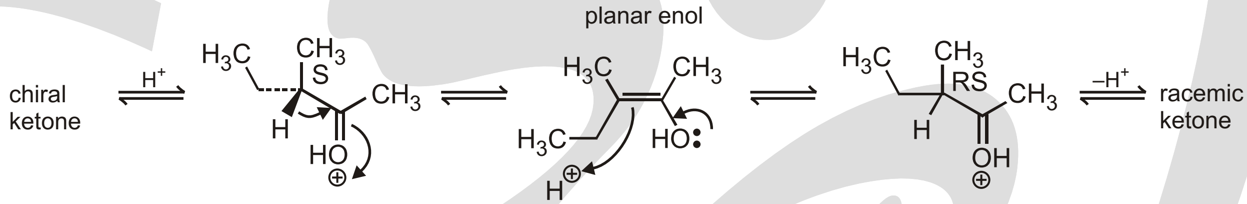
The process of hydrogen exchange shown above has implications if the a-carbon is chiral and has a hydrogen attached. Removal of the proton will generate a planar enol or enolate anion, and regeneration of the keto form may then involve supply of protons from either face of the double bond, so changing a particular enantiomer into its racemic form. Reacquiring a proton in the same stereochemical manner that it was lost will generate the original substrate, but if it is acquired from the other face of the double bond it will give the enantiomer, *i.e.,* together making a racemate. Note that removal and replacement of protons at the other a-carbon, *i.e.,* the methyl, will occur, but has no stereochemical consequences.



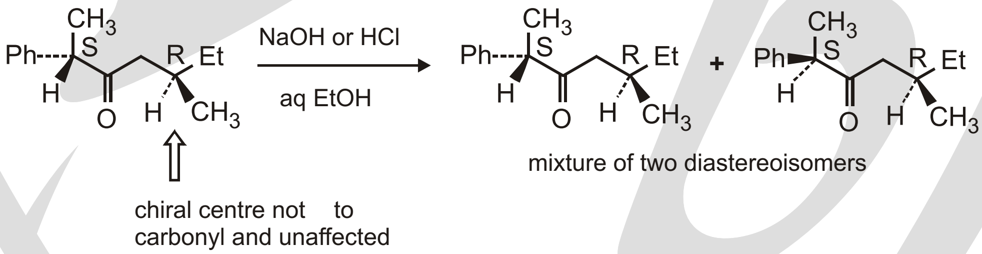
in base:



in acid:



The chiral centre must be   to the carbonyl and must contain a hydrogen substituent. If there is more than one chiral centre in the molecule with only one centre   to the carbonyl, then the other centres will not be affected by enolization, so the product will be a mixture of **diastereoisomers** of the original compound rather than the racemate.



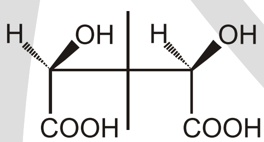
Solved Example

- Which of the following symmetry(s) are present in any conformer of meso tarteric acid ?

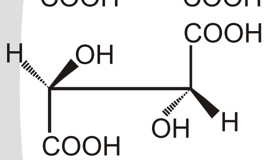
(A) POS (Plane of symmetry) (B) COS (Centre of symmetry)

(C) AAOS (Alternate axis of symmetry) (*D) All

Sol.



(POS = S_1)



(COS = S_2)

Solved Example

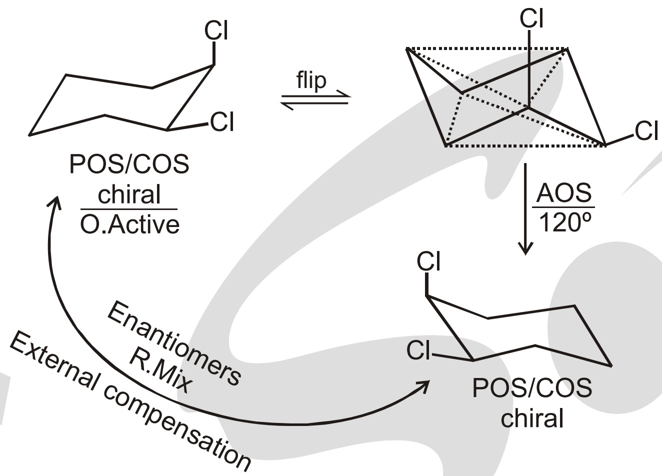
- Cis-1,2-dichloro cyclohexane is optically inactive due to

(A) Plane of symmetry (B) Internal compensation

(C) External compensation (D) All

Ans. (C)

Sol.



Be economical

When we draw organic structures we try to be as realistic as we can be without putting in superfluous detail. Look at these three pictures.

1

2

3

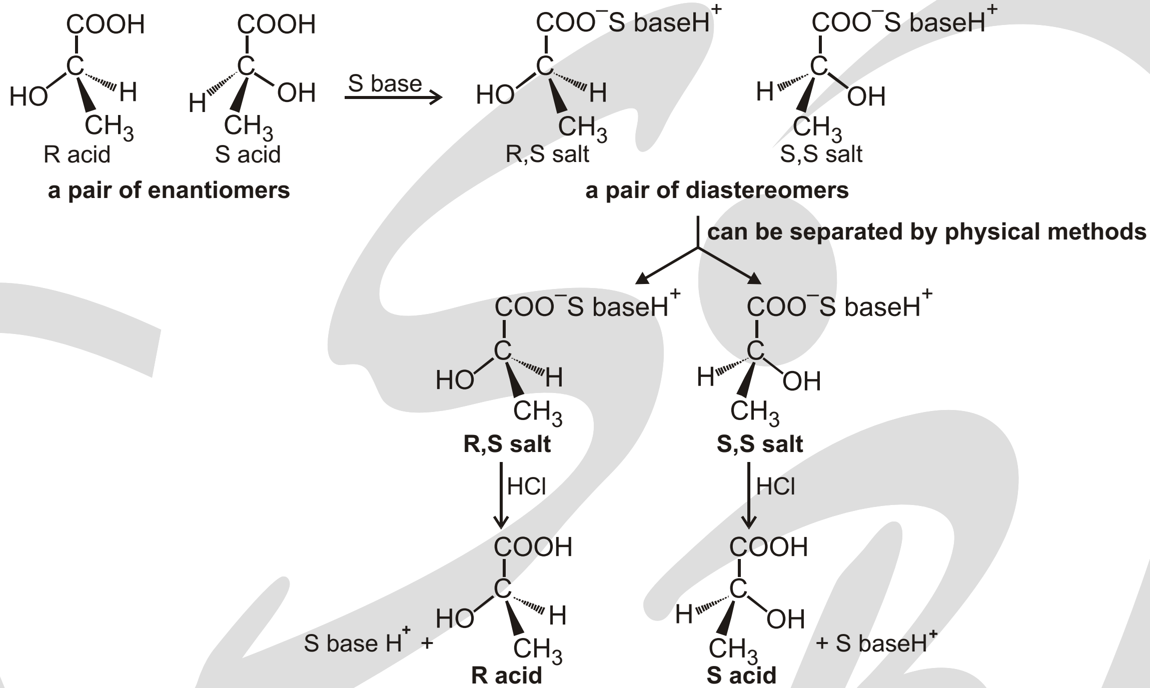
(1) is immediately recognizable as Leonardo da Vinci's Mona Lisa. You may not recognize (2)—it's also Leonardo da Vinci's Mona Lisa—this time viewed from above. The frame is very ornate, but the picture tells us as much about the painting as our rejected linear and 90° angle diagrams did about our fatty acid. They're both correct—in their way—but sadly useless. What we need when we draw molecules is the equivalent of (3). It gets across the idea of the original, and includes all the detail necessary for us to recognize what it's a picture of, and leaves out the rest. And it was quick to draw—this picture was drawn in less than 10 minutes: we haven't got time to produce great works of art!

☑ SPECIAL TOPIC

SEPARATION OF ENANTIOMERS

Enantiomers cannot be separated by the usual separation techniques such as fractional distillation or crystallization because their identcal boiling points and solubilities cause them to distill or crystallize simultaneously. Louis Pasteur was the first to separate a pair of enantiomers successfully.

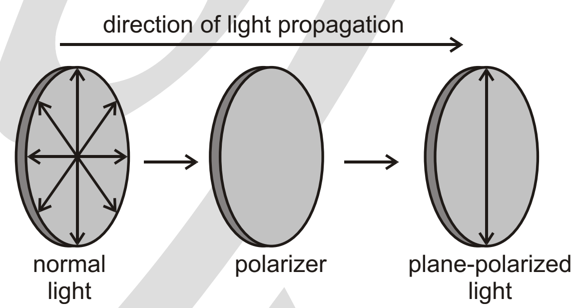
"Separation of enantiomers is called the **resolution of a racemic mixture**.



☑ SPECIAL TOPIC

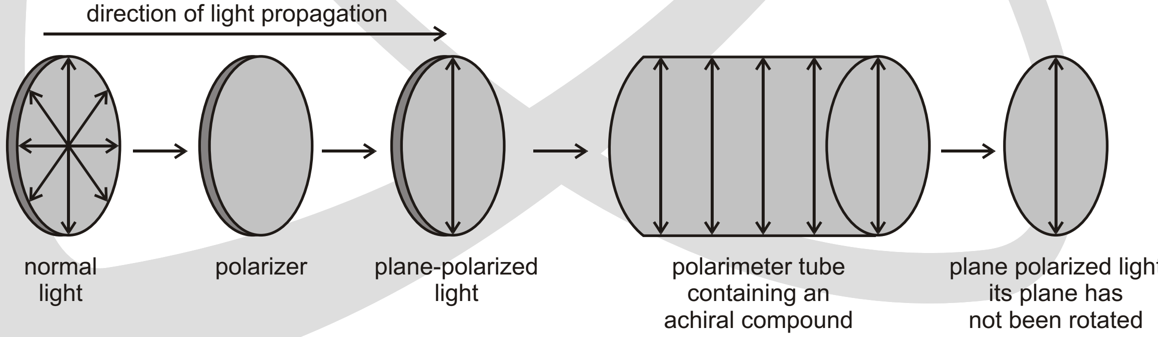
OPTICAL ROTATION

Normal light consists of electromagnetic waves that oscillate in all planes passing through the direction the light travels. Plane-polarized light oscillates only in a single plane passing through the direction the light travles. Plane-polarized light is produced by passing normal light through a polarizer such as a polarized lens or a Nicol prism.

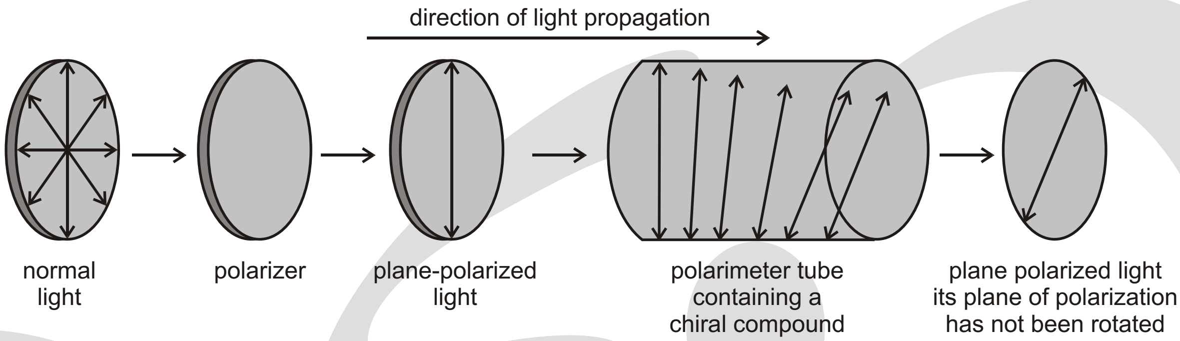


Some compounds rotated the plane of polarization in a clockwise direction and some in a counterclockwise direction, while others did not rotate the plane of polarization at all. Ability to rotate the plane of polairzation was attributable to some asymmetry that existed in the molecule.

When plane-polarized light passes through a solution of achiral molecules, the light emerges from the solution with its plane of polarization unchanged because there is no asymmetry in the molecules. An achiral compound does not rotate the plane of polarization. It is optically inactive.



If one enantiomer rotates the plane of polarization in a clock-wise direction, its mirror image will rotate the plane of polarization exactly the same amount in a counterclockwise direction.



A compound that rotates the plane of polarization is said to be **optically active**. in other words, chiral compounds are optically active and achiral compounds are **optically inactive**.

If an optically active compound rotates the plane of polarization in a clockwise direction, it is called **dextrorotatory**, indicated by (+). If an optically active compound rotates the plane of polarization in a counterclockwise direction, it is called **Laevorotatory**, indicated by (–). Dextro and levo are Latin prefixes for "to the right" and "to the left," respectively.

Do not confuse (+) and (–) with *R* and *S*. The (+) and (–) symbols indicate the direction in which an optically active compound rotates plane-polarized light, where as *R* and *S* indicate the arrangement of the groups about chirality center. Some compounds with the *R* configuration are (+) and some are (–).

The amount that an optically active compound rotates the plane of polarization can be measured with an instrument called a polarimeter.

☞ **Specific rotation**

Since optical rotation of the kind we are interested in is caused by individual molecules of the active compound, the amount of rotation depends upon how many molecules the light encounters in passing through the tube.

Specific rotation is the number of degrees of rotation observed if a 1-dm (10-cm) tube is used, and the compound being examined is present to the extent of 1 g/mL. This is usually calculated from observations with tubes of other lengths and at different concentrations by means of the equation

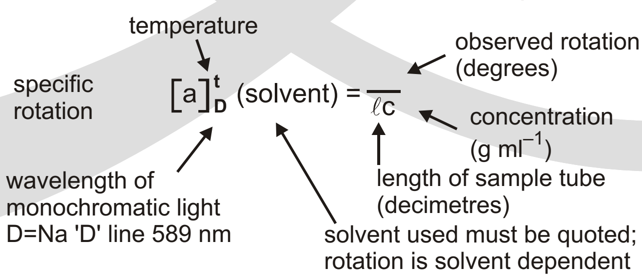
$$[\quad]_D^T \quad \frac{}{l \quad d}$$

$$\text{specific rotation} \quad \frac{\text{observed rotation (degrees)}}{\text{length (dm)} \quad \text{g / mL}}$$

where *d* represents density for a pure liquid or concentration for a solution and $[\quad]_D^T$ is specific rotation at constant temperature and D-Line of sodium.

Optically active compounds are those compounds which are not superimposable on their mirror images. One method to identify optically active compounds is to make separate models of the molecule and its mirror image, and check the superimposition of the molecule on its mirror image.

☞ **Optical activity** is the ability of a compound to rotate the plane of polarized light. This property arises from an interaction of the electromagnetic radiation of polarized light with the unsymmetric electric fields generated by the electrons in a chiral molecule. The rotation observed will clearly depend on the number of molecules exerting their effect, *i.e.*, it depends upon the concentration. Observed rotations are thus converted into specific rotations that are a characteristic of the compound according to the formula below.



- **Enantiomers** have equal and opposite rotations. The (+) – or **dextrorotatory** enantiomer is the one that rotates the plane of polarization clockwise (as determined when facing the beam), and the (–)- or **laevorotatory** enantiomer is the one that rotates the plane anticlockwise. In older publications, *d* and *l* were used as abbreviations for dextrorotatory and laevorotatory respectively, but these are not now employed, thus avoiding any possible confusion with D and L.
- **Ibuprofen** is an interesting case, in that the (S)-(+)-form is an active analgesic, but the (R)-(–)-enantiomer is inactive. However, in the body there is some metabolic conversion of the inactive (R)-isomer into the active (S)-isomer, so that the potential activity from the racemate is considerably more than 50%. Box 10.11 shows a mechanism to account for this isomerism.

  There are two approaches to producing drugs as a single enantiomer. If a synthetic route produces a racemic mixture, then it is possible to separate the two enantiomers by a process known as resolution (see Section 3.4.8). This is often a tedious process and, of course, half of the product is then not required. The alternative approach, and the one now favoured, is to design a synthesis that produces only the required enantiomer, *i.e.*, a chiral synthesis.

Solved Example

- A solution prepared by mixing 10 mL of a 0.10 M solution of the *R* enatiomer of a compound and 30 mL of a 0.10 of a 0.10 M solution of the *S* enantiomer was found to have an observed specific rotation of + 4.8. What is the specific rotation of each of the enantiomers? (**Hint :** mL × M = millimole, abbreviated as mmol)

Sol. One mmol (10 mL × 0.10 M) of the *R* enantiomer is mixed with 3 mmol (30 mL × 0.10 M) of the S enantiomer ; 1 mmol of the *R* enantiomer plus 1 mmol of the *S* enantiomer will form 2 mmol of a racemic mixture, so there will be 2 mmol of *S* enantiomer left over. Because 2 out of 4 mmol is excess *S* enantiomer, the solution has a 50% enantiomeric excess. Knowing the enantiomeric excess and the observed specific rotation allows us to calculate the specific rotation

$$\text{enantiomeric excess} \quad \frac{\text{observed specific rotation}}{\text{specific rotation of the pure enantiomer}} \quad 100\%$$

$$50\% \quad \frac{4.8}{x} \quad 100\%$$

$$\frac{50}{100} \quad \frac{4.8}{x}$$

$$\frac{1}{2} \quad \frac{4.8}{x}$$

$$x \quad 2(\ 4.8)$$

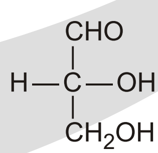
$$x \quad 9.6$$

The *S* enantiomer has a specific rotation of + 9.6, so the *R* enantiomer has a specific rotation of – 9.6.

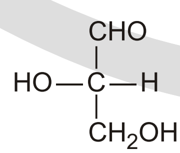
☑ SPECIAL TOPIC

D, L-CONFIGURATION

Another example is :



(+) Glyceraldehyde

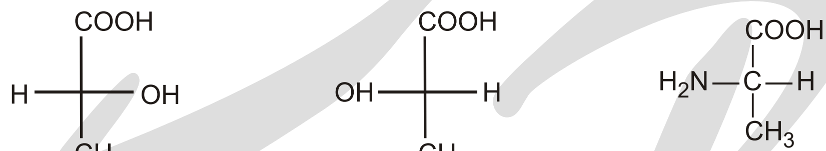


(–) Glyceraldehyde

In a molecule, the configuration is the attachment of various groups in space. Here, we will describe a method of specifying the relative spatial position of the four groups attached to a chiral carbon. An optically active compound may have a *D*– or *L*– configuration. In any configuration, in which at the chiral carbon, the –OH or any such group is on the right and H atom is on the left-hand side and the more oxidised carbon atom at the top and the less oxidised carbon at the bottom (but the chiral carbon should be next to less oxidised carbon which is written at the bottom) is assigned a *D* configuration. The configuration involving H and OH at reverse positions is assigned *L* configuration. For example, consider glyceraldehyde; the two configurations are

D(+) Glyceraldehyde L(–) Glyceraldehyde

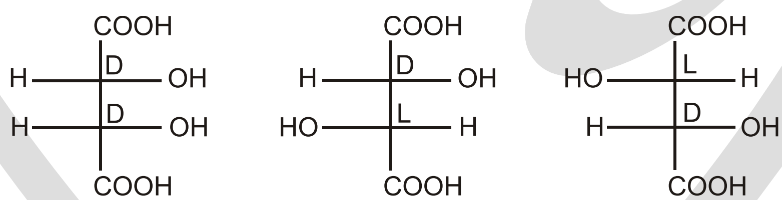
It is not always that the *D* configuration is (+) rotatory. In lactic acid, the *D* configuration is (–) laevo-rotatory and the terms *D* and *d* are different from each other.



D(–) Lactic Acid L(+) Lactic Acid or L-Serine (Amino acid)

Any assymetric compound prepared or derived form *D*-glyceraldehyde will have *D*-configuration. Similar is the case with *L*-glyceraldehyde.

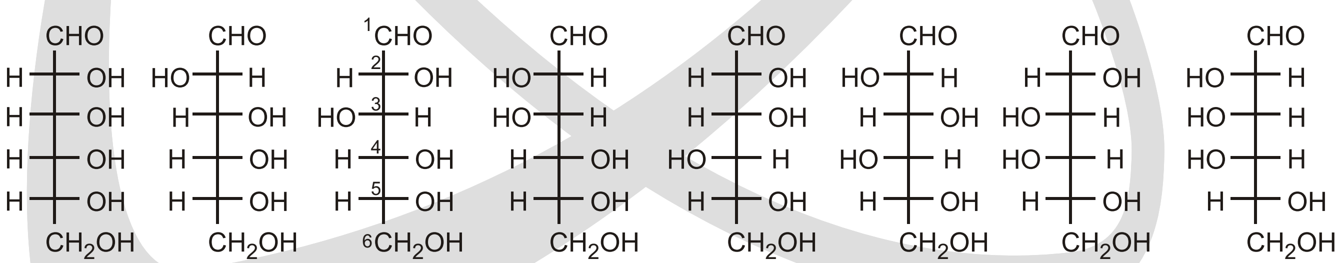
If the molecule has more than one asymmetric atom, there can be different configurations at different chiral carbon.



This, however, is an older method of designating the configuration of enantiomers.

Fischer projections of glucose and stereoisomers

The sugar glucose has four chiral centres; therefore, 2^4 16 different stereoisomers of this structure may be considered. These are shown below as Fischer projections.

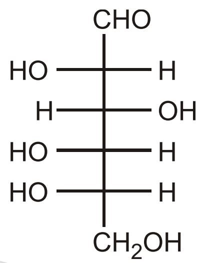


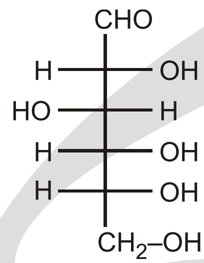
D-(+)-allose D-(+)-altrose D-(+)-glucose D-(+)-mannose D-(–)-gulose D-(+)-idose D-(+)-galactose D-(+)-talose (C-4 epimer of D-glucose)

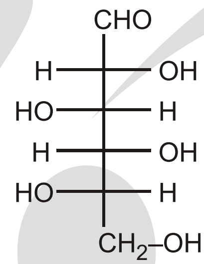
❑ **NOTE :** Mirror images of the above 8 *D*-isomers are *L*-isomers.

Solved Example

- Which of the following is the structure of L-Glucose?

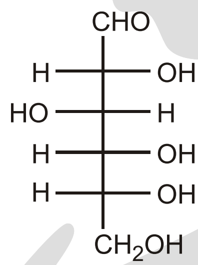
(A) 

(B) 

(C) 

(D) None of these

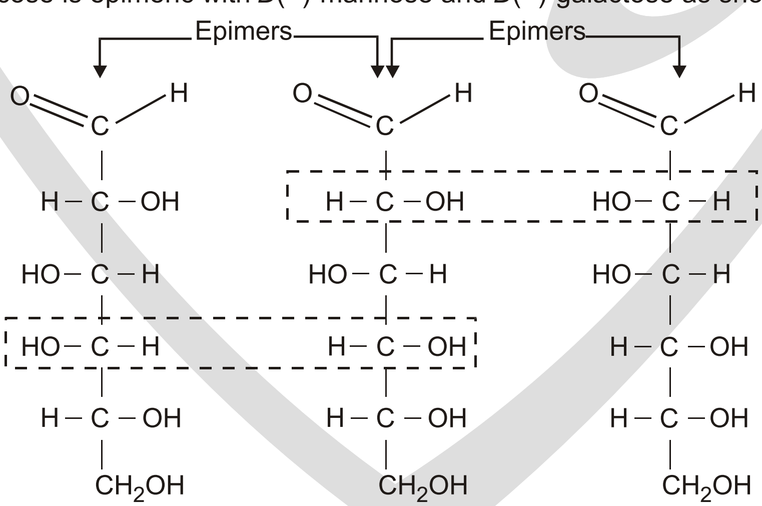
Sol. (A) *L*-Glucose is enantiomer of *D*-Glucose.

D-Glucose is 

☑ SPECIAL TOPIC

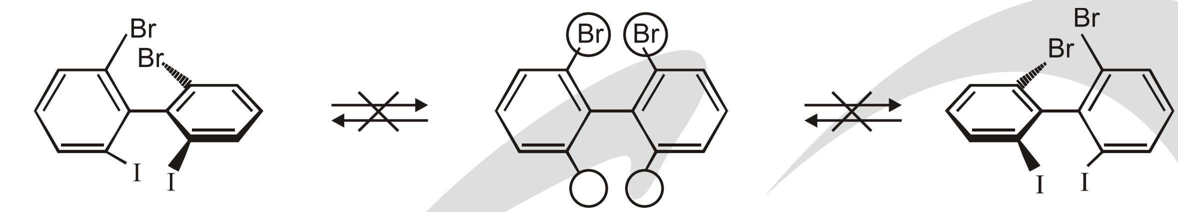
EPIMERS

Epimers A pair of diastereomers that differ only in the configuration about of a single carbon atom are said to be epimers. *D*(+)-glucose is epimeric with *D*(+)-mannose and *D*(+)-galactose as shown below.



STEREOCHEMISTRY OF BIPHENYL

Three conformations of a sterically crowded derivative of biphenyl. The center drawing shows the molecule in its most symmetric conformation. This conformation is planar, and it has a mirror plane of symmetry. If the molecule could achieve this conformation, or even pass through it for an instant, it would not be optically active. This planar conformation is very high in energy, however, because the iodine and bromine atoms are too large to be forced so close together. The molecule is conformationally locked. It can exist only in one of the two staggered conformations shown on the left and right.



staggered conformation (chiral) | eclipsed conformation (symmetric, achiral) | staggered conformation (chiral)

HOW TO FIND TOTAL STEREOISOMER

Stereochemistry of Molecules with Two or More Asymmetric Carbons

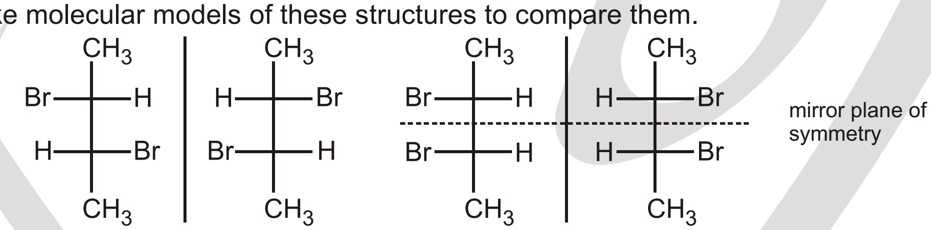
In the preceding section, we saw there are four stereoisomers (two pairs of enantiomers) of 2-bromo-3-chlorobutane. These four isomers are simply all the permutations of (*R*) and (*S*) configurations at the two asymmetric carbon atoms, C2 and C3:

diastereomers

(2R, 3R) (2S, 3S) enantiomers | (2R, 3S) (2S, 3R) enantiomers

A compound with *n* asymmetric carbon atoms might have as many as has 2^n stereoisomers. This formula is called the **2^n rule**, where *n* is the number of chirality centers (usually asymmetric carbon atoms). The 2^n rule suggests we should look for a maximum 2^n of stereoisomers. We may not always find 2^n isomers, especially when two of the asymmetric carbon atoms have identical substituents.

2,3-Dibromobutane has fewer than 2^n stereoisomers. It has two asymmetric carbons (C2 and C3), so the 2^n rule predicts a maximum of four stereoisomers. The four permutations of (*R*) and (*S*) configurations at C2 and C3 are shown next. Make molecular models of these structures to compare them.

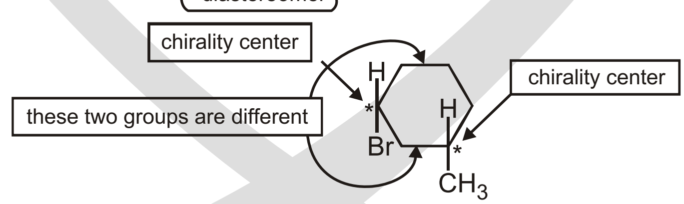


(2R, 3R) (2S, 3S) (2R, 3S) (2S, 3R)

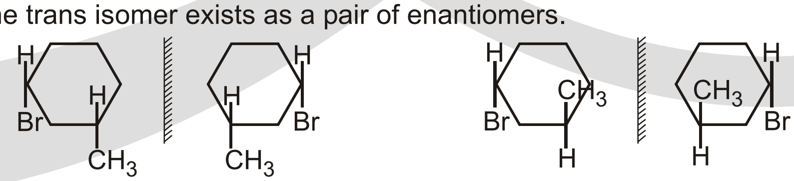
enantiomers
the (±) diastereomer

same compound!
the meso diastereomer

diastereomer



Because the compound has two chirality centers, it has four stereoisomers. The cis isomer exists as a pair of enantiomers, and the trans isomer exists as a pair of enantiomers.



cis-1-bromo-3-methylexane and trans-1-bromo-3-methylexane

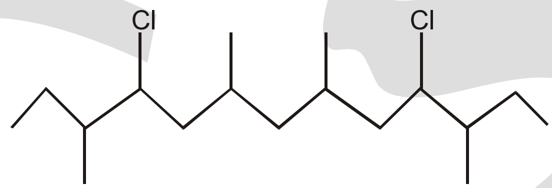
Calculation of total number of optical isomerism

(a) For unsymmetrical compound, total O.I. 2^n (where 'n' is number of chiral center)

(b) For symmetrical compound, total O.I.

No. of Chiral centre n	Optically active	Meso	Total optical isomerism
Even no. (n)	$2^{n\ 1}$	$2^{n/2\ 1}$	$2^{n\ 1}\quad 2^{n/2\ 1}$
Odd no. (n)	$2^{n\ 1}\quad 2^{n\ 1/2}$	$2^{n\ 1/2}$	$2^{n\ 1}$

Solved Example



How many stereoisomers (enantiomers and diastereomers). are there of this molecule. Note the possible symmetry of the stereoisomers is a function of the absolute configurations.

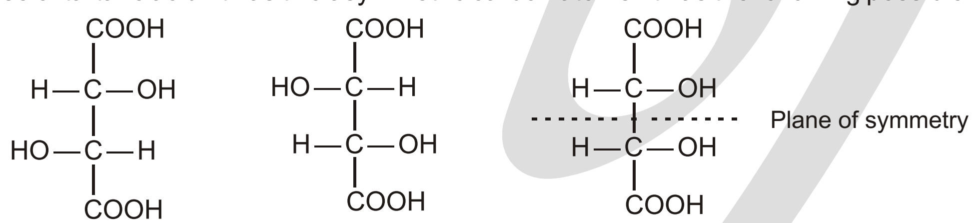
Ans. Total optical isomer $2^{n\ 1}\quad 2^{n/2\ 1}$

$2^{6\ 1}\quad 2^{6/2\ 1}$

32 (Optical active) 4 (Meso)

36

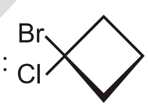
Consider the case of tartaric acid. It has two asymmetric carbon atoms. It has the following possible isomers:



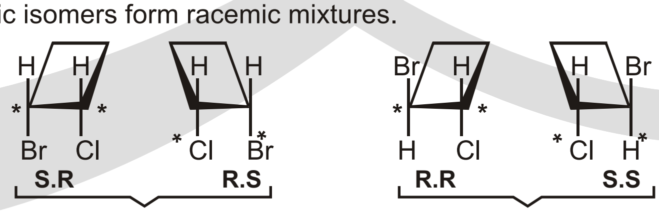
(+) tartaric acid (I) | (–) tartaric acid (II) | meso-tartaric acid (III)

Solved Example

Give the names, structural formulas and stereochemical designations of the isomers of (a) bromochlorocyclobutane, (b) dichlorocyclobutane, (c) bromochlorocyclopentane, (d) diiodocyclopentane, (e) dimethylcyclohexane. Indicate chiral C's.

(a) There is only one structure for 1-bromo-l-chlorocyclobutane : 

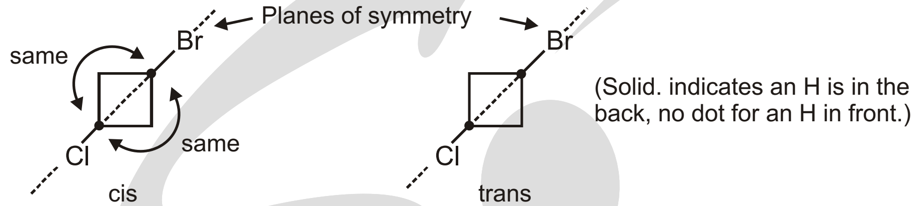
With l-bromo-2-chlorocyclobutane there are cis and trans isomers and both substituted C's are chiral. Both geometric isomers form racemic mixtures.



cis, racemic | trans, racemic

1-Bromo-2-chlorocyclobutane

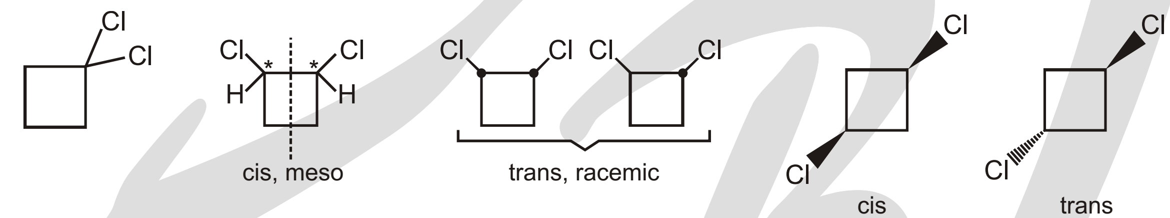
(a) In 1-bromo-3-chlorocyclobutane there are cis and trans isomers, but no enantiomers; C^1 and C^3 are not chiral, because a plane perpendicular to the ring bisects them and their four substituents. The sequence of atoms is identical going around the ring clockwise or counterclockwise from C^1 to C^3.



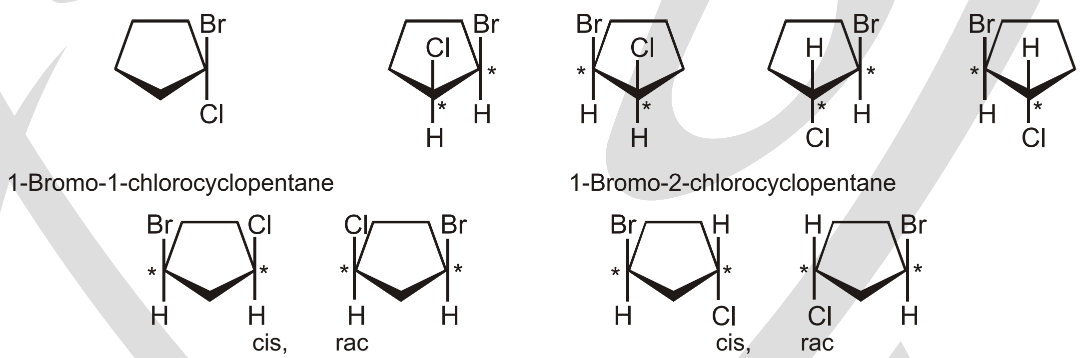
1-Bromo-3-chlorocyclobutane

In these structural formulas, the other atoms on C^1 and C^3 are directly in back of those shown and are bisected by the indicated plane.

(b) Same as (a) except that the cis-1,2-dichlorocyclobutane has a plane of symmetry (dashed line below) and is meso.

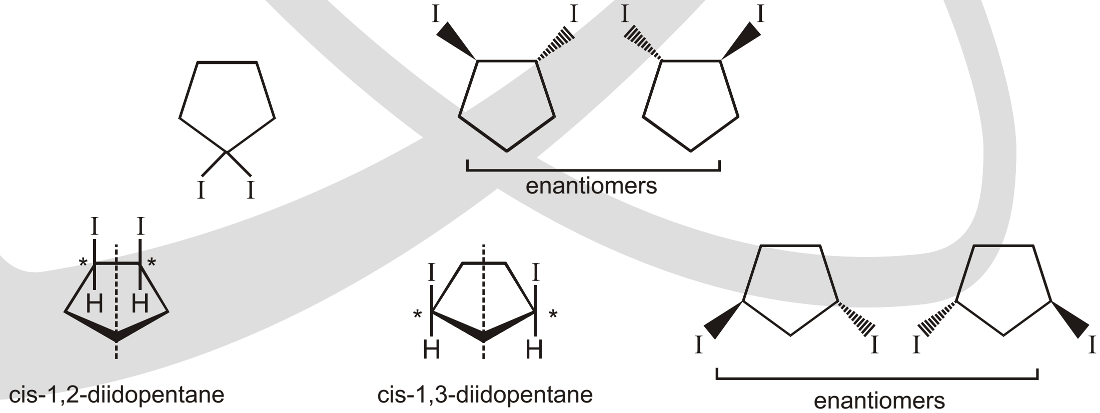


(c) There are nine isomers because both 1,2- and 1,3-isomers have cis and trans geometric isomers, and these have enantiomers.



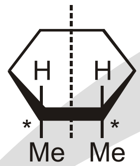
1-Bromo-3-chlorocyclopentane

(d) The diiodocyclopentanes are similar to the bromochloro derivative, except that both the cis-1,2- and the cis-1,3-diiodo derivatives are meso. They both have planes of symmetry.

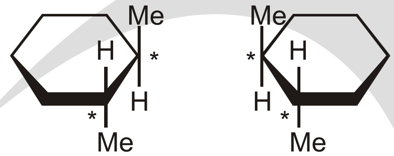


(e) There are nine isomeric dimethylcyclohexanes.

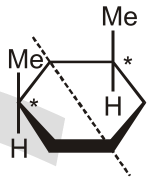
1,1-Dimethyl



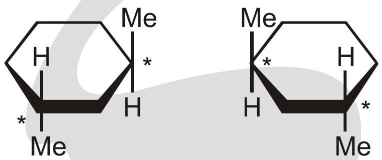
cis, meso-1,2-Dimethyl



trans, rac-1,2-Dimethyl

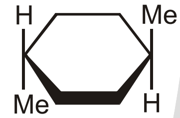


cis, meso-



trans, rac-1,3-Dimethyl

cis

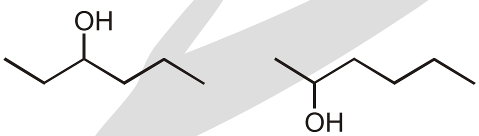


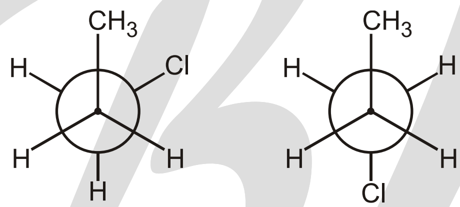
trans-1,4-Dimethyl

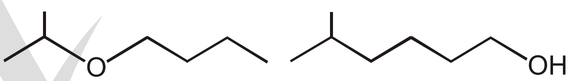
(no chiral centers)

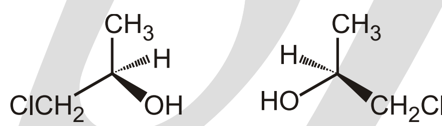
Solved Example

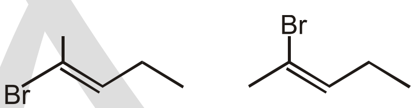
- Find relation between given pair?

(1) 

(2) 

(3) 

(4) 

(5) 

(6) 

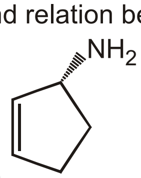
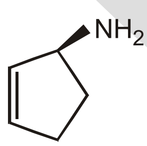
Ans. (1) constitutional isomer (positional isomer) (2) Conformers

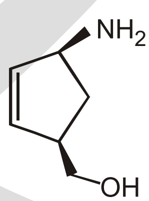
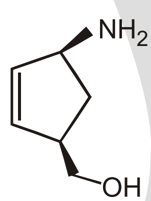
(3) Constitutional isomer (Functional isomer) (4) Enantiomer

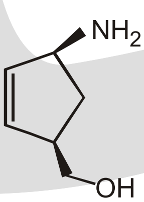
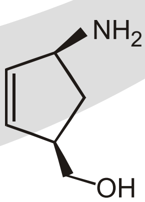
(5) Geometrical or Diastereomers (6) Conformer (By ring flip)

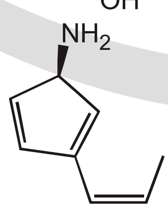
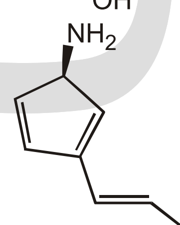
Solved Example

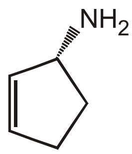
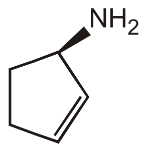
- Find relation between given pair?

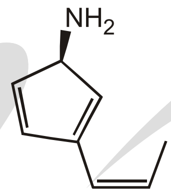
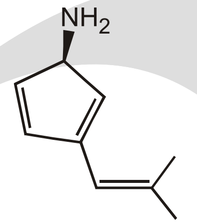
(a)  and 

(b)  and 

(c)  and 

(d)  and 

(e)  and 

(f)  and 

Ans. (a) Enantiomers (b) Indenticle

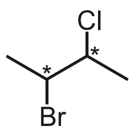
(c) Positional isomer (d) Geometrical isomers

(e) Identical (f) Homologas compounds

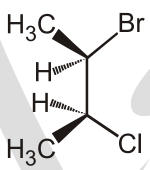
Solved Example

- Find total stereoisomers and relationship between them in 2-bromo-3-chlorobutane.

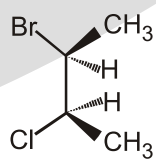
Ans. 2 stereocenters



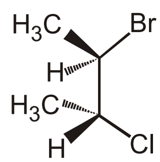
4 stereoisomers



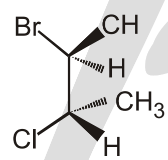
A



B



C



D

(i) A and B are enantiomers.

(ii) C and D aare enantiomers.

(iii) A and C, or A and D, or B and C, or B and D are diastereomers.

(iv) Diastereomers are stereoisomers which are not mirror images of other.

Solved Example

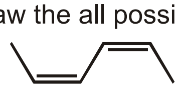
- How many pair of diastereomer are possible for given compound

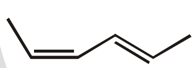
$CH_3 — CH \quad CH — CH \quad CH — CH_3$

(A) 0 (B) 2 (C) 3 (D) 4

Ans. (C)

Draw the all possible configuration of given compound

(A)  cis, cis

(B)  cis, trans

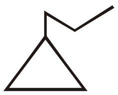
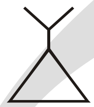
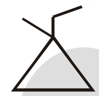
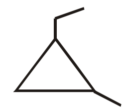
(C)  trans, trans

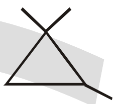
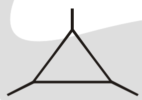
pair of diastereomars ((A) (B), (A) (C) and (B) (C))

Solved Example

- The total number of structural isomers possible for compound with the molecular formula C_6H_{12} having cyclopropane ring only.

Ans. 6

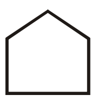
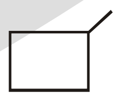
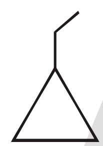
(a) 
(b) 
(c) 
(d) 

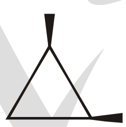
(e) 
(f) 

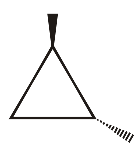
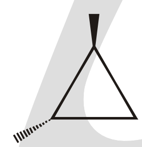
Solved Example

- The total number of cyclic structural as well as stereo isomers possible for compound with the molecular formula C_5H_{10} is :

Ans. 7

Sol. (a) 
(b) 
(c) 
(d) 

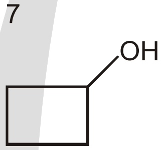
(e) 
Meso

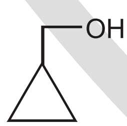
(f) 
(g) 

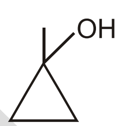
Solved Example

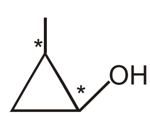
- The total number of cyclic structural as well as stereo isomers possible for compound with the molecular formula C_4H_8O (only alcohol) will be.

Ans. 7







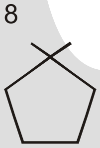


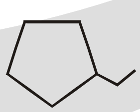
1 + 1 + 1 + 4 = 7 total

Solved Example

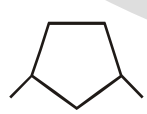
- The total number of five membered cyclic structural as well as stereoisomers possible for compound with the molecular formula C_7H_{14} is :

Ans. 8


1

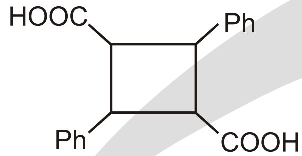

1


3


3

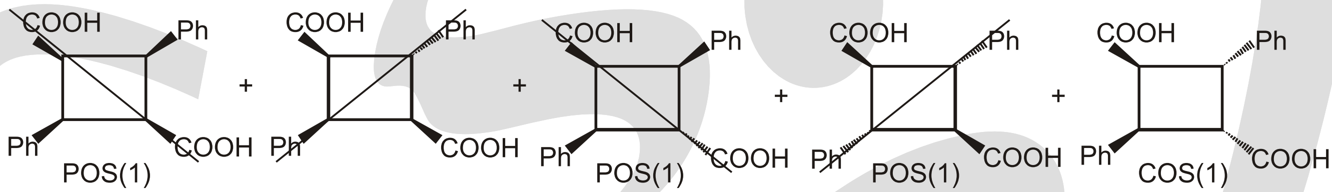
Solved Example

- Find the number of optical isomers of sires compound



(A) 3 (B) 2 (C) 6 (*D) Zero

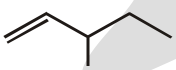
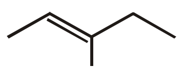
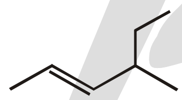
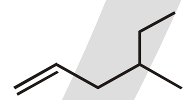
Sol.



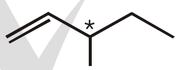
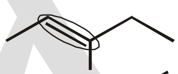
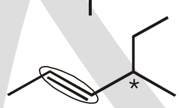
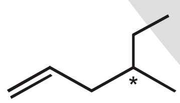
All are optically inactive
So number of optical isomers = Zero

Solved Example

- Identify Alkene showing both geometrical as well as optical isomerism.

(A)  (B)  (*C)  (D) 

Sol.

	Geometrical Isomerism	Optical Isomerism
(A) 	✗	✔
(B) 	✔	✗
(C) 	✔	✔
(D) 	✗	✔

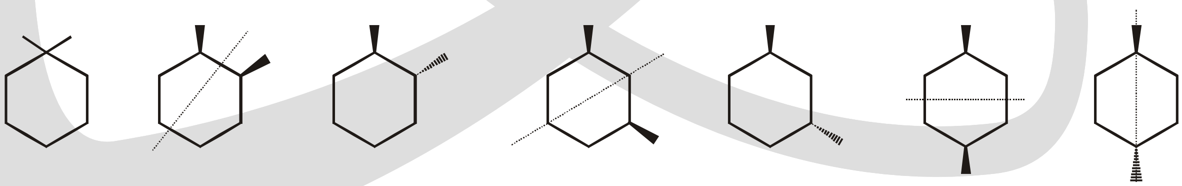
(* : Chiral centre)

Solved Example

- *X* Total possible isomers of dimethylcyclohexane which are chiral. Find the value of *X*?

Ans. 4

Sol.

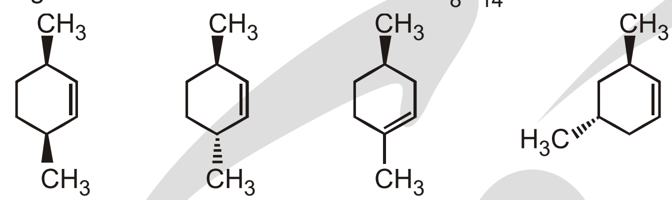


(achiral) (+M.I) (achiral) (+M.I) (achiral) (achiral)

There are 4 isomers which are chiral.

Solved Example

- Consider the following structures of molecular formula C_8H_{14}



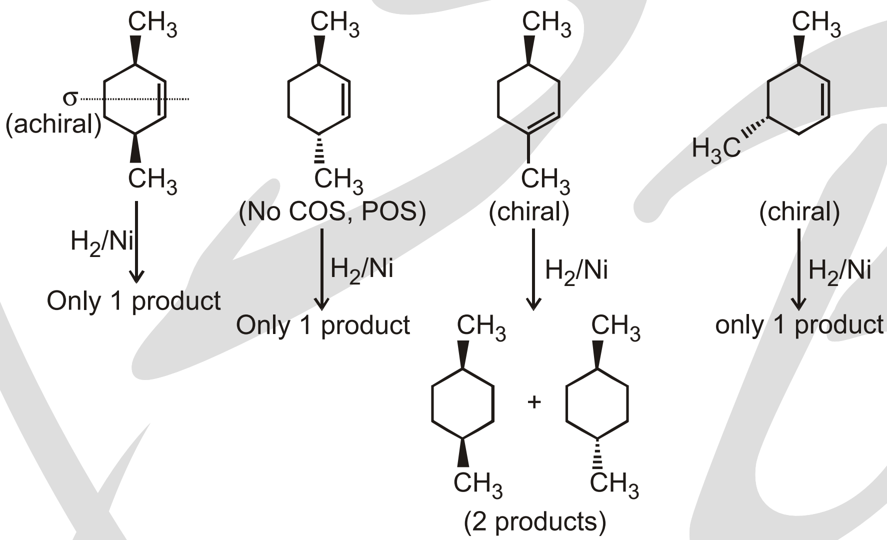
X Number of compounds which are optically active

Y Sum of total number of products obtained when each compound undergoes catalytic hydrogenation.

Find the sum of X Y ?

Ans. 8

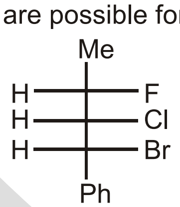
Sol.



X 3; Y 5 $\qquad$ X Y 8

Solved Example

- How many pair of diastereomer are possible for Given compound



Ans. 6

Sol. Total stereoisomers will be 8. Given compound form 7 pairs with other sterioisomers in which expect 1 enantiomers all other 6 pairs will be distereomeric pairs.

Solved Example

- Total number of isomer has formula C_5H_8 with 2sp^3 carbon, 2sp^2 carbon, 1sp carbon.

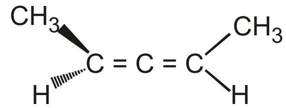
Ans. 4

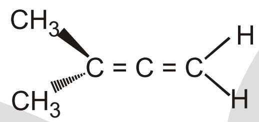
Sol. **1.** *sp* carbon *i.e.,* 1C connected with 2 -bonds

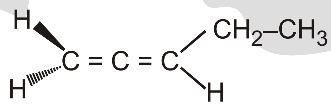
2. sp^2 carbon *i.e.,* 2C connected with 1 bond each

3. *sp*^{3} carbon *i.e.,* 2C connected with only bonds.

Possible structures are –


+ enantiomer = 2

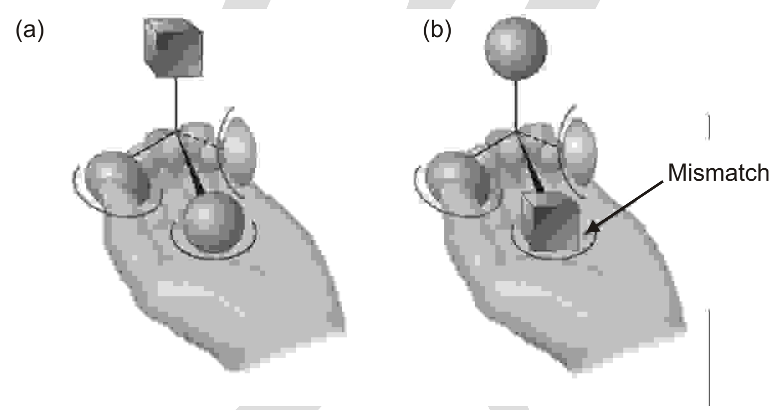

= 1


= 1

total = 4

WHY DO DIFFERENT ENANTIOMERS HAVE DIFFERENT BIOLOGICAL PROPERTIES?

Why do different enantiomers have different biological properties? To have a biological effect, a substance typically must fit into an appropriate receptor that has an exactly complementary shape. But because biological receptors are chiral, only one enantiomer of a chiral substrate can fit in, just as only a right hand can fit into right-handed glove. The mirror-image enantiomer will be a misfit, like a left hand in a right-handed glove. A representation of the interaction between a chiral molecule and a chiral biological receptor is shown in Figure : one enantiomer fits the receptor perfectly, but the other does not.



Imagine that a left hand interacts with a chiral object, much as chiral object, much as chiral molecule (a) One enantiomer fits a biological receptor interacts with a chiral molecule. One enantiomer fits into the hand perfectly, thumb, palm and finger, with the substituent exposed (b) The other enantiomer, however, can't fit into the hand. When the thumb and finger interact appropriately, the palm holds a substituent rather than a one, with the substituent exposed.

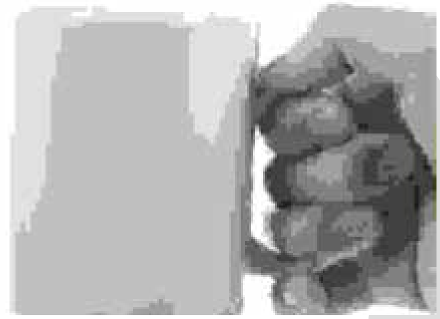
The hand-in-glove fit of a chiral substrate into a chiral receptor is relatively straightforward, but it's less obvious how a prochiral substrate can undergo a selective reaction. Take the reaction of ethanol with NAD catalyzed by yeast alcohol dehydrogenase. As we saw at the end of Section, the reaction occurs with exclusive removal of the pro-R hydrogen from ethanol and with addition only to the Re face of the NAD carbon.

We can understand this result by imagining that the chiral enzyme receptor again has three binding sites, as was previously the case in. When substituents of a prochiral substrate are held appropriately, however, only one of the two substituents—say, the pro-S one—is also held while the other, pro-R, substituent is exposed for reaction.

We describe the situation by saying that the receptor provides a chiral environment for the substrate. In the absence of a chiral environment, the two substituents are chemically identical, but in the presence of the chiral environment, they are chemically distinctive. The situation is similar to what happens when you pick up a coffee mug. By itself, the mug has a plane of symmetry and is achiral. When you pick up the mug, however, your hand provides a chiral environment so that one side becomes much more accessible and easier to drink from than the other.

(a) When a prochiral molecule is held in a chiral environment, the two seemingly identical substituents are distinguishable. (b) similarly, when an achiral coffee mug is held in the chiral environment of your hand, it's much easier to drink from one side than the other because the two sides of the mug are now distinguishable.

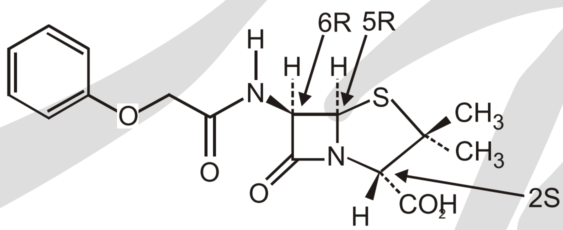
(b)



CHIRAL DRUGS

The hundreds of different pharmaceutical agents approved for use by the U.S. Food and Drug Administration come from many sources. Many drugs are isolated directly from plants or bacteria, and others are made by chemical modification of naturally occurring compounds. An estimated 33%, however, are made entirely in the laboratory and have no relatives in nature.

Those drugs that come from natural sources, either directly or after chemical modification, are usually chiral and are generally found only as a single enantiomer rather than as a racemate. Penicillin V, for example, an antibiotic isolated from the Penicillium mold, has the 2S,5R,6R configuration. Its enantiomer, which does not occur naturally but can be made in the laboratory, has no antibiotic activity.



Penicillin V(2S, 5R, 6R configuration)

EXERCISE

MATCH THE COLUMN

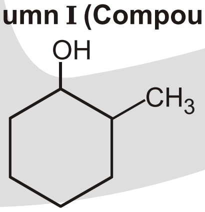
1. Match List-I with List-II and select the correct answer using the codes given below

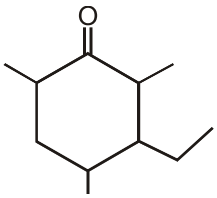
List-I	List-II
(P) Distereomers	(1) Internal compansation
(Q) Meso compound	(2) External compansation
(R) Conformers	(3) Different reaction under chiral medium
(S) Racemic mixture	(4) Results by the free rotation about C–C bond
(T) Enantiomers	(5) Cis-Trans isomerism

Codes :

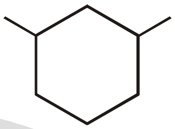
	P	Q	R	S	T
(A)	5	2	4	1	3
(B)	3	1	4	2	5
(C)	3	2	5	5	4
(D)	5	1	4	2	3

2. Column I (Compound) **Column II (Number of streocenter)**

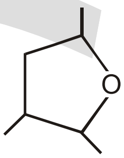
(A)  (P) 4

(B) 

(Q) 1

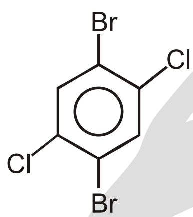
(C) 

(R) 3

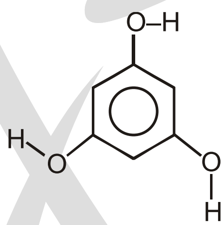
(D) 

(S) Shows G.I.

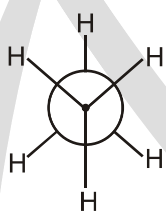
3. Column I | **Column II**

(A) 

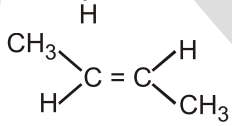
(P) C_2-axis of symmetry is present

(B) 

(Q) C_3-axis of symmetry is present

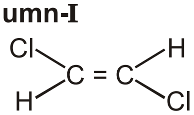
(C) 

(R) Ci (Center of symmetry) is present

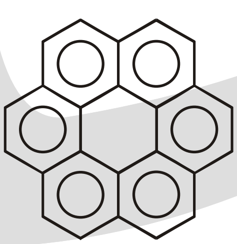
(D) 

(S) Plane of symmetry is present

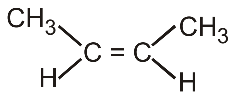
4. Column-I | **Column-II**

(A) 

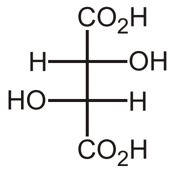
(P) C_2 – axis of symmetry is present

(B) 

(Q) C_3–axis of symmetry is present

(C) 

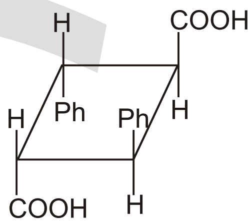
(R) Plane of symmetry is present

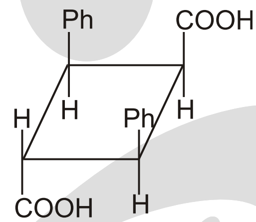
(D) 

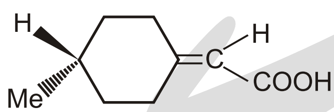
(S) Center of symmetry is present

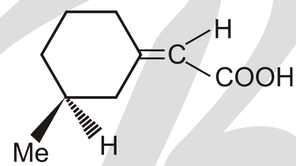
(T) S_4 alternative axis of symmetry

5.

(I) 

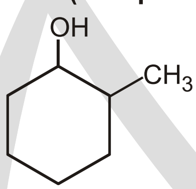
(II) 

(III) 

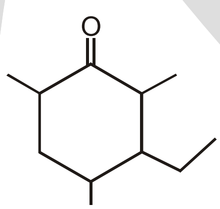
(IV) 

Column I	**Column II**
(A) Plane of symmetry	(P) I
(B) Centre of symmetry	(Q) II
(C) Show geometrical isomerism	(R) III
(D) Show optical isomerism	(S) IV

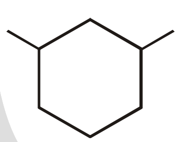
6. ColumnI (Compound) **Column II (Number of streocenter)**

(A) 

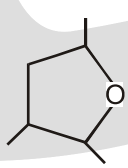
(P) 4

(B) 

(Q) 1

(C) 

(R) 3

(D) 

(S) Shows G.I.

FIND THE RELATIONSHIP

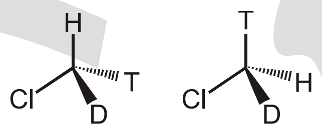
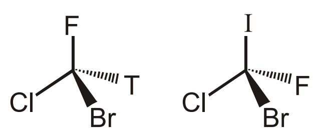
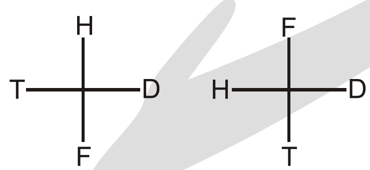
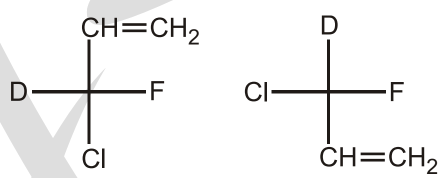
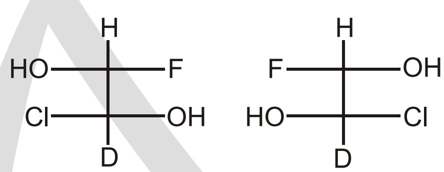
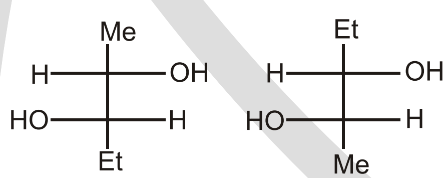
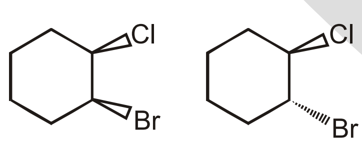
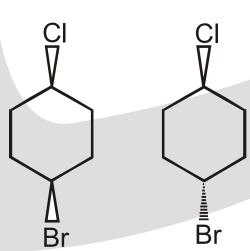
Relatoinship between compounds

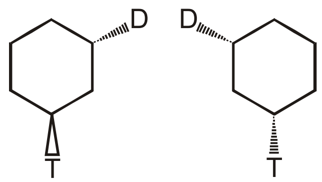
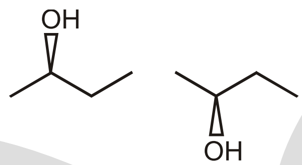
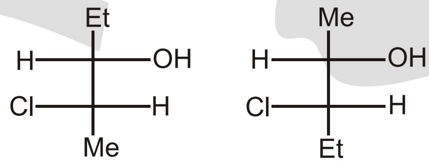
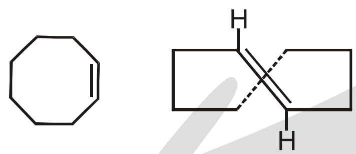
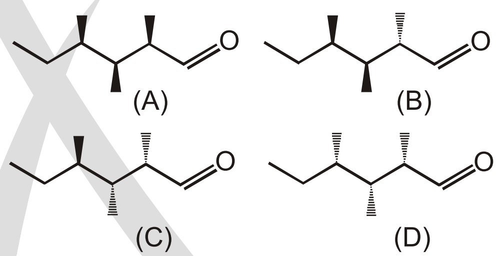
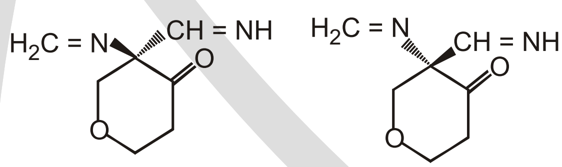
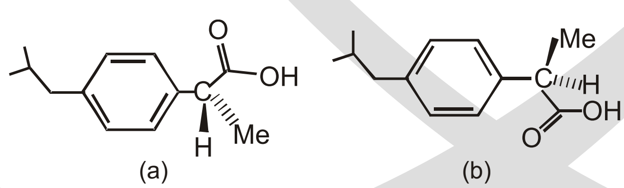
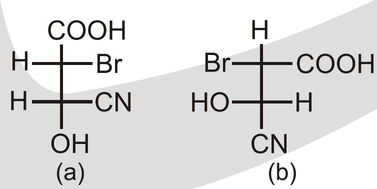
Non-superimposable mirror images compounds Enantiomer

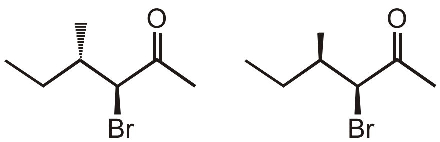
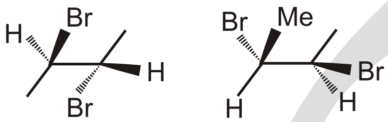
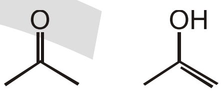
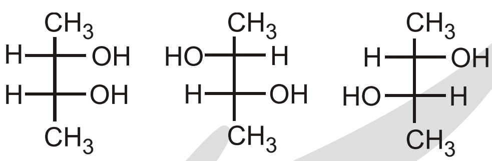
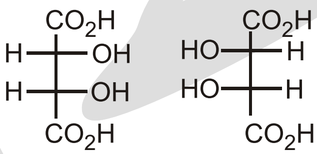
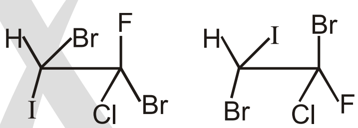
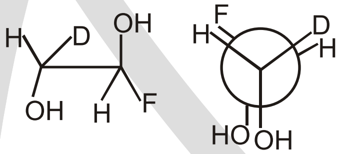
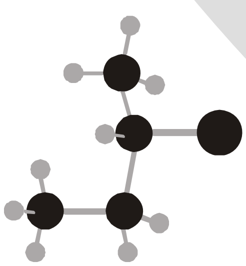
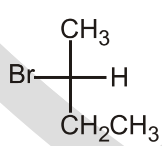
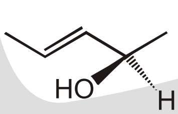
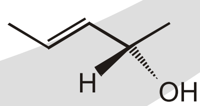
Non-superimposable non-mirror images compounds Diastereomers

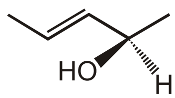
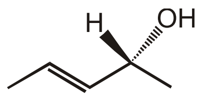
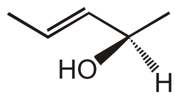
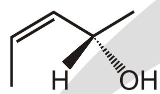
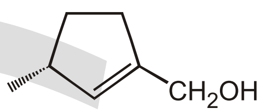
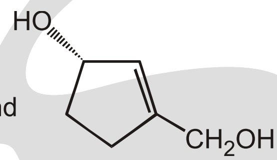
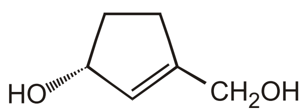
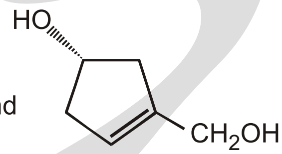
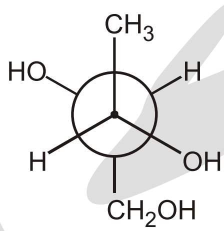
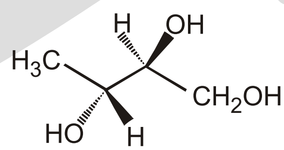
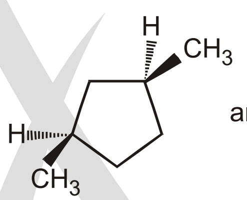
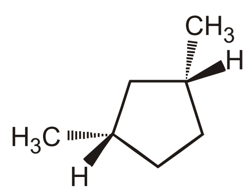
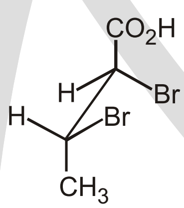
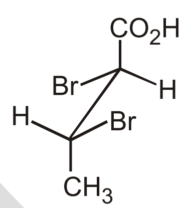
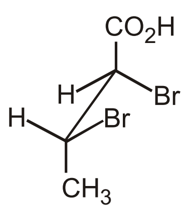
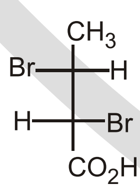
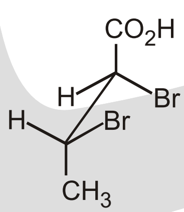
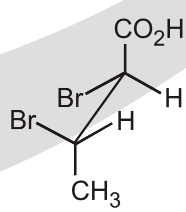
Problem: Indicate whether each of the following pairs of compounds are identical or are enantiomers, diastereomers, or constitutional isomers:

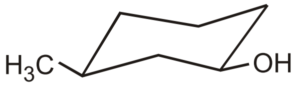
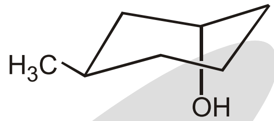
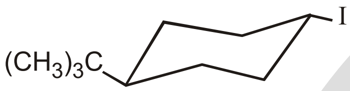
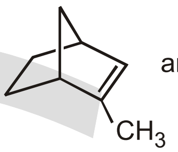
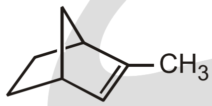
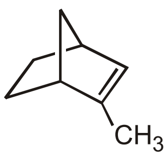
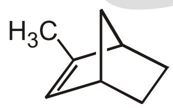
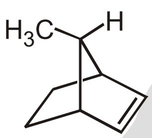
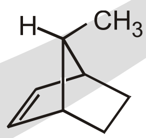
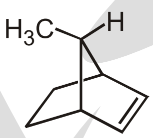
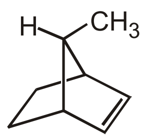
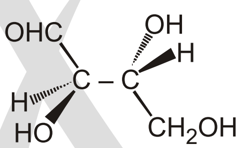
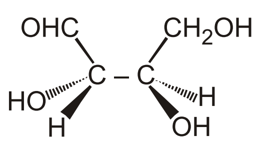
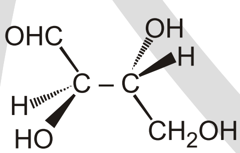
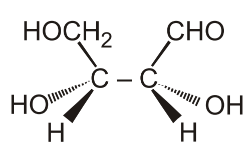
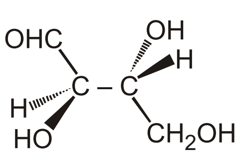
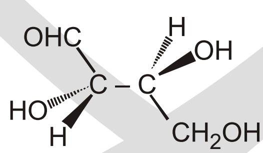
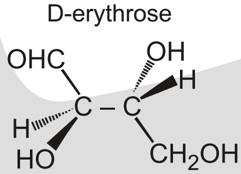
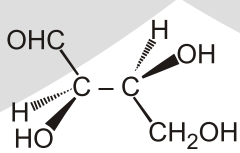
1. Find relationship between given pairs as enantiomer, diastereomer or other :

S.No.	Identify the relation	Enantiomer	Diastereomer	Other
1.				
2.				
3.				
4.				
5.				
6.				
7.				
8.				

9.				
10.				
11.				
12.				
13.				
14.				
15.				
16.				
17.				

18.	 (a) (b)			
19.	 (a) (b)			
20.	 (a) (b)			
21.	 (a) (b) (c)			
22.	 (a) (b)			
23.				
24.				
25.	 and 			
26.	 and 			

27.	 and 			
28.	 and 			
29.	 and  HO			
30.	 and 			
31.	 and 			
32.	 and 			
33.	 and 			
34.	 and 			
35.	 and 			

36.	 and 			
37.	 and 			
38.	 and 			
39.	 and 			
40.	 and 			
41.	 and 			
42.	 D-erythrose and 			
43.	 D-erythrose and 			
44.	 D-erythrose and 			
45.	 D-erythrose and 			

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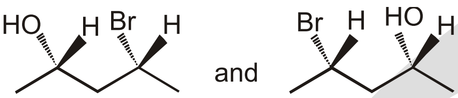
47.				
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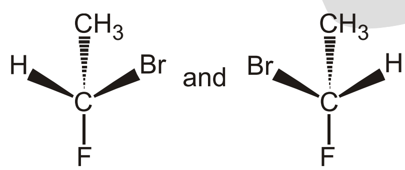
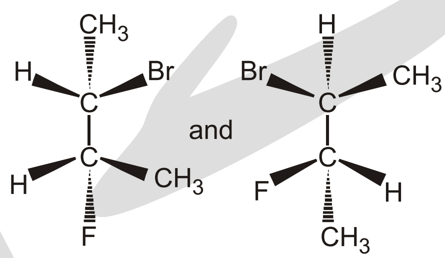
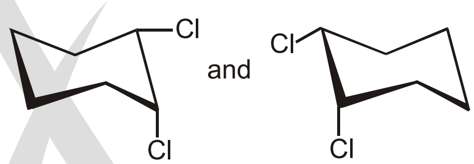
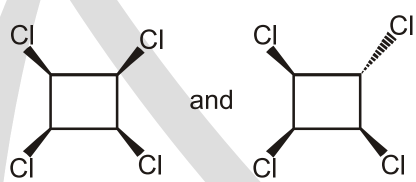
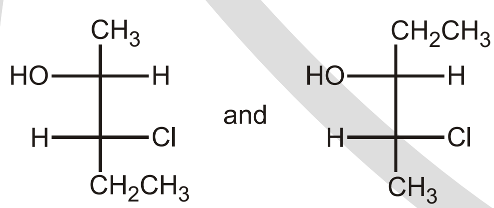
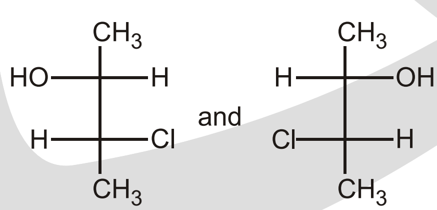
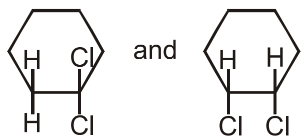
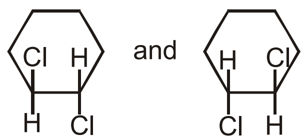
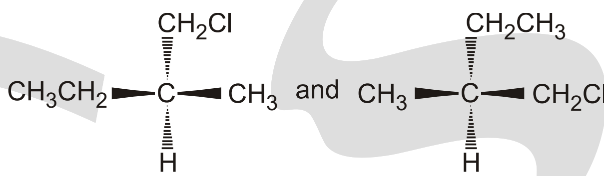
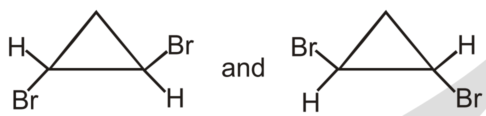
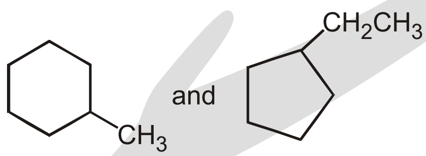
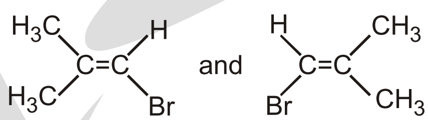
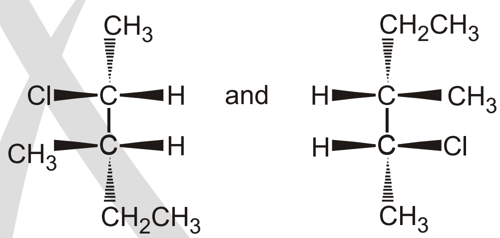
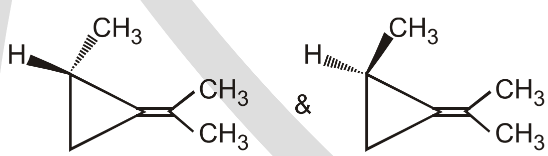
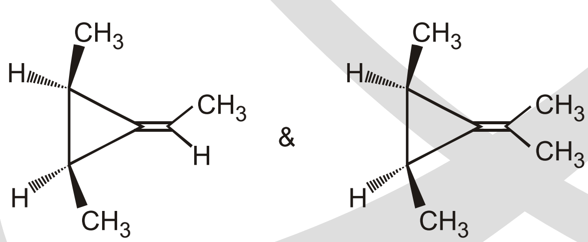
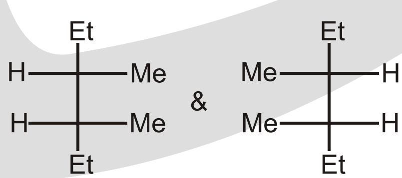
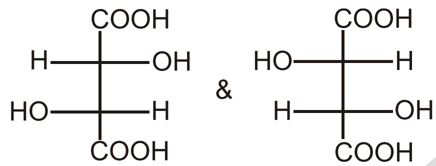
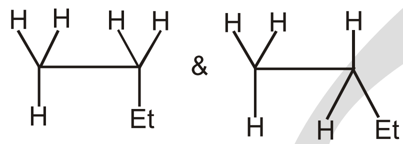
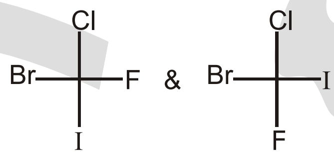
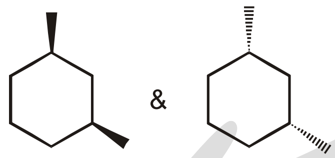
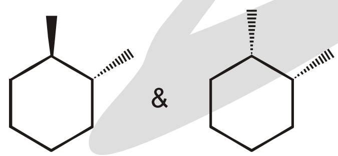
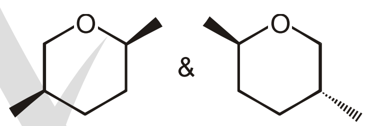
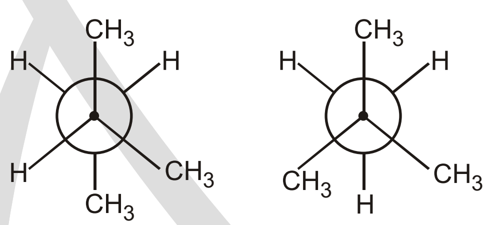
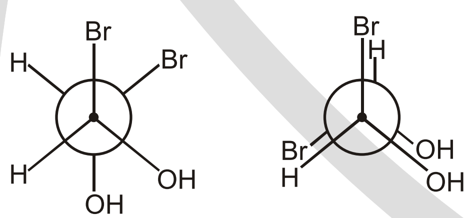
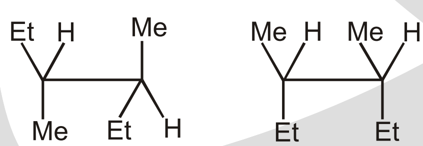
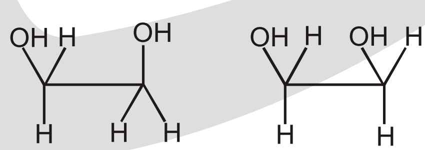
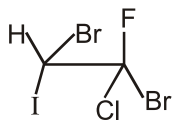
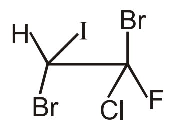
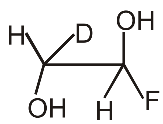
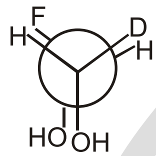
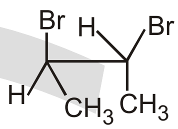
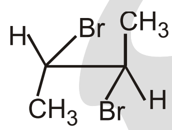
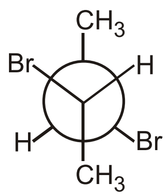
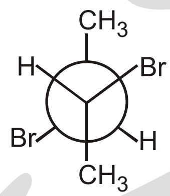
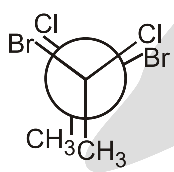
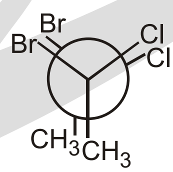
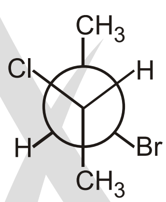
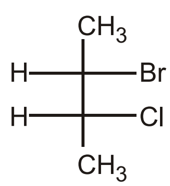
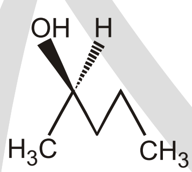
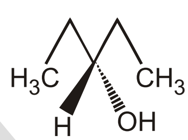
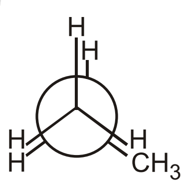
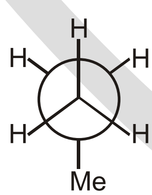
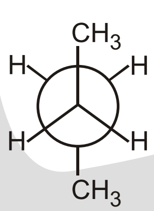
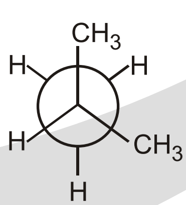
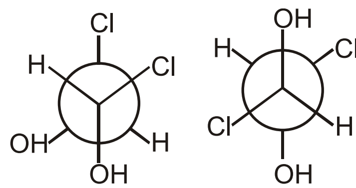
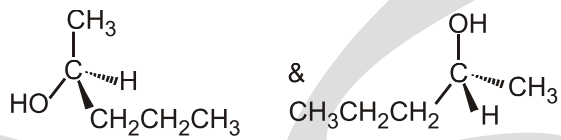
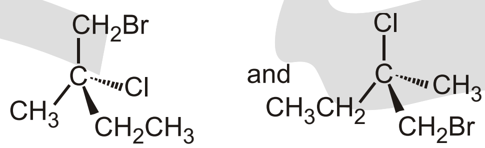
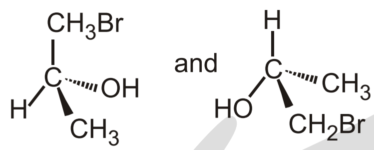
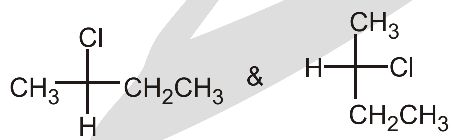
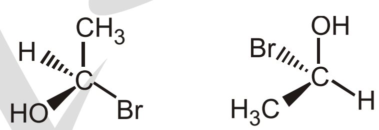
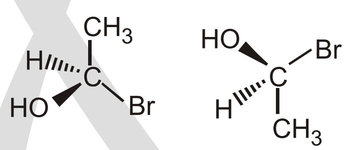
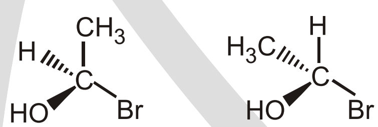
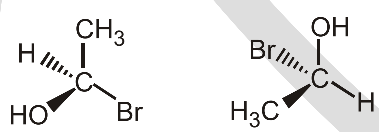
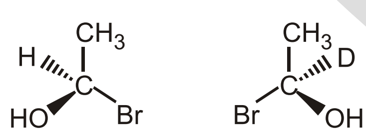
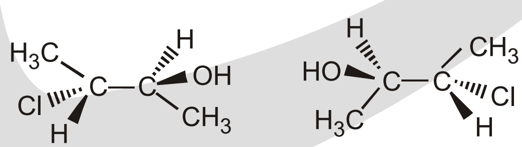
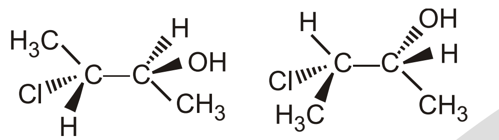
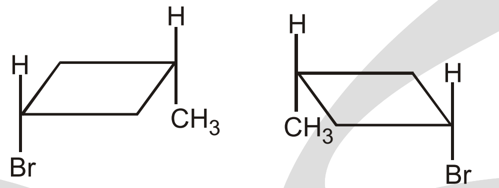
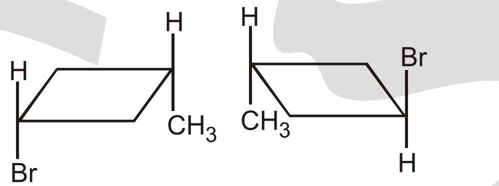
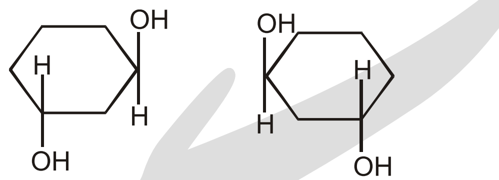
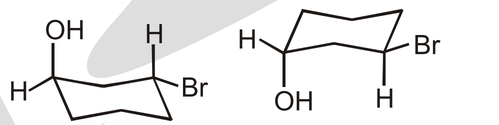
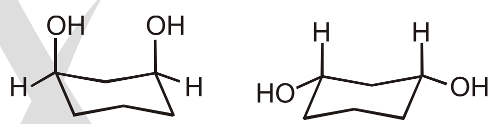
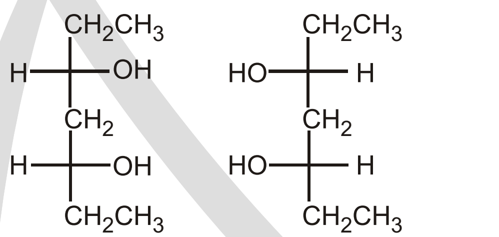
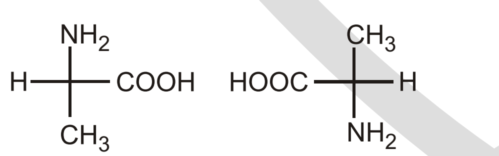
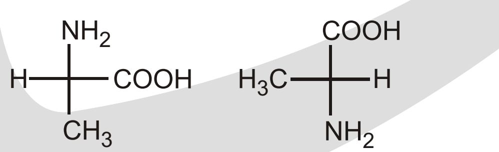
50.				
51.				
52.				
53.				
54.				
55.				

56.				
57.				
58.				
59.				
60.				
61.				
62.				
63.				
64.				
65.				

66.				
67.				
68.				
69.				
70.				
71.				
72.				
73.				
74.				
75.				

76.	 			
77.	 			
78.	 			
79.	 			
80.	 			
81.	 			
82.	 			
83.	 			
84.	 			

85.				
86.				
87.				
88.				
89.				
90.				
91.				
92.				
93.				
94.				
95.				

96.				
97.				
98.				
99.				
100.				
101.				
102.				
103.				
104.				

UNSOLVED EXAMPLES

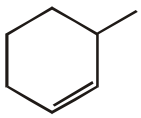
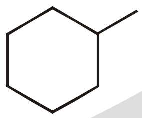
1. Which of the following compounds has an asymmetric center?

(a) $CH_3CH_2\underset{Cl}{\underset{|}{C}}HCH_3$ (b) $CH_3CH_2\underset{CH_3}{\underset{|}{C}}HCH_3$

(c) $CH_3CH_2\overset{CH_3}{\overset{|}{\underset{Br}{\underset{|}{C}}}}CH_2CH_2CH_3$ (d) CH_3CH_2OH

(e) $CH_3CH_2\underset{Br}{\underset{|}{C}}HCH_2CH_3$ (f) $CH2\quad CH\underset{NH_2}{\underset{|}{C}}HCH_3$

2. Which of the following compounds has one or more asymmetric centers?


A

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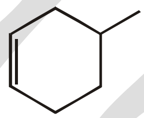
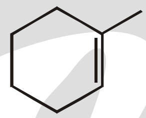
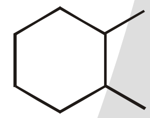
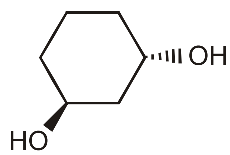
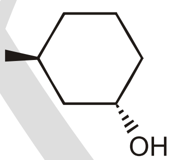

C

D

E

3. Indicate whether each of the structures in the second row is an enantiomer of, is a diastereomer of, or is identical to the structure in the top row.





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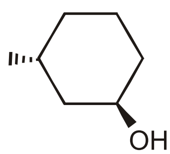

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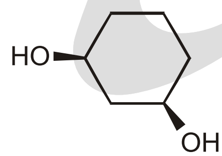

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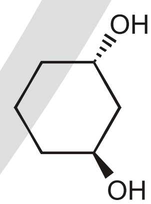
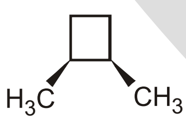
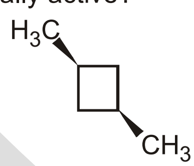
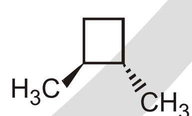
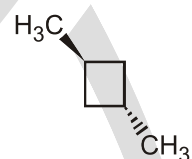


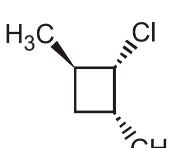
4. Which of the following are optically active?

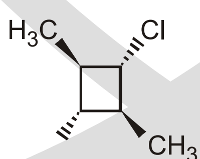


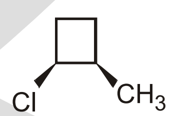


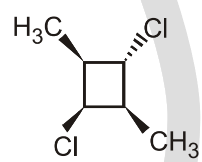




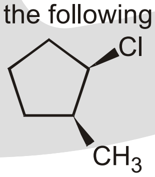


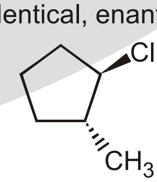


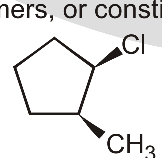


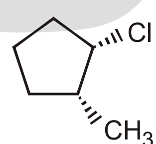


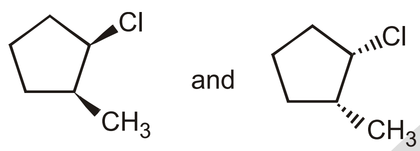
5. Are the following pairs identical, enantiomers, diastereomers, or constitutional isomers?

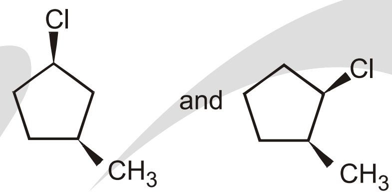
(a) 

and 

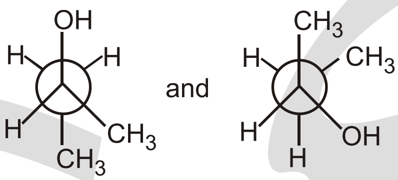
(b) 

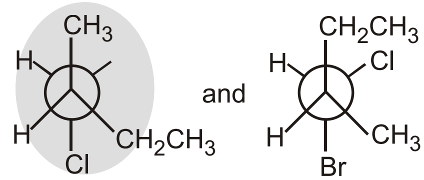
and 

(c) 

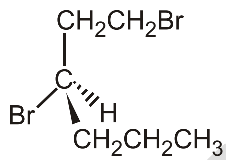
(d) 

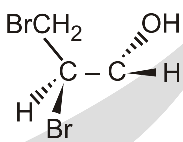
6. Are the following pairs identical, enantiomers, diastereomers, or constitutional isomers?

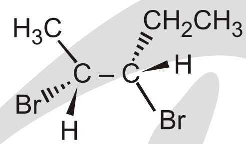
(a) 

(b) 

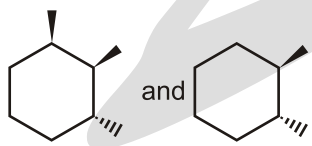
7. What is the configuration of the asymmetric centers in the following structures?

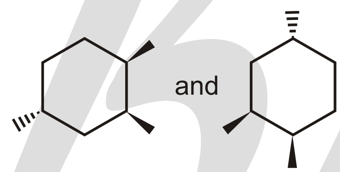
(a) 

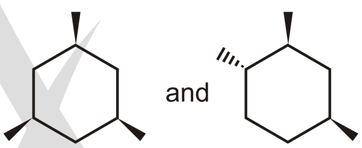
(b) 

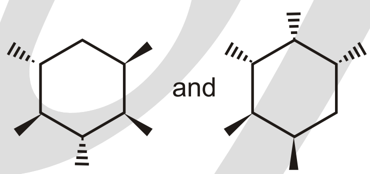
(c) 

8. Are the following pairs identical, enantiomers, diastereomers, or constitutional isomers?

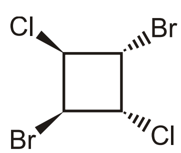
(a) 

(b) 

(c) 

(d) 

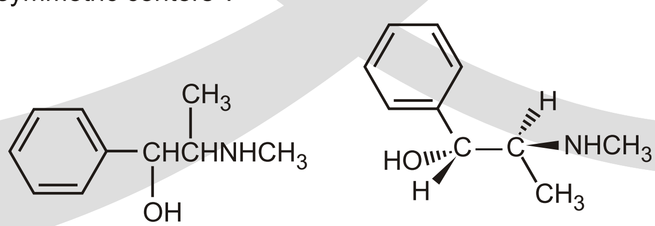
9. Is the following compound optically active?



10. For any centuries, the Chinese have used extracts from a group of herbs known as ephedra to treat asthma. A compound named ephedrine has been isolated from these herbs and found to be a potent dilator of air passages in the lungs.

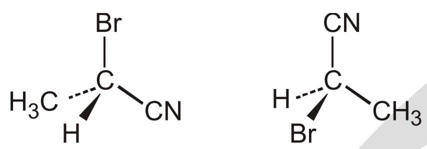
(a) How many stereoisomers does ephedrine have ?

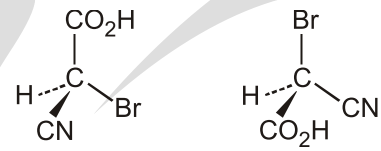
(b) The stereoisomer shown here is the one that is pharmacologically active. What is the configuration of each of the asymmetric centers ?

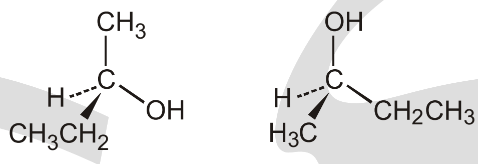


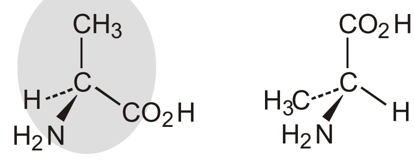
ephedrine

11. Which of the following pairs of structures represent the same enantiomer, and which represent different enantiomers?

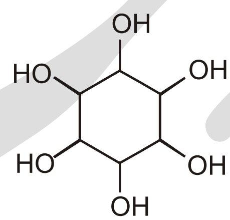
(a) 

(b) 

(c) 

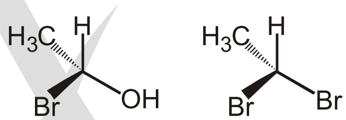
(d) 

12. Just for fun, you might like to try and work out just how many diastereoisomers inositol has and how many of them are meso compounds.

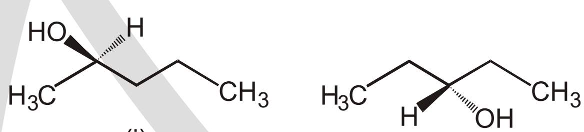


inositol

13. Identify chiral and achiral molecules in each of the following pair of compounds. (Wedge and Dash representations).

(I) 

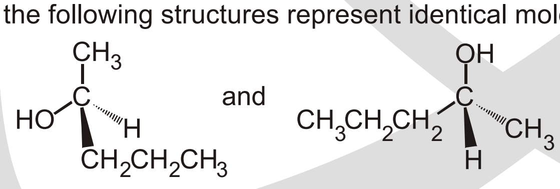
(i) (ii)

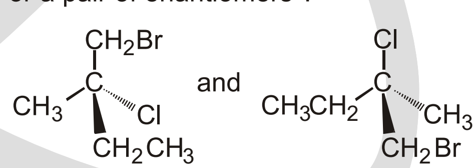
(II) 

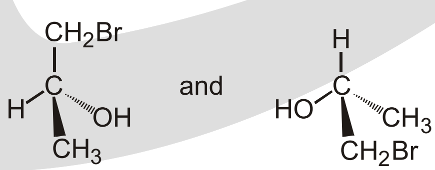
(i) (ii)

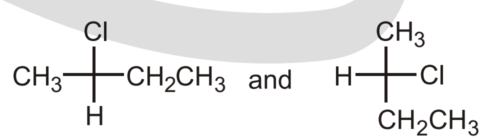
(III) $\underset{Br}{CH_3CHCH_2CH_3}$ (i) $CH_3CH_2CH_2CH_2Br$ (ii)

14. Do the following structures represent identical molecules or a pair of enantiomers ?

(a) 

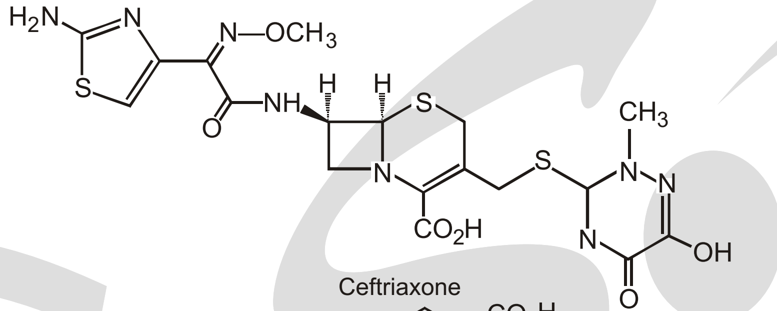
(b) 

(c) 

(d) 

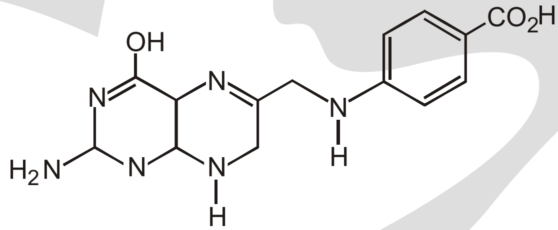
15. For each of the following compounds, identify any centers of chirality, and calculate the number of possible optical isomers :

(a)



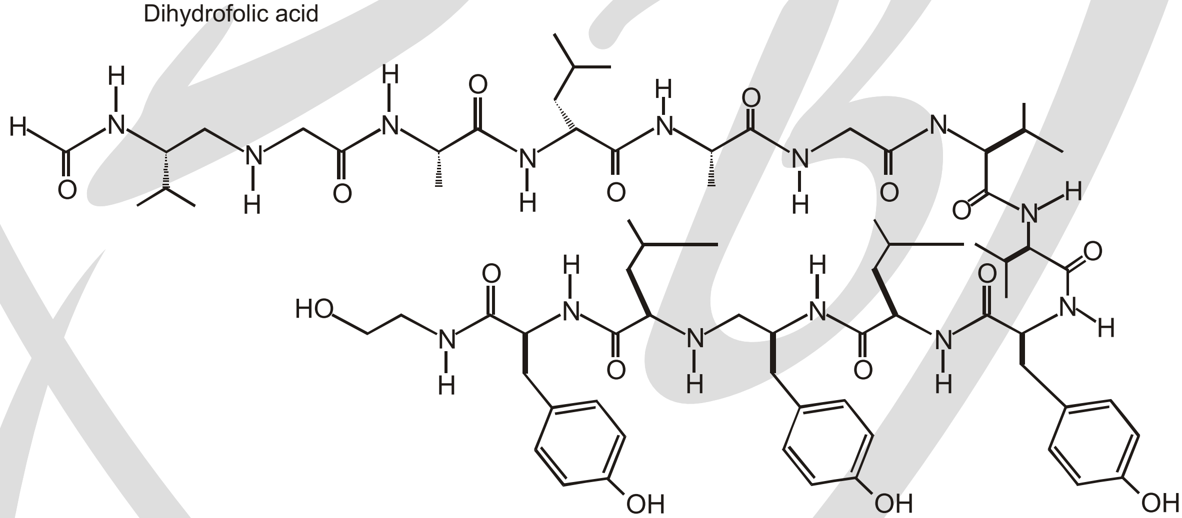
Ceftriaxone

(b)



Dihydrofolic acid

(c)



Gramicidin A

16. If molecule is pyramidal, **X** stereoisomers are possible for :

Cabde

find the value of **X.**

17. Total number of plane of symmetry in all conformations of 1, 2-dibromo of ethane

18. Total number of isomers of molecular formula of C6H12 having cyclobutane ring are.

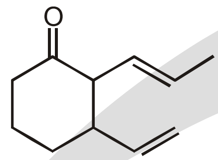
19. (i) Total stereoisomers of 2, 3-di-chlorobutane are (a).

(ii) $CH_3-CH=CH-CH=CH-\underset{Cl}{\overset{H}{C}}-CH_3$

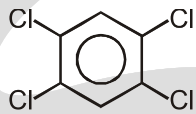
Total number of stereocentre in given compound are (b).

Value of a + b will be.

20. How many stereoisomers of given compound are possible when alkenyl substituent have a cis configuration



21. Total number of plane of symmetry present in given compound is



22. Find out the total number of stereocentre in the given compound.

$CH_3—CH=CH—CH(Br)—CH(Cl)—CH_3$

23. Find out the total number of stereoisomers of the given following compound.

$Cl—CH=CH—C(Cl)(H)—CH=CH—Cl$

24. Find the total number of isomers of C_7H_{14} (only 5-member ring).

◎ Special Problems

1. Number of stereocenters in cis-2-butene are:
(A) 0 (B) 1 (C) 2 (D) 3

2. Number of Carbon needed by Ester to show optical isomerism are:
(A) 4 (B) 5 (C) 6 (D) 7

3. Number of stereocenters in D-glucose :
(A) 1 (B) 5 (C) 4 (D) 7

4. Number of chiral centers in alpha-D-Glucopyranose are:
(A) 4 (B) 5 (C) 6 (D) 7

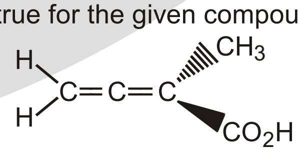
5. Number of enantiomers of 2-butanol are:
(A) 0 (B) 1 (C) 2 (D) 3

6. Number of diastereomers of Gammaxene are:
(A) 6 (B) 8 (C) 16 (D) 24

7. Number of diastereomers in 3-methyl-5-propylcyclohexene are:
(A) 2 (B) 4 (C) 6 (D) 3

8. Number of Carbon needed by a cycloalkane to show optical isomerism are:
(A) 4 (B) 5 (C) 6 (D) 7

9. Which of the following is true for the given compound?



(A) can show optical isomerism
(B) can show geometrical isomerism
(C) no POS or COS present in the compound
(D) None of these above is true

10. Number of chiral centers in beta-D-Glucofuranose are:

(A) 4 (B) 5 (C) 6 (D) 7

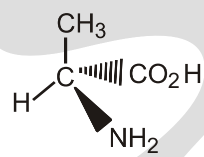
11. Number of diastereomers of 2-cyclohexyl pent-2-ene are:

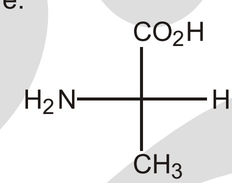
(A) 0 (B) 1 (C) 2 (D) 3

12. Number of diastereomer pairs of Tartaric acid are:

(A) 1 (B) 2 (C) 3 (D) 0

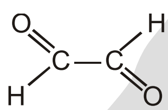
13. D,L configurations of given isomers respectively are:





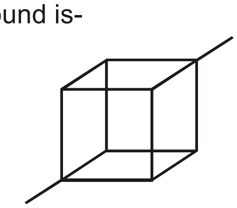
(A) D,D (B) L,L (C) D,L (D) L,D

14. Which of the following molecules have P.O.S. and n-fold perpendicular Axis of Symmetry?

(A) 

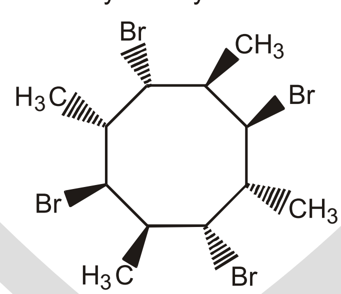
(B) 2-chlorobutane (C) s-cis-1,3-butadiene (D) Lactic acid

15. Number of P.O.S. in given compound is-



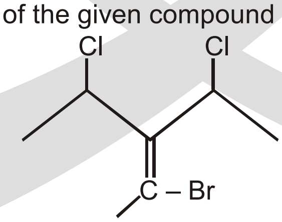
(A) 1 (B) 3 (C) 6 (D) 9

16. The given molecule has which Axis of Symmetry?



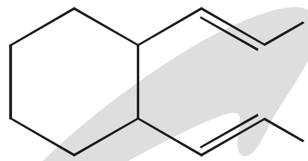
(A) S2 (B) S4 (C) POS (D) S6

17. Total number of stereoisomers of the given compound is:



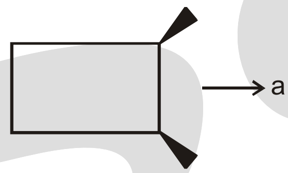
(A) 2 (B) 4 (C) 6 (D) 8

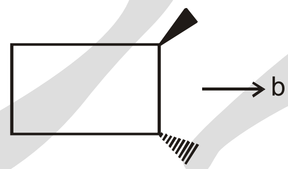
18. Total number of stereoisomers of the given compound is:



(A) 8 (B) 16 (C) 10 (D) 32

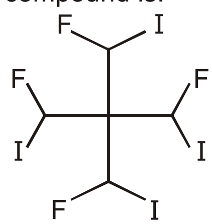
19. a and b are diastereomers of the given compouds. What are the values of a and b respectively:





(A) 1,1 (B) 1,2 (C) 2,1 (D) 2,2

20. Number of meso isomers of the given compound is:



(A) 5 (B) 2 (C) 0 (D) 1

21. Which of the following Axis of symmetry is present in Trans-2-butene?

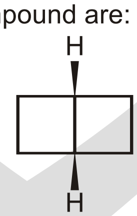
(A) 2 (B) 3 (C) 4 (D) 6

22. Which of the following is chiral?

(A) Flask (B) Chair

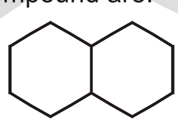
(C) Aishwarya Rai (D) Book

23. Number of stereoisomers for the given compound are:



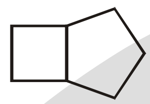
(A) 1 (B) 2 (C) 3 (D) 4

24. Number of stereoisomers for the given compound are:



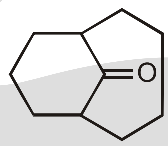
(A) 1 (B) 2 (C) 3 (D) 4

25. Number of stereoisomers for the given compound are:



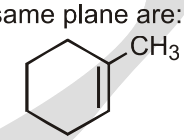
(A) 1 (B) 2 (C) 3 (D) 4

26. Number of stereoisomers for the given compound are:



(A) 1 (B) 2 (C) 3 (D) 4

27. Number of Carbon atoms that lie in same plane are:



(A) 3 (B) 4 (C) 5 (D) 7

28. Which of the following statements can be used to prove that carbon is tetrahedral?

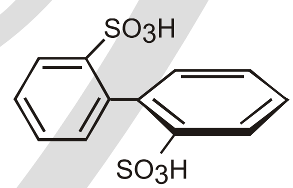
(A) Methyl bromide does not have constitutional isomers.

(B) Tetrachloromethane does not have a dipole moment.

(C) Dibromomethane does not have constitutional isomers.

(D) None of these.

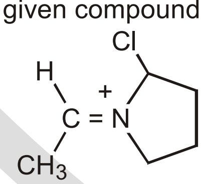
29. What happens when the given compound is heated?



(A) Racemic mixture (B) Diastereomers

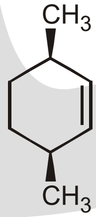
(C) 100% pure enantiomer (D) No change

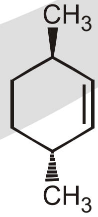
30. Number of stereocenters in the given compound are:

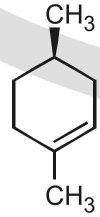


(A) 0 (B) 1 (C) 2 (D) 3

31. An optically active compound A with molecular formula C_8H_{14} undergoes catalytic hydrogenation to give an optically inactive product. Which of the following can be the structure of A?

(A) 

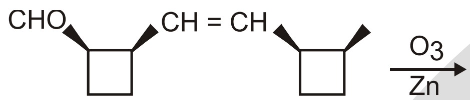
(B) 

(C) 

(D) B & C

32. Which of the following compounds can be resolved?

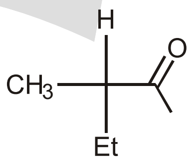
(A) cis-1,3-dimethylcyclohexane
(B) 1,1-dimethylcyclohexane
(C) cis-1,4-dimethylcyclohexane
(D) cis-1-ethyl-3-methylcyclohexane

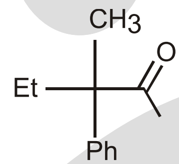
33. 

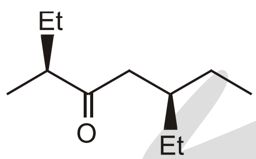
Number of chiral products are:

(A) 0 (B) 1 (C) 2 (D) 3

34. Which of the following is chiral and gives racemic mixture on treatment with trace of base?

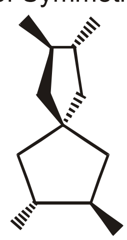
(A) 

(B) 

(C) 

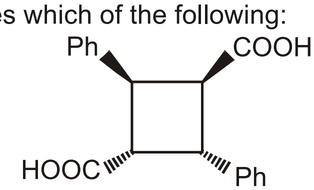
(D) All of these

35. The given compound has which Axis of Symmetry (S_n):



(A) S_2 (B) S_3 (C) S_4 (D) S_6

36. The given compound possesses which of the following:

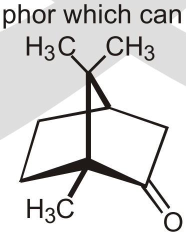


(A) Mirror symmetry
(B) Center of inversion
(C) n-fold alternating axis of symmetry
(D) All of these

37. Which of the following is the correct expression for maximum number of configurational isomers?
n = number of stereocenters m= number of stereogenic double bonds

(A) 2(n–m) (B) 2m + 2n (C) 2m + 2n/2 (D) 2(n+m)/2

38. The number of stereoisomers of Camphor which can exist are:

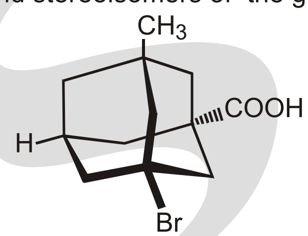


(A) 1 (B) 2 (C) 3 (D) 4

39. The number of stereoisomers of Twistane which can exist are:

(A) 2 (B) 3 (C) 4 (D) 8

40. The number of Stereocenters and stereoisomers of the given Adamantane derivative which can exist are:



(A) 2, 2 (B) 2, 4 (C) 4, 2 (D) 4, 4

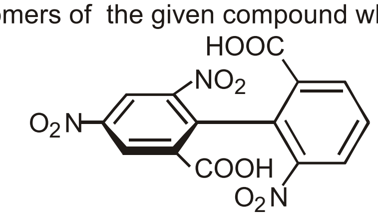
41. The number of stereoisomers of Penta-2,3-diene (consider only cumulated dienes) which can exist are:

(A) 2 (B) 3 (C) 4 (D) 6

42. The number of Stereocenters and stereoisomers of cyclooctene are:

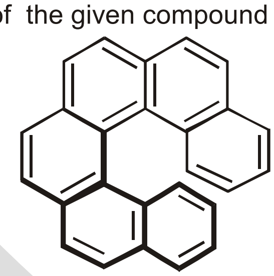
(A) 0, 1 (B) 1, 1 (C) 2, 3 (D) 2, 4

43. The number of stereoisomers of the given compound which can exist are:



(A) 0 (B) 1 (C) 2 (D) 4

44. The number of stereoisomers of the given compound which can exist are:

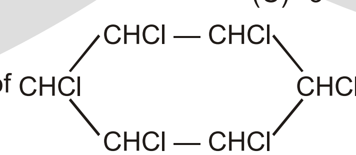


(A) 0 (B) 1 (C) 2 (D) 4

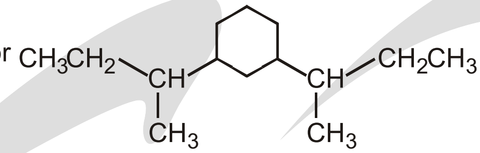
45. $CH_3 - \underset{\displaystyle Br}{\underset{|}{CH}} - CH \quad CH - \underset{\displaystyle Br}{\underset{|}{CH}} - CH_3$

A mixture of all stereoisomers possible from this structure is subjected to fractional distillation. Then how many fractions will be obtained ?

(A) 5 (B) 6 (C) 3 (D) 4

46. Find number of meso-isomers of 

(A) 6 (B) 9 (C) 8 (D) 7

47. How many meso-isomers are possible for 

(A) 1 (B) 2 (C) 3 (D) 4

48. Which one of the following statements is not true ?

(A) Diastereomers are a pair of stereoisomers that are not mirror images of one another.

(B) A pair of enantiomeric compounds has identical melting points.

(C) Diastereomers do not have equal specific rotations.

(D) Diastereomers are superimposable mirror images of one another.

49. The amount of the major enantiomer in a solution which is 80% optically pure is?

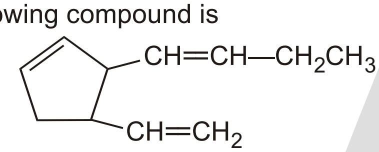
(A) 90% (B) 60% (C) 70% (D) 80%

50. Consider the following compound :

What is the number of stereoisomers possible for the above compound?

(A) 2 (B) 3 (C) 4 (D) 5

51. Stereoisomers possible for following compound is

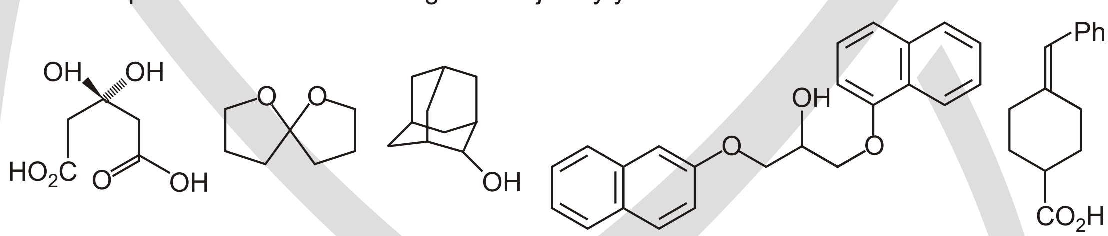


(A) 8 (B) 16 (C) 32 (D) 64

52. (+)-Tartaric acid has a specific rotation of +12.0°. Calculate the specific rotation of a mixture of 68% (+)-tartaric acid and 32% (–)- tartaric acid.

SUBJECTIVE TYPE QUESTIONS

1. Are these compounds chiral ? Draw diagrams to justify your answer.

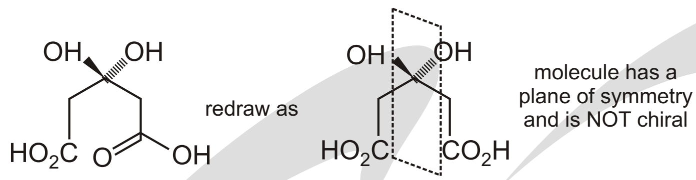


Purpose of the problem

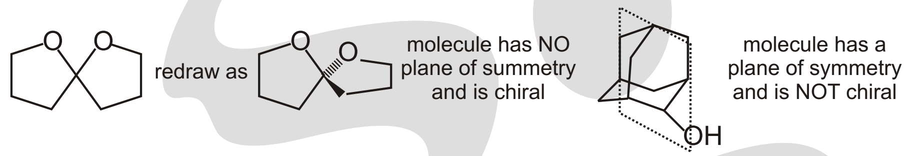
Reinforcement of the very important criterion for chirality. The previous problems were almost childish compared with this one; make sure you understand the answer.

Suggested solution

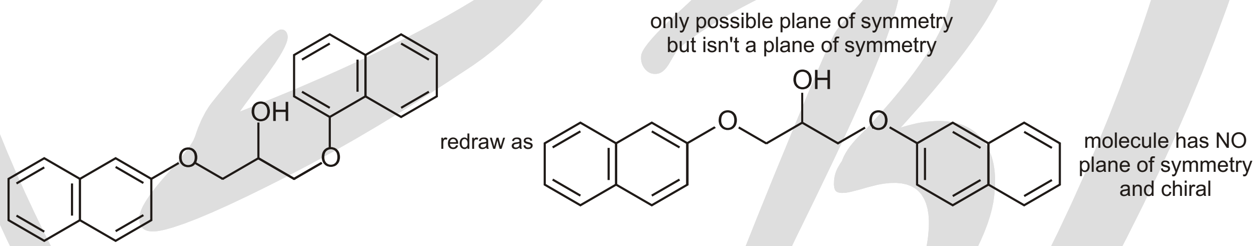
Only one thing matters – does the molecule have a plane of symmetry ? We need to redraw some of them to see if they do. On an account look for chiral centres of carbon atoms with four different groups or whatever-just look for a plane of symmetry (POS).



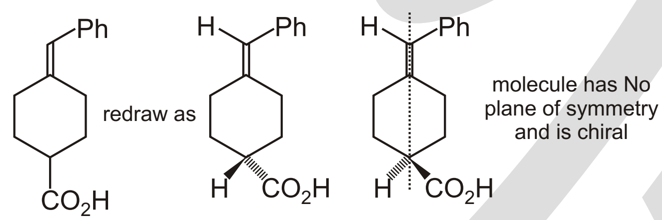
The second molecule is a 'spiro' compound having two rings joined at a tetrahedral carbon atom. These two rings are orthogonal so there is no plane of symmetry. The third molecule does have a plane of symmetry. It is much easier to see this if you make a model.



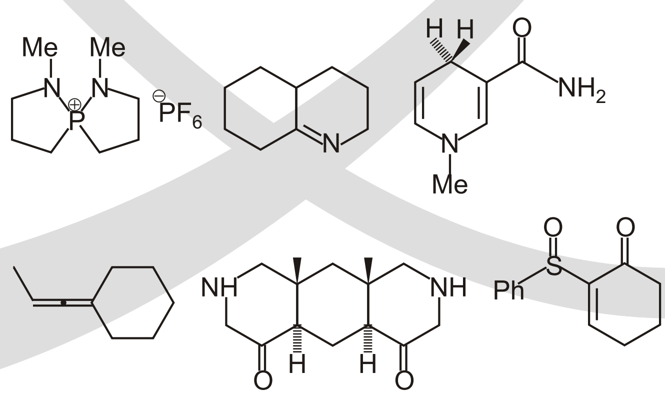
The fourth molecule is a bit of a trick. It needs to be redrawn to see if it has a plane of symmetry but when you did the redrawing your might not have noticed that the two naphithalene rings were joined at different positions. The molecule is chiral.



The last molecule is an interesting case. It is chiral but, if you got this one wrong, don't be too disappointed. Again, making a model will help but the vital thing is to realise that the CO_2H group is on a tetrahedral centre so the ring itself is not a plane of symmetry. The alkene puts the phenyl group to one side and a hydrogen atom to the other so the plane at right angles to the ring (dotted line) isn't a plane of symmetry either.



2. What makes molecule chiral ? Give three examples of different types of chirality. State with explanations whether the following compounds are chiral.



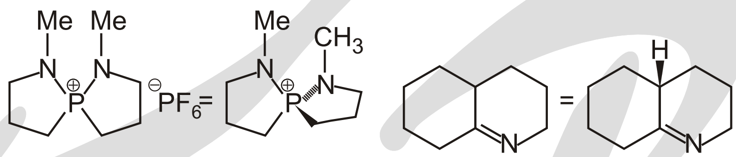
Purpose of the problem

Revision of the criterion for chirality with examples of the main classes of chiral molecules. Examstyle question.

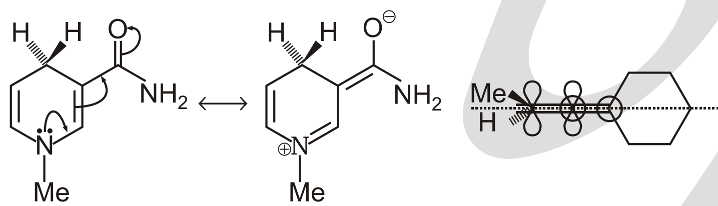
Suggested solution

Molecules are chiral if they have no plane of symmetry. This may arise form a tetrahedral atom with four different substituents or from a molecule that is forced to adopt shape that lacks a plane of symmetry. Examples include spiro compounds, axial chirality in allenes, chirality in allenes, chiral C, P, S, etc. You should give definite examples in this part of the answer, which are different from those given in the question, Ask someone to check if yours are all right.

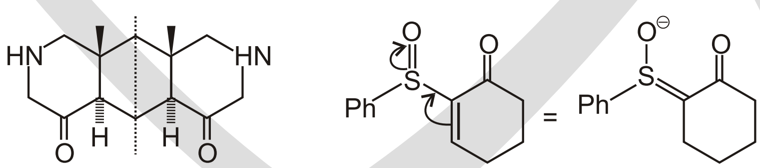
The phosphorus compound does not have a chiral phosphorus atom but the molecule is chiral because it is a spiro compound like the second molecule in the last question. The second molecule is nearly planar but the combination of a double bond and a tetrahedral centre at the other ring junction removes all possibility of a plane of symmetry. This too is chiral.



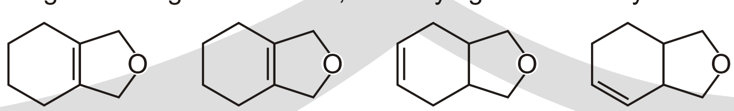
The third molecule tries to look chiral but it is almost planar because of conjugation, and the hydrogen atom above the plane reflects the hydrogen atom below it. The plane of the ring is a plane of symmetry and the molecule is not chiral. The fourth molecule is an allene with the two alkenes orthogonal to each other. It needs to be drawn more realistically to show the there is a plane of symmetry cutting the cyclohexane ring at right angles and passing through the methyl group on other end of the allene. Not chiral either.



The last two molecules are more straight forward. The tricyclic compound has a plane of symmetry vertically down the middle and is not chiral. The sulfoxide is a simple example of a stereogenic atom other than carbon. Sulfoxides are tetrahedral with the oxygen atom and the lone pair above and before the plane as drawn. This one is chiral.



3. Discuss the stereochemistry of these compounds. (Hint. This means saying how many diastereoisomers there are, drawing clear diagrams of each, and saying whether they are chiral or Not.)

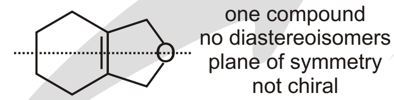


Motive of the Problem

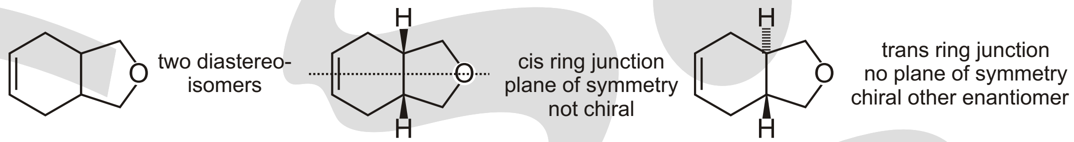
Making sure that you can handle this important approach to the stereochemistry of molecules.

Suggested solution

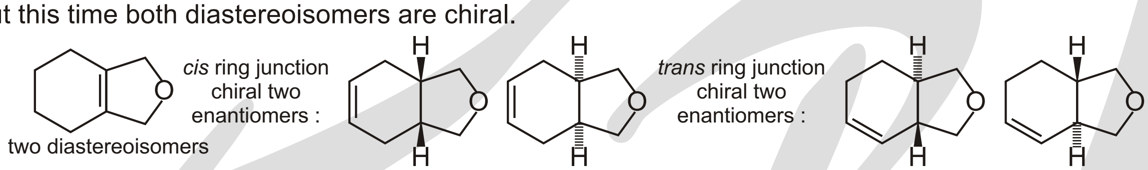
Just follow the hint given in the question! Diastereoisomers are different compounds so they must be distinguished first. Then it is easy to say if each diastereoisomer is chiral or not. The first two are simple.



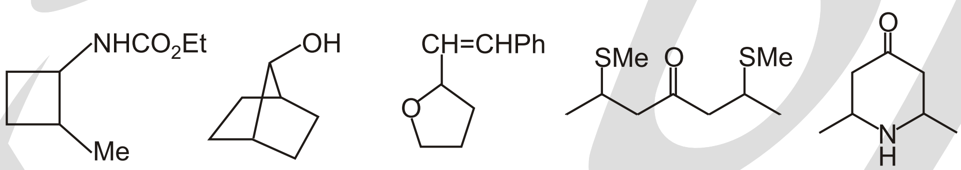
The third structure may exist as two diastereoisomers: one has a plane of symmetry (a meso compound) but the other is chiral (it has C_2 symmetry).



The last compound is most complicated as it has no symmetry. Again we can have a cis or trans ring junction but this time both diastereoisomers are chiral.



4. Discuss the stereochemistry of these compounds. The diagrams are deliberately poor ones that are ambiguous about stereochemistry – your answer should use good diagrams that shown the stereochemistry clearly.

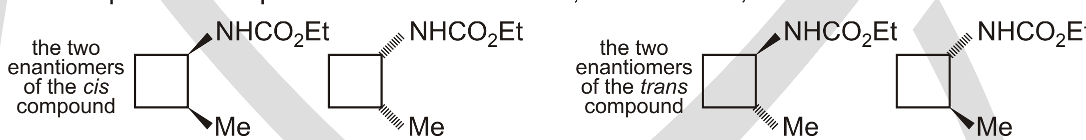


Purpose of the problem

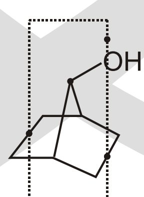
Practice at spotting stereochemistry and unravelling the different possible stereochemical relationships.

Suggested solution

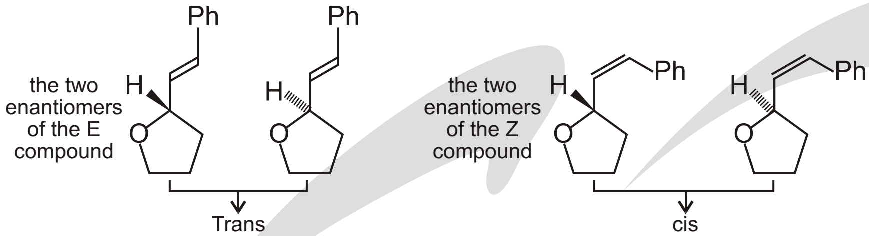
The first compound is simple : two diastereoisomers, cis and trans, both are chiral.



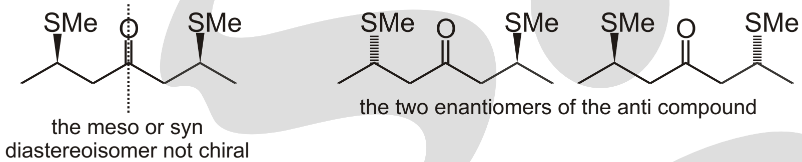
The second is simple too – the molecule has a plane of symmetry passing through the black dots and is not chiral. No diastereoisomers.



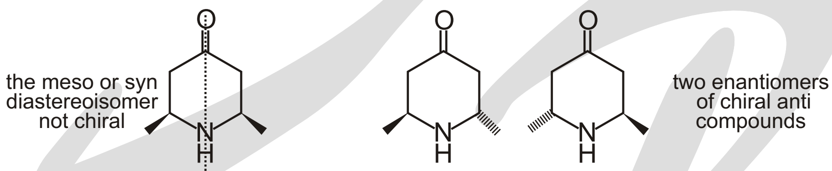
The third compound has two stereochemical units : an alkene that can be Z (*cis*) or E (*trans*) and the provides two different compounds, or diastereoisomers. There is also a chiral centre as each allene isomer has two enantiomers.



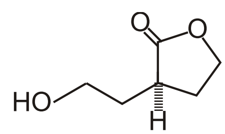
The fourth compound has some symmetry. There are two diastereoisomers with the MeS groups arranged *syn* or *anti* (as drawn). One has a plane of symmetry and is a *meso* compound while the other is chiral.



The fifth compound is similar : two diastereoisomers; one is chiral.



5. This compound racemizes in base. Why is that ?

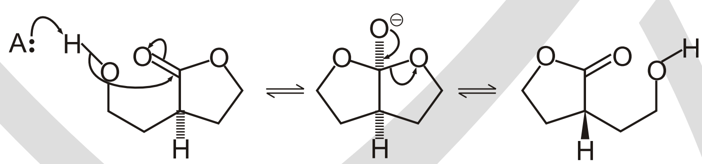


Purpose of the problem

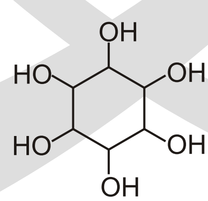
To draw your attention to the dangers in nearly symmetrical molecules and revision of ester exchange.

Suggested solution

Ester exchange in base goes through a symmetrical tetrahedral intermediate with a plane of symmetry. Loss of the right-hand leaving group gives one enantiomer of the ester and loss of the left-hand leaving group gives the other.



6. Just for fun, you might like to try and work out just how many diastereoisomers inosital has and how many of them are meso compounds,



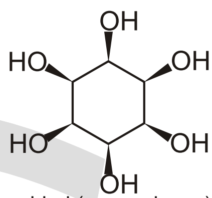
Purpose of the problem

Fun, it says! There is a more serious purpose in that the relationship between symmetry and stereochemistry is interesting and, in this human brain chemical, important to understand.

Suggested solution

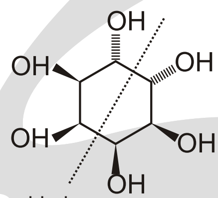
If we start with all the OH group on one side and gradually move them over, we should get the right answer. If you got too many diastereoisomers, check that some of yours aren't the same as other. There are either diasteroisomers altogether and, incredibly, all except one are achiral. Some have one, some two, and two of the most synmetrical have many planes of symmetry.

all OHs up



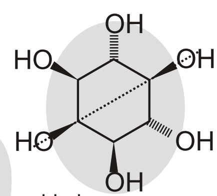
achiral (many planes)

four OHs up



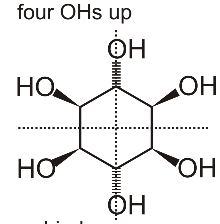
achiral

four OHs up



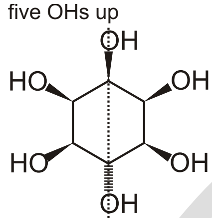
achiral

four OHs up



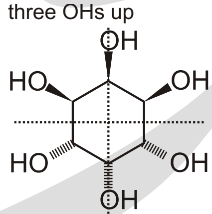
achiral

five OHs up



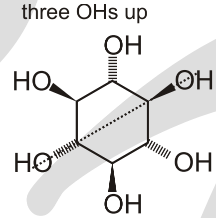
achiral

three OHs up



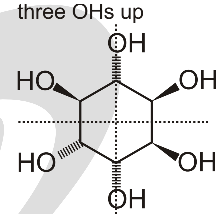
achiral

three OHs up



achiral (many planes)

three OHs up



chiral

7. What is meant by operators in terms of elements of symmetry ? What is order of symmetry operation ? Show the symmetry operators of chair form of cyclohexane.

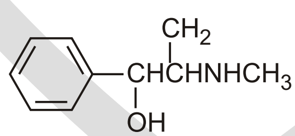
8. How many stereoisomers are possible for a molecule having the formula CA_4^* where A^* represent an asymmetric centre. Are all of them optically active?

9. If the compound C_{abcd} is assumed to be square-planar, then how many stereoisomers are possible? What are the stereochemical relationships among them? If each of the square planer structure assumes pyramidal structure then how many stereoisomers are possible and what is their stereochemical relationship.

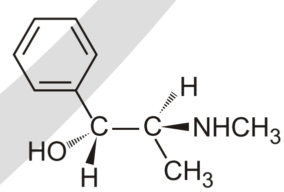
10. For many centuries, the chinese have used extracts from a group of herbs known as ephedra to treat asthma. A compound named ephedrine has been isolated from these herbs and found to be a potent dilator of air passage in the lungs.

(a) How many stereoisomers does ephedrine have?

(b) The stereoisomers shown here is the one that is pharmacologically active. What is the configuration of each of the asymmetric centres?



ephedrine

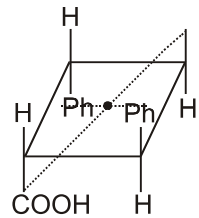


Answers

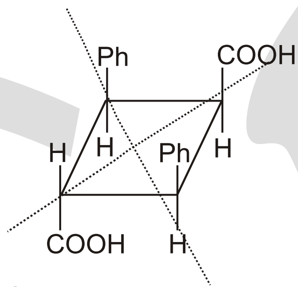
Match the Column

1. **D**
2. **A- S B-PS C-S D-RS**
3. A P, R, S ; B Q, S ; C P, Q, R, S ; D P, R, S
4. A – P,R,S ; B – P, Q, R, S ; C – P, R ; D – P

5. (A) Q; (B) P; (C) P,Q,S; (D) R,S]

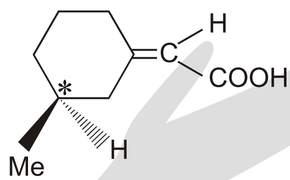
Sol. (I) 

Show G.I. no O.I. center of symmetry

(II) 

P.O.S. Show G.I. No. O.I.

(III) Chiral molecular No symmetry show O.I.

(IV) 

Chiral molecular No symmetry show O.I. Me is on the side of COOH show G.I.

6. A S ; B P, S ; C S ; D RS

FIND THE RELATIONSHIP

1. Enantiomer	**2.** Enantiomer	**3.** Identical	**4.** Identical
5. Enantiomer	**6.** Identical	**7.** Diastereomer	**8.** Diastereomer
9. Diastereomers	**10.** Enantiomer	**11.** Positional isomers	**12.** Diastereomers
13. Diastereomers	**14.** A, B diastereomers ; B, C diastereomers A, D Enantiomers ; A and C diastereomers		
15. Enantiomer	**16.** Identical	**17.** Identical	**18.** Diastereomer
19. Diastereomer	**20.** Constitutional isomer	**21.** Enantiomer (bc) ; Diastereomer (ab) & (ca)	
22. Identical	**23.** Enantiomer	**24.** Identical	**25.** Enantiomer
26. Enantiomer	**27.** Identical	**28.** Diastereomer	**29.** Enantiomer
30. Positional Isomer	**31.** Enantiomer	**32.** Identical	**33.** Diastereomer
34. Identical	**35.** Enantiomer	**36.** Diastereomer	**37.** Diastereomer
38. Enantiomer	**39.** Identical	**40.** Identical	**41.** Diastereomer
42. Diastereomer	**43.** Identical	**44.** Enantiomer	**45.** Diastereomer
46. Diastereomer	**47.** Enantiomer	**48.** Identical	**49.** Enantiomer
50. Enantiomers	**51.** Identical	**52.** Identical	**53.** Diastereomers
54. Consitutional isomers	**55.** Enantiomers	**56.** Consitutional isomers	
57. Enantiomers	**58.** Identical	**59.** Enantiomers	**60.** Consitutional isomers
61. Identical	**62.** Identical	**63.** Enantiomers	**64.** Different Compound
65. Identical	**66.** Enantiomers	**67.** Identical	**68.** Enantiomers
69. Identical	**70.** Diastereomers	**71.** Diastereomers	**72.** Consitutional isomers

73. Diastereomers **74.** Diastereomers **75.** Identical **76.** Enantiomer
77. Identical **78.** Identical **79.** Identical **80.** Diastereomer
81. Enantiomer **82.** Constitutional isomer **83.** Other
84. Constitution alisomer **85.** Diastereomer **86.** Enantiomer **87.** Enantiomer
88. Enantiomer **89.** Enantiomer **90.** enantiomers **91.** enantiomers.
92. enantiomers. **93.** Identical **94.** Different **95.** enantiomers.
96. enantiomers. **97.** Identicla **98.** diastereomers. **99.** enantiomers.
100. Identical **101.** Identical **102.** Identical **103.** Identical
104. enantiomers.

Unsolved Question

1. a, c, f **2.** A, C, E

3. A -Identical B- Enantiomer C- Diastereomers D- Identical **4.** iii, iv, vi

5. (a) Diastereomers (b) Enantiomers (c) Enantiomers (d) Constitutional isomers

6. (a) Enantiomers (b) Diastereomers

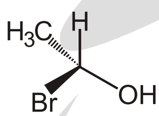
7. (a) R (b) 2R, 3S (c) 2R, 3S

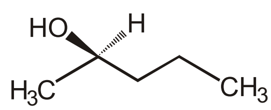
8. (a) Diastereomers (b) Constitutional isomers
(c) Constitutional isomers (d) Diastereomers

9. Nobecause it has C.O.S. **10.** (a) 2 (b) 1R, 2S

11. (a) Enantiomer (b) Identical (c) Identical (d) Identical

12. Total SI = 9 Meso =7

13. (i) 

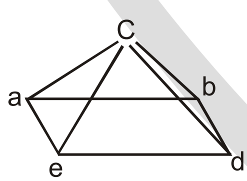
(ii) 

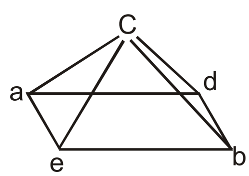
(iii) $CH_3\underset{\underset{Br}{|}}{C}HCH_2CH_3$

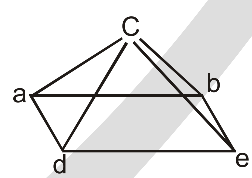
14. The first structure shown in part (a) has the S configuration and the second structure has the R configuration. Because they have opposite configurations, the structures represent a pair of enantiomers. All pairs are Eantiomers.

15. (a) chiral centers= 2 No. of O.I. = 22 (b) chiral centers= 3 No. of O.I. = 23
(c) chiral centers= 11 No. of O.I. = 211

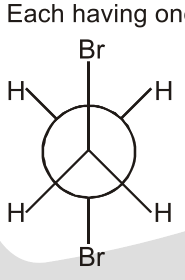
16. 6 or 8

Sol. 

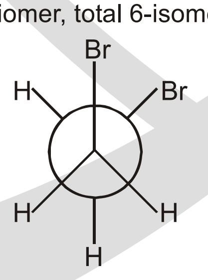




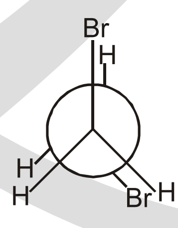
Each having one enantiomer, total 6-isomers.

17. 

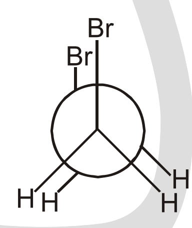
(1)



(0)



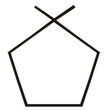
(0)



(2)

Ans. 3

18. 7 **19.** 8 **20.** 4 **21.** 3 **22.** 4 **23.** 4 **24.** 8

Sol. 
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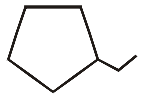
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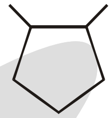
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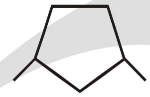
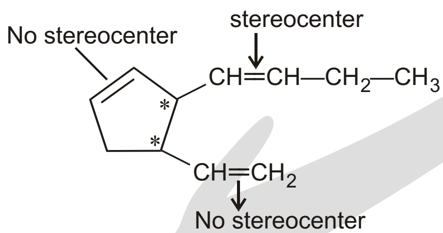

3

Special Problems

1. (C)	**2.** (B)	**3.** (C)	**4.** (B)	**5.** (C)	**6.** (B)	**7.** (A)	**8.** (B)
9. (D)	**10.** (B)	**11.** (C)	**12.** (B)	**13.** (B)	**14.** (A)	**15.** (B)	**16.** (B)
17. (B)	**18.** (C)	**19.** (C)	**20.** (D)	**21.** (A)	**22.** (C)	**23.** (A)	**24.** (B)
25. (C)	**26.** (C)	**27.** (C)	**28.** (C)	**29.** (A)	**30.** (D)	**31.** (D)	**32.** (D)
33. (B)	**34.** (D)	**35.** (C)	**36.** (B)	**37.** (A)	**38.** (B)	**39.** (A)	**40.** (C)
41. (A)	**42.** (C)	**43.** (B)	**44.** (C)	**45.** (D)	**46.** (D)	**47.** (B)	**48.** (D)
49. (A)	**50.** (C)	**51.** (A)					

51. 

Hence, total stereoisomer = 23 = 8]

52. Ans. 4.32°]

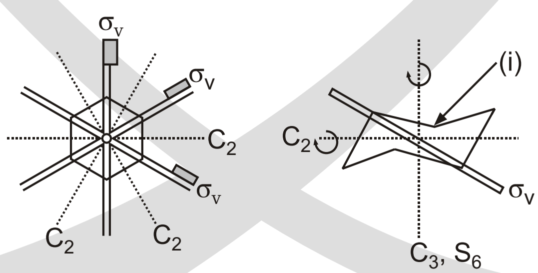
Racemic 32%, (+) 36%]

Subjective Type Questions

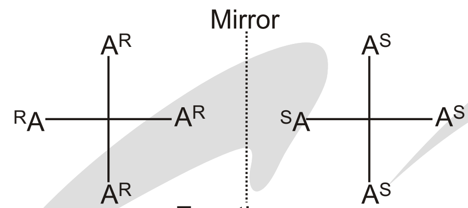
7. If a structure of a molecule can be transformed into an identical or indistinguishable structure by the physical movement based on an element of symmetry, without breaking O , deforming any part of the molecule, then that manipulation is called a symmetry operation.

The order of a symmetry operation is the number of total operations that can be done to convert a structure to its equivalent/identical structure.

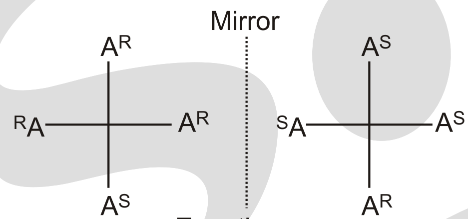
The chair form of cyclohexane belongs to the point group D_{3d}. Its elements of symmetry are C_3, $3C_2$, 3 , (diagonal) planes, i, S_6. Its operation of identity $E(C_1)$ is also a symmetry operation. Moreover two times C_3 operation is an identify operation (E). It can also be confirmed that five times operation of S_6 is also an operation of identify. Therefore, the order of symmetry operation of chair form of cyclohexane is 12. They are E, $C_3^1, C_3^2, 3C_2, 3S_v i, S_6^1, S_6^5$. The structure of the chair form of cyclohexane is as follows.



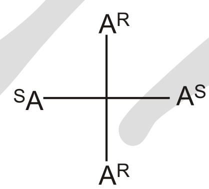
8. The molecule represented as CA_4* can have two pairs of enantiomers and one meso-compound, the asymmetric substituents can have both R and S configurations. The Fisher projection of stereoisomers in a perpendicular mirror plane can be shown as follows.



Enantiomers

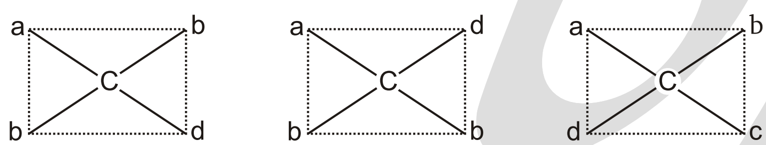


Enantiomers



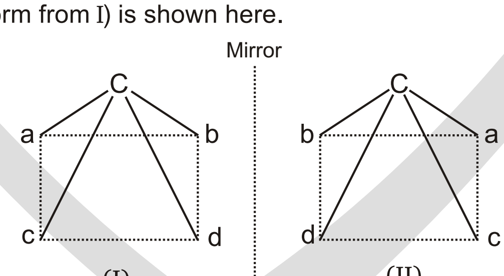
meso form

9. When the compound C_{abcd} assumed square-planar structure then three stereoisomers are possible. Since all of them have planar structure, none of them is chiral. Stereochemically they are diastereoisomers. Structures are shown here.



(I) (II) (III)

When each of these square-planar structures is converted to pyramidal structure with C at the apex then three pairs of enantiomers are formed, that is, six stereoisomers are formed. Only one pair of enantiomers (transformed to pyramid form from I) is shown here.



(I) (II)

A pair of enantiomers

❑❑❑

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CHAPTER

27

Basic Organic Chemistry

WHY CHEMICAL REACTION TAKE PLACE ?

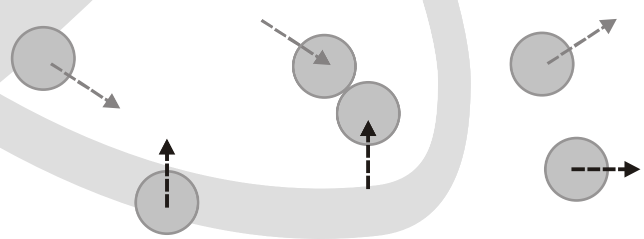
Chemical reactions

Most molecules are at peace with themselves. Bottles of water, or acetone (propanone, Me_2C O), or methyl iodide (iodomethane CH_3I) can be stored for years without any change in the chemical composition of the molecules inside. Yet when we add chemical reagents, say, HCl to water, sodium cyanide (NaCN) to acetone, or sodium hydroxide to methyl iodide, chemical reactions occur. This chapter is an introduction to the reactivity of organic molecules: why they don't and why they do react; how we can understand reactivity in terms of charges and orbitals and the movement of electrons; how we can represent the detailed movement of electrons—the mechanism of the reaction—by a special device called the curly arrow.

Molecules react because they move. When two molecules bump into each other, they may combine with the formation of a new bond, and a chemical reaction occurs. We are first going to think about collisions between molecules.

Not all collisions between molecules lead to chemical change

All organic molecules have an outer layer of many electrons, which occupy filled orbitals, bonding and nonbonding. Charge–charge repulsion between these electrons ensures that all molecules repel each other. Reaction will occur only if the molecules are given enough energy (the activation energy for the reaction) for the molecules to pass the repulsion and get close enough to each other. If two molecules lack the required activation energy, they will simply collide, each bouncing off the electrons on the surface of the other and exchanging energy as they do so, but remain chemically unchanged. This is rather like a collision in snooker or pool. Both balls are unchanged afterwards but are moving in different directions at new velocities.



on collision course impact after collision

Electron flow is the key to reactivity

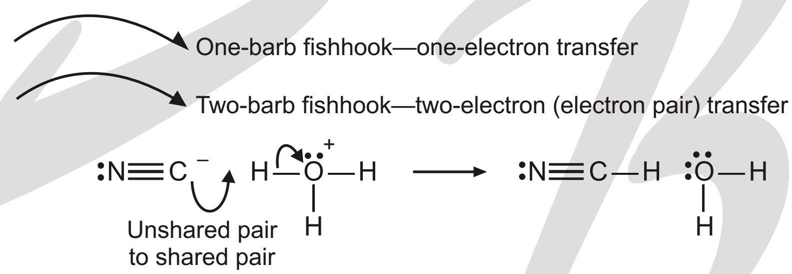
The vast majority of organic reactions are polar in nature. That is to say, electrons flow from one molecule to another as the reaction proceeds. The electron donor is called a nucleophile (nucleus loving) while the electron acceptor is called the electrophile (electron-loving). These terms come from the idea of charge attraction as a dominating force in reactions. The nucleophile likes nuclei because nucleus is positively charged and the electrophile likes electrons because electrons are negatively charged. Though we no longer regard reactions as controlled only by charge interactions, these names have stuck.

► **Molecules repel each other because of their outer coatings of electrons.**
Molecules attract each other because of :
- attraction of opposite charges
- overlap of high-energy filled orbitals with low-energy empty orbitals

For reaction, molecules must approach each other so that they have :
- enough energy to overcome the repulsion
- the right orientation to use any attraction

CURVED ARROWS

FISHHOOKERY



Mechanisms are your key to success in this course. If you can master the mechanisms, you will do very well in this class. If you don't master mechanisms, you will do poorly in this class. What are mechanisms and why are they so important?

When two compounds react with each other to form new and different products, we try to understand how the reaction occurred. Every reaction involves the flow of electron density—electrons move to break bonds and form new bonds. Mechanisms illustrate how the electrons move during a reaction. The flow of electrons is shown with curved arrows.

These arrows show us how the reaction took place. For most of the reactions that you will see this semester, the mechanisms are well understood (although there are some reactions whose mechanisms are still being debated today). You should think of a mechanism as ''bookkeeping of electrons.'' Just as an accountant will do the bookkeeping of a company's cash flow (money coming in and money going out), the mechanism of a reaction is the bookkeeping of the flow of electrons.

When you understand a mechanism, you will understand why the reaction took place, why the stereocenters turned out the way they did, and so on. If you do not understand the mechanism, then you will find yourself memorizing the exact details of every single reaction.

In this chapter, we will not learn every mechanism that you need to know. Rather, we will focus on the tools that you need to properly read a mechanism and abstract the important information. You will learn some of the basic ideas behind arrow pushing in mechanisms, and these ideas will help you conquer the early mechanisms that you will learn.

CURVED ARROWS

We have already gotten quite a bit of experience with curved arrows in chapter 2 (Resonance). The curved arrows that we use in mechanisms refer to the actual movement of electrons. Electrons are moving to break and form bonds (hence the term chemical reaction).

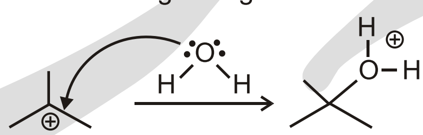
Let's just have a quick review of curved arrows, and the different types of arrows that you can draw. Every curved arrow has a head and a tail. It is essential that the head and tail of every arrow be drawn in the proper place. The tail shows where the electrons are coming from, and the head shows where the electrons are going :

Therefore, there are only two things that you have to get right when drawing each arrow. The tail needs to be in the right place and the head needs to be in the right place. Remember that electrons exist in orbitals, either as lone pairs or as bonds. So the tail of an arrow can only come from a bond or from a lone pair: The head of an arrow can only be drawn to make a bond or to make a lone pair: In total, this gives us four possibilities :

1. Lone pair bond
2. Bond lone pair
3. Bond bond
4. Lone pair lone pair

From a Lone Pair to a Bond

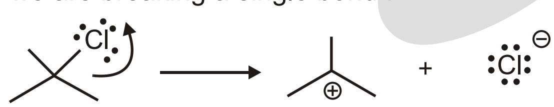
Consider the step below, where we are forming a single bond :



The tail of the arrow is coming from a lone pair on the oxygen atom, and the head of the arrow is going to form a bond between oxygen and carbon. Since the head of the arrow is placed on an atom, it might seem like the electrons are going from a lone pair to a lone pair, but they are not. The electrons are going from the oxygen lone pair to form a bond to the carbon atom. If this makes you unhappy, there is an alternative way of drawing the arrow that shows it more clearly:

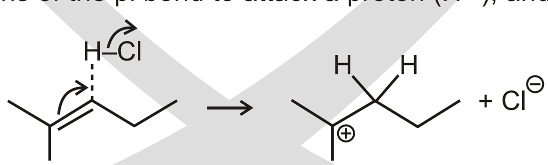
From a Bond to a Lone Pair

Consider the step below, where we are breaking a single bond :



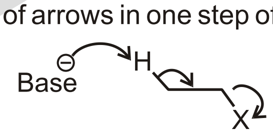
The tail of the arrow is on a bond, and the head of an arrow is forming a lone pair on the chlorine atom. The two electrons of the bond used to be shared between the carbon and the chlorine atoms. But now, both electrons are going on the chlorine. So the carbon has lost an electron, and the chlorine has gained one. This is why the carbon ends up with a positive charge, and the chlorine gets a negative charge.

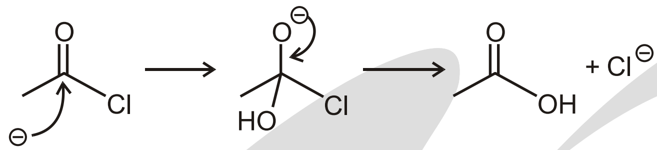
By the way, a chlorine atom with a negative charge is called a chloride ion (-ide- implies the negative charge). So in this reaction chloride is popping off of the molecule to form a carbocation (a carbon with a positive charge). Where we are using the electrons of the pi bond to attack a proton (H), and kicking off Cl in the process:



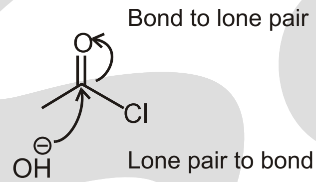
The first arrow has its tail on the pi bond and the head is being used to form a bond between a carbon atom and the proton.

In fact, it is possible to have all three types of arrows in one step of a mechanism. Consider the example below :

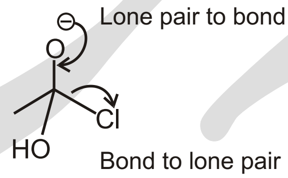




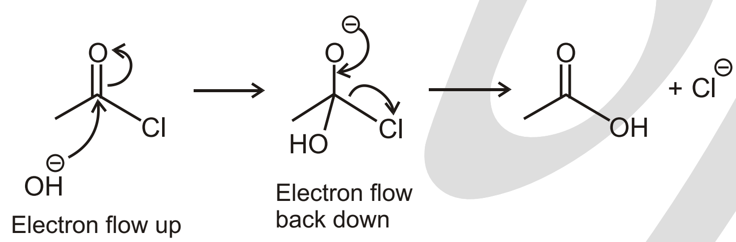
This type of reaction will be covered much later on in your course, but let's use it now as an example. Notice that there are two steps to this mechanism. In the first step, we have two arrows: from a lone pair to form a bond, and then from a bond to form a lone pair:



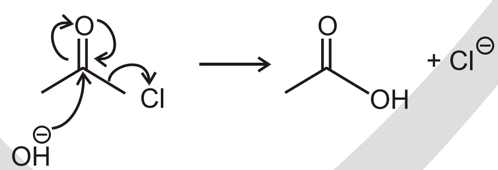
In the second step of the mechanism, we also have two arrows: from a lone pair to form a bond, and then from a bond to form a lone pair:



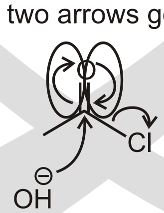
If we consider the overall reaction, we notice that the OH is replacing the Cl. If we look at how the electrons flowed, we see that it all started at the negative charge of the attacking OH . This charge flowed up temporarily on to the oxygen atom of the C O in step I of the mechanism, and then the charge flowed back down to kick off Cl :



When we consider how the charge flowed throughout the whole reaction, it might be tempting to draw it all in one step, like this :

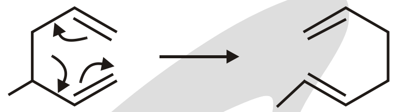


However, this is no good, because we have two arrows going in opposite directions:



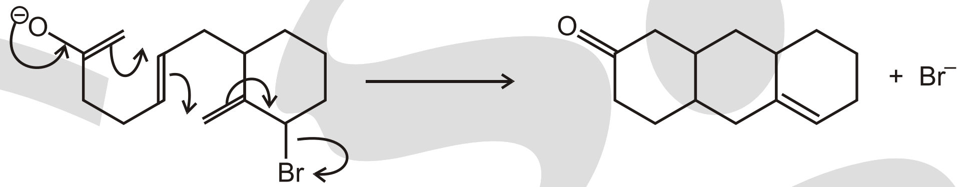
Never draw arrows in opposite directions. That would imply that the electrons were flowing in opposite directions at the same time. That is not possible. In this reaction,

INTRODUCTION



- Illustration of arrow pushing applied to the Cope rearrangement.
- Application of arrow pushing to homolytic cleavage using single-barbed arrows.
- Application of arrow pushing to heterolytic cleavage using double-barbed arrows.

(a)



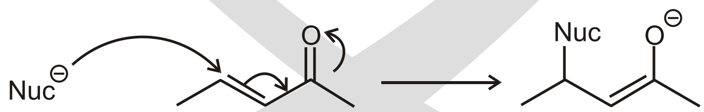
INTERMEDIATES

Drawing Intermediates

We have seen the different types of arrows and how to draw them. Now we need to get practice drawing intermediates when we are given the arrows. Intermediates are compounds that exist for a very short time before reacting further. Let's consider an analogy. Imagine that you are trying to climb a mountain and it is very cold (below freezing). You are wearing a hat that keeps your ears warm, but it is loose and keeps slipping off. Your friend offers you a spare hat that he brought, and you borrow it. Now you need to take your old hat off to replace it with the new hat. If someone were to take a picture of you while you have nothing on your head, the picture would look very strange. There you are, in the freezing cold, with no hat on. You were only like that for 3 seconds, but it was long enough for someone to take a picture. Intermediates of reactions are similar.

Intermediates are intermediate structures in going from the starting material to the product. They do not live for very long, and it is rare that you can isolate one and store it in a bottle, but they do exist for very short periods of time. Their structures are often critical in understanding the next step of the reaction. Going back to the analogy, if I saw the picture of you without your hat on, and I knew how cold it was on that mountain, then I would have been able to predict that you put on a hat right after the picture was taken. I would have known this because I would have been able to immediately identify an uncomfortable situation, and I could have predicted what resolution must have taken place to alleviate the problem. The same is true of intermediates. If we can look at an intermediate and determine which part of the intermediate is unstable, and we also know what options are available to alleviate the instability, then we can predict the products of the reaction based on an analysis of the intermediate. That's why they are so important.

Let's read the arrows. The first arrow is from a lone pair to form a bond. The arrow shows electrons in a lone pair on a nucleophile (anything that is electron rich) forming a bond with a carbon atom. The second arrow is from a bond to a bond. The third arrow goes from a bond to form a lone pair. All in all, these arrows serve as a road map for drawing the intermediate:



Solved Example

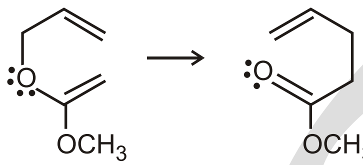
- Use arrow pushing to explain the following reactions :

  When drawing arrows to illustrate movement of electrons, it is important to remember that electrons form the bonds that join atoms. The following represent heterolytic-type reaction mechanisms:

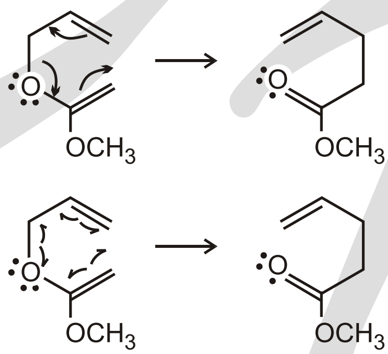
  (a) $N \equiv C^{\ominus} + H_3C - I \longrightarrow N \equiv C - CH_3 + I^{\ominus}$

Sol. This is an example of an S_N2 reaction mechanism converting an alkyl iodide (iodomethane) to an alkyl nitrile (acetonitrile). Arrow pushing is illustrated below :

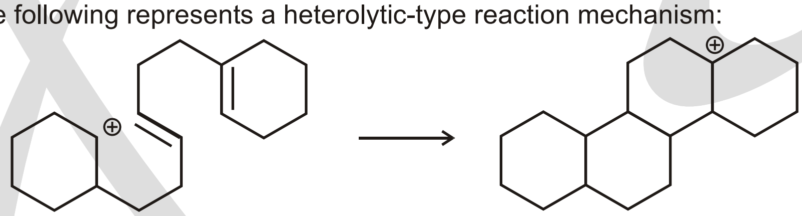
$$N\equiv \overset{\ominus}{C} + H_3C-I \longrightarrow N\equiv C-CH_3 + \overset{\ominus}{I}$$

(b) 

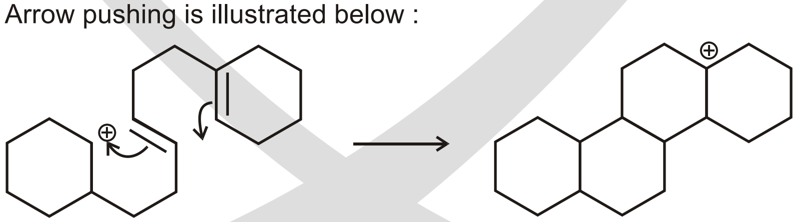
Sol. This is an example of a Claisen rearrangement and occurs through a concerted reaction mechanism. As illustrated, concerted mechanisms can be described either by movement of electron pairs or by movement of single electrons. However, these mechanisms are generally represented by movement of electron pairs using double-barbed arrows as is done for heterolytic reaction mechanisms. Although, mechanistically, the movement of electron pairs is preferred over the movement of single electrons, both processes are illustrated below using arrow pushing :



The following represents a heterolytic-type reaction mechanism:

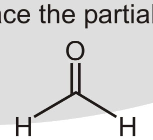
(c) 

Sol. This is an example of a cation-*p* cyclization. Note that unlike the previously described heterolytic reaction mechanisms, this reaction is influenced by a positive charge. Also, please note that this reaction shares some characteristics with concerted mechanisms in that formation of the new bonds occurs almost simultaneously. Arrow pushing is illustrated below :

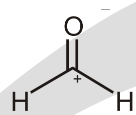


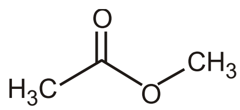
Solved Example

- Place the partial charges on the following molecules.

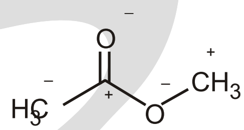
(a) 

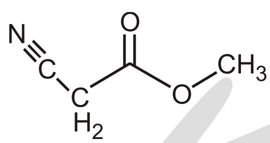
Sol. Carbonyls are polarized such that a partial negative charge resides on the oxygen and a partial positive charge resides on the carbon.



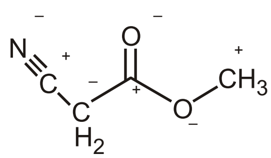
(b) 

Sol. Because of the polarity of the carbonyl, adjacent groups are also polarized. In general, where a partial positive charge rests, an adjacent atom will bear a partial negative charge. This can occur on more than one adjacent atom or heteroatom.



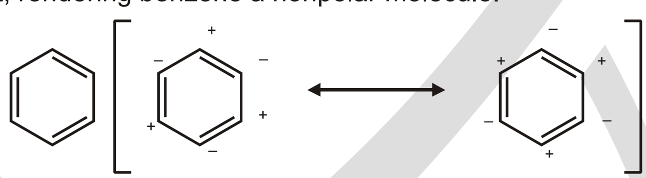
(c) 

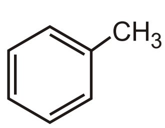
Sol. Nitriles, like carbonyls, are polarized with the nitrogen bearing a partial negative charge and the carbon possessing a partial positive charge.



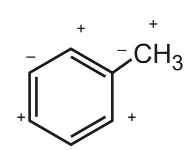
(d) 

Sol. Benzene has no localized positive or negative charges because of its symmetry. The two illustrated resonance forms are equivalent, rendering benzene a nonpolar molecule.

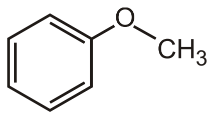


(e) 

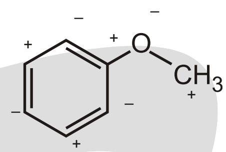
Sol. Methyl groups are electron donating. This is not due to any defined positive charges on the carbon atom and is more the result of hyperconjugation. Hyperconjugation, in this case, relates to the ability of the carbon-hydrogen s bonds of the methyl group to donate electrons into the conjugated system of benzene. While this effect will be discussed in more detail later, let us, for now, define methyl groups as possessing a formal partial positive charge. This resulting positive charge thus polarizes each double bond in the ring.

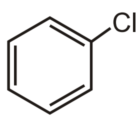


❑ **NOTE :** This is not a hybrid.

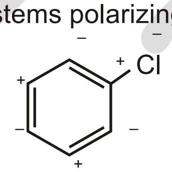
(f) 

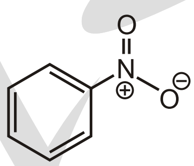
Sol. As with the previous example, groups possessing partial negative charge characteristics donate electrons into conjugated systems and polarize the double bonds. This effect is generally noted with heteroatoms such as oxygen. Also, while in the previous example a methyl group was argued to possess a partial negative charge, the partial positive charge illustrated here is due to the overriding partial negative characteristics of the oxygen atom.



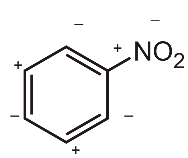
(g) 

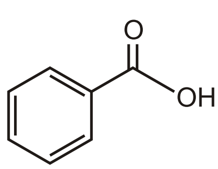
Sol. As with the previous example, heteroatoms such as chlorine possess partial negative charge characteristics and donate electrons into conjugated systems polarizing the double bonds.



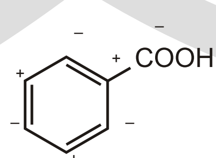
(h) 

Sol. As with groups possessing negative charge characteristics, when a positive charge is present on an atom connected to a conjugated system, the double bonds are polarized. This polarization is opposite of that observed for negatively charged groups.



(i) 

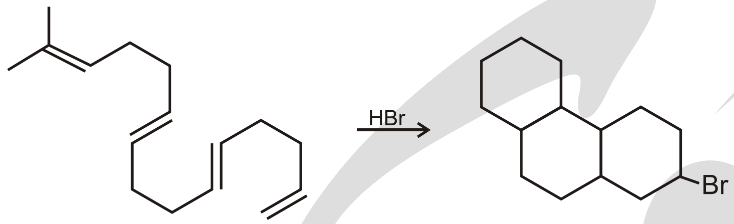
Sol. As with groups possessing negative charge characteristics, when a partial positive charge is present on an atom connected to a conjugated system, the double bonds are polarized. This polarization is opposite that observed for negatively charged groups.



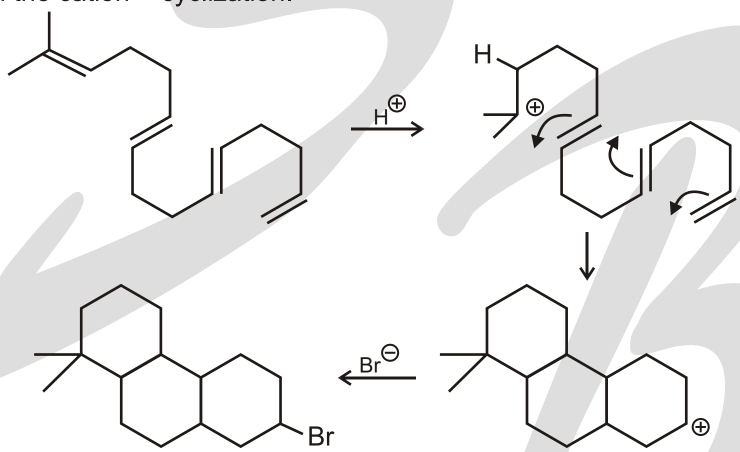
Solved Example

- Explain the following reactions in mechanistic terms. Show arrow pushing.

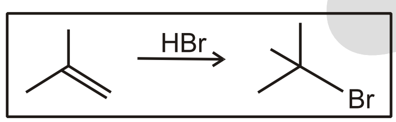
(a)



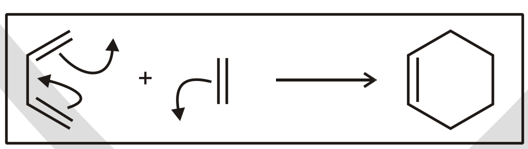
Sol. As presented in this chapter, olefins can become protonated under acidic conditions, leading to the formation of electrophilic and cationic carbon atoms. Furthermore, because olefins have nucleophilic character, they can add to sites of positive charge. The cascading of this mechanism, illustrated below, generates polycyclic systems through the cation- cyclization.



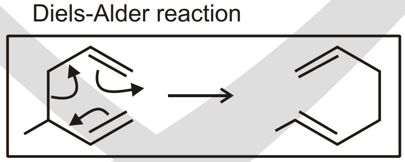
MOVING FORWARD



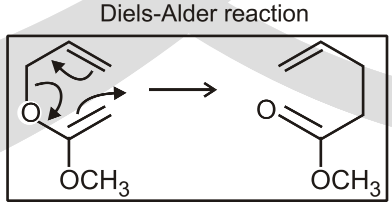
Markovnikov addition of hydrobromic acid across a double bond.



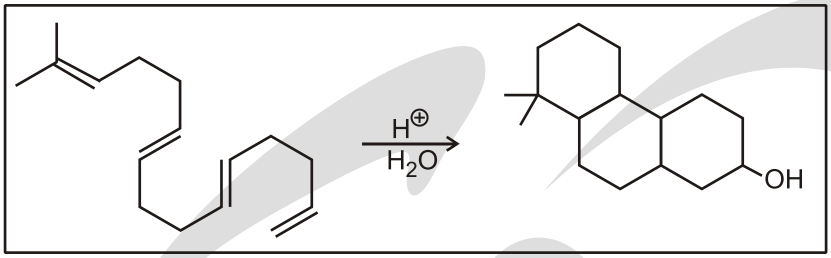
Diels-Alder reaction



Diels-Alder reaction

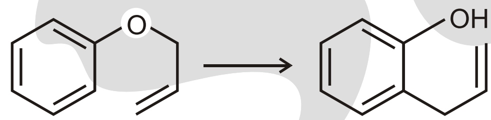


Diels-Alder reaction

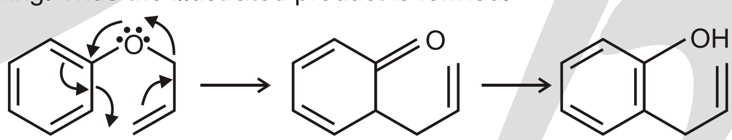


Cation- cyclization

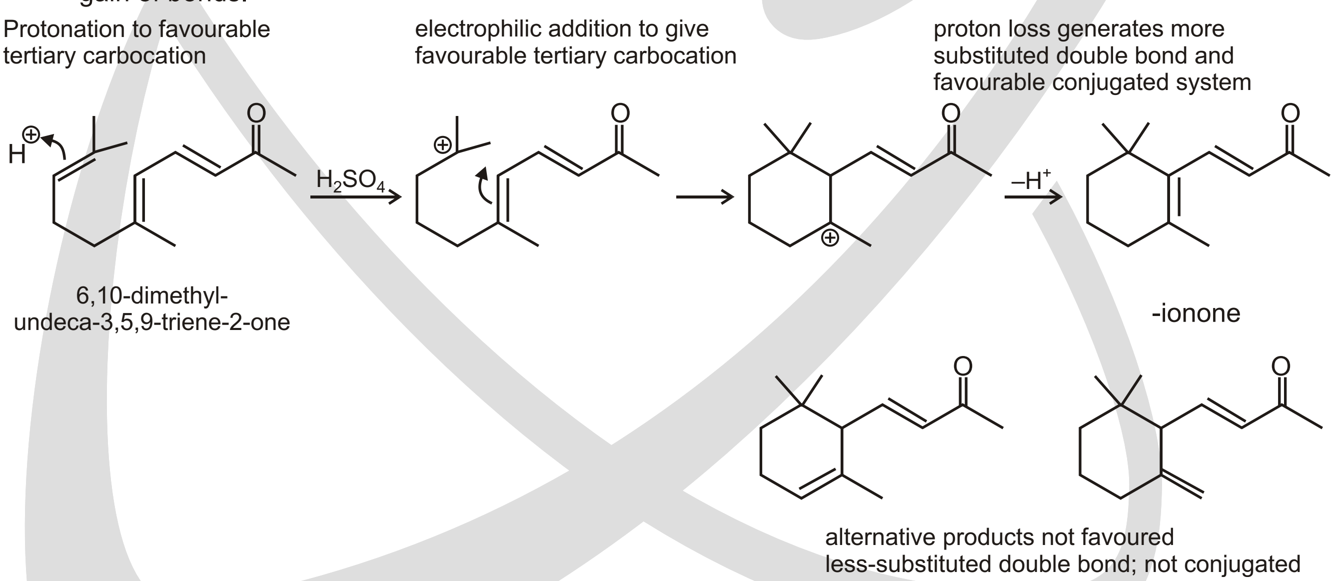
Solved Example



Sol. This is an example of a Claisen rearrangement which is an electrocyclic reaction where no charges are involved. While no charges are involved, like the Diels-Alder reaction, electron pairs do move and their movement can be illustrated using arrow pushing. The mechanism, illustrated below, involves moving a lone pair of electrons from the oxygen into the aromatic ring. The aromatic ring then adds electrons to the double bond. The double bond then migrates and the carbon-oxygen bond is cleaved. While the expected product may be the illustrated ketone, spontaneous conversion to the enol form is facilitated by the stability of the resulting aromatic ring. Thus the illustrated product is formed.

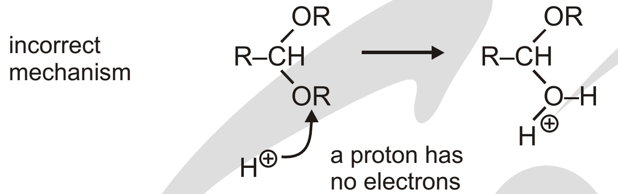


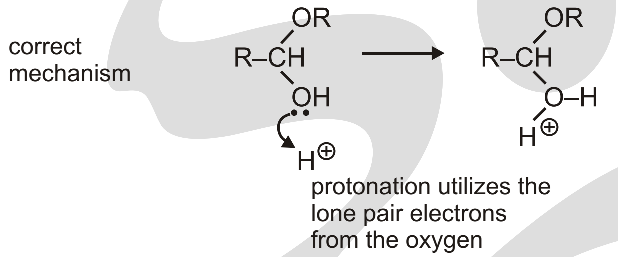
When considering the above mechanistic description, it is important to recognize that all of these steps occur concurrently. Furthermore, like the Diels-Alder reaction (and all electrocyclic reactions), there is no net loss or gain of bonds.



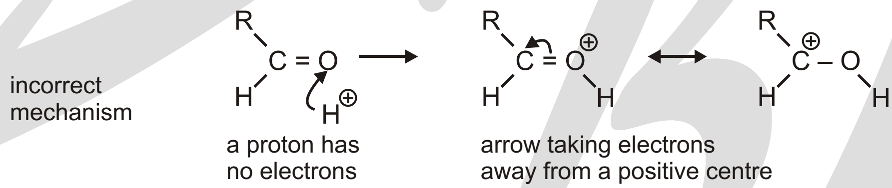
SOME COMMON MISTAKES IN DRAWING MECHANISMS

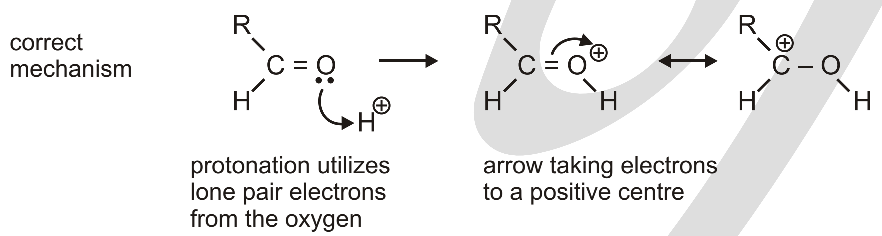
Arrows from protons Ask yourself how many electrons are there in a proton ? We trust the answer is none, and you will thus realize that arrows representing movement of electrons can never ever start from a proton. It seems that this mistake is usually made because, if one thinks of protonation as addition of a proton, it is tempting to show the proton being put on via an arrow. With curly arrows, we must always think in terms of electons.



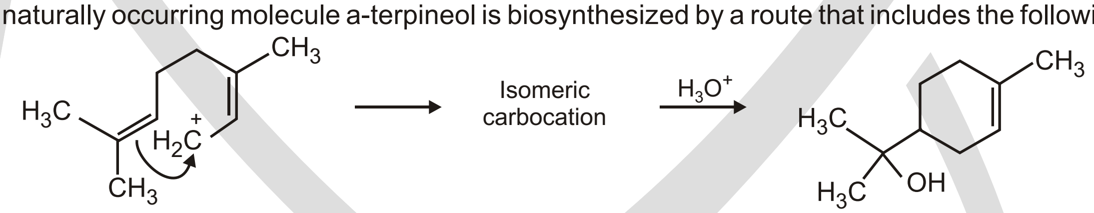


We were even less keen on the second example, where, in the resonance delocalization step, an arrow is shown taking electrons away from a positive charge and creating a new positive centre.





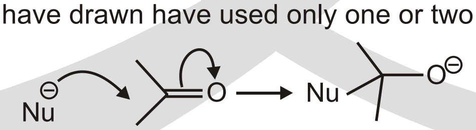
- The naturally occurring molecule a-terpineol is biosynthesized by a route that includes the following step :



Carbocation -Terpineol

Curly arrows also show movement of electrons within molecules

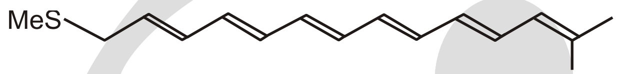
So far all the mechanisms we have drawn have used only one or two



arrows in each step. In fact, there is no limit to the number of arrows that might be involved and we need to look at some mechanisms with three arrows. The third arrow in such mechanismsusually represents movement of electrons inside of the reacting molecules. Some pages back we drew out the addition of a nucleophile to a carbonyl compound.

Mostly for entertainment value we shall end this section with a mechanism involving no fewer than eight arrows. See if you can draw the product of this reaction without looking at the result.

The first arrow forms a new C — S bond and the last arrow breaks a C — Br bond but all the rest just move bonds along the molecule. The product is therefore:



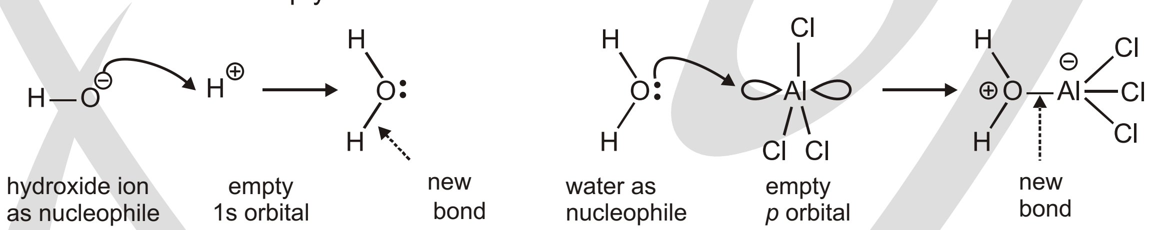
Now for a real test: can you draw a mechanism for this reaction?

ELECTROPHILE AND NUCLEOPHILE

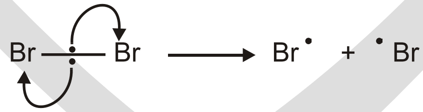
Organic chemists use curly arrows to represent reaction mechanisms

You have seen several examples of curly arrows so far and you may already have a general idea of what they mean. The representation of organic reaction mechanisms by this means is so important that we must now make quite sure that you do indeed understand exactly what is meant by a curly arrow, how to use it, and how to interpret mechanistic diagrams as well as structural diagrams.

A curly arrow represents the *actual movement of a pair of electrons* from a filled orbital into an empty orbital. You can think of the curly arrow as representing a pair of electrons thrown, like a climber's grappling hook, across from where he is standing to where he wants to go. In the simplest cases, the result of this movement is to form a bond between a nucleophile and an electrophile. Here are two examples we have already seen in which lone pair electrons are transferred to empty atomic orbitals.



These three examples all have the leaving group taking both electrons from the old s bond. This type of decomposition is sometimes called **heterolytic fission** or simply heterolysis and is the most common in organic chemistry. There is another way that a bond can break. Rather than a pair of electrons moving to one of the atoms, one electron can go in either direction. This is known as **homolytic fission** as two species of the same charge (neutral) will be formed. It normally occurs when similar or

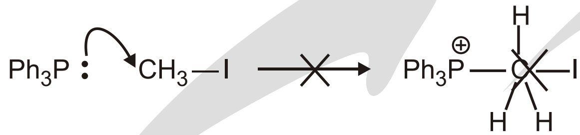


indeed identical atoms are at each end of the bond to be broken. Both fragments have an unpaired electron and are known as radicals. This type of reaction occurs when bromine gas is subjected to sunlight.

The weak Br–Br bond breaks to form two bromine **radicals**. This can be represented by two single-headed curly arrows, **fish hooks**, to indicate that only one electron is moving. This is virtually all you will see of this special type of curly arrow until we consider the reactions of radicals in more detail. When you meet a new reaction you should assume that it is an ionic reaction and use two-electron arrows unless you have a good reason to suppose otherwise.

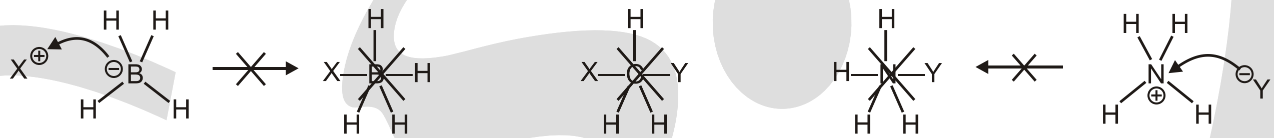
☞ **Warning! Eight electrons is the maximum for B, C, N, or O**

We now ought to spell out one thing that we have never stated but rather assumed. Most organic atoms, if they are not positively charged, have their full complement of electrons (two in the case of hydrogen, eight in the cases of carbon, nitrogen, and oxygen) and so, if you make a new bond to one of those elements, you must also break an existing bond. Suppose you just 'added' Ph_3P to MeI in this last example without breaking the C — I bond: what would happen?



wrong mechanism

impossible structure
carbon has five bonds



impossible reaction | impossible structure boron has five bonds | impossible structure carbon has five bonds | impossible structure nitrogen has five bonds | impossible reaction

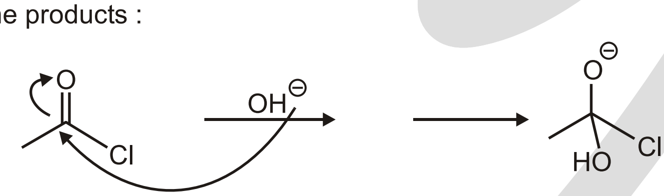
Nucleophiles and Electrophiles

Whenever one compound uses its electrons to attack another compound, we call the attacker a nucleophile, and we call the compound being attacked an electrophile. It is very simple to tell the difference between an electrophile and a nucleophile. You just look at the arrows and see which compound is attacking the other. A nucleophile will always use a region of high electron density (either a lone pair or a bond) to attack the electrophile (which, by definition, has a region of low electron density that can be attacked). These are important terms, so let's make sure we know how to identify nucleophiles and electrophiles.

Solved Example

- Look at the arrows below, and draw the intermediate that you get after pushing the arrows:

Ans. We need to read the arrows like a road map: the first arrow is going from a lone pair on HO to form a bond with the carbon of the C O. The second arrow goes from the C O bond to form a lone pair on oxygen. We use this info to draw the products :



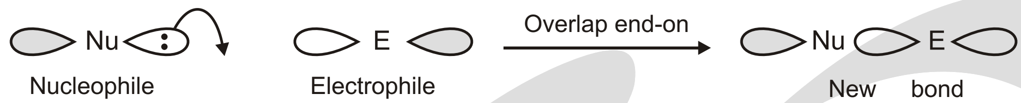
The hard part was assigning formal charges. Notice that we had two arrows moving in a flow. We had a negative charge in the beginning, so we must have a negative charge in the end. It started off on the first atom in the flow of arrows, and it ended on the last atom of the flow (the oxygen).

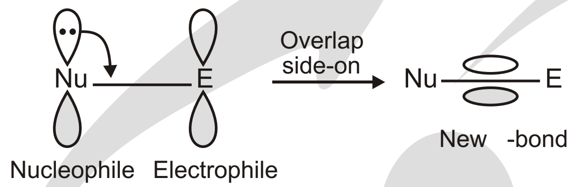
Orbital overlap and energy

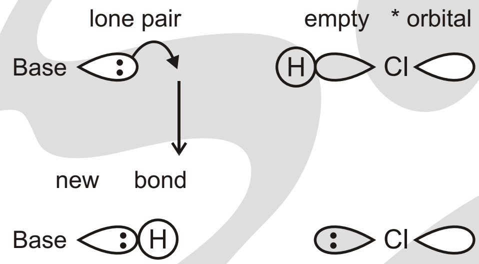
Two atomic orbitals can combine to give two molecular orbitals – one bonding molecular orbital (lower in energy than the atomic orbitals) and one antibonding molecular orbital (higher in energy than the atomic orbitals). Orbitals that combine in-phase form a bonding molecular orbital and for best orbital overlap, the orbitals should be of the same size.

The orbitals can overlap end-on (as for s-bonds) or side-on (as for -bonds). The empty orbital of an electrophile (which accepts electrons) and the filled orbital of a nucleophile (which donates electrons) will point in certain directions in space. For the two to react, the filled and empty orbital must be correctly aligned; for end on overlap, the filled orbital should point directly at the empty orbital.

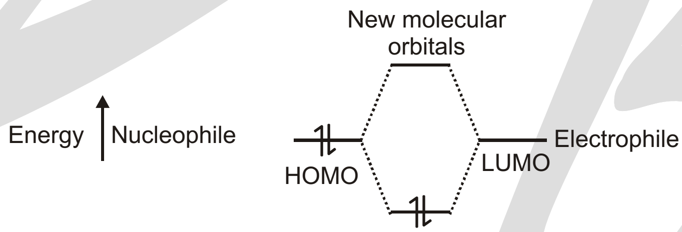
Molecules must approach one another so that the filled orbital of the nucleophile can overlap with the empty orbital of the electrophile







The orbitals must also have a similar energy. For the greatest interaction, the two orbitals should have the same energy. Only the highest-energy occupied orbitals (or HOMOs) of the nucleophile are likely to be similar in energy to only the lowest-energy unoccupied orbitals (or LUMOs) of the electrophile.

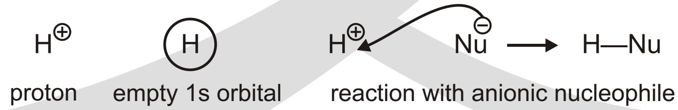


The two electrons enter a lower energy molecular orbital.
There is therefore a gain in energy and a new bond is formed.
The further apart the HOMO and LUMO, the lower the gain in energy.

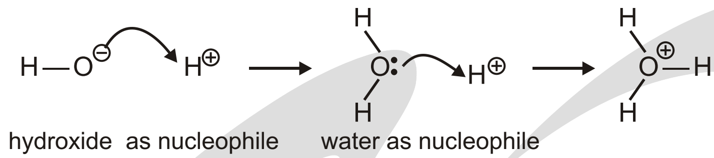
- The HOMO of a nucleophile is usually a (non-bonding) lone pair or a (bonding) -orbital. (These are higher in energy than a -orbital).
- The LUMO of an electrophile is usually an (antibonding) *-orbital. (This is lower in energy than a * orbital.)
- In reaction mechanisms
- **Nucleophiles donate electrons**
- **Electrophiles accept electrons**

Electrophiles have a low-energy vacant orbital

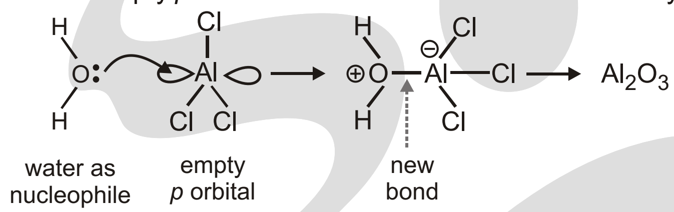
Electrophiles are neutral or positively charged species with an empty atomic orbital (the opposite of a lone pair) or a low-energy antibonding orbital. The simplest electrophile is the proton, H , a species without any electrons at all and a vacant 1s orbital. It is so reactive that it is hardly ever found and almost any nucleophile will react with it.



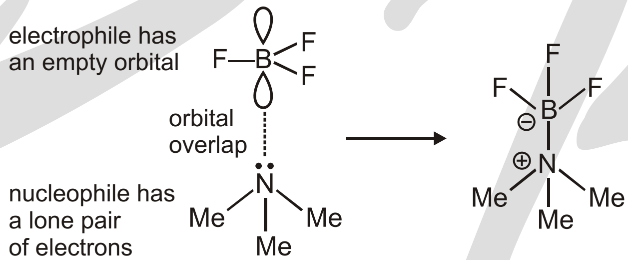
Each of the nucleophiles we saw in the previous section will react with the proton and we shall look at two of them together. Hydroxide ion combines with a proton to give water. This reaction is governed by charge control. Then water itself reacts with the proton to give H_3O , the true acidic species in all aqueous strong acids.



We normally think of protons as acidic rather than electrophilic but an acid is just a special kind of electrophile. In the same way, Lewis acids such as BF_3 or $AlCl_3$ are electrophiles too. They have empty orbitals that are usually metallic *p* orbitals. We saw above how BF_3 reacted with Me_3N. In that reaction BF_3 was the electrophile and Me_3N the nucleophile. Lewis acids such as $AlCl_3$ react violently with water and the first step in this process is nucleophilic attack by water on the empty *p* orbital of the aluminium atom. Eventually alumina (Al_2O_3) is formed.



More often, reaction occurs when electrons are transferred from a lone pair to an empty orbital as in the reaction between an amine and BF_3. The amine is the nucleophile because of the lone pair of electrons on nitrogen and BF_3 is the electrophile because of the empty *p* orbital on boron.



The kind of bond formed in these two reactions used to be called a 'dative

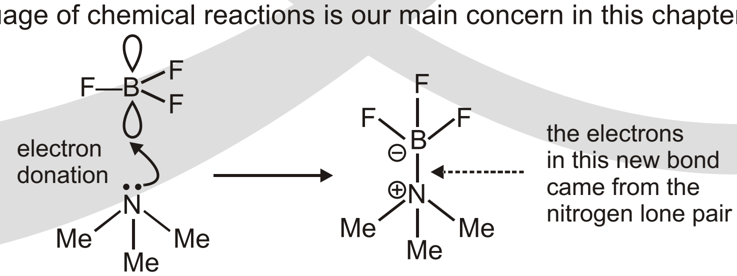
► A 'dative covalent bond' is just an ordinary bond whose electrons happen to come from one atom. Most bonds are formed by electron donation from one atom to another and a classification that makes it necessary to know the history of the molecule is not useful. Forget 'dative bonds' and stick to bonds or bonds.

Covalent bond' because both electrons in the bond were donated by the same atom. We no longer classify bonds in this way, but call them bonds or bonds as these are the fundamentally different types of bonds in organic compounds. Most new bonds are formed by donation of both electrons from one atom to another.

These simple charge or orbital interactions may be enough to explain simple inorganic reactions but we shall also be concerned with nucleophiles that supply electrons out of bonds and electrophiles that accept electrons into antibonding orbitals. For the moment accept that polar reactions usually involve electrons flowing from a nucleophile and towards an electrophile.

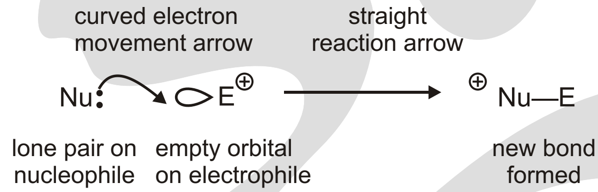
Since we are describing a dynamic process of electron movement from one molecule to another in this last reaction, it is natural to use some sort of arrow to represent the process. Organic chemists use a curved arrow (called a 'curly arrow') to show what is going on. It is a simple and eloquent symbol for chemical reactions.

The curly arrow shows the movement of a pair of electrons from nitrogen into the gap between nitrogen and boron to form a new bond between those two atoms. This representation, what it means, and how it can be developed into a language of chemical reactions is our main concern in this chapter.



The orbitals must also have about the right amount of energy to interact profitably. Electrons are to be passed from a full to an empty orbital. Full orbitals naturally tend to be of lower energy than empty orbitals—that is after all why they are filled! So when the electrons move into an empty orbital they have to go up in energy and this is part of the activation energy for the reaction. If the energy gap is too big, few molecules will have enough energy to climb it and reaction will be bad. The ideal would be to have a pair of electrons in a filled orbital on the nucleophile and an empty orbital on the electrophile of the same energy. There would be no gap and reaction would be easy. In real life, a small gap is the best we can hope for.

Now we shall discuss a generalized example of a neutral nucleophile, Nu, with a lone pair donating its electrons to a cationic electrophile, E, with an empty orbital. Notice the difference between the curly arrow for electron movement and the straight reaction arrow. Notice also that the nucleophile has given away electrons so it has become positively charged and that the electrophile has accepted electrons so it has become neutral.



REACTION REAGENTS

Organic reagents can be classified in two categories :

(a) Electrophile : Electron deficient species or electron acceptor is electrophile.

(b) Nucleophile : Electron rich species or electron or electron donor is nucleophile.

(a) Electrophiles

It can be classified into two categories :

(i) Charged electrophiles (E)

(ii) Neutral electrophiles (E)

(i) Charged electrophiles : Positively charged species in which central atom has incomplete octet is charged electrophile

$$H^{\oplus}, X^{\oplus}, R^{\oplus}\ N\begin{smallmatrix}\nearrow\!\!\!=O\\ \searrow O\end{smallmatrix},\ \overset{\oplus}{\underset{\bullet\bullet}{N}}=O,\ \overset{\oplus}{S}O_3H$$

Note : All cations are charged electrophiles except cations of IA, IIA group elements, Al and $\overset{\oplus}{N}H_4$.

(ii) Neutral electrophiles : It can be classified into three categories :

- Neutral covalent compound in which central atom has incomplete octet is neutral electrophile,

$$BrCl_2,\ BH_3,\ ZnCl,\ AlX_3,\ FeX_3,\ CH_3, CH_2, CX_2$$

- Neutral covalent compound in which central atom has complete or expended octet and central atom has unfilled -d-shell is neutral electrophile

$$SnCl_4, SiCl_4, PCl_5, SF_6, IF_7$$

- Neutral covalent compound in which central atom is bonded only with two or more than two electronegative atoms is neutral electrophile.

$$BeCl_2, BX_3, AlX_3, FeX_3, SnCl_4, PCl_3$$

$$PCl_5, NF_3, CX_2, CO_2, SO_3, CS_3, CX_4$$

❑ **NOTE :**

(i) Cl_2, Br_2 and I_2 also behave as neutral electrophiles.

(ii) Electrophiles are Lewis acids.

(b) Nucleophiles

Nucleophiles can be classified into three categories :

(i) Charged nucleophiles : Negatively charged species are charged nucleophiles.

$$\overset{\ominus}{H},\ \overset{\ominus}{OH},\ R-\overset{\ominus}{O},\ \overset{\ominus}{CH_3},\ \overset{\ominus}{X},\ \overset{\ominus}{S}H,\ R-\overset{\ominus}{S}$$

(ii) Neutral nucleophiles : It can be classified into two categories :

- Neutral covalent compound, in which central atom has complete octet, has at least one lone pair of electrons and all atoms present on central atom should not be electronegative, is neutral nucleophile.

$$NH_3,\ R-NH_2,\ R_2NH,\ R_3N,\ NH_2-NH_2$$

$$H-O-H,\ R-O-H,\ R-O-R$$

$$H-S-H,\ R-S-H,\ R-S-R$$

$$PH_3,\ RPH_2,\ R_2PH,\ R_3PH_3$$

- Organic compound containing carbon, carbon multiple bond/bonds behave as nucleophile.

  Alkenes, Alkynes, Benzene, Pyrole, Pyridine

(iii) Ambident nucleophile : Species having two nucleophilic centres, one is neutral (complete octet and has at least one lone pair of electrons) and other is charged (negative charge) behaves as ambident nucelophile.

$$\overset{\ominus}{C}\quad N,\ \overset{\ominus}{O}-\ddot{N}=O,\quad \overset{\ominus}{O}-\overset{\overset{O}{\uparrow}}{\underset{\underset{O}{\downarrow}}{S}}-OH$$

❑ **NOTE :**

(A) Organometallic compounds are nucleophiles.

(B) Nucleophiles are Lewis bases.

Organic compounds which behave as electrophile as well as nucleophile :

Organic compound in which carbon is bonded with electronegative atom (O, N, S) by multiple bond/bonds behaves as electrophile as well as nucleophile.

$$R-\overset{\overset{O}{\|}}{C}-H,\quad R-\overset{\overset{O}{\|}}{C}-R,\quad R-\overset{\overset{O}{\|}}{C}-OH,\quad R-\overset{\overset{O}{\|}}{C}-Cl$$

$$R-\overset{\overset{O}{\|}}{C}-OR,\quad R-\overset{\overset{O}{\|}}{C}-NH_2,\quad R-C\quad N,$$

$$R-\overset{\oplus}{N}\ \overset{\ominus}{C}$$

❑ **NOTE :** During the course of chemical reaction electrophile reacts with nucleophile.

SUMMARY

Orbitals of Nucleophiles and Electrophiles

Nucleophiles

1. Filled nonbonding orbital

n 

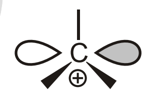
2. Filled   bonding orbital

π 

3. Filled   bonding orbital

Electrophiles

1. Empty nonbonding atomic orbital

a 

2. Empty pi antibonding orbital

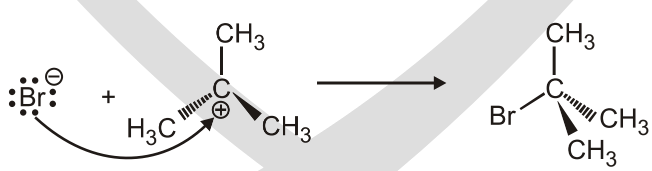
π^* 

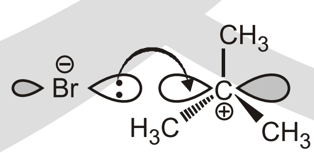
3. Empty sigma antibonding orbital

* 

Filled Nonbonding + Empty Nonbonding

n + a

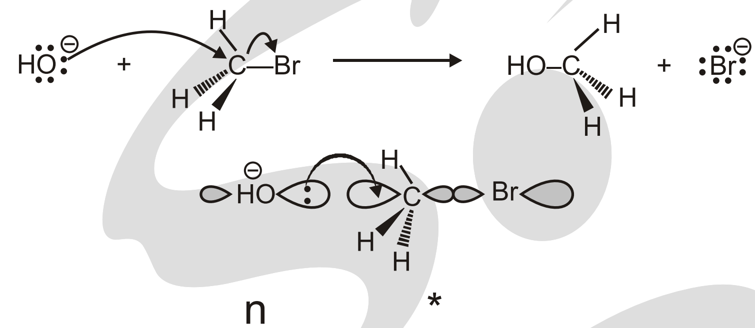




n   a

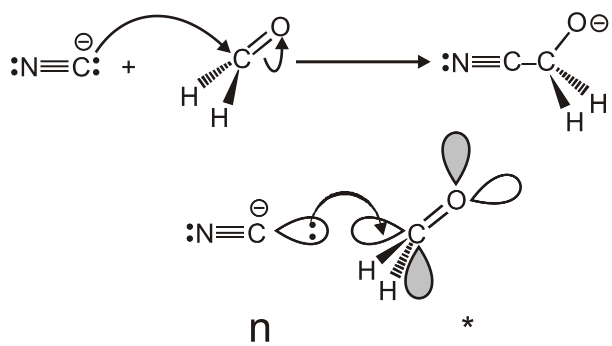
Filled Nonbonding + Sigma Antibonding

n + *



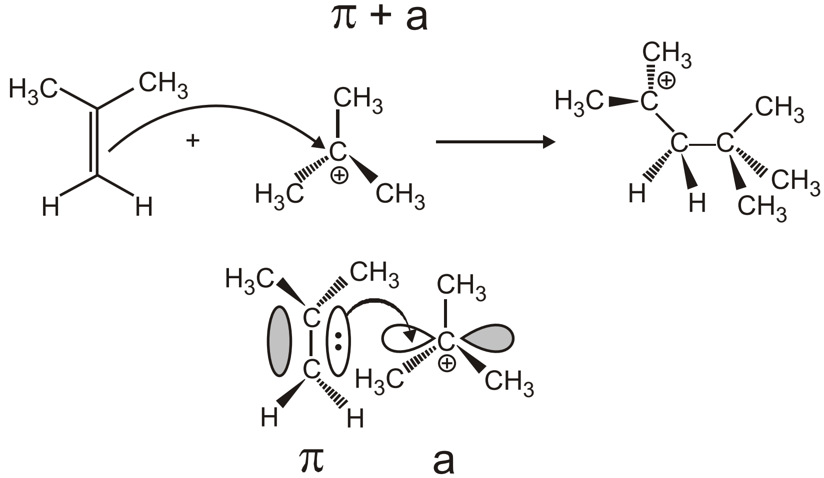
Filled Nonbonding + Pi Antibonding

n + *

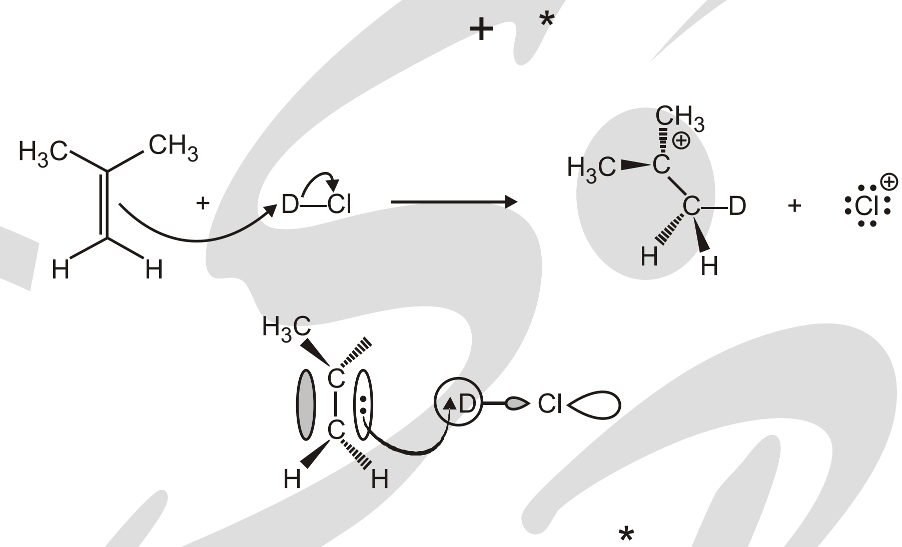


Pi Bonding + Empty Nonbonding

π + a



Pi Bonding + Sigma Antibonding



Still Having Trouble Identifying Nu: and E +?

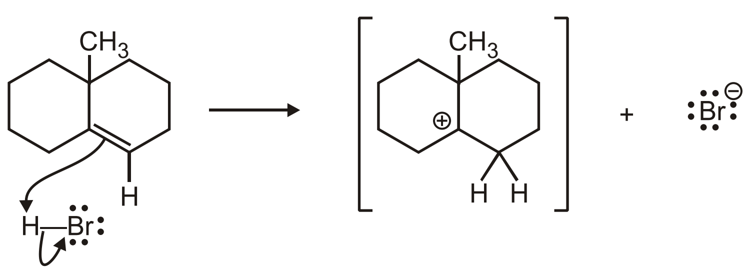
1. Look for regions of electron density and electron deficiency
2. Draw in all lone pairs
3. Draw as many resonance structures as you can

(Often, the 2nd best resonance structure shows the electrophilic and nucleophilic sites in a molecule)

The terms ''nucleophile'' and ''electrophile'' can mean the entire molecule or specific atoms and functional groups.

Don't let the dual meaning confuse you!

A Few Notes About Electron Pushing

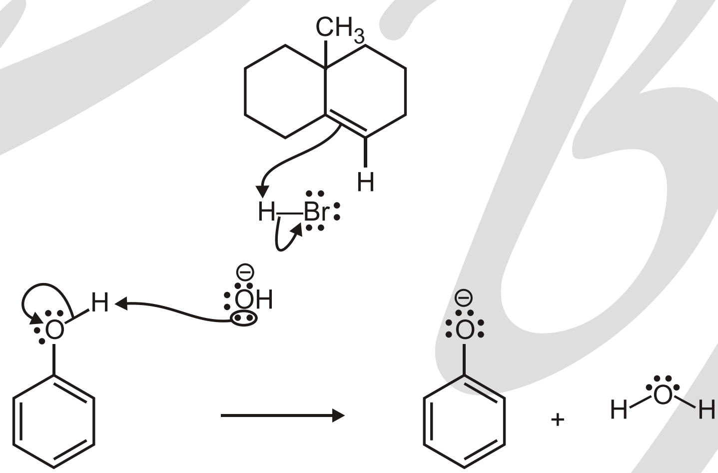


- A curved arrow show the **"movement"** of an electron pair.
- The tail of the arrow shows the **source** of the electron pair, which is a **filled** orbital. This will be a :
  - lone pair
  - -bond
  - -bond
- The head of the arrow indicates the **destination (sink)** of the electron pair which will be :
  - An **electronegative atom** able to support negative charge
  - An **empty** orbital when a new bond is formed
- Overall **charge is conserved.** Check that your products obey this rule.

Courtesy of Jefrey S. Moore, Department of Chemistry, University of Illinois at Urbana-Champaign Used with permission. Adapted by Kimberly Berkowski.

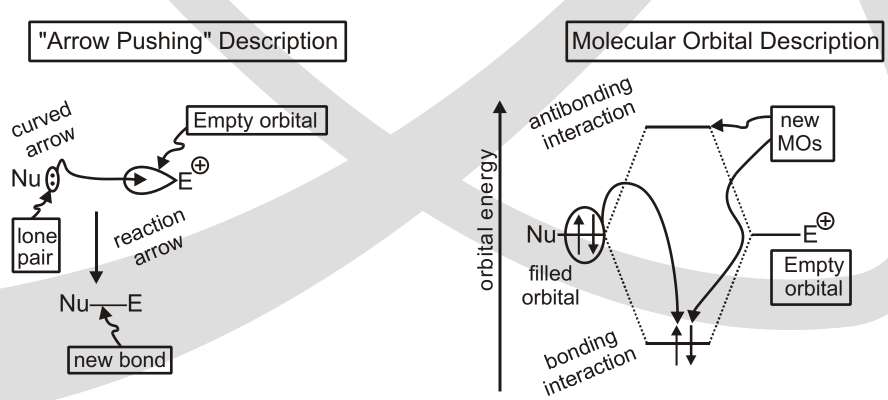
Electron Pushing to Uncharged C, H, N, or O

If you make a new bond by pushing an arrow to an uncharged
C, H, N, or O, you must also break one of the existing bonds in the same step.



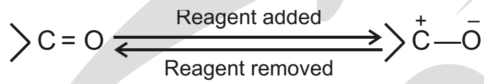
Courtesy of Jefrey S. Moore, Department of Chemistry, University of Illinois at Urbana-Champaign Used with permission. Adapted by Kimberly Berkowski.

Addition of Nucleophiles to Electrophiles

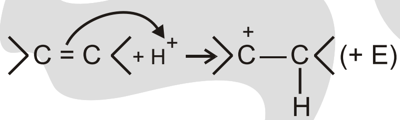


ELECTROMERIC EFFECT

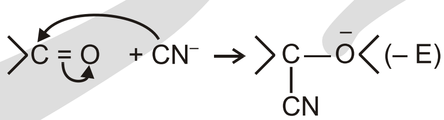
This involves a transfer of electrons of a multiple bond (double or triple) to one of the bonded atoms (usually more electronegative) in the presence of an attacking reagent. The effect is temporary and takes place only in the presence of a reagent. As soon as the reagent is removed, the molecule reverts back to its original position.



If the electrons are transferred to the atom of the double bond to which the reagent gets finally attached, it is +E effect. Consider addition of acid to alkenes.



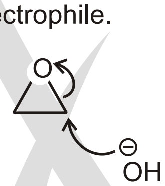
If, however, the electrons of the double bond are transferred to an atom of the double bond other than the one to which the reagent gets finally attached, the effect is called –E effect. Consider the addition of CN to the carbonyl group.



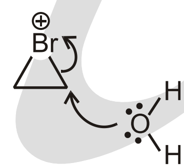
Solved Example

- In each of the reactions below, determine which compound is the nucleophile and which compound is the electrophile.

(a)

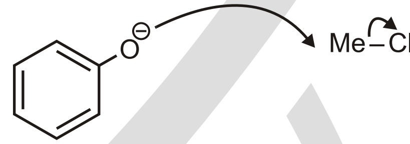


(b)



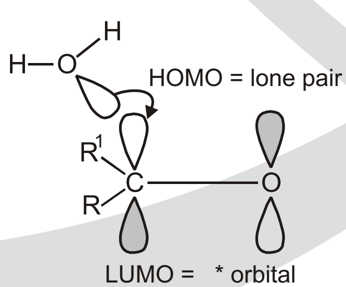
(c)

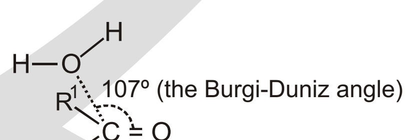
(d)



Solved Example

- Addition of water to a ketone

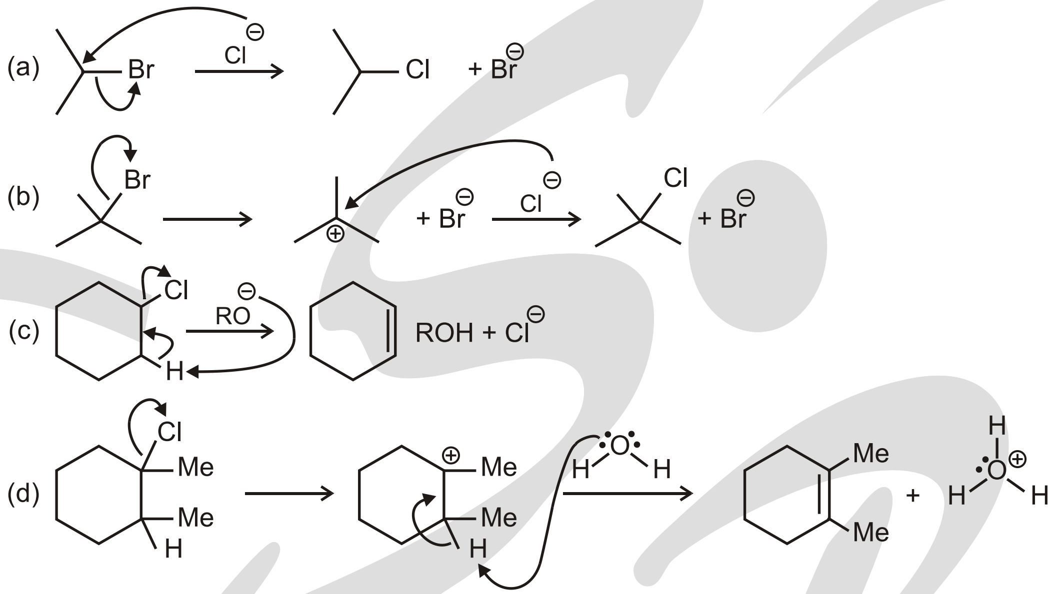


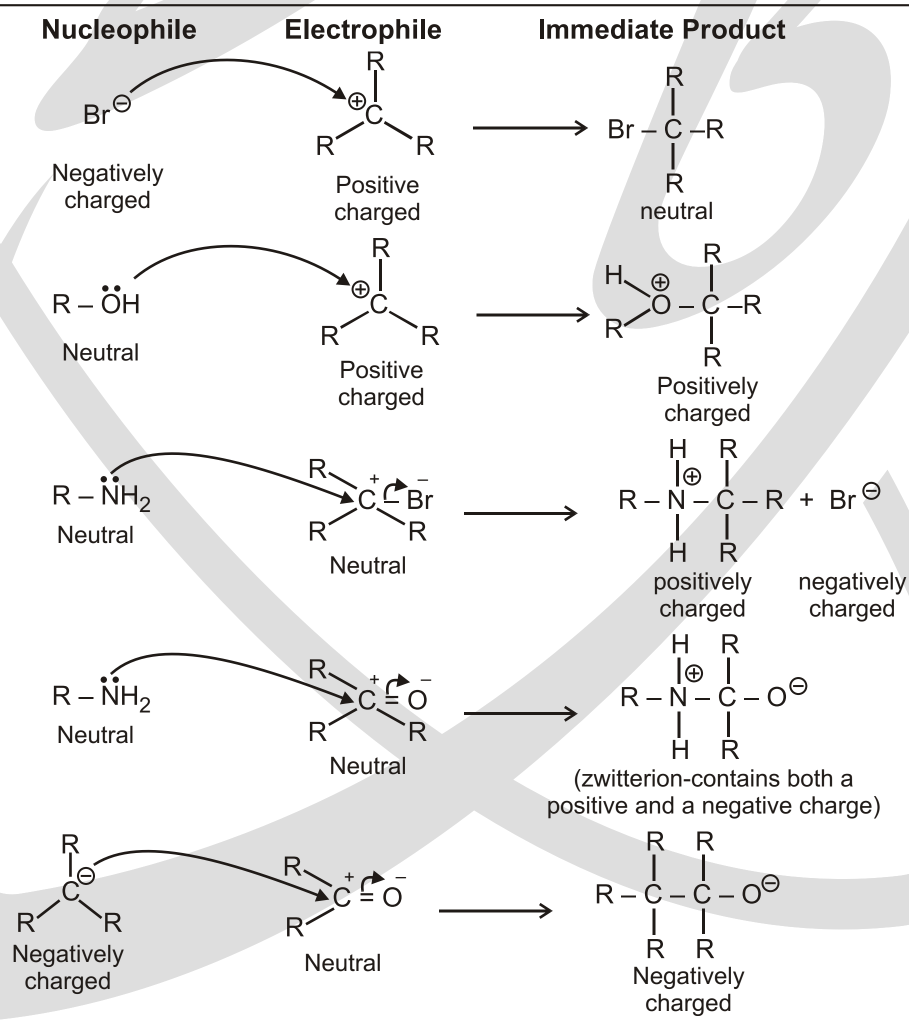


For maximum orbital overlap attack at 90º is required. However, attack at – 107º is observed because of the greater electron density on the carbonyl oxygen atom, which repels the lone pair on HO_2

Solved Example

- For each transformation below, complete the mechanism by drawing the proper arrows.





EXERCISE

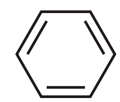
SINGLE CHOICE QUESTIONS

1. Which of the following is not an electrophile?

(A) H (B) BF_3 (C) NO_2 (D) Fe^{3}

(E) CH_2CH_2

2. Which of the following is not a nucleophile?

(A) $FeBr_3$ (B) Br (C) NH_3 (D) 

(E) CH_3OCH_3

3. Which is the MOST basic nucleophile in the following series?

(A) F (B) CH_3CH_2OH (C) H_2O (D) CH_3CH_2O

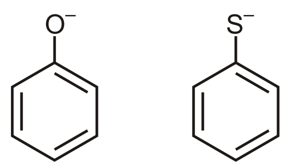
4. Which is an electrophile :

(A) BCl_3 (B) CH_3OH (C) NH_3 (D) None

5. Which of the following is an electrophile :

(A) H_2O (B) SO_3 (C) NH_3 (D) ROR

6. Consider the following two anionic molecules. Which of the following statements is TRUE ?



I II

(A) I is more basic and more nucleophilic than II.

(B) I is less basic and less nucleophilic than II.

(C) I is more basic but less nucleophilic than II.

(D) I is less basic but more nucleophilic than II.

7. How are basicity and leaving group ability related?

(A) They are not related to each other

(B) Good leaving groups are strong bases

(C) Good leaving groups are weak bases

(D) Leaving group Basic strength

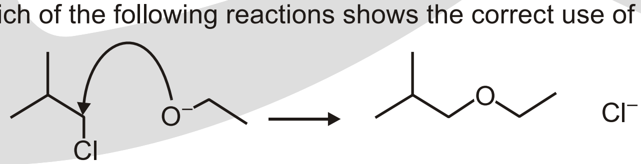
8. Ethers can act as :

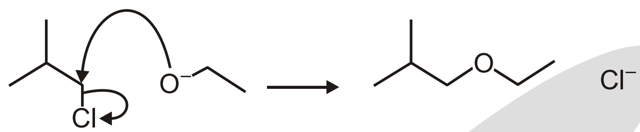
(A) Bronsted acids (B) Bronsted bases

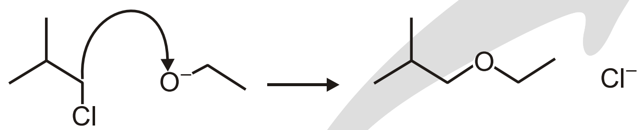
(C) Lewis acids (D) Lewis bases

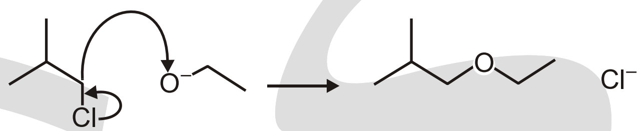
(E) An amphoteric species

9. Which of the following reactions shows the correct use of "curly arrows"?

(A) 

(B) 

(C) 

(D) 

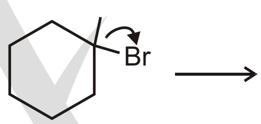
10. There are a number of definitions for acids and bases. Match the following definitions to the correct theory.

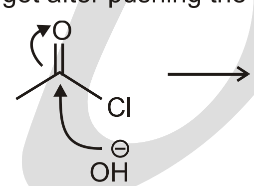
Theory	Definition
A. Arrhenius	I. Donates or accepts protons
B. Bronsted-Lowry	II. Donates or accepts a lone pair of electrons
C. Lewis	III. Donates a proton or a hydroxide

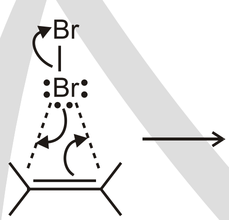
(A) A(I), B(II), C(III) (B) A(I), B(III), C(II) (C) A(II), B(III), C(II) (D) A(II), B(I), C(III)
(E) A(III), B(I), C(II) (F) A(III), B(II), C(I)

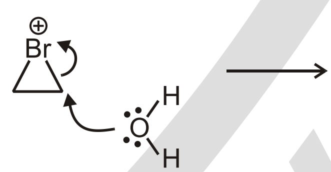
UNSOLVED EXAMPLE

1. For each problem below, draw the intermediate that you get after pushing the arrows.

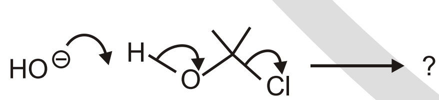
(a) 

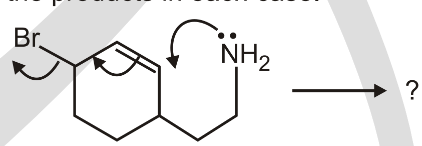
(b) 

(c) 

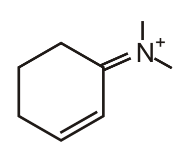
(d) 

2. Complete these mechanisms by drawing the structure of the products in each case.

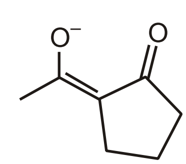
(a) 

(b) 

3. Circle all the electrophilic carbon atoms in the following structure.
Explain your answer with resonance contributors.



4. Draw in all lone pairs. Circle all the nucleophilic atoms in the following structure. Explain your answer with resonance contributors.



SUBJECTIVE TYPE QUESTIONS

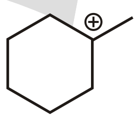
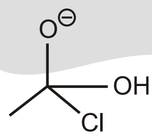
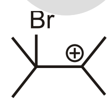
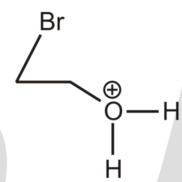
1. What are isohyptic and non-isohyptic reactions? Explain with reactions.

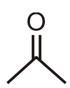
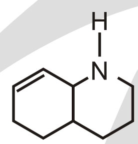
Answers

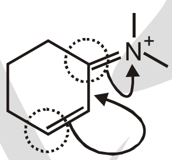
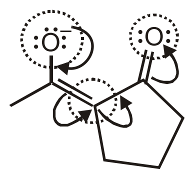
Single Choice Questions

1. (E) **2.** (A) **3.** (D) **4.** (A) **5.** (B) **6.** (C) **7.** (C) **8.** (D)
9. (B) **10.** (E)

Unsolved Examples

1. (a)  Br^- (b)  (c)  + $Br^{\ominus}$ (d) 

2. (a)  + H–O–H + Cl^- (b)  + HBr

3.  **4.** 

Subjective Type Questions

1. Isohyptic reactions are those reactions in which there is no change in the oxidation states of the carbon atoms participating in the reactions. Substitution reactions are found to be isohyptic reactions.

$$HO^- \; CH_3-Cl \xrightarrow{S_N2} HO-\overset{2-}{C}H_2$$

In this reaction, the oxidation state of the carbon atom has not changed after the reaction. Therefore, it can be identified as an ‘isohyptic reaction’.

In the case of addition and elimination reactions, there are changes in the oxidation state of the carbon atoms involved in the reaction.

$$H_2C = CH_2 + Cl_2 \xrightarrow[\text{reaction}]{\text{Addition}} H_2\overset{1-}{C}(Cl)-\overset{1-}{C}H_2(Cl)$$

$$\overset{3-}{C}H_2(H)-\overset{1-}{C}H_2(Br) \xrightarrow[\text{E2 reaction}]{OH^-} H_3\overset{1-}{C} = \overset{1-}{C}H_3 + H_2O + Br$$

The above mentioned two reactions, namely addition and elimination, may be considered as isohyptic reactions because they involve a change in the oxidation levels of the carbon atoms involved in the reactions.

❑❑❑

CHAPTER

28

Alkane

HYDROCARBON

◎ Introduction

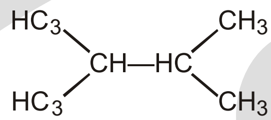
Hydrocarbons play a key role in our daily life. The term 'hydrocarbon' is self-explanatory which means compounds of carbon and hydrogen only. They are mainly obtained from petroleum, natural gas and coal, e.g. ethane, methane etc. You must be familiar with the terms 'LPG' and 'CNG' used as fuels. LPG is the abbreviated form of liquified petroleum gas whereas CNG stands for compressed natural gas. Another term 'LNG' (liquified natural gas) is also in news these days. This is also a fuel and is obtained by liquifaction of natural gas. Petrol, diesel and kerosene oil are obtained by the fractional distillation of petroleum found under the earth's crust. Coal gas is obtained by the destructive distillation of coal. The gas after compression is known as compressed natural gas.

In an innovative attempt to tackle the rising air pollution in the city, New Delhi implemented the odd-even rule from 1st January, 2016 to 15th January, 2016. According to it, based on the registration number, vehicles with odd and even number would be allowed to run on alternate days. Interestingly, CNG vehicles were exempted from the rule. Inspite of difficulty in maintainence, storage and higher costs of CNG, as compared to petrol, it is advantageous in the aspects of environmental friendly fuel, usage safety and also mileage.

LPG is used as a domestic fuel with the least pollution. Kerosene oil is also used as a domestic fuel but it causes some pollution. Automobiles need fuels like petrol, diesel and CNG. CNG operated automobiles cause less pollution. All these fuels contain mixture of hydrocarbons, which are sources of energy. Hydrocarbons are also used for the manufacture of polymers like polythene, polypropene, polystyrene etc. Higher hydrocarbons are used as solvents for paints. They are also used as the starting materials for manufacture of many dyes and drugs.

Alkanes or paraffins are saturated hydrocarbons with general molecules formula C_nH_{2n+2}. Series of alkanes in which the members differ in composition from one another by $-CH_2$ group is known as homologous series, the individual members being known as homologues. The nomenclature of alkanes is according to IUPAC rules.

e.g. CH_4 — Methane



2,2-dimethyl butane

◎ METHODS OF PREPARATION

Alkanes are prepared by the following methods

(I) Reductions Methods:

Alkanes can be prepared by the reduction of various organic compounds as follows :

(a) By hydrogenation of unsaturated hydrocarbons

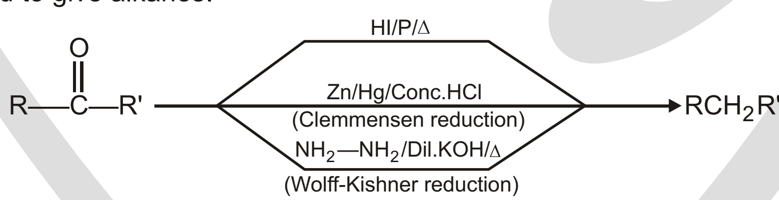
Alkanes are obtained by hydrogenation of unsaturated hydrocarbons (alkenes & alkynes) in presence of finelly divided catalyst e.g. Ni, Pt, Pd etc.

The following reaction in known as Sabatier & Senderen's reaction

$$R-CH\quad CH-R\quad H_2 \xrightarrow{Ni} R-CH_2-CH_2-R$$

$$R-Cr\quad C-R\quad 2H_2 \xrightarrow{Pt} R-CH_2-CH_2-R$$

(b) By reduction of Carbonyl Compounds: By following three methods aldehydes & ketones are reduced to give alkanes.



(c) By reduction of Carboxylic acid: Reduction by HI/P/D gives alkane.

$$\underset{\text{(No. of C n)}}{R-COOH} \xrightarrow{HI/P/D} \underset{\text{(No. of C n)}}{R-CH_3}$$

$$CH_3-COOH \xrightarrow{HI/P/} CH_3-CH_3$$

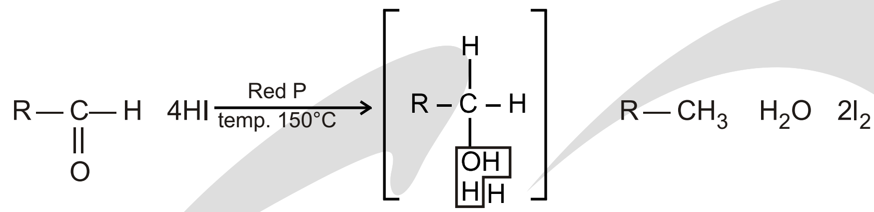
$$C_6H_5-COOH \xrightarrow{HI/P/} C_6H_5-CH_3$$

(d) From Alcohol, Aldehyde, Ketone and Acids :

When all are reduced with HI acid in presence of red phosphorus then respective alkanes are formed.

$$R-OH\quad 2HI \xrightarrow[150\ C]{Red\ P} RH\quad H_2O\quad I_2$$

★ If red P would have been absent in this reaction then product would be alkyl iodide. Red P neutralises the released iodine in form of PI_3 otherwise this iodine further reacts with alkane to form alkyl iodide.



$$R-\underset{\underset{O}{\|}}{C}-R \quad 4HI \quad ^{\text{Red P}} \quad R-CH_2-R \quad 2I_2 \quad H_2O$$

$$R-\underset{\underset{O}{\|}}{C}- \quad 4H \quad ^{\text{Zn–Hg/HCl}} \quad R-CH_2-R \quad H_2O$$

(II) From Organometallic Compounds

(a) Grignard reagents

Grignards' reagents and alkyl lithium reacts with water and other compounds having acidic hydrogen to give hydrocarbon corresponding to the alkyl group of the organometallic compounds.

$$R-MgX \quad H-OH \qquad R-H \quad MgOHX$$

$$R-\overset{+}{Li} \quad H-\overset{-}{O}H \qquad R-H \quad LiOH$$

Grignard's reagents reacts with alkyl halides to give alknaes

$R-MgX \quad R\,X \qquad R-R$ (Alkanes with higher number of carbons atoms)

(b) Corey – House Synthesis

Step (i) : Alkyl halides react with lithium in dry ether to form alkyl lithium

$$R-X \quad 2Li \quad ^{\text{Dry ether}} \quad R-Li \quad LiX$$

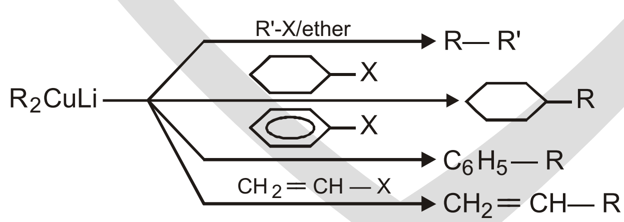
Step (ii) : This alkyl lithium reacts wtih CuI to give dialkyl lithium cuprate knwon as Gillman reagent

$$2R-Li \quad CuI \qquad R_2CuLi \quad LiI$$

Step (iii) : This further reacts with alkyl halides to give alkanes.

$$R_2CuLi \quad R-X \qquad R-R \quad R-Cu \quad LiX$$

Some examples are



This method is suitable for the preparation of unsymmetrical alkanes of the type $R-R$

(III) Decarboxylation of acids

Sodium salts of carboxylic acids undergo decarboxylation when heated with soda lime (mixture of sodium hydroxide and calcium oxide). This process of elimination of carbon dioxide from a carboxylic acid is known as decarboxylation.

$$\underset{C\,=\,n}{R-COONa} \quad \overset{\text{Soda lime /D}}{\underset{(NaOH\ \ CaO)}{}} \quad \underset{C\,=\,n-1}{R-H} \quad Na_2CO_3$$

This reaction takes place as follows

$$R-\overset{\overset{O}{\|}}{C}-\overset{\ominus}{O}\overset{\oplus}{Na} \longrightarrow \overset{\ominus}{R} + CO_2 \xrightarrow{HOH} R-H$$

$$\underset{\text{Sodium ethanoate}}{CH_3COO-Na} \quad NaOH \quad ^{CaO} \quad CH_4 \quad Na_2CO_3$$

☞ **Note:** Sodium formate gives hydrogen gas instead of R–H.

Solved example

- Sodium salt of which acid will be needed for the preparation of propane ? Write chemical equation for the reaction.

Sol. Butanoic acid,

$$CH_3CH_2CH_2COO-Na \quad NaOH \quad ^{CaO} \quad CH_3CH_2CH_3 \quad Na_2CO_3$$

(IV) Miscellaneous Methods

(a) Metal carbides (aluminium and beryllium carbides) on hydrolysis gives methane.

$$Al_4C_3 \quad ^{H_2O} \quad Al(OH)_3 \quad CH_4$$

$$Be_2C \quad ^{H_2O} \quad Be(OH)_2 \quad CH_4$$

(b) Berthelot Synthesis can be used for the preparation of methane and ethane.

$$C \quad H_2 \quad \overset{\text{Eelectric arc}}{\underset{120\ \ C}{}} \quad CH_4$$

$$2C \quad 3H_2 \quad ^{\text{Eelectric arc}} \quad CH_3-CH_3$$

(c) Kolbe's electrolytic method An aqueous solution of sodium or potassium salt of a carboxylic acid on electrolysis gives alkane containing even number of carbon atoms at the anode.

$$\underset{\text{Sodium acetate}}{2CH_3COO—Na^+} + 2H_2O$$

$$\Big\downarrow \text{Elecrolysis}$$

$$CH_3-CH_3 + 2CO_2 + H_2 + 2NaOH$$

The reaction is supposed to follow the following path :

$$2CH_3COO-Na \rightleftharpoons 2CH_3-\overset{\overset{O}{\|}}{C}-O \quad 2Na$$

At anode :

$$\underset{\text{Acetate ion}}{2CH_3-\overset{\overset{O}{\|}}{C}-\bar{O}} \xrightarrow{-2e^-} \underset{\text{Acetate free radical}}{2CH_3-\overset{\overset{O}{\|}}{C}-\ddot{\underset{\cdot\cdot}{O}}\!:} \longrightarrow \underset{\text{Methyl free radical}}{2\dot{C}H_3 + 2CO_2}$$

$$H_3C \quad CH_3 \qquad H_3C-CH_3$$

At cathode :

$$H_2O \quad e \qquad {}^{-}OH \quad H$$

$$2H \qquad H_2$$

The Kolbe electrolysis of propanoic acid, leads to the formation of a mixture of products due the involvement of the radicals in the reaction mechanism. Formation of the compounds, other than just n-butane (major), can be understood by the following reactions.

$$C_2H_5CO_2 \qquad C_2H_5CO_2 \quad e \qquad \text{... (i)}$$

$$C_2H_5CO_2 \qquad C_2H_5 + CO_2 \qquad \text{... (ii)}$$

$$\underset{\text{(i)}}{C_2H_5CO_2} \quad \underset{\text{(ii)}}{C_2H_5} \qquad \underset{\text{(Ester)}}{C_2H_5CO_2C_2H_5}$$

$$C_2H_5 + C_2H_5 \qquad C_2H_5 - C_2H_5$$

$$C_2H_5^{\bullet} + C_2H_5^{\bullet} \xrightarrow[\text{Hydrogenation}]{\text{Intermolecular}} C_2H_6 + C_2H_4$$

Reduction

Oxidation

(d) Wurtz Reaction

This reaction involves the condensation of two molecules of alkyl halides in the presence of sodium and dry ether.

Case-I : When both alkyl halides are same :

$$\underset{C=n}{R-X} + \underset{C=n}{R-X} \xrightarrow{\text{Na/dry ether}} \underset{C=2n}{R-R} + 2NaX$$

Coupling

$$CH_3-X \quad CH_3-X \quad \overset{\text{Na/dryether}}{} \quad CH_3-CH_3$$

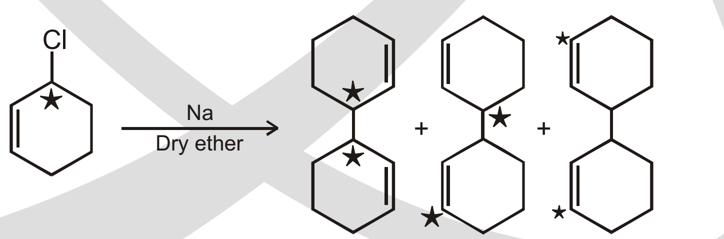
Case-II : When both alkyl halides are different :

$$R-X \quad R-X \quad \overset{\text{Na/dry ether}}{} \quad R-R \quad R-R \quad R-R \quad \text{(3-products wil be obtained)}$$

$$CH_3-CH_2-Br \quad CH_3-CH_2-CH_2-CH_2-CH_2-Br \quad \overset{\text{Na/dry ether}}{}$$

n-butane + *n*-octane + *n*-hexane

Case-III :



☞ **Note :**

(a) Methane cannot be prepared by this method.

(b) Tertiary alkyl halides do not give this reaction.

(c) Best result is obtained when both alkyl halides are same, i.e., this method is suitable for preparation of alkanes having even number of carbons.

◎ Physical Properties

1. The first four alkanes (from methane to butane) are colourless and odourless gases. The next thirteen (from pentane to heptadecane) are colourless and odourless liquids. And, the rest of higher alkanes (having 18 carbon atoms or more) are colourless solids at ordinary temperature.

Physical state:

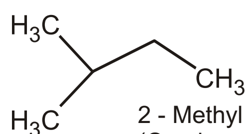
Alkanes		
$C_1 — C_4$	Gaseous state	
$C_5 — C_{17}$	Liquid state	(except neo pentane)
C_{18} & above	Solid like wax	

2. Alkanes being non-polar molecules, are soluble in non-polar solvents like benzene, ether, and chloroform. However, they are insoluble in polar solvents like water. Their solubility decreases with increase in their molecular weight.

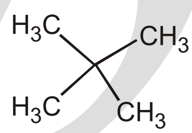
3. Intermolecular forces in alkanes are weak van der Waal's forces for isomer (i.e. London dispersion interactions); the greater the surface area the stronger the intermolecular forces. Thus, the boiling point of alkanes increases by 20°C to 30°C for each carbon atom that is added to the chain.

Among isomeric alkanes, straight chain isomers have a higher boiling point than the branched chain isomers. With an increase in branching, the shape of the molecule approaches that of a sphere and there is a reduction in the surface area. This renders the intermolecular forces weak and the same can be overcome at a relatively lower temperature.

n - pentane
(no. branching)
b.p. = 309.1 K



2 - Methyl butane
(One branch)
b.p. = 300.9 K



2, 2-dimethyl propane
(Neopentane) (two branches)
b.p. = 282.5 K

4. The density of alkanes increases with the size of the molecule and approaches a constant specific gravity of 0.8 for n-hexadecane, that is, all alkanes are lighter than water.

5. The melting point of alkanes increase with an increase in the number of carbon atoms. However, the rise is not uniform. The rise in melting point is more as one moves to higher alkanes having an even number of carbon atoms as compared to those having an odd number of carbon atoms. This is because the melting point depends not only on the size of molecules but also on how well they can fit into crystal lattice, that is, how symmetrical is a molecule. Alkanes with an even number of carbon atoms are more symmetrical and thus have a higher melting point.

◎ Chemical Properties

1. **(I) Halogenation of alkanes:** Alkanes react with halogens in the presence of light to give alkyl halides.

$$R—H \quad \underset{(excess)}{Cl_2} \quad \overset{U.V.light}{\quad} \quad R—Cl \quad HCl$$

$$CH_4 \quad Cl_2 \quad \overset{h}{\quad} \quad CH_3Cl \quad HCl$$

Alkyl halides formed further react with halogen to give di, tri and tetra halogen

(CH_3Cl, CH_2Cl_2, $CHCl_3$, CCl_4) compounds.

Mechanism of given reaction is free-radical substitution. If Cl_2 is taken in excess amount than CH_3Cl, CH_2Cl_2, $CHCl_3$ and CCl_4 also produced.

☞ Mechanism

Step-I : Chain initiating step

$$Cl_2 \quad \text{heat or light} \quad 2Cl$$

Step-II : Chain-propagating step

$$Cl + CH_4 \qquad HCl \quad CH_3$$

$$CH_3 \quad Cl_2 \qquad CH_3Cl \quad Cl$$

Step-III : Chain-terminating steps

$$Cl + Cl \qquad Cl_2$$

$$CH_3 \quad CH_3 \qquad CH_3CH_3$$

$$CH_3 \quad Cl \qquad CH_3Cl$$

- In this reaction the reactivity order of halogen is

$$F_2 \quad Cl_2 \quad Br_2 \quad I_2$$

- Fluorine can react in dark. Cl_2 and Br_2 require light energy. I_2 does not show any reaction at room temperature, but on heating it shows iodination.
- When chlorine reacts with methane in presence of Benzoyl peroxide the product is CCl_4, because in presence of Benzoyl peroxide chlorine does not require light energy.
- When methane reacts with chlorine with excess O_2, the reaction is stopped because oxygen reduces the reactivity of alkyl group and changes it into peroxide radical.
- The reaction is reversible with iodine because by product HI is a strong reducing agent.

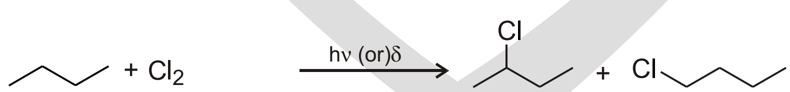
$$CH_4 \quad I_2 \rightleftharpoons CH_3I \quad HI$$

☞ Iodination of methane is done in presence of oxidising agents such as HNO_3 / HIO_3 / HgO which neutralises HI.

Relative rates of alkyl radical formation by a chlorine radical at room temperature :

tertiary	>	secondary >	primary
5.0		3.8	1.0

Increasing rate of formation



Relative amount of 1-chlorobutane
number of hydrogens × reactivity
6 × 1.0 = 6.0

per cent yield = $\frac{6.0}{21}$ = 29%

Relative amount of 2-chlorobutane
number of hydrogens × reactivity
4 × 3.8 = 15

per cent yield = $\frac{15}{21}$ = 71%

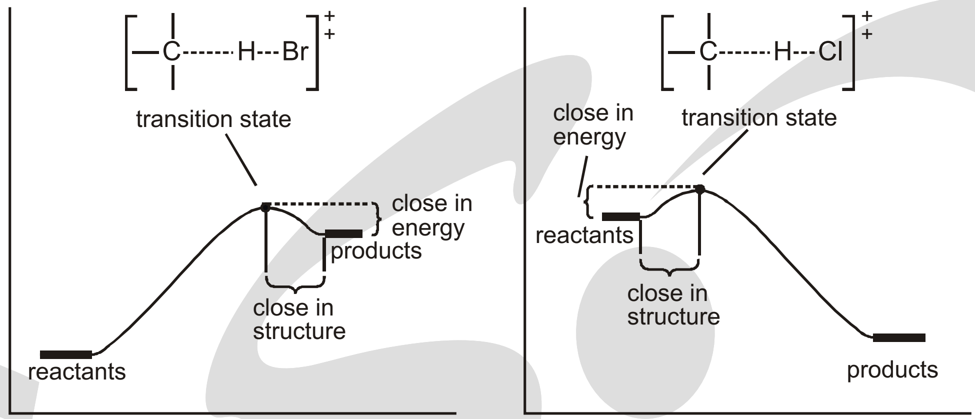


Figure: Comparison of transition states for bromination and chlorination. In the endothermic bromination, the transition state resembles the products (the free radical and HBr). In the exothermic chlorination, the free radical has just begun to form in the transition state, so the transition state resembles the reactants.

Figure: Compares the transition states for bromination and chlorination. In the product-like transition state for bromination, the C—H bond is nearly broken and the carbon atom has a great deal of radical character. The energy of this transition state reflects most of the energy difference of the radical products. In the reactant-like transition state for chlorination, the C—H bond is just beginning to break, and the carbon atom has little radical character. This transition state reflects only a small part (about a third) of the energy difference of the radical products. Therefore, chlorination is less selective.

These reactions are examples of a more general principle called the Hammond postulate.

Faster: CH_4 + Cl. CH_3Cl + HCl relative rate = 12

Slower. CD_4 + Cl. CD_3Cl + DCl relative rate = 1

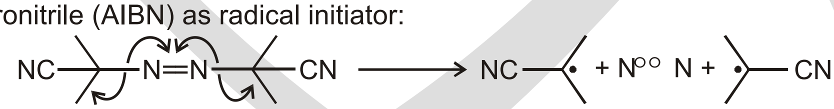
This effect, called a kinetic isotope effect, is clearly seen in the chlorination of methane. Methane undergoes free-radical chlorination 12 times as fast as tetradeuteriomethane (CD_4).

◎ Radical Initiators

Only for some of the radical reactions discussed in is the initiating radical produced immediately from the starting material or the reagent. In all other radical substitution reactions an auxiliary substance, the radical initiator, added in a substoichiometric amount, is responsible for producing the initiating radical.

Radical initiators are thermally labile compounds, which decompose into radicals upon moderate heating.These radicals initiate the actual radical chain through the formation of the initiating radical. The most frequently used radical initiators are azobisisobutyronitrile (AIBN) and dibenzoyl peroxide. After AIBN has been heated for only 1 h at 80ºC, it is half-decomposed, and after dibenzoyl peroxide has been heated for only 1 h at 95ºC, it is half-decomposed as well.

Azobisisobutyronitrile (AIBN) as radical initiator:



Dibenzoyl peroxide as radical initiator:

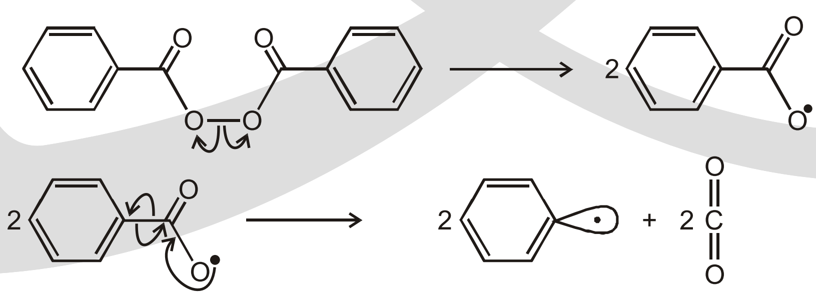


Figure: Radical initiators and their mode of action (in the “arrow formalism” for showing reaction mechanisms used in organic chemistry, arrows with half-heads show where single electrons are shifted, whereas arrows with full heads show where electron pairs are shifted).

◎ Simple and Multiple Chlorinations

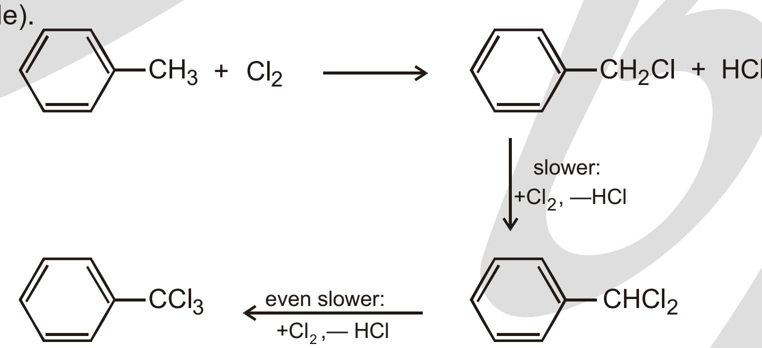
Presumably you are already familiar with the mechanism for the thermal chlorination of methane.We will use Figure to review briefly the net equation, the initiation step, and the propagation steps of the monochlorination of methane. Figure shows the energy profile of the propagation steps of this reaction.

$$CH_4 \text{ (large excess)} + Cl_2 \xrightarrow{400\ C} CH_3Cl \quad HCl$$

◎ Regioselectivity

A given molecular transformation, for example, the reaction C — H   C — Cl, is called regioselective when it takes place preferentially or exclusively at one place on a substrate. Resonance-stabilized radicals are produced regioselectively as a consequence of product-development control in the radical-forming step.

In the industrial synthesis of benzyl chloride (Figure), only the H atoms in the benzyl position are replaced by Cl because the reaction takes place via resonancestabilized benzyl radicals as intermediates. At a reaction temperature of 100ºC, the first H atom in the benzyl position is substituted a little less than 10 times faster (benzyl chloride) than the second (benzal chloride) and this is again 10 times faster than the third (benzotrichloride).



Incidentally, this reaction of chlorine with propene is also chemoselective.

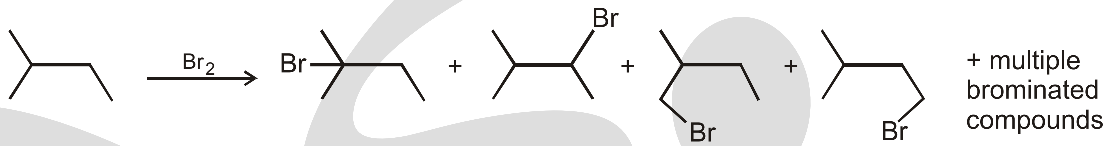
◎ Regioselectivity of Radical Brominations Compared to Chlorinations :

In sharp contrast to monochlorination, monobromination products with pronounced regioselectivity (**Table**). Monobromination of isopentane gives the products 2-Bromo-2-methylbutane, 2-Bromo-3-methyl-butane and other products with relative yields of 92.2%, 7.4% and 0.4% respectively. The analysis of these regioselectivities illustrated in **Table** gives relative rates of 2000, 79, and 1 for the bromination of C_{tert} — H, C_{sec} — H, and C_{prim} — H, respectively.

The low regioselectivity of the radical chain chlorination in Table and the high regioselectivity of the analogous radical chain bromination in Table are typical. In the following we will explain mechanistically why the regioselectivity for chlorination is so much lower than for bromination.

How the enthalpy H of the substrate / reagent pair changes when R and H — Cl are produced from it is plotted for four radical chlorinations in **Figure**. Hammond’s postulate can be applied to this series of the selectivity-determining steps of the radical chlorination and bromination shown in **Figure**. Chlorination takes place via early transition states, where as bromination takes place via late transition states. As early transition states are similar to the reactants, which in the case of chlorination indicates that the difference in the energy of the transition states (Alkyl radical HX) is very small. This leads to formation of the different products in chlorination with similar rates of reaction. Bromination proceeds via late transition states that are similar to the products, which are position isomers of each other, indicates that there is a high difference in energy of the transition states generated. This leads to a high difference in the rates of formation of different products via bromination.

Table: Regioselectivity of Radical Bromination of Isopentane



The relative yields of the above monobromination products are...	... 92.2%	7.38%	0.28%	0.14%
In order to produce the above compounds in the individual case...	... 1	2	6	3 ...
... H atoms were available for the substitution. Yields on a per-H-atom basis were...	... 92.2%	3.69%	0.047%	0.047% ...
... for the monobromination product above. In other words: $k_{C—H,\ C—Br}$ rel in the position concerned is ...	... 2000	79	1	1
..., that is, generally for ...	... C_{tert}—H	C_{sec}—H	C_{prim}—H	

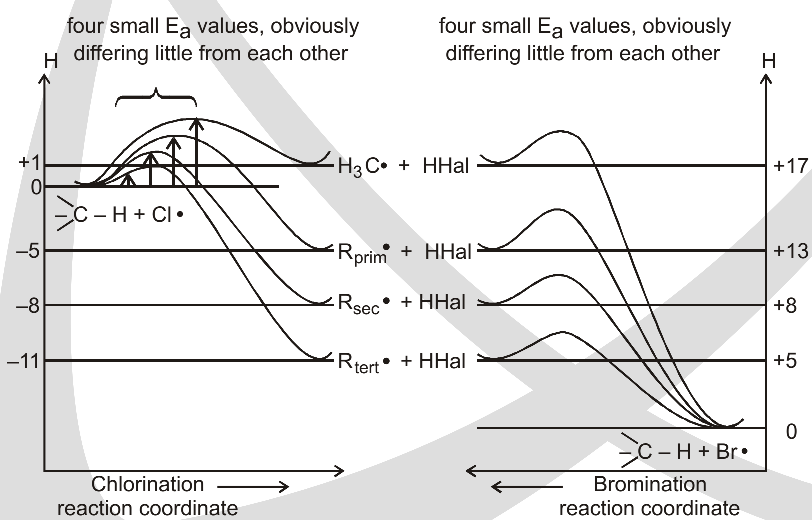
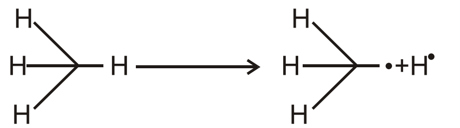
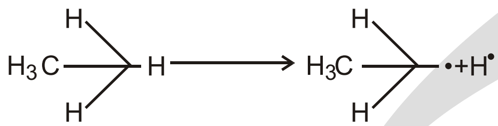


Figure : Thermochemical analysis of that propagation step of radical chlorination (left) and bromination (right) of alkanes that determines the regioselectivity of the overall reaction. The H^r values were determined experimentally; the DH^{++} values are estimated

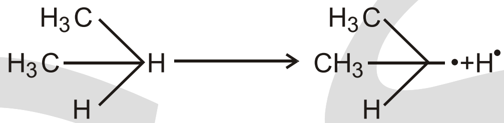
HHal = halogen



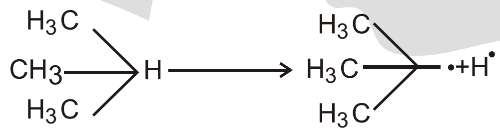
$\Delta H = 105\ \text{kcal mol}^{-1}$



$\Delta H = 100\ \text{kcal mol}^{-1}$



$\Delta H = 96\ \text{kcal mol}^{-1}$



$\Delta H = 93\ \text{kcal mol}^{-1}$

Bond energy of extraction of 1°H > 2°H > 3° H, hence the stability order of alkyl radical free is 1° < 2° < 3°H

Solved Example

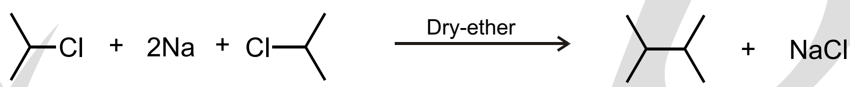
- Which alkane can be prepared by Wurtz's reaction in good yield.

(A) n-pentane (B) iso-pentane

(C) 2,3-dimethyl butane (D) 2,2-dimethyl butane

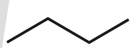
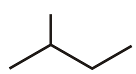
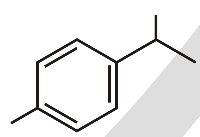
Sol. (C) $R + \boxed{Cl + 2Na + Cl - R} \xrightarrow{\text{Dry-ether}} R - R + 2NaCl$

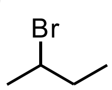
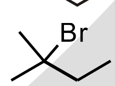
(Symmetric Hydrocarbon)

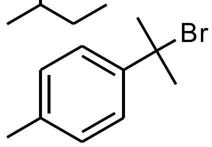
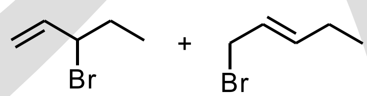
(C) 

Solved Example

- Predict which hydrogen will be preferentially substituted in the free-radical bromination of each of the following compounds by drawing the expected product :

(a)  (b)  (c)  (d) 

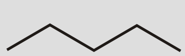
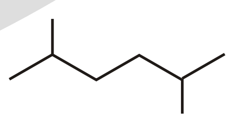
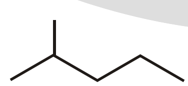
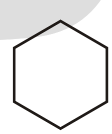
Sol. (a)  (b) 

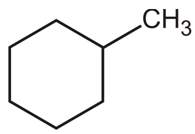
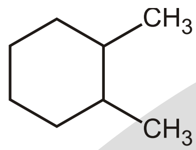
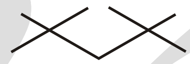
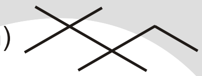
(c)  (d) 

Solved Example

- How many alkyl chlorides can be obtained from monochlorination of the following alkanes?

(Disregard stereoisomers.)

(a)  (b)  (c)  (d) 

(e)  (f)  (g)  (h) 

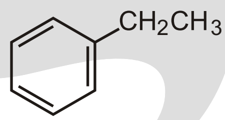
(i)

Sol. (a) 3 (b) 3 (c) 5 (d) 1 (e) 5 (f) 4 (g) 2 (h) 4 (i) 4

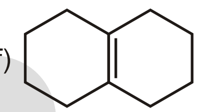
Solved Example

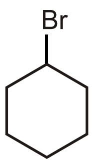
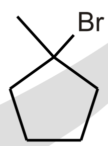
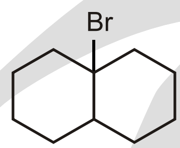
- For each compound, predict the major product of free-radical bromination. Remember that bromination is highly selective, and only the most stable radical will be formed.

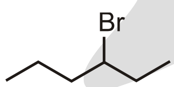
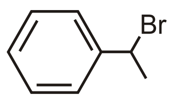
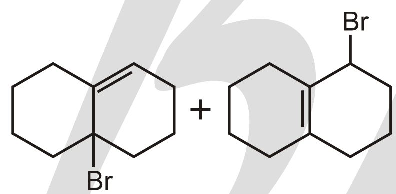
(a) cyclohexane (b) methylcyclopentane (c) decalin

(d) hexane (e) 

ethylbenzene

(f)  (2 products)

Sol. (a)  (b)  (c) 

(d)  (e)  (f) 

Solved Example

- Explain why

(a) H_2O (100 C) has a higher boiling point than CH_3OH(65 C).

(b) H_2O (100 C) has a higher boiling point than NH_3 (–33 C).

(c) H_2O (100 C) has a higher boiling point than HF (20 C)

(d) HF (20 C) has a higher boiling point than NH_3 (–33 C)

Solved Example

- The deuterium kinetic isotope effect for the halogenation of an alkane is defined in the following equation, where X Cl or Br

$$\text{Deuterium kinetic isotope effect } \frac{\text{rate of homolytic cleavage of a C — H bond by X}}{\text{rate of homolytic cleavage of a C — D bond by X}}$$

Predict whether chlorination of bromination would have a greater deuterium kinetic isotope effect.

❑❑❑

CHAPTER
29

Alkene

◎ Introduction

Unsaturated hydrocarbons are hydrocarbons that contain one or more carbon carbon double or triple bonds. There are three classes of unsaturated hydrocarbons : alkenes, alkynes and aromatic hydrocarbons.

Alkenes are the unsaturated hydrocarbons that contain one double bond. They have the general formula C_nH_{2n}, and the double bond is known as the 'olefinic bond' or 'ethylenic bond' (*i.e.* oleum, oil + fines, forming) because lower alkene react with halogens to form oily substances.

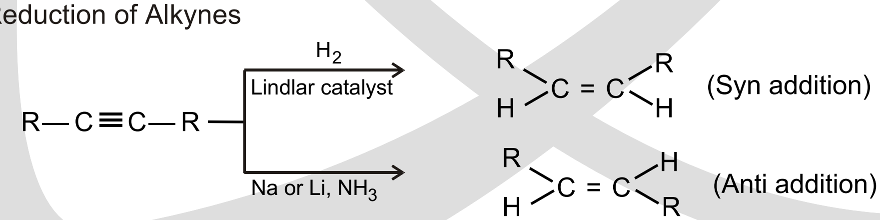
◎ PHYSICAL PROPERTIES

At room temperature alkenes differs in their physical state depending upon the number of carbon atom.

C_2 — C_4	:	Gases
C_5 — C_{17}	:	Liquids
C_{18}— Onwa rds	:	Solids

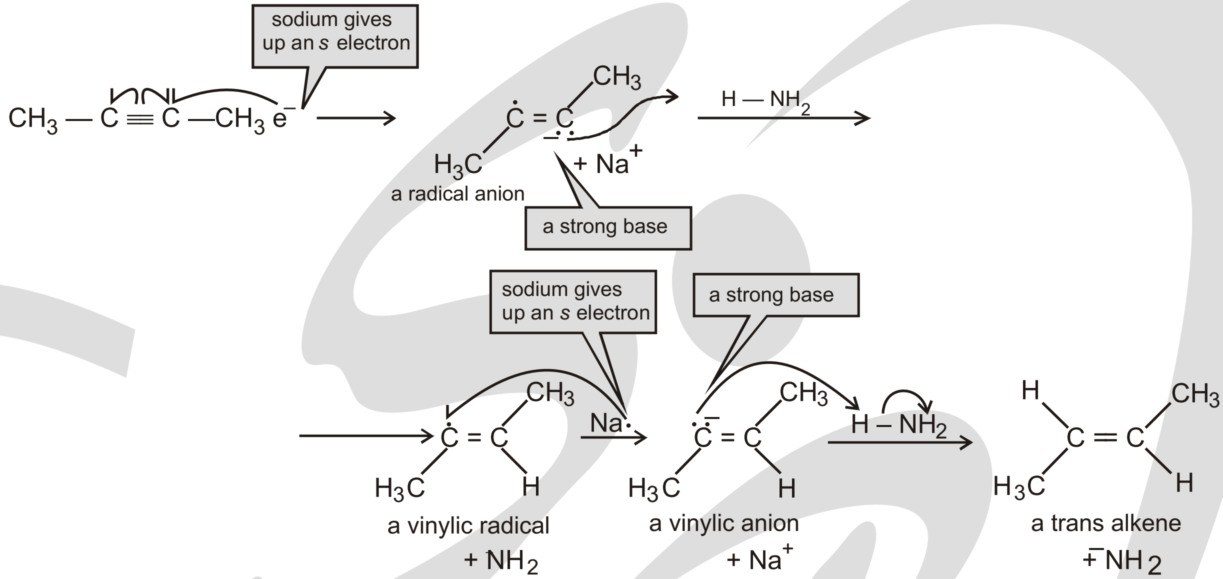
◎ Methods of Preparation of Alkenes

1. Reduction of Alkynes



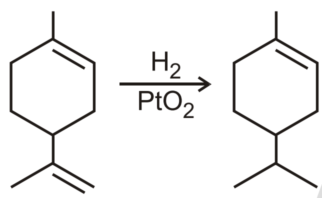
Metallic palladium deposited conditioned with lead acetate and quinoline is *Lindlar's catalyst*. It reduces alkynes to cis alkenes whereas trans alkenes are obtained predominantly by reduction with sodium or lithium in liquid ammonia.

MECHANISM FOR THE CONVERSION OF AN ALKYNE TO A TRANS ALKENE



An arrowhead with a double barb signifies the movement of two electrons.

In polyenes that contain differently substituted C C double bonds, it is often possible to hydrogenate chemoselectively the least hindered C C double bond or unstable double bond:



2. Dehydrohalogenation of Alkyl Halides

$$-\underset{H}{\overset{|}{\underset{|}{C}}}-\underset{X}{\overset{|}{\underset{|}{C}}}- +KOH \xrightarrow{\text{alcohol}} -\overset{|}{C}\quad\overset{|}{C}- \quad KX \quad H_2O$$

Dehydrohalogenation belongs to a general class of reactions called 1,2 - elimination reactions.

Such elimination reactions are characterised by the following:

(a) The substrate should contain a leaving group (an atom or group that leaves the molecule, taking its electron pair with it).

(b) The substrate should have an atom or a group in a position beta to leaving group (nearly always hydrogen) that can be extracted by a base, leaving its electron pair behind.

(c) The reaction is brought about by a base. It can be a basic anion like hydroxide or an alkoxide derived from alcohol like ethoxide, $C_2H_5O^-$ and tert-butoxide, $(CH_3)_3CO$.

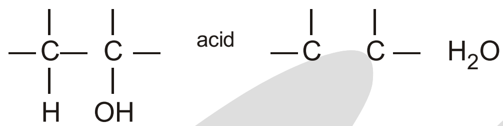
3. Dehydration of Alcohols

Alcohols form alkenes on dehydration by

(a) reaction with conc. H_2SO_4 at 100 C

(b) reaction with phosphoric acid at 200 C

(c) passing alcohol vapour over a lumina (Al_2O_3), P_2O_5 or anhydrous $ZnCl_2$ at 350 — 400 C H_2O



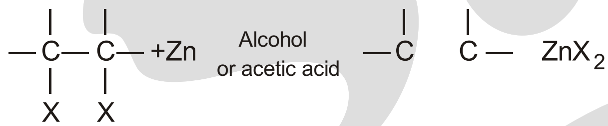
Note : *The carbo-cation intermediate may undergo rearrangement (if possible) before forming an alkene.*

Reactivity of Alcohols in Dehydration

3° alcohol > 2° alcohol > 1° alcohol

Experimental conditions like temperature and acid concentration are harshest for primary alcohols and extremely mild for tertiary alcohols.

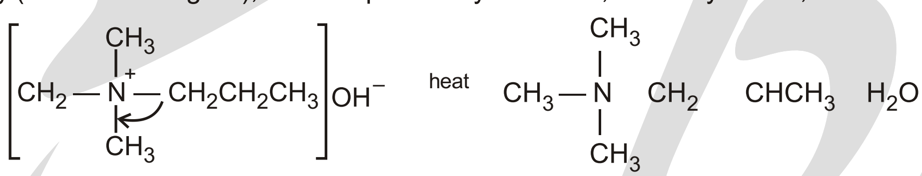
4. Dehalogenation of Vicinal Dihalides



Vicinal dihalides are compounds having two halide atoms on adjacent carbon atoms. Compounds having two halide atoms on same carbon atom are called geminal (or gem) dihalides.

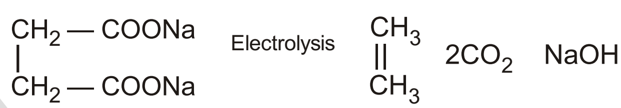
5. Hoffmann's Elimination of Quaternary Ammonium Hydroxides

A compound having the structure R_4N+ OH– is called a quaternary ammonium hydroxide (the four alkyl groups attached to nitrogen may be same or different). When a quaternary ammonium hydroxide is heated strongly (to 125°C or higher), it decomposes to yield water, a tertiary amine, and an alkene. For example,



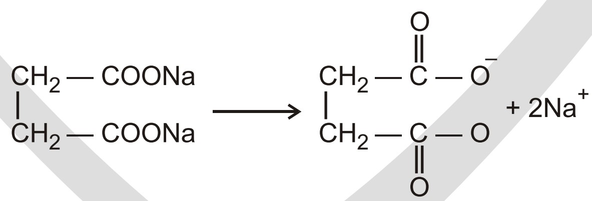
Trimethyl-*n*-propylammonium hydroxide

6. Kolbe's Electrolytic Decarboxylation: Electrolysis of aqueous solutions of sodium salts of vicinal dicarboxylic acids leads to the decarboxylation of the compound leading to the formation of alkenes.



Sod. Succinate

Mechanism

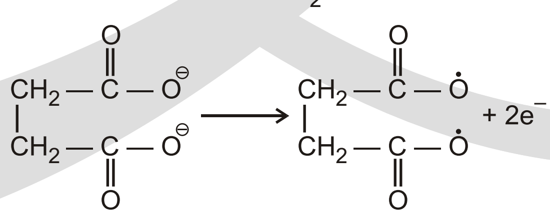


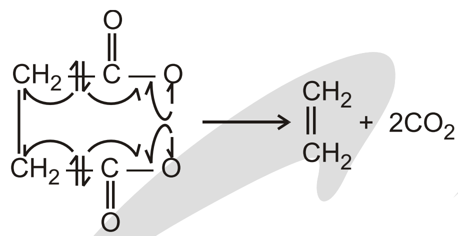
At Cathode

$H_2O \quad H \quad OH$

$2H \quad 2e \quad H_2$

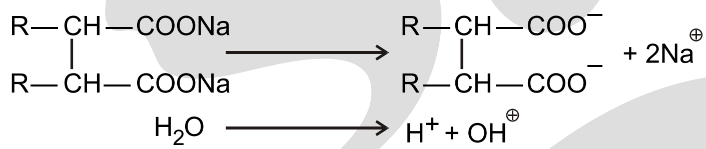
At Anode



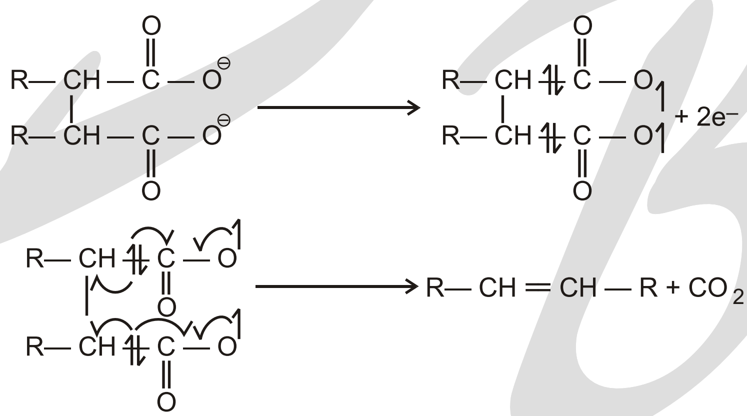


The solution hence contains Na and OH there by becoming more and more basic. Here no other side product is formed because the elimination of CO_2 takes place without the formation of new free Radical.

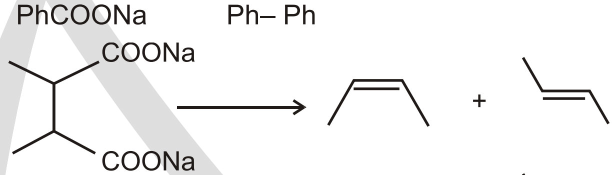
Solved Example



At Anode



(i) PhCOONa Ph– Ph

(ii) 

Hint: 

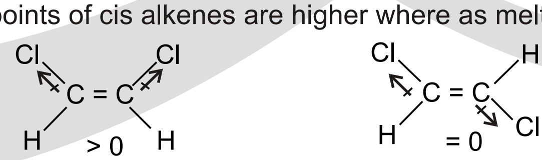
Structure And Physical Properties

1. Physical Properties

(a) Alkenes containing two to four carbon atoms are gases, those containing five to seventeen are liquids and higher alkenes are solids

(b) These are insoluble in water but soluble in organic Solvents.

(c) The boiling points of cis alkenes are higher where as melting points of trans-alkenes are more.

e.g. 

As Cis isomers being more polar, boils at a higher temperature, where as trans isomers being more symmetrical fits well into the crystal lattice and have higher melting point.

2. Structure

(a) The characteristic feature of the alkene structure, is the carbon carbon double bond. It is thus the functional group of alkenes and as the functional group, it determines the characteristic reactions that alkenes undergo.

These reactions are of two kinds.

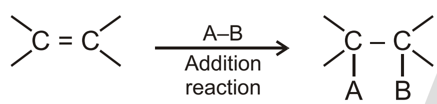
(b) First, there are those that take place at the double bond itself and, in doing this, destroy the double bond.

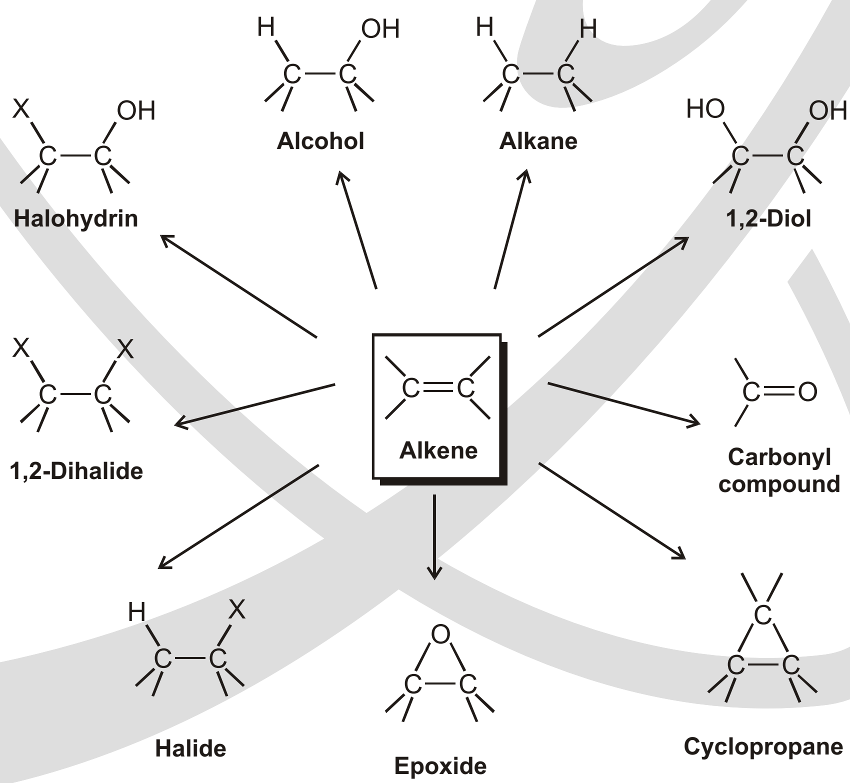
(c) There are the reactions that take place, not at the double bond, but at certain positions having special relationships to the double bond. Outwardly the double bond is not involved; it is found intact in the product. Yet it plays an essential, though hidden, role in the reaction : it determines how fast reactions take place and by which mechanism or whether it takes place at all.

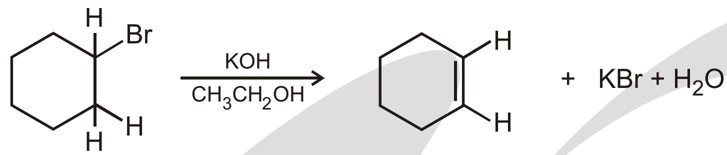
◎ CHEMICAL PROPERTIES

Alkene addition reactions occur widely, both in the laboratory and in living organisms. Although we've studied only the addition of HX thus far, many closely related reactions also take place. In this chapter, we'll see briefly how alkenes are prepared and we'll discuss further examples of alkene addition reactions. Particularly important are the addition of a halogen to give a 1,2-dihalide, addition of a hypohalous acid to give a halohydrin, addition of water to give an alcohol, addition of hydrogen to give an alkane, addition of a single oxygen to give a three-membered cyclic ether called an epoxide, and addition of two hydroxyl groups to give a 1,2-diol.

General reaction of alkene :



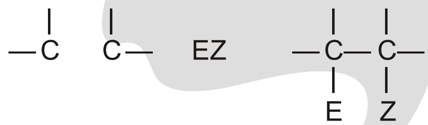




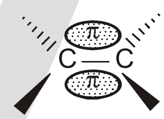
Bromocyclohexane Cyclohexene (81%)

Addition Reactions :

Reactions at the carbon–carbon double bond. : The double bond consists of a strong bond and a weak bond; we expect, therefore, that reaction would involve breaking of this weaker bond. This expectation is correct; the typical reactions of the double bond are of the sort where the bond is broken and two strong bonds are formed in its place.



A reaction in which two molecules combine to yield a single molecule or product is called an addition reaction. The reagent is simply added to the substrate. Addition reactions are necessarily limited to compounds that contain atoms sharing more than one pair of electrons, that is, to compounds that contain multiple bonded atoms. Formally, addition is the opposite of elimination; just as elimination generates a multiple bond, addition destroys it.



In the structure of the bond there is a cloud of electrons above and below the plane of the atoms. These electrons are less involved than the electrons in holding together the carbon nuclei.

As a result, they are themselves held less tightly. These loosely held electrons are particularly available to a reagent that is seeking electrons. It is not surprising, then, that in many of its reactions the carbon–carbon double bond serves as a source of electrons : that is, it acts as a base. The compounds with which it reacts are those that are deficient in electrons, that is, are acids. These acidic reagents that are seeking a pair of electrons are called electrophilic reagents. The typical reaction of an alkene is electrophilic addition, or, in other words, addition of acidic reagents.

Reagents of another kind *i.e.*, free radicals also seek electrons–or, rather, seek an electron. And so we find that alkenes also undergo free-radical addition.

Electrophilic Addition Reaction – Mechanism

Addition of the acidic reagent, HZ, involves two steps : (Z may be — Cl, — Br, — I, — CN, — OH, — OSO_3H etc.)

Step (1) the first step involves the addition of H leading to the formation of carbocation.

Step (2) is the combining of the carbocation with the base : Z.

The evidence for this mechanism includes.

(a) The rate of reaction depends upon the concentration of both the alkene and the reagent HZ.

(b) Where the structure permits, reaction is accompanied by rearrangements. In addition, the mechanism is consistent with structures :

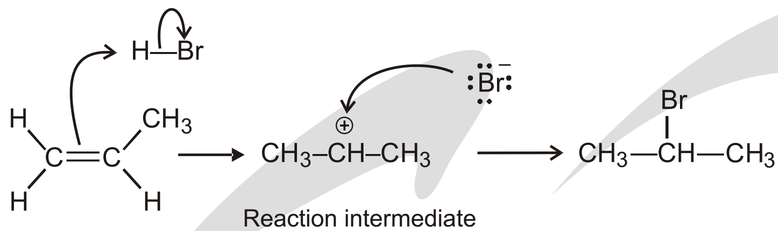
(c) The orientation of addition; and

(d) The relative reactivities of alkenes.

1. Hydrohalogenation (addition of H–X) :

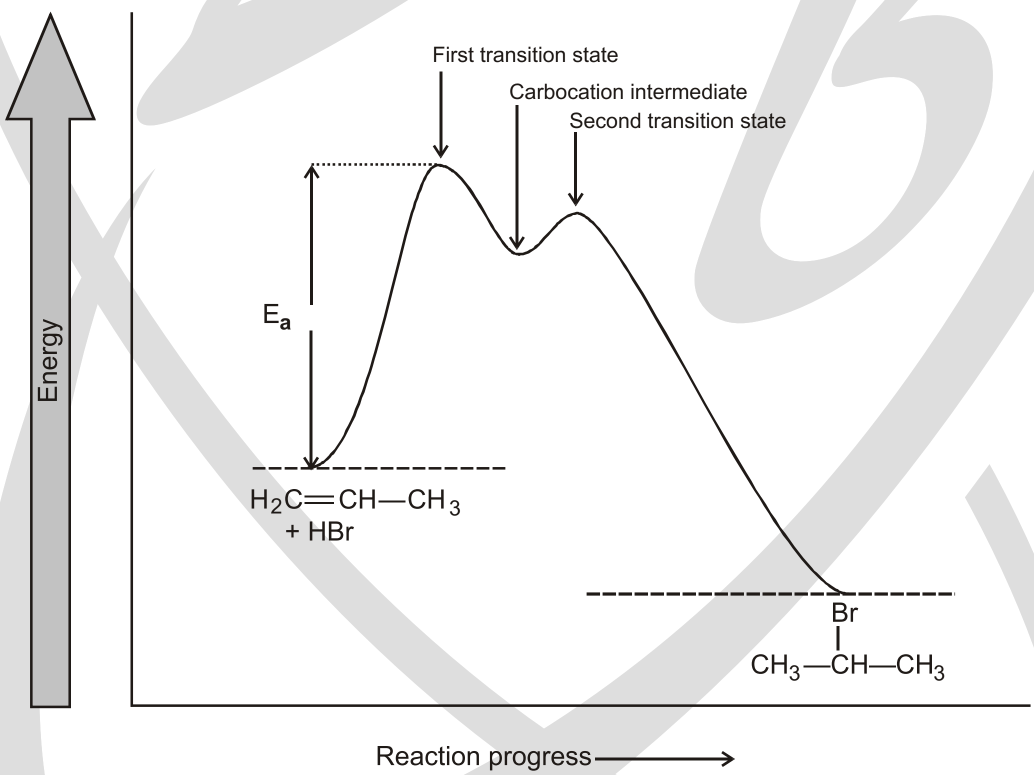
Describing a Reaction: Intermediates :

How can we describe the carbocation formed in the first step of the reaction of ethylene with HBr? The carbocation is clearly different from the reactants, yet it isn't a transition state and it isn't a final product.



We call the carbocation, which exists only transiently during the course of the multistep reaction, a **reaction intermediate**. As soon as the intermediate is formed in the first step by reaction of ethylene with H , it reacts further with Br in a second step to give the final product, bromoethane. We can picture the second transition state as an activated complex between the electrophilic carbocation intermediate and the nucleophilic bromide anion, in which Br donates a pair of electrons to the positively charged carbon atom as the new C–Br bond just starts to form.

A complete energy diagram for the overall reaction of ethylene with HBr is shown. In essence, we draw a diagram for each of the individual steps and then join them so that the carbocation product of step 1 is the reactant for step 2. As indicated, the reaction intermediate lies at an energy minimum between steps. Because the energy level of the intermediate is higher than the level of either the reactant that formed it or the product it yields, the intermediate can't normally be isolated. It is, however, more stable than the two transition states that neighbour it.

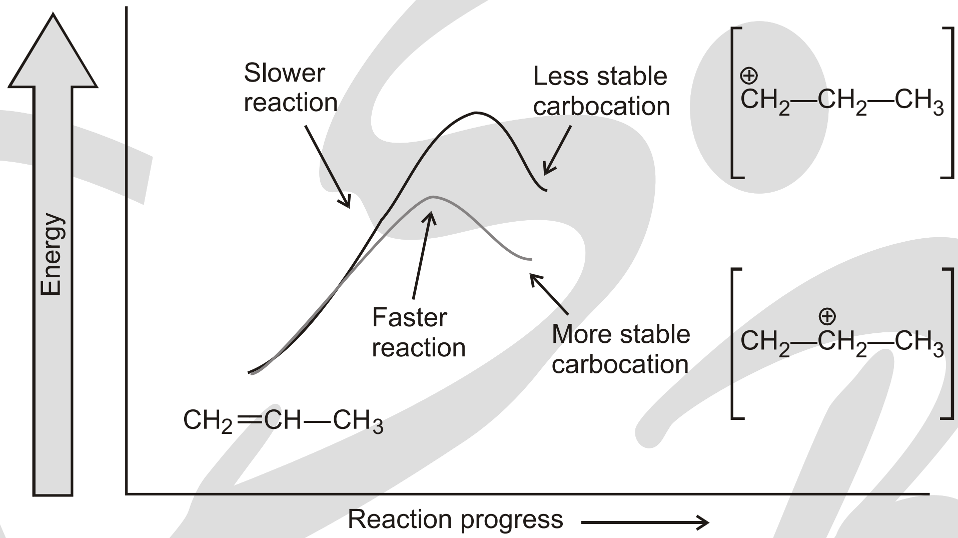


An energy diagram for the reaction of propylene with HBr.

Hammond postulate

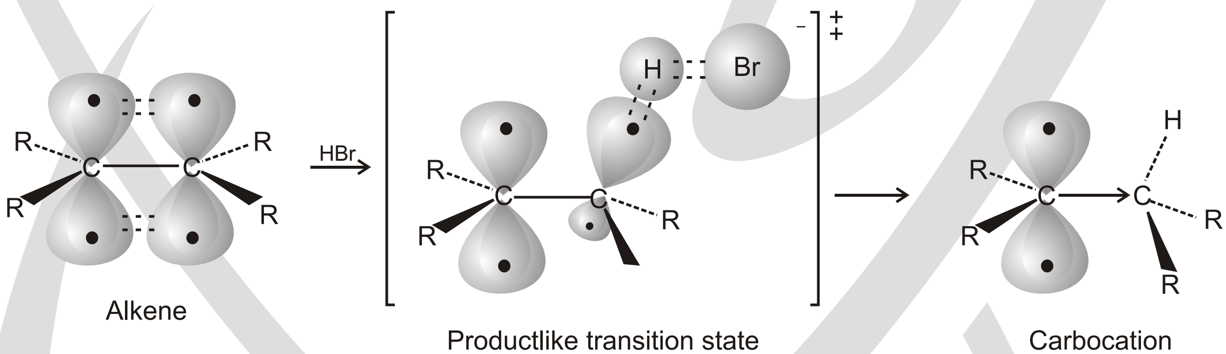
The structure of a transition state resembles the structure of the nearest stable species. Transition states for endergonic steps structurally resemble products, and transition states for exergonic steps structurally resemble reactants.

How does the Hammond postulate apply to electrophilic addition reactions? The formation of a carbocation by protonation of an alkene is an endergonic step. Thus, the transition state for alkene protonation structurally resembles the carbocation intermediate, and any factor that stabilizes the carbocation will also stabilize the nearby transition state. Since increasing alkyl substitution stabilizes carbocations, it also stabilizes the transition states leading to those ions, thus resulting in a faster reaction. More stable carbocations form faster because their greater stability is reflected in the lower-energy transition state leading to them



Energy diagrams for carbocation formation.

The more stable tertiary carbocation is formed faster because its increased stability lowers the energy of the transition state leading to it.



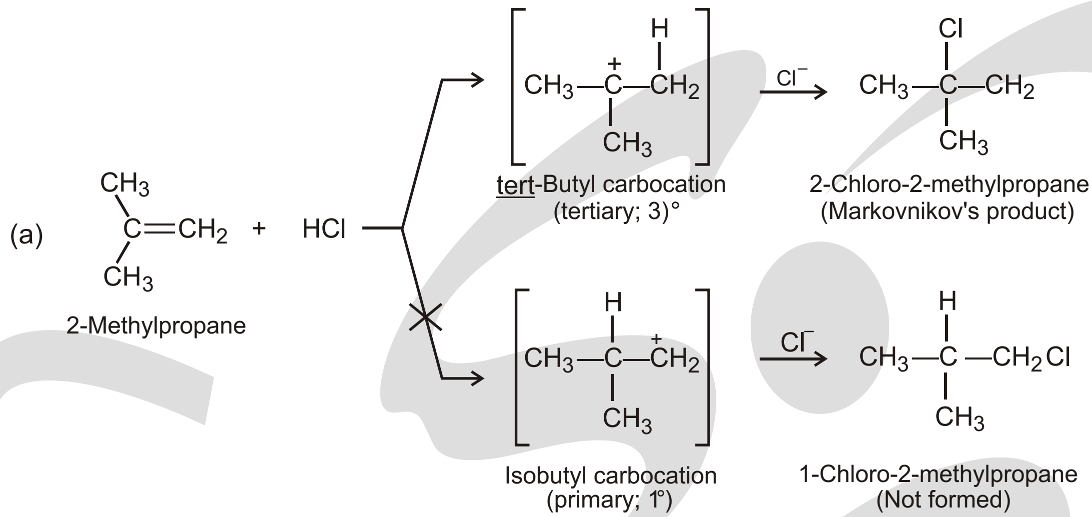
The hypothetical structure of a transition state for alkene protonation. The transition state is closer in both energy and structure to the carbocation than to the alkene. Thus, an increase in carbocation stability also causes an increase in transition-state stability, thereby increasing the rate of its formation.

Markovnikov's Rule and Regioselectivity :

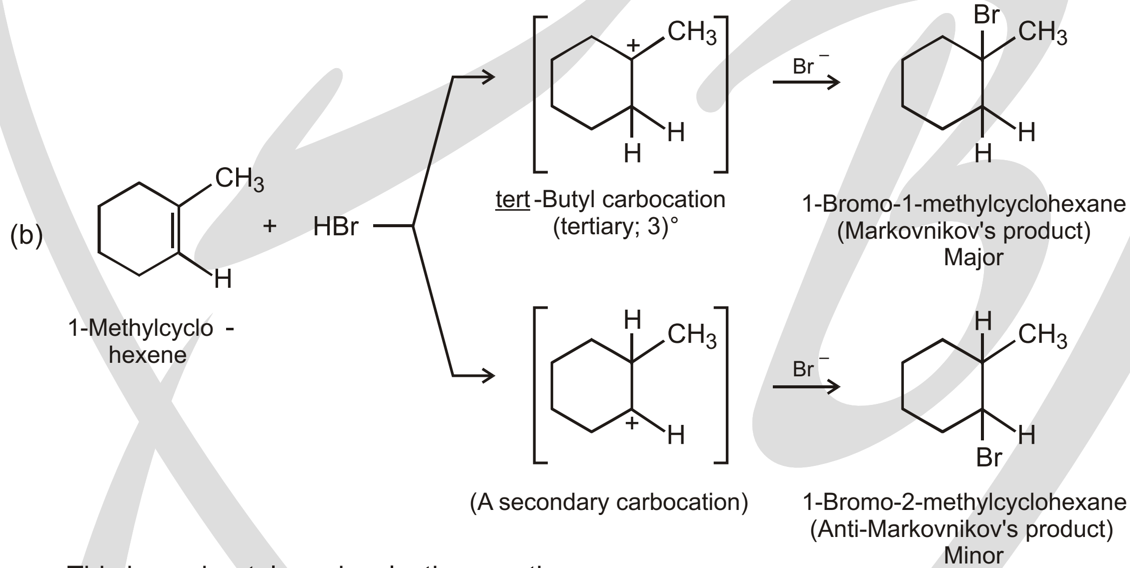
Markovnikov's rule is applicable on unsymmetric alkenes and alkynes. According to it, in electrophilic addition reactions, H+(an electrophile) attacks on least substituted carbon or the carbon having more number of hydrogens present. Product obtained in accordance with above rule, is called Markovnikov product.

A reaction in which two or more constitutional isomers could be obtained as products, but one of them predominates, is called a **regioselective reaction.**

There are degrees of **regioselectivity** : a reaction can be moderately regioselective, highly regioselective, or completely regioselective. For example,

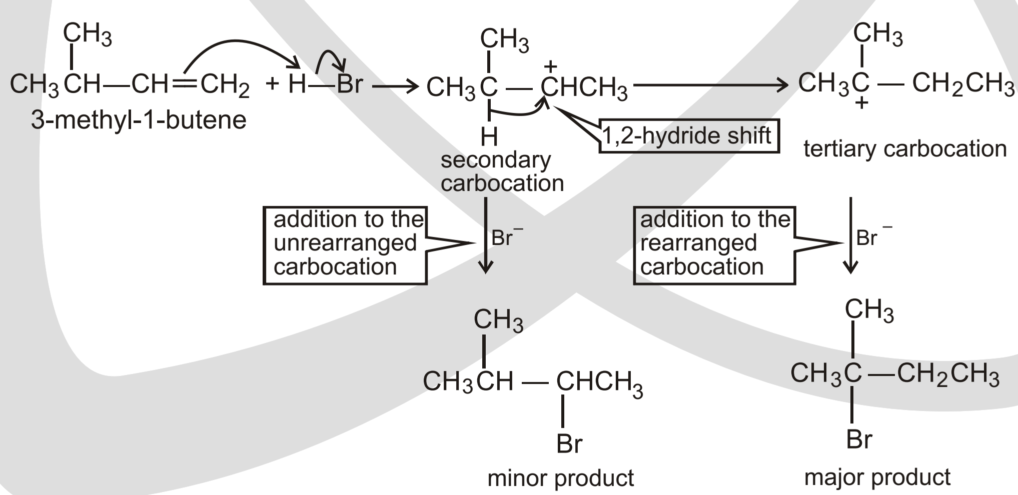


This is highly regioselective reaction.



This is moderately regioselective reaction.

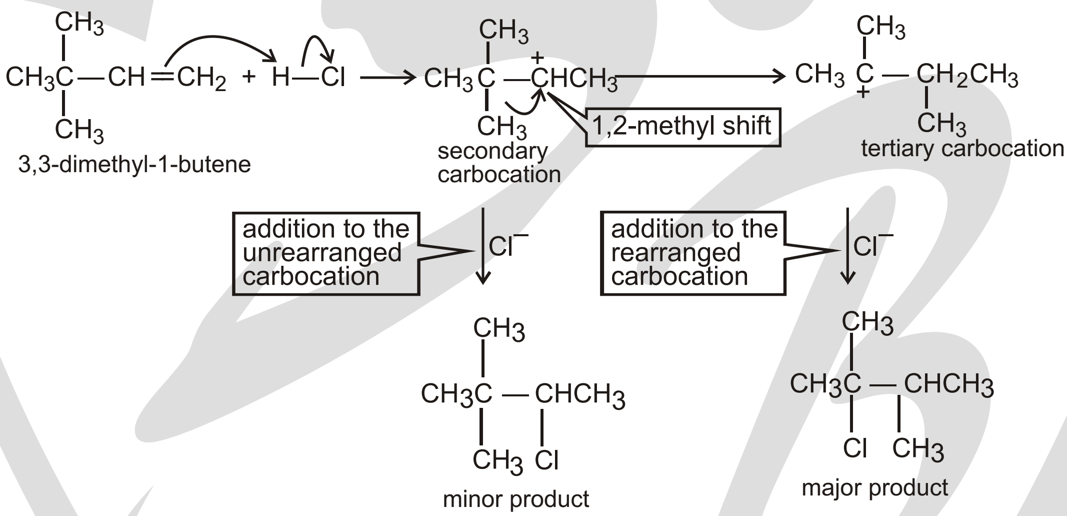
Rearrangement in electrophilic addition reactions :



Because a hydrogen shifts with its pair of electrons, the rearrangement is called a hydride shift. (Recall that - is a hydride ion.) More specifically, it is called a **1,2- hydride shift** because the hydride ion moves from one carbon to an adjacent carbon.

As a result of the **carbocation rearrangement**, two alkyl halides are formed, one from adding the nucleophile to the unrearranged carbocation and one from adding the nucleophile to the rearranged carbocation. The major product results from adding the nucleophile to the rearranged carbocation.

In the second reaction, again a secondary carbocation is formed initially. Then one of the methyl groups, with its pair of electrons, shifts to the adjacent positively charged carbon to form a more stable tertiary carbocation. This kind of rearrangement is called a **1,2-methyl shift** —the methyl group moves with its electrons from one carbon to an adjacent carbon. Again, the major product is the one formed by adding the nucleophile to the rearranged carbocation.



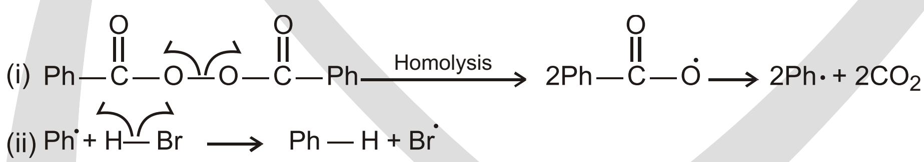
◎ Peroxide Effect or Kharash effect or Anti Markownikov's rule

In presence of peroxide Anti Markownikov's addition takes place

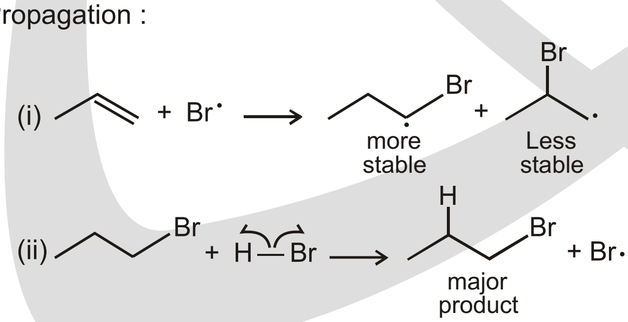
$$CH_3 - CH \quad CH_2 \xrightarrow[\text{Peroxide}]{HBr} CH_3CH_2 - CH_2 - Br$$

Mechanism: Peroxide effect proceeds via free radical mechanism as given below.

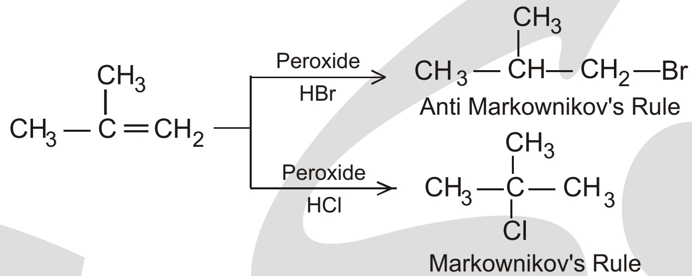
Initiation



Propagation :



SPECIAL PROBLEM



Peroxide effect is not observed in addition of HCl, HI, HF. This may be due to the fact that both. Propagation step must be exothermic for continuous chain reaction which is only observed by HBr not by HF, HCl, HI.

1. Consider the reaction of an alkene with HBr:

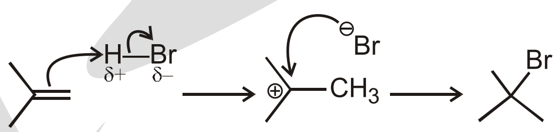
(a) Write the mechanism for the reaction.

(b) Why do the bond electrons attack the hydrogen end of HBr ?

(c) Briefly explain why the addition of HBr gives the product shown instead of the primary alkyl halide.

Sol.

(a)



(b) The hydrogen was attacked because it is less electronegative than bromine and bears the in the HBr bond (Vacant anti-bonding of H–Br).

(c) In looking at the mechanism in 3a, we can see that the formation of the carbocation is the first step. The formation of a more stable carbocation has a lower energy of activation (E_{act}). A tertiary carbocation is much more stable than an isomeric primary carbocation, so the E_{act} for the formation of the tertiary carbocation is lower than the Eact for the formation of the primary carbocation.

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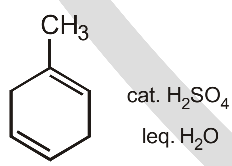
2. Consider the reaction of a nonoconjugated diene with aqueous sulfuric acid :



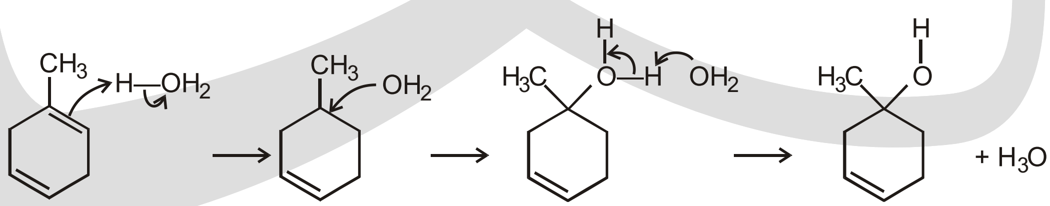
(a) Show the major product for the following reaction and provide a detailed mechanism for the reaction.

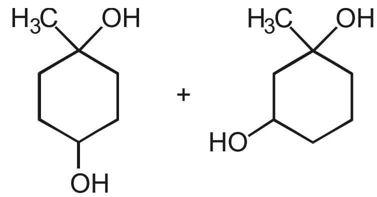
(b) Show the product if a second equivalent of H_2O is added.

(c) What happens if no acid catalyst is added ?

Sol.

(a)



(b) 

(c) Water by itself is a poor electrophile, as the magnitude of charge on the hydrogen atoms is not great enough for the reaction to proceed. Protonation of the water oxygen makes this oxygen atom more electron poor, thus amplifying the magnitude of the on hydrogen. Hydronium ion is sufficiently electrophilic to undergo the energetically expensive reaction with the alkene p bond to form a carbocation. In the absence of acid catalyst, the reaction does not proceed at a useful rate.

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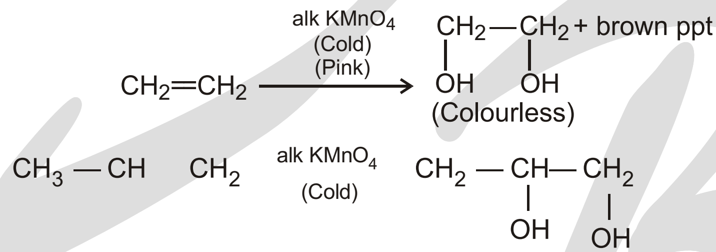
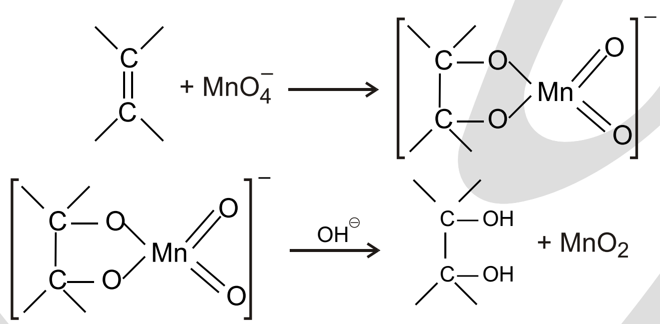
2. Hydroxylation :

(i) With Bayers Reagent : When an alkene is reacted with dilute alkaline $KMnO_4$ solution in cold condition then the alkene is converted to vicinal dihydroxy compound.



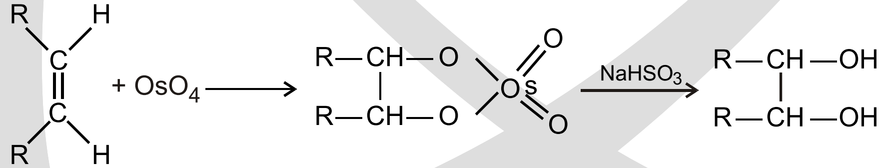
Mechanism



This reaction which gives rise to vicinal diols and is a SYN–ADDITION reaction.

This is supported by the mechanism that the oxygen atoms of OH group in the diol formed are from the permanganate ions which add to the alkene molecule from the same side.

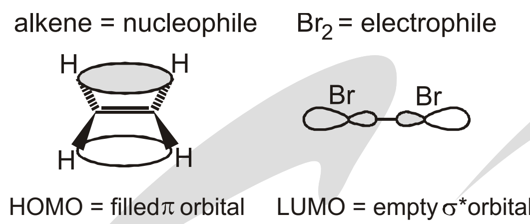
(ii) With OsO_4 (Osmium tetraoxide)



This is again a SYN–ADDITION reaction

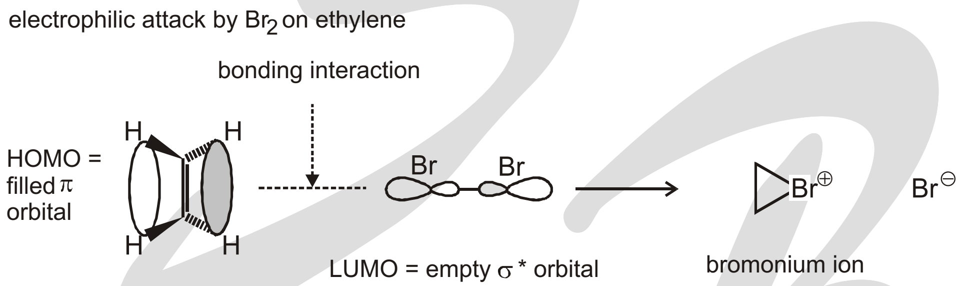
3. Halogenation :

When it reacts with Br_2, the alkene's filled orbital (the HOMO)



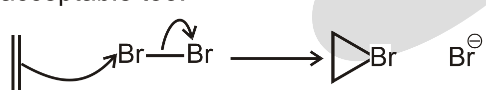
will interact with the bromine's empty * orbital to give a product. But what will that product be? Look at the orbitals involved.

The highest electron density in the orbital is right in the middle, between the two carbon atoms, so this is where we expect the bromine to attack. The only way the HOMO can interact in a bonding manner with the * LUMO is if the Br_2 approaches end-on—and this is how the product forms. The symmetrical three-membered ring product is called a bromonium ion.



How shall we draw curly arrows for the formation of the bromonium ion? We have a choice. The simplest is just to show the middle of the p bond attacking Br–Br, mirroring what we know happens with the orbitals.

But there is a problem with this representation: becauseonly one pair of electrons is moving, we can't form two new C–Br bonds. We should really then represent the C–Br bonds as partial bonds. Yet the bromonium ion is a real intermediate with two proper C–Br bonds So an alternativeway of drawing the arrows is to involve a lone pair on bromine. We think the first way represents more accurately the key orbital interaction involved, and we shall use that one, but the second is acceptable too.



Of course, the final product of the reaction isn't the bromonium ion. The second step of the reaction follows on at once: the bromonium ion is an electrophile, and it reacts with the bromide ion lost from the bromine in the addition step. We can now draw the correct mechanism for the whole reaction, which is termed electrophilic addition to the double bond, because bromine is an electrophile. Overall, the molecule of bromine adds across the double bond of the alkene.

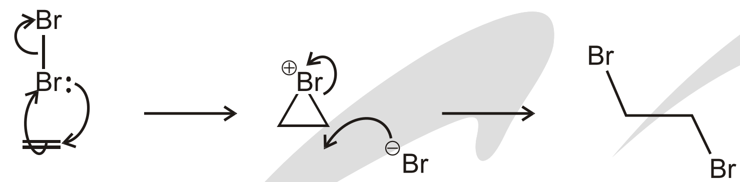
Alkenes react with halogens (x_2 / CCl_4) to give vicinal-dihalide, an oily liquid. Halogen (Cl_2, Br_2, I_2) acts as an electrophilc and halogen molecule adds on alkene and p-bond is destroyed. It is anti-addition-

$$\underset{}{CH_2{=}CH_2} + \underset{\text{(red-brown colour)}}{Br_2} \xrightarrow{CCl_4} \underset{\text{(oily liquid)}}{\underset{Br}{}\overset{Br}{\overset{|}{C}H_2}-\underset{\underset{Br}{|}}{CH_2}} + \text{Red-Brown colour disappears}$$

This reaction is used as a tool to detect whether a p-bond is present or not in an unknown organic compound (Test of unsaturation).

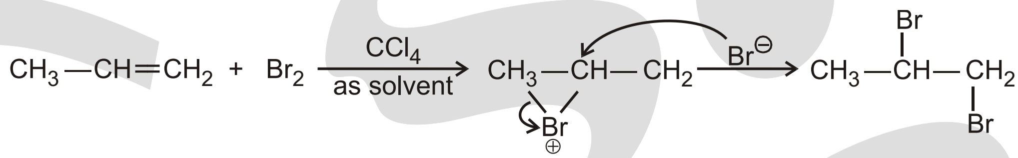
Mechanism :

electrophilic addition of bromine to ethylene

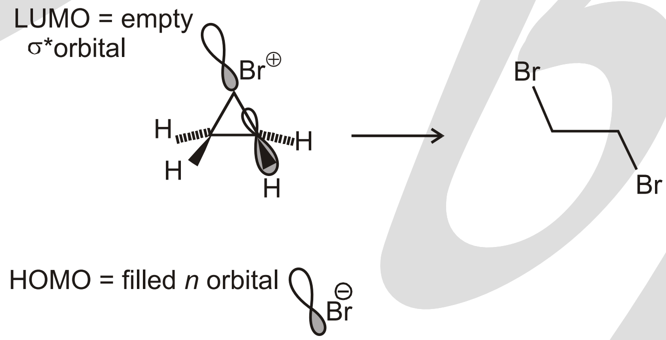


(Cyclic Bromonium ion)

Reaction intermediate is non-classical carbocation in which charge is delocalized over all three atoms.In unsymmetric non-classical carbocation, Br attacks on that carbon where charge is relatively more stabilized.



Attack of Br on a bromonium ion is a normal S_N2 substitution—the key orbitals involved are the HOMO of the bromide and the * of one of the two carbon–bromine bonds in the strained three-membered ring. As with all S_N2 reactions, the nucleophile maintains maximal overlap with the * by approaching in line with the leaving group but from the opposite side, resulting in inversion at the carbon that is attacked. The stereochemical outcome of more complicated reactions (discussed below) is important evidence for this overall reaction mechanism.

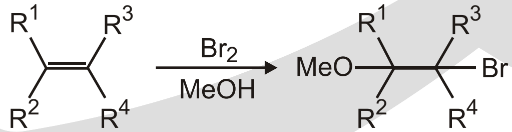


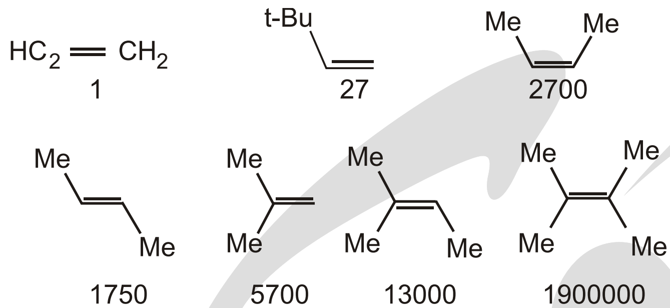
Why doesn't the bromine simply attack the positive charge and re-form the bromine molecule ? Well, in fact, it does and the first step is reversible.

Rates of bromination of alkenes

The pattern you saw for epoxidation with peroxyacids (more substituted alkenes react faster) is followed by bromination reactions too. The bromonium ion is a reactive intermediate, so the rate-determining step of the brominations is the bromination reaction itself. The chart shows the effect on the rate of reaction with bromine in methanol of increasing the number of alkyl substituents from none (ethylene) to four. Each additional alkene substituent produces an enormous increase in rate. The degree of branching (Me versus n-Bu versus t-Bu) within the substituents has a much smaller, negative effect (probably of steric origin) as does the geometry (E versus Z) and substitution pattern (1,1-disubstituted versus 1,2-disubstituted) of the alkene.

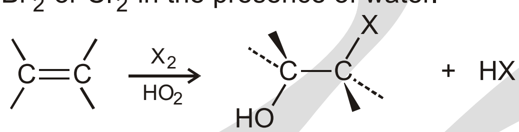
Relative rates of reaction of alkenes with bromine in methanol solvent.



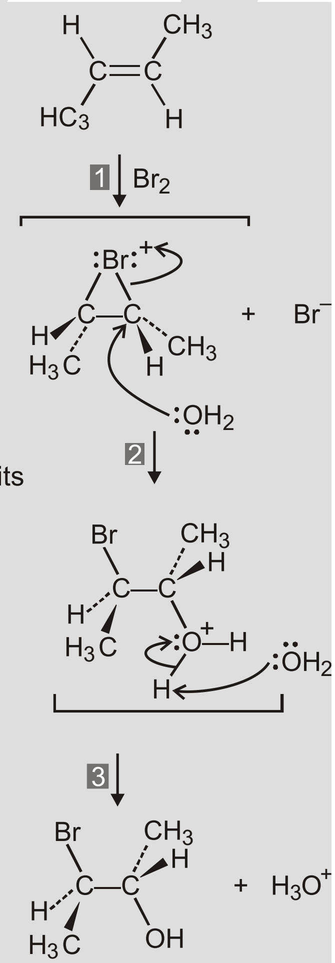


4. Halohydrins from Alkenes : Addition of HOX :

Another example of an electrophilic addition is the reaction of alkenes with the hypohalous acids HO–Cl or HO–Br to yield 1,2-halo alcohols, called halohydrins. Halohydrin formation doesn't take place by direct reaction of an alkene with HOBr or HOCl, however. Rather, the addition is done indirectly by reaction of the alkene with either Br_2 or Cl_2 in the presence of water.



An alkene A halohydrin



1. Reaction of the alkene with Br yields a bromonium ion intermediate, as previouslydiscussed.

2. Water acts as a nucleophile, using a lone pair of electrons to open the bromonium ion ring and form a bond to carbon. Since oxygen donates its electrons in this step, it now has the positive charge.

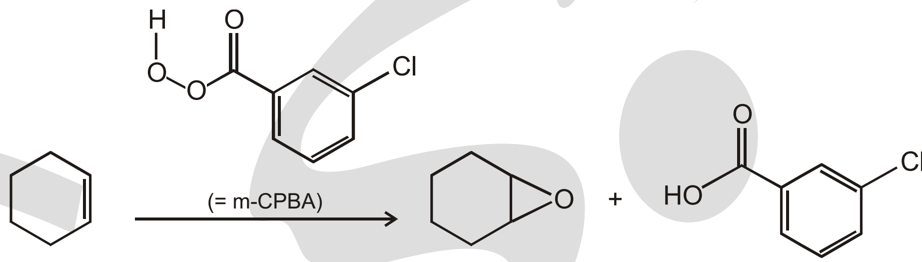
3. Loss of a proton (H) from oxygen then gives H_3O and the neutral bromohydrin addition product.

3-Bromo-2-butanol
(a bromohydrin)

5. Epoxidation :

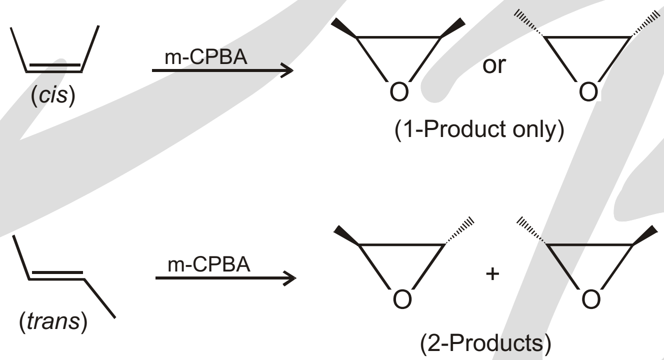
An alkene on reaction with peroxy-acid gives 3-membered cyclic ether (*i.e.* epoxide).

The most commonly used peroxy-acid is known as m-CPBA, or meta-Chloro Peroxy Benzoic Acid. m-CPBA is a safely crystalline solid. Here it is, reacting with cyclohexene, to give the epoxide in 95% yield. It is syn-additon.



95% yield

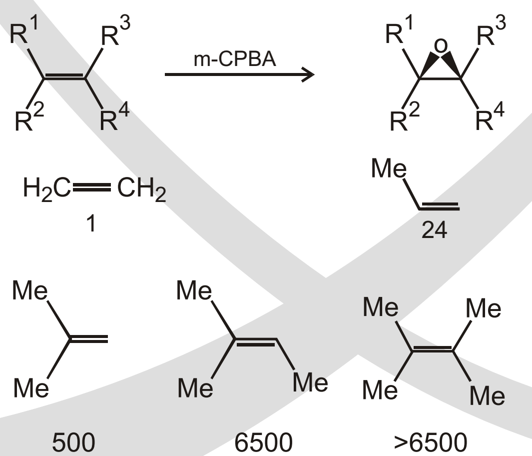
Because both new C–O bonds are formed on the same face of the alkene's p bond, the geometry of the alkene is reflected in the stereochemistry of the epoxide. The reaction is therefore stereospecific. Here are two examples demonstrating this: cis-alkene gives cis-epoxide and trans-alkene gives transepoxide.



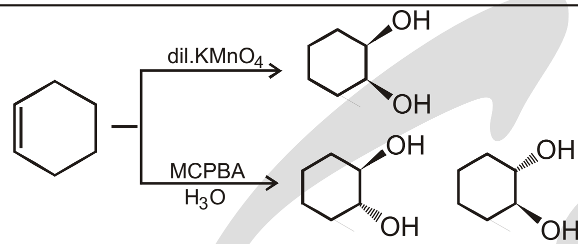
More substituted alkenes epoxidize faster

Peracids give epoxides from alkenes with any substitution pattern but the chart alongside shows how the rate varies according to the number of substituents on the double bond. Not only are more substituted double bonds more stable , but they are more nucleophilic. We showed you that alkyl groups are electrondonating because they stabilize carbocations.

Relative rates of reaction of alkenes with m-CPBA



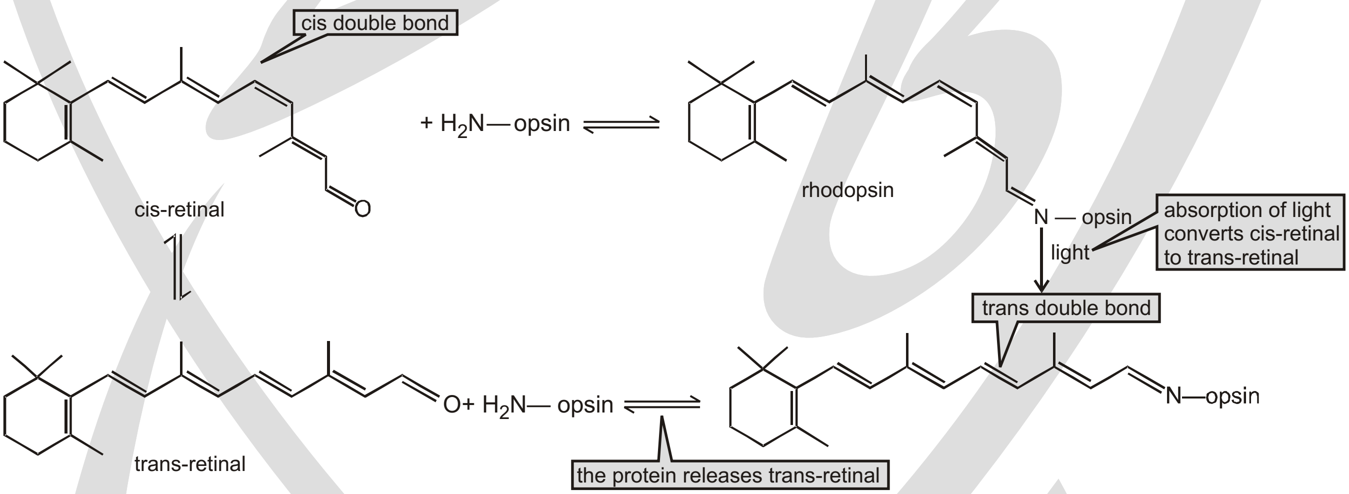
Solved Example



SPECIAL PROBLEM

***cis-trans* Interconversion in Vision :**

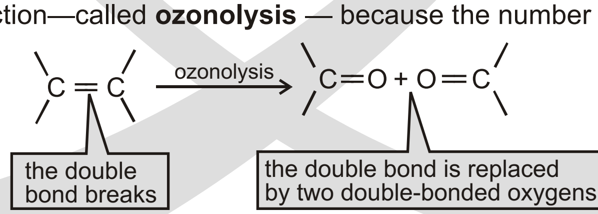
Our ability to see depends in part on an interconversion of cis and trans isomers that takes place in our eyes. A protein called opsin binds to cis-retinal (formed from vitamin A) in photoreceptor cells (called rod cells) in the retina to form rhodopsin. When rhodopsin absorbs light, a double bond interconverts between the cis and trans configurations, triggering a nerve impulse that plays and important role in vision. trans-Retinal is then released from opsin. trans-Retinal isomerizes back to cis-retinal and another cycle begins. To trigger the nerve impulse, a group of about 500 rod cells must register five to seven rhodopsin isomerizations per cell within a few tenths of a second.



◎ 6. THE ADDITION OF OZONE TO AN ALKENE: OZONOLYSIS

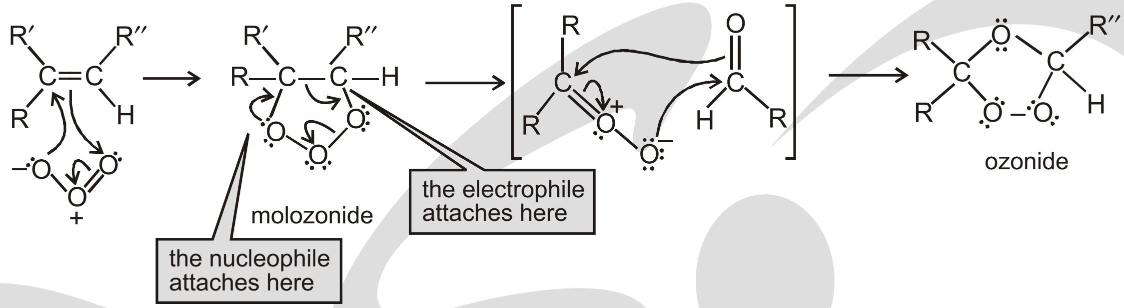
When an alkene is treated with ozone (O_3) at a low temperature, both the sigma and pie bonds of the double bond break and the carbons that were doubly bonded to each other are now doubly bonded to oxygens instead.

This is an oxidation reaction—called **ozonolysis** — because the number of C—O bonds increases.



Ozonolysis is an example of **oxidative cleavage** —an oxidation reaction that cleaves the reactant into pieces (***lysis*** is Greek for “breaking down”).

MECHANISM FOR OZONIDE FORMATION

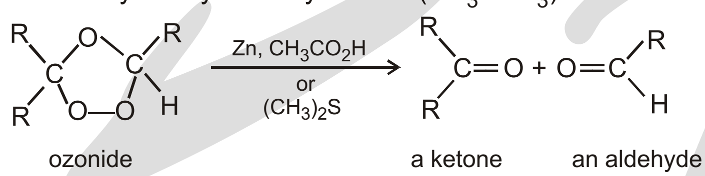


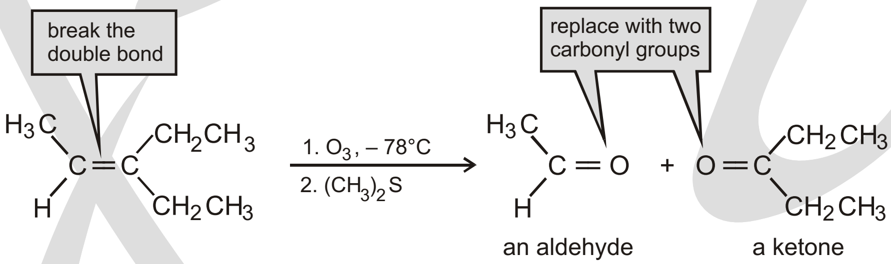
The electrophile (an oxygen at one end of the ozone molecule) adds to one of the sp^2 carbons, and a nucleophile (the oxygen at the other end) adds to the other sp^2 carbon.

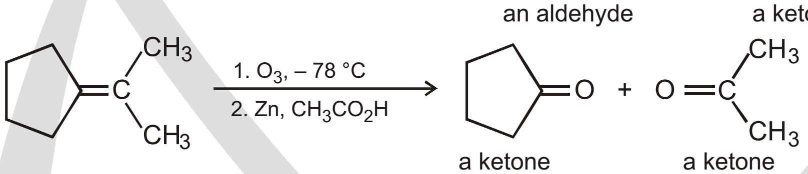
The product is a molozonide.

- The molozonide is unstable because it has two O—O bonds; it immediately rearranges to a more stable ozonide.

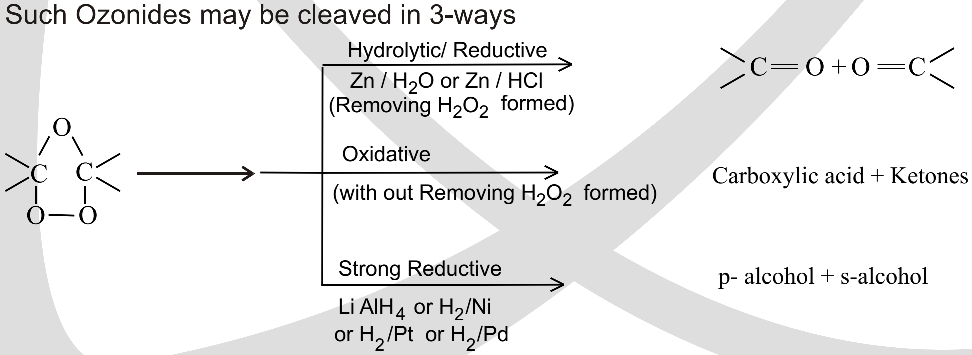
Because ozonides are explosive, they are not isolated. Instead, they are immediately conver ted to ketones and/or aldehydes by dimethyl sulfide (CH_3SCH_3) or zinc in acetic acid (CH_3CO_2H).





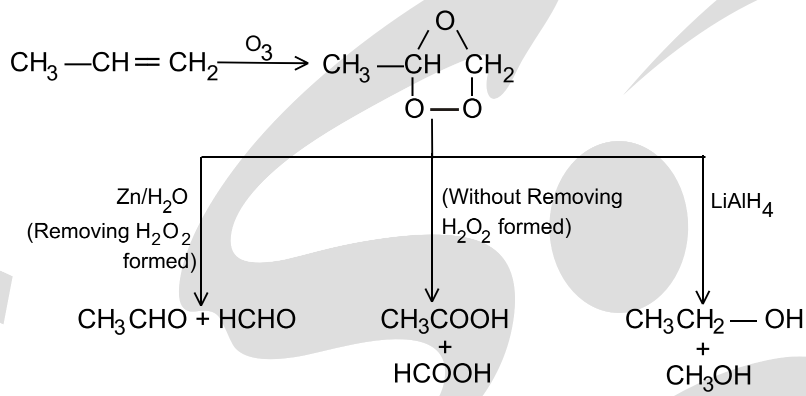


Such Ozonides may be cleaved in 3-ways

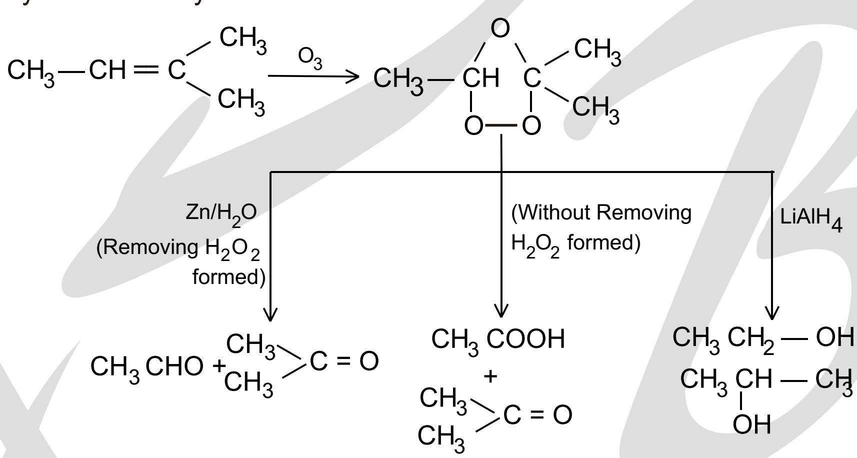


Solved example

(a) Ozonolysis of Propene

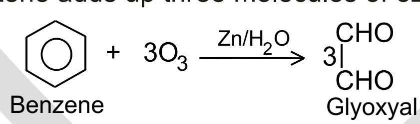


(b) Ozonolysis of 2-Methyl but-2-ene



Note Only
CH_3CHO gets converted to acid
i.e. aldehydic part gets converted to carboxylic acid

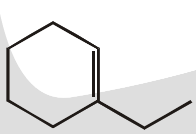
(c) Addition of ozone – Benzene adds up three molecules of ozone forming glyoxal.



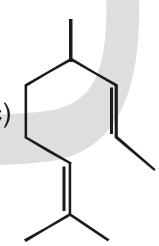
PROBLEM-SOLVING STRATEGY

Determing the Products of Oxidative Cleavage

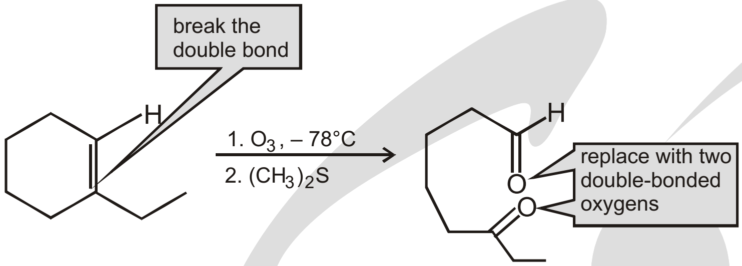
What products would you would expect to obtain when the following compounds react with ozone and then with dimethylsulfide?

(a) 

(b) $CH_3CH_2CH=CHCH_2CH_3$

(c) 

Sol. (a) Break the double bond and replace it with two double-bonded oxygens.



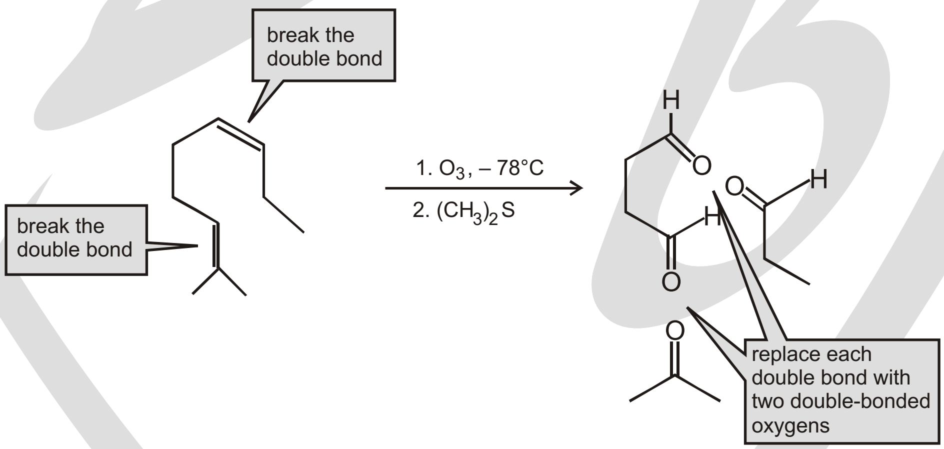
Sol. (b) Break the double bond and replace it with two double-bonded oxygens. Because the alkene is symmetrical, only one product is formed.

$$CH_3CH_2CH = CHCH_2CH_3 \xrightarrow[2.\ (CH_3)_2S]{1.\ O_3\ -\ 78°C} CH_3CH_2CH = O \quad O = CH_2CH_3CH$$

break the double bond

replace the double bond with = O and O =

Sol. (c) Since the reactant has two double bonds, each one must be replaced with two double-bonded oxygens.



Sol. Because only one product is obtained, the reactant must be a cyclic alkene.

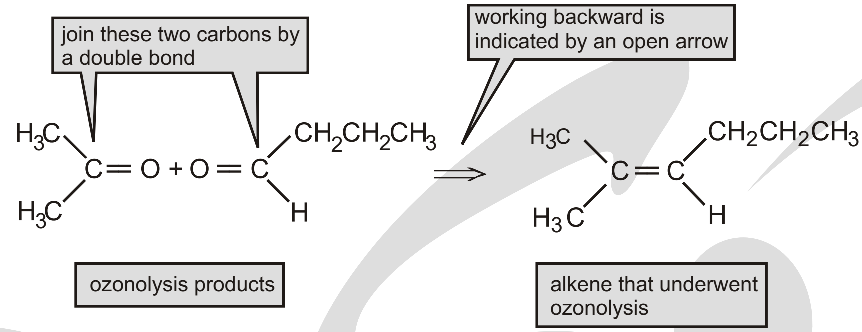
Numbering the product shows that the carbonyl groups are at C-1 and C-6, so the double bonds must be between C-1 and C-6.

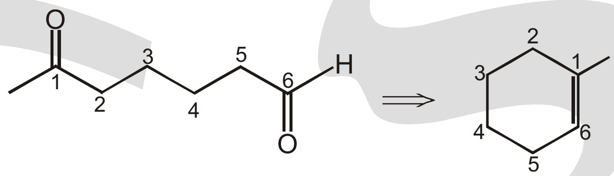
Predicting the Reactant in an Ozonolysis Reaction

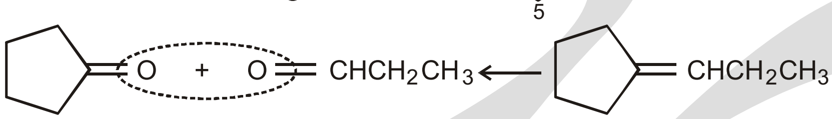
Ozonolysis can be used to determine the structure of an unknown alkene. If we know what carbonyl compounds are formed by ozonolysis, we can mentally work backward to deduce the structure of the alkene. In other words, delete the “O and O“ and join the carbons by a double bond. (Recall that working backward is indicated by an open arrow.)

Strategy :

Reaction of an alkene with ozone, followed by reduction with zinc, cleaves the C C bond and gives two carbonyl-containing fragments. That is, the C C bond becomes two C O bonds. Working backward from the carbonyl-containing products, the alkene precursor can be found by removing the oxygen from each product and joining the two carbon atoms to form a double bond.

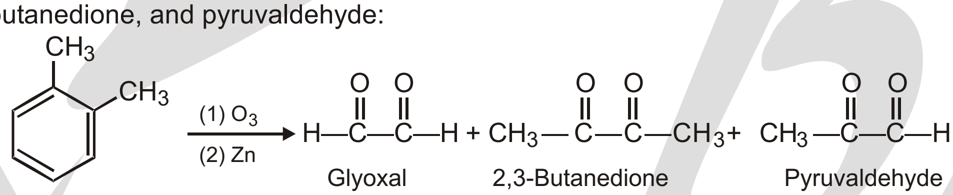
(a) 

(b) 

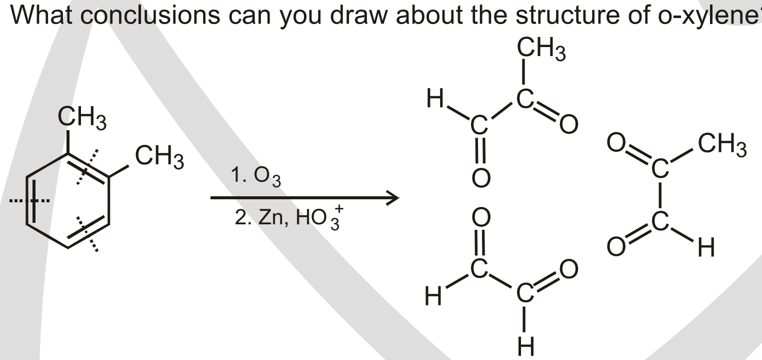
(c) 

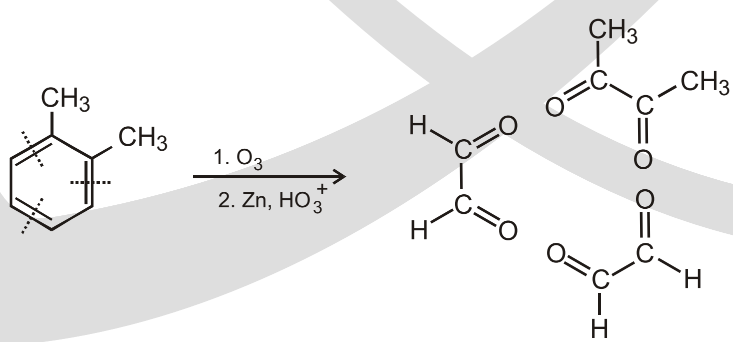
Solved example

- In 1932, A. A. Levine and A. G. Cole studied the ozonolysis of o-xylene and isolated three products: glyoxal, 2,3-butanedione, and pyruvaldehyde:



In what ratio would you expect the three products to be formed if o-xylene is a resonance hybrid of two structures? The actual ratio found was 3 parts gly-oxal, 1 part 2,3-butanedione, and 2 parts pyruvaldehyde. What conclusions can you draw about the structure of o-xylene?

Sol. 



Solved example

- With HOT $KMnO_4$: Alkenes on reaction with hot alkaline $KMnO_4$ give a mixture of carboxylic acid and ketones or only ketones or carboxylic acids.

$$CH_2 \quad CH_2 \xrightarrow[\text{Hot KMnO}_4]{(O)} 2HCOOH \xrightarrow{(O)} 2$$

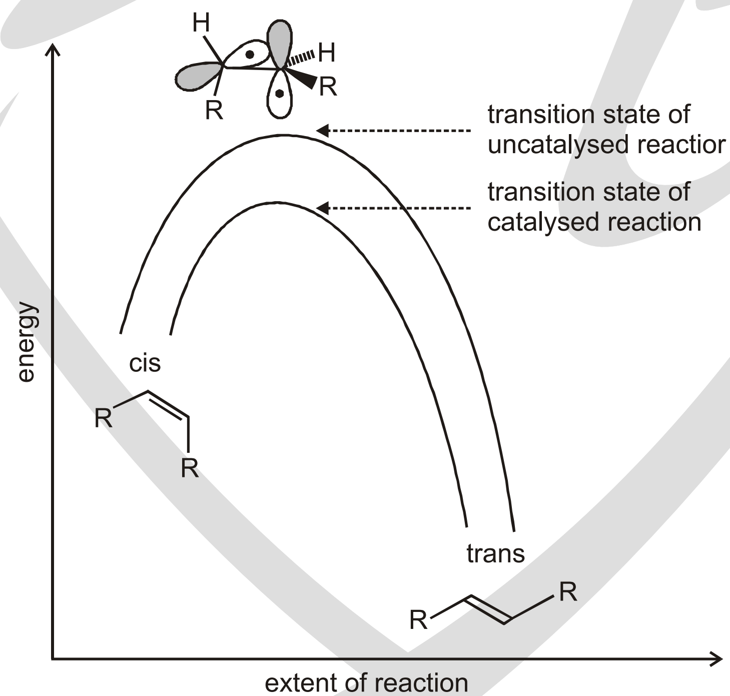
$$CH_3 - CH \quad CH_2 \xrightarrow[\text{Hot KMnO}_4]{(O)} CH_3COOH + CO_2 \quad H_2O$$

$$\underset{\displaystyle CH_3 - C}{\overset{CH_3}{|}} \quad CH_2 \xrightarrow[\text{Hot KMnO}_4]{(O)} \underset{\displaystyle CH_3 - C}{\overset{CH_3}{|}} \quad O + CO_2 + H_2O$$

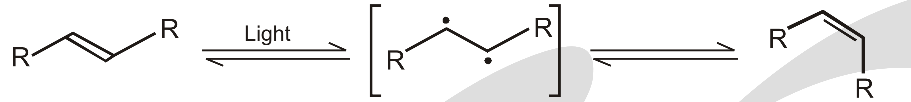
SPECIAL PROBLEM

◎ How to catalyse the isomerization of alkenes

The rate at which a reaction occurs depends on its activation energy—quite simply, if we can decrease this, then the reaction rate will speed up. There are two ways by which the activation energy may be decreased: one way is to raise the energy of the starting materials; the other is to lower the energy of the transition state. In the cis/trans isomerization of alkenes, the transition state will be halfway through the twisting operation—it has *p* orbitals on each carbon at right angles to each other. It is the most unstable point on the reaction pathway.

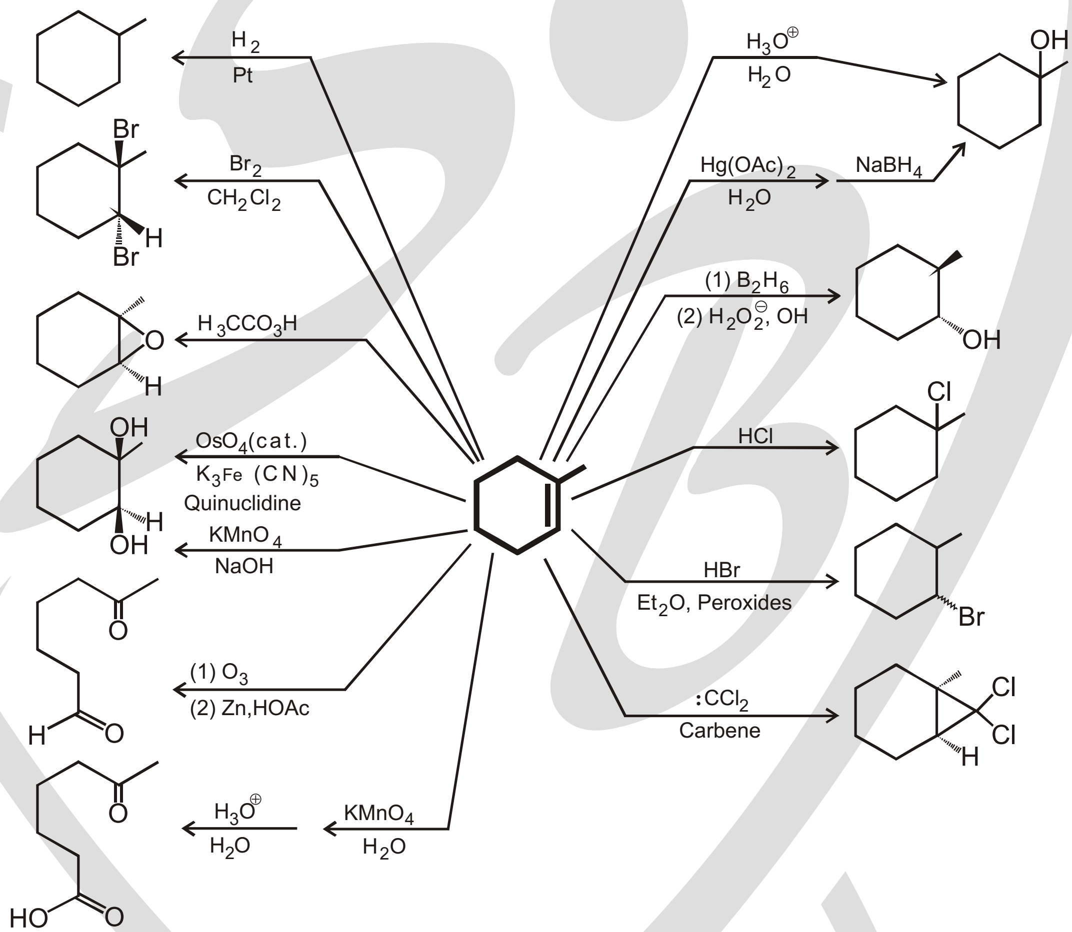


Lowering the energy of the transition state means stabilizing it in some way or other. For example, if there is a separation of charge in the transition state, then a more polar solvent that can solvate this will help to lower the energy of the transition state. Catalysts generally work by stabilizing the transition states or intermediates in a reaction. It can be catalyzed by the light.



Another approach to alkene isomerization would be to use a catalyst. Base catalysis is of no use as there are no acidic protons in the alkene. Acid catalysis can work if a carbocation is formed by protonation of the alkene.

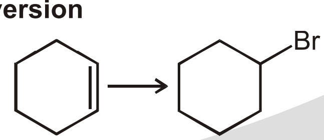
Summary :



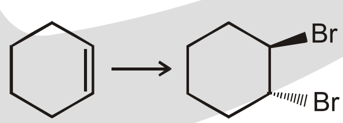
EXERCISE

MATCH THE COLUMN

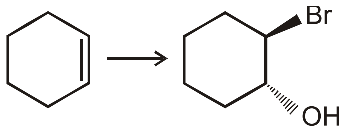
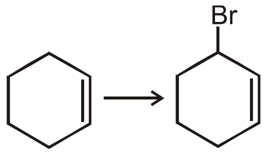
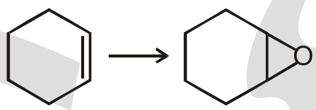
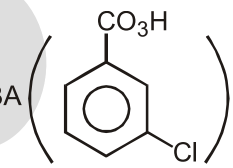
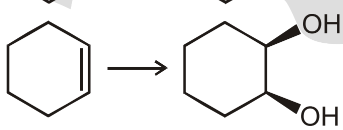
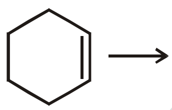
1. **Conversion** | **Reagent is used**

(A) 

(P) Br_2 / H_2O

(B) 

(Q) NBS

(C) 	(R) Br_2/CCl_4
(D) 	(S) HBr/CCl_4
(E) 	(T) MCPBA 
(F) 	(U) cold. dil. KMnO4
(G)  Adipic acid	(V) hot KMnO4

Answers

MATCH THE COLUMN

1. (A) S ; (B) R ; C P ; D Q ; E T ; F U ; G V

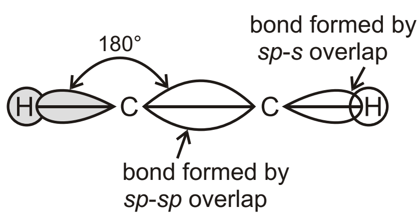
❑❑❑

CHAPTER

30

Alkyne

INTRODUCTION



$$H-C \overset{180°}{\equiv} C-H$$

A triple bond consist of a -bond formed by (*sp* *sp*) overlap *p* and 2*p*-bonds formed by *p*-*p* overlap.

- These are the acyclic hydrocarbons which contains carbon-carbon triple bond are called alkyne.
- Hybridization state of triply bonded carbon in alkyne is sp or also called as diagonal hybridisation.
- Geometry of carbon is linear in alkynes.
- Bond angle in alkyne is 180 .
- Their general formula is $C_nH_{2n\ 2}$.
- C — C triple bond length is 1.20 Å.
- C — H bond length is 1.08 Å.
- C — C triple bond energy is 190 kcal./mol.
- C — H bond energy is 102.38 kcal./mol.
- Alkyne shows chain, position and functional isomerism. They are functional isomer with cycloalkene and alkadiene.

(i) $C_1 — C_4$ compounds do not show chain isomerism.

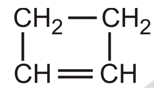
(ii) Functional isomers of C_4H_6

e.g.,

$$\underset{\text{1-Butyne}}{CH_3-CH_2-C\equiv CH} \qquad \underset{\text{1, 2-Butadiene}}{CH_3-CH=C=CH_2}$$

$CH_2=CH-CH=CH_2$

1,3-Butadiene



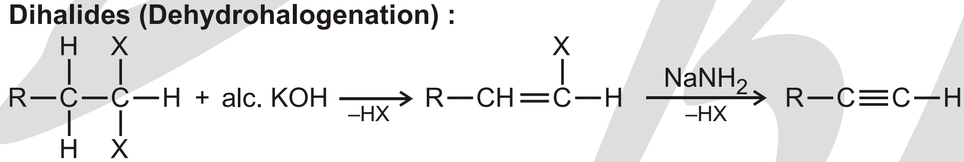
Cyclobutene

Physical Properties

- Alkynes are colourless, odourless and tasteless.
- Lower alkynes are partially soluble in H_2O. (It is due to its polarisibility).
- Higher alkynes are insoluble in water due to more % of covalent character.
- Completely soluble in organic solvents.
- Melting point and boiling point increases with molecular mass and decreases with number of branches.
- Upto C_4 alkynes are gaseous.C_5 — C_{11} are liquid, C_{12} & above are solids.
- Pure acetylene is odourless and impure acetylene has odour like garlic. It is due to impurities of Arsene (AsH_3) & Phosphine (PH_3).
- Acetylene & 1- alkyne are acidic in nature. It is due to greater electronegativity of *sp* hybridised 'C'.
- Acetylene has two acidic hydrogen atoms. It can neutralise two equivalents of base at the same time. So it is also called as dibasic acid. But the base should be very stronger as $-NH_2$ or $-CH_3$ etc.

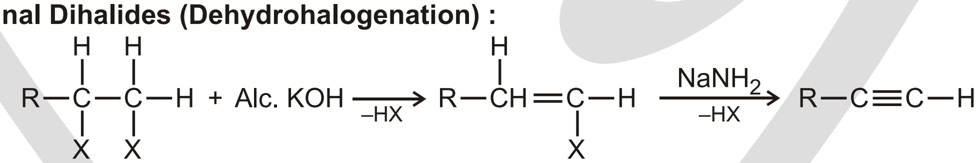
Preparation Methods :

1. **From Gem Dihalides (Dehydrohalogenation) :**



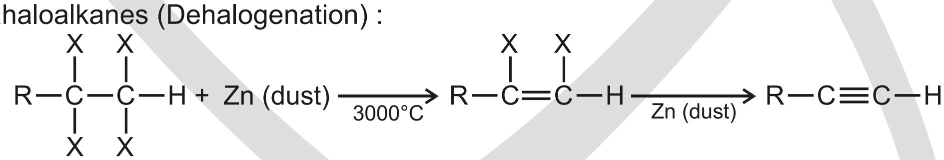
❑ **NOTE :** Alc.KOH is not used for elimination in second step because in this case elimination takes place from doubly bonded carbon atom which is stable due to resonance so strong base $NaNH_2$ is used for elimination of HX.

2. **From Vicinal Dihalides (Dehydrohalogenation) :**



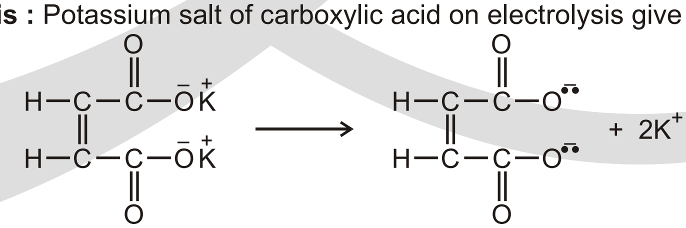
❑ **NOTE :** In the above reaction if the reactant secondary butylene chloride is taken then the products are 2-butyne, 1, 2-butadiene and 1, 3- butadiene in which 2-butyne is the chief product.

3. From Tetrahaloalkanes (Dehalogenation) :



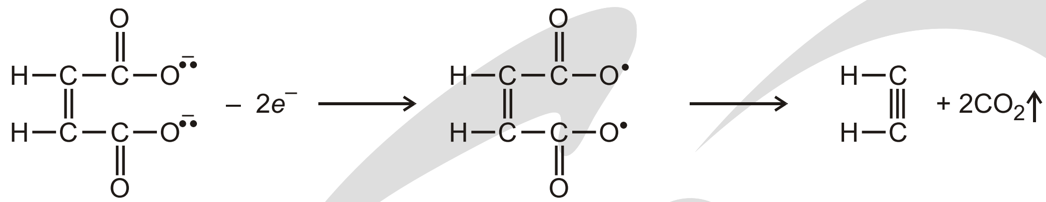
In the above reaction it is necessary that the four halogen atoms must be attached at vicinal carbons. If they are attached at the two ends then the product cyclo alkene is obtained.

4. **From Kolbe's Synthesis :** Potassium salt of carboxylic acid on electrolysis give hydrocarbon:



Potassium Malaete

At Anode :

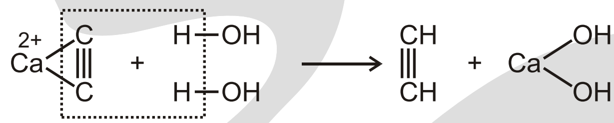


At Cathode :

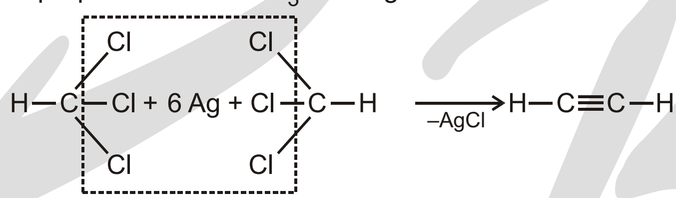
$2K \quad 2e \quad 2K$; $\quad 2K \quad 2H_2O \quad 2KOH \quad H_2$

5. Laboratory method of preparation of Acetylene :

(a) In laboratory acetylene is prepared by hydrolysis of calcium carbide.



(b) It can also be prepared from $CHCl_3$ with Ag dust.



6. From Lower Homologues to Higher Homologues

Because of the acidic characters of H-atom in acetylens.

$H—C \quad C—H \quad NaNH_2 \qquad H—C \quad C\ Na \quad NH_3$

or $\quad H—C \quad C—H \quad 2NaNH_2 \qquad Na\bar{C} \quad \bar{C}Na \quad 2NH_3$

$H—C \quad \bar{C}Na \quad R—X \qquad H—C \quad C—R \quad NaX$

$Na\bar{C} \quad \bar{C}Na \quad R—X \qquad R—C \quad C—R \quad NaX$

Chemical Properties :

1. Alkyne Acidity : Formation of Acetylide Anions

The most striking difference between alkenes and alkynes is that terminal alkynes are relatively acidic. When a terminal alkyne is treated with a strong base, such as sodium amide, Na NH$_2$, the terminal hydrogen is removed and an **acetylide anion** is formed.

$$\underset{\text{A terminal alkyne}}{R—C\equiv C—H} \xrightarrow{^{+}Na^{-}NH_2} \underset{\text{An acetylide anion}}{R—C\equiv C:^{-}\ Na^{+}} + :NH_3$$

According to the Bronsted–Lowry definition, an acid is a substance that donates H . Although we usually think of oxyacids (H_2SO_4, HNO_3) or halogen acids (HCl, HBr) in this context, any compound containing a hydrogen atom can be an acid under the right circumstances. By measuring dissociation constants of different acids and expressing the results as pK_a values, an acidity order can be established. Recall that a lower pK_a corresponds to a stronger acid and a higher pK_a corresponds to a weaker acid.

Where do hydrocarbons lie on the acidity scale? As the data in show, both methane (pK_a 60) and ethylene (pK_a = 44) are very weak acids and thus do not react with any of the common bases. Acetylene, however, has pK_aa = 25 and can be deprotonated by the conjugate base of any acid whose pKa is greater than 25. Amide ion (NH_2), for example, the conjugate base of ammonia (pK_a = 35), is often used to deprotonate terminal alkynes.

Acidity of Simple Hydrocarbons

Family	Example	K_a	pK_a	
Alkyne	HC CH	10^{25}	25	Stronger acid
Alkene	H_2C CH_2	10^{44}	44	↑
Alkane	CH_4	10^{60}	60	Weaker acid

Alkyne alkylation is not limited to acetylene itself. Any terminal alkyne can be converted into its corresponding anion and then alkylated by treatment with an alkyl halide, yielding an internal alkyne. For example, conversion of 1-hexyne into its anion, followed by reaction with 1-bromobutane, yields 5-decyne.

$$\underset{\text{1-Hexyne}}{CH_3CH_2CH_2CH_2C{\equiv}CH} \xrightarrow[\text{2. } CH_3CH_2CH_2CH_2Br]{\text{1. } NaNH_2,\ NH_3} \underset{\text{5-Decyne (76\%)}}{CH_3CH_2CH_2CH_2C{\equiv}CCH_2CH_2CH_2CH_3}$$

Because of its generality, acetylide alkylation is a good method for preparing substituted alkynes from simpler precursors. A terminal alkyne can be prepared by alkylation of acetylene itself, and an internal alkyne can be prepared by further alkylation of a terminal alkyne.

$$\underset{\text{Acetylene}}{H{-}C{\equiv}CH} \xrightarrow{NaNH_2} [H{-}C{\equiv}C{:}^-\,Na^+] \xrightarrow{RCH_2Br} \underset{\text{A terminal alkyne}}{H{-}C{\equiv}C{-}CH_2Br}$$

$$\underset{\text{A terminal alkyne}}{R{-}C{\equiv}C{-}H} \xrightarrow{NaNH_2} [R{-}C{\equiv}C{:}^-\,Na^+] \xrightarrow{R'CH_2Br} \underset{\text{An internal alkyne}}{R{-}C{\equiv}C{-}CH_2R'}$$

The alkylation reaction is limited to the use of primary alkyl bromides and alkyl iodides because acetylide ions are sufficiently strong bases to cause elimination instead of substitution when they react with secondary and tertiary alkyl halides. For example, reaction of bromocyclohexane with propyne anion yields the elimination product cyclohexene rather than the substitution product 1-propynylcyclohexane.

$$CH_3CH_2C{\equiv}\ddot{C}^- + CH_3CH_2\overset{+}{CH_2}{-}\overset{-}{Br} \longrightarrow CH_3CH_2C{\equiv}CCH_2CH_2CH_3 + Br^-$$

$$\underset{\text{1-pentyne}}{CH_3CH_2CH_2C{\equiv}CH} \xrightarrow[\text{2. } CH_3CH_2CH_2CH_2CH_2Cl]{\text{1. } NaNH_2} \underset{\text{4-decyne}}{CH_3CH_2CH_2C{\equiv}CCH_2CH_2CH_2CH_2CH_3}$$

Distinction Between Terminal And Non-Terminal Alkynes :

All terminal alkynes contain acidic H-atom. Hence such H-atom may be replaced by metals while non-terminal alkynes do not have acidic H-atoms hence they will not react with metals

(a) Tollen's Reagent

Ammoniacal Silver nitrate solution is called Tollen's reagent.

Alkynes containing acidic H-atoms will react with Ammoniacal silver nitrate solution

$$HC \quad CH \quad 2[Ag(NH_3)_2] \quad OH \qquad \underset{\text{White ppt (Silver acetylide)}}{GC \quad CAg}$$

$$R{-}C \quad CH \quad [Ag(NH_3)_2] \quad OH^{\ominus} \qquad \underset{\text{White ppt}}{R{-}C \quad CAg}$$

(b) Ammoniacal Cuprous chloride

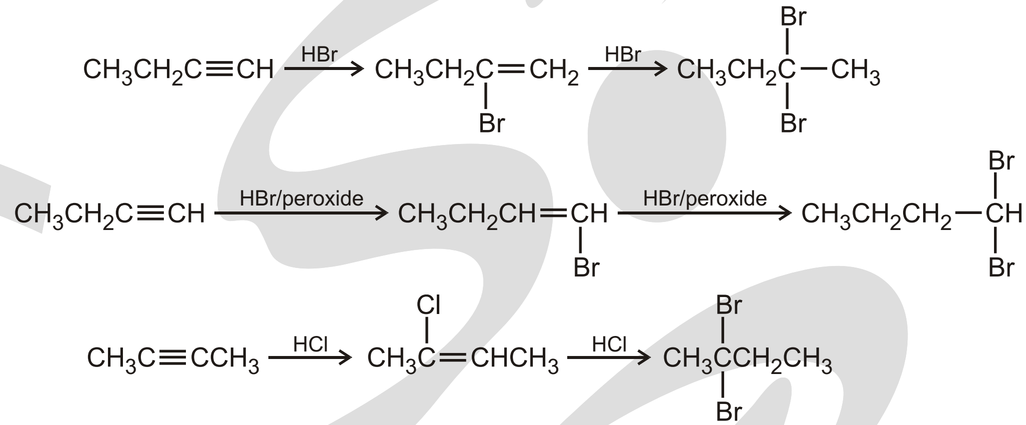
$$CuCl \quad NH_4OH \qquad NH_4Cl \quad CuOH$$

$$CuOH \quad NH_4OH \qquad (Cu(NH_3)_2)OH \quad 2H_2O$$

HC CH 2[Cu(NH$_3$)$_2$]OH CuC CCu (Cuprous acetylide) Red ppt.

RC CH [CH(NH$_3$)$_2$OH R — C C — Cu

2. Reaction with halogen acid : Halogen acid (H-X) adds on unsymmetric alkyne to give product according to Markovnikov's rule where as in the presence of peroxide, anti Markovnikov's rule is followed :

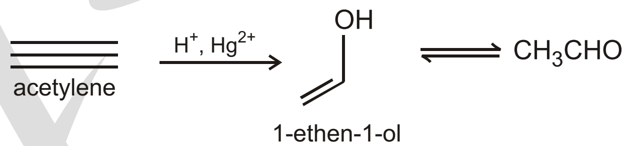


3. Acid catalyzed hydration :

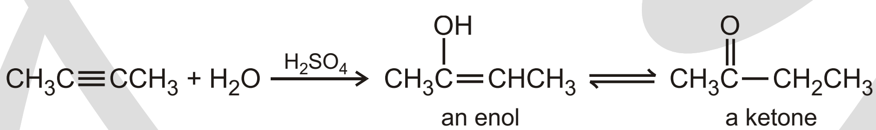
Alkyne hydration : Kucherov-reaction :

Alkyne readily combine with water in the presence of acid (usually sulfuric acid) and mercury (II) salts (usually the sulfate is used) to form carbonyl compounds, in a process known as Kucherov's reaction. In the case of acetylene (ethyne) the product is acetaldehyde (ethanal), while other alkynes form ketones.

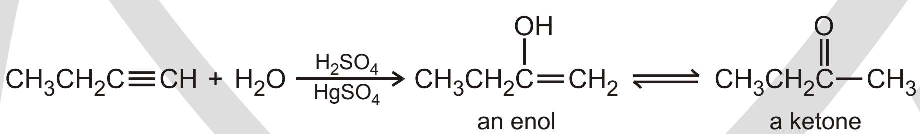
Ex. : (a)



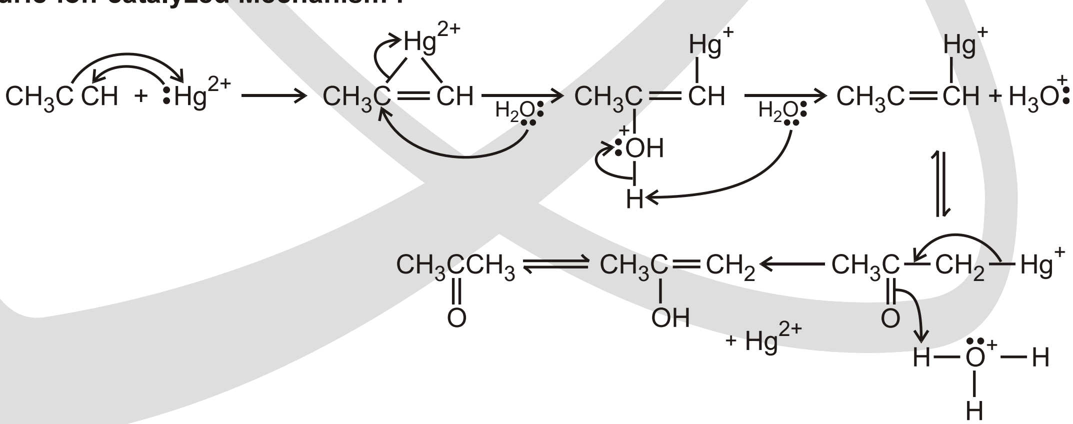
(b)



(c)



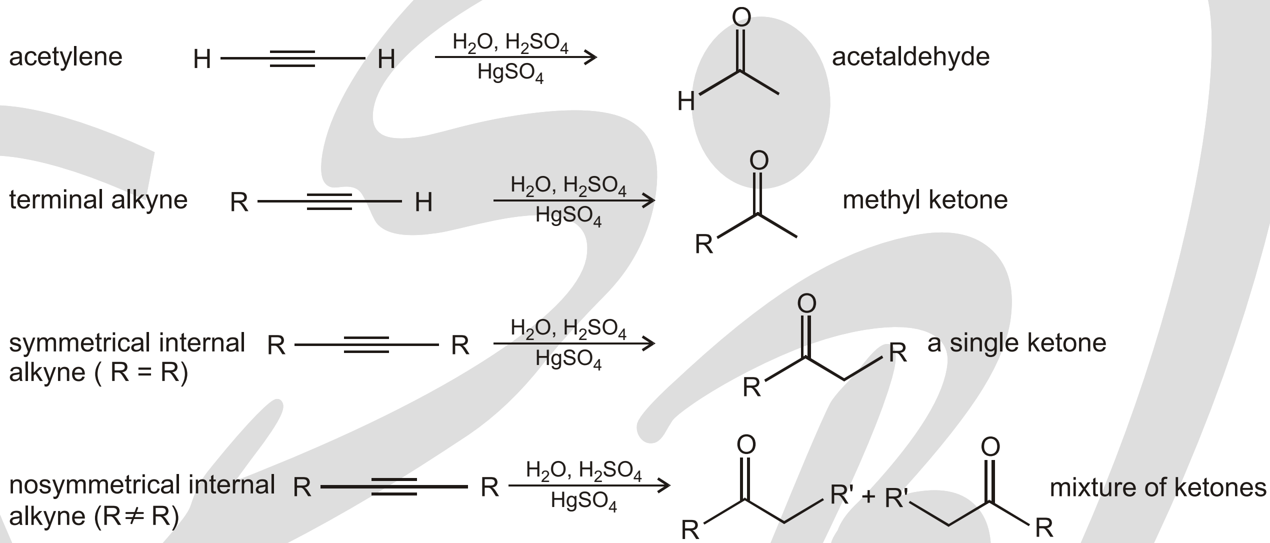
Mercuric-ion-catalyzed Mechanism :



The intermediate product is an enol (an alkene with a hydroxyl group attached to a doubly bonded carbon), which then transforms to its keto tautomer through keto-enol tautomerization.

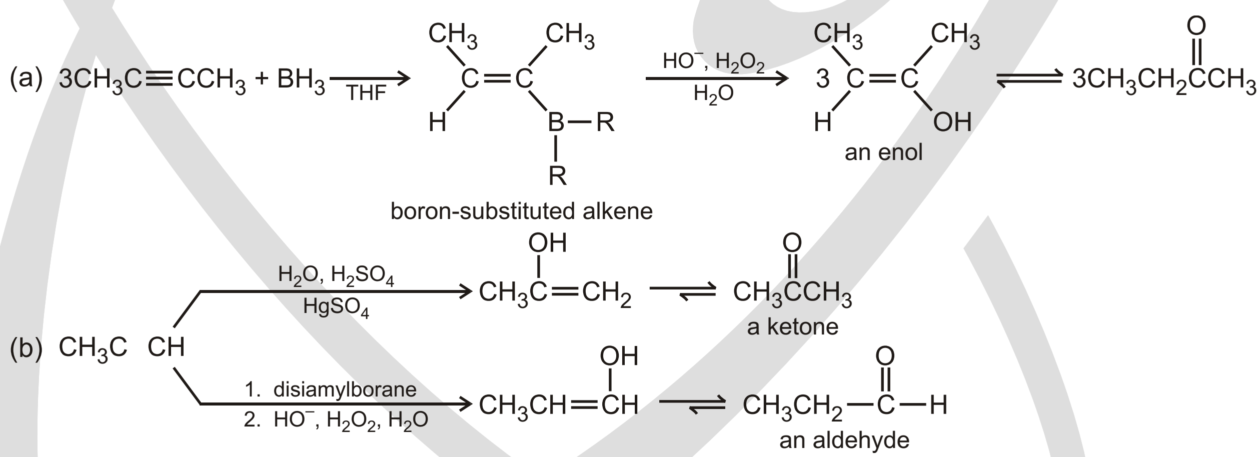
Alkene hydration only requires acid catalysis and proceeds through the most stable carbocation available Terminal alkynes selectively produce a methyl ketone. But unsymmetrical internal alkynes can produce two different vinyl cations, whose stabilities may be similar if the alkyne's substituents are different.

Different results occur for this reaction depending on the substitution pattern of the alkyne.

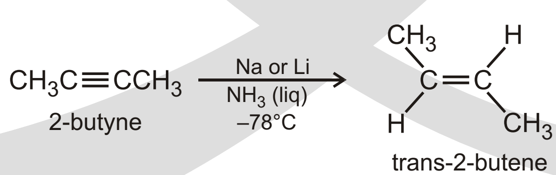


4. **Hydroboronation oxidation of alkyne :**

It is a two-step organic reaction that converts an alkene into a neutral alcohol by the net addition of water across the double bond. The hydrogen and hydroxyl group are added in a syn addition leading to cis stereochemistry. It is an antiMarkovnikov reaction, with the hydroxyl group attaching to the less-substituted carbon.



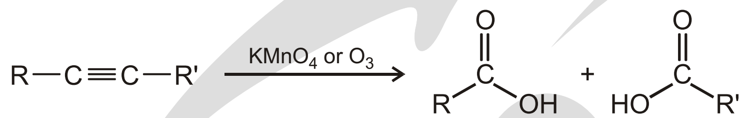
5. **Birch Reduction :**



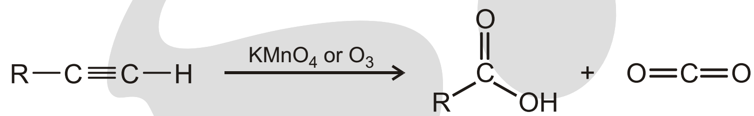
6. **Oxidative Cleavage of Alkynes :**

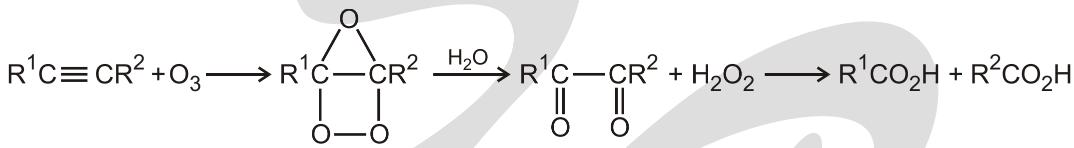
Alkynes, like alkenes, can be cleaved by reaction with powerful oxidizing agents such as ozone or $KMnO_4$, although the reaction is of little value and we mention it only for completeness. A triple bond is generally less reactive than a double bond, and yields of cleavage products are sometimes low. The products obtained from cleavage of an internal alkyne are carboxylic acids; from a terminal alkyne, CO_2 is formed as one product.

An internal alkyne



A terminal alkyne



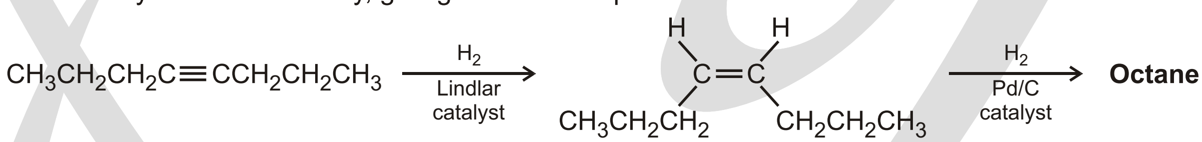


Acetylene is exceptional in that it gives glyoxal as well as formic acid

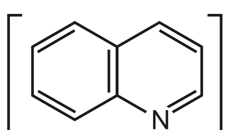
$$CH\equiv CH \xrightarrow[\text{(ii) } H_2O]{\text{(i) } O_3} OCHCHO + HCO_2H$$

7. Reduction :

Complete reduction to the alkane occurs when palladium on carbon (Pd/C) is used as catalyst, but hydrogenation can be stopped at the alkene stage if the less active Lindlar catalyst is used. The Lindlar catalyst is a finely divided palladium metal that has been precipitated onto a calcium carbonate support and then deactivated by treatment with lead acetate and quinoline, an aromatic amine. The hydrogenation occurs with syn stereochemistry, giving a cis alkene product.

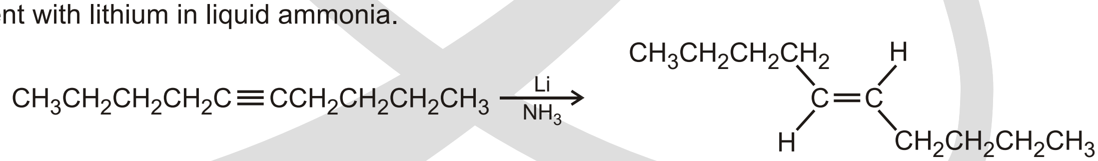


cis-4-Octene



Quinoline

An alternative method for the conversion of an alkyne to an alkene uses sodium or lithium metal as the reducing agent in liquid ammonia as solvent. This method is complementary to the Lindlar reduction because it produces trans rather than cis alkenes. For example, 5-decyne gives trans-5-decene on treatment with lithium in liquid ammonia.



5-Decyne trans-5-Decene (78%)

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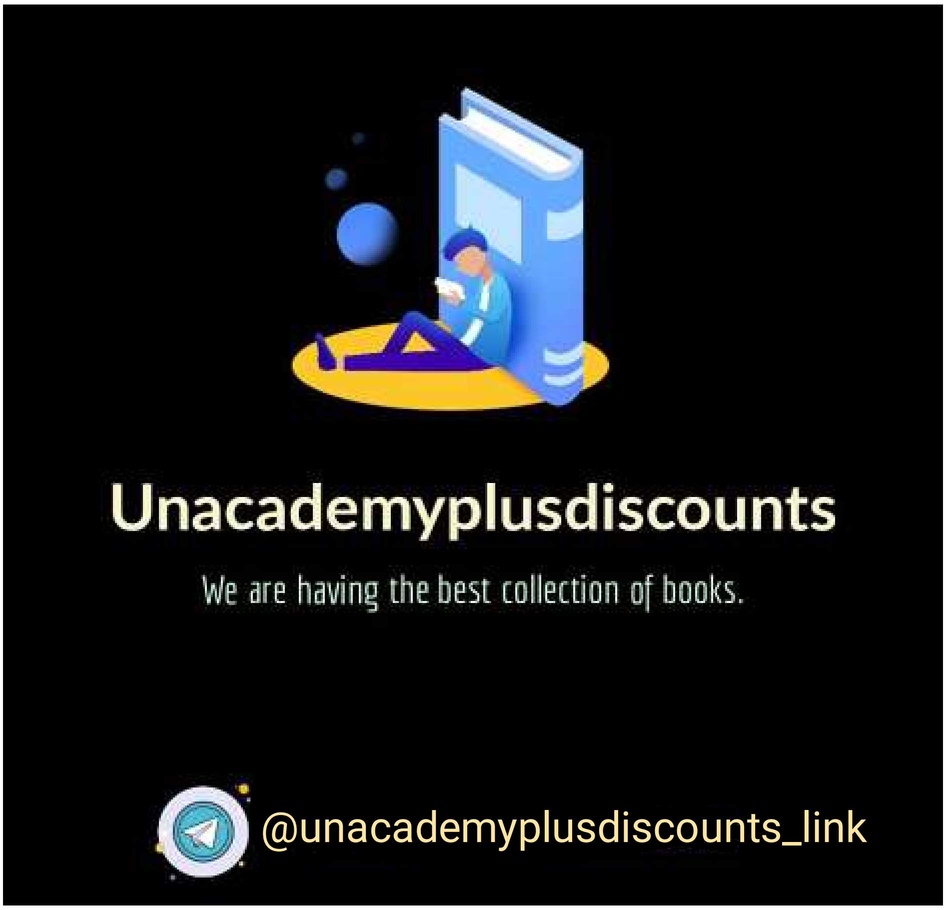
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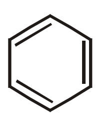
CHAPTER

31

Benzene

(A) INTRODUCTION

Aromatic Hydrocarbon

Benzene C_6H_6 = 

The compound with pleasant smell are called aromatic compounds.

These compounds contain Benzene having ring of six carbon atoms, later on it was found that many compounds we known resemble benzene in their structure but did not have any pleasant smell. So now the term aromatic is not applied to compounds with pleasant smell but instead any compound which is benzene or which resembles benzene in their chemical behaviour are called aromatic compounds.

(B) PREPERATION OF BENZENE

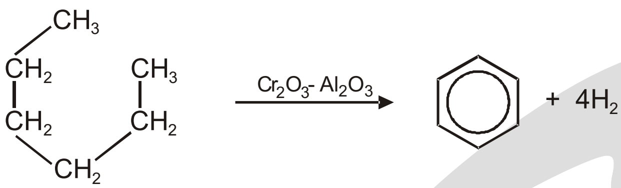
Preparation

1. **Catalytic reforming :**

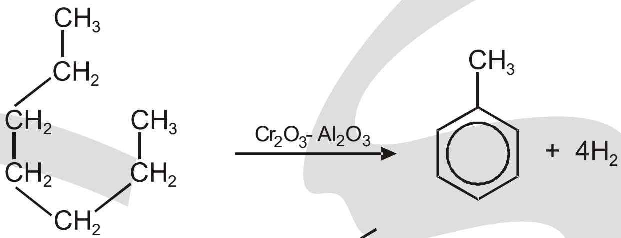
   What is reforming ?

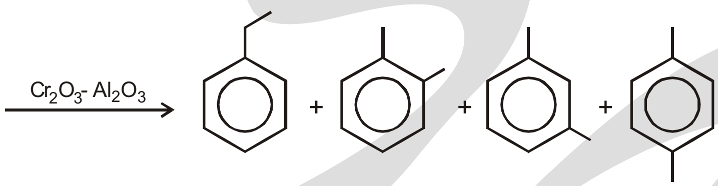
   Reforming takes straight chain hydrocarbons in the C_6 to C_8 range from the gasoline or naphtha fractions and rearranges them into compounds containing benzene rings. Hydrogen is produced as a by-product of the reactions.

   For example, hexane, C_6H_{14}, loses hydrogen and turns into benzene. As long as you draw the hexane bent into a circle, it is easy to see what is happening.

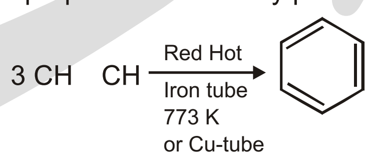
(a) 

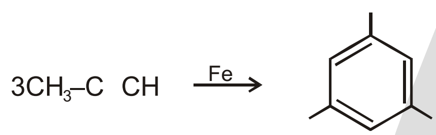
Similarly, methylbenzene (toluene) is made from heptane :

(b) 

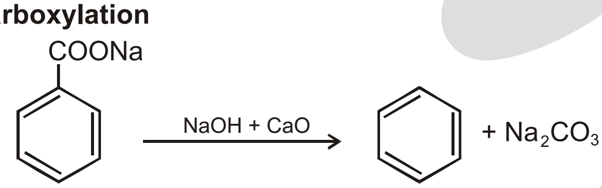
(c) Octane 

2. From Acetylene : Berthalot prepared benzene by passing acetylene through red hot tube.

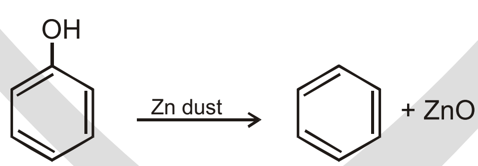




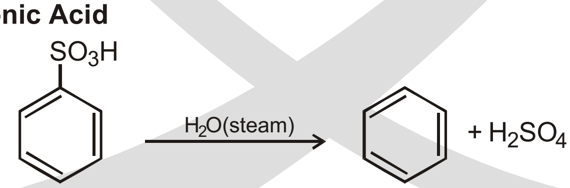
3. From Sodalime Decarboxylation



4. From Phenol



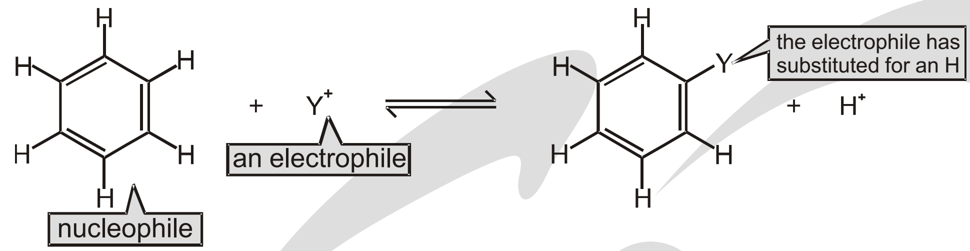
5. From Benzene Sulfonic Acid



(C) REACTION OF BENZENE

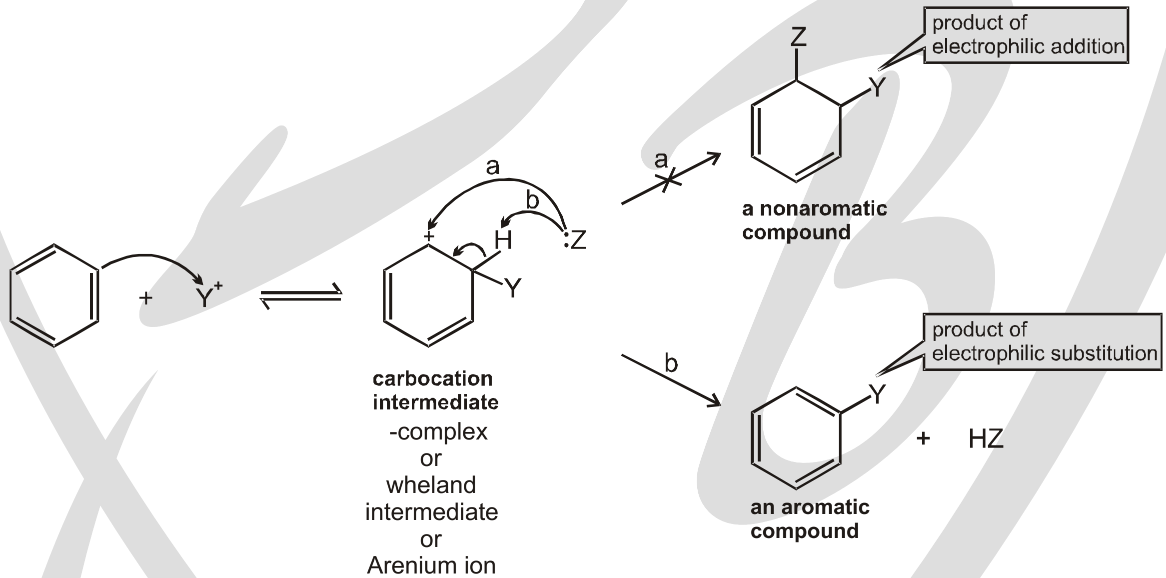
HOW BENZENE REACTS

Aromatic compounds such as benzene undergo **electrophilic aromatic substitution reactions :** an electrophile substitutes for one of the hydrogens attached to the benzene ring.

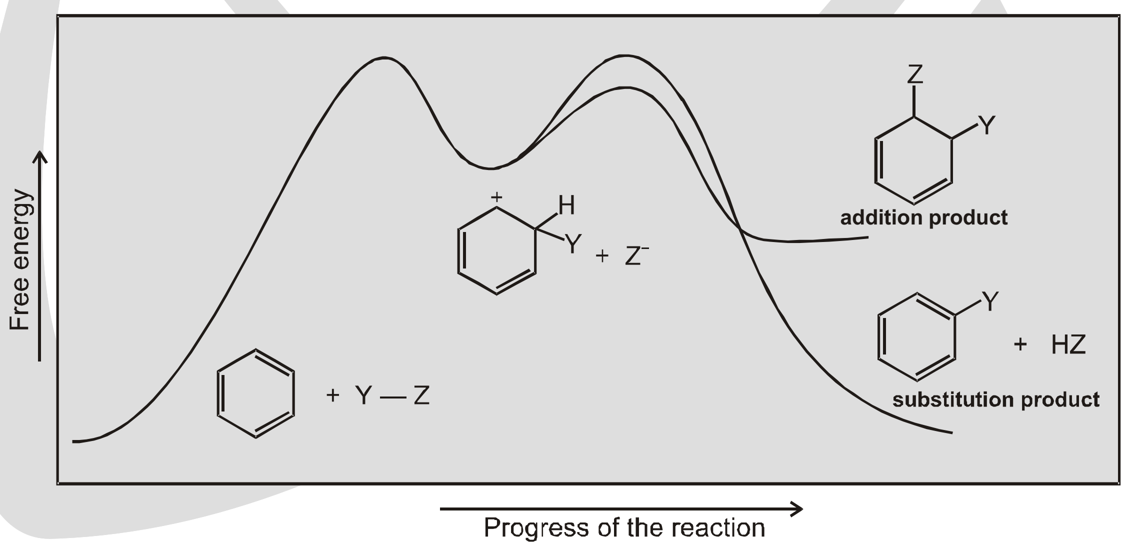


Now let's look at why this substitution reaction occurs. The cloud of p electrons above and below the plane of its ring makes benzene a nucleophile, so it reacts with an electrophile (Y^+). When an electrophile attaches itself to a benzene ring, a carbocation intermediate is formed.

If the carbocation intermediate that is formed from the reaction of benzene with an electrophile were to react similarly with a nucleophile (depicted as path a in Figure 1), then the addition product would not be aromatic. But, if the carbocation instead were to lose a proton from the site of electrophilic addition and form a substitution product (depicted as path b in Figure 1), then the aromaticity of the benzene ring would be restored.



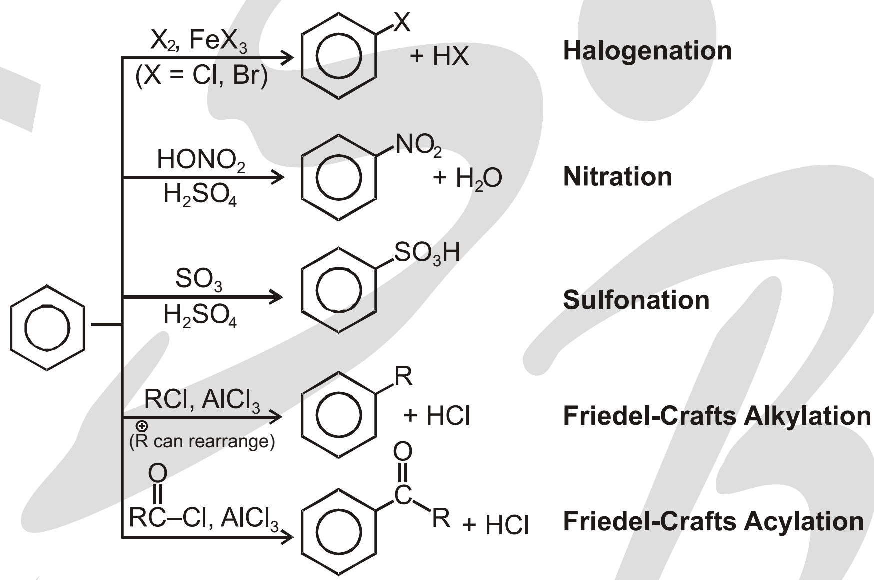
characteristic of alkenes—that would destroy aromaticity. The substitution reaction is more accurately called an **electrophilic aromatic substitution reaction,** since the electrophile substitutes for a hydrogen of an aromatic compound.



(A) The five most common electrophilic aromatic substitution reactions are the following :

1. **Halogenation :** a bromine, chlorine, or iodine (Br, Cl, I) substitutes for a hydrogen.
2. **Nitration :** a nitro (NO_2) group substitutes for a hydrogen.
3. **Sulfonation :** a sulfonic acid (SO_3H) group substitutes for a hydrogen.
4. **Friedel-Crafts acylation :** an acyl (RC = O) group substitutes for a hydrogen.
5. **Friedel-Crafts alkylation :** a alkyl (R) group substitutes for a hydrogen.

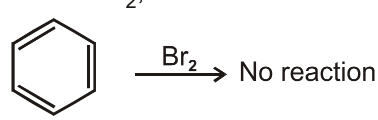
(B)



HALOGENATION AND THE ROLE OF LEWIS ACIDS

GENERAL MECHANISM FOR ELECTROPHILIC AROMATIC SUBSTITUTION

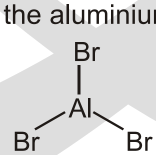
When benzene is heated in the presence of Br_2, no reaction is observed.



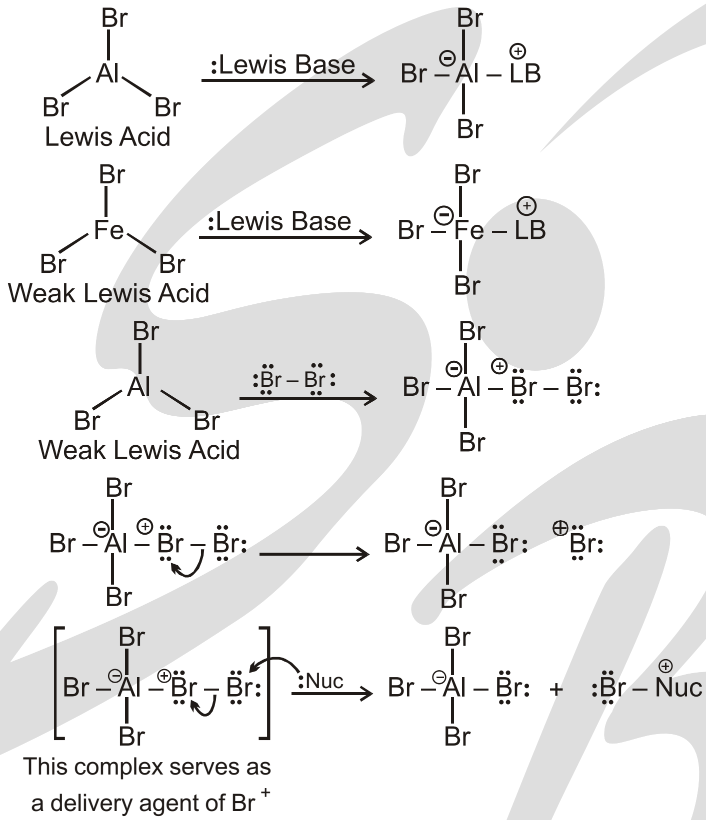
We can understand this because benzene is an aromatic compound. It has a special stabiliy due to its aromaticity. If we add Br_2 across bezene, aromaticity will be lost. And that is why the reaction does not take place it would be going "uphill" in energy. But is it possible to try to "force" the reaction to happen ?

Halogenation and the role of lewis acids :

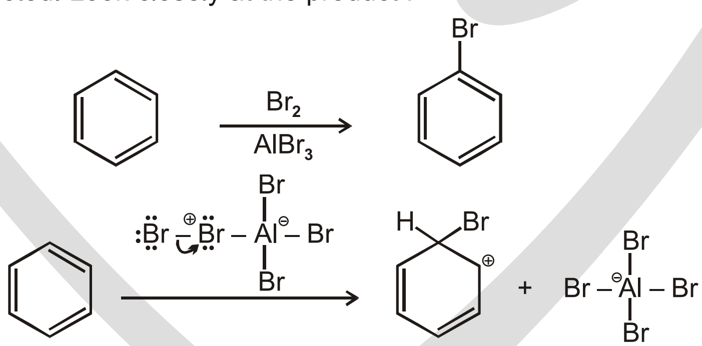
Consider the compound $AlBr_3$. The central atom in this structure is aluminium. Aluminium is in column 3A of the periodic table, and therefore, it has three valence aluminium electrons. It uses each of these electrons to form a bond, which is why we see three bonds to the aluminium atom in $AlBr_3$:



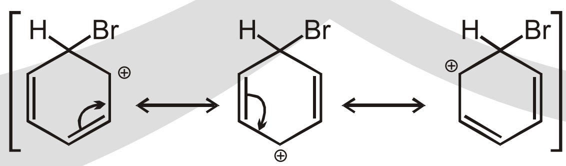
But you should notice that the aluminium atoms does not have an octet. If you count the electrons around the aluminium atom, there are only six electrons. That means that aluminum has one empty orbital. That empty orbitals ale to accept electrons. In fact, it will exhibit a tendency accept electrons because that would give aluminium an octet of electrons :



The important point is that this complex can function as a delivery agent of Br^+, and that is what we needed in order to force a reaction between benzene and bromine. So, now let's try our reaction again. When we treat benzene with bromine in the presence of a Lewis acid, such as $AlBr_3$, a reaction is indeed observed. BUT it is not the reaction that we expected. Look closely at the product :

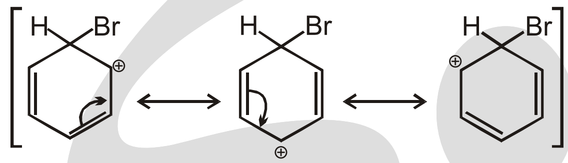


This step generates an intermediate that is not aromatic. It is true that aromaticity has been temporarily destroyed, but it will soon be reestablished in the second step (final step) of the mechanism. In this first step of the mechanism (shown above), the ring attacks Br^+ to form an intermediate that has three important resonance structures:



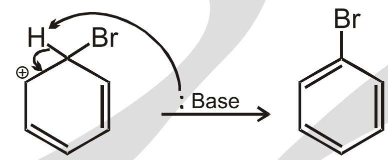
It is important to remember what resonance structures represent. Recall from the first semester that resonance is NOT a molecule flipping back and forth between different states. Rather, resonance is the way we deal with the inadequacy of our drawings. There is no one single drawing that adequately captures the essence of the intermediate, so we draw three drawings, and we meld them all together in our minds to gain a better understanding of the intermediate.

This intermediate has some special names. We often call it a sigma complex, or sometimes, we call it an Arenium ion. These are just two different names for the same intermediate. From now on, in this book, we will call it a **-complex :**

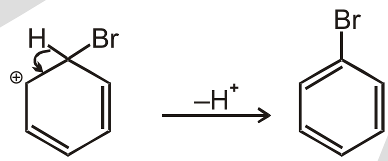


SIGMA COMPLEX

The second step (last step) of the mechanism involves transfer of a proton to re-form aromaticity:

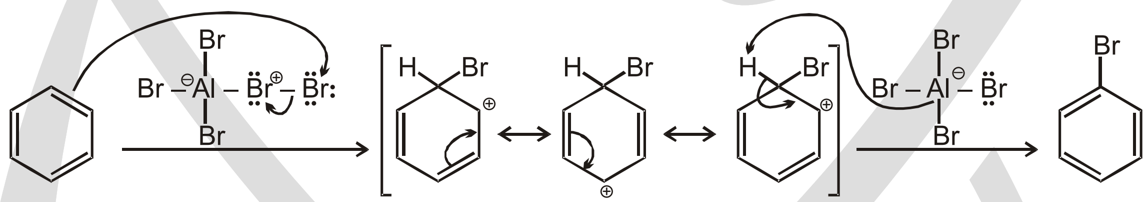


Notice that we are using a base to remove the proton. Technically, it is not correct to just let a proton fall off into space by itself, like this:



Notice that, in the end, the Lewis acid ($AlBr_3$) is regenerated. So the Lewis acid is actually not being consumed by the reaction. It is only there to help the reaction along, which is why we call it a catalyst in this case. That is why the presence of even a small quantity of the Lewis acid will suffice.

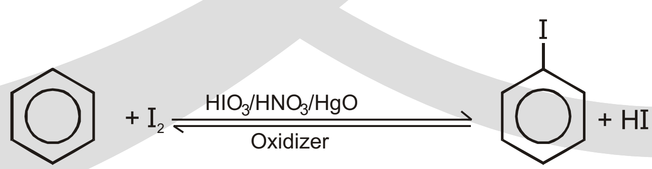
Now that we have seen both steps of the mechanism, let's take a close look at the mechanism all at once:



SIGMA COMPLEX

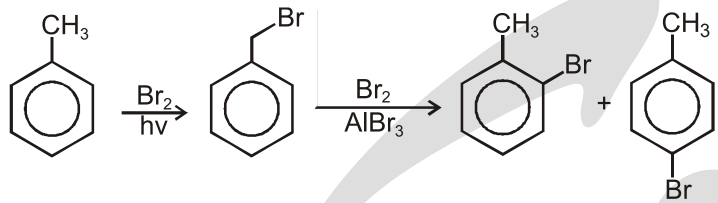
☑ SPECIAL TOPIC

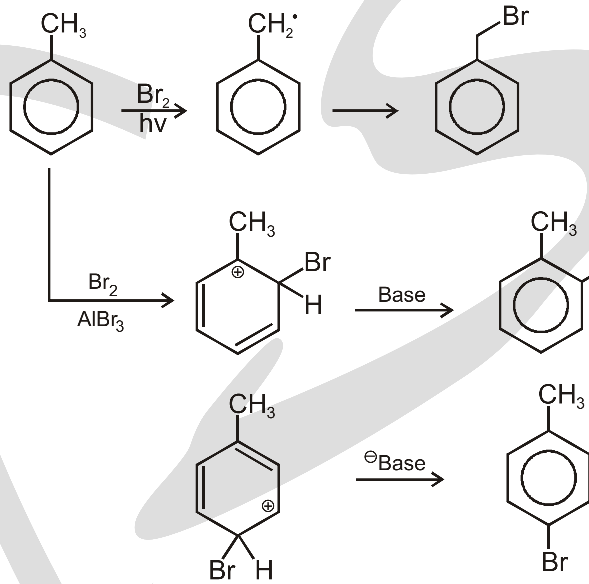
Iodination is a reversible reaction which can be forward in presence of oxidiser which consume side product HI.



Ex. $5HI + HIO_3 \quad 3I_2 + 3H_2O$

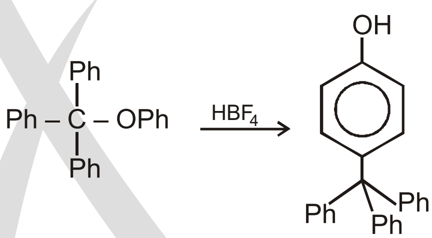
Solved Example

(a) 

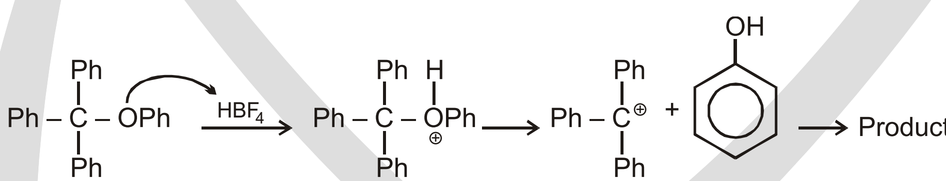
Sol. 

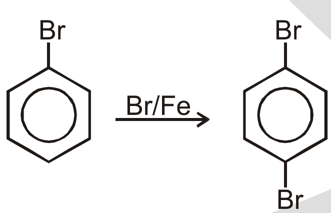
(Free radical substitution reaction)

(In proxence of lewis acid electrophilic aromatic substitution)

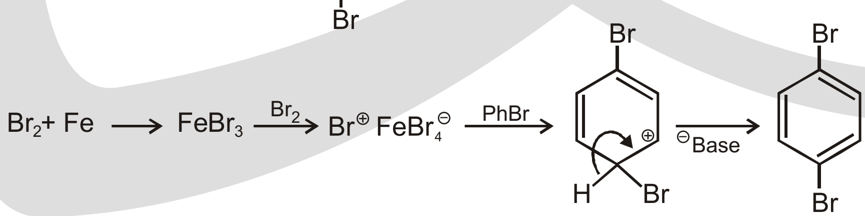
(b) 

(rearrangement reaction)

Sol. 

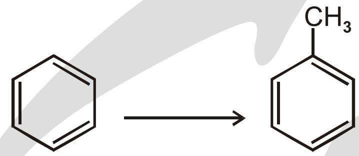
(c) 

major product but rate is less than benzene. Explain ?

Sol. 

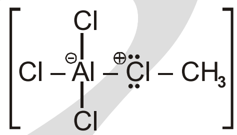
FRIEDEL-CRAFTS ALKYLATION AND ACYLATION

Let's start with the simplest of all alkyl groups: a methyl group. So, the question is: what reagents would we need to achieve the following transformation:



If we take methyl chloride and we mix it with a pinch of $AlCl_3$, we will have a source of CH_3 :

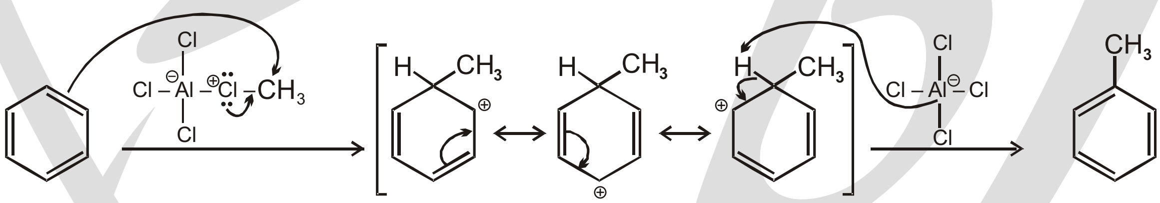
The truth is that we are NOT really forming a free methyl carbocation that can float off into solution. A free CH_3 would be too unstable to form. So, instead, we must view this as a complex that can serve as a "source" of CH_3



Source of CH_3^+

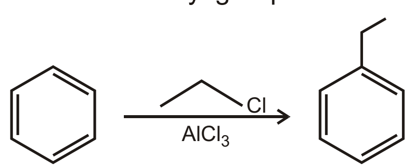
This provides us with a method for methylating a benzene ring :

And the mechanism is, once again, the same mechanism that we have seen over and over again. It is an electrophilic aromatic substitution : CH_3 on the ring and then H+ off :

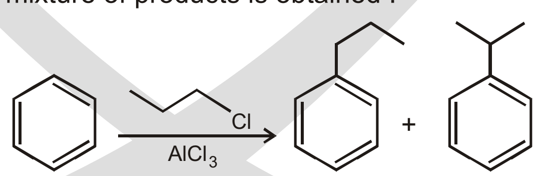


SIGMA COMPLEX

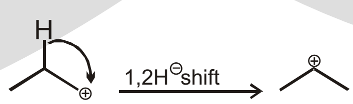
We can use the exact same process to install an ethyl group on an aromatic ring :



This process (installing an alkyl group onto an aromatic ring) is called a Friedel-Crafts alkylation. It works very well for installing a methyl group or an ethyl group on the ring. BUT we run into problems when we try to install a propyl group on the ring, because a mixture of products is obtained :



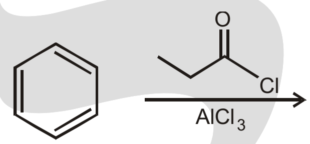
Similarly, an ethyl carbocation cannot rearrange to become any more stable. But a propyl carbocation CAN rearrange (via a hydride shift) :



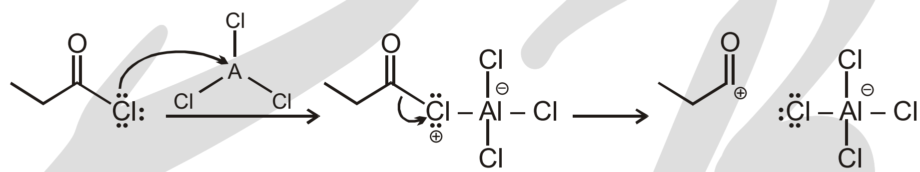
Clearly, we need a trick. And there is a trick. To see how it works, we need to take a close look at a similar reaction that also bears the name Freidel-Crafts. But this reaction is not an alkylation.

Rather, it is called an acylation. To see the difference, let's quickly compare an alkyl group with an acyl group.

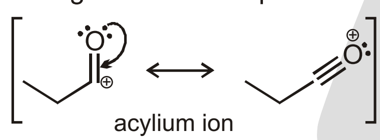
We can install an acyl group on a benzene ring in exactly the same way that we installed an alkyl group on the ring. We simply use the following reagents:



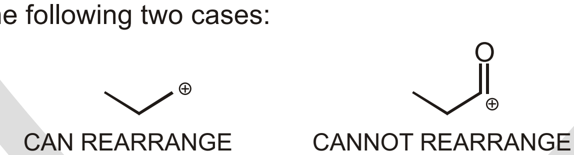
The first reagent is called an acyl chloride (or acid chloride), and we are already familiar with the role of $AlCl_3$ (the Lewis acid). The Lewis acid interacts with the Cl atom of the acyl chloride, generating a resonance-stabilized acylium ion:



The term **acylium** ion should make sense—"acyl" because we can see that this electrophile has an acyl group; and "ium" because there is a positive charge. This electrophile actually has an important resonance structure :



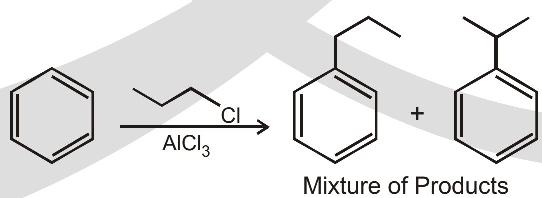
These resonance structures are important because they indicate that an acylium ion is stabilized by resonance, and therefore, it will NOT undergo a carbocation rearrangement. (If it did, it would lose this resonance stabilization.) Compare the following two cases:



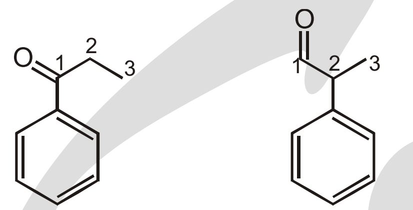
Using a Friedel-Crafts acylation, we can cleanly install an acyl group on a benzene ring (without any rearrangements) :

We don't observe any side products that would result from a carbocation rearrangement. Once again, compare a Friedel-Crafts alkylation with a Friedel-Crafts acylation :

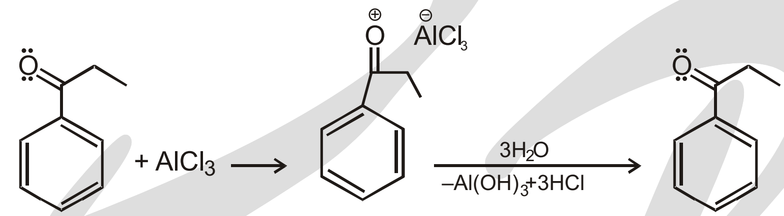
ALKYLATION



Now take a close look at the acylation above, and let's point out a very important feature. Notice that we have installed a three-carbon chain on the ring, with the chain being attached by the first carbon of the chain, as opposed to the middle carbon of the chain :

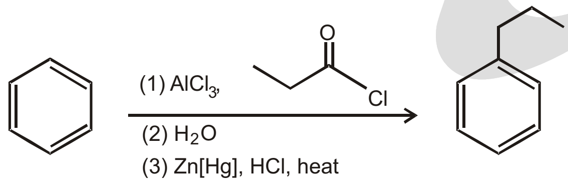


Before we move on, there is one subtle point to mention about a Friedel-Crafts acylation (step 1 of the synthesis above). Remember that this reaction takes place in the presence of a Lewis acid ($AlCl_3$), which is constantly on the lookout for electrons with which it can interact. Well, the product of the acylation step is a ketone. And a ketone has electrons that the Lewis acid is seeking:

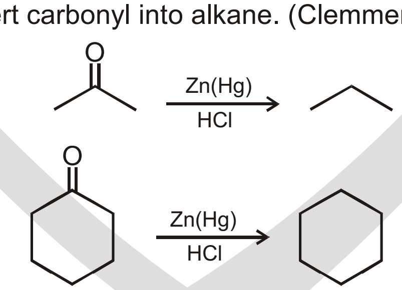


Solved example

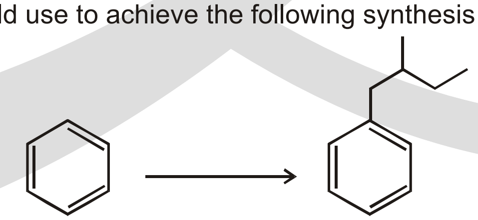
- Therefore, whenever we perform an acylation reaction, we need to pull the Lewis acid off our product in the end. And there is a simple way to do this. We just give the Lewis acid some other source of electrons to interact with instead. The easiest thing to use is water (H_2O). So, whenever you use a Friedel-Crafts acylation in a synthesis problem, it is appropriate to include water in your list of reagents (immediately after the acylation) and more than 1 mole of lewis acid is required.



Zn(Hg), HCl is used to convert carbonyl into alkane. (Clemmension reduction)

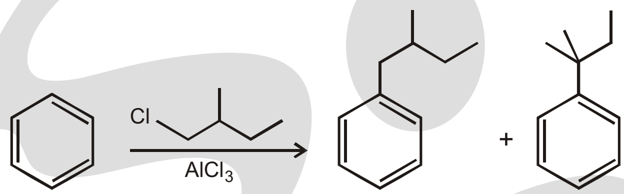


- Show the reagents you would use to achieve the following synthesis :

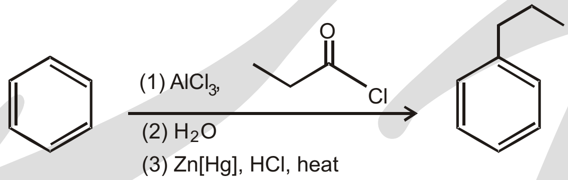


Ans. In this problem, we need to install an alkyl group on a benzene ring. So, we first look to see if we could do this one step, using a Friedel-Crafts alkylation. In this case, we cannot do it in one step because we have to worry about a carbocation rearrangement. If we think about the electrophile that we would need to make, we will see that it could rearrange:

And this would give us a mixture of products:

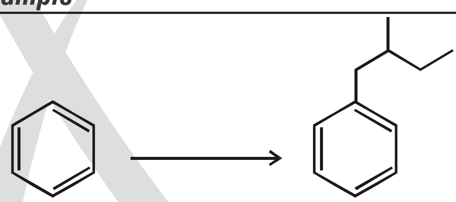


So, instead we will have to use a Friedel-Crafts acylation followed by a Clemmensen reduction:

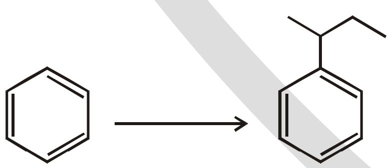


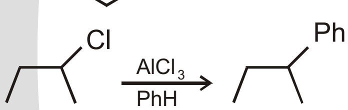
For each of the following problems, show what reagents you would use to accomplish the transformation. In some situations, you will want to use a Friedel-Crafts alkylation, while in other situations, you will want to use a Friedel-Crafts acylation.

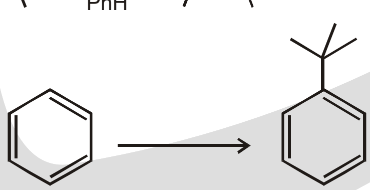
Solved Example

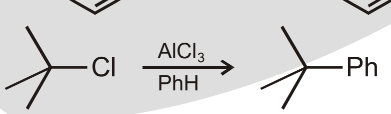
(a) 

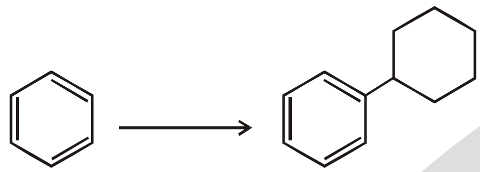
Sol. $CH_3CH_2CH_2COCl \xrightarrow[PhH]{AlCl_3} CH_3CH_2CH_2COPh \xrightarrow{ZnHg\ \ HCl} CH_3CH_2CH_2CH_2Ph$

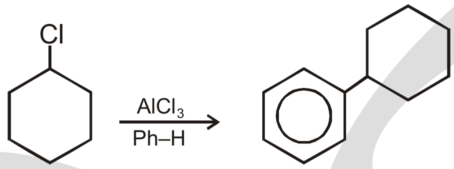
(b) 

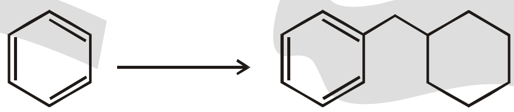
Sol. 

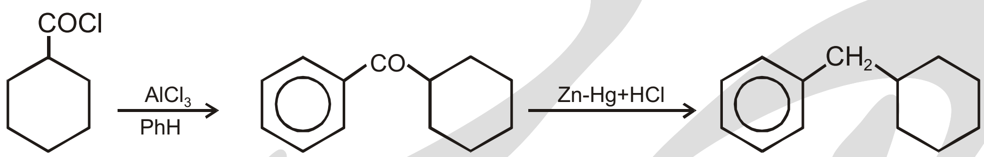
(c) 

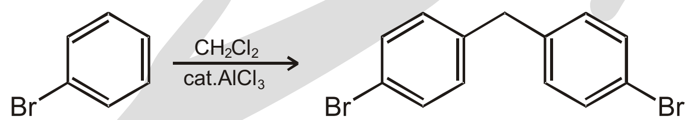
Sol. 

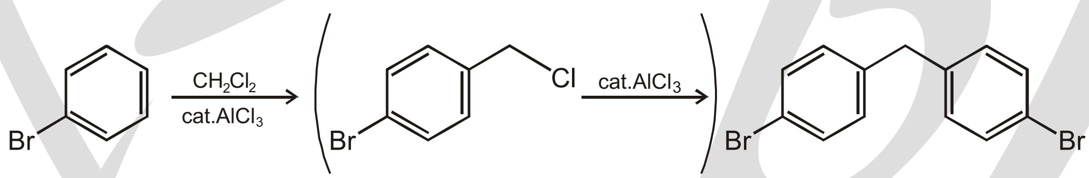
▸ (d) 

Sol. 

▸ (e) 

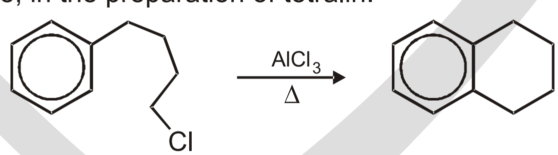
Sol. 

▸ (f) 

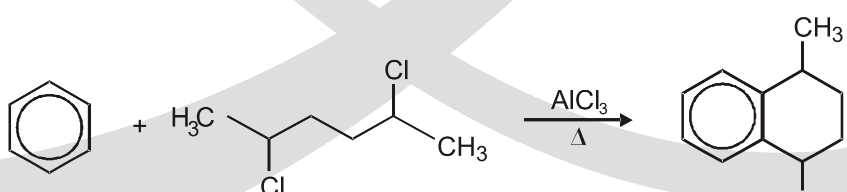
Sol. 

Intramolecular Friedel–Crafts alkylation :

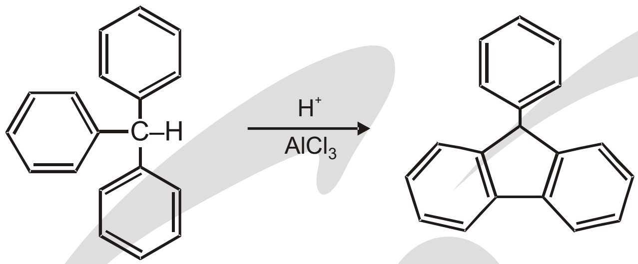
(a) An important use of the Friedel–Crafts alkylation reaction is to effect ring closure. The most common method is to heat with aluminium chloride an aromatic compound having a halogen, hydroxy, or alkene group in the proper position, as, for example, in the preparation of tetralin.



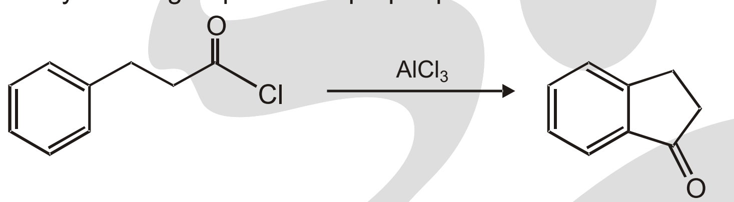
(b) Another way of effecting ring closure through Friedel–Crafts alkylation is to use a reagent containing two groups.



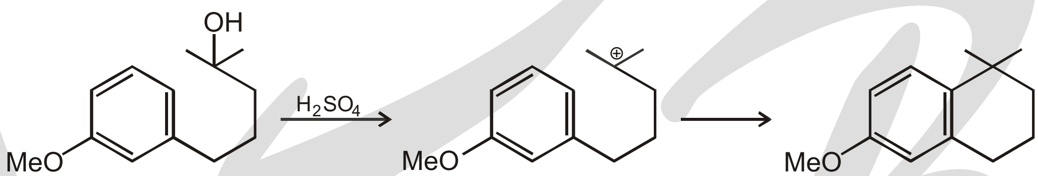
(c) Intramolecular Scholl reactions, such as formation of product from triphenylmethane, are much more successful than the intermolecular reaction.



(d) An important use of the Friedel–Crafts acylation is to effect ring closure. This can be done if an acyl halide, anhydride, or carboxylic acid group is in the proper position.

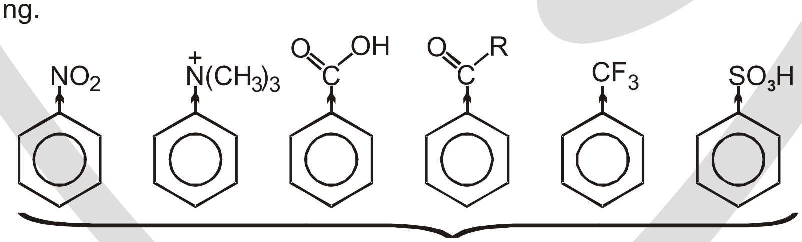


(e)



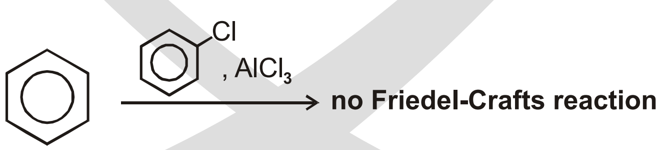
LIMITATIONS OF FRIEDEL CRAFTS Alkylation

1. When the carbocation formed from an alkyl halide, alkene, or alcohol can rearrange to a more stable carbocation, it usually does so, and the major product obtained from the reaction is usually the one from the more stable carbocation.
2. Friedel-Crafts reactions usually give poor yields when powerful electron-withdrawing groups are present on the aromatic ring.



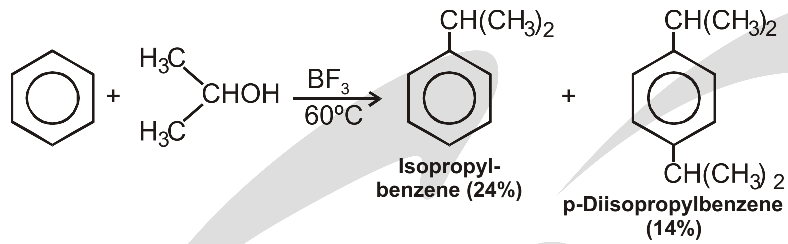
Aromatic ring is too electron deficient that Friedel craft is not possible

3. Aryl and vinylic halides cannot be used as the halide component because they do not form carbocations readily.

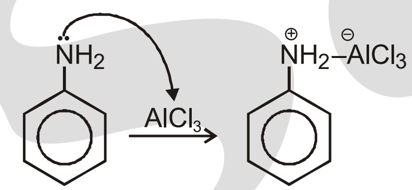


;

4. Polyalkylations often occur. Alkylgroups are electron-releasing groups, and once one is introduced into the benzene ring, it activates the ring toward further substitution.



5. Aniline will not undergo Friedel-Crafts reaction because of lewis acid base reaction interconvert into deactivating group.



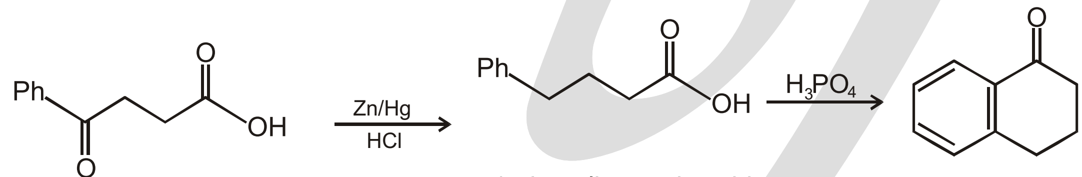
The advantages of acylation over alkylation :

Two problems in Friedel - Crafts alkylation do not arise with acylation.

1. The acyl group in the product withdraws electrons from the p system making multiple substitutions harder. Indeed, the ring is too deactivated to start off with, Friedel-Crafts acylation may not be possible at all-Nitrobenzene is inert to Friedel-Crafts acylation and is often used as a solvent for these reactions.

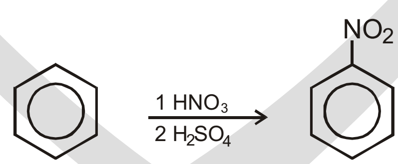
2. Rearrangements are also no longer a problem because the electrophile, the acylium cation, is already relatively stable.

Because the acylation reaction is so much more reliable than Friedel-Crafts alkylation, a common method to alkylate is actually to acylate first and then reduce the carbonyl to a methylene group ($–CH_2–$).



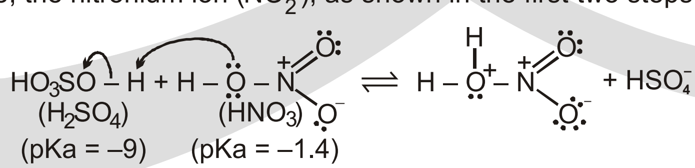
NITRATION OF BENZENE

Benzene reacts slowly with hot concentrated nitric acid to yield nitrobenzene. The reaction is much faster if it is carried out by heating benzene with a mixture of concentrated nitric acid and concentrated sulfuric acid. In this reaction HNO_3 will act as a base.



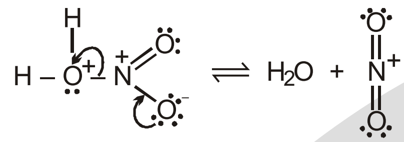
Concentrated sulfuric acid increases the rate of the reaction by increasing the concentration of the electrophile, the nitronium ion (NO_2), as shown in the first two steps of the following mechanism.

Step-I



In this step nitric acid accepts a proton from the stronger acid, sulfuric acid.

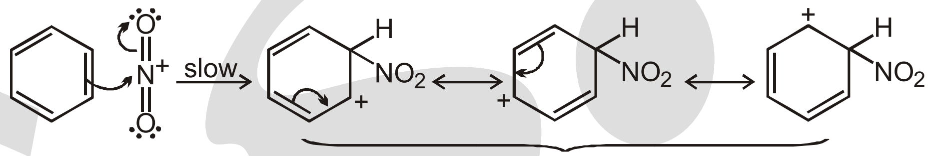
Step-II



Nitronium ion

Now that it is protonated, nitric acid can dissociate to from a nitronium ion

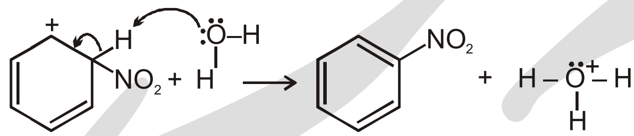
Step-III



Arenium ion

The nitronium ion is the actual electrophile in nitration; it reacts with benzene to form a resonance-stabilized arenium ion.

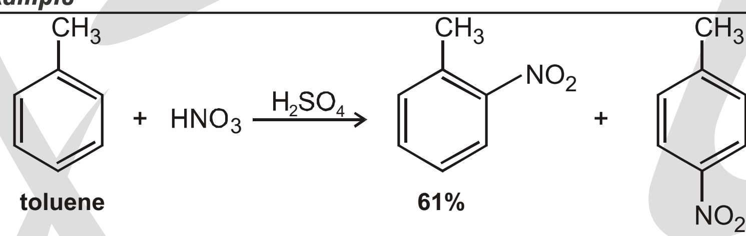
Step-IV



The arenium ion then loses a proton to a Lewis base and becomes nitrobenzene.

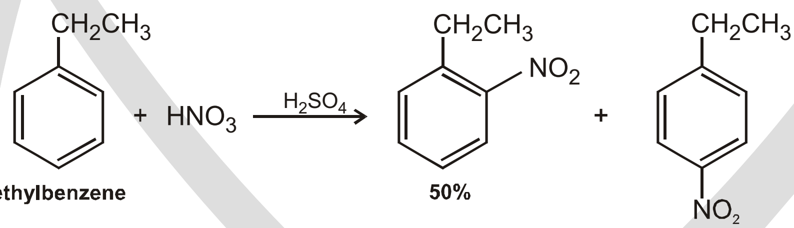
Solved Example

- **(a)**



toluene **61%** **39%**

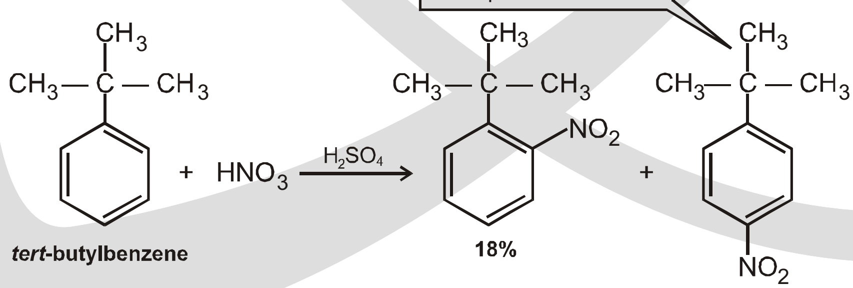
- **(b)**



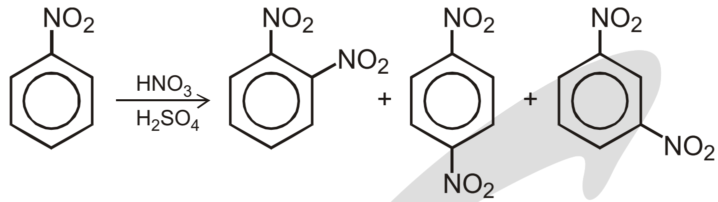
ethylbenzene **50%** **50%**

a bulky substituent increases the percentage of the para isomer

- **(c)**



***tert*-butylbenzene** **18%** **82%**

(d) 

(6%) (1%) (93%)

In (d) the nitro group is a very strong deactivating group. Nitrobenzene undergoes nitration at a rate only 10–4 times that of benzene. The nitro group is a meta director. When nitrobenzene is nitrated with nitric and sulfuric acids, 93% of the substitution occurs at the meta position :

SULFONATION

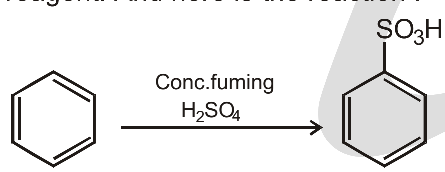
When we do this analysis for each of the three double bonds in SO_3, we begin to see that the sulfur atom is VERY electron-poor:

In fact, the sulfur atom is so electron-poor that it is an excellent electrophile, even though the compound is overall neutral (no net charge). Now we are going to see a reaction that uses SO_3 as an electrophile. But first, let's see where SO_3 comes from.

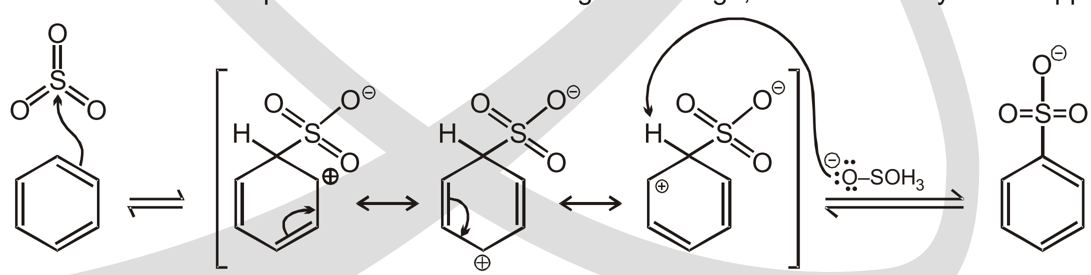
Sulfuric acid is constantly in equilibrium with SO_3 and water:

$$H_2SO_4 \rightleftharpoons SO_2 \quad H_2O$$

That means that any bottle of sulfuric acid will have some SO_3 in it. At room temperature, SO_3 is a gas, and it is possible to add extra SO_3 gas to sulfuric acid (which shifts the equilibrium). When we do this, we call the mixture fuming sulfuric acid. So, from now on, whenever you see concentrated, **fuming** sulfuric acid, you should realize that we are talking about SO_3 as the reagent. And here is the reaction :

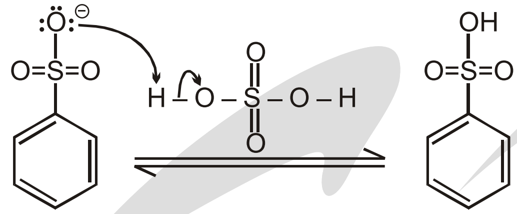


Notice that, in the end, we have installed the SO_3H group on the ring. The obvious question is: why is the H attached to the SO_3 in the end? To see why, let's take a closer look at the mechanism. Remember our two steps for any electrophilic aromatic substitution: E^+ goes on the ring, and then H^+ comes off. But wait a second. . . . In this case, we are not using an electrophile with a net positive charge. The electrophile in this case has no net charge. In all of the reactions we have seen so far, we put something positively charged onto the ring, and then we took something positively charged off of the ring. So in the end, our ring never gained or lost any charges. But in this case, we are putting something neutral (SO_3) onto the ring, and then we are removing something positively charged (H^+). That should leave our product with an overall negative charge, which is exactly what happens:

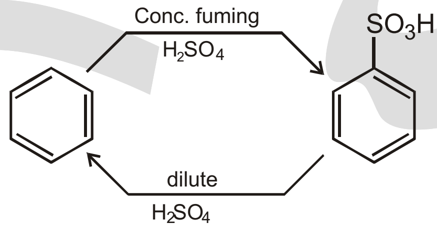


SIGMA COMPLEX

So, we need to add one more step to our mechanism. The negatively charged oxygen atom removes a proton from sulfuric acid :

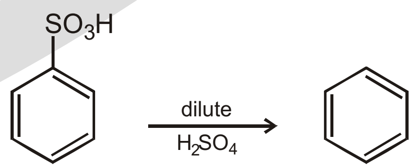


An important feature of this reaction (and this is the feature that will make this reaction so important for synthesis problems) is how easily the reaction can be reversed. The amount of product is equilibrium controlled, and it is very sensitive to the conditions. So, if you use dilute sulfuric acid instead, the equilibrium leans the other way :



We can use this to our advantage because this provides a way to remove the SO_3H group whenever we want. We would just use dilute sulfuric acid to remove it:

Desulfonation reaction



In order to force a reaction to occur, we introduced a Lewis acid into the reaction mixture, which generated a better electrophile (Br+ is a better electrophile than Br_2). In fact, all of the reactions we have explored thus far have been examples of benzene reacting with powerful electrophiles (Cl^+, NO_2, alkyl+, $acyl^+$ and SO_3).

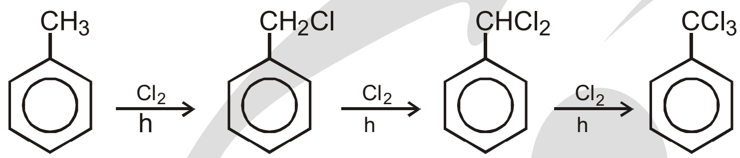
Summary :

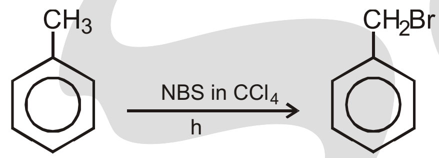
	Reaction	Reagent	Catalyst	Product	E or E
(a)	Halogenation	X_2(X Cl, Br)	FeX_3 (from Fe + X_2)	AlCl, AgBr	X
(b)	Nitrogen	HNO_3	H_2SO_4	$ArNO_2$	NO_2
(c)	Sulfonation	H_2SO_4 or $H_2S_2O_7$	none	$ArSO_3H$	SO_3
(d)	Friedel-Crafts alkylation	RX, $ArCH_2X$ ROH H H RC CH	$AlCl_3$ HF, H_2SO_4, BF_3 H_3PO_4 or HF	Ar — R, Ar — CH_2Ar Ar — R Ar — CHCH$_3$ R	R
(e)	Friedel-Crafts alkylation	RCOCl	$AlCl_3$	O Ar — C— R	RC O

Substitution in Side-Chain

When halogenation is carried out in the absence of a Lewis acid under conditions that favour radical formation, side chain halogenation predominates.

For example,





Benzyl bromide

Hydrogen atoms attached to the carbon joined directly to an aromatic ring are called benzylic hydrogens.

Benzylic hydrogen

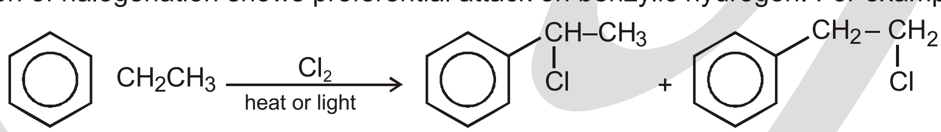
Benzylic hydrogens are extremely easy to abstract (like allylic hydrogen) owing to resonance stabilisation of benzylic radical formed.

Ease of abstraction of H atoms : allylic, benzylic > 3° > 2° > 1° > CH_4 > vinylic

Ease of formation and stability of

free radicals : allyl, benzyl > 3° > 2° > 1° > CH_3• > vinyl

Thus, the orientation of halogenation shows preferential attack on benzylic hydrogen. For example,



1-chloro-1-phenyl ethane (Major product 91%)

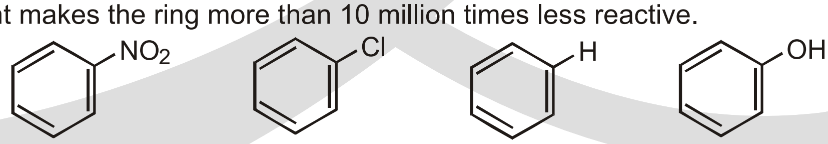
2-chloro-1-phenyl ethane (9%)

ORTHO/PERA GROUPS

Substituent Effects in Substituted Aromatic Rings

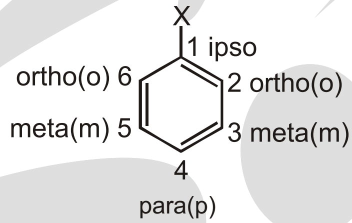
Only one product can form when an electrophilic substitution occurs on benzene, but what would happen if we were to carry out a reaction on an aromatic ring that already has a substituent? A substituent already present on the ring has two effects.

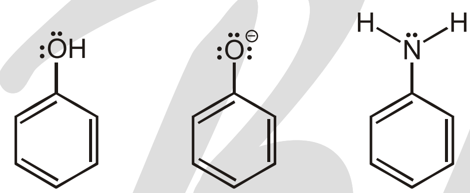
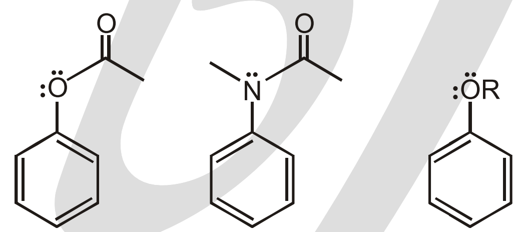
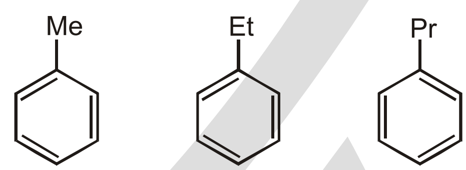
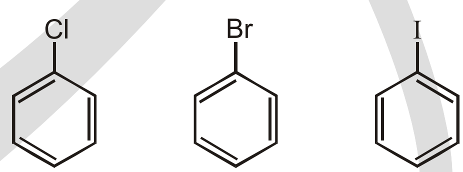
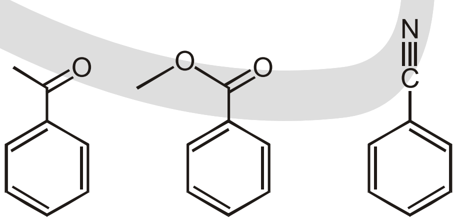
- Substituents affect the reactivity of the aromatic ring. Some substituents activate the ring, making it more reactive than benzene, and some deactivate the ring, making it less reactive than benzene. In aromatic nitration, for instance, an –OH substituent makes the ring 1000 times more reactive than benzene, while an $–NO_2$ substituent makes the ring more than 10 million times less reactive.



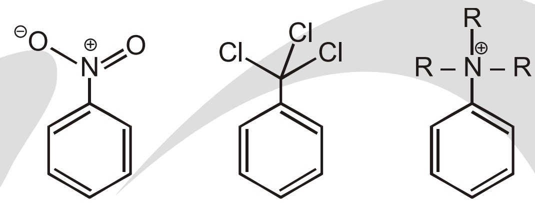
Relative rate of nitration 6×10^{-8} 0.033 1 1000

Substituents affect the orientation of the reaction. The three possible disubstituted products-ortho, meta, and para-are usually not formed in equal amounts. Instead, the nature of the substituent already present on the benzene ring determines the position of the second substitution. An –OH group directs substitution toward the ortho and para positions, for instance, while a carbonyl group such as –CHO directs substitution primarily toward the meta position.



Common Feature		Some Examples
ACTIVATORS		
(Increase electron density in the ring)		
Strong	Lone pair next to ring	
Moderate	Lone pair next to ring But tied up in resonance outside of ring as well	
Weak	Alkyl groups	
DEACTIVATORS		
(Decrease electrondensity in the ring)		
Weak	Halogens	
Moderate	Pi bond to an electronegative atom (next to ring)	

Strong Very powerfully electron withdrawing

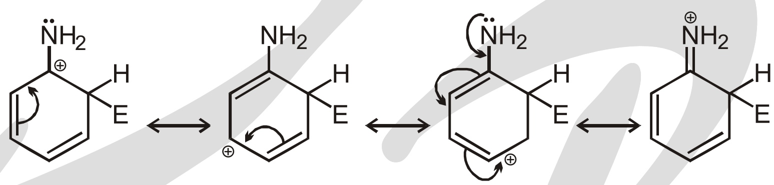


Effect of Substituent on Benzene Rings on Electrophilic Substitution :

When further substitution is done on mono substituted benzene some groups. (alkyl group, halogen group, –OH group) direct the new entrant to ortho as well as para positions, other groups like, $-NO_2$, –CHO, –COOH direct new entrant to meta position. Let us study this orientation and the directive influence of the groups already present on benzene ring.

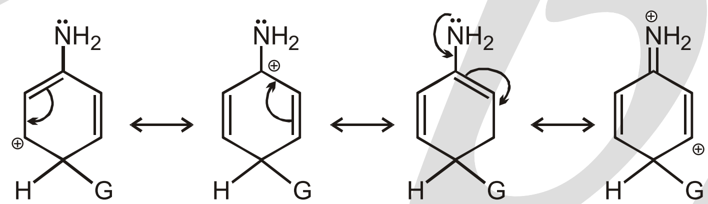
Let us consider those compounds, in which benzene carries a group, containing a lone pair of electrons e.g., aniline. The carbonium ions formed in these cases can be written as :

1. Ortho substitution



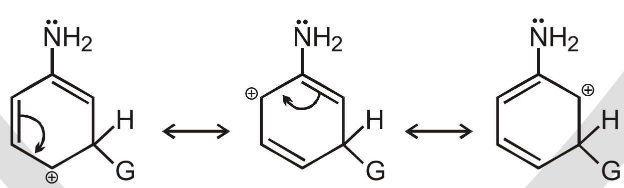
I II III IV

2. Para substitution



V VI VII VIII

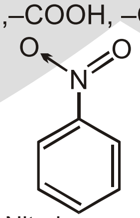
3. Meta substitution



IX X XI

Out of these structures (IV) and (VIII) are particularly stable because in these cases the octet of each carbon atom is complete. In case of carbonium ion for meta substitution such a structure is not formed. Hence we say that the resonance hybrid for carbonium ion in ortho and para substitution are more stable and are formed more easily. The further substitution in such compounds always occur at ortho and para positions and not at meta positions.

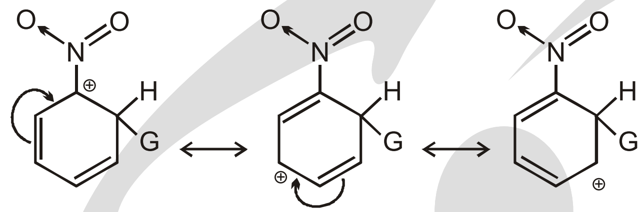
Electron withdrawing groups i.e., $-NO_2$,–COOH, –CN etc. Consider case of nitro benzene.



Nitrobenzene

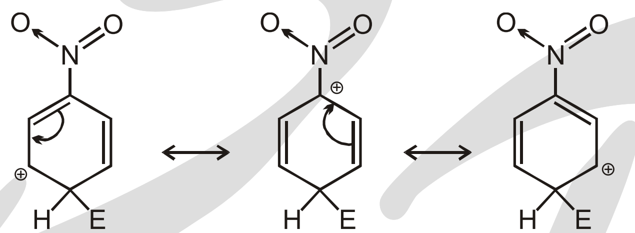
When this compound, is attacked by an electrophile the carbonium ions which are formed for the ortho, meta or para position can be written as follows :

1. Ortho substitution



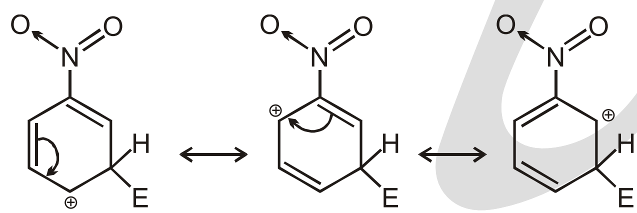
I II III

2. Para substitution



IV V VI

3. Meta substitution



VIII IX X

Structure I and V are particularly unstable, because the $-NO_2$ group which is attached to the C with positive charge helps to intensify the charge. Greater the intensification lesser is the stability. Hence in the resonating structure for ortho and para substitution the resonance hybrid will be less stable. The resonating structures, for meta substitution, make the hybrid more stable thus meta substitution will be preferred. Whenever the electron withdrawing groups like $-NO_2$, $-COOH$,$-CN$ $-SO_3H$ are present on the ring then the further substitution, will always occur at meta position.

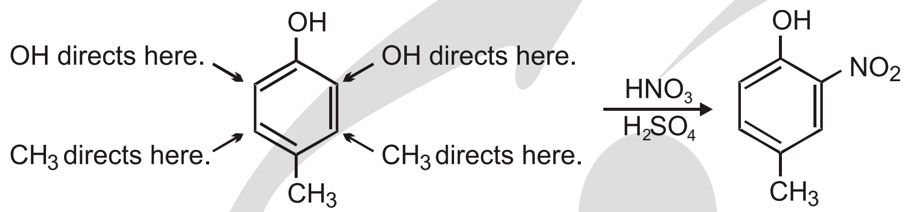
(F) REACTION OF SUBSTITUTED BENZENE

Additivity of Effects

Electrophilic substitution of a disubstituted benzene ring is governed by the same resonance and inductive effects that affect monosubstituted rings. The only difference is that it's necessary to consider the additive effects of two different groups. In practice, this isn't as difficult as it sounds; three rules are usually sufficient.

1. If the directing effects of the two groups reinforce each other, the situation is straightforward. In p-nitrotoluene, for example, both the methyl and the nitro group direct further substitution to the same position (ortho to the methyl = meta to the nitro). A single product is thus formed on electrophilic substitution.

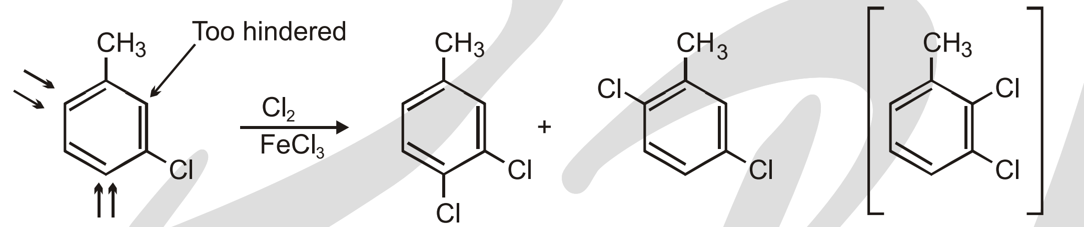
2. If the directing effects of the two groups oppose each other, the more powerful activating group has the dominant influence, but mixtures of products are often formed. For example, bromination of p-methylphenol yields primarily 2-bromo-4-methylphenol because –OH is a more powerful activator than — CH_3.



4-Methylphenol **4-Methyl-2-nitrophenol**

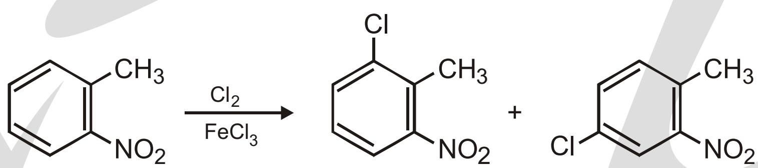
3. Further substitution rarely occurs between the two groups in a metadisubstituted compound because this site is too hindered. Aromatic rings with three adjacent substituents must therefore be prepared by some other route, such as by substitution of an ortho-disubstituted compound.

(A)



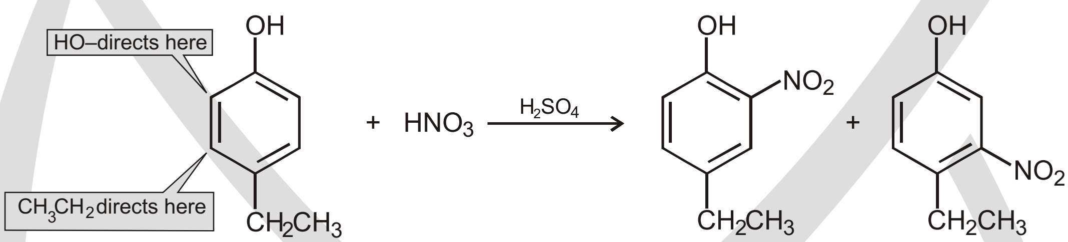
m-Chlorotoluene **3,4-Dichlorotoluene** **2,5-Dichlorotoluene** **Least formation of product**

(B)

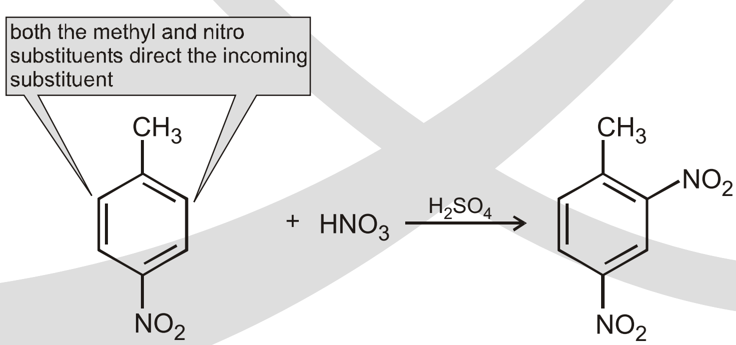


o-Nitrotoluene **2-Chloro-6-nitrotoluene** **4-Chloro-2-nitrotoluene**

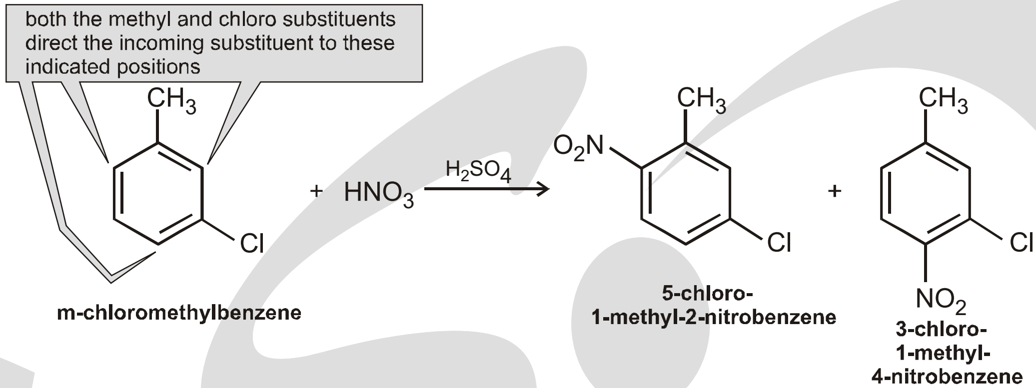
(C)

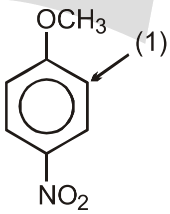


(D)

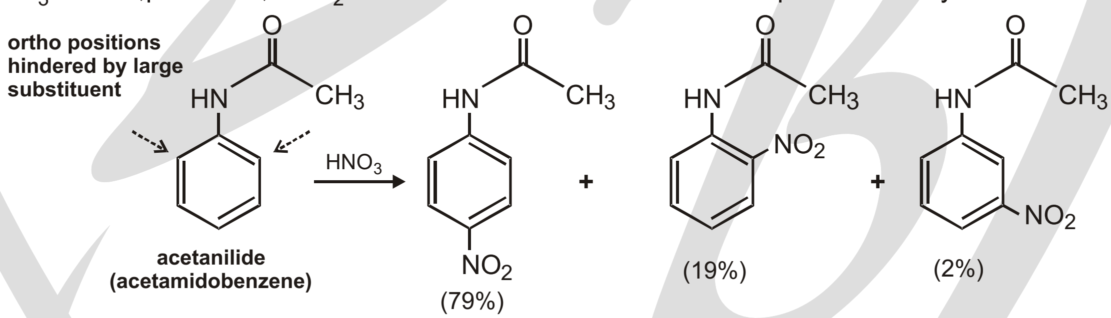


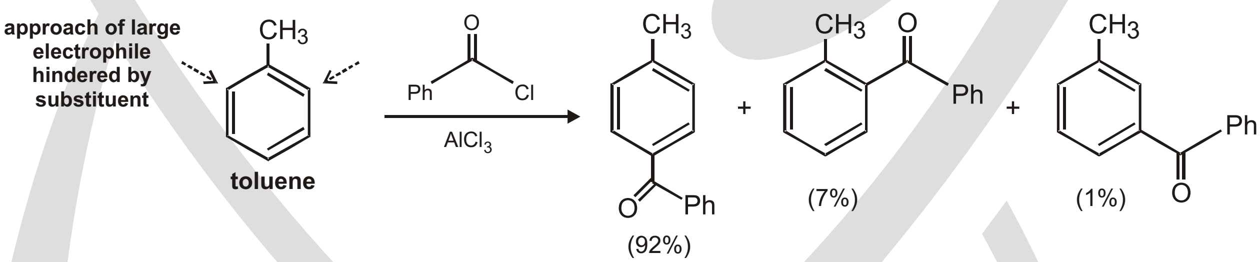
p-methylnitrobenzene **1-methyl-2,4-dinitrobenzene**

(E) 

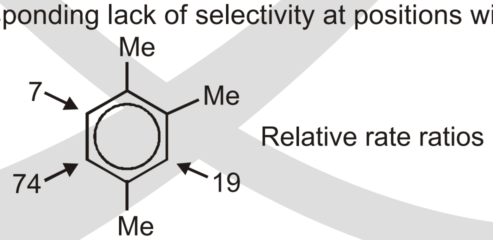
(F) 

$-OCH_3$ is an o,p-director, $-NO_2$ is a m-director. Attack will occur at position 1 only.



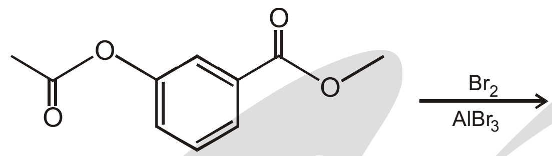
(G) 

(H) Given this behaviour (little selectivity in distinguishing between different substrate molecules), the selectivity relationship would predict that positional selectivity should also be very small. However, it is not. For example, under conditions where nitration of p-xylene and 1,2,4-trimethylbenzene takes place at about equal rates, there was no corresponding lack of selectivity at positions within the latter.97 Though



Solved Example

- Predict the product of the following reaction:

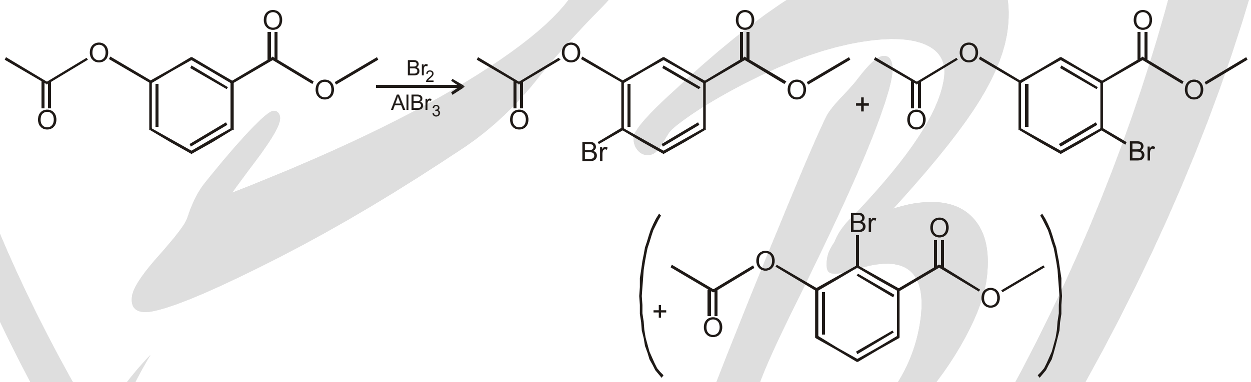


Ans. We look at the reagents to see what kind of reaction we expect. The reagents are bromine and aluminum tribromide. These reagents generate Br^+, which is an excellent electrophile. So, we know that we will be installing a Br^+ on the ring. But the question is: where?

To answer this question, we must predict the directing effects of the two groups that are currently on the ring. The group on the left is a moderate activator (make sure that you know why), and therefore, it directs ortho-para to itself:

The group on the right is a moderate deactivator (make sure you know why), so it directs meta to itself:

There is a competition between these two groups. Remember our first rule for determining which group wins: ortho-para directors beat meta directors. So, we expect the following products:



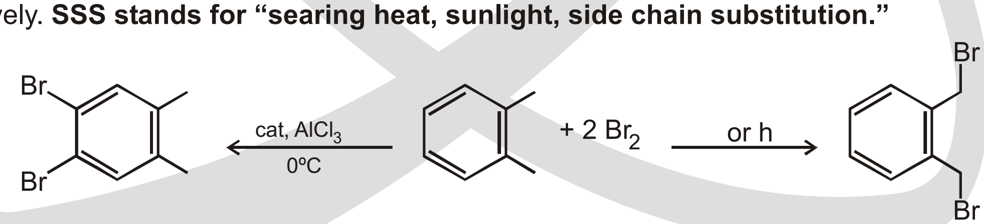
Notice that I placed the last product in parentheses. This product is actually a very minor product. We will see why in the next section. For now, we will just write that we expect three products. Then, we will fine-tune this prediction in the next section.

Special Problem

Chemoselectivity of Radical Brominations

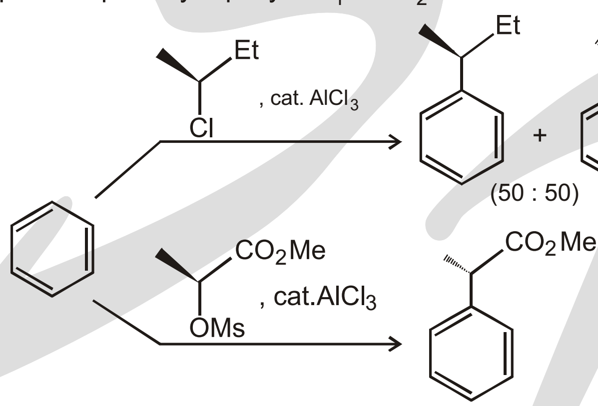
Let us go back to radical brominations. The bromination of alkyl aromatics takes place completely regioselectively: only the benzylic position is brominated. The intermediates are the most stable radicals that are available from alkyl aromatics, namely, benzylic radicals. Refluxing ortho-xylene reacts with 2 equiv. of bromine to give one monosubstitution per benzylic position. The same transformation occurs when the reactants are irradiated at room temperature in a 1:2 ratio (Figure, right).

The rule of thumb "SSS" applies to the reaction conditions that afford these benzylic substitutions chemoselectively. **SSS stands for "searing heat, sunlight, side chain substitution."**



Special Problem

Secondary alkyl halides and sulfonates can react with aromatic compounds via an SN_1 mechanism. The pair of reactions in Figure shows that the balance between SN_2 and SN_1 mechanisms can be subject to subtle substituent effects. S-2-Chlorobutane and $AlCl_3$ alkylate benzene with complete racemization. This means that the chlorobutane is substituted by the aromatic substrate through an SN_1 mechanism, which takes place via a solventseparated ion pair.On the other hand, the mesylate of S-methyl lactate and $AlCl_3$ alkylate benzene with complete inversion of the configuration of the stereocenter. This means that the mesylate group is displaced by the aromatic compound in an SN_2 reaction. There are two reasons why these substitution mechanisms are so different and so clearly preferred in each case. On the one hand, the lactic acid derivative cannot undergo an SN_1 reaction. The carbenium ion that would be generated, $H_3C-CH^+-CO_2Me$, would be strongly destabilized by the electron-withdrawing CO_2Me group (cf. discussion of figure). On the other hand, SN_2 reactions in the a position to an ester group take place especially rapidly. SN_1 & SN_2 will be discus in next chapter.

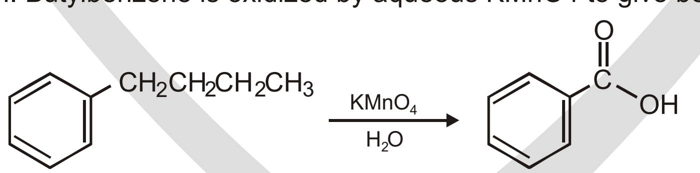


(G) OXIDATION REACTION

Oxidation of Aromatic Compounds

Oxidation of Alkyl Side Chains

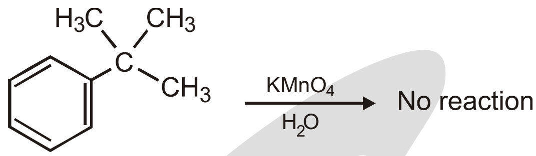
Despite its unsaturation, the benzene ring is inert to strong oxidizing agents such as $KMnO_4$ and $Na_2Cr_2O_7$, reagents that will cleave alkene carbon-carbon bonds. It turns out, however, that the presence of the aromatic ring has a dramatic effect on alkyl side chains. Alkyl side chains react rapidly with oxidizing agents and are converted into carboxyl groups, —CO_2H.The net effect is conversion of an alkylbenzene into a benzoic acid, Ar R Ar CO_2H. Butylbenzene is oxidized by aqueous KMnO4 to give benzoic acid, for instance.



Butylbenzene **Benzoic acid (85%)**

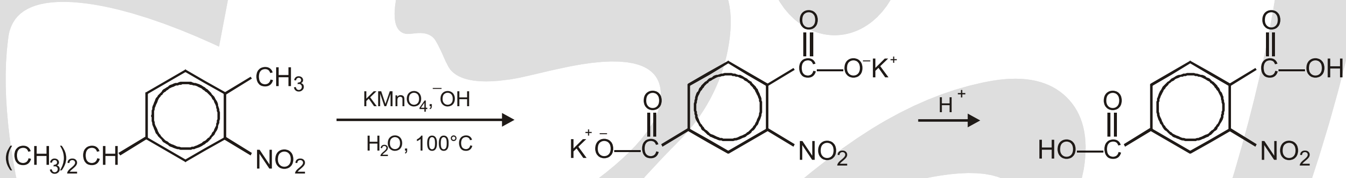
A similar oxidation is employed industrially for the preparation of the terephthalic acid used in the production of polyester fibers. Worldwide, approximately 20 million tons per year of terephthalic acid is produced by oxidation of p-xylene, using air as the oxidant and Co(III) salts as catalyst.

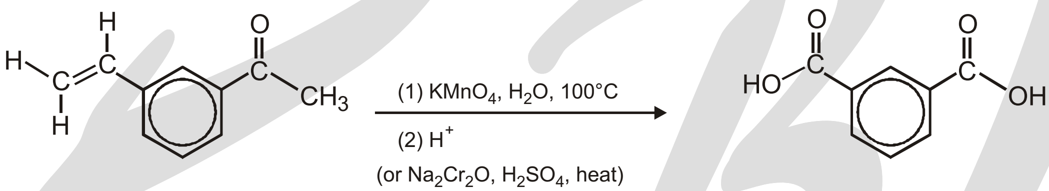
The mechanism of side-chain oxidation is complex and involves reaction of C–H bonds at the position next to the aromatic ring to form intermediate benzylic radicals. tert-Butylbenzene has no benzylic hydrogens, however, and is therefore inert.



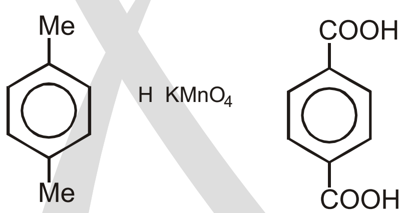
tert-Butylbenzene

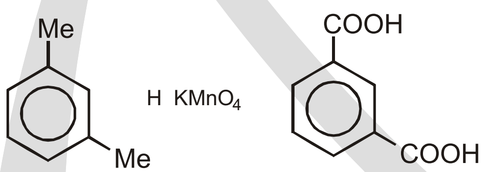
An aromatic ring imparts extra stability to the nearest carbon atom of its side chains. The aromatic ring and one carbon atom of a side chain can survive a vigorous permanganate oxidation. The product is a carboxylate salt of benzoic acid. (Hot chromic acid can also be used for this oxidation.) This oxidation is occasionally useful for making benzoic acid derivatives, as long as any other functional groups are resistant to oxidation. Functional groups such as —NO_2, halogens, —COOH, and —SO_3H usually survive this brutal oxidation.

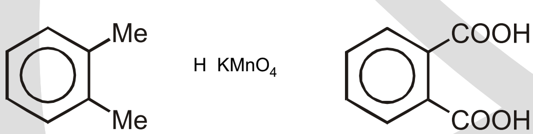




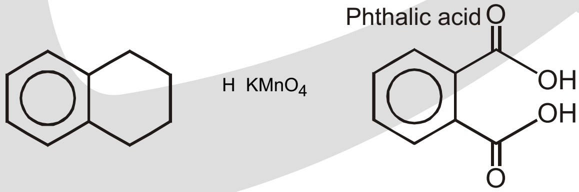
OXIDATION OF AROMATIC COMPOUNDS

(c)  terephthalic acid

(d)  isophthalic acid

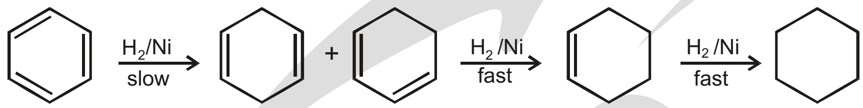
(e) 

Phthalic acid

(f) 

Reduction of aromatic compounds :

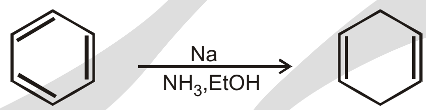
Hydrogenation of benzene under pressure using a metal catalyst such as nickel results in the addition of three molar equivalents of hydrogen and the formation of cyclohexane. The intermediate cyclohexadienes and cyclohexene cannot be isolated because these undergo catalytic hydrogenation faster than benzene does.



Benzene **Cyclohexadienes** **Cyclohexene** **Cyclohexane**

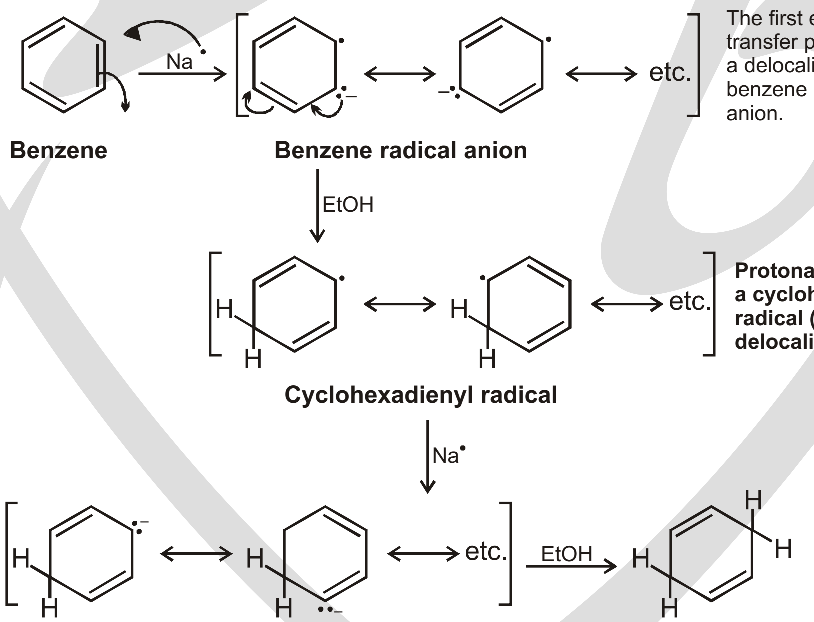
BIRCH REDUCTION

Benzene can be reduced to 1,4-cyclohexadiene by treating it with an alkali metal (sodium, lithium, or potassium) in a mixture of liquid ammonia and an alcohol :



Benzene **1,4-Cyclohexadiene**

Birch reduction :



Benzene **Benzene radical anion**

The first electron transfer produces a delocalized benzene radical anion.

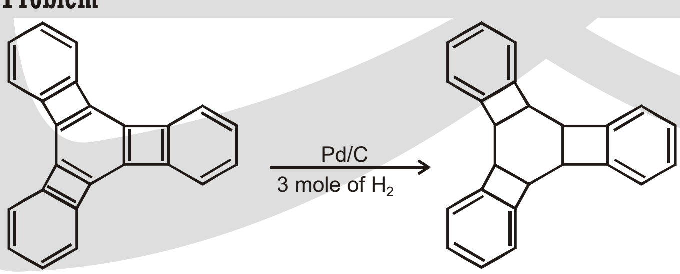
Cyclohexadienyl radical

Protonation prodcues a cyclohexadienyl radical (also a delocalized species).

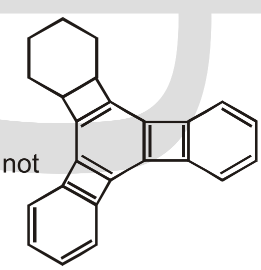
Cyclohexadienyl anion **1,4-Cyclohexadiene**

Transfer of another electron leads to the formation of a delocalized cyclohexadienyl anion, and protonation of this produces the 1,4-cyclohexadiene.

Special Problem



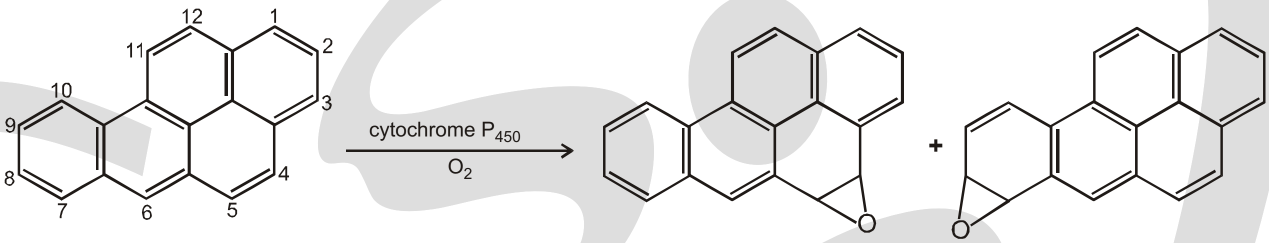
is formed but not



Sol. Anti-aromatic are highly unstable that's why it's reduction prefer.

Benzo[a]pyrene and Cancer

Benzo[a]pyrene is one of the most carcinogenic arenes. It is formed whenever an organic compound is not completely burned. For example, benzo[a]pyrene is found in cigarette smoke, automobile exhaust, and charcoal-broiled meat. Several arene oxides can be formed from benzo[a]pyrene. The two most harmful are the 4,5-oxide and the 7, 8-oxide.

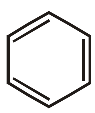
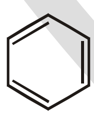
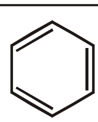
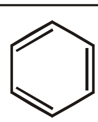
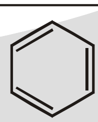
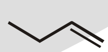


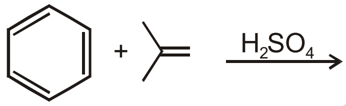
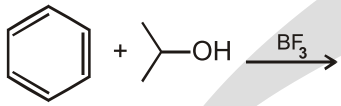
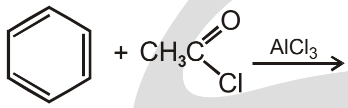
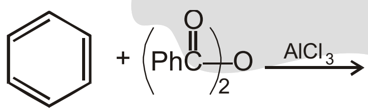
benzo[a]pyrene 4-5-benzo[a]pyrene oxide 7-8-benzo[a]pyrene oxide

EXERCISE

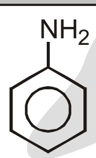
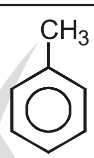
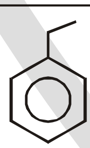
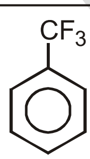
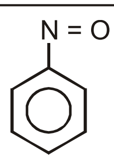
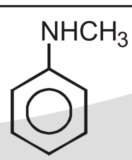
WORK SHEET - 1

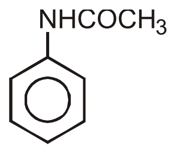
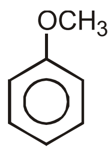
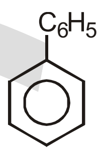
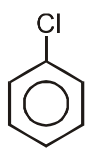
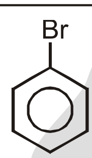
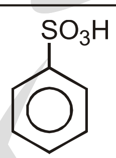
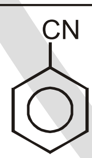
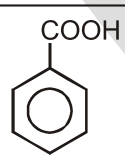
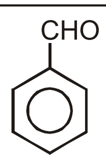
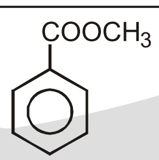
1. Each of the following are classified as reactions that occur by an electrophilic aromatic substitutionmechanism. Write a general two-step mechanistic scheme (use curved arrows to show movement of electron pairs) for these reactions. Construct a table showing the elctrophile and the electrophilic substitution intermediate for each reaction.

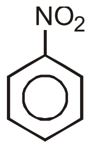
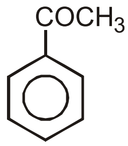
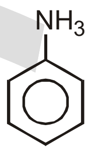
(a)	 + Cl_2 $\xrightarrow{FeCl_3}$
(b)	 + Br_2 $\xrightarrow{FeCl_3}$
(c)	 + HNO_3 $\xrightarrow{H_2SO_4}$
(d)	 $\xrightarrow{SO_3, H_2SO_4}$
(e)	 + CH_3CH_2Cl $\xrightarrow{AlCl_3}$
(f)	 +  $\xrightarrow{HF}$

(g)	
(h)	
(i)	
(j)	

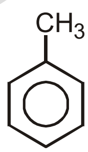
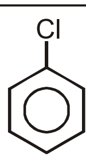
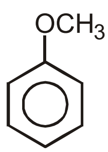
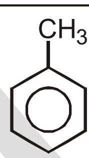
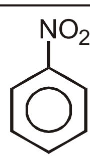
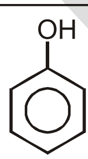
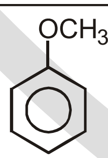
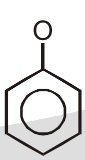
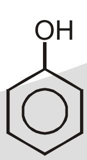
WORK SHEET - 2

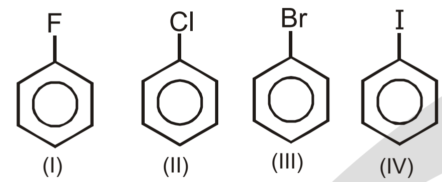
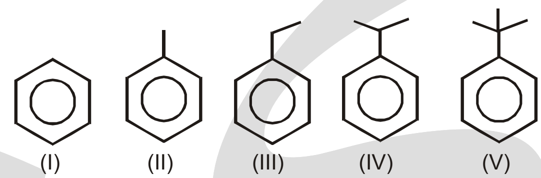
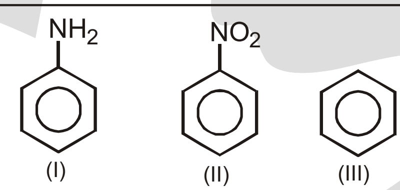
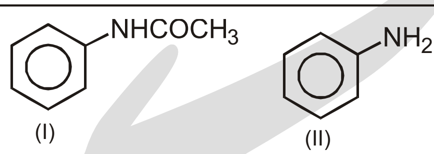
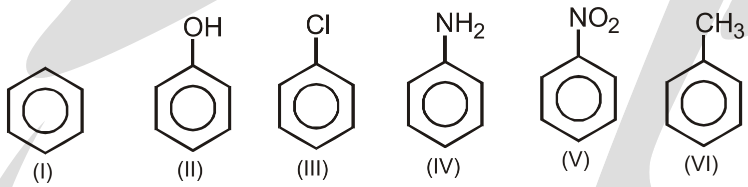
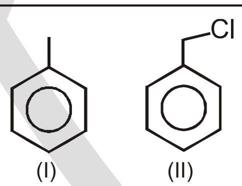
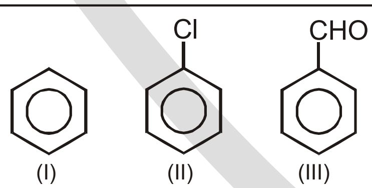
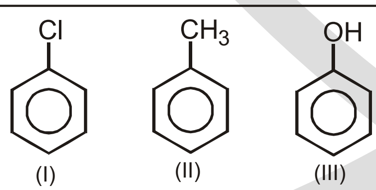
S.No.	Compound	Ortho para directing	Meta director directing	Activating	Deactivating
1.					
2.					
3.					
4.					
5.					
6.					

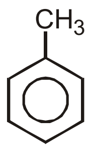
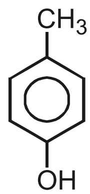
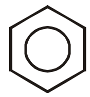
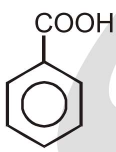
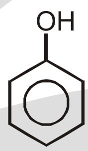
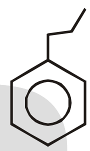
7.					
8.					
9.					
10.					
11.					
12.					
13.					
14.					
15.					
16.					

17.					
18.					
19.					

WORK SHEET - 3

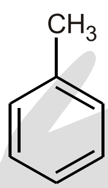
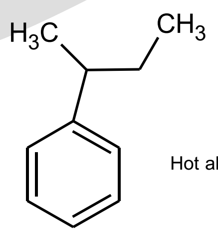
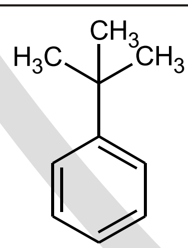
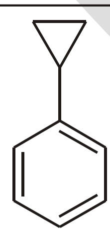
S.No.	Compound	Rate of Electrophilic Aromatic Subtitution
1.	 (I)  (II)  (III)	
2.	 (I)  (II)  (III)	
3.	 (I)  (II)  (III)	
4.	 (I)  (II)  (III)	
5.	 (I)  (II)	

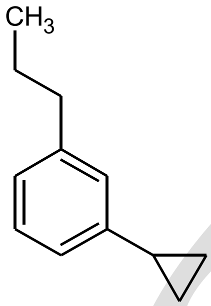
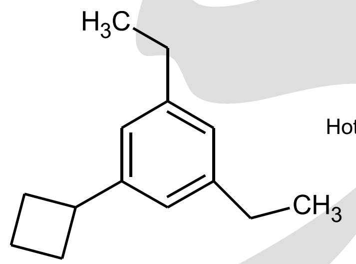
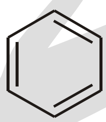
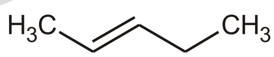
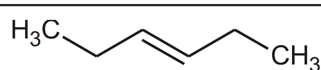
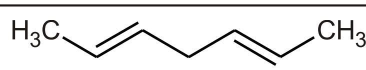
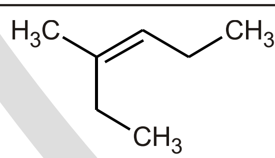
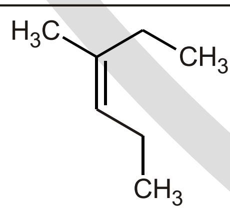
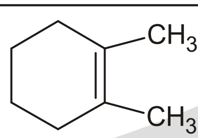
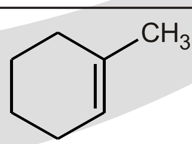
6.		
7.		
8.		
9.		
10.		
11.		
12.		
13.		

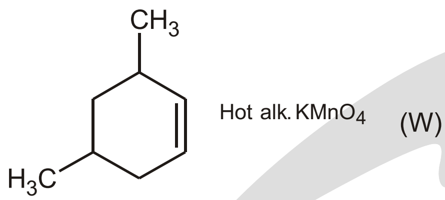
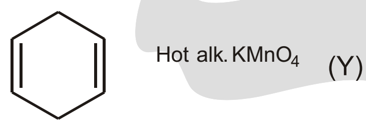
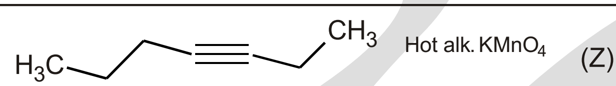
14.	 (I)  (II)  (III)  (IV)	
15.	 (I)  (II)  (III)  (IV)	

WORK SHEET - 4

Q. Identify the Product

(1)	 Hot alk. $KMnO_4$ (A)
(2)	 Hot alk. $KMnO_4$ (D)
(3)	 Hot alk. $KMnO_4$ (E)
(4)	 Hot alk $KMnO_4$ (F)

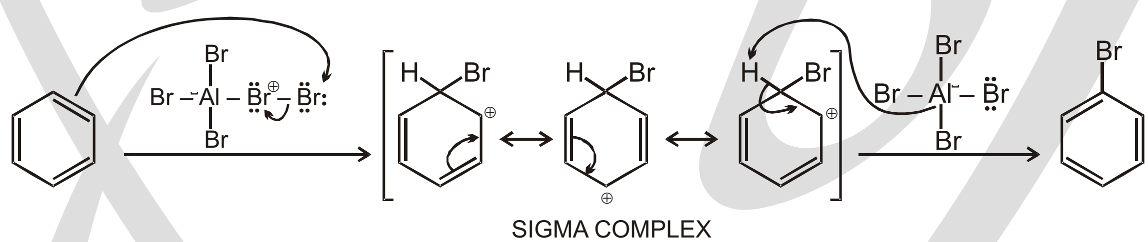
(5)	 Hot alk. $KMnO_4$ (G)
(6)	 Hot alk. $KMnO_4$ (J)
(7)	 Hot alk. $KMnO_4$ (K)
(8)	 Hot alk. $KMnO_4$ (L)
(9)	 Hot alk. $KMnO_4$ (M)
(10)	 Hot alk. $KMnO_4$ (M)
(11)	 Hot alk. $KMnO_4$ (O)
(12)	 Hot alk. $KMnO_4$ (P)
(13)	 Hot alk. $KMnO_4$ (U)
(14)	 Hot alk. $KMnO_4$ (V)

(15)	
(16)	
(17)	
(18)	

Answers

Work sheet - 1

Similar reaction mechanism :



Work sheet - 2

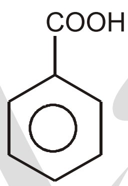
S.No.	Ortho para directing	Meta director directing	Activating	Deactivating
1.	✓		✓	
2.	✓		✓	
3.	✓		✓	
4.	✓	✓		✓
5.	✓			✓
6.	✓		✓	
7.	✓		✓	
8.	✓		✓	
9.	✓		✓	
10.	✓			
11.	✓			✓

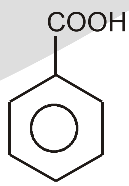
12.		✓		✓
13.		✓		✓
14.		✓		✓
15.		✓		✓
16.		✓		✓
17.		✓		✓
18.		✓		✓
19.		✓		✓

Work sheet - 3

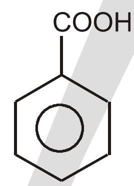
1. III > I > II
2. III > I > II
3. II > I > III
4. II > III > I
5. I > II
6. I > II > III > IV
7. II > III > IV > V > I
8. I > III > II
9. II > I
10. IV > II > VI > I > III > V
11. I > II
12. I > II > III
13. III > II > I
14. II > IV > I > III
15. III > IV > I > II

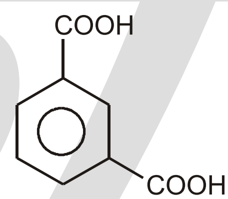
Work sheet - 4

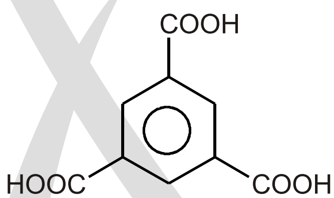
1. 

2. 

3. No reaction

4. 

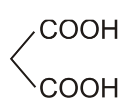
5. 

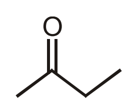
6. 

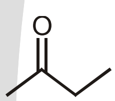
7. No reaction

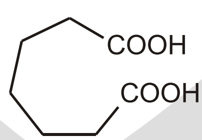
8. CH_3COOH CH_3CH_2COOH

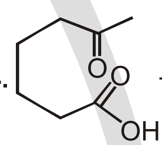
9. CH_3CH_2COOH

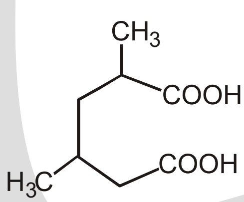
10.  + CH_3COOH

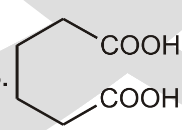
11. CH_3CH_2COOH 

12.  CH_3CH_2COOH

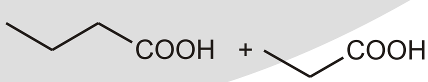
13. 

14.  + HCOOH

15. 

16. 

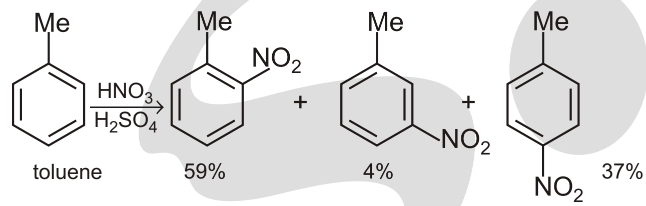
17. 

18. 

Advance Problem 2016

SUBJECTIVE TYPE QUESTIONS

1. How reactive are the different sites in toluene ? Nitration of toluene produces the three possible products in the ratios shown. What would be the ratio of products if all sites were equally reactive ? What is the actual relative reactivity of the three sites ? (You could express this as x:y:1 or as a:b:c where a b c 100). Comment on the ratio you deduce.

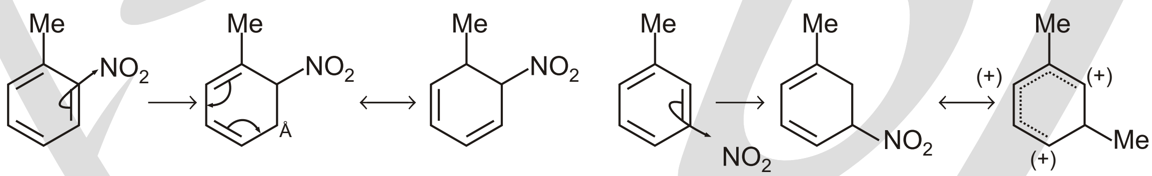


Purpose of the problem

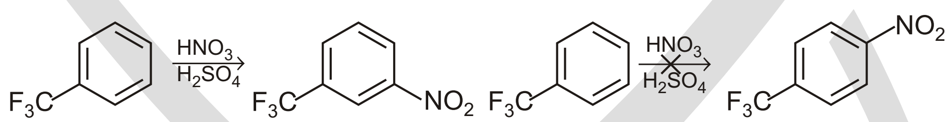
A more quantitative assessement of the relative reactivities of the various positions in an aromatic ring

Suggested solution

There are two ortho and two meta sites but only one para site so the ratio would be 2:2:1 o/m/p. If they were equally reactive. It is actually 30:2:37 or 15:1:18 or 43:3:54 depending on how you expressed it. The ortho and para positions are roughly equally reactive because the alkyl group is electron-donating. The para is slightly more reactive than the ortho because of steric hindrance. The meta is an order of magnitude less reactive because the intermediate cation is not stabilized by -conjugation from the methyl group.



2. In the chapter, we established that electron-withdrawing groups direct meta. Among such reactions is the nitration of trifluoromethyl benzene. Draw out the detailed mechanism for this reaction and also for a reaction that does not happen – the nitration of the same compound in the para position. Draw all the delocalized structures of the intermediates and convince yourself that the intermediate for para substitution is destabilized by the CF_3 group while that for meta substitution is not.

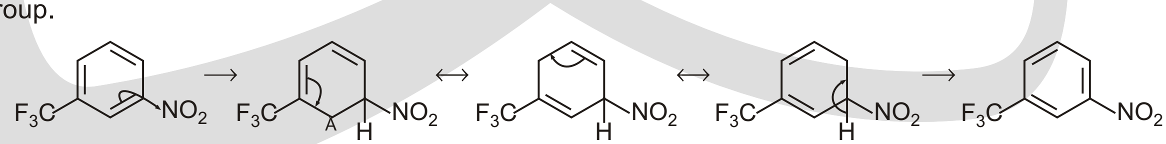


Purpose of the problem

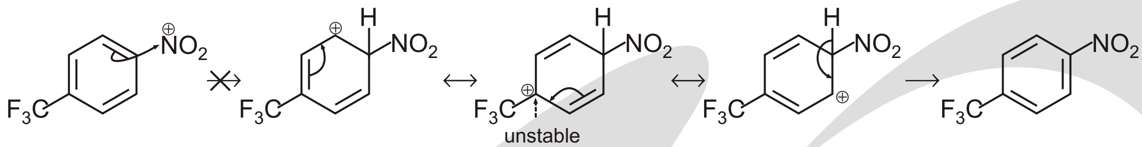
Making sure that you really do understand why groups like CF_3 are meta-directing.

Suggested solution

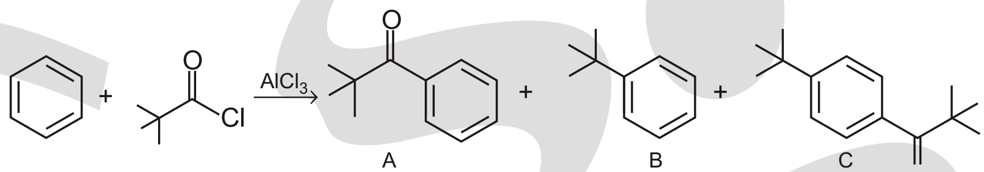
Just do what the question says! You need to do this once to convince yourself. In the mechanism for meta substitution, the positive charge in the intermediate is not delocalized to the carbon atom carrying the CF_3 group.



In the mechanism for para substitution, the positive charge in the intermediate is delocalized to the carbon atom carrying the electron-withdrawing CF_3 group. This intermediate is unstable and is not formed.



3. Attempted Friedel-Crafts acylation of benzene with *t*-BuCOCl gives some of the expected ketone, as a minor product, and also some *t*-butyl benzene, But the major product is the disubstituted compound C. Explain how these compounds are formed and suggested the order in which the two substituents are added to form compound C.



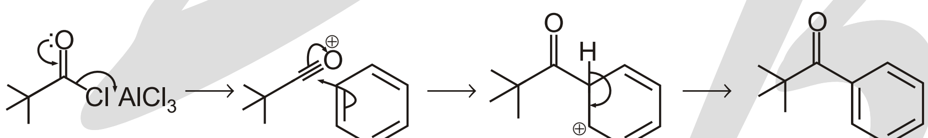
A B C

Purpose of the problem

Detailed analysis of a revealing example of a Friedel-Crafts acylation.

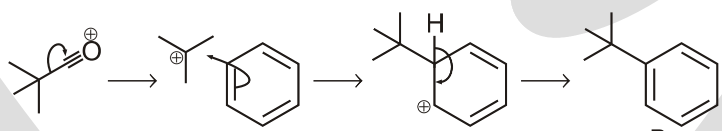
Suggested solution

The expected reaction to give A is a simple Friedel-Crafts acylation with the usual acylium ion as the reactive intermediate.



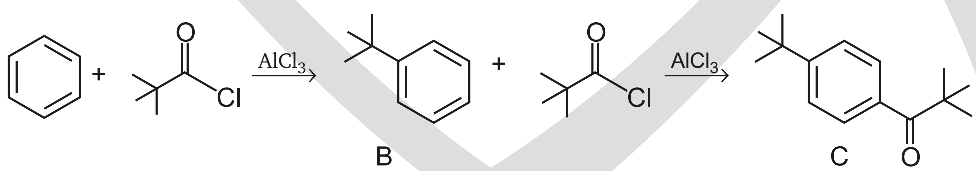
A

Product B must arise from a Friedel-Crafts when t-butyl cation acts as intermediate. This comes from the loss of carbon monoxide from the acylium ion. Such a reaction happens only when the simple carbocation is stable.



B

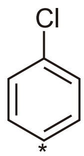
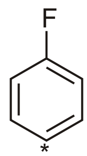
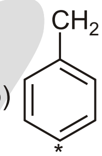
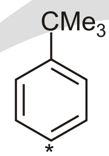
The main product, *C*, comes from the addition of both these electrophiles, but which adds first ? The ketones in A is meta-directing but the *t*-butyl group in *B* is para-directing. Product C has a para relationship and must come from Friedel-Crafts acylation of *B* with the acylium cation.

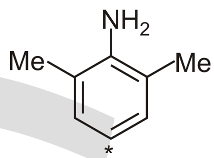
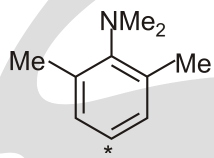


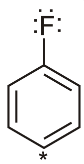
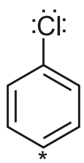
B C

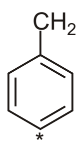
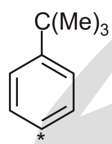
We have now answered the question but you might like to a stage further. Both A and C are formed with alkylation of benzene as the first step. The decomposition of the acylium ion it evidently faster than the acylation of benzene. However, when B reacts further, it is mainly acylated (only a small amount of di-*t*-butyl benzene is formed). Evidently, the decomposition of the acylium ion is slower than the acylation of B! This is not, in fact, unreasonable as the *t*-Bu group in B accelerates electrophilic attack – it is just a dramatic demonstration of that acceleration.

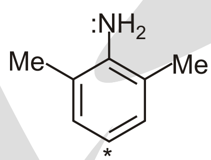
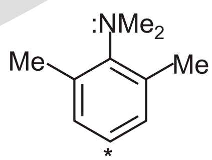
111. Which compound amount the following pairs will have a higher electron density at the marked carbon atom ?

(a)  and  (b)  and 

(c)  and 

Sol. (a)  >  Fluorine has greater M effect compound to chlorine

(b)  >  C — H hyperconjugation is more effective than C — C hyperconjugation

(c)  >  Steric inhibition of resonance decrease the electron density in N, N 2,6- tetramethlyl aniline

❑❑❑

GLOSSARY

-Carbon

The carbon atom directly attached to the functional group of interest, typically a carbonyl group (C O).

-Hydrogen

A hydrogen atom directly bonded to the a-carbon.

Achiral

An achiral molecule has a superimposable mirror image.

Activating group

A substituent that increases the reactivity of an aromatic ring to electrophilic energy substitution.

Activation energy

The difference in energy between the reactants and the transition state.

Acyclic

A molecule, or part of a molecule, whose atoms are not part of a ring.

Acyl

A group containing C O bonded to an alkyl group or an aryl group (e.g. $COCH_3$,—COPh).

Addition reaction

Two groups added to opposite ends of a bond.

Alicyclic

Cyclic compounds that are not aromatic.

Aliphatic

Compounds that are not cyclic and not aromatic.

Alkyl group

Formed on removal of a hydrogen atom from an alkane (e.g. methane, CH_4, gives methyl, CH_3).

Alkylation

A reaction that adds an alkyl group to a reactant.

Angle strain

Strain due to deviation from one or more ideal bond angles.

Anhydrous

Without water.

Anti

When two atoms or groups point in opposite directions; when the atoms or groups lie in the same plane they are antiperiplanar.

Anti addition

An addition reaction in which two substituents add to the opposite sides of the molecule.

Anti elimination

An elimination reaction in which two substituents are eliminated from opposite sides of the molecule.

Antibonding molecular orbital

A molecular orbital formed by out-of-phase overlap of atomic orbitals.

Aprotic solvent

A solvent that is not a hydrogen bond donor.

Aromatic

Cyclic compounds such as benzene and related ring systems.

Aryl

A phenyl group (—C_6H_5), or a substituted phenyl group.

Asymmetric centre

An atom bonded to four different atoms or groups.

Basicity

The tendency of a molecule to share its electrons with a proton.

-Elimination

A reaction in which two atoms/groups on adjacent atoms are lost to form a bond (e.g. elimination of HBr from $BrCH_2CH_3$ to form H_2C CH_2).

Bimolecular

A reaction whose rate depends on the concentration of two reactants (e.g. an SN_2 reaction).

Bond angle

The angle formed by three contiguous bonded atoms.

Bonding molecular orbital

A molecular orbital formed by in-phase overlap of atomic orbitals.

Carbanion

A species with a negative charge on carbon.

Carbene

A species with a nonbonded pair of electrons on carbon and an empty orbital (e.g. H_2C:).

Carbocation

A species with a positive charge on carbon.

Catalyst

A species that increases the rate at which a reaction occurs without being consumed in the reaction.

Catalytic

The reactant is a catalyst, and is present in a small amount.

Chemoselectivity

Preferential reaction of one functional group in the presence of others.

Chiral

A chiral molecule has a nonsuperimposable mirror image.

Concerted reaction

A single-step reaction in which reactants are directly converted into products without the involvement of any intermediates.

Configuration

The arrangement of atoms or groups in a molecule to give configurational isomers that cannot be inter converted without breaking a bond.

Conformation

The three-dimensional arrangement of atoms that result from rotation of a single bond.

Conformer

A specific conformation of a molecule that is relatively stable.

Conjugate acid

The acid formed on protonation of a base.

Conjugate base

The base formed on deprotonation of an acid.

Conjugation

Stability associated with molecules containing alternating single and double bonds, due to overlapping orbitals and electron delocalisation.

Constitutional isomers

Molecules that have the same molecular formula but differ in the way their atoms are connected.

Coordinate (dative) bond

A covalent bond where one of the atoms provides both electrons.

Covalent bond

A bond formed as a result of sharing electrons.

Cyclisation

A reaction leading to the formation of a ring.

Cycloaddition

An addition reaction that forms a ring.

Deactivating group

A substituent that decreases the reactivity of an aromatic ring to electrophilic substitution.

Decarboxylation

Loss of carbon dioxide.

Dehydration

Loss of water.

Delocalisation

When lone pairs or electrons in bonds are spread over several atoms.

Deprotonate

To remove a proton (H).

Diastereoisomer

Stereoisomers that are not mirror images of each other.

Dielectric constant

A measure of how well a solvent can insulate opposite charges from one another (polar solvents have relatively high dielectric constants).

Dihedral angle

The angle between two planes, where each plane is defined by three atoms.

Dipole moment

A measure of the separation of charge in a bondc or in a molecule.

Disubstituted

Typically an alkene or substituted benzene that contains two groups, or atoms other than hydrogen.

Electron donating group

An atom or group that releases electron density to neighbouring atoms.

Electron withdrawing group

An atom or group that draws electron density from neighbouring atoms towards itself.

Electronegative

An atom that attracts electrons toward itself.

Electronic effect

The reactivity of a part of the molecule is affected by electron attraction or repulsion.

Electrophile

Electron-deficient reactant that accepts two electrons to form a covalent bond.

Electrophilicity

The relative reactivity of an electrophilic reagent.

Elimination

A reaction that involves the loss of atoms or molecules from the reactant, typically from adjacent atoms.

Enantiomer (optical isomer)

One of a pair of molecules which are mirror images of each other and non superimposable.

Endothermic

A reaction in which the enthalpy of the products is greater than the enthalpy of the reactants; the overall standard enthalpy change is positive.

Energy profile

A plot of the conversion of reactants into products versus energy(Gibbs free energy or enthalpy).

Enolisation

The conversion of a keto form into an enol form.

Enthalpy

The heat given off or the heat absorbed during the course of a reaction.

Entropy

A measure of the disorder or randomness in a closed system.

Equilibrium constant

The ratio of products to reactants at equilibrium or the ratio of the rate constants for the forward and reverse reactions.

Exothermic

A reaction in which the enthalpy of the products is smaller than the enthalpy of the reactants; the overall standard enthalpy change is negative.

Formal charge

A positive or negative charge assigned to atoms that have an apparent abnormal number of bonds.

Functional group

An atom, or a group of atoms that has similar chemical properties whenever it occurs in different compounds.

Geometric isomers

cis/trans or E/Z isomers.

Heteroaromatic

An aromatic molecule that contains at least one heteroatom as part of the aromatic ring (e.g. pyridine)

Heteroatom

Any atom other than carbon or hydrogen.

Heterocycle

A cyclic molecule where at least one atom in the ring is not carbon.

Heterolysis

Breaking a bond unevenly, so that both electrons stay with one of the atoms.

Homolysis

Breaking a bond evenly, so that each atom gets one electron.

Hybridisation

Mixing atomic orbitals to form new hybrid orbitals.

Hydration

Addition of water to the reactant.

Hydrogen bond

A noncovalent attractive force caused when the partially positive hydrogen of one molecule interacts with the partially negative heteroatom of another molecule.

Hydrogenation

Addition of hydrogen.

Hydrolysis

A reaction in which water is a reactant.

Hydrophilic

A molecule or part of a molecule with high polarity ('water loving').

Hydrophobic

A molecule or part of a molecule with low polarity ('water-fearing').

Hyperconjugation

Donation of electrons from C — H or C — C sigma bonds to an adjacent empty p orbital.

Inductive effect

The polarisation of electrons in sigma bonds.

Intermediate

A species with a lifetime appreciably longer than a molecular vibration (corresponding to a local potential energy minimum) that is formed from the reactants and reacts further to give (either directly or indirectly) the products of a reaction.

Intermolecular

A process that occurs between two or more separate molecules.

Intramolecular

A process that occurs within a single molecule.

Ionic bond

A bond formed through the attraction of two ions of opposite charges.

Isomerisation

The process of converting a molecule into its isomer.

Isomers

Non-identical compounds with the same molecular formula.

Kinetic control

A reaction in which the product ratio is determined by the rate at which the products are formed.

Leaving group

An atom or group (charged or uncharged) that becomes detached from the main part of a reactant, that takes a pair of electrons with it.

Lewis acid

Accepts a pair of electrons.

Lewis base

Donates a pair of electrons.

Lone pair

Two (paired) electrons in the valence shell of a single atom that are not part of a covalent bond.

Mechanism

A step-by-step description of the bond changes in a reaction, often shown using curly arrows.

Mesomeric effect

Delocalisation of electron density through bonds.

Molecular orbital

An orbital which extends over two or more atoms.

Monomer

A repeating unit in a polymer.

Nucleophile

Electron-rich reactant that donates two electrons to form a covalent bond.

Nucleophilicity

The relative reactivity of a nucleophilic reagent.

One equivalent

Amount of a substance that reacts with one mole of another substance.

Orbital

The volume of space around the nucleus in which an electron is most likely to be found.

Oxidising agent

In a reaction, the reactant that causes the oxidation (and becomes reduced).

Oxidation reaction

A reaction that increases the number of covalent bonds between carbon and a more electronegative atom (e.g. O, N, a halogen atom), and decreases the number of C–H bonds.

Pi bond

A covalent bond formed by side-on overlapping between atomic orbitals.

pKa

The tendency of a compound to lose a proton (a quantitative measure of acidity).

Polar bond

A covalent bond formed by the unequal sharing of electrons.

Polar reaction

Reaction of a nucleophile with an electrophile.

Polarisability

Indicates the ease with which the electron cloud of an atom can be distorted.

Polymer

A molecule in which one or more subunits (called monomers) is repeated many times.

Protic solvent

A solvent that is a hydrogen bond donor.

Racemate

A mixture of equal amounts of a pair of enantiomers.

Radical (or free radical)

An atom or molecule containing an unpaired electron.

Radical anion

A species with a negative charge and an unpaired electron.

Radical cation

A species with a positive charge and an unpaired electron.

Rate-determining step

The step in a reaction that has the transition state with the highest energy.

Reducing agent

In a reaction, the reactant that causes the reduction (and becomes oxidised).

Reduction reaction

A reaction that decreases the number of covalent bonds between carbon and a more electronegative atom (e.g. O, N, a halogen atom), and increases the number of C — H bonds.

Regioselective

A reaction that favours bond formation at a particular atom over other possible atoms.

Resonance

When a molecule with delocalised electrons is more accurately described by two or more structures.

Resonance energy

The extra stability gained by electron delocalisation due to resonance.

Resonance hybrid

The actual structure of a compound with delocalised electrons; it is the average of two or more resonance forms.

Resonance structure

One of a set of structures that differ only in the distribution of electrons in covalent bonds and lone pairs.

Saturated

A compound with no double or triple bond.

Sigma bond

A covalent bond formed by head-on overlapping between atomic orbitals.

Skeletal structure

Represents the C — C bonds as lines and does not show the C — H bonds.

Solvation

The interaction between a solvent and anotherc molecule or ion.

Solvolysis

Reaction of the reactant with the solvent.

Stereochemistry

The study of the spatial relationship of atoms within a molecule.

Stereoisomers

Isomers that differ in the way their atoms are arranged in space (enantiomers and diastereoisomers are stereoisomers).

Stereoselectivity

A reaction that leads to preferential formation of one stereoisomer (enantiomer, diastereoisomer, or alkene isomer) over another.

Stereospecific

A reaction in which the stereochemistry of the reactant controls the outcome of the reaction (e.g. an E2 elimination is stereospecific).

Steric effect

Any effect on a molecule or reaction due to the size of atoms.

Steric hindrance

Describes bulky groups at the site of a reaction that make it difficult for the reactants to approach each other.

Substituent

An atom or group, other than hydrogen, in a molecule.

Substitution reaction

A reaction in which an atom or group is replaced by another atom or group.

Syn

When two atoms or groups point in the same direction; when the atoms or groups lie in the same plane they are synperiplanar.

Syn addition

An addition reaction in which two substituents add to the same side of the molecule.

Syn elimination

An elimination reaction in which two substituents are eliminated from the same side of the molecule.

Tautomers

Rapidly equilibrating isomers that differ in the location of their bonding electrons (e.g. keto and enol forms of an aldehyde).

Thermodynamic control
A reaction in which the product ratio is determined by the relative stability of the products.

Torsional strain

On bond rotation, the strain caused by repulsion of electrons when different groups pass one another.

Transition state

The highest energy structure along the reaction coordinate between reactants and products for every step of a reaction mechanism.

Unimolecular

A reaction whose rate depends on the concentration of one reactant (e.g. an SN1 reaction)

Unsaturated

A compound with one or more double or triple bonds.

Valence electron

An electron in the outermost shell of an atom.

Ylide
A compound with both a negative charge and a positive charge on adjacent atoms.

Zwitterion
A compound with both a negative charge and a positive charge that are on nonadjacent atoms.

TABLE 2.2 PAULING ELECTRONEGATIVITY VALUES

H						
2.1						
Li	Be	B	C	N	O	F
1.0	1.6	2.0	2.5	3.0	3.5	4.0
Na	Mg	Al	Si	P	S	Cl
0.9	1.2	1.5	1.8	2.1	2.5	3.0
K						Br
0.8						2.8
						I
						2.5

5.6 Arrows

reaction (one step)

or reaction (several steps)

$\rightleftharpoons$ equilibrium

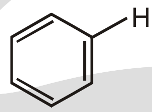
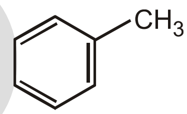
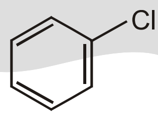
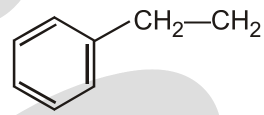
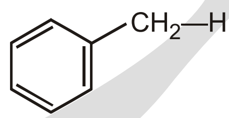
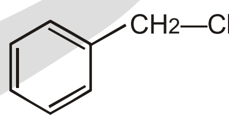
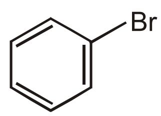
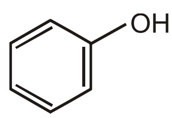
$\rightleftharpoons$	equilibrium (right-hand product favoured)
$\rightleftarrows$	transformation in either direction (but not equilibrium)
$\longleftrightarrow$	resonance
$\curvearrowright$	curly arrow-movement of two electrons
$\curvearrowright$	curly arrow-movement of one electron

Information on arrows

A a B	reaction with a converts A into B
A a/b B	reaction with a in the represence of b converts A into B
A a/solvent B	reaction with a in suitable solvent converts A into B
A a t ,h,hr B	reaction with a at tºC, for h hours converts A into B
A a b B or A (i) a (ii) b B	reaction with a first, then with b converts A into B
$A \rightleftarrows B$	reagent a achieves conversion A B, reagent b achieves conversion B A,

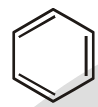
TABLE 2.4 SOLUBILITIES OF ALCOHOLS IN WATER

Formula	Name	Solubility in water (g/100 g)
CH_3OH	Methanol	Infinitely soluble
CH_3CH_2OH	Ethanol	Infinitely soluble
$CH_3(CH_2)_2OH$	Propanol	Infinitely soluble
$CH_3(CH_2)_3OH$	Butanol	9
$CH_3(CH_2)_4OH$	Pentanol	2.7
$CH_3(CH_2)_5OH$	Hexanol	0.6
$CH_3(CH_2)_6OH$	Heptanol	0.18
$CH_3(CH_2)_7OH$	Octanol	0.054
$CH_3(CH_2)_9OH$	Decanol	Infinitely soluble

Bond	D (kJ/mol)	Bond	D (kJ/mol)	Bond	D (kJ/mol)
H — H	436	$(CH_3)_3C$ — I	227	$(CH_3)_2CH$ — CH_3	369
H — F	570	H_2C CH — H	464	$(CH_3)_3C$ — CH_3	363
H — Cl	431	H_2C CH — Cl	396	H_2C CH — CH_3	426
H — Br	366	H_2C $CHCH_2$ — H	369	H_2C $CHCH_2$ — CH_3	318
H — I	298	H_2C $CHCH_2$ — Cl	298	H_2C CH_2	728
Cl — Cl	242		472		427
Br — Br	194		400		325
I — I	152		375	$CH_3\overset{\overset{O}{\Vert}}{C}$ — H	374
CH_3 — H	439		300	HO — H	497
CH_3 — Cl	350		336	HO — OH	211
CH_3 — Br	294		464	CH_3O — H	440
CH_3 — I	239	HC C — H	558	CH_3S — H	336
CH_3 — OH	385	CH_3 — CH_3	377	C_2H_5O — H	441
CH_3 — NH_2	386	C_2H_5 — CH_3	370	C_2H_5O — H	352
C_2H_5 — H	421			$CH_3\overset{\overset{O}{\Vert}}{C}$ — CH_3	355
C_2H_5 — Cl	352			NH_2 — H	450
C_2H_5 — Br	293			H — CN	528
C_2H_5 — I	233				
C_2H_5 — OH	391				
$(CH_3)_2CH$ — H	410				
$(CH_3)_2CH$ — Cl	354				
$(CH_3)_2CH$ — Br	299				
$(CH_3)_3C$ — H	400				
$(CH_3)_3C$ — Cl	352				
$(CH_3)_3C$ — Br	293				

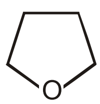
In other words, exothermic reactions are favored by products with strong bonds and by reactants with weak, easily broken bonds.

Table Dipole Moments of Some Compounds

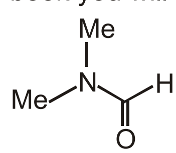
Compound	Dipole moment(D)	Compound	Dipole moment(D)
NaCl	9.00	NH_3	1.47
CH_2O	2.33	CH_3NH_3	1.31
CH_3Cl	1.87	CO_2	0
H_2O	1.85	CH_4	0
CH_3OH	1.70	CH_3CH_3	0
CH_3CO_2H	1.70	 **Benzne**	0
CH_3SH	1.52		

COMPOUNDS NAMED AS ACRONYMS

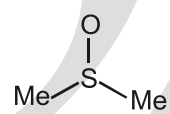
Some compounds are referred to by acronyms, shortened versions of either their systematic or their trivial name. We just saw TNT as an abbreviation for TriNitroToluene but the commoner use for acronyms is to define solvents and reagents in use all the time. Later in the book you will meet these solvents.



THF
(TetraHydroFuran)

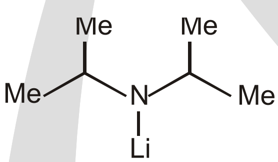


DMF
(DiMethylFormamide)

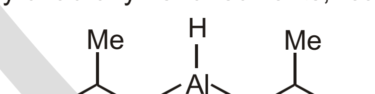


DMSO
(DiMethylSulfOxide)

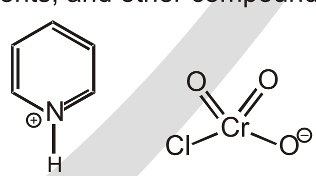
The following reagents are usually referred to by acronym and their functions will be introduced in other chapters so you do not need to learn them now. You may notice that some acronyms refer to trivial and some to systematic names. There is a glossary of acronyms for solvents, reagents, and other compounds on p. 000.



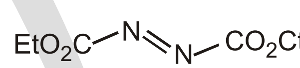
LDA
Lithium**D**i-isopropy **A**mide



DIBAL
Di-**I**so**B**utyl**AL**uminiumhydride



PCC
Pyridinium **C**hloro**C**hromate



DEAD
Di**E**thyl **A**zo-**D**icarboxylate

Summary

Asymmetric carbon

The carbon atom which is attached with four different groups of atoms is called asymmetric carbon.

Asymmetric molecule

If all the four aubstituents attached to carbon are different, the resulting molecule will lack symmetry. Such a molecule is called asymmetric molecule. Asymmetry of molecule is responsible for opitcal activity in such organic compounds.

Achiral molecule

A molecule that is superposable on its mirror image. Achiral molecules lack handedness and are incapable of existing as a pair of enantiomers.

Axial bond

The six bonds of a cyclohexane ring (below) that are perpendicular to the general plane of the ring, and that alternate up and down around the ring.

Boat conformation

A conformation of cyclohexane that resembles a boat and that has eclipsed bonds along its two sides.

Chair conformation

The all-staggered conformation of cyclohexane that has no angle strain or torsional strain and is, therefore, the lowest energy conformation.

Chiral molecule

A molecule that is not superposable on its mirror image. Chiral molecules have handedness and are capableof existing as a pair of enantiomers.

Chirality

The property of having handeness.

Configuration

The particular arrangement of atoms (or groups) in space that is characteristic of a given stereoisomer.

Conformation

A particular temporary orientation of a molecule that results from rotations about its single bonds.

Conformational anylysis

An analysis of the energy changes that a molecule undergoes as its groups undergo rotation (sometimes only partial) about the single bonds that join them.

Conformer

A particular staggered conformation of a molecule.

Connectivity

The sequence, or order, in which the atoms of a molecule are attached to each other.

Dextrorotatory

Those substances which rotate the plane of polarisation of light towards right are called dextrorotatory. Currently, dextro and laevo rotations are represented by algebraic signs of (+) and (–) respectively.

Leavorotatory

A compound that rotates plane polarized light in a counterclockwise direction.

Eclipsed conformation

A temporary orientation of groups around two atoms joined by a single bond such that the groups directly oppose each other.

Enantiomers

Stereoisomers that are mirror images of each other. enantiomers rotate the plane of polarised light to the same extent but in opposite direction.

Equatorial bond

The six bonds of a cyclohexane ring that lie generally around the "equator" of the molecule.

Meso compound

An optically inactive compound whose molecules are achiral even though they contain tetrahedral atoms with four different attached groups. A meso-compound is optically inactive due to internal compensation.

Optically active substances

Those substances which rotate the plane of polarisation of plane-polarised light when it is passed through their solutions are called optically active substances. This phenomenon is called optical activity.

Plane of symmetry

An imaginary plane that bisects a molecule in a way such that the two halves of the molecule are mirror images of each other. Any molecule with a plane of symmetry will be achiral.

Plane-polarized light

Ordinary light in which the oscillations of the electrical field occur only in one plane. It is obtained by passing a monochromatic light (light of single wavelength) through a Nicol prism.

Polarimeter

A device used for measuring optical activity.

(R–S) System

A method for designating the configuration of tetrahedral stereogenic centres.

Racemic form (racemate or racemic mixture)

An equimolar mixture of enantiomers. A racemic mixture is optically inactive due to external compensation.

Racemisation

The process of conversion of an enantiomer into racemic mixture is known as an racemisation.

Retention

If in an optically active molecule that relative configuration of the atoms groups around a chiral centre remains the same before and after the reaction, the reaction is said to proceed with retention of configuration.

Relative configuration

The relationship between the configuration of two chiral molecules. Molecules are said to have the same relative configuration when similar or identical groups in each occupy the same position in space. The configurations of molecules can be related to each other through reactions of known stereochemistry, for example, through reactions that cause no bonds to a stereogenic center to be broken.

Resolution

The process by which the enantiomers of a recemic form are separated.

Ring flip

The change in a cyclohexane ring (resulting from partial bond rotations) that converts one ring conformation to another. A chair-chair ring flip converts any equatorial substitutent to an axial substituent and vice versa.

Ring strain

The increased potential energy of the cyclic form of a molcule (usually measured by heats of combustion) when compared to its acyclic form.

Specific rotation

Specific rotation is defined as the number of degrees of rotation observed when the concentration of optically active substance is 1 g cm ^{3} and length of polarimeter tube is 1 decimetre (dm) for D-line of sodium vapour lamp at 25ºC.

Stereogenic center

An atom bearing group of such nature that an interchange of any two groups will produce a stereoisomer.

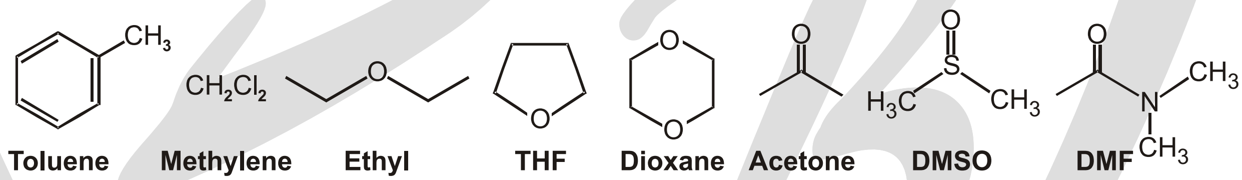
Steric hindrance

An effect on relative reaction rates caused when the spatial arrangement of atoms or groups at or near the reacting site hinders or retards a reaction.

Torsional strain

The strain associated with an eclipsed conformation of a molecule; it is caused by repulsions between the aligned electron pairs of the eclipsed bonds.

(a) Aprotic Solvents



	Toluene	Methylene chloride	Ethyl ether	THF	Dioxane	Acetone	DMSO	DMF
bp (ºC) :	111	40	35	66	101	56	189	153

MATCHING LIST (2)

1. **List-I** / **List-II**

	List-I		List-II
(P)	Homologoues	(1)	May be constitutional isomers
(Q)	Duma's method	(2)	For estimation, Nitrogen collected as NH_3
(R)	Kjeldal's method	(3)	For estimation, nitrogen collected as N_2
(S)	Dimethyl ether & n-propyl ether	(4)	Differ by 14 amu in terms of molecular mass

Codes:

	P	Q	R	S
(A)	4	2	3	1
(*B)	4	3	2	1
(C)	1	3	2	4
(D)	1	2	3	4

Sol. (P) Homologous may be constitutional isomers

(Q) Duma's method nitrogen collected as N_2

(R) Kjeldal method Nitrogen collected as NH_3

(S) $CH_3 — O — CH_3$ and $CH_3CH_2CH_2 — OH$ differ by 14 amu.

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